

Fundamental algebra and analysis

Huayi Chen

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Chapter 1

Basic logic

The purpose of this chapter is to introduce the mathematical logic language used throughout the book, including the fundamentals of propositional logic and predicate logic. By studying this chapter, students will master the basic methods of mathematical reasoning, laying a solid foundation for learning algebra and analysis.

1.1 Mathematical statements

1.1.1 Definition. We call *statement* a declarative sentence without free variables in a given mathematical theory, whose truth value (that is, *true* or *false*) can in principle be determined. In a consistent mathematical theory, any statement is either true or false, but cannot be true and false at the same time.

1.1.2 Example. “2 is an even number”, “ $1 > 2$ ” are both statements. The former one is true, the latter is false.

1.1.3 Example. “ $1 + 2$ ” is a computational formula, it is *not* a statement. The inequality “ $2x + 1 > 0$ ” contains a free variable x , so judging its truth value is meaningless. Therefore, it is *not* a statement. In this book, we use the symbol $:=$ to interpret the notation on the left-hand side of the symbol as the expression on the right-hand side. For example, “ $a := 3 + 5$ ” means that we denote by a the value of $3 + 5$. This expression is *not* a statement.

1.1.4 Definition. In a mathematical theory, statements admitted as true without justification are called the *axioms*. All statements that can be rigorously deduced from axioms are true. A statement that is confirmed to be true through rigorous mathematical proof is called a *theorem*. A statement that can be deduced from a theorem via straightforward reasoning is usually referred to as a *corollary* of that *theorem*.

In mathematical literature, a statement is often called a *proposition*. In this book, the term “*proposition*” is rather used to label theorems that are relatively simple or not repeatedly applied in the book.

1.1.5 Remark. Some mathematical conjectures have neither been proved nor disproved. However, these conjectures have similar forms with the statements the truth values of which have been determined. In principle the truth values of these conjectures could be determined in the future. They should be classified as statements. Furthermore, there exist declarative sentences that are neither provable nor disprovable within a certain mathematical theory. However, in an enriched axiomatic mathematical theory, their truth values can be in principle determined. Such declarative sentences should also be recognised as statements in the enriched mathematical theory.

1.2 Negation

Starting from given statements, compound statements can be constructed through the syntax of construction. This section introduces the negation of statements.

1.2.1 Definition. Let P be a statement. Then the sentence “*not P*” is also a statement, called the *negation* of P . It has the opposite true value of that of P . Sometimes we denote the statement “*not P*” as $\neg P$.

1.2.2 Notation. When a statement is expressed by a linking verb of judgement, its negation can be expressed by negating the linking verb. For example, the negation of the statement “*2 is an even number*” can be expressed as “*2 is not an even number*”.

If a statement is expressed as a relation linked by a relation symbol, its negation could be expressed by overlaying the relation symbol with a diagonal line from the upper right to the lower left. For example, the negation of “ $1 > 2$ ” can be expressed as “ $1 \not> 2$ ”.

1.2.3 Remark. The double negation of a statement P has the same truth value as that of the statement P . This point can be seen by the table of truth values as follows, where T stands for “true” and F stands for “false”.

P	$\neg P$	$\neg\neg P$
T	F	T
F	T	F

1.3 Conjunction and disjunction

1.3.1 Definition. Let P and Q be statements. Then “ P and Q ” is a statement, called the *conjunction* of P and Q , often denoted as $P \wedge Q$. When both P and Q are true, the statement $P \wedge Q$ is true; otherwise it is false.

Similarly, the sentence “ P or Q ” is a statement, called the *disjunction* of P and Q , often denoted as $P \vee Q$. When both P and Q are false, the statement $P \vee Q$ is false; otherwise it is true.

1.3.2 Remark. We describe the truth values of conjunction and disjunction in the following table.

P	Q	$P \wedge Q$	$P \vee Q$	$Q \wedge P$	$Q \vee P$
T	T	T	T	T	T
T	F	F	T	F	T
F	T	F	T	F	T
F	F	F	F	F	F

We observe from the table that $P \wedge Q$ and $Q \wedge P$ have the same truth value, and $P \vee Q$ and $Q \vee P$ have the same truth value.

1.3.3 Proposition. Let P and Q be statements. The statements $\neg(P \wedge Q)$ and $(\neg P) \vee (\neg Q)$ have the same truth value, and $\neg(P \vee Q)$ and $(\neg P) \wedge (\neg Q)$ have the same truth value.

Proof. This can be observed from the following tables.

P	Q	$P \wedge Q$	$\neg(P \wedge Q)$	$\neg P$	$\neg Q$	$(\neg P) \vee (\neg Q)$
T	T	T	F	F	F	F
T	F	F	T	F	T	T
F	T	F	T	T	F	T
F	F	F	T	T	T	T

P	Q	$P \vee Q$	$\neg(P \vee Q)$	$\neg P$	$\neg Q$	$(\neg P) \wedge (\neg Q)$
T	T	T	F	F	F	F
T	F	T	F	F	T	F
F	T	T	F	T	F	F
F	F	F	T	T	T	T

□

1.4 Conditional statement

1.4.1 Definition. Let P and Q be statements. The sentence “if P , then Q ” is a statement, often denoted as $P \Rightarrow Q$. It has the same truth value as that of $(\neg P) \vee Q$. A statement of this form is called a *conditional statement*. We describe its true value in the following table.

P	Q	$\neg P$	$P \Rightarrow Q$
T	T	F	T
T	F	F	F
F	T	T	T
F	F	T	T

1.4.2 Remark. We observe from the table that, only in the case where P is true, and Q is false, the statement $P \Rightarrow Q$ is false, otherwise it is true. Therefore, if one can prove Q under the assumption that P is true, then the statement $P \Rightarrow Q$ is true. However, the statement $P \Rightarrow Q$ itself does not signify that we are considering a proof of the statement Q from P .

We also observe from the table that, in the case where both P and $P \Rightarrow Q$ are true, the statement Q must be true. This type of reasoning often appears in a mathematical proof.

1.4.3 Proposition. Let P , Q and R be statements. If both $P \Rightarrow Q$ and $Q \Rightarrow R$ are true, then $P \Rightarrow R$ is true.

Proof. It suffices to treat the case where P is true, since otherwise $P \Rightarrow R$ is automatically true. In the case where P is true, since $P \Rightarrow Q$ is true, we deduce that Q is true. Furthermore, since $Q \Rightarrow R$ is true, we deduce that R is true. Therefore, $P \Rightarrow R$ is true. $\square$

1.4.4 Proposition. Let P and Q be statements. The statements $P \Rightarrow Q$ and $(\neg Q) \Rightarrow (\neg P)$ have the same truth value.

Proof. We could conclude from the following table.

P	Q	$\neg P$	$\neg Q$	$(\neg Q) \Rightarrow (\neg P)$
T	T	F	F	T
T	F	F	T	F
F	T	T	F	T
F	F	T	T	T

$\square$

1.4.5 Definition. Let P and Q be two statements. The statement $(\neg Q) \Rightarrow (\neg P)$ is called the *contrapositive* of the statement $P \Rightarrow Q$. If one proves the contrapositive statement $(\neg Q) \Rightarrow (\neg P)$, then, by Proposition 1.4.4, we obtain that the statement $P \Rightarrow Q$ is also true. This method is called *proof by contraposition*.

1.4.6 Example. Let n be an integer. We prove by contraposition that, if n^2 is an even number, then n is an even number. Note that the contrapositive of this statement says that, if n is not an even number, then n^2 is not an even number. Since n is an integer, if n is not an even number, it must be an odd number, namely it is of the form $2k + 1$, with k being an integer. Therefore,

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1$$

is an odd number. Thus, we have proved the contrapositive statement. Hence, the initial statement is also true.

1.5 Biconditional statement

1.5.1 Definition. Let P and Q be statements. Then “ P if and only if Q ” is also a statement, often denoted as $P \Leftrightarrow Q$. When P and Q have the same truth value, the statement $P \Leftrightarrow Q$ is true; otherwise, $P \Leftrightarrow Q$ is false. By definition, the statements $P \Leftrightarrow Q$ and $Q \Leftrightarrow P$ have the same true value.

1.5.2 Proposition. Let P and Q be statements. Then $(P \Rightarrow Q) \wedge (Q \Rightarrow P)$ and $P \Leftrightarrow Q$ have the same truth value.

Proof. We could conclude from the following table.

P	Q	$P \Rightarrow Q$	$Q \Rightarrow P$	$P \Leftrightarrow Q$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

□

1.5.3 Remark. The above proposition shows that, given two statements P and Q , to prove that they have the same truth value, it suffices to prove conditional statements of both directions $P \Rightarrow Q$ and $Q \Rightarrow P$. The statement $Q \Rightarrow P$ is called the *converse* of $P \Rightarrow Q$. The contraposition of $Q \Rightarrow P$, namely $(\neg P) \Rightarrow (\neg Q)$, is called the *inverse* of $P \Rightarrow Q$. Proposition 1.4.4 shows that the converse and

the inverse of a conditional statement have the same truth value. Therefore, to justify the biconditional statement $P \Leftrightarrow Q$, it also suffices to prove the conditional statement $P \Rightarrow Q$ and its inverse $(\neg P) \Rightarrow (\neg Q)$.

1.5.4 Example. Let n be an integer. Then n^2 is an even number if and only if n is an even number. In fact, we have proved in Example 1.4.6 that, if n^2 is an even number, then n is an even number. It remains to check that, if n is an even number, then n^2 is also an even number. Assume that n is of the form $2k$, where k is an integer. Then $n^2 = 4k^2$ is divisible by 2. Hence n^2 is an even number.

1.6 Proof by contradiction

1.6.1 Definition. In a consistent mathematical theory, a statement cannot be true and false at the same time. Let P be a statement. If we assume that $\neg P$ is true and deduce that a certain statement is both true and false, then we say that a *contradiction* happens and the assumption $\neg P$ is false. Thus the statement P is true. Such a reasoning is called *proof by contradiction*.

1.6.2 Example. We prove by contradiction that the equation $x^2 = 2$ does not have any rational solution. Suppose by contradiction that p/q is a solution of the equation $x^2 = 2$, where p is an integer and q is a positive integer, which do not have common prime divisor. By definition, one has $p^2 = 2q^2$. Hence p^2 is an even number. By Example 1.5.4, we obtain that p is an even number. Hence, there exists $p_1 \in \mathbb{Z}$ such that $p = 2p_1$. Hence, we deduce from $p^2 = 2q^2$ that $2q_1^2 = q^2$. This shows that q^2 is an even number and hence q is an even number. Thus p and q have 2 as a common prime divisor, which leads to a contradiction.

Exercises

1. Let P and Q be statements. Use truth tables to determine the truth values of the following statements according to the truth values of P and Q :

$$P \wedge \neg P, P \vee \neg P, (P \vee Q) \Rightarrow (P \wedge Q), (P \Rightarrow Q) \Rightarrow (Q \Rightarrow P)$$

2. Let P and Q be statements.

- (1) Show that $P \Rightarrow (Q \wedge \neg Q)$ has the same truth value as $\neg P$.
- (2) Show that $(P \wedge \neg Q) \Rightarrow Q$ has the same truth value as $P \Rightarrow Q$.

3. Consider the following statements:

$P :=$ “Little Bear is happy”,
 $Q :=$ “Little Bear has done her math homework”,
 $R :=$ “Little Rabbit is happy”.

Express the following statements using P , Q , and R , along with logical connectives:

- (1) If Little Bear is happy and has done her math homework, then Little Rabbit is happy.
- (2) If Little Bear has done her math homework, then she is happy.
- (3) Little Bear is happy only if she has done her math homework.

4. Does the following reasoning hold? Justify your answer.

- It is known that Little Bear is both smart and lazy, or Little Bear is not smart.
- It is also known that Little Bear is smart.
- Therefore, Little Bear is lazy.

5. Does the following reasoning hold? Justify your answer.

- It is known that at least one of the lion or the tiger is guilty.
- It is also known that either the lion is lying or the tiger is innocent.
- Therefore, the lion is either lying or guilty.

6. An explorer arrives at a cave with three closed doors, numbered 1, 2, and 3. Exactly one door hides treasure, while the other two conceal deadly traps.

- Door 1 states: “*The treasure is not here*”;
- Door 2 states: “*The treasure is not here*”;
- Door 3 states: “*The treasure is behind Door 2*”.

Only one of these statements is true. Which door should the explorer open to find the treasure?

7. The Kingdom of Truth sent an envoy to the capital of the Kingdom of Lies. Upon entering the border, the envoy encountered a fork with three paths: dirt, stone, and concrete. Each path had a signpost:

- The concrete path's sign: "*This path leads to the capital, and if the dirt path leads to the capital, then the stone path also does.*"
 - The stone path's sign: "*Neither the concrete nor the dirt path leads to the capital.*"
 - The dirt path's sign: "*The concrete path leads to the capital, but the stone path does not.*" All signposts lie. Which path should the envoy take?
8. Let a and b be real numbers. Prove that, if $a \neq -1$ and $b \neq -1$, then $ab + a + b \neq -1$.
9. Let a , b , and c be positive real numbers such that $abc > 1$ and

$$a + b + c < \frac{1}{a} + \frac{1}{b} + \frac{1}{c}.$$

Prove the following:

- (1) None of a , b , or c equals 1.
 - (2) At least one of a , b , or c is greater than 1.
 - (3) At least one of a , b , or c is less than 1.
10. Let $a \neq 0$ and b be real numbers. For real numbers x and y , prove that if $x \neq y$, then $ax + b \neq ay + b$.
11. Let $n \geq 2$ be an integer. Prove that if n is composite, then there exists a prime number p dividing n such that $p \leq \sqrt{n}$.
12. Let n be an integer. Prove that either 4 divides n^2 or 4 divides $n^2 - 1$.
13. Let n be an integer. Prove that 12 divides $n^2(n^2 - 1)$.
14. Prove that any integer divisible by 4 can be written as the difference of two perfect squares.
15. Let x and y be non-zero integers. Prove that $x^2 - y^2 \neq 1$.
16. A plane has 300 seats and is fully booked. The first passenger ignores their assigned seat and chooses randomly. Subsequent passengers take their assigned seat if available; otherwise, they choose randomly. What is the probability that the last passenger sits in their assigned seat?

17. Little Bear, Little Goat, and Little Rabbit are all wearing hats. A parrot prepared four red feathers and four blue feathers to decorate their hats. The parrot selected two feathers for each hat-wearing animal to place on their hats. Each animal cannot see the feathers on their own hat but can see the feathers on the other animals' hats. Here is their conversation:

- Little Bear: *I don't know what color the feathers on my hat are, but I know the other animals also don't know what color the feathers on their hats are.*"
- Little Goat: *Haha, now even without looking at Little Bear's hat, I know what color the feathers on my hat are.*"
- Little Rabbit: *Now I know what color the feathers on my hat are.*"
- Little Bear: *Hmm, now I also know what color the feathers on my hat are..*"

Question: What color are the feathers on Little Goat's hat?

18. The Sphinx tells the truth on one fixed weekday and lies on the other six. Cleopatra visits the Sphinx for three consecutive days:

- Day 1: The Sphinx declared, *"I lie on Monday and Tuesday."*
- Day 2: The Sphinx declared, *"Today is either Thursday, or Saturday, or Sunday."*
- Day 3: The Sphinx declared, *"I lie on Wednesday and Friday."*

On which day does the Sphinx tell the truth? On which days of the week did Cleopatra visit the Sphinx?

Chapter 2

Sets

This book presents the fundamentals of algebra and analysis based on set theory. Set theory, proposed initially by Cantor, is the cornerstone of modern mathematics. Before the emergence of set theory, the progress of mathematics was often built upon intuition and visual imagery. Moreover, due to the limitations of available tools, the scope of mathematical research was relatively narrow—for instance, analysis was often confined to studying functions with analytic expressions. Cantor’s set theory introduced a new language and new tools to mathematics, greatly advancing the development of modern mathematics.

Naive set theory regards a set as a collection of distinct objects that are clearly defined. However, treating any such collection as a set without restriction leads to paradoxes. For example, Russell’s paradox considers the collection of all sets that do not contain themselves. If this collection is treated as a set, one arrives at a contradiction regardless of whether the set contains itself or not. Determining which types of collections should be considered as sets is a subtle issue that lacks universal agreement. Axiomatic set theory treats set theory itself as a structure governed by a system of axioms. These axioms ensure the existence of constructions needed in mathematics while avoiding known paradoxes, thus providing a sound foundation for mathematics.

The goal of this chapter is to introduce some fundamental concepts of set theory and basic structures to prepare for the chapters that follow. Since our aim is not to give a systematic introduction to mathematical logic, we will explain ideas as much as possible using natural language rather than formal language.

2.1 Roster notation

2.1.1 Definition. In naive set theory, a set refers to a certain collection of *distinct* objects. The objects in a set are called *elements* of it. Two sets A and B are said

to be *equal* if they have the same elements. We denote by $A = B$ the statement “ A and B are equal”.

If A is a set and a is an object, we denote by $a \in A$ the statement

“ a is an element of A ”;

we denote by $a \notin A$ the statement

“ a is not an element of A ”.

If a is an element of A , we also say that a *belongs to* A .

2.1.2 Notation. The *roster method* of representing a set refers to a notation where all elements of the set are explicitly enumerated in a single row within a pair of curly braces. For example the set consisting of the elements 1, 2 and 3 can be denoted as $\{1, 2, 3\}$. The represented set is independent of the enumeration order or element repetition. For example, one has

$$\{1, 2, 3\} = \{3, 2, 1\} = \{1, 1, 2, 3\}.$$

When representing a set using the roster method, an ellipsis (...) may be used to indicate elements with an obvious pattern. For example, the set of positive integers less than 100 can be written as:

$$\{1, 2, \dots, 100\}.$$

Similarly, if n is a natural number the set of natural numbers not exceeding n can be written as

$$\{0, 1, \dots, n\}.$$

When no ambiguity arises, an ellipsis may also be used to omit infinitely many elements with a clear pattern. For example, the set of all even integers can be expressed in roster notation as:

$$\{\dots, -6, -4, -2, 0, 2, 4, 6, \dots\}.$$

When representing sets using the roster method, if the elements are indexed, an ellipsis can be used to summarise elements corresponding to indices with obvious patterns. For example, if n is a positive integer and for each $i \in \{1, \dots, n\}$, a mathematical object x_i is given, then the set of these objects can be written as:

$$\{x_1, \dots, x_n\}.$$

Similarly, if, for every natural number n , a mathematical object y_n , then the set of these objects can be written as:

$$\{y_0, y_1, y_2, \dots\}.$$

More generally, if I is a set and if, for any $i \in I$, a mathematical object x_i is given, then the set of these objects can be written as

$$\{x_i \mid i \in I\},$$

where I is called the *index set* of this roster writing. For example,

$$\{2n \mid n \in \mathbb{Z}\}$$

denotes the set of all even numbers,

$$\{2n + 1 \mid n \in \mathbb{Z}\}$$

denotes the set of all odd numbers.

2.1.3 Definition. Let A and B be sets. We denote by $A \times B$ the following set of ordered pairs

$$\{(x, y) \mid x \in A, y \in B\},$$

and call it the *Cartesian product* of sets A and B .

More generally, if n is a positive integer and $A_1, \dots, A_n$ be sets, we denote by

$$A_1 \times \dots \times A_n$$

the set of all n -tuples $(x_1, \dots, x_n)$, where $x_1 \in A_1, \dots, x_n \in A_n$.

2.2 Subsets and powersets

2.2.1 Definition. Let A and B be sets. If any element of A is an element of B , we say that A is a *subset* of B . We denote by $A \subseteq B$ or by $B \supseteq A$ the statement

“ A is a subset of B ”.

If A is a subset of B and A is *not* equal to B , we say that A is a *proper subset* of B , denoted by $A \subset B$ or by $B \supset A$.

If A is a subset (resp. proper subset) of B , we also say that A is *contained* (resp. *strictly contained*) in B , or that B *contains* (resp. *contains strictly*) A .

2.2.2 Remark. By definition, any set A is a subset of itself. Moreover, for any sets A and B , if $A \subseteq B$ and $B \subseteq A$, then the sets A and B have the same elements, namely $A = B$.

2.2.3 Definition. We denote by $\emptyset$ the set that does not contain any element, and we call it the *empty set*. By definition, the empty set is a subset of any set.

2.2.4 Proposition. Let A , B and C be sets. If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.

Proof. Let x be an element of A . Since $A \subseteq B$, x is an element of B . Since $B \subseteq C$, x is an element of C . Therefore, one has $A \subseteq C$. $\square$

2.2.5 Definition. Let X be a set. We denote by $\mathcal{P}(X)$ the set of all subsets of X , called the *power set* of X . Note that $\{\emptyset, X\} \subseteq \mathcal{P}(X)$.

2.2.6 Example. The only subset of the empty set is itself, namely $\mathcal{P}(\emptyset) = \{\emptyset\}$. Moreover, $\mathcal{P}(\{\emptyset\}) = \{\emptyset, \{\emptyset\}\}$.

2.3 Set-builder notation

2.3.1 Definition. Let A be a set. If for any $x \in A$, we fix a statement $P(x)$, then we say that $P(\cdot)$ is a *condition* on A . For example, “ x is an odd number” is a condition on $\mathbb{Z}$. If $P(x)$ is true, then we say that x *satisfies* the condition $P(\cdot)$.

Let A be a set and $P(\cdot)$ be a condition on A . We denote by

$$\{x \in A \mid P(x)\}$$

the set of elements $x \in A$ such that $P(x)$ is true. This is a subset of A . Such representation of a new set by a condition on a given set is called *set-builder notation*.

2.3.2 Notation. Sometimes we combine the roster notation and the set-builder notation to describe a set. Let I be a set. For any $i \in I$, let x_i be a mathematical object. Let $P(\cdot)$ be a condition on I , and I_P be the set $\{i \in I \mid P(i)\}$. Then we use the expression

$$\{x_i \mid i \in I, P(i)\}$$

to denote the set

$$\{x_i \mid i \in I_P\}.$$

2.3.3 Example. In the set-builder notation, it is important to fix an environmental set on which we consider a condition. If, for any set A , we consider a statement $P(A)$ (having a determined truth value), then the collection of all sets A such that $P(A)$ is true does not necessarily form a set.

For example, Russell’s paradox considers the collection of all sets A such that $A \notin A$. If we consider this collection as a set, whether it belongs to itself or not will lead to a contradiction.

Moreover, the collection of all sets should not be considered as a set. In fact, if the collection $\mathcal{S}$ of all sets is a set, then by the set-builder notation, the construction in Russell’s paradox should also be a set, which would be expressed as

$$\{A \in \mathcal{S} \mid A \notin A\}.$$

This leads to a contradiction.

2.4 Set difference

2.4.1 Definition. Let A and B be sets. Then the set

$$\{x \in B \mid x \notin A\}$$

is called the *set difference* of B and A . In the case where A is a subset of B , we also call $B \setminus A$ as the *complement* of A in B .

2.4.2 Example. Let A be a set, $P(\cdot)$ be a condition on A . Then the following equality holds:

$$\{x \in A \mid \neg P(x)\} = A \setminus \{x \in A \mid P(x)\}. \quad (2.1)$$

2.4.3 Proposition. Let A and B be sets. Then $B \setminus A = \emptyset$ if and only if $B \subseteq A$. In particular, in the case where A is a subset of B , the set $B \setminus A$ is empty if and only if $A = B$.

Proof. Suppose that $B \setminus A = \emptyset$. For any element x of B , one has $x \in A$ since otherwise $x \in B \setminus A$, which leads to a contradiction. Hence $B \subseteq A$.

Suppose that $B \setminus A \neq \emptyset$. Let x be an arbitrary element of $B \setminus A$. It is hence an element of B that does not belong to A . Therefore, B is not included in A .

Suppose that $A \subseteq B$. If $B \setminus A = \emptyset$, by the first conclusion we obtain $B \subseteq A$, hence we deduce $A = B$ by the condition $A \subseteq B$. Conversely, if $A = B$, by definition one has $B \setminus A = \emptyset$. $\square$

2.4.4 Proposition. Let A and B be sets. Then

$$B \setminus (B \setminus A) = \{x \in B \mid x \in A\}.$$

When $A \subseteq B$, one has

$$B \setminus (B \setminus A) = A.$$

In particular, $B \setminus \emptyset = B$.

Proof. By definition,

$$B \setminus (B \setminus A) = \{x \in B \mid x \notin B \setminus A\}.$$

Let x be an element of $B \setminus (B \setminus A)$. By definition, $x \in B$ and $x \notin B \setminus A$. If x does not belong to A , then $x \in B \setminus A$, which leads to a contradiction. Hence, we obtain $x \in A$.

Conversely, let x be a common element of A and B . By definition, $x \in B$, and x is not an element of $B \setminus A$ since $x \in A$. Hence x belongs to $B \setminus (B \setminus A)$.

In the case where $A \subseteq B$, by definition one has

$$\{x \in B \mid x \in A\} = A.$$

Hence

$$B \setminus (B \setminus A) = A.$$

Apply this equality to the case where $A = B$, by Proposition 2.4.3 we get

$$B \setminus \emptyset = B \setminus (B \setminus B) = B.$$

□

2.5 Quantifiers

2.5.1 Definition. Let A be a set and $P(\cdot)$ be a condition on A .

We use the expression

$$\forall x \in A, P(x)$$

to denote the statement

$$\{x \in A \mid P(x)\} = A.$$

We often read it as

“for any $x \in A$, x satisfies the condition $P(\cdot)$ ”.

We use the expression

$$\exists x \in A, P(x)$$

to denote the statement

$$\{x \in A \mid P(x)\} \neq \emptyset.$$

We often read it as

“there exists $x \in A$ which satisfies the condition $P(\cdot)$ ”.

2.5.2 Example. Let $P(\cdot)$ be any condition on the empty set $\emptyset$. Note that

$$\{x \in \emptyset \mid P(x)\}$$

is a subset of $\emptyset$. Hence, it is equal to $\emptyset$. Therefore,

$$\forall x \in \emptyset, P(x)$$

is true, and

$$\exists x \in \emptyset, P(x)$$

is false.

2.5.3 Proposition. (1) *The statements*

$$“\exists x \in A, \neg P(x)” \text{ and } “\forall x \in A, P(x)”$$

have opposite truth values.

(2) *The statements*

$$“\forall x \in A, \neg P(x)” \text{ and } “\exists x \in A, P(x)”$$

have opposite truth values.

Proof. (1) Note that

$$\{x \in A \mid \neg P(x)\} = A \setminus \{x \in A \mid P(x)\}.$$

By Proposition 2.4.3, $\{x \in A \mid P(x)\} = A$ if and only if $\{x \in A \mid \neg P(x)\} = \emptyset$. Hence the statements

$$“\exists x \in A, \neg P(x)” \text{ and } “\forall x \in A, P(x)”$$

have opposite truth values.

(2) We apply (1) to the condition $\neg P(\cdot)$ to obtain that

$$“\forall x \in A, \neg P(x)” \text{ and } “\exists x \in A, \neg \neg P(x)”$$

have opposite truth values. Since, for any $x \in A$, $\neg \neg P(x)$ and $P(x)$ have the same truth value, we obtain that “ $\exists x \in A, \neg \neg P(x)$ ” and “ $\exists x \in A, P(x)$ ” have the same truth value. The statement is thus proved. $\square$

2.6 Sufficient and necessary conditions

2.6.1 Definition. Let A be a set, $P(\cdot)$ and $Q(\cdot)$ be conditions on A . If

$$\{x \in A \mid P(x)\} \subseteq \{x \in A \mid Q(x)\},$$

we say that $P(\cdot)$ is a *sufficient condition* of $Q(\cdot)$, and $Q(\cdot)$ is a *necessary condition* of $P(\cdot)$. If

$$\{x \in A \mid P(x)\} = \{x \in A \mid Q(x)\},$$

we say that $P(\cdot)$ is a *necessary and sufficient condition* of $Q(\cdot)$, or that the conditions $P(\cdot)$ and $Q(\cdot)$ are *equivalent*.

2.6.2 Proposition. Let A be a set, $P(\cdot)$ and $Q(\cdot)$ be conditions on A .

(1) $P(\cdot)$ is a sufficient condition of $Q(\cdot)$ if and only if

$$\forall x \in A, P(x) \Rightarrow Q(x).$$

(2) $P(\cdot)$ is a necessary condition of $Q(\cdot)$ if and only if

$$\forall x \in A, Q(x) \Rightarrow P(x).$$

(3) $P(\cdot)$ is a necessary and sufficient condition of $Q(\cdot)$ if and only if

$$\forall x \in A, P(x) \Leftrightarrow Q(x).$$

Proof. (1) By 2.4.3, we obtain that $P(\cdot)$ is a sufficient condition of $Q(\cdot)$ if and only if

$$\{x \in A \mid P(x)\} \setminus \{x \in A \mid Q(x)\} = \emptyset.$$

Moreover,

$$\begin{aligned} \{x \in A \mid P(x)\} \setminus \{x \in A \mid Q(x)\} &= \{x \in A \mid P(x) \wedge (\neg Q(x))\} \\ &= A \setminus \{x \in A \mid (\neg P(x)) \vee Q(x)\} = A \setminus \{x \in A \mid P(x) \Rightarrow Q(x)\}, \end{aligned}$$

where the second equality comes from (2.1) and Proposition 1.3.3. By Proposition ??,

$$\{x \in A \mid P(x)\} \setminus \{x \in A \mid Q(x)\} = \emptyset.$$

if and only if

$$\{x \in A \mid P(x) \Rightarrow Q(x)\} = A,$$

namely

$$\forall x \in A, P(x) \Rightarrow Q(x).$$

(2) follows from (1) by switching $P(\cdot)$ and $Q(\cdot)$.

(3) follows from (1), (2), and Proposition 1.5.2. $\square$

2.7 Union

2.7.1 Definition. Let I be a set. For any $i \in I$, let A_i be a set. We say that $(A_i)_{i \in I}$ is a *family of sets* parametrised by I .

We denote by $\bigcup_{i \in I} A_i$ the set consisting of all elements of all A_i . It is called the *union* of the sets A_i , $i \in I$. By definition, a mathematical object x belongs to $\bigcup_{i \in I} A_i$ if and only if

$$\exists i \in I, x \in A_i.$$

In particular, $\bigcup_{i \in I} A_i$ is empty when $I = \emptyset$.

2.7.2 Notation. Let n be a positive integer, $A_1, \dots, A_n$ be sets. We denote $\bigcup_{i \in \{1, \dots, n\}} A_i$ as

$$A_1 \cup \dots \cup A_n.$$

Note that it does not depend on the order of $A_1, \dots, A_n$.

2.7.3 Proposition. Let I be a set and $(A_i)_{i \in I}$ be a family of sets parametrised by I . Let B be a set. Then $\bigcup_{i \in I} A_i \subseteq B$ if and only if

$$\forall i \in I, A_i \subseteq B.$$

Proof. Let $A := \bigcup_{i \in I} A_i$.

For any $i \in I$ one has $A_i \subseteq A$. If $A \subseteq B$, then by Proposition ?? we deduce that $A_i \subseteq B$.

Conversely, we suppose that, for any $i \in I$, one has $A_i \subseteq B$. Let x be an element of A . By definition, there exists $i \in I$ such that $x \in A_i$. Since $A_i \subseteq B$, one has $x \in B$. Therefore, $A \subseteq B$. $\square$

2.7.4 Corollary. Let B and I be sets. For any $i \in I$, let $P_i(\cdot)$ be a condition on B . Then

$$\{x \in B \mid \exists i \in I, P_i(x)\} = \bigcup_{i \in I} \{x \in B \mid P_i(x)\}.$$

Proof. Let

$$A := \{x \in B \mid \exists i \in I, P_i(x)\}.$$

For any $i \in I$, let

$$A_i := \{x \in B \mid P_i(x)\}.$$

For any $x \in A$, there exists $i \in I$ such that $P_i(x)$ is true. Hence $x \in \bigcup_{i \in I} A_i$.

Conversely, by definition, for any $i \in I$, one has $A_i \subseteq A$. Hence, by Proposition 2.7.3, one has

$$\bigcup_{i \in I} A_i \subseteq A.$$

$\square$

2.7.5 Proposition. Let $(A_i)_{i \in I}$ be a family of sets, and B be a set. Then

$$\left(\bigcup_{i \in I} A_i \right) \setminus B = \bigcup_{i \in I} (A_i \setminus B).$$

Proof. Let $A := \bigcup_{i \in I} A_i$.

For any $i \in I$, one has $A_i \subseteq A$. Hence $A_i \setminus B \subseteq A \setminus B$. By Proposition 2.7.3, we obtain $\bigcup_{i \in I} (A_i \setminus B) \subseteq A \setminus B$.

Conversely, if $x \in A \setminus B$, then $x \in A$ and $x \notin B$. By definition, there exists $i \in I$ such that $x \in A_i$, and hence $x \in A_i \setminus B$. This leads to $A \setminus B \subseteq \bigcup_{i \in I} (A_i \setminus B)$. $\square$

2.8 Intersection

2.8.1 Definition. Let I be a non-empty set and $(A_i)_{i \in I}$ be a family of sets parametrised by I . We denote by $\bigcap_{i \in I} A_i$ the set of all common elements of A_i , $i \in I$. This set is called the *intersection* of A_i , $i \in I$. Note that, if i_0 is an arbitrary element of I , the set-builder notation ensures that

$$\{x \in A_{i_0} \mid \forall i \in I, x \in A_i\}$$

is a set. This set is the intersection of $(A_i)_{i \in I}$.

By definition, an mathematical object x belongs to $\bigcap_{i \in I} A_i$ if and only if

$$\forall i \in I, x \in A_i.$$

2.8.2 Notation. Let n be a positive integer, $A_1, \dots, A_n$ be sets. We denote $\bigcap_{i \in \{1, \dots, n\}} A_i$ as

$$A_1 \cap \dots \cap A_n.$$

Note that it does not depend on the order of $A_1, \dots, A_n$. In particular, if A and B are two sets, then

$$\{x \in B \mid x \in A\} = B \cap A.$$

Therefore, the first statement of Proposition 2.4.4 becomes

$$\text{for any sets } A \text{ and } B, \text{ one has } B \setminus (B \setminus A) = B \cap A.$$

2.8.3 Remark. In set theory, it does not make sense to consider the intersection of an empty family of sets. In fact, if such an intersection existed as a set, for any mathematical object x , since the statement

$$\forall i \in \emptyset, x \in A_i$$

is true (see Example 2.5.2), we would obtain that x belongs to $\bigcap_{i \in \emptyset} A_i$. By Russell's paradox, this is impossible.

2.8.4 Proposition. Let I be a non-empty set and $(A_i)_{i \in I}$ be a set parametrised by I . Let B be a set. Then $B \subseteq \bigcap_{i \in I} A_i$ if and only if

$$\forall i \in I, B \subseteq A_i.$$

Proof. Let $A = \bigcap_{i \in I} A_i$.

Suppose that $B \subseteq A$. For any $x \in B$, one has $x \in A$, and hence

$$\forall i \in I, x \in A_i.$$

Therefore, for any $i \in I$, B is contained in A_i .

Suppose that, for any $i \in I$, $B \subseteq A_i$. Then, for any $x \in B$ and any $i \in I$, one has $x \in A_i$. Hence, for any $x \in B$, one has $x \in A$. Therefore, $B \subseteq A$. $\square$

2.8.5 Corollary. *Let B be a set, I be a non-empty set. For any $i \in I$, let $P_i(\cdot)$ be a condition on B . Then*

$$\{x \in B \mid \forall i \in I, P_i(x)\} = \bigcap_{i \in I} \{x \in B \mid P_i(x)\}.$$

Proof. Let

$$A := \{x \in B \mid \forall i \in I, P_i(x)\}.$$

For any $i \in I$, let

$$A_i := \{x \in B \mid P_i(x)\}.$$

For any $x \in A$ and any $i \in I$, $P_i(x)$ is true. Hence $A \subseteq A_i$. By Proposition 2.8.4, we obtain

$$A \subseteq \bigcap_{i \in I} A_i.$$

Conversely, if $x \in \bigcap_{i \in I} A_i$, then for any $i \in I$, one has $x \in A_i$. Hence $x \in B$, and for any $i \in I$, $P_i(x)$ is true. Thus $x \in A$. $\square$

2.8.6 Proposition. *Let B be a set, $(A_i)_{i \in I}$ be a non-empty family of sets. The following equality holds*

$$\left(\bigcap_{i \in I} A_i \right) \setminus B = \bigcap_{i \in I} (A_i \setminus B).$$

Proof. Let $A := \bigcap_{i \in I} A_i$. For any $i \in I$, one has $A \subseteq A_i$. Hence

$$A \setminus B = \{x \in A \mid x \notin B\} \subseteq \{x \in A_i \mid x \notin B\}.$$

By Proposition 2.8.4, we get

$$A \setminus B \subseteq \bigcap_{i \in I} (A_i \setminus B).$$

Conversely, if $x \in \bigcap_{i \in I} (A_i \setminus B)$, then, for any $i \in I$, one has $x \in A_i \setminus B$, namely $x \in A_i$ and $x \notin B$. Thus $x \in \bigcap_{i \in I} A_i$ and $x \notin B$. Therefore $x \in A \setminus B$. $\square$

2.8.7 Proposition. *Let I be a set and $(A_i)_{i \in I}$ be a family of sets parametrised by I . For any set B , the following statements hold.*

- (1) $B \cap (\bigcup_{i \in I} A_i) = \bigcup_{i \in I} (B \cap A_i)$.
- (2) If $I \neq \emptyset$, $B \cup (\bigcap_{i \in I} A_i) = \bigcap_{i \in I} (B \cup A_i)$,

(3) If $I \neq \emptyset$, $B \setminus \bigcup_{i \in I} A_i = \bigcap_{i \in I} (B \setminus A_i)$,

(4) If $I \neq \emptyset$, $B \setminus \bigcap_{i \in I} A_i = \bigcup_{i \in I} (B \setminus A_i)$.

Proof. (1) By Corollary 2.7.4, we obtain

$$B \cap \left(\bigcup_{i \in I} A_i \right) = \{x \in B \mid \exists i \in I, x \in A_i\} = \bigcup_{i \in I} \{x \in B \mid x \in A_i\} = \bigcup_{i \in I} (B \cap A_i).$$

(2) Let $A := \bigcap_{i \in I} A_i$. By definition, for any $i \in I$, one has $A \subseteq A_i$ and hence $B \cup A \subseteq B \cup A_i$. Thus, by Proposition 2.8.4, we obtain

$$B \cup \left(\bigcap_{i \in I} A_i \right) \subseteq \bigcap_{i \in I} (B \cup A_i).$$

Conversely, let $x \in \bigcap_{i \in I} (B \cup A_i)$. For any $i \in I$, one has $x \in B \cup A_i$. If $x \in B$, then $x \in B \cup (\bigcap_{i \in I} A_i)$; otherwise one has

$$\forall i \in I, x \in A_i,$$

and we still get $x \in B \cup (\bigcap_{i \in I} A_i)$.

(3) By Proposition 2.5.3,

$$B \setminus \bigcup_{i \in I} A_i = \{x \in B \mid \neg(\exists i \in I, x \in A_i)\} = \{x \in B \mid \forall i \in I, x \notin A_i\}.$$

By Corollary 2.8.5, this is equal to

$$\bigcap_{i \in I} \{x \in B \mid x \notin A_i\} = \bigcap_{i \in I} (B \setminus A_i).$$

(4) By Proposition 2.5.3,

$$B \setminus \bigcap_{i \in I} A_i = \{x \in B \mid \neg(\forall i \in I, x \in A_i)\} = \{x \in B \mid \exists i \in I, x \notin A_i\}.$$

By Corollary 2.7.4, this is equal to

$$\bigcup_{i \in I} \{x \in B \mid x \notin A_i\} = \bigcup_{i \in I} (B \setminus A_i).$$

□

Exercises

1. Let $A = \{1, 2, 3, \text{Pikachu}\}$, $B = \{a, b, c, \text{Pikachu}\}$. Determine $A \cup B$ and $A \cap B$.

2. For any $k \in \mathbb{N}$, let

$$k\mathbb{N} = \{kn \mid n \in \mathbb{N}\}.$$

- (1) Determine $2\mathbb{N} \cap 3\mathbb{N}$.
 (2) Does the equality $2\mathbb{N} \cup 3\mathbb{N} = \mathbb{N}$ hold?
 (3) Determine

$$\bigcap_{k \in \mathbb{N}} k\mathbb{N}.$$

3. Are the following sets equal:

$$A = \{(x, y) \in \mathbb{R}^2 \mid 4x - y = 1\}, \quad B = \{(t + 1, 4t + 3) \mid t \in \mathbb{R}\}?$$

Prove your conclusion.

4. Is the statement $0 \in \{\{0\}\}$ true?
 5. Let A and B be two sets such that $A \subseteq B$. Prove that $\mathcal{P}(A) \subseteq \mathcal{P}(B)$.
 6. Let A , B and C be sets. Prove the following equalities:

$$A \cup (B \cap C) = (A \cup B) \cap C, \quad A \cap (B \cup C) = (A \cap B) \cup C.$$

7. Let I and J be non-empty sets, $(A_{i,j})_{(i,j) \in I \times J}$ be a family of sets. The goal of this exercise is to compare the following sets:

$$M = \bigcup_{i \in I} \bigcap_{j \in J} A_{i,j}, \quad N = \bigcap_{j \in J} \bigcup_{i \in I} A_{i,j}.$$

For any $i \in I$, let

$$M_i = \bigcap_{j \in J} A_{i,j}.$$

For any $j \in J$, let

$$N_j = \bigcup_{i \in I} A_{i,j}.$$

Thus,

$$M = \bigcup_{i \in I} M_i, \quad N = \bigcap_{j \in J} N_j.$$

- (1) Let $(i, j) \in I \times J$. Prove that $M_i \subseteq A_{i,j} \subseteq N_j$.
- (2) Prove that $M \subseteq N$.
- (3) Consider the case $I = J = \mathbb{N}$. For any $(i, j) \in \mathbb{N} \times \mathbb{N}$, let $A_{i,j} = \{|i - j|\}$. Determine the sets M and N . Are they equal?

8. Let X and Y be sets.

- (1) Prove that $\mathcal{P}(X \cap Y) = \mathcal{P}(X) \cap \mathcal{P}(Y)$.
- (2) Prove that $\mathcal{P}(X) \cup \mathcal{P}(Y) \subseteq \mathcal{P}(X \cup Y)$.
- (3) Construct two sets X and Y such that the equality

$$\mathcal{P}(X) \cup \mathcal{P}(Y) = \mathcal{P}(X \cup Y)$$

does not hold.

9. Let A and B be sets. Prove that

$$B \setminus A = (B \cup A) \setminus A = B \setminus (A \cap B).$$

10. Let A , B and C be sets. Prove that

$$C \setminus (B \setminus A) = (C \cap A) \cup (C \setminus B).$$

11. If X and Y are two sets, we denote by $X \Delta Y$ the symmetric difference

$$(X \setminus Y) \cup (Y \setminus X).$$

- (1) Prove that, for any sets X and Y ,

$$X \Delta Y = (X \cup Y) \setminus (X \cap Y).$$

- (2) Prove that, if $X \supseteq Y$, then $X \Delta Y = X \setminus Y$.

12. Let A , B and C be sets.

- (1) Compute $A \Delta A$ and $A \Delta \emptyset$.
- (2) Prove that $(A \Delta B) \cap C = (A \cap C) \Delta (B \cap C)$.

13. For each of the following statements, determine a statement of the opposite truth value, where no quantifier is preceded by a negation symbol. Determine their truth values.

- (1) $\exists x \in \mathbb{R}, \forall y \in \mathbb{R}, x + y \geq 0$;
- (2) $\exists x \in \mathbb{R}, \exists y \in \mathbb{R}, x + y \geq 0$;
- (3) $\forall x \in \mathbb{R}, \forall y \in \mathbb{R}, x + y \geq 0$;
- (4) $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, x + y \geq 0$;

14. By using quantifiers, rewrite the following statements.

- (1) Any natural number has a real square root.
- (2) Any natural number is strictly smaller than a certain real number.
- (3) There is a real number that is smaller than or equal to any natural number.

15. Consider the condition on $\mathbb{R}$:

$$P(x) := “\forall y \in [0, 1], x \geq y \Rightarrow x \geq 2y”, \quad x \in \mathbb{R}.$$

Determine all real numbers satisfying $P(\cdot)$.

16. For each of the following conditions on $\mathbb{R}$, describe the sets of $x \in \mathbb{R}$ that satisfy the condition.

- (1) $((x > 0) \wedge (x < 1)) \vee (x = 0)$,
- (2) $(x > 3) \wedge (x < 5) \wedge (x \neq 4)$,
- (3) $((x \leq 0) \wedge (x > 1)) \vee (x = 4)$,
- (4) $(x \geq 0) \Rightarrow (x \geq 2)$.

17. Consider the following statement:

$$\exists a \in \mathbb{R}_{<0}, \forall x \in \mathbb{R}_{\geq 1}, x > a \Rightarrow \left((x^2 > \frac{a^2}{4}) \vee (x \leq 0) \right).$$

Determine a statement of the opposite truth value, where no quantifier is preceded by a negation symbol. Is this statement true?

18. Recall that a prime number is by definition a natural number that is ≥ 2 , the only divisors of which are 1 and itself. Let $\mathbb{P}$ be the set of all prime numbers. Express the following statements as formulas by using quantifiers.

- (1) If a prime number divides the product of two integers, then it divides at least one of them.
- (2) Any integer that is greater than or equal to 2 is divisible by a prime number.

- (3) The natural number 2 is the only prime number that is even.
- (4) Any primer number that is ≥ 5 is congruent to 1 or -1 modulo 6.
- (5) Any even number that is > 2 can be written as the sum of two prime numbers.
- (6) Any prime number that is congruent to 1 modulo 4 can be written as the sum of two squares.
- (7) For any natural number n such that $n \geq 2$, there is always a primer number that lies between n and $2n$.
- (8) There exist infinitely many prime numbers.
- (9) Let p be an integer such that $p \geq 2$. Then p is a prime number if and only if p divides $(p-1)! + 1$.

19. Let x, y, x' , and y' be mathematical objects. Prove that

$$\{\{x\}, \{x, y\}\} = \{\{x'\}, \{x', y'\}\}$$

if and only if $x = x'$ and $y = y'$.

Chapter 3

Correspondences

3.1 Correspondence and its inverse

3.1.1 Definition. We call a *correspondence* any triplet of the form

$$f = (\mathcal{D}_f, \mathcal{A}_f, \Gamma_f),$$

where $\mathcal{D}_f$, $\mathcal{A}_f$ are two sets, called respectively the *departure set* and the *arrival set* of f , and Γ_f is a subset of $\mathcal{D}_f \times \mathcal{A}_f$, called the *graph* of f .

If X and Y are two sets and f is a correspondence of the form (X, Y, Γ_f) , we say that f is a correspondence *from* X *to* Y .

3.1.2 Definition. Let f be a correspondence. We denote by f^{-1} the correspondence defined as follows:

$$\begin{aligned}\mathcal{D}_{f^{-1}} &:= \mathcal{A}_f, & \mathcal{A}_{f^{-1}} &:= \mathcal{D}_f, \\ \Gamma_{f^{-1}} &:= \{(y, x) \in \mathcal{A}_f \times \mathcal{D}_f \mid (x, y) \in \Gamma_f\}.\end{aligned}$$

The correspondence f^{-1} is called the *inverse correspondence* of f . Clearly one has

$$(f^{-1})^{-1} = f, \tag{3.1}$$

namely f is the inverse correspondence of f^{-1} .

3.1.3 Example. Let X be a set. Denote by Δ_X the following subset of $X \times X$:

$$\{(x, x) \mid x \in X\},$$

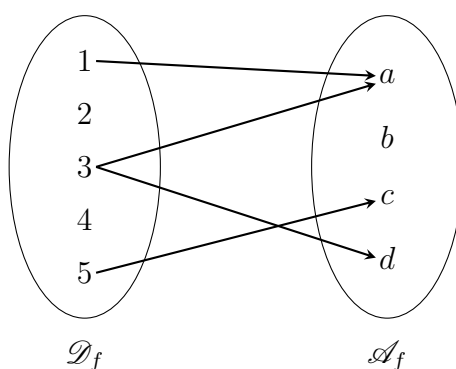
called the *diagonal subset* of $X \times X$. The correspondence (X, X, Δ_X) is called the *identity correspondence* of X , denoted as Id_X . By definition, one has $\text{Id}_X^{-1} = \text{Id}_X$.

3.1.4 Example. Let X and Y be two sets. There is a correspondence from X to Y whose graph is the empty set. This correspondence is called the *empty correspondence* from X to Y . Note that the inverse correspondence of the empty correspondence from X to Y is the empty correspondence from Y to X .

3.2 Illustration of a correspondence

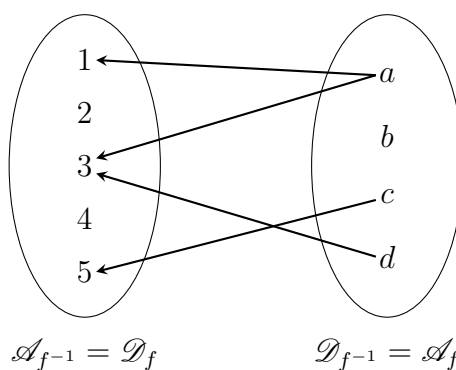
3.2.1 Remark. A correspondence can be viewed as an oriented graph. The elements of $\mathcal{D}_f$ and $\mathcal{A}_f$ are illustrated by two groups of vertices. For any ordered pair in the graph Γ_f , we link the corresponding vertices by an arrow. In the following figure, we illustrate a correspondence from $\{1, 2, 3, 4, 5\}$ to $\{a, b, c, d\}$.

Figure 3.1: Visualization of a correspondence f



The inverse correspondence f^{-1} can be visualised by inverting the direction of the arrows in the above figure.

Figure 3.2: Visualization of the inverse correspondence f^{-1}



3.2.2 Remark. We can also represent a correspondence f by a table, whose rows are labelled by elements of $\mathcal{D}_f$ and whose columns are labelled by elements of $\mathcal{A}_f$. For each pair in Γ_f we mark the cell of corresponding coordinates by a $\checkmark$. For

example, the correspondence described by Figure ?? can be represented by the following table.

Table 3.1: Table representation of the correspondence f

	a	b	c	d
1	✓			
2				
3	✓			✓
4				
5			✓	

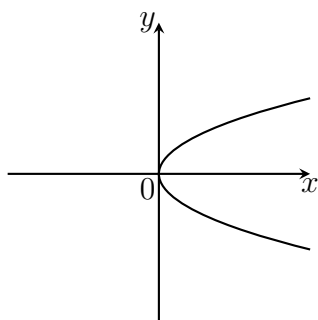
Its inverse correspondence can be represented by the transposed table.

Table 3.2: Table representation of the inverse correspondence f^{-1}

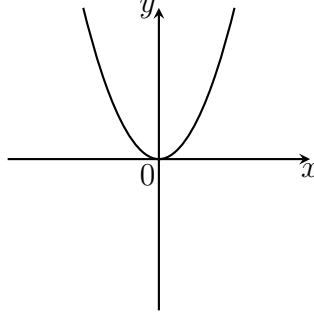
	1	2	3	4	5
a	✓		✓		
b					
c					✓
d			✓		

3.2.3 Remark. A correspondence from $\mathbb{R}$ to $\mathbb{R}$ can be illustrated by its graph in the coordinate plane. Consider for example such a correspondence f , with

$$\Gamma_f := \{(x, y) \in \mathbb{R} \times \mathbb{R} \mid x = y^2\}.$$



The inverse correspondence is illustrated by a parabola that opens upward.



3.3 Image and preimage

3.3.1 Definition. Let X and Y be sets, and f be a correspondence from X to Y . If (x, y) is an element of Γ_f , we say that x is a *preimage* of y under f , and y is an *image* of x under f .

If A is a set, we denote by $f(A)$ the set

$$\{y \in \mathcal{A}_f \mid \exists x \in A, (x, y) \in \Gamma_f\},$$

called the *image* of A by the correspondence f .

If B is a set, the set $f^{-1}(B)$ is called the *preimage* of B by the correspondence f . Note that it is by definition the image of B by the inverse correspondence f^{-1} .

3.3.2 Definition. Let f be a correspondence. The set $f(\mathcal{D}_f)$ is called the *range* of f , denoted as $\text{Im}(f)$. The set $f^{-1}(\mathcal{A}_f)$ is called the *domain of definition* of f , denoted as $\text{Dom}(f)$. Note that the domain of definition of a correspondence f is the projection of the graph Γ_f to the departure set $\mathcal{D}_f$, and the range of a correspondence f is the projection of the graph Γ_f to the arrival set $\mathcal{A}_f$.

For any sets A and B ,

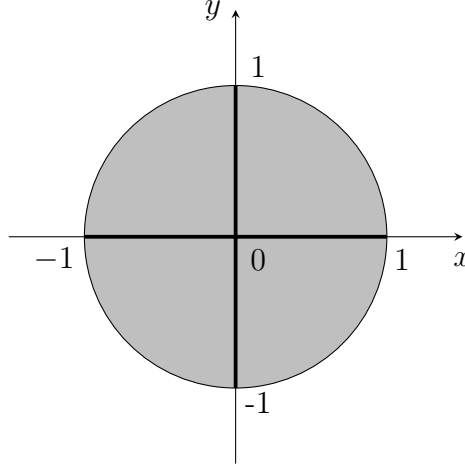
$$f(A) \subseteq \text{Im}(f), \quad f^{-1}(B) \subseteq \text{Dom}(f).$$

Moreover,

$$\text{Dom}(f) = \text{Im}(f^{-1}), \quad \text{Im}(f) = \text{Dom}(f^{-1}).$$

3.3.3 Example. The following figure illustrates a correspondence from $\mathbb{R}$ to $\mathbb{R}$, the image of which is

$$\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}.$$



The domain of definition and the range of this correspondence are both $[-1, 1]$.

3.3.4 Proposition. *Let f be a correspondence.*

- (1) *If A and A' are two sets such that $A' \subseteq A$, then one has $f(A') \subseteq f(A)$.*
- (2) *If B and B' are two sets such that $B' \subseteq B$, then one has $f^{-1}(B') \subseteq f^{-1}(B)$.*

Proof. It suffices to prove the first statement. Let y be an element of $f(A')$. By definition there exists $x \in A'$ such that $(x, y) \in \Gamma_f$. Since $A' \subseteq A$, one has $x \in A$ and hence $y \in f(A)$. $\square$

3.3.5 Proposition. *Let f be a correspondence. The following equalities hold:*

$$\text{Im}(f) = f(\text{Dom}(f)), \quad \text{Dom}(f) = f^{-1}(\text{Im}(f)).$$

Proof. Since $\text{Dom}(f) \subseteq \mathcal{D}_f$, by Proposition 3.3.4, one has

$$f(\text{Dom}(f)) \subseteq f(\mathcal{D}_f) = \text{Im}(f).$$

Let y be an element of $\text{Im}(f)$. There exists $x \in \mathcal{D}_f$ such that $(x, y) \in \Gamma_f$. By definition one has $x \in \text{Dom}(f)$ and hence $y \in f(\text{Dom}(f))$. Therefore the equality $\text{Im}(f) = f(\text{Dom}(f))$ is true. Applying this equality to f^{-1} , we obtain the second equality. $\square$

3.3.6 Proposition. *Let f be a correspondence.*

- (1) *Let A be a set and y be an mathematical object. Then y belongs to $f(A)$ if and only if $A \cap f^{-1}(\{y\}) \neq \emptyset$.*

- (2) Let B be a set and x be a mathematical object. Then x belongs to $f^{-1}(B)$ if and only if $B \cap f(\{x\}) \neq \emptyset$.

Proof. (1) By definition, $y \in f(A)$ if and only if there exists $x \in A$ such that $(x, y) \in \Gamma_f$, or equivalently $x \in A \cap f^{-1}(\{y\})$

Applying (1) to f^{-1} , we obtain (2). $\square$

3.3.7 Proposition. Let f be a correspondence.

- (1) Let I be a set and $(A_i)_{i \in I}$ be a family of sets parametrised by I . Then the equality

$$f\left(\bigcup_{i \in I} A_i\right) = \bigcup_{i \in I} f(A_i)$$

holds. Moreover, if I is not empty, then

$$f\left(\bigcap_{i \in I} A_i\right) \subseteq \bigcap_{i \in I} f(A_i)$$

- (2) Let I be a set and $(B_i)_{i \in I}$ be a family of sets parametrised by I . Then the equality

$$f^{-1}\left(\bigcup_{i \in I} B_i\right) = \bigcup_{i \in I} f^{-1}(B_i)$$

holds. Moreover, if I is not empty, then

$$f^{-1}\left(\bigcap_{i \in I} B_i\right) \subseteq \bigcap_{i \in I} f^{-1}(B_i).$$

Proof. (1) By Propositions 3.3.6 and 2.8.7, we obtain

$$\begin{aligned} f\left(\bigcup_{i \in I} A_i\right) &= \left\{ y \in Y \mid \left(\bigcup_{i \in I} A_i\right) \cap f^{-1}(y) \neq \emptyset \right\} \\ &= \left\{ y \in Y \mid \bigcup_{i \in I} (A_i \cap f^{-1}(\{y\})) \neq \emptyset \right\} \\ &= \left\{ y \in Y \mid \exists i \in I, A_i \cap f^{-1}(\{y\}) \neq \emptyset \right\} = \bigcup_{i \in I} f(A_i), \end{aligned}$$

where the last equality comes from Corollary 2.7.4.

Let $A = \bigcap_{i \in I} A_i$. For any $i \in I$, one has $A \subseteq A_i$ and hence, by Proposition 3.3.4, one has $f(A) \subseteq f(A_i)$. By Proposition 2.8.4, we get

$$f(A) \subseteq \bigcap_{i \in I} f(A_i).$$

Applying (1) to f^{-1} , we obtain (2). $\square$

3.4 Composition

3.4.1 Definition. Let f and g be correspondences. We define the *composite* of g and f as the correspondence $g \circ f$ from $\mathcal{D}_f$ to $\mathcal{A}_g$ whose graph $\Gamma_{g \circ f}$ is composed of the elements (x, z) of $\mathcal{D}_f \times \mathcal{A}_g$ such that there exists some object y satisfying $(x, y) \in \Gamma_f$ and $(y, z) \in \Gamma_g$ (note that the object y should be an element of $\mathcal{A}_f \cap \mathcal{D}_g$). In other words,

$$\Gamma_{g \circ f} = \{(x, z) \in \mathcal{D}_f \times \mathcal{A}_g \mid \exists y \in \mathcal{A}_f \cap \mathcal{D}_g, (x, y) \in \Gamma_f \text{ and } (y, z) \in \Gamma_g\}.$$

3.4.2 Proposition. Let f and g be correspondences. The following equality holds:

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}. \quad (3.2)$$

Proof. Let $x \in \mathcal{A}_{f^{-1}} = \mathcal{D}_f$, $z \in \mathcal{D}_{g^{-1}} = \mathcal{A}_g$. Then $(z, x) \in \Gamma_{(g \circ f)^{-1}}$ if and only if $(x, z) \in \Gamma_{g \circ f}$, namely there exists y such that $(x, y) \in \Gamma_f$ and $(y, z) \in \Gamma_g$. This is also equivalent to the existence of y such that $(z, y) \in \Gamma_{g^{-1}}$ and $(y, x) \in \Gamma_{f^{-1}}$, namely $(z, x) \in \Gamma_{f^{-1} \circ g^{-1}}$. $\square$

3.4.3 Proposition. Let f , g and h be correspondences. The following equality holds:

$$h \circ (g \circ f) = (h \circ g) \circ f. \quad (3.3)$$

Proof. It suffices to verify that $h \circ (g \circ f)$ and $(h \circ g) \circ f$ have the same graph. By definition, an element $(x, w) \in \mathcal{D}_f \times \mathcal{A}_h$ belongs to the graph of $h \circ (g \circ f)$ if and only if there exists z such that $(x, z) \in \Gamma_{g \circ f}$ and $(z, w) \in \Gamma_h$, which is equivalent to the existence of y and z such that $(x, y) \in \Gamma_f$, $(y, z) \in \Gamma_g$ and $(z, w) \in \Gamma_h$. A similar argument shows that this condition is also equivalent to $(x, w) \in \Gamma_{(h \circ g) \circ f}$. Hence, the equality (3.3) holds. $\square$

3.4.4 Proposition. Let X and Y be sets, f be a correspondence from X to Y . Then the following equalities hold:

$$f \circ \text{Id}_X = f = \text{Id}_Y \circ f.$$

Proof. Let $(x, y) \in X \times Y$. By definition, (x, y) belongs to the graph of $f \circ \text{Id}_X$ if and only if there exists x' such that $(x, x') \in \Delta_X$ and $(x', y) \in \Gamma_f$, that is, $x = x'$ and $(x, y) \in \Gamma_f$. Therefore $f \circ \text{Id}_X = f$. Applying this equality to f^{-1} , we obtain

$$f^{-1} \circ \text{Id}_Y = f^{-1}.$$

Taking the inverse correspondences, by (3.1) and (3.2) we deduce

$$\text{Id}_Y \circ f = f.$$

$\square$

3.4.5 Proposition. *Let f and g be correspondences.*

(1) *For any set A , one has*

$$(g \circ f)(A) = g(f(A)).$$

In particular,

$$\text{Im}(g \circ f) = g(\text{Im}(f)) \subseteq \text{Im}(g).$$

If in addition $\text{Dom}(g) \subseteq \text{Im}(f)$, then the equality $\text{Im}(g \circ f) = \text{Im}(g)$ holds.

(2) *For any set B , one has*

$$(g \circ f)^{-1}(B) = f^{-1}(g^{-1}(B)).$$

In particular,

$$\text{Dom}(g \circ f) = f^{-1}(\text{Dom}(g)) \subseteq \text{Dom}(f).$$

If in addition $\text{Im}(f) \subseteq \text{Dom}(g)$, then the equality $\text{Dom}(g \circ f) = \text{Dom}(f)$ holds.

Proof. (1) By definition

$$\begin{aligned} (g \circ f)(A) &= \{z \in \mathcal{A}_g \mid \exists x \in A, (x, z) \in \Gamma_{g \circ f}\} \\ &= \{z \in \mathcal{A}_g \mid \exists x \in A, \exists y \in \mathcal{A}_f, (x, y) \in \Gamma_f, (y, z) \in \Gamma_g\} \\ &= \{z \in \mathcal{A}_g \mid \exists y \in f(A), (y, z) \in \Gamma_g\} = g(f(A)). \end{aligned}$$

Applying this equality to the case where $A = \mathcal{D}_f$, we obtain

$$\text{Im}(g \circ f) = (g \circ f)(\mathcal{D}_f) = g(f(\mathcal{D}_f)) = g(\text{Im}(f)) \subseteq \text{Im}(g).$$

In the case where $\text{Dom}(g) \subseteq \text{Im}(f)$, by Propositions 3.3.5 and 3.3.4 we obtain

$$\text{Im}(g) = g(\text{Dom}(g)) \subseteq g(\text{Im}(f)) = \text{Im}(g \circ f).$$

Hence the equality $\text{Im}(g \circ f) = \text{Im}(g)$ holds.

Applying (1) to g^{-1} and f^{-1} , by (3.2) we obtain (2). □

3.5 Surjectivity

3.5.1 Definition. Let f be a correspondence. If $\mathcal{A}_f = \text{Im}(f)$, we say that f is *surjective*. If f^{-1} is surjective, or equivalently $\text{Dom}(f) = \mathcal{D}_f$, we say that f is a *multivalued mapping*.

3.5.2 Proposition. *Let f be a correspondence.*

- (1) Assume that f is surjective. Then, for any subset B of $\mathcal{A}_f$, one has $B \subseteq f(f^{-1}(B))$.
- (2) Assume that f is a multivalued mapping. Then, for any subset A of $\mathcal{D}_f$, one has $A \subseteq f^{-1}(f(A))$.

Proof. (1) Let y be an element of B . Since f is surjective, there exists $x \in \mathcal{D}_f$ such that $(x, y) \in \Gamma_f$. Therefore, $x \in f^{-1}(B)$ and hence $y \in f(f^{-1}(B))$.

Applying (1) to f^{-1} , we obtain (2). $\square$

3.5.3 Proposition. *Let f and g be correspondences.*

- (1) If $g \circ f$ is surjective, so is g .
- (2) If $g \circ f$ is a multivalued mapping, so is f .

Proof. (1) By (1) of Proposition 3.4.5, one has

$$\text{Im}(g \circ f) \subseteq \text{Im}(g) \subseteq \mathcal{A}_g = \mathcal{A}_{g \circ f}. \quad (3.4)$$

If $g \circ f$ is surjective, namely $\text{Im}(g \circ f) = \mathcal{A}_{g \circ f}$, then we deduce from (3.4) that $\text{Im}(g) = \mathcal{A}_g$, namely g is surjective.

Applying (1) to g^{-1} and f^{-1} , we obtain (2). $\square$

3.5.4 Proposition. *Let f and g be correspondences.*

- (1) If g is surjective and $\text{Dom}(g) \subseteq \text{Im}(f)$, then $g \circ f$ is also surjective.
- (2) If f is a multivalued mapping and $\text{Im}(f) \subseteq \text{Dom}(g)$, then $g \circ f$ is a multivalued mapping.

Proof. (1) Since $\text{Dom}(g) \subseteq \text{Im}(f)$, by (1) of Proposition 3.4.5 we obtain

$$\text{Im}(g \circ f) = \text{Im}(g).$$

Since g is surjective,

$$\text{Im}(g) = \mathcal{A}_g = \mathcal{A}_{g \circ f}.$$

Hence $g \circ f$ is also surjective.

Applying (1) to g^{-1} and f^{-1} , we obtain (2). $\square$

3.6 Injectivity

3.6.1 Definition. Let f be a correspondence. If each element of $\mathcal{D}_f$ has at most one image under f , we say that f is a *function*. If f^{-1} is a function, or equivalently, each element of $\mathcal{A}_f$ has at most one preimage under f , we say that f is *injective*.

3.6.2 Notation. Functions form a special case of correspondences. The defining feature of functions is that, corresponding to each element in the domain of definition, there is a unique element in the range set of the function.

Let f be a function, and let $x \in \text{Dom}(f)$. We denote the unique image of x under f as $f(x)$, and we say that f sends $x \in \text{Dom}(f)$ to $f(x)$ or $f(x)$ is the *value* of f at x . We can also use the notation

$$x \longmapsto f(x)$$

to indicate the correspondence of x to its image under f .

3.6.3 Proposition. *Let f be a correspondence.*

- (1) *Assume that f is injective. For any set A one has $f^{-1}(f(A)) \subseteq A$.*
- (2) *Assume that f is a function. For any set B one has $f(f^{-1}(B)) \subseteq B$.*

Proof. (1) Let x be an element of $f^{-1}(f(A))$. By definition, there exists $y \in f(A)$ such that $(x, y) \in \Gamma_f$. Since $y \in f(A)$ there exist $x' \in A$ such that $(x', y) \in \Gamma_f$. Since y admits at most one preimage, we obtain $x = x'$. Hence $x \in A$.

Applying (1) to f^{-1} , we obtain (2). □

3.6.4 Proposition. *Let f and g be correspondences.*

- (1) *If f and g are functions, so is $g \circ f$. Moreover, for any $x \in \text{Dom}(g \circ f)$, one has $(g \circ f)(x) = g(f(x))$.*
- (2) *If f and g are injective, so is $g \circ f$.*

Proof. Let x be an element of $\text{Dom}(g \circ f)$. Assume that z and z' are images of x under $g \circ f$. Let y and y' be such that

$$(x, y) \in \Gamma_f, \quad (y, z) \in \Gamma_g, \quad (x, y') \in \Gamma_f, \quad (y', z') \in \Gamma_g.$$

Since f is a function, one has $y = y' = f(x)$. Since g is a function, we deduce that $z = z' = g(f(x))$. Therefore $g \circ f$ is a function, and the equality $(g \circ f)(x) = g(f(x))$ holds for any $x \in \text{Dom}(g \circ f)$.

Applying (1) to g^{-1} and f^{-1} , we obtain (2). □

3.6.5 Proposition. *Let f and g be correspondences.*

- (1) *If $g \circ f$ is injective and $\text{Im}(f) \subseteq \text{Dom}(g)$, then f is also injective.*
- (2) *If $g \circ f$ is a function and $\text{Dom}(g) \subseteq \text{Im}(f)$, then g is also a function.*

Proof. (1) Let y be an element of the image of f . Let x and x' be preimages of y under f . Since $\text{Im}(f) \subseteq \text{Dom}(g)$, one has $y \in \text{Dom}(g)$. Hence there exists $z \in \mathcal{A}_g$ such that $(y, z) \in \Gamma_g$. We then deduce that (x, z) and (x', z) are elements of $\Gamma_{g \circ f}$. Since $g \circ f$ is injective, we obtain $x = x'$. Therefore, f is injective.

Applying (1) to g^{-1} and f^{-1} , we obtain (2). □

3.6.6 Proposition. *Let f be a correspondence, and I be a non-empty set.*

- (1) *Suppose that f is a function. For any family $(B_i)_{i \in I}$ of sets parametrised by I , one has*

$$f^{-1}\left(\bigcap_{i \in I} B_i\right) = \bigcap_{i \in I} f^{-1}(B_i).$$

- (2) *Suppose that f is injective. For any family $(A_i)_{i \in I}$ of sets parametrised by I , one has*

$$f\left(\bigcap_{i \in I} A_i\right) = \bigcap_{i \in I} f(A_i).$$

Proof. (1) Let x be an element of $\bigcap_{i \in I} f^{-1}(B_i)$. For any $i \in I$, one has $f(x) \in B_i$. Hence $x \in f^{-1}(\bigcap_{i \in I} B_i)$. Therefore we obtain

$$f^{-1}\left(\bigcap_{i \in I} B_i\right) \supseteq \bigcap_{i \in I} f^{-1}(B_i).$$

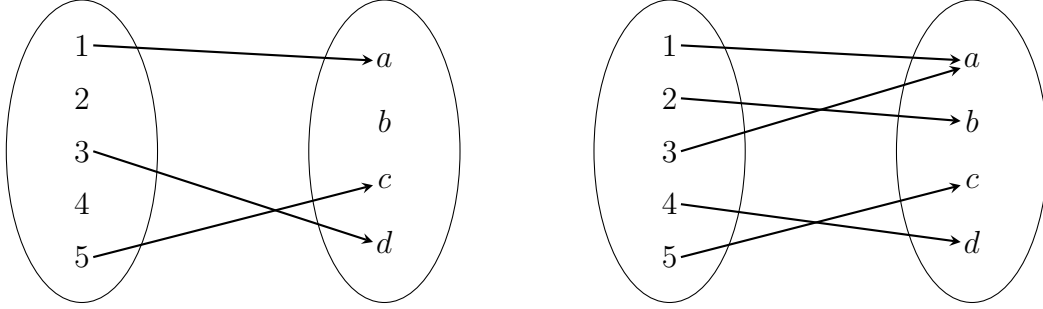
Combing with (2) of Proposition 3.3.7, we obtain the equality

$$f^{-1}\left(\bigcap_{i \in I} B_i\right) = \bigcap_{i \in I} f^{-1}(B_i).$$

Applying (1) to f^{-1} , we obtain (2). □

3.6.7 Example. In the following, we illustrate two functions.

Figure 3.3: Visualization of two functions



The function illustrated on the left-hand side of the figure is injective and not surjective; the function illustrated on the right-hand side is surjective and not injective.

3.7 Mappings

3.7.1 Definition. A correspondence f is said to be a *mapping* if any element of $\mathcal{D}_f$ has a unique image, or equivalently, f is a function and $\mathcal{D}_f = \text{Dom}(f)$. Note that f is a mapping if and only if f^{-1} is both injective and surjective.

3.7.2 Notation. Let X and Y be sets. We denote by Y^X the set of all mappings from X to Y . An element $u \in Y^X$ is often written in the form of a family of elements of Y parametrised by X as follows

$$(u(x))_{x \in X}.$$

In the case where $X = \{1, \dots, n\}$, where n is a positive integer, the set $Y^{\{1, \dots, n\}}$ is also denoted as Y^n . An element u of Y^n is often written as an n -tuple

$$(u(1), \dots, u(n)).$$

3.7.3 Example. (1) Let X be a set. The identity correspondence Id_X is a mapping. It is also called the *identity mapping* of X .

(2) Let X and Y be sets and y be an element of Y . The mapping from X to Y sending any $x \in X$ to y is called the *constant mapping with value y* .

3.7.4 Remark. Let $f : X \rightarrow Y$ be a mapping, and I be a set.

(1) By (1) of Proposition 3.3.7, for any family of sets $(A_i)_{i \in I}$, one has

$$f\left(\bigcup_{i \in I} A_i\right) = \bigcup_{i \in I} f(A_i).$$

By (2) of Proposition 3.3.7, for any family of sets $(B_i)_{i \in I}$, one has

$$f^{-1}\left(\bigcup_{i \in I} B_i\right) = \bigcup_{i \in I} f^{-1}(B_i).$$

(2) Assume that I is not empty. By (1) of Proposition 3.3.7, for any family of sets $(A_i)_{i \in I}$, one has

$$f\left(\bigcap_{i \in I} A_i\right) \subseteq \bigcap_{i \in I} f(A_i).$$

By (1) of Proposition 3.6.6, for any family of sets $(B_i)_{i \in I}$, one has

$$f^{-1}\left(\bigcap_{i \in I} B_i\right) = \bigcap_{i \in I} f^{-1}(B_i).$$

(3) By (2) of Proposition 3.6.3, for any set B , one has $f(f^{-1}(B)) \subseteq B$. Since $\text{Dom}(f) = X$, by (2) of Proposition 3.5.2, for any subset A of X one has $A \subseteq f^{-1}(f(A))$. If moreover f is injective, by (1) of Proposition 3.6.3, one has $f^{-1}(f(A)) \subseteq A$ and hence the equality $f^{-1}(f(A)) = A$ holds.

3.7.5 Proposition. *Let f and g be mappings. Suppose that $\text{Im}(f) \subseteq \mathcal{D}_g$. Then $g \circ f$ is also a mapping. Moreover, for any $x \in \mathcal{D}_f = \mathcal{D}_{g \circ f}$ one has*

$$(g \circ f)(x) = g(f(x)).$$

Proof. Note that $\mathcal{D}_g = \text{Dom}(g)$ since g is a mapping. Hence, the statement is a direct consequence of Propositions 3.6.4 and 3.5.4. $\square$

3.7.6 Remark. Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be mappings.

- (1) By Proposition 3.5.4, if f and g are both surjective, so is $g \circ f$. By Proposition 3.5.3, if $g \circ f$ is surjective, so is g .
- (2) By Proposition 3.6.4, if f and g are both injective, so is $g \circ f$. By Proposition 3.6.5, if $g \circ f$ is injective, so is f .

3.8 Bijection

3.8.1 Definition. Let f be a mapping, that is, a correspondence such that f^{-1} is injective and surjective. If f is injective and surjective, we say that f is a *bijection*, or a *one-to-one correspondence*. Note that a correspondence is a bijection if and only if its inverse is a bijection.

3.8.2 Proposition. Let X and Y be sets, f be a correspondence from X to Y . If f is a bijection, then $f^{-1} \circ f = \text{Id}_X$ and $f \circ f^{-1} = \text{Id}_Y$. Conversely, if there exists a correspondence g such that $g \circ f = \text{Id}_X$ and $f \circ g = \text{Id}_Y$, then f is a bijection and $g = f^{-1}$.

Proof. If f is a bijection, then f and f^{-1} are both mappings. By Proposition 3.7.5, one has

$$\begin{aligned}\forall x \in X, \quad (f^{-1} \circ f)(x) &= f^{-1}(f(x)) = x, \\ \forall y \in Y, \quad (f \circ f^{-1})(y) &= f(f^{-1}(y)) = y.\end{aligned}$$

Hence $f^{-1} \circ f = \text{Id}_X$ and $f \circ f^{-1} = \text{Id}_Y$.

Assume that g is a correspondence such that $g \circ f = \text{Id}_X$ and $f \circ g = \text{Id}_Y$. Since identity correspondences are surjective mappings, by Proposition 3.5.3, we deduce from the equality $g \circ f = \text{Id}_X$ (which can also be written as $f^{-1} \circ g^{-1} = \text{Id}_X$) that g and f^{-1} are surjective. In particular, $\text{Dom}(f) = X = \text{Im}(g)$.

Similarly, we deduce from the equality $f \circ g = \text{Id}_Y$ (which can also be written as $g^{-1} \circ f^{-1} = \text{Id}_Y$) that f and g^{-1} are surjective. In particular, $\text{Dom}(g) = Y = \text{Im}(f)$.

Since identity correspondences are injective, by Proposition 3.6.5, we deduce from $g \circ f = \text{Id}_X$ that f is injective. Similarly, we deduce from $f \circ g = \text{Id}_Y$ that f is a function. Therefore, f is a mapping which is injective and surjective, namely a bijection.

Finally, by Propositions 3.4.4 and 3.4.3, we obtain

$$g = g \circ \text{Id}_Y = g \circ (f \circ f^{-1}) = (g \circ f) \circ f^{-1} = \text{Id}_X \circ f^{-1} = f^{-1}.$$

□

3.8.3 Proposition. Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be bijections. Then the composite correspondence $g \circ f$ is also a bijection.

Proof. This is a direct consequence of Propositions 3.7.5, 3.6.4, and 3.5.4. □

3.8.4 Proposition. Let X and Y be sets, f be a correspondence from X to Y , and g be a correspondence from Y to X . If $f \circ g$ and $g \circ f$ are bijections, then f and g are both bijections.

Proof. By Proposition 3.5.3, f and g are surjective and are multivalued mappings. In particular,

$$\text{Dom}(f) = X, \quad \text{Im}(f) = Y, \quad \text{Dom}(g) = Y, \quad \text{Im}(g) = X.$$

Therefore, by Proposition 3.6.5, we deduce that f and g are injective and are functions. Hence f and g are both bijections. $\square$

3.9 Direct product

3.9.1 Definition. Let I be a set and $(A_i)_{i \in I}$ be a family of sets parametrised by I . We denote by

$$\prod_{i \in I} A_i$$

the set of all mappings from I to $\bigcup_{i \in I} A_i$ which send any $i \in I$ to an element of A_i . This set is called the *direct product* of $(A_i)_{i \in I}$. Using Notation 3.7.2, we often write an element of the direct product in the form of a family $x := (x_i)_{i \in I}$ parametrised by I , where each x_i is an element of A_i , called the i -th *coordinate* of x . In the case where I is the empty set, the union $\bigcup_{i \in I} A_i$ is empty. Therefore, the direct product contains a unique element (identity mapping of $\emptyset$).

For each $j \in I$, we denote by

$$\text{pr}_j : \prod_{i \in I} A_i \longrightarrow A_j$$

the mapping which sends each element $(a_i)_{i \in I}$ of the direct product to its j -th coordinate a_j . This mapping is called the *projection to the j -th coordinate*.

3.9.2 Notation. Let n be a non-zero natural number. If $(A_i)_{i \in \{1, \dots, n\}}$ is a family of sets parametrised by $\{1, \dots, n\}$, then the set

$$\prod_{i \in \{1, \dots, n\}} A_i$$

is often denoted as

$$A_1 \times \cdots \times A_n.$$

3.9.3 Axioms (Axiom of choice). In this book, we adopt the following axiom. If I is a non-empty set and if $(A_i)_{i \in I}$ is a family of non-empty sets, then the direct product $\prod_{i \in I} A_i$ is not empty.

3.9.4 Proposition. *Let I be a set and $(A_i)_{i \in I}$ be a family of sets parametrised by I . For any set X , the mapping*

$$\left(\prod_{i \in I} A_i \right)^X \longrightarrow \prod_{i \in I} A_i^X, \quad (3.5)$$

which sends f to $(\text{pr}_i \circ f)_{i \in I}$, is a bijection.

In other words, for any family $(f_i : X \rightarrow A_i)_{i \in I}$ of mappings, there exists a unique mapping $f : X \rightarrow \prod_{i \in I} A_i$ such that

$$\forall i \in I, \text{pr}_i \circ f = f_i$$

Proof. Let $(f_i)_{i \in I}$ be an element of

$$\prod_{i \in I} A_i^X,$$

where each f_i is a mapping from X to A_i . Let $f : X \rightarrow \prod_{i \in I} A_i$ be the mapping which sends $x \in X$ to $(f_i(x))_{i \in I}$. By definition, for any $i \in I$ one has

$$\forall x \in X, \text{pr}_i(f(x)) = f_i(x).$$

Therefore the mapping (3.5) is surjective.

If f and g are two mappings from X to $\prod_{i \in I} A_i$ such that $\text{pr}_i \circ f = \text{pr}_i \circ g$ for any $i \in I$, then, for any $x \in X$ one has

$$\forall i \in I, \text{pr}_i(f(x)) = \text{pr}_i(g(x)).$$

Hence $f(x) = g(x)$ for any $x \in X$, namely $f = g$. Therefore the mapping (3.5) is injective. $\square$

3.9.5 Notation. Let I be a set, $(A_i)_{i \in I}$ be a family of sets parametrised by I .

Let X be a set. For any $i \in I$, let $f_i : X \rightarrow A_i$ be a mapping from X to A_i . By Proposition 3.9.4, there exists a unique mapping $f : X \rightarrow \prod_{i \in I} A_i$ such that $\text{pr}_i \circ f = f_i$ for any $i \in I$. By abuse of notation, we denote by $(f_i)_{i \in I}$ this mapping.

Let $(B_i)_{i \in I}$ be a family of sets parametrised by I . For any $i \in I$, let $g_i : B_i \rightarrow A_i$ be a mapping from B_i to A_i . We denote by

$$\prod_{i \in I} g_i : \prod_{i \in I} B_i \longrightarrow \prod_{i \in I} A_i$$

the mapping which sends $(b_i)_{i \in I}$ to $(g_i(b_i))_{i \in I}$. In the case where $I = \{1, \dots, n\}$, where n is a non-zero natural number, the mapping $\prod_{i \in \{1, \dots, n\}} g_i$ is also denoted as

$$g_1 \times \cdots \times g_n.$$

3.9.6 Proposition. *Let $f : X \rightarrow Y$ be a mapping.*

- (1) *If f is surjective, then there exists an injective mapping $g : Y \rightarrow X$ such that $f \circ g = \text{Id}_Y$.*
- (2) *If f is injective and X is not empty, then there exists a surjective mapping $h : Y \rightarrow X$ such that $h \circ f = \text{Id}_X$.*

Proof. (1) The case where $Y = \emptyset$ is trivial since in this case $X = \emptyset$ and f is the identity mapping of $\emptyset$. In the following, we assume that Y is not empty. Since f is surjective, for any $y \in Y$, the set $f^{-1}(\{y\})$ is not empty. Hence, the direct product

$$\prod_{y \in Y} f^{-1}(\{y\})$$

is not empty. In other words, there exists a mapping g from Y to X such that $f(g(y)) = y$ for any $y \in Y$, that is $f \circ g = \text{Id}_Y$. By (2) of Remark 3.7.6, g is injective.

- (2) Let x_0 be an element of X . We define a mapping $h : Y \rightarrow X$ as follows:

$$h(y) := \begin{cases} f^{-1}(y), & \text{if } y \in \text{Im}(f), \\ x_0, & \text{else.} \end{cases}$$

Then, by construction, one has $h \circ f = \text{Id}_X$. By (1) of Remark 3.7.6, h is surjective. □

3.10 Restriction and extension

3.10.1 Definition. Let f and g be correspondences. If $\Gamma_f \subseteq \Gamma_g$, we say that f is a *restriction* of g and that g is an *extension* of f . By definition, f is a restriction of g if and only if f^{-1} is a restriction of g^{-1} .

Let X and Y be sets, h be a correspondence from X to Y , and A be a subset of X . Denote by $h|_A$ the correspondence from A to Y such that

$$\Gamma_{h|_A} = \Gamma_h \cap (A \times Y).$$

We call it the *restriction of h to A* .

3.10.2 Proposition. *Let g be a correspondence.*

- (1) *If g is a function, then all its restrictions are functions.*
- (2) *If g is injective, then all its restrictions are injective.*

(3) If g is a function and A is a subset of $\text{Dom}(g)$, then $g|_A$ is a mapping.

Proof. (1) Let f be a restriction of g . Then one has $\text{Dom}(f) \subseteq \text{Dom}(g)$. For any $x \in \text{Dom}(f)$, if $(x, y) \in \Gamma_f$, then $(x, y) \in \Gamma_g$. Since g is a function, x admits at most one image under g . Therefore, it also admits at most one image under f . Hence f is a function.

Applying (1) to g^{-1} , we obtain (2).

(3) By (1), $g|_A$ is a function. Moreover, for any $x \in A$,

$$(x, g(x)) \in (A \times \mathcal{A}_g) \cap \Gamma_g = \Gamma_{g|_A},$$

which implies that $x \in \text{Dom}(g|_A)$. Hence $g|_A$ is a mapping. $\square$

3.10.3 Definition. Let X be a set and A be a subset of X . The restriction of the identity mapping Id_X to A is called the *inclusion mapping* of A into X , denoted by $j_A : A \rightarrow X$. Note that, if h is a correspondence from X to a set Y , then $h|_A$ identifies with $h \circ j_A$, the composite correspondence of h with the inclusion mapping $j_A : A \rightarrow X$.

3.11 Equivalence relation

3.11.1 Definition. Let X be a set. We call *binary relation* on X any correspondence from X to itself. If R is a binary relation on X , for any $(x, y) \in X \times X$, we denote by $x R y$ the statement “ $(x, y) \in \Gamma_R$ ”, and denote by $x \not R y$ the statement “ $(x, y) \notin \Gamma_R$ ”. If Y is a subset of X , then there is a unique binary relation on Y , the graph of which is $\Gamma_R \cap (Y \times Y)$. We call it the *restriction of R to Y as a binary relation*. We emphasize that it differs from the restriction of R to Y as a correspondence.

3.11.2 Notation. Let X be a set, n be a positive integer, $R_1, \dots, R_n$ be binary relations on X . If $x_0, \dots, x_n$ are elements of X , then

$$x_0 R_1 x_1 R_2 \dots R_{n-1} x_{n-1} R_n x_n$$

denotes

$$\forall i \in \{0, \dots, n-1\}, \quad x_i R_{i+1} x_{i+1}.$$

3.11.3 Definition. Let X be a set and R be a binary relation on X .

- (a) If for any $x \in X$, one has $x R x$, then we say that the binary relation R is *reflexive*. Note that R is reflexive if and only if $\Delta_X \subseteq \Gamma_R$.
- (b) If for any $(x, y) \in X \times X$, $x R y$ implies $y R x$, then we say that the binary relation R is *symmetric*. Note that R is symmetric if and only if $R = R^{-1}$.

- (c) If for all elements x, y and z of X , $x R y$ and $y R z$ imply $x R z$, then we say that the binary relation R is *transitive*. Note that R is transitive if and only if $\Gamma_{R \circ R} \subseteq \Gamma_R$.
- (d) If R is reflexive, symmetric, and transitive, then we say that R is an *equivalence relation*.

3.11.4 Lemma. Let X be a set and $\sim$ be a symmetric and transitive binary relation on X . For any $x \in X$, denote by $[x]$ the subset

$$\{y \in X \mid y \sim x\}.$$

- (1) If x_1 and x_2 are elements of X such that $x_1 \sim x_2$, then one has $[x_1] = [x_2]$.
- (2) If x_1 and x_2 are elements of X such that $x_1 \not\sim x_2$, then one has $[x_1] \cap [x_2] = \emptyset$.

Proof. (1) Let y be an element of $[x_1]$. By definition, $y \sim x_1$. Since $x_1 \sim x_2$, by the transitivity we obtain $y \sim x_2$, namely $y \in [x_2]$. Therefore, we obtain $[x_1] \subseteq [x_2]$. Moreover, by the symmetry, one has $x_2 \sim x_1$, and hence, by what has been proved above, we obtain $[x_2] \subseteq [x_1]$. Therefore $[x_1] = [x_2]$.

(2) We reason by contraposition. Assume that z is a common element of $[x_1]$ and $[x_2]$. By definition, one has $z \sim x_1$ and $z \sim x_2$. Hence, by the symmetry and the transitivity, we get $x_1 \sim x_2$. $\square$

3.11.5 Definition. Let X be a set, $\sim$ be an equivalence relation on X . A subset of X of the form

$$[x] := \{y \in X \mid y \sim x\}, \text{ where } x \in X$$

is called an *equivalence class* under the equivalence relation $\sim$, and the element x is called a *representative* of the equivalence class. We denote by $X/\sim$ the set of all equivalence classes under $\sim$, called the *quotient set* of X by the equivalence relation $\sim$. This is a subset of the power set $\mathcal{P}(X)$. The mapping from X to $X/\sim$ which sends $x \in X$ to the equivalence class $[x]$ represented by x is called the *projection mapping*. Note that the projection mapping is surjective.

3.11.6 Proposition. Let X be a set and $\sim$ be an equivalence relation on X . Then the elements of $X/\sim$ are pairwise disjoint sets and one has

$$X = \bigcup_{A \in X/\sim} A. \quad (3.6)$$

In other words, X is the disjoint union of the equivalence classes under $\sim$.

Proof. By Lemma 3.11.4, two equivalence classes under $\sim$ are either disjoint or equal, which shows that the sets in $X/\sim$ are pairwise disjoint. Moreover, by the reflexivity of $\sim$, for any $x \in X$, one has $x \in [x]$. Hence

$$X \subseteq \bigcup_{A \in X/\sim} A.$$

Conversely, for any $A \in X/\sim$, one has $A \subseteq X$. Hence, by Proposition 2.7.3, one has

$$\bigcup_{A \in X/\sim} A \subseteq X.$$

Hence the equality (3.6) holds. $\square$

3.11.7 Example. Let p be a natural number. Consider the binary relation $\sim_p$ on $\mathbb{Z}$ defined as follows:

$$x \sim_p y \text{ if and only if there exists } n \in \mathbb{Z} \text{ such that } x - y = pn.$$

One can check that $\sim_p$ is an equivalence relation on $\mathbb{Z}$. We call it the *relation of congruence modulo p* . For any $a \in \mathbb{Z}$, we denote by $a + p\mathbb{Z}$ the equivalence class of a under $\sim_p$.

3.11.8 Proposition. *Let X be a set, $\sim$ be an equivalence relation on X , and $f : X \rightarrow Y$ be a mapping. Denote by $\pi : X \rightarrow X/\sim$ the projection mapping. Assume that, for any $(x, x') \in X \times X$ such that $x \sim x'$, one has $f(x) = f(x')$. There exists a unique mapping $\tilde{f} : X/\sim \rightarrow Y$ such that $\tilde{f} \circ \pi = f$ (namely, such that $\tilde{f}([x]) = f(x)$ for any $x \in X$). Moreover, the equality $\text{Im}(f) = \text{Im}(\tilde{f})$ holds.*

Proof. By (1) of Proposition 3.9.6, since π is surjective, there exists an injective mapping $g : X/\sim \rightarrow X$ such that $\pi \circ g = \text{Id}_{X/\sim}$. Therefore, if $\tilde{f} : X/\sim \rightarrow Y$ is a mapping such that $\tilde{f} \circ \pi = f$, then one has

$$f \circ g = (\tilde{f} \circ \pi) \circ g = \tilde{f} \circ (\pi \circ g) = \tilde{f} \circ \text{Id}_{X/\sim} = \tilde{f}.$$

This shows the uniqueness of $\tilde{f}$. Moreover, since $f = \tilde{f} \circ \pi$ and since π is surjective, by (1) of Proposition 3.4.5, we obtain $\text{Im}(\tilde{f}) = \text{Im}(f)$.

It remains to check that, if we define $\tilde{f}$ as $f \circ g$, then the mapping $\tilde{f}$ satisfies the equality $\tilde{f} \circ \pi = f$. Note that, for any $\alpha \in X/\sim$, one has $\pi(g(\alpha)) = \alpha$, namely $g(\alpha)$ is a representative of α . Therefore, for any $x \in X$, one has $g(\pi(x)) \sim x$ and hence

$$\tilde{f}(\pi(x)) = f(g(\pi(x))) = f(x).$$

$\square$

3.11.9 Definition. Let X be a set, $\sim$ be an equivalence relation on X , and $f : X \rightarrow Y$ be a mapping. If, for any $(x, x') \in X \times X$ such that $x \sim x'$, one has $f(x) = f(x')$, then we say that the mapping f is *compatible* with the equivalence relation $\sim$.

Assume that $f : X \rightarrow Y$ is a mapping compatible with the equivalence relation $\sim$. The mapping $\tilde{f} : X/\sim \rightarrow Y$ described in Proposition 3.11.8 is said to be *induced by f by passing to the quotient*.

3.11.10 Corollary. Let $f : X \rightarrow Y$ be a mapping of sets. The binary relation $\sim_f$ on X defined by

$$x \sim_f x' \text{ if and only if } f(x) = f(x')$$

is an equivalence relation. Moreover, f induces by passing to quotient an injective mapping $\tilde{f} : X/\sim_f \rightarrow Y$, the range of which is $f(X)$.

Proof. For any $x \in X$ one has $f(x) = f(x)$ and hence $x \sim_f x$. If x and x' are elements of X such that $x \sim_f x'$, then one has $f(x') = f(x)$ and hence $x' \sim_f x$. If x, x' and x'' are elements of X such that $x \sim_f x'$ and $x' \sim_f x''$, then the equalities $f(x) = f(x') = f(x'')$ hold, and hence $x \sim_f x''$.

By Proposition 3.11.8, the mapping f induces by passing to quotient a mapping $\tilde{f} : X/\sim_f \rightarrow Y$. If α and β are two elements of $X/\sim_f$, which are represented by elements x and y of X respectively, and such that $\tilde{f}(\alpha) = \tilde{f}(\beta)$, then one has $f(x) = f(y)$ and hence $x \sim_f y$. This leads to $\alpha = \beta$. Therefore $\tilde{f}$ is injective.

Finally, by Proposition 3.11.8, f and $\tilde{f}$ have the same range, namely $\text{Im}(\tilde{f}) = \text{Im}(f) = f(X)$. $\square$

3.11.11 Corollary. Let I be a set and $(X_i)_{i \in I}$ be a family of sets parametrised by I . For any $i \in I$, let R_i be an equivalence relation on X_i and $\pi_i : X_i \rightarrow X_i/R_i$ be the projection mapping. Let $X = \prod_{i \in I} X_i$ and let $\sim$ be the binary relation on X defined as follows

$$(x_i)_{i \in I} \sim (y_i)_{i \in I} \text{ if and only if } \forall i \in I, x_i R_i y_i.$$

(1) The binary relation $\sim$ is an equivalence relation on X .

(2) The mapping

$$\prod_{i \in I} \pi_i : X \longrightarrow \prod_{i \in I} (X_i/R_i)$$

induces by passing to quotient a bijection from $X/\sim$ to $\prod_{i \in I} (X_i/R_i)$.

Proof. Note that, for any elements $(x_i)_{i \in I}$ and $(y_i)_{i \in I}$ of X , $(x_i)_{i \in I} \sim (y_i)_{i \in I}$ if and only if $(\pi_i(x_i))_{i \in I} = (\pi_i(y_i))_{i \in I}$. Therefore, by Corollary 3.11.10, the binary relation $\sim$ is an equivalence relation, and the mapping $\prod_{i \in I} \pi_i$ induces by passing

to quotient an injective mapping from $X/\sim$ to $\prod_{i \in I} (X_i/R_i)$. Moreover, by Proposition 3.11.8, this mapping has the same image as that of $\prod_{i \in I} \pi_i$. Since $\prod_{i \in I} \pi_i$ is surjective, we obtain that the induced mapping is also surjective. $\square$

Exercises

1. Here is a table that shows courses chosen by 6 students, followed by a list of the assigned instructors.

	Analysis	Physics	Chemistry	Programming	Calculus	Phys. Lab	Chem. Lab
Charles	✓	✓		✓			
Charlie		✓		✓	✓		
Carlo			✓		✓		✓
Karl	✓	✓				✓	
Charuzu		✓	✓		✓		
Charlot	✓	✓	✓				

- Analysis: Maicon,
- Physics: Mikkell,
- Chemistry: Michelle,
- Programming: Mikhail,
- Calculus: Micheal,
- Physics Lab: Mitchell,
- Chemistry Lab: Mihai.

Let X denote the set that consists of these 6 students, Y denote the set of all the courses, Z denote the set of all the instructors, f denote the correspondence from X to Y based on the above table, g denote the correspondence from Y to Z based on the assignment of the course instructors.

- (1) Draw a table of g .
- (2) Is the correspondence f injective? surjective? a function? a multi-valued mapping? What about g ?
- (3) Draw a table of the composite $g \circ f$. What can we conclude from it?

- (4) Draw a table of the composite $f^{-1} \circ f$. What can we conclude from it?
2. For each of the following mappings, determine if it is injective or surjective. Justify your answer.
- (1) $f : \mathbb{Z} \rightarrow \mathbb{Z}, f(x) = 3x + 4$.
 - (2) $f : \mathbb{N} \rightarrow \mathbb{N}, f(x) = 3x + 4$.
 - (3) $f : \mathbb{Z} \rightarrow \mathbb{Z}, f(x) = 2x + 4$.
 - (4) $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2, f(x, y) = (x + y, x - y)$.
3. For each of the following functions from $\mathbb{R}$ to $\mathbb{R}$, determine its domain of definition and its range.
- (1) $\cos \circ \sin$.
 - (2) $\exp \circ \ln$.
 - (3) $\ln \circ \exp$.
 - (4) ι defined as $\iota(x) = x^{-1}$.
 - (5) $\iota \circ \iota$.
4. Consider the correspondence φ from $\mathbb{R}$ to $\mathbb{R}$ whose graph is given by

$$\{(x, y) : x^2 + y^2 \leq 1\}.$$

- (1) Draw the graph of the correspondence φ .
 - (2) Determine the domain of definition and the range of φ .
 - (3) We view φ as a binary relation on $\mathbb{R}$. Is this binary relation reflexive, symmetric, transitive?
 - (4) Draw the graph of $\varphi \circ \varphi$.
5. Consider the mapping $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = x^2$. Let

$$A = \{x \in \mathbb{R} \mid -1 \leq x \leq 4\}.$$

Determine $f(A)$ and $f^{-1}(A)$.

6. Let f and g be functions from $\mathbb{R}$ to $\mathbb{R}$ defined as

$$f(x) = 3x + 1, \quad g(x) = x^2 - 1.$$

Determine the functions $f \circ g$ and $g \circ f$. Are they equal?

7. For each of the following function h from $\mathbb{R}$ to $\mathbb{R}$. Determine two functions f and g such that $h = f \circ g$.

- (1) $h(x) = \sqrt{3x - 1}$,
- (2) $h(x) = \sin(x + \pi/2)$,
- (3) $h(x) = 1/(x + 1)$.

8. Let f and g be mappings from $\mathbb{N}$ to $\mathbb{N}$ defined as

$$f(n) = 2n, \quad g(n) = \begin{cases} n/2, & \text{if } n \text{ is even,} \\ 0, & \text{if } n \text{ is odd.} \end{cases}$$

Determine $g \circ f$ and $f \circ g$. Is the mapping f injective, surjective, a bijection? Is the mapping g injective, surjective, a bijection?

9. Let $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ be the mapping sending $(n, p) \in \mathbb{N} \times \mathbb{N}$ to $2^n(2p + 1)$.

- (1) Show that f is injective.
- (2) Determine the range of f .

10. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a mapping. For each of the following statements, construct a statement of the opposite truth value, where no quantifier is preceded by a negation symbol.

- (1) $\forall x \in \mathbb{R}, f(x) \neq 0$,
- (2) $\forall M \in \mathbb{R}_{>0}, \exists A \in \mathbb{R}_{>0}, \forall x \geq A, f(x) > M$.
- (3) $\forall x \in \mathbb{R}, f(x) > 0 \Rightarrow x \leq 0$.
- (4) $\forall \varepsilon \in \mathbb{R}_{>0}, \exists \eta \in \mathbb{R}_{>0}, \forall (x, y) \in \mathbb{R} \times \mathbb{R}, |x - y| < \eta \Rightarrow |f(x) - f(y)| < \varepsilon$.

11. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a mapping. By using the quantifiers, express the following statements:

- (1) f is constant.
- (2) 0 lies in the range of f .
- (3) f only takes non-negative values.

12. We consider the following condition $P(\cdot)$ on the set $\mathbb{R}^{\mathbb{R}}$ of mappings from $\mathbb{R}$ to $\mathbb{R}$

$$P(f) := "\exists M \in \mathbb{R}, \forall x \in \mathbb{R}, f(x) \leq M".$$

- (1) Construct a condition $Q(\cdot)$ on $\mathbb{R}^{\mathbb{R}}$ without quantifier proceeded by a negation symbol, such that, for any $f \in \mathbb{R}^{\mathbb{R}}$, $Q(f)$ and $\neg P(f)$ have the same truth value.
 - (2) Construct an example of mapping in $\mathbb{R}^{\mathbb{R}}$ that satisfies the condition $P(\cdot)$.
 - (3) Construct an example of mapping in $\mathbb{R}^{\mathbb{R}}$ that does not satisfies the condition $P(\cdot)$.
 - (4) For each of the following conditions, determine if it is satisfied by all mappings in $\mathbb{R}^{\mathbb{R}}$, if it is not satisfied by none of the mappings in $\mathbb{R}^{\mathbb{R}}$, or if it is satisfied by some but not all of the mappings in $\mathbb{R}^{\mathbb{R}}$.
 - a) $\exists x \in \mathbb{R}, \forall M \in \mathbb{R}, f(M) \leq x$.
 - b) $\exists x \in \mathbb{R}, \forall M \in \mathbb{R}, f(x) \leq M$.
 - c) $\forall M \in \mathbb{R}, \exists x \in \mathbb{R}, f(x) \leq M$.
 - d) $\forall M \in \mathbb{R}, \exists x \in \mathbb{R}, f(M) \leq x$.
- 13.** Let n and p be two natural numbers, $X = \{1, \dots, n\}$ and $Y = \{1, \dots, p\}$.
- (1) How many correspondences are there from X to Y ?
 - (2) How many injective correspondences are there from X to Y ?
 - (3) How many surjective correspondences are there from X to Y ?
 - (4) How many functions are there from X to Y ?
 - (5) How many mappings are there from X to Y ?
 - (6) How many bijections are there from X to Y .
- 14.** For any natural number n , let E_n be the set $\{1, \dots, n\}$ (by convention E_0 denotes the empty set). For any $(n, p) \in \mathbb{N}^2$, let $S_{n,p}$ be the number of surjective mappings from E_n to E_p .
- (1) Determine $S_{n,p}$ for $(n, p) \in \mathbb{N}^2$ such that $p > n$.
 - (2) For any $n \in \mathbb{N}$, determine $S_{n,0}$.
 - (3) Show that, for any non-zero natural number p , one has

$$\sum_{i=0}^p (-1)^i \binom{p}{i} = 0,$$

where

$$\binom{n}{p} := \frac{n!}{p!(n-p)!}$$

- (4) Let i, q and p be natural numbers such that $i \leq q \leq p$. Show that

$$\binom{p}{q} \binom{q}{i} = \binom{p}{i} \binom{p-i}{q-i}.$$

- (5) Show that, for any $(i, p) \in \mathbb{N}^2$ such that $i < p$, one has

$$\sum_{q=i}^p (-1)^q \binom{p}{q} \binom{q}{i} = 0.$$

- (6) Let p and q be natural numbers such that $q \leq p$. Show that the number of mappings from E_n to E_p whose range has exactly q elements is equal to $S_{n,q} \binom{p}{q}$.

- (7) Let p and n be natural numbers such that $p \leq n$ and $n \geq 1$. Show that

$$p^n = \sum_{q=1}^p \binom{p}{q} S_{n,q}.$$

- (8) Let p and n be natural numbers such that $p \leq n$ and $n \geq 1$. Show that

$$(-1)^p S_{n,p} = \sum_{i=1}^p (-1)^i \binom{p}{i} i^n.$$

- (9) Let p and n be natural numbers such that $1 \leq p < n$. Show that

$$S_{n,p} = p(S_{n-1,p} + S_{n-1,p-1}).$$

- (10) Let $p \in \mathbb{N}$. Determine $S_{p,p}$ and $S_{p+1,p}$.

- (11) Let p and n be natural numbers such that $p \leq n$. Show that the number of surjective functions from E_n to E_p is equal to

$$\sum_{j=p}^n \binom{n}{j} S_{j,p}.$$

- (12) Prove the equality

$$\sum_{j=p}^n \binom{n}{j} S_{j,p} = \frac{1}{p+1} S_{n+1,p+1}.$$

- 15.** For any natural number n , let E_n be the set $\{1, \dots, n\}$ (by convention E_0 denotes the empty set).

- (1) Let n and p be natural numbers such that $n \leq p$. Show that the number of injective mappings from E_n to E_p is equal to $p!/(p-n)!$.
- (2) Show that the number of injective functions from E_n to E_p is

$$\sum_{k=0}^{\min\{n,p\}} \binom{n}{k} \binom{p}{k} k!.$$

16. Let X be a set and $\mathcal{P}(X)$ be the power set of X , namely $\mathcal{P}(X)$ is the set of all subsets of X . The purpose of this exercise is to prove that there does not exist any injective mapping from $\mathcal{P}(X)$ to X . We reason by contradiction, assuming that $\Phi : \mathcal{P}(X) \rightarrow X$ is an injective mapping.

- (1) Show that Φ^{-1} is a function from X to $\mathcal{P}(X)$.
- (2) Let A be the set of $n \in \text{Im}(\Phi)$ such that $n \notin \Phi^{-1}(n)$. Show that $\Phi^{-1}(\Phi(A)) = A$.
- (3) Examine the truth value of the statement $\Phi(A) \in A$ and obtain a contradiction.

Chapter 4

Ordering

4.1 Partially ordered set

4.1.1 Definition. Let X be a set and $\underline{R}$ be a binary relation (see Definition 3.11.3) on X . If, for any $(x, y) \in X \times X$, $x \underline{R} y$ and $y \underline{R} x$ imply $x = y$, then we say that the binary relation $\underline{R}$ is *antisymmetric*. If the binary relation $\underline{R}$ is reflexive, transitive (see Definition 3.11.3) and antisymmetric, then we say that $\underline{R}$ is a *partial order* and that $(X, \underline{R})$ is a *partially ordered set*. Note that, if $\underline{R}$ is a partial order on X , then its inverse correspondence (see Definition 3.1.2) is also a partial order on X .

Let $\underline{R}$ be a partial order on X . If, for any $(x, y) \in X \times X$, $x \underline{R} y$ or $y \underline{R} x$ holds, then we say that $\underline{R}$ is a *total order* and that $(X, \underline{R})$ is a *totally ordered set*.

4.1.2 Notation. Given a binary relation represented by a certain symbol, we often use the mirror-symmetric symbol to represent the inverse of this binary relation. For instance, if we use the symbol $\leq$ to represent a partial order on a set, then the symbol $\geq$ represents the inverse of the partial order $\leq$, that is, $x \leq y$ if and only if $y \geq x$.

In the case where the partial order is denoted by $\leq$, it is common to refer to the partially ordered set by its underlying set, without explicitly mentioning the partial order.

4.1.3 Example. Let X be a set, $\mathcal{P}(X)$ be the power set of X . Then the inclusion relation $\subseteq$ is a partial order on $\mathcal{P}(X)$.

4.1.4 Example. Consider a partially ordered set $(X, \leq)$ and let A be a subset of X . The restriction of the binary relation $\leq$ to A defines a partial order on A , referred to as the *induced partial order* on A . We denote by $(A, \leq)$ the partially ordered set obtained by endowing A with this induced partial order.

4.1.5 Proposition. Let $(X, \leq)$ be a partially ordered set. We denote by $<$ the binary relation on X such that

$$\forall (x, y) \in X \times X, \quad x < y \text{ if and only if } (x \leq y \text{ and } x \neq y).$$

Then the binary relation $<$ has the following properties.

- (1) Irreflexivity: for any $x \in X$, $x < x$ does not hold.
- (2) Asymmetry: for any $(x, y) \in X \times X$, if $x < y$, then $y < x$ does not hold.
- (3) Transitivity: for any elements x, y and z of X , if $x < y$ and $y < z$, then $x < z$.

Proof. (1) By definition, $x < x$ is equivalent to $x \leq x$ and $x \neq x$. Since $x \neq x$ is never true, the condition $x < x$ could not hold.

(2) If $x < y$ and $y < x$ hold at the same time, then $x \leq y$ and $y \leq x$ also hold at the same time, which leads to $x = y$ by the symmetry of $\leq$. Then by (1) $x < y$ could not be true, which is a contradiction.

(3) If $x < y$ and $y < z$ hold, then, by the transitivity of $\leq$ one obtains $x \leq z$. Note that, by (2), $x = z$ could not hold. Therefore, we get $x < z$. $\square$

4.1.6 Notation. Let $(X, \leq)$ be a partially ordered set. For any $a \in X$, we denote by $X_{\leq a}$ the set

$$\{x \in X \mid x \leq a\}.$$

Similarly, we denote by $X_{< a}$ the set

$$\{x \in X \mid x < a\}.$$

4.1.7 Definition. We call *strict partial order* a binary relation which is irreflexive, asymmetric, and transitive. The binary relation $<$ in Proposition 4.1.5 is a strict partial order, which is called the *strict partial order associated with $\leq$* .

4.1.8 Notation. In this book, we always use underlined symbols to denote partial orders and use the corresponding ununderlined symbol to denote the associated strict partial order. For instance, if $\leq$ denotes a partial order on a set, then $<$ denotes the associated strict partial order on the same set.

4.2 Monotonic functions

4.2.1 Definition. Let I and X be partially ordered sets, f be a function from I to X .

- (a) If, for any elements a and b of $\text{Dom}(f)$ such that $a < b$, one has $f(a) \leq f(b)$, we say that f is *increasing*.
- (b) If, for any elements a and b of $\text{Dom}(f)$ such that $a < b$, one has $f(a) < f(b)$, we say that f is *strictly increasing*.
- (c) If, for any elements a and b of $\text{Dom}(f)$ such that $a < b$, one has $f(a) \geq f(b)$, we say that f is *decreasing*.
- (d) If, for any elements a and b of $\text{Dom}(f)$ such that $a < b$, one has $f(a) > f(b)$, we say that f is *strictly decreasing*.

A function f is said to be *monotonic* if it is either increasing or decreasing. If f is strictly increasing or strictly decreasing, then it is said to be *strictly monotonic*. A restriction of a monotonic (resp. strictly monotonic) function is also monotonic (resp. strictly monotonic) and has the same monotonicity direction.

4.2.2 Example. Let X be a partially ordered set, and let A be a subset of X equipped with the induced partial order. Then, the inclusion mapping $\iota_A : A \rightarrow X$ is strictly increasing.

4.2.3 Proposition. *Let f and g be functions between partially ordered sets.*

- (1) *Assume that f and g are both increasing or are both decreasing, then $g \circ f$ is increasing.*
- (2) *Assume that f and g are both strictly increasing or are both strictly decreasing, then $g \circ f$ is strictly increasing.*
- (3) *Assume that the one of g and f is increasing and the other is decreasing, then $g \circ f$ is decreasing.*
- (4) *Assume that one of g and f is strictly increasing and the other is strictly decreasing, then $g \circ f$ is strictly decreasing.*

Proof. Let x and y be elements of the domain of definition of $g \circ f$, such that $x < y$. If f is increasing, then $f(x) \leq f(y)$, if moreover g is increasing, then $g(f(x)) \leq g(f(y))$. Similarly, if both f and g are strictly increasing, then one deduces from $x < y$ the relation $f(x) < f(y)$ and furthermore $g(f(x)) < g(f(y))$. Therefore, if both f and g are increasing (resp. strictly increasing), then $g \circ f$ is increasing (resp. strictly increasing).

If one replaces one or both of the partial orders of $\text{Dep}(f)$ and $\text{Arr}(g)$ by the corresponding inverse partial order(s), we deduce from what we have proved above the other statements.

□

4.2.4 Proposition. *Let I and X be partially ordered sets, and f be a function from I to X .*

- (1) *If f is monotonic and is injective, then it is strictly monotonic.*
- (2) *If I is a totally ordered set and f is strictly monotonic, then it is injective.*

Proof. (1) Let a and b be elements of $\text{Dom}(f)$, such that $a < b$. Assume that f is increasing. Then one has $f(a) \leq f(b)$. Since f is injective, $f(a)$ and $f(b)$ are different. Hence, one has $f(a) < f(b)$. Therefore, the function f is strictly increasing. The case where f is decreasing is similar.

(2) Let a and b be different elements of $\text{Dom}(f)$. Since I is totally ordered, either $a < b$ or $b < a$. Since f is strictly monotone, one deduces that, either $f(a) < f(b)$, or $f(b) < f(a)$. Hence f is injective. $\square$

4.2.5 Proposition. *Let X be a totally ordered set, Y be a partially ordered set, f be an injective function from X to Y (in this case f^{-1} is also an injective function). If f is monotonic, then so is f^{-1} . Moreover, f^{-1} and f have the same direction of monotonicity.*

Proof. Without loss of generality, we may assume that f is increasing. Let a and b be two elements of $\text{Im}(f) = \text{Dom}(f^{-1})$. Since X is totally ordered, either $f^{-1}(a) < f^{-1}(b)$, or $f^{-1}(b) < f^{-1}(a)$. Since f is increasing and injective, by (1) of Proposition 4.2.4, f is strictly increasing. If $f^{-1}(b) < f^{-1}(a)$, then

$$b = f(f^{-1}(b)) < f(f^{-1}(a)) = a,$$

which contradicts the hypothesis $a < b$. Therefore, one should have $f^{-1}(a) < f^{-1}(b)$. Hence f^{-1} is strictly increasing. $\square$

4.3 Supremum and infimum

4.3.1 Definition. Let $(X, \leq)$ be a partially ordered set, A be a subset of X .

- (a) Let M be an element of X . If for every $x \in A$, we have $x \leq M$, then we say that M is an *upper bound* of A with respect to $\leq$. If Y is a subset of X which contains A and if A admits an upper bound that belongs to Y , we say that A is *bounded from above in Y* . If A is bounded from above in A , then it has a unique upper bound in A (by the antisymmetry of $\leq$). We denote this unique upper bound by $\max_{\leq} A$ and call it the *greatest element*¹ of A .

¹In some literature, it is also called the *maximum* of A , which explains the expression $\max$.

- (b) Let m be an element of X . If for every $x \in A$, we have $m \leq x$, then we say that m is a *lower bound* of A with respect to $\leq$. If Y is a subset of X which contains A and if A admits a lower bound that belongs to Y , we say that A is *bounded from below in Y* . If A is bounded from below in A , then it has a unique lower bound in A . We denote this unique lower bound by $\min_{\leq} A$ and call it the *least element*² of A .
- (c) Let f be a function from a set I to X . If the image of f is bounded from above (resp. bounded from below) in X , we say that f is *bounded from above* (resp. *bounded from below*). If f is bounded from above and bounded from below at the same time, we say that f is *bounded*. Similarly, if a subset A of X is bounded from above and bounded from below at the same time, we say that A is *bounded*.
- (d) Let Y be a subset of X such that $A \subseteq Y$. If the set of all upper bounds of A in Y admits a least element with respect to $\leq$, then we call this least element the *supremum* of A in Y with respect to $\leq$, denoted by $\sup_{(Y, \leq)} A$.
- (e) Let Y be a subset of X such that $A \subseteq Y$. If the set of all lower bounds of A in Y admits a greatest element with respect to $\leq$, then we call this greatest element the *infimum* of A in Y with respect to $\leq$, denoted by $\inf_{(Y, \leq)} A$.

4.3.2 Notation. Let $(X, \leq)$ be a partially ordered set, A and Y be subsets of X such that $A \subseteq Y$. We denote by A_Y^u the set of all upper bounds of A in Y , and denote by A_Y^ℓ the set of all lower bounds of A in Y . In the case where $Y = X$, the sets A_X^u and A_X^ℓ are denoted as A^u and A^ℓ respectively.

The greatest element, least element, supremum, and the infimum introduced in Definition 4.3.1 are specific to the choice of partial order on a set X . It is important to note that the greatest and least elements of a subset A of X only depend on the induced partial order of A . On the contrary, the supremum and infimum of a subset A of X depend on both the partial order and the choice of the ambient subset Y that contains A and the induced order relation on Y .

When there is no ambiguity about the partial order and ambient subset, we often omit the expressions “relatively to $\dots$ ” and “in $\dots$ ” when referring to these concepts, and omit the subscripts specifying the choices of the partial order and the ambient subset. For example, the expressions $\min_{\leq} A$, $\max_{\leq} A$, $\sup_{(X, \leq)} A$, and $\inf_{(X, \leq)} A$ are often abbreviated as $\min A$, $\max A$, $\sup A$, and $\inf A$, respectively.

4.3.3 Notation. Let $(X, \leq)$ be a partially ordered set and let I be a set.

²In some literature, it is also called the *minimum* of A , which explains the expression $\min$.

- (1) For a function f from I to X , we denote the greatest element, the least element, the supremum, and the infimum of the image of f by $\max f$, $\min f$, $\sup f$, and $\inf f$, respectively.
- (2) For a family $(x_i)_{i \in I}$ of elements of X parametrised by I , we denote the greatest element, the least element, the supremum, and the infimum of the set

$$\{x_i \mid i \in I\}$$

as

$$\max_{i \in I} x_i, \quad \min_{i \in I} x_i, \quad \sup_{i \in I} x_i, \quad \inf_{i \in I} x_i,$$

respectively. We also refer to them as the greatest element, the least element, the supremum, and the infimum of the family $(x_i)_{i \in I}$.

- (3) Let $(x_i)_{i \in I}$ be a family of elements of X parametrised by I and $\mathbb{P}$ be a statement depending on a parameter in I . we denote the greatest element, the least element, the supremum, and the infimum of the set

$$\{x_i \mid i \in I, \mathbb{P}(i)\}$$

as

$$\max_{i \in I, \mathbb{P}(i)} x_i, \quad \min_{i \in I, \mathbb{P}(i)} x_i, \quad \sup_{i \in I, \mathbb{P}(i)} x_i, \quad \inf_{i \in I, \mathbb{P}(i)} x_i,$$

respectively.

4.3.4 Example. Consider the set of natural numbers $\mathbb{N}$ equipped with the usual total order. Note that 0 is the least element of $\mathbb{N}$ but $\mathbb{N}$ does not have any greatest element. In fact, for any $n \in \mathbb{N}$ one has $n < n + 1$. Hence n could not be the greatest element of $\mathbb{N}$.

4.3.5 Proposition. *Let $(X, \leq)$ be a partially ordered set, and let A , Z and Y be subsets of X such that $A \subseteq Z \subseteq Y$.*

- (1) *If A has a greatest element, then $\max A$ is also the supremum of A in Y .*
- (2) *If A has a least element, then $\min A$ is also the infimum of A in Y .*
- (3) *If A has a supremum in Y that belongs to Z , then $\sup_{(Y, \leq)} A$ is also the supremum of A in Z . In particular, if $\sup_{(Y, \leq)} A$ belongs to A , then it is the greatest element of A .*
- (4) *If A has an infimum in Y that belongs to Z , then $\inf_{(X, \leq)} A$ is also the infimum of A in Z . In particular, if $\inf_{(Y, \leq)} A$ belongs to A , then it is the least element of A .*

Proof. (1) By definition, $\max A$ is an upper bound of A in Y . If M is an upper bound of A , since $(\max A) \in A$, one obtains $M \geq \max A$. So $\max A$ is the least upper bound of A in Y , namely $\max A = \sup_{(Y, \leq)} A$. By inverting the partial order of X , we deduce (2) from (1).

(3) Let $M \in Z$ be an upper bound of A . By definition one has $\sup_{(Y, \leq)} A \leq M$. If $\sup_{(Y, \leq)} A$ belongs to Z , then it is the least upper bound of A in Z , namely the following equality holds:

$$\sup_{(Y, \leq)} A = \sup_{(Z, \leq)} A.$$

By inverting the partial order of X , we deduce (4) from (3). $\square$

4.3.6 Proposition. *Let $(X, \leq)$ be a partially ordered set, and let A, B and Y be subsets of X such that $A \subseteq B \subseteq Y$.*

(1) *If both A and B have suprema in Y , then $\sup_{(Y, \leq)} A \leq \sup_{(Y, \leq)} B$.*

(2) *If both A and B have infima in Y , then $\inf_{(Y, \leq)} A \geq \inf_{(Y, \leq)} B$.*

Proof. (1) Suppose that A and B have suprema in Y . By definition $\sup_{(Y, \leq)} B$ is an upper bound of B in Y , hence it is also an upper bound of A in Y since $A \subseteq B$. Therefore, $\sup_{(Y, \leq)} A \leq \sup_{(Y, \leq)} B$.

Inverting the partial order of X , we deduce (2) from (1). $\square$

4.3.7 Proposition. *Let $(X, \leq)$ be a partially ordered set, I be a set, f and g be two functions from I to X .*

(1) *Assume that the domain of definition of g is contained in that of f , and, for any element t of the domain of definition of g , one has $f(t) \leq g(t)$. If both $\inf f$ and $\inf g$ exist, then the following inequality holds:*

$$\inf f \leq \inf g.$$

(2) *Assume that the domain of definition of f is contained in that of g , and for any element t of the domain of definition of f , one has $f(t) \leq g(t)$. If both $\sup f$ and $\sup g$ exist, then the following inequality holds:*

$$\sup f \leq \sup g.$$

Proof. (1) For any s in the domain of definition of g , one has

$$\inf f \leq f(s) \leq g(s).$$

Hence $\inf f$ is a lower bound of $g(I)$. Therefore

$$\inf f \leq \inf g.$$

By inverting the partial order of X , we deduce (2) from (1). $\square$

4.3.8 Proposition. *Let I be a totally ordered set, X be a partially ordered set, $f : I \rightarrow X$ be a mapping, and J be a subset of I . Assume that J does not have any upper bound in I .*

(1) *If f is increasing, then $f(I)^u = f(J)^u$.*

(2) *If f is decreasing, then $f(I)^\ell = f(J)^\ell$.*

Proof. (1) Since $f(J) \subseteq f(I)$, any upper bound of $f(I)$ is also an upper bound of $f(J)$. Conversely, suppose that M is an upper bound of $f(J)$. Let $x \in I$. Since I is totally ordered and J does not have any upper bound in I , there exists $y \in J$ such that $x < y$. Since f is increasing, one obtains $f(x) \leq f(y) \leq M$. Hence M is also an upper bound of $f(I)$.

By inverting the partial order of X , we deduce (2) from (1). $\square$

4.3.9 Proposition. *Let $(X, \leq)$ be partially ordered set, Y be a subset of X , $(A_i)_{i \in I}$ be a family of subsets of Y , and A be the union of $(A_i)_{i \in I}$.*

(1) *Suppose that each A_i has a supremum y_i in Y . Then*

$$A^u = \{y_i \mid i \in I\}^u.$$

In particular, the set A has a supremum in Y if and only if $\{y_i \mid i \in I\}$ has a supremum in Y . Moreover, in the case where these suprema exist, the following equality holds:

$$\sup_{(Y, \leq)} A = \sup_{(Y, \leq)} \{y_i \mid i \in I\}.$$

(2) *Suppose that each A_i has a infimum z_i in Y . Then*

$$A^\ell = \{z_i \mid i \in I\}^\ell.$$

In particular, the set A has an infimum in Y if and only if $\{z_i \mid i \in I\}$ has an infimum in Y . Moreover, in the case where these infima exist, the following equality holds:

$$\inf_{(Y, \leq)} A = \inf_{(Y, \leq)} \{z_i \mid i \in I\}.$$

Proof. It suffices to treat the first statement. Let y be an element of Y . Note that the following statement are equivalent:

- (a) y is an upper bound of A ,
- (b) for any $i \in I$, y is an upper bound of A_i ,
- (c) $\forall i \in I, y_i \leq y$.

Therefore, the sets A and $\{y_i \mid i \in I\}$ have the same upper bounds in Y . The statement is thus proved. $\square$

4.4 Intervals

4.4.1 Notation. Let $(X, \leq)$ be a totally ordered set. For $(a, b) \in X \times X$, we denote the set

$$\{x \in X \mid a \leq x \leq b\}$$

as $[a, b]$. Note that $[a, b]$ is empty when $a \not\leq b$.

4.4.2 Definition. Let $(X, \leq)$ be a partially ordered set, and I be a subset of X . If, for any $(x, y) \in I \times I$, one has $[x, y] \subseteq I$, then we say that I is an *interval* in X .

4.4.3 Example. Let $(X, \leq)$ be a partially ordered set. For $(a, b) \in I \times I$, we define the sets $]a, b[$, $[a, b[$, $]a, b]$ and $[a, b]$ as follows.

$$\begin{aligned}]a, b[&:= \{x \in X \mid a < x < b\}, \\ [a, b[&:= \{x \in X \mid a \leq x < b\}, \\]a, b] &:= \{x \in X \mid a < x \leq b\}, \\ [a, b] &:= \{x \in X \mid a \leq x \leq b\}. \end{aligned}$$

These sets are all intervals in X .

Moreover, for any $x \in X$, the sets

$$\begin{aligned} X_{<a} &:= \{x \in X \mid x < a\}, \\ X_{\leq a} &:= \{x \in X \mid x \leq a\}, \\ X_{>a} &:= \{x \in X \mid x > a\}, \\ X_{\geq a} &:= \{x \in X \mid x \geq a\} \end{aligned}$$

are all intervals in X .

Finally the set X itself is also an interval in X .

4.4.4 Proposition. Let $(X, \leq)$ be a partially ordered set and $(I_\alpha)_{\alpha \in \Lambda}$ be a family of intervals in X parametrised by a non-empty set Λ .

- (1) The intersection $\bigcap_{\alpha \in \Lambda} I_\alpha$ is an interval.
- (2) Suppose that $(X, \leq)$ is a totally ordered set. If $\bigcap_{\alpha \in \Lambda} I_\alpha$ is not empty, then $\bigcup_{\alpha \in \Lambda} I_\alpha$ is an interval.

Proof. (1) Let a and b be elements of $\bigcap_{\alpha \in \Lambda} I_\alpha$. For any $\alpha \in \Lambda$, one has $\{a, b\} \subseteq I_\alpha$ and hence $[a, b] \subseteq I_\alpha$ since I_α is an interval. Therefore we obtain $[a, b] \subseteq \bigcap_{\alpha \in \Lambda} I_\alpha$, as required.

(2) Let x be an element of $\bigcap_{\alpha \in \Lambda} I_\alpha$. Let a and b be two elements of $\bigcup_{\alpha \in \Lambda} I_\alpha$ such that $a \leq b$. Let α and β be elements of Λ such that $a \in I_\alpha$ and $b \in I_\beta$. We will show that $[a, b] \subseteq I_\alpha \cup I_\beta$.

If $b \leq x$ then one has $[a, b] \subseteq [a, x] \subseteq I_\alpha$, where the second inclusion follows from the hypothesis that I_α is an interval and the condition $\{a, x\} \subseteq I_\alpha$. Similarly, if $x \leq a$, then one has $[a, b] \subseteq [x, b] \subseteq I_\beta$. If $a < x < b$, then $[a, b] = [a, x] \cup [x, b]$. Since $[a, x] \subseteq I_\alpha$ and $[x, b] \subseteq I_\beta$, we obtain $[a, b] \subseteq I_\alpha \cup I_\beta$. $\square$

4.4.5 Definition. Let $(X, \leq)$ be a partially ordered set and I be a non-empty interval in X . If I has a supremum in X , then $\sup I$ is called the *right endpoint* of X . If I has an infimum in X , then $\inf I$ is called the *left endpoint* of X .

4.4.6 Proposition. Let $(X, \leq)$ be a totally ordered set and I be an interval in X .

- (1) Assume that I has a supremum b in X . For any $x \in I$, one has $[x, b[\subseteq I$.
- (2) Assume that I has an infimum a in X . For any $x \in I$, one has $]a, x] \subseteq I$.

Proof. (1) Let y be an element of X such that $x \leq y < b$. Since b is the supremum of I , y cannot be an upper bound of I , namely, there exists $z \in I$ such that $y < z$. Since I is an interval, one has $[x, z] \subseteq I$. Hence $y \in I$.

If we apply (1) to $(X, \geq)$, then we obtain (2). $\square$

4.4.7 Proposition. Let $(X, \leq)$ be a totally ordered set and I be a non-empty interval in X . Assume that I has a supremum b and an infimum a in X . Then I is equal to one of the following intervals: $[a, b]$, $]a, b]$, $[a, b[$, $]a, b[$.

Proof. Since I is non-empty, one has $a \leq b$. For any $x \in I$ one has $a \leq x \leq b$. Hence $I \subseteq [a, b]$. If $\{a, b\} \subseteq I$, then by the hypothesis that I is an interval, one has $[a, b] \subseteq I$. Therefore $I = [a, b]$.

If $a \in I$ but $b \notin I$, then $I \subseteq [a, b[$. Moreover, any element $t \in [a, b[$ is not an upper bound of I , otherwise b cannot be the least upper bound of I . Hence, there exists $x \in I$ such that $x > t$. By the hypothesis of the proposition, one has $[a, x] \subseteq I$ and hence $t \in I$. Thus we obtain $[a, b[\subseteq I$. By the same method, one can prove that, if $b \in I$ and $a \notin I$, then $I =]a, b]$.

Suppose that $\{a, b\} \cap I = \emptyset$. Then $I \subseteq]a, b[$. For any $t \in]a, b[$, there exists elements x and y of I such that $x < t < y$. By the hypothesis of the proposition, one has $[x, y] \subseteq I$ and hence $t \in I$. Therefore $I =]a, b[$. $\square$

4.4.8 Definition. Let $(X, \leq)$ be a totally ordered set. If for any couple (x, z) of elements of X such that $x < z$, there exists $y \in X$ such that $x < y < z$, we say that the totally ordered set $(X, \leq)$ is *dense*.

4.4.9 Example. The set $\mathbb{Q}$ equipped with the usual partial order $\leq$ is dense. However, the set $\mathbb{Z}$ equipped with the usual partial order $\leq$ is not dense.

4.4.10 Proposition. *Let $(X, \leq)$ be a dense totally ordered set, and $(a, b) \in X \times X$ such that $a \leq b$. If I is one of the intervals $[a, b]$, $]a, b]$, $[a, b[$ and $]a, b[$, then the following equalities hold:*

$$\inf I = a, \quad \sup I = b.$$

Proof. By definition, b is an upper bound of I . Since $(X, \leq)$ is a totally ordered set, if b is not the supremum of I , then there exists an upper bound M of I such that $M < b$. Pick an element $x \in I$. By definition one has $x \leq M < b$. Hence by (1) of Proposition 4.4.6 one obtains $M \in I$. Therefore M is the greatest element of I . However, since X is dense, there exists M' such that $M < M' < b$. Still by (1) of Proposition ?? one has $M' \in I$, which leads to a contradiction. Hence b is the supremum of I in X . Applying the above argument to $(X, \geq)$, we obtain that a is the infimum of I in $(X, \leq)$. $\square$

4.5 Order-completeness

4.5.1 Definition. Let $(X, \leq)$ be a partially ordered set. If any subset of X has a supremum in X , we say that $(X, \leq)$ is *order-complete*. Note that an order-complete partially ordered set is never empty.

4.5.2 Proposition. *Let $(X, \leq)$ be an order-complete partially ordered set. Then any subset of X has an infimum in X . Moreover, X has a greatest element and a least element.*

Proof. Let A be a subset of X and L_A be the set of all lower bounds of A in X . Since $(X, \leq)$ is an order-complete partially ordered set, L_A admits a supremum in X , which we denote by m . By definition, any element a of A is an upper bound of L_A . Hence $m \leq a$. This shows that m is a lower bound of A , namely $m \in L_A$. By (3) of Proposition 4.3.5, one obtains that m is actually the greatest element of L_A , namely $m = \inf A$. By definition, the infimum of $\emptyset$ is the greatest element of X , the supremum of $\emptyset$ is the least element of X . $\square$

4.5.3 Example. Let A be a set. The power set $\mathcal{P}(A)$ equipped with the relation of inclusion forms a partially ordered set that is order-complete. Specifically, if $(A_i)_{i \in I}$ is a family of subsets of A , then the supremum of the family in $\mathcal{P}(A)$ is given by $\bigcup_{i \in I} A_i$. If I is non-empty, then the infimum of the family is $\inf_{i \in I} A_i = \bigcap_{i \in I} A_i$. If I is empty, then the infimum of the empty family in $\mathcal{P}(A)$ is A .

4.5.4 Definition. Let X be a set and f be a function from X to X . If x is an element of X such that $f(x) = x$, we say that x is a *fixed point* of f .

4.5.5 Theorem (Knaster-Tarski fixed point theorem). *Let $(X, \leq)$ be an order-complete partially ordered set and $f : X \rightarrow X$ be an increasing mapping. Then the set of fixed points*

$$F = \{x \in X \mid f(x) = x\},$$

equipped with the induced partial order, forms an order-complete partially ordered set. In particular, f has at least one fixed point.

Proof. Let A be a subset of F . We consider the following subset of X :

$$S_A = \{y \in X \mid y \text{ is an upper bound of } A \text{ and } f(y) \leq y\}.$$

Let m be the infimum of S_A . Since S_A is a subset of the set of upper bounds of A , by (2) of Proposition 4.3.6, one obtains $\sup A \leq m$, namely m is an upper bound of A . Since f is increasing, for any $y \in S_A$, one has

$$f(m) \leq f(y) \leq y.$$

Therefore $f(m) \leq m$. Moreover, since f is increasing, for any $x \in A$ one has

$$x = f(x) \leq f(m),$$

and one has $f(f(m)) \leq f(m)$. Therefore, $f(m)$ belongs to S_A and hence $m \leq f(m)$. Thus m is a fixed point of f . Finally, if y is a fixed point of f and is an upper bound of A , then $y \in S_A$ and hence $m \leq y$. Therefore, m is the least upper bound of A in F , namely the supremum of A in F . $\square$

4.5.6 Theorem (Cantor-Bernstein). *Let X and Y be sets. Assume that there exists an injective mapping from X to Y and an injective mapping from Y to X , then there exists a bijection from X to Y .*

Proof. Let $f : X \rightarrow Y$ and $g : Y \rightarrow X$ be injective mappings. We intend to find a subset C of X such that the restriction of g to $Y \setminus f(C)$ has $X \setminus C$ as its image. Then the mapping $h : X \rightarrow Y$ such that

$$\forall x \in X, \quad h(x) = \begin{cases} f(x), & \text{if } x \in C, \\ g^{-1}(x), & \text{if } x \in X \setminus C \end{cases}$$

is a bijection. By Example 4.5.3, the partially ordered set $(\mathcal{P}(X), \subseteq)$ is set-complete. We consider the mapping $\varphi : \mathcal{P}(X) \rightarrow \mathcal{P}(X)$ defined as follows:

$$\forall A \subseteq X, \quad \varphi(A) = X \setminus g(Y \setminus f(A)).$$

If A and B are two subsets of X such that $A \subseteq B$, then one has $f(A) \subseteq f(B)$ and furthermore

$$(Y \setminus f(B)) \subseteq (Y \setminus f(A)), \quad g(Y \setminus f(B)) \subseteq g(Y \setminus f(A)),$$

which leads to $\varphi(A) \subseteq \varphi(B)$. Thus φ is increasing. By Theorem 4.5.5, the mapping φ admits a fixed point C , as required. $\square$

4.5.7 Lemma. *Let $(X, \leq)$ be a partially ordered set.*

- (1) *Let A and B be subsets of X such that $A \subseteq B$. Then $B^u \subseteq A^u$, $B^\ell \subseteq A^\ell$.*
- (2) *For any $A \in \mathcal{P}(X)$, we have $A \subseteq (A^u)^\ell \cap (A^\ell)^u$.*

Proof. (1) Since $A \subseteq B$, it follows that every upper bound of B is an upper bound of A , and every lower bound of B is a lower bound of A .

(2) Let $a \in A$. For any $x \in A^u$, we have $a \leq x$, so a is a lower bound of A^u . Similarly, for any $y \in A^\ell$, we have $y \leq a$, so a is an upper bound of A^ℓ . $\square$

4.5.8 Theorem (MacNeille). *Let $(X, \leq)$ be a partially ordered set. Define*

$$\widehat{X} := \{A \in \mathcal{P}(X) \mid (A^u)^\ell = A\}.$$

- (1) *$(\widehat{X}, \subseteq)$ forms an order complete partially ordered set.*
- (2) *For any $A \in \mathcal{P}(X)$, $A^\ell \in \widehat{X}$.*
- (3) *The mapping*

$$X \longrightarrow \widehat{X}, \quad (x \in X) \longmapsto \widehat{x} := \{x\}^\ell$$

is strictly increasing.

- (4) *For any $A \in \widehat{X}$, the following equality holds:*

$$A = \bigcup_{x \in A} \widehat{x}.$$

In particular, A is the supremum of $\{\widehat{x} \mid x \in A\}$ in $\widehat{X}$.

- (5) *Let $A \in \widehat{X}$. If $A^u = \emptyset$, then $A = X$; if A^u is nonempty, then*

$$A = \bigcap_{x \in A^u} \widehat{x}.$$

In particular, A is the infimum of $\{\widehat{x} \mid x \in A^u\}$ in $\widehat{X}$.

Proof. (1) By Lemma 4.5.7 (1), the map

$$\varphi : \mathcal{P}(X) \longrightarrow \mathcal{P}(X), \quad A \longmapsto (A^u)^\ell$$

is increasing. By definition, $\widehat{X}$ is the set of all fixed points of φ . Since the poset $(\mathcal{P}(X), \subseteq)$ is order complete (Example 4.5.3), by Theorem 4.5.5 we obtain the order completeness of $(\widehat{X}, \subseteq)$.

(2) Apply the first inclusion of Lemma 4.5.7 (2) to A^ℓ to get $A^\ell \subseteq ((A^\ell)^u)^\ell$. Then apply the second inclusion of Lemma 4.5.7 (2) to A to get $A \subseteq (A^\ell)^u$. Finally, by Lemma 4.5.7 (1), we have $((A^\ell)^u)^\ell \subseteq A^\ell$.

(3) If $x < y$, then every lower bound of x is a lower bound of y , i.e., $\hat{x} \subseteq \hat{y}$. Moreover, y is a lower bound of itself but not of x , so $\hat{x} \neq \hat{y}$.

(4) First, for any $x \in A$, we have $x \in \hat{x}$. Hence

$$A \subseteq \bigcup_{x \in A} \hat{x}.$$

Conversely, for any $x \in A$, x is the least element of $\{x\}^u$, so

$$\{x\}^\ell = (\{x\}^u)^\ell \subseteq (A^u)^\ell = A,$$

where the inclusion follows from Lemma 4.5.7 (1). This yields

$$\bigcup_{x \in A} \hat{x} \subseteq A,$$

and thus the equality $A = \bigcup_{x \in A} \hat{x}$ holds. By Example 4.5.3, A is the supremum of $\{\hat{x} \mid x \in A\}$ in $\mathcal{P}(X)$. By Proposition 4.3.5, we deduce that A is the supremum of $\{\hat{x} \mid x \in A\}$ in $\hat{X}$.

(5) If $A^u = \emptyset$, then

$$A = (A^u)^\ell = \emptyset^\ell = X.$$

Now assume A^u is nonempty. By definition,

$$\bigcap_{x \in A^u} \hat{x} = \bigcap_{x \in A^u} \{a \in X \mid a \leq x\} = (A^u)^\ell = A.$$

By Example 4.5.3, A is the infimum of $\{\hat{x} \mid x \in A^u\}$ in $\mathcal{P}(X)$. By Proposition 4.3.5, we deduce that A is the infimum of $\{\hat{x} \mid x \in A^u\}$ in $\hat{X}$. $\square$

4.5.9 Remark. Let $(X, \leq)$ be a partially ordered set, $A \in \hat{X}$, and x be an element of X . If $\hat{x} \subseteq A$, then by $x \in \hat{x}$ we obtain $x \in A$. Conversely, from the proof of Theorem 4.5.8 (4), we observe that, for any $x \in A$, we have $\hat{x} \subseteq A$. Hence

$$A = \{x \in X \mid \hat{x} \subseteq A\}.$$

Similarly, $x \in X$ is an upper bound of A if and only if $A \subseteq \hat{x}$, namely

$$A^u = \{x \in X \mid A \subseteq \hat{x}\}.$$

4.5.10 Definition. Let $(X, \leq)$ be a partially ordered set. Then the set

$$\widehat{X} = \{A \in \mathcal{P}(X) \mid (A^u)^\ell = A\}$$

under the inclusion relation $\subseteq$ forms an order complete partially ordered set, called the *Dedekind-MacNeille order completion* of X . Theorem 4.5.8 shows that the mapping $X \rightarrow \widehat{X}$, $x \mapsto \widehat{x}$ is strictly increasing. Via this mapping, we can identify X with a subset of $\widehat{X}$ and regard the inclusion relation on $\widehat{X}$ as an extension of the partial order $\leq$.

4.6 Mathematical induction

4.6.1 Definition. Let $(X, \leq)$ be a partially ordered set. If any non-empty subset of X has a least element, we say that $\leq$ is a *well-order* and that $(X, \leq)$ is a *well-ordered set*.

4.6.2 Proposition. Let $(X, \leq)$ be a well-ordered set and Y be a subset of X . Then $(Y, \leq)$ is also a well-ordered set.

Proof. Let A be a non-empty subset of Y . It is also a non-empty subset of X . Hence it has a least element. $\square$

4.6.3 Proposition. A well-ordered set is totally ordered.

Proof. Let $(X, \leq)$ be a well-ordered set. If x and y are two elements of X , then $\{x, y\}$ has a least element. If this least element is x , then $x \leq y$, otherwise $y \leq x$. Hence $(X, \leq)$ is a totally ordered set. $\square$

4.6.4 Axioms. We equip $\mathbb{N}$ with the usual partial order $\leq$ defined as follows:

$$a \leq b \text{ if and only if } \exists c \in \mathbb{N}, b = a + c.$$

Then $(\mathbb{N}, \leq)$ forms a well-ordered set. Proposition 4.6.2 shows that any subset of $\mathbb{N}$ equipped with the restricted partial order is a well-ordered set.

4.6.5 Theorem (Induction). Let $(X, \leq)$ be a well-ordered set and $P(\cdot)$ be a condition on X . Suppose that

$$\forall x \in X, (\forall y \in X_{<x}, P(y)) \Rightarrow P(x). \quad (4.1)$$

Then the following statement holds:

$$\forall x \in X, P(x).$$

Proof. Let

$$A = \{x \in X \mid \neg P(x)\}.$$

If A is not empty, then it admits a least element α . For any $\beta \in X_{<\alpha}$, one has $\beta \notin A$ and hence $P(\beta)$ is true. By the hypothesis of the theorem, we obtain that $P(\alpha)$ is true, which leads to a contradiction. Therefore, for any $x \in X$, $P(x)$ is true. □

4.6.6 Remark. We keep the notation and the assumptions of the previous theorem. The statement $\forall y \in X_{<x}, P(y)$ in (4.1) is called *induction hypothesis* for x . If for any $x \in X$ we could deduce $P(x)$ from the induction hypothesis for x , then Theorem 4.6.5 proves that the condition $P(\cdot)$ is satisfied by any element of X . Let α be the least element of X . The statement

$$(\forall y \in X_{<\alpha}, P(y)) \Rightarrow P(\alpha)$$

is true if and only if $P(\alpha)$ is true. Therefore, to check the condition (4.1), it is necessary to check that the statement $P(\alpha)$ is true.

4.6.7 Proposition. *Let $(X, \leq)$ be a totally ordered set and n be a non-zero natural number. For any family $(x_i)_{i \in \{1, \dots, n\}}$ of elements of X , the set $\{x_1, \dots, x_n\}$ admits a greatest element and a least element.*

Proof. We apply Theorem 4.6.5 to the well ordered set $\mathbb{N}_{>0}$.

In the case where $n = 1$, we observe that, for any $x_1 \in X$, the element x_1 is both greatest and least element of $\{x_1\}$.

Let n be a natural number such that $n \geq 2$. Assume that, for any $m \in \mathbb{N}_{>0}$ such that $m < n$, any family $(x_i)_{i \in \{1, \dots, m\}}$ of elements of X admits a greatest element and a least element.

Let $(x_i)_{i \in \{1, \dots, n\}}$ be a family of elements of X . By the induction hypothesis, the set $\{x_1, \dots, x_{n-1}\}$ admits a least element a and a greatest element A . Since X is totally ordered, either $a \leq x_n$ or $x_n \leq a$. If $a \leq x_n$, then a is the least element of $\{x_1, \dots, x_n\}$, otherwise x_n is the least element of $\{x_1, \dots, x_n\}$. Therefore $\{x_1, \dots, x_n\}$ admits a least element. A similar argument shows that either A or x_n is the greatest element of $\{x_1, \dots, x_n\}$.

By Theorem 4.6.5, we deduce the current proposition. □

4.6.8 Proposition. *Let $(X, \leq)$ be a partially ordered set and $x_1, \dots, x_n$ be elements of X where n is positive integer. Assume that $x_1 \leq \dots \leq x_n$ (see Notation 3.11.2). Then $x_1 \leq x_n$.*

Proof. We reason by induction on n . The case where $n = 1$ is trivial. In the following, we consider the case where $n \geq 2$ in assume that the statement is true

for less than n elements. The induction hypothesis then leads to $x_1 \leq x_{n-1}$. Since $x_{n-1} \leq x_n$, by the transitivity of the partial order $\leq$, we obtain $x_1 \leq x_n$, as required. $\square$

4.6.9 Corollary. *Let $(X, \leq)$ be a partially ordered set and $x_1, \dots, x_n$ be elements of X where n is a natural number which is greater than or equal to 2. Assume that $x_1 \leq \dots \leq x_n$ and $x_n \leq x_1$. Then all elements $x_1, \dots, x_n$ are equal.*

Proof. By Proposition 4.6.8, for any $i \in \{2, \dots, n\}$, from $x_1 \leq \dots \leq x_i$, we deduce $x_1 \leq x_i$, and, from $x_i \leq \dots \leq x_n \leq x_1$, we deduce $x_i \leq x_1$. Hence $x_1 = x_i$. $\square$

4.6.10 Definition. Let $(X, \leq)$ be a partially ordered set. If a subset I of X satisfies

$$\forall a \in I, X_{<a} \subseteq I,$$

then we say that I is an *initial segment* of X .

4.6.11 Example. Let $(X, \leq)$ be a partially ordered set. For any $a \in X$, the sets $X_{<a}$ is an initial segment of X . In fact, if x is an element of $X_{<a}$ and if y is an element of X such that $y < x$, then the inequality $y < a$ holds. Similarly, the set $X_{\leq a}$ is an initial segment of X , too.

4.6.12 Proposition. *Let $(X, \leq)$ be a totally ordered set, and I and J are initial segments of X . Then, either $I \subseteq J$ or $J \subseteq I$.*

Proof. Suppose that x is an element of $I \setminus J$. We claim that, for any $y \in J$ one has $y \leq x$ and hence $y \in I$ since I is a initial segment. We reason by contradiction in assuming that $y \not\leq x$. Since $\leq$ is a total order, one has $x < y$. Since J is an initial segment, one has $x \in X_{<y} \subseteq J$, which leads to a contradiction. $\square$

4.6.13 Proposition. *Let $(X, \leq)$ be a well-ordered set. If I is an initial segment of X such that $I \neq X$, then there exists a unique element $a \in X$ such that $I = X_{<a}$.*

Proof. Since $I \neq X$, the set $X \setminus I$ is not empty. Since $(X, \leq)$ is well ordered, $X \setminus I$ has a least element which we denote as a .

We claim that $I = X_{<a}$. In fact, for any $x \in I$ one has $x < a$ since otherwise by the hypothesis that I is an initial segment we obtain $a \in I$, which leads to a contradiction. Hence $I \subseteq X_{<a}$. Conversely, if y is an element of $X_{<a}$, then by the hypothesis that a is the least element of $X \setminus I$ we obtain $y \in I$.

Since a well-ordered set is totally ordered, for any $b \in X$ one has $X_{\geq b} = X \setminus X_{<b}$ and thus $b = \min(X \setminus X_{<b})$. Hence we obtain the uniqueness of $a \in X$ such that $I = X_{<a}$. $\square$

4.6.14 Proposition. *Let $(X, \leq)$ be a partially ordered set. If $(I_\lambda)_{\lambda \in \Lambda}$ is a family of initial segments of X , then the union $I = \bigcup_{\lambda \in \Lambda} I_\lambda$ is an initial segment of X .*

Proof. Let x be an element of I . There exists $\lambda \in \Lambda$ such that $x \in I_\lambda$. Since I_λ is an initial segment, we obtain

$$X_{<x} \subseteq I_\lambda \subseteq I.$$

Hence I is also an initial segment. $\square$

4.6.15 Theorem. *Let $(X, \leq)$ be a well-ordered set, and Y be a set. For any $x \in X$ and any mapping $h : X_{<x} \rightarrow Y$, we fix an element $\Phi(h)$ of Y . Then, there exists a unique mapping $f : X \rightarrow Y$ such that*

$$\forall x \in X, \quad f(x) = \Phi(f|_{X_{<x}}).$$

Proof. We first prove the uniqueness. Suppose that f and g are two mappings from X to Y such that

$$\forall x \in X, \quad f(x) = \Phi(f|_{X_{<x}}), \quad g(x) = \Phi(g|_{X_{<x}}).$$

Then, for any $x \in X$, one has

$$(\forall y \in X_{<x}, \quad f(y) = g(y)) \implies f(x) = g(x).$$

By Theorem 4.6.5 we obtain

$$\forall x \in X, \quad f(x) = g(x).$$

In the following, we prove the existence. Let $\mathcal{S}$ be the set of all initial segments S of X that satisfies the following condition:

$$\text{there exists } f_S : S \rightarrow Y \text{ such that } \forall x \in S, \quad f_S(x) = \Phi(f_S|_{X_{<x}}).$$

Note that the uniqueness part applied to $(S, \leq)$ shows that $f_S : S \rightarrow Y$ is the unique mapping that satisfies

$$\forall x \in S, \quad f_S(x) = \Phi(f_S|_{X_{<x}}).$$

Let $X_0 = \bigcup_{S \in \mathcal{S}} S$. By Proposition 4.6.14, X_0 is an initial segment of X .

For any $x \in X_0$, there exists $S \in \mathcal{S}$ such that $x \in S$. We claim that the value of $f_S(x)$ does not depend on the choice of S . In fact, if S and S' are two elements of $\mathcal{S}$ such that $x \in S \cap S'$. By Proposition 4.6.12, either $S \subseteq S'$ and $f_{S'}|_S = f_S$ (by the uniqueness of f_S), or $S' \subseteq S$ and $f_S|_{S'} = f_{S'}$. We denote by $f(x)$ the value $f_S(x)$. We thus obtain a mapping $f : X_0 \rightarrow Y$, that satisfies

$$\forall x \in X_0, \quad f(x) = \Phi(f|_{X_{<x}}).$$

In particular, one has $X_0 \in \mathcal{S}$. Hence it is the greatest element of $\mathcal{S}$.

Suppose that $X_0 \neq X$. By Proposition 4.6.13, there exists $a \in X$ such that $X_0 = X_{<a}$. We extend f to a mapping from $X_{\leq a}$ to Y that sends a to $\Phi(f)$. This construction shows that $X_{\leq a} \in \mathcal{S}$, which contradicts the fact that X_0 is the greatest element of $\mathcal{S}$. Therefore, one has $X_0 = X$, which concludes the theorem. $\square$

4.7 Finiteness and countability

4.7.1 Definition. Let A be a set. If there is an injective mapping from A to $\mathbb{N}$, we say that A is *countable*. If there is an injective mapping from A to $\mathbb{N}$ which is bounded from above (that is, the image of the mapping is bounded from above in $\mathbb{N}$, see (c) of Definition 4.3.1), we say that A is *finite*. By definition, if a subset I of $\mathbb{N}$ is bounded from above, then it is finite (one can consider the inclusion mapping from I to $\mathbb{N}$).

4.7.2 Lemma. (1) Let n be a natural number and let $x_0, \dots, x_n$ be elements of $\mathbb{N}$ such that $x_0 < \dots < x_n$. Then, for any $i \in \{0, \dots, n\}$, one has $i \leq x_i$.

(2) Let $(x_i)_{i \in \mathbb{N}}$ be a family of natural numbers. Assume that

$$x_0 < x_1 < \dots < x_i < x_{i+1} < \dots$$

Then, for any $i \in \mathbb{N}$, one has $i \leq x_i$.

Proof. We reason by induction on $i \in \{0, \dots, n\}$ in (1), and on $i \in \mathbb{N}$ in (2).

Since $x_0 \in \mathbb{N}$, one has $0 \leq x_0$. Assume that

$$\forall j \in \{0, \dots, i-1\}, j \leq x_j.$$

By $x_i > x_{i-1}$, we obtain $x_i \geq x_{i-1} + 1$ since x_i and x_{i-1} are natural numbers. Hence $x_i \geq (i-1) + 1 = i$. By Theorem 4.6.5, we obtain the statement announced in the lemma. $\square$

4.7.3 Proposition. Let $f : A \rightarrow B$ be a mapping.

- (1) If f is injective and B is countable (resp. finite), then A is also countable (resp. finite). In particular, subsets of a countable (resp. finite) set are all countable (resp. finite).
- (2) If f is surjective and A is countable (resp. finite), then B is also countable (resp. finite). In particular, the quotient set of a countable (resp. finite) set by an equivalence relation is a countable (resp. finite) set.

Proof. (1) Assume that $\varphi : B \rightarrow \mathbb{N}$ is an injective mapping. Consider the composed mapping $\varphi \circ f : A \rightarrow \mathbb{N}$. By (2) of Proposition 3.6.4, $\varphi \circ f$ is injective. Moreover, by Proposition 3.4.5 one has $\text{Im}(\varphi \circ f) \subseteq \text{Im}(\varphi)$. If $\text{Im}(\varphi)$ is bounded from above, then $\text{Im}(\varphi \circ f)$ is also bounded from above. Therefore the countability (resp. finiteness) of B implies that of A . By applying this statement to an inclusion mapping, we obtain the second statement.

(2) By (1) of Proposition 3.9.6, there exists an injective mapping from B to A . Hence the first statement of (2) follows from the first statement of (1). By applying this statement to a quotient mapping, we obtain the second statement. $\square$

4.7.4 Proposition. *Let X and Y be sets.*

- (1) *If both X and Y are countable (resp. finite), then $X \cup Y$ is countable (resp. finite).*
- (2) *If X is uncountable (resp. infinite) and Y is countable (resp. finite), then $X \setminus Y$ is uncountable (resp. infinite).*

Proof. (1) Let $f : X \rightarrow \mathbb{N}$ and $g : Y \rightarrow \mathbb{N}$ be injective mappings. We construct a mapping $h : X \cup Y \rightarrow \mathbb{N}$ as follows:

$$h(x) := \begin{cases} 2f(x), & \text{if } x \in X, \\ 2g(x) + 1, & \text{if } x \in Y \setminus X. \end{cases}$$

Let x and y be two different elements of $X \cup Y$. If $\{x, y\} \subseteq X$, then, by the injectivity of f we deduce that

$$h(x) = 2f(x) \neq 2f(y) = h(y).$$

If $\{x, y\} \subseteq Y \setminus X$, then by the injectivity of g we deduce that

$$h(x) = 2g(x) + 1 \neq 2g(y) + 1 = h(y).$$

If one of x and y belongs to X and the other belongs to $Y \setminus X$, then one of $h(x)$ and $h(y)$ is even and the other is odd, and hence $h(x) \neq h(y)$. Therefore h is injective. Moreover, if both f and g are bounded, then h is also bounded. Therefore the statement holds.

(2) Assume that $X \setminus Y$ is countable (resp. finite). By (1) of Proposition 4.7.3, $X \cap Y$ is countable (resp. finite) since Y is countable (resp. finite). Therefore, we deduce from (1) that

$$X = (X \setminus Y) \cup (X \cap Y),$$

is countable (resp. finite). □

4.7.5 Notation. Let X be a set and $f : X \rightarrow X$ be a mapping. We denote by f^0 the identity mapping Id_X . In a recursive way, for any non-zero natural number n , we denote by f^n the composite mapping $f \circ f^{n-1}$.

4.7.6 Proposition. *Let X be a subset of $\mathbb{N}$.*

- (1) *The identity mapping Id_X is the only increasing bijection from X to itself.*
- (2) *If X is bounded from above, then the identity mapping Id_X is the only strictly increasing mapping from X to itself. In particular, for any proper subset Y of X , there does not exist any strictly increasing mapping from X to Y .*

Proof. Let $f : X \rightarrow X$ be a strictly increasing mapping and let

$$A = \{n \in \mathbb{N} \mid f(n) \neq n\}.$$

Note that, by (2) of Proposition 4.2.4, the mapping f is injective. To show (1) and the first statement of (2), it suffices to prove that, if $f \neq \text{Id}_X$, namely the set A is not empty, then f is not surjective and X is not bounded from above.

Assume that A is not empty. Since $(\mathbb{N}, \leq)$ is well-ordered, A admits a least element n_0 . By definition one has $f(n_0) \neq n_0$. Since $\mathbb{N}$ is totally ordered, either $f(n_0) < n_0$ or $n_0 < f(n_0)$. If $f(n_0) < n_0$, then $f(n_0) \notin A$ and hence $f(f(n_0)) = f(n_0)$, this contradicts the injectivity of f . Thus we should have $n_0 < f(n_0)$. For any $n \in X$, if $n_0 \leq n$, then

$$n_0 < f(n_0) \leq f(n);$$

if $n < n_0$, then

$$f(n) = n < n_0.$$

Hence in both cases $f(n) \neq n_0$. In particular, f is not surjective. Moreover, since f is strictly increasing, from $n_0 < f(n_0)$ we deduces that

$$n_0 < f(n_0) < f(f(n_0)) < \dots,$$

which implies that $f^k(n_0) \geq k$ for any $k \in \mathbb{N}$ by Lemma 4.7.2. Therefore X is not bounded from above.

It remains to prove the second statement of (2). Assume that there is a strictly increasing mapping $g : X \rightarrow Y$. Let $j_Y : Y \rightarrow X$ be the inclusion mapping. Then the composite mapping $j_Y \circ g$ is a strictly increasing mapping from X to X (see Example 4.2.2 and (2) of Proposition 4.2.3). However, the image of $j_Y \circ g$ is contained in Y and hence $j_Y \circ g$ can not be equal to Id_X . This contradicts the first statement of (2). $\square$

4.7.7 Proposition. *Let I be a non-empty subset of $\mathbb{N}$.*

- (1) *If I is bounded from above, then there exists a unique pair (N, f) , where N is a natural number and $f : \{0, \dots, N\} \rightarrow I$ is an increasing bijection.*
- (2) *If I is not bounded from above, then there exists a unique increasing bijection from $\mathbb{N}$ to I .*

Proof. We construct in a recursive way a family of distinct elements of I as follows. Let x_0 be the least element of I . If $x_0, \dots, x_n$ has been constructed and if $I \setminus \{x_0, \dots, x_n\}$ is not empty, we choose x_{n+1} to be the least element of $I \setminus \{x_0, \dots, x_n\}$.

By definition one has $x_n \neq x_{n+1}$. Moreover, since x_{n+1} belongs to $I \setminus \{x_0, \dots, x_n\}$ (in the case where $n = 0$, by convention $\{x_0, \dots, x_{n-1}\}$ denotes the empty set), one has $x_n \leq x_{n+1}$. Hence $x_n < x_{n+1}$. Therefore, by Lemma 4.7.2, we obtain that $n \leq x_n$ for any n . In particular, if I is bounded from above, then the recursive procedure terminates at a certain step, namely there exists $N \in \mathbb{N}$ such that $I = \{x_0, \dots, x_N\}$. The mapping from $\{0, \dots, N\}$ to I sending n to x_n is an increasing bijection. Suppose that M is another natural number and $g : \{0, \dots, M\} \rightarrow I$ is an increasing bijection. Note that

$$f^{-1} \circ g : \{0, \dots, M\} \longrightarrow \{0, \dots, N\} \text{ and } g^{-1} \circ f : \{0, \dots, N\} \longrightarrow \{0, \dots, M\}$$

are both strictly increasing maps. By (2) of Proposition 4.7.6, we obtain that $M = N$ and $f = g$.

If I is not bounded from above, then $I \setminus \{x_0, \dots, x_n\}$ is never bounded from above and the recursive procedure does not terminate. We then obtain a family $(x_n)_{n \in \mathbb{N}}$ of elements of I parametrised by $\mathbb{N}$, which corresponds to a strictly increasing mapping $f : \mathbb{N} \rightarrow I$ sending $n \in \mathbb{N}$ to x_n . If f is not surjective, the set $I \setminus f(\mathbb{N})$ admits a least element N . Then there exists $n \in \mathbb{N}$ such that $f(n) < N < f(n+1)$, this contradicts the way of construction of $f(n+1)$.

Suppose that $g : \mathbb{N} \rightarrow I$ is another increasing bijection. By Propositions 3.8.3 and 4.2.3, $g^{-1} \circ f$ is an increasing bijection from $\mathbb{N}$ to $\mathbb{N}$. By Proposition 4.7.6, one has $g^{-1} \circ f = \text{Id}_{\mathbb{N}}$ and hence $g = f$. $\square$

4.7.8 Corollary. *A set X is finite if and only if there exist a natural number N and a family $(x_n)_{n \in \{0, \dots, N\}}$ of elements of X such that $X = \{x_0, \dots, x_N\}$. In particular, a subset of $\mathbb{N}$ is finite if and only if it is bounded from above.*

Proof. Assume that $X = \{x_0, \dots, x_N\}$ where $N \in \mathbb{N}$. Then the mapping from $\{0, \dots, N\}$ to X sending $n \in \{0, \dots, N\}$ to x_n is surjective. Since $\{0, \dots, N\}$ is finite, by (2) of Proposition 4.7.3, the set X is finite.

Conversely, assume that $f : X \rightarrow \mathbb{N}$ is an injective mapping such that $f(X)$ is bounded. By (1) of Proposition 4.7.7, there exists $N \in \mathbb{N}$ and a bijection

$$g : \{0, \dots, N\} \rightarrow f(X).$$

Thus $f^{-1} \circ g : \{0, \dots, N\} \rightarrow X$ is a bijection and hence

$$X = \{f^{-1}(g(0)), \dots, f^{-1}(g(N))\}.$$

It remains to prove that a finite subset I of $\mathbb{N}$ is bounded from above (by definition a subset of $\mathbb{N}$ which is bounded from above is always finite). By the first statement, there exists a natural number N and a family $(x_n)_{n \in \{0, \dots, N\}}$ of elements of I such that $I = \{x_0, \dots, x_N\}$. By Proposition 4.6.7, I admits a greatest element, and hence is bounded from above. $\square$

4.7.9 Proposition. *Let X be a set. The following statements are equivalent.*

- (1) X is infinite.
- (2) There exists an injective mapping from $\mathbb{N}$ to X .
- (3) There exists an injective mapping from X to X the image of which is a proper subset of X .

Proof. “(1) $\implies$ (2)”: We construct in a recursive way a family of distinct elements $(x_n)_{n \in \mathbb{N}}$ as follows. Since X is infinite, it is not empty. We pick an arbitrary element x_0 of X . Suppose that $x_0, \dots, x_n$ are chosen. By Corollary 4.7.8, $\{x_0, \dots, x_n\}$ is finite. Therefore, by (2) of Proposition 4.7.4, the set $X \setminus \{x_0, \dots, x_n\}$ is infinite, and hence is non-empty. We pick an arbitrary element x_{n+1} in $X \setminus \{x_0, \dots, x_n\}$. Thus we obtain an injective mapping from $\mathbb{N}$ to X , sending $n \in \mathbb{N}$ to x_n .

“(2) $\implies$ (3)”: Let $f : \mathbb{N} \rightarrow X$ be an injective mapping. We define an injective mapping $g : X \rightarrow X$ as follows:

$$g(x) = \begin{cases} f(n+1), & \text{if } x = f(n), \\ x, & \text{if } x \notin f(\mathbb{N}). \end{cases}$$

Note that $g(X)$ is a proper subset of X since it does not contain $f(0)$.

“(3) $\implies$ (2)”: Let $g : X \rightarrow X$ be an injective mapping and x_0 be an element of $X \setminus g(X)$. For any $n \in \mathbb{N}$, let $x_n = g^n(x_0)$. Since g is injective, one has $x_n \in g^n(X) \setminus g^{n+1}(X)$. Hence the elements $(x_n)_{n \in \mathbb{N}}$ are distinct. Therefore one obtains an injective mapping from $\mathbb{N}$ to X .

“(2) $\implies$ (1)” follows from Corollary 4.7.8 (which shows that $\mathbb{N}$ is infinite) and Proposition 4.7.3. $\square$

4.7.10 Definition. Let X and Y be sets. If there exists a bijection from X to Y , we say that X and Y are *equipotent*.

4.7.11 Proposition. *Let X be an infinite set and Y a countable set. Then $X \cup Y$ is equipotent to X .*

Proof. Note that the equality $X \cup Y = X \cup (Y \setminus X)$ holds. Since $Y \setminus X$ is a countable set, we may assume without loss of generality that X and Y are disjoint. By Proposition 4.7.9, X has a subset A equipotent to $\mathbb{N}$. By Proposition 4.7.4, $A \cup Y$ is an infinite countable set, so there exists a bijection $\varphi : A \rightarrow A \cup Y$. Consider the map $\Phi : X \rightarrow X \cup Y$ defined by sending $x \in X \setminus A$ to x , and $a \in A$ to $\varphi(a)$. Since Φ is a bijection, the sets X and $X \cup Y$ are equipotent. $\square$

4.7.12 Lemma. *The set $\mathbb{N} \times \mathbb{N}$ is equipotent to $\mathbb{N}$.*

Proof. Consider the map from $\mathbb{N} \times \mathbb{N}$ to $\mathbb{N}$ that sends (a, b) to $2^a(2b+1) - 1$. First, we show this map is injective. Suppose (a, b) and (x, y) are two elements of $\mathbb{N} \times \mathbb{N}$ such that

$$2^a(2b+1) - 1 = 2^x(2y+1) - 1.$$

Assume without loss of generality that $a \leq x$. Then

$$2b+1 = 2^{x-a}(2y+1).$$

Hence $x = a$, otherwise $2b+1$ would be odd while $2^{x-a}(2y+1)$ would be even, a contradiction. This gives $2b+1 = 2y+1$ and thus $b = y$.

Let n be a natural number. Then $n+1$ is a positive integer, so it can be written as a power of 2 multiplied by an odd number. Therefore, the map is surjective. $\square$

4.7.13 Proposition. *For any positive integer k , the set $\mathbb{N}^k$ is equipotent to $\mathbb{N}$.*

Proof. The case where $k = 1$ is trivial. Assume that $\mathbb{N}^k$ is equipotent to $\mathbb{N}$ and let $f : \mathbb{N}^k \rightarrow \mathbb{N}$ be a bijection. Then the mapping

$$\mathbb{N}^{k+1} \longrightarrow \mathbb{N} \times \mathbb{N}, \quad (x_1, \dots, x_{k+1}) \longmapsto (f(x_1, \dots, x_k), x_{k+1})$$

is also a bijection. Its inverse is given by $(a, b) \longmapsto (f^{-1}(a), b)$. Therefore, $\mathbb{N}^{k+1}$ is equipotent to $\mathbb{N} \times \mathbb{N}$, and hence is equipotent to $\mathbb{N}$, by Lemma 4.7.12. $\square$

4.7.14 Proposition. *Let X be a set. There does not exist any surjective mapping from X to $\mathcal{P}(X)$. In particular, X and $\mathcal{P}(X)$ are not equipotent.*

Proof. We reason by contradiction that there exists a surjective mapping $f : X \rightarrow \mathcal{P}(X)$. Let

$$A = \{x \in X \mid x \notin f(x)\}.$$

Since f is surjective, there exists $y \in X$ such that $f(y) = A$. If $y \in A$, then by the definition of A one has $y \notin f(y) = A$, which leads to a contradiction; if $y \notin A$, then by the definition of A one has $y \in f(y) = A$, which is also contradictory. $\square$

4.8 Zorn's lemma

4.8.1 Lemma. *Let $(X, \leq)$ be a partially ordered set, $\mathcal{S}$ be a subset of $\mathcal{P}(X)$ which satisfies the following conditions:*

- (a) *For any $A \in \mathcal{S}$, $(A, \leq)$ is well ordered.*
- (b) *For any $(A, B) \in \mathcal{S} \times \mathcal{S}$, either A is an initial segment of B , or B is an initial segment of A .*

Let Y be the union of sets in $\mathcal{S}$. Then the following assertion holds:

- (1) Any $A \in \mathcal{S}$ is an initial segment of Y .
- (2) $(Y, \leq)$ is well ordered.

Proof. (1) Let x be an element of A and y be an element of Y such that $y < x$. Let B be an element of $\mathcal{S}$ such that $y \in B$. Assume by contradiction that y does not belong to A . Then $B \not\subseteq A$ and hence B is not an initial segment of A . By the condition (b), A is an initial segment of B . Hence $y \in A$, which leads to a contradiction.

(2) Let Z be a non-empty subset of Y . There then exists $A \in \mathcal{S}$ such that $A \cap Z \neq \emptyset$. Let m be the least element of $A \cap Z$. We show that m is also the least element of Z . Let z be an element of Z and B be an element of $\mathcal{S}$ such that $z \in B$. If z belongs to A , then by definition $m \leq z$. Otherwise A is an initial segment of B . Since B is well-ordered, by Proposition 4.6.3 we obtain that B is totally ordered. If $m \not\leq z$, then $z < m$, which implies that $z \in A$. This leads to a contradiction. $\square$

4.8.2 Definition. Let $(X, \leq)$ be a partially ordered set and A be a subset of X . We say that an element M of A is a *maximal element* of A if there does not exist any element $x \in A$ such that $M < x$. We say that an element m of A is a *minimal element* of A if there does not exist any element $x \in A$ such that $x < m$.

4.8.3 Remark. Let $(X, \leq)$ be a partially ordered set and A be a subset of X . If A has a greatest element (resp. least element), then its greatest element (resp. least element) is the unique maximal element (resp. minimal element) of A .

4.8.4 Theorem (Zorn's lemma). *Let $(X, \leq)$ be a non-empty partially ordered set. If any well-ordered subset of X has an upper bound in X , then X admits a maximal element.*

Proof. Suppose by contradiction that X does not have any maximal element.

Let $\mathcal{W}$ be the set of all well-ordered subsets of X . By the axiom of choice, there exists a mapping $g : \mathcal{W} \rightarrow X$ which sends any $A \in \mathcal{W}$ to an upper bound of A in X which does not belong to A (in the case where A has a greatest element, we could take an element of X strictly greater than $\max A$ by the assumption that X does not have any maximal element). If A is an element of $\mathcal{W}$ such that

$$\forall a \in A, \quad a = g(A_{<a}), \quad (4.2)$$

we say that A is a g -set. Let $\mathcal{S}$ be the subset of $\mathcal{W}$ that is composed of all g -sets. It is a non-empty subset of $\mathcal{W}$ since $\emptyset \in \mathcal{S}$. Moreover, by definition, if $A \in \mathcal{S}$, then $A \cup \{g(A)\} \in \mathcal{S}$.

Let A and B be two g -sets. We claim that, either A is an initial segment of B , or B is an initial segment of A . Let I be the union of subsets of $A \cap B$ which are common initial segments of A and B . By Proposition 4.6.14, I is also a common initial segment of A and B , and hence is the greatest (with respect to the relation of inclusion) common initial segment. If $I \subset A$ and $I \subset B$, by Proposition 4.6.13 there exist $a \in A$ and $b \in B$ such that

$$A_{<a} = I = B_{<b}.$$

Since A and B are both g -sets, by (4.2) we obtain

$$a = g(I) = b.$$

Therefore $I \cup \{a\}$ is a common initial segment of A and B which contains strictly I . This contradicts the maximality of I . Therefore, either $I = A$ and A is an initial segment of B , or $I = B$ and B is an initial segment of A .

By Lemma 4.8.1, the union $Y = \bigcup_{A \in \mathcal{S}} A$ is well ordered, and any $A \in \mathcal{S}$ is an initial segment of Y . For any $a \in Y$, there exists $A \in \mathcal{S}$ such that $a \in A$. Since A is an initial segment of Y , one has $Y_{<a} = A_{<a}$ and hence $g(Y_{<a}) = g(A_{<a}) = a$. Therefore Y belongs to $\mathcal{S}$ and hence it is the greatest element of $\mathcal{S}$ with respect to $\subseteq$. However this is not possible since $Y \cup \{g(Y)\}$ is also an element of $\mathcal{S}$. This leads to a contradiction. $\square$

4.8.5 Corollary. *Let $(X, \leq)$ be a non-empty partially ordered set. If any totally ordered subset of X has an upper bound in X , then X has a maximal element.*

Proof. This follows immediately from Theorem 4.8.4 and the fact that all well-ordered sets are totally ordered (see Proposition 4.6.3). $\square$

4.8.6 Theorem (Well-ordering principle). *Let X be a set. There exists a partial order $\leq$ on X such that $(X, \leq)$ is well-ordered.*

Proof. Let $\mathcal{S}$ be the set of all pairs $(A, \leq_A)$, where A is a subset of X and $\leq_A$ is a partial order on A such that $(A, \leq_A)$ is well-ordered. We consider the following binary relation $\preceq$ as follows: $(A, \leq_A) \preceq (B, \leq_B)$ if and only if $A \subseteq B$, $\leq_B$ extends $\leq_A$ and A is an initial segment of B . This is a partial order on $\mathcal{S}$. If $\mathcal{S}_0$ is a totally ordered subset of $\mathcal{S}$, by Lemma 4.8.1, the set $\mathcal{S}_0$ has an upper bound in $(\mathcal{S}, \preceq)$. Therefore, by Zorn's lemma the partially ordered set $(\mathcal{S}, \preceq)$ has a maximal element $(Y, \leq_Y)$. If $X \setminus Y$ is not empty and z is an element of $X \setminus Y$, we take $Z = Y \cup \{z\}$ and extends $\leq_Y$ to a partial order $\leq_Z$ on Z such that $y <_Z z$ for any $y \in Y$.

We claim that $(Z, \leq_Z)$ is well ordered. Let A be a non-empty subset of Z . If $A \cap Y \neq \emptyset$, then the least element of $A \cap Y = A \setminus \{z\}$ is also a least element of A .

If $A \cap Y = \emptyset$, then $A = \{z\}$ and z is the least element of A . Hence any non-empty subset of Z has a least element, namely $(Z, \leq_Z)$ is well ordered.

Note that Y is an initial segment of Z , which contradicts the maximality of Y . Therefore, one has $X = Y$ and $(X, \leq_Y)$ is well-ordered. $\square$

Exercises

1. Compare $7/13$ and $6/11$.
2. Let a and b be real numbers. Compare $a^2 + b^2$ and ab .
3. Compare $a = 1\,000\,000\,002^2$ and

$$b = 999\,999\,996 \times 1\,000\,000\,008.$$

4. Compare

$$\frac{2 + \sqrt{3}}{2 - \sqrt{3}} \text{ and } 7 + 4\sqrt{3}$$

5. Resolve in $\mathbb{R}$ the following inequalities

$$|x + 3| \leq 5, \quad |2x - 4| \leq |x + 2|, \quad |x + 1| + |x - 3| \leq 6.$$

6. Resolve in $\mathbb{R}$ the inequality $x - 1 \leq \sqrt{x + 2}$.
7. Does each of the following subsets of $\mathbb{R}$ have the least element? If it is the case, determine it.
 - (a) The set of even natural numbers.
 - (b) $\{-1, 0, 1\}$.
 - (c) The image of the function $\exp : \mathbb{R} \rightarrow \mathbb{R}$.

8. Determine if each of the following set has an upper bound in $\mathbb{R}$, a lower bound in $\mathbb{R}$? Determine their supremum and infimum.

$$A = \{x \in \mathbb{R} \mid x^2 < 2\}, \quad B = \{\frac{1}{n} \mid n \in \mathbb{N}, n \geq 1\}.$$

9. Determine the infimum of

$$\left\{ \frac{a}{b} + \frac{b}{a} \mid (a, b) \in \mathbb{R}_{>0}^2 \right\}$$

10. Let n be a positive integer. Determine

$$\inf \{ (x_1 + \cdots + x_n)(x_1^{-1} + \cdots + x_n^{-1}) \mid (x_1, \dots, x_n) \in \mathbb{R}_{>0}^n \}.$$

11. Let X be a set.

- (1) Prove that the relation of inclusion $\subseteq$ is an order relation on $\mathcal{P}(X)$.
- (2) Let I be a set and $(A_i)_{i \in I}$ be a family of elements of $\mathcal{P}(X)$. Prove that $\sup_{i \in I} A_i$ and $\inf_{i \in I} A_i$ exist.

12. In this exercise, we fix a partially ordered set $(X, \leq)$. We assume that X is *well-ordered*, namely, any non-empty subset of X has a least element. If a subset I of X satisfies

$$\forall a \in I, \quad X_{<a} := \{x \in X \mid x < a\} \subseteq I,$$

then we say that I is an *initial segment* of X .

- (1) Let a be an element of X . Prove that

$$X_{<a} = \{x \in X \mid x < a\}$$

is an initial segment of X .

- (2) Let I be an initial segment of X . Assume that $I \neq X$. Prove that there exists a unique element $a \in X$ such that $I = X_{<a}$.
- (3) Let Θ be a set and $(I_\lambda)_{\lambda \in \Theta}$ be a family of initial segments of X parametrised by Θ . Prove that $I = \bigcup_{\lambda \in \Theta} I_\lambda$ is an initial segment of X .
- (4) Let $(\mathbb{P}(x))_{x \in X}$ be a family of mathematical statements. Let 0 be the least element of X . Assume that
 - (a) the statement $\mathbb{P}(0)$ is true.
 - (b) for any $a \in X$, if $\mathbb{P}(b)$ is true for any $b \in X_{<a}$, then $\mathbb{P}(a)$ is true.
 Prove that, for any $x \in X$, $\mathbb{P}(x)$ is true.

13. Prove that, for any $n \in \mathbb{N}$, one has

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}.$$

14. Prove that, for any positive integer n , one has

$$\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}.$$

15. Let $(u_n)_{n \in \mathbb{N}}$ be the sequence in $\mathbb{R}$ defined as

$$u_0 = 1, \quad u_{n+1} = \sqrt{2 + u_n} \text{ for } n \in \mathbb{N}.$$

Prove that, for any $n \in \mathbb{N}$, $0 < u_n < 2$.

16. Let $(u_n)_{n \in \mathbb{N}}$ be the sequence in $\mathbb{R}$ defined as

$$u_0 = 1, \quad u_{n+1} = \sqrt{1 + u_n} \text{ for } n \in \mathbb{N}.$$

Prove that the sequence $(u_n)_{n \in \mathbb{N}}$ is increasing.

17. Prove that, for any positive integer n ,

$$2^{n-1} \leq n! \leq n^n.$$

18. Let x be a real number such that $x \geq -1$. Prove that, for any positive integer n , one has $(1 + x)^n \geq 1 + nx$.

19. Let $(u_n)_{n \in \mathbb{N}}$ be the sequence in $\mathbb{R}$ defined as

$$u_0 = 2, \quad u_1 = 3, \quad u_{n+2} = 3u_{n+1} - 2u_n \text{ for } n \in \mathbb{N}.$$

Prove that, for any $n \in \mathbb{N}$, $u_n = 2^n + 1$.

20. Consider the Fibonacci sequence $(u_n)_{n \in \mathbb{N}}$ defined as

$$u_0 = u_1 = 1, \quad u_{n+2} = u_n + u_{n+1} \text{ for } n \in \mathbb{N}.$$

(1) Prove that, for any $n \in \mathbb{N}$,

$$u_n u_{n+2} - u_{n+1}^2 = (-1)^n.$$

(2) Prove that the mapping

$$(n \in \mathbb{N}) \longrightarrow \frac{u_{2n+1}}{u_{2n}}$$

is increasing.

(3) Prove that the mapping

$$(n \in \mathbb{N}) \longrightarrow \frac{u_{2n+2}}{u_{2n+1}}$$

is decreasing.

(4) Let

$$\alpha = \frac{1 + \sqrt{5}}{2}, \quad \beta = \frac{1 - \sqrt{5}}{2}.$$

Prove that, for any $n \in \mathbb{N}$,

$$\alpha^{n+2} = \alpha^{n+1} + \alpha^n, \quad \beta^{n+2} = \beta^{n+1} + \beta^n.$$

(5) Find real numbers λ and μ such that

$$\lambda + \mu = 1, \quad \lambda\alpha + \mu\beta = 1.$$

21. Recall that $\sqrt{2}$ is irrational. Deduce that $\cos(1^\circ)$ is irrational. We could assume by contradiction that $\cos(1^\circ)$ is rational and prove by induction that $\cos(n^\circ)$ is rational for any $n \in \mathbb{N}$.

22. We denote by $\mathbb{N}$ the set of all natural numbers. We define a binary relation $|$ on $\mathbb{N}$ as follows: for any $(d, n) \in \mathbb{N} \times \mathbb{N}$, $d | n$ if and only if there exists $m \in \mathbb{N}$ such that $n = dm$. For any $d \in \mathbb{Z}$, we denote by $d\mathbb{Z}$ the set

$$\{dn \mid n \in \mathbb{Z}\}.$$

In this exercise, we could use without justification the following Euclidean division principle: if $d \in \mathbb{N} \setminus \{0\}$ and $n \in \mathbb{Z}$, there exists a unique element $(m, r) \in \mathbb{Z} \times \{0, \dots, d-1\}$ such that $n = dm + r$.

- (1) Prove that $|$ is a partial order on $\mathbb{N}$.
- (2) Let d and n be natural numbers such that $d | n$. Prove that, if $n \neq 0$, then $d \leq n$.
- (3) Prove that the partially ordered set $(\mathbb{N}, |)$ has a least element. Determine this element.
- (4) Prove that the partially ordered set $(\mathbb{N}, |)$ has a greatest element. Determine this element.
- (5) Let A be an infinite subset of $\mathbb{N}$. Prove that A has a supremum. Determine $\sup_{(\mathbb{N}, |)} A$.
- (6) Let A be a non-empty and finite subset of $\mathbb{N} \setminus \{0\}$. Let

$$M(A) := \{n \in \mathbb{N} \setminus \{0\} \mid \forall d \in A, d | n\}$$

- (a) Prove that $M(A)$ is not empty.
- (b) Let n_0 be the least element of $M(A)$ with respect to the usual partial order $\leq$. Prove that, for any $n \in M(A)$, one has $n_0 | n$.

- (c) Deduce that A has a supremum in $(\mathbb{N}, |)$.
- (7) Let A be a non-empty subset of $\mathbb{N} \setminus \{0\}$. We denote by $A\mathbb{Z}$ the set
- $$\{a_1n_1 + \dots + a_kn_k \mid k \in \mathbb{N} \setminus \{0\}, (a_1, \dots, a_k) \in A^k, (n_1, \dots, n_k) \in \mathbb{Z}^k\}.$$
- (a) Let x and y be elements of $A\mathbb{Z}$. Prove that $x + y \in A\mathbb{Z}$.
- (b) Let $x \in A\mathbb{Z}$ and $y \in \mathbb{Z}$. Prove that $xy \in A\mathbb{Z}$.
- (c) Prove that $A \subseteq A\mathbb{Z}$ and deduce that $(A\mathbb{Z}) \cap (\mathbb{N} \setminus \{0\})$ is not empty.
- (d) Let d be the least element of $(A\mathbb{Z}) \cap (\mathbb{N} \setminus \{0\})$ with respect to the usual partial order $\leq$. Prove that $A\mathbb{Z} = d\mathbb{Z}$.
- (e) Deduce that d is the infimum of A in $(\mathbb{N}, |)$.
- (8) Let A be a subset of $\mathbb{N}$. Prove that A has a supremum and an infimum in $(\mathbb{N}, |)$. We denote by

$$\gcd(A) := \inf_{(\mathbb{N}, |)} A, \quad \text{lcm}(A) := \sup_{(\mathbb{N}, |)} A.$$

- (9) Let a and b be positive integers, and $d = \gcd(\{a, b\})$. Prove that, there exists $(n, m) \in \mathbb{Z} \times \mathbb{Z}$ such that

$$d = an + bm.$$

- (10) Let a and b be positive integers. Prove that

$$\gcd(\{a, b\}) \text{lcm}(\{a, b\}) = ab.$$

23. If x and y are two sets, we denote by $x \subseteq y$ the statement: “either $x = y$, or $x \in y$ ”. We say that a set α is an ordinal if the elements of α are all subsets of α , and $(\alpha, \subseteq)$ forms a well-ordered set.

- (1) Let x and y be sets. Prove that $x \subseteq y$ if and only if $x \in y \cup \{y\}$.
- (2) Prove that $\emptyset$ is an ordinal.
- (3) Let α be a set. Assume that its elements are all sets and

$$(\alpha \cup \{\alpha\}, \subseteq)$$

forms a well ordered set, then α is an ordinal.

- (4) Let α be an ordinal. Prove that $\alpha \cup \{\alpha\}$ is an ordinal and

$$\alpha = \bigcup_{x \in \alpha \cup \{\alpha\}} x.$$

In the following questions, if α is an ordinal, we denote by $\alpha + 1$ the ordinal $\alpha \cup \{\alpha\}$.

- (5) Let α and β be ordinals. Prove that, if $\alpha + 1 = \beta + 1$, then $\alpha = \beta$.
- (6) Let α be an ordinal, and x and y be two elements of $\alpha + 1$. Prove that one and only one of the following three conditions holds:

$$x \in y, \quad x = y, \quad y \in x.$$

- (7) Let α be an ordinal. Prove that all elements of α are ordinals.
- (8) Let α and β be ordinals. Prove that if $\beta \subseteq \alpha$, then β is an initial segment of α .
- (9) Let α be an ordinal. Prove that $\alpha + 1$ identifies with the set of all initial segments of α . Deduce that all initial segments of α are ordinals.
- (10) Let α and β be ordinals. Prove that the following conditions are equivalent:
- (a) $\beta \in \alpha$,
 - (b) β is an initial segment of α and $\beta \neq \alpha$
 - (c) $\beta \subsetneq \alpha$.
- (11) Let I be a non-empty set and $(\alpha_i)_{i \in I}$ be a family of ordinals. Prove that

$$\bigcap_{i \in I} \alpha_i$$

is an ordinal.

- (12) Let α and β be ordinals. Prove that, either $\alpha \subseteq \beta$, or $\beta \subsetneq \alpha$.
- (13) Let A be a family of ordinals. Prove that $(A, \subseteq)$ is a well-ordered set and that the binary relations $\subseteq$ and $\subseteq$ coincide on A .
- (14) Let I be a set and $(\alpha_i)_{i \in I}$ be a family of ordinals. Prove that $\bigcup_{i \in I} \alpha_i$ is an ordinal.

24. Let α be an ordinal. The ordinal $\alpha + 1$ is called the *successor* of α . If an ordinal β is neither $\emptyset$ nor *successor* of any ordinal, we say that β is a *limit ordinal*. Let n be an ordinal, if none of the elements of $n + 1$ is a limit ordinal, we say that n is a *natural number*.

- (1) Let α be an ordinal. Prove that it is a successor of an ordinal if and only if

$$\bigcup_{x \in \alpha} x \subsetneq \alpha.$$

- (2) Let α be an ordinal. Prove that the following conditions are equivalent:

- (a) α is a limit ordinal.
 - (b) $\bigcup_{x \in \alpha} x = \alpha$.
 - (c) $\alpha \subseteq \bigcup_{x \in \alpha} x$.
- (3) We denote by 0 the ordinal number $\emptyset$. Prove that 0 is a natural number.
- (4) Prove that, if n is a natural number, then so is $n + 1$.
- (5) Let α be an ordinal number. Prove that α is a natural number if and only if any non-empty subset of α has a greatest element.
- (6) Prove that, if n is a natural number, then all of its elements are natural numbers.
- (7) Suppose that there exists a set A consisting of some sets such that $\emptyset \in A$ and that

$$\forall x \in A, \quad x \cup \{x\} \in A.$$

Prove that the set $\mathbb{N}$ of all natural numbers forms an ordinal. Moreover, prove that, if an ordinal α is not a natural number, then $\mathbb{N} \subseteq \alpha$.

- (8) Let $(\mathbb{P}(n))_{n \in \mathbb{N}}$ be a family of statements parametrized by natural numbers. We assume that $\mathbb{P}(0)$ holds and that $\mathbb{P}(n)$ implies $\mathbb{P}(n + 1)$ for any $n \in \mathbb{N}$. Prove that, for any $n \in \mathbb{N}$, $\mathbb{P}(n)$ holds.
- (9) Let $f : \mathbb{N} \rightarrow \mathbb{N}$ is an increasing bijection. Prove that f is the identity mapping of $\mathbb{N}$.

Chapter 5

Groups

5.1 Composition laws

5.1.1 Definition. Let X be a set. By *composition law* on X , we mean a mapping from $X \times X$ to X . If $*$ is a composition law on X , for $(x, y) \in X \times X$, we usually denote the image of (x, y) by the mapping $*$ as $x * y$.

5.1.2 Definition. Let X be a set and $*$ be a composition law on X . If Y is a subset such that $x * y \in Y$ for any $(x, y) \in Y \times Y$, we say that Y is *closed* under the composition law $*$. In this case the mapping $*$ induces by restriction a composition law on Y , called the *induced composition law* of $*$. We often use the symbol representing the composition law on X to denote the induced composition law on Y .

5.1.3 Definition. Let X be a set and $*$ be a composition law on X .

- (1) If, for any $(x, y) \in X \times X$, one has

$$x * y = y * x,$$

then we say that $*$ is *commutative*.

- (2) If, for any elements x, y and z of X , one has

$$(x * y) * z = x * (y * z),$$

then we say that $*$ is *associative*.

- (3) Assume that $*$ is associative. If n is a natural number such that $n \geq 2$ and $x_1, \dots, x_n$ are elements of X , we define $x_1 * \dots * x_n$ in a recursive way as

$$(x_1 * \dots * x_{n-1}) * x_n.$$

5.1.4 Definition. Let G be a set and $*$ be a composition law on G . If $*$ is associative, we say that $(G, *)$ is a *semigroup*. If in addition the composition law $*$ is commutative, we say that the semigroup $(G, *)$ is commutative.

5.1.5 Example. Let d be a natural number and let $\mathbb{N}_{\geq d}$ be the set of natural numbers which are $\geq d$. This set equipped with the additive law or with the multiplicative law, forms in both cases a commutative semigroup.

Similarly, the set

$$d\mathbb{N} := \{dn \mid n \in \mathbb{N}\},$$

equipped with the additive law or with the multiplicative law, forms in both cases a commutative semigroup.

5.1.6 Example. Let X be a set, $*$ be a composition law on X . Then the mapping

$$X \times X \longrightarrow X, \quad (x, y) \longmapsto y * x$$

defines a composition law on X , denoted as $\check{*}$, and called the *opposite* of $*$. Clearly, one has

$$\check{\check{*}} = *.$$

By definition $* = \check{*}$ if and only if $*$ is commutative. Moreover, if $*$ is associative, then, for any $(x, y, z) \in X \times X \times X$, one has

$$x \check{*} (y \check{*} z) = (z * y) * x = z * (y * x) = (x \check{*} y) \check{*} z.$$

Hence $\check{*}$ is also associative. In this case $(X, \check{*})$ is called the *opposite semigroup* of $(X, *)$.

5.1.7 Example. Let X be a set, $\mathfrak{M}_X$ be the set of all mappings from X to itself. The composition of mappings defines a composition law on $\mathfrak{M}_X$ as

$$\mathfrak{M}_X \times \mathfrak{M}_X \longrightarrow \mathfrak{M}_X, \quad (f, g) \longmapsto f \circ g.$$

By Proposition 3.4.3, the composition law $\circ$ is associative.

Suppose that X has two different elements a and b . Let $f : X \rightarrow X$ and $g : X \rightarrow X$ be constant mappings taking values a and b , respectively. Then $f \circ g = f$, $g \circ f = g$. This observation shows that $\circ$ is not commutative.

5.1.8 Notation. When considering a semigroup, if there is no ambiguity regarding the composition law of the semigroup, the symbol representing the underlying set can be used to denote the semigroup. Additionally, unless otherwise specified, multiplication is commonly used to represent the composition law of the semigroup. For example, if we denote by G a semigroup without specifying its composition law, then by convention its composition law is written as

$$((a, b) \in G \times G) \longmapsto a \cdot b.$$

Sometimes $a \cdot b$ is also written in an abbreviated way as ab .

5.1.9 Proposition. Let X be a set and $*$ be an associative composition law on X . If n is a non-zero natural number and $x_1, \dots, x_n$ are elements of X , then for any $i \in \{1, \dots, n-1\}$, the following equality holds:

$$x_1 * \dots * x_n = x_1 * \dots * x_{i-1} * (x_i * x_{i+1}) * x_{i+2} * \dots * x_n. \quad (5.1)$$

Proof. When $i = 1$, by definition

$$x_1 * \dots * x_n = (x_1 * x_2) * \dots * x_n.$$

We suppose $i \geq 2$. By the associativity of $*$, we get

$$\begin{aligned} x_1 * \dots * x_{i+1} &= ((x_1 * \dots * x_{i-1}) * x_i) * x_{i+1} \\ &= (x_1 * \dots * x_{i-1}) * (x_i * x_{i+1}) = x_1 * \dots * x_{i-1} * (x_i * x_{i+1}). \end{aligned}$$

Hence

$$x_1 * \dots * x_n = (x_1 * \dots * x_{i+1}) * \dots * x_n = x_1 * \dots * x_{i-1} * (x_i * x_{i+1}) * \dots * x_n.$$

□

5.1.10 Definition. Let $(G, *)$ be a set equipped with a composition law, g be an element of G .

If

$$\forall (x, y) \in G \times G, (g * x = g * y) \Rightarrow (x = y),$$

we say that g is *left cancellative*.

If

$$\forall (x, y) \in G \times G, (x * g = y * g) \Rightarrow (x = y),$$

then we say that g is *right cancellative*.

If $*$ is commutative, g is left cancellative if and only if it is right cancellative. In this case we just say that g is *cancellative*.

5.1.11 Example. In the additive semigroup $(\mathbb{N}, +)$, any element is cancellative. In the multiplicative semigroup $(\mathbb{N}, \cdot)$, any positive natural number is cancellative.

5.2 Neutral element, invertible elements

5.2.1 Definition. Let X be a set, $*$ be a composition law on X and e be an element of X . If, for any $x \in X$, one has

$$x * e = e * x = x,$$

then we say that e is a *neutral element* of $(X, *)$.

5.2.2 Proposition. *Let X be a set and $*$ be a composition law on X . Assume that $(X, *)$ admits a neutral element, then its neutral element is unique.*

Proof. Let e and e' be neutral elements of $(X, *)$. Then by definition one has

$$e = e * e' = e'.$$

□

5.2.3 Definition. Let $(G, *)$ be a semigroup. If $(G, *)$ has a neutral element, then we say that $(G, *)$ is a *monoid*. By definition, if $(G, *)$ admits a neutral element e , then e is also the neutral element of the opposite semigroup $(G, \check{*})$. In this case $(G, \check{*})$ is a monoid, called the opposite monoid of $(G, *)$.

5.2.4 Example. Let X be a set. Proposition 3.4.4 shows that Id_X is the neutral element of the monoid $(\mathfrak{M}_X, \circ)$. Hence $(\mathfrak{M}_X, \circ)$ forms a monoid.

5.2.5 Example. Let $d \in \mathbb{N}$, the semigroup $(d\mathbb{N}, +)$ has 0 as its neutral element. Thus $(d\mathbb{N}, +)$ forms a monoid. Moreover, $(\mathbb{N}, \cdot)$ also forms a monoid. Its neutral element is 1.

5.2.6 Definition. Let $(G, *)$ be a monoid, e be its neutral element, x and y be elements of G . If $x * y = e$, we say that x is a *left inverse* of y , and y is a *right inverse* of x .

Let $g \in G$. If g has a left inverse, we say that g is *left invertible*; if g has a *right inverse*, we say that g is *right invertible*. The equality $e * e = e$ shows that, the neutral element e is a left inverse and right inverse of itself.

5.2.7 Remark. Let $(G, *)$ be a monoid, $x \in G$. Then x is left invertible in $(G, *)$ if and only if it is right invertible in the opposite monoid $(G, \check{*})$; x is right invertible in $(G, *)$ if and only if it is left invertible in $(G, \check{*})$.

5.2.8 Proposition. *Let $(G, *)$ be a monoid, and $g \in G$. If g is left invertible and right invertible, then g has a unique left inverse and a unique right inverse, which actually coincide.*

Proof. Let e be the neutral element of $(G, *)$. Let x be a left inverse of g , y be a right inverse of g . By the associativity of $*$ we obtain

$$x = x * e = x * (g * y) = (x * g) * y = e * y = y.$$

This implies that any left inverse of g is equal to y , namely g has a unique left inverse. Similarly, any right inverse of g is equal to x , namely g has the unique right inverse. The above formula also shows that the left inverse and the right inverse of g are equal. □

5.2.9 Definition. Let $(G, *)$ be a monoid and $g \in G$. If g is left invertible and right invertible, then we simply say that g is *invertible*. In this case, Proposition 5.2.8 shows that the left inverse and the right inverse of g are equal. We just call it the *inverse* of g , usually denoted as $\iota(g)$.

5.2.10 Proposition. Let $(G, *)$ be a monoid, $g \in G$.

- (1) If g is right invertible, then it is right cancellative.
- (2) If g is left invertible, then it is left cancellative.

Proof. The second statement follows from the first one applied to the opposite monoid $(G, \check{*})$. Below we prove the first statement. Let e be the neutral element of $(G, *)$. Let h be the right inverse of g . If $x * g = y * g$, then the following inequalities hold

$$x = x * e = x * (g * h) = (x * g) * h = (y * g) * h = y * (g * h) = y * e = y.$$

□

5.2.11 Notation. Let G be a monoid.

If the composition law of G is written multiplicatively, unless otherwise specified, the neutral element of G is written as 1. Moreover, if an element g of $(G, \cdot)$ is invertible, then we denote by g^{-1} its inverse.

If the composition law of G is written additively, unless otherwise specified, the neutral element of G is written as 0. Moreover, if an element a of $(G, +)$ is invertible, then the inverse of a is often written as $-a$. If b is another element of G , then $b - a$ denotes $b + (-a)$.

5.2.12 Proposition. Let $(G, *)$ be a monoid.

- (1) If x is an invertible element of G , then $\iota(x)$ is also an invertible element of G , and $\iota(\iota(x)) = x$.
- (2) If x and y are invertible elements of G , then $x * y$ is also an invertible element of G , and

$$\iota(x * y) = \iota(y) * \iota(x).$$

Proof. Let e be the neutral element of $(G, *)$.

- (1) comes from the equality $x * \iota(x) = \iota(x) * x = e$.
- (2) By Proposition 5.1.9, we obtain

$$\begin{aligned} (x * y) * (\iota(y) * \iota(x)) &= x * (y * \iota(y)) * \iota(x) = x * e = x, \\ (\iota(y) * \iota(x)) * (x * y) &= \iota(y) * (\iota(x) * x) * y = \iota(y) * e = \iota(y). \end{aligned}$$

□

5.2.13 Definition. Let $(G, *)$ be a monoid. If any element of G is invertible, we say that $(G, *)$ is a *group*. A commutative group is also called an *Abelian group*.

5.2.14 Example. $(\mathbb{Z}, +)$ forms a group.

5.2.15 Example. Let X be a set. Denote by $\mathfrak{S}_X$ the set of all bijections from X to itself. Proposition 3.8.3 shows that the composition of mappings $\circ$ defines a structure of monoid on $\mathfrak{S}_X$ with Id_X as its neutral element. Proposition 3.8.2 shows that any element f of $\mathfrak{S}_X$ is invertible, and f^{-1} is its inverse. Hence $(\mathfrak{S}_X, \circ)$ forms a group, called the *symmetric group* of X . If n is a natural number, the symmetric group of $\{1, \dots, n\}$ is denoted as $\mathfrak{S}_n$.

5.2.16 Notation. Let $X = \{x_1, \dots, x_n\}$ be a finite set, where n is a natural number, $x_1, \dots, x_n$ are distinct element. Sometimes we denote an element f of $\mathfrak{S}_X$ as a table

$$\begin{pmatrix} x_1 & x_2 & \dots & x_n \\ f(x_1) & f(x_2) & \dots & f(x_n) \end{pmatrix}.$$

For example, the element

$$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

of $\mathfrak{S}_3$ sends 1 to 3, 2 to 1, and 3 to 2.

5.3 Substructure

5.3.1 Definition. Let $(G, *)$ be a semigroup, H be a subset of G . If H is stable by the composition law $*$, namely for any $(a, b) \in H \times H$ one has $a * b \in H$, then we say that H is a *subsemigroup* of $(G, *)$. Note that H equipped with the restriction of $*$ forms a semigroup.

Let $(G, *)$ be a monoid. If a subsemigroup H of $(G, *)$ contains the neutral element of $(G, *)$, then we say that H is a *submonoid* of $(G, *)$. Note that H equipped with the restriction of $*$ forms a monoid.

5.3.2 Example. Let $d \in \mathbb{N}$. Then $d\mathbb{N}$ forms a submonoid of $(\mathbb{N}, +)$. Note that $d\mathbb{N}$ is a subsemigroup of $(\mathbb{N}, \cdot)$. However, if $d \neq 1$, one has $1 \notin d\mathbb{N}$. In this case $d\mathbb{N}$ is not a submonoid of $(\mathbb{N}, \cdot)$.

5.3.3 Example. The set $\mathfrak{S}_X$ forms a submonoid of $(\mathfrak{M}_X, \circ)$.

5.3.4 Proposition. Let $(M, *)$ be a monoid, H be a non-empty subset of M . Suppose that any element of H is invertible in M , and

$$\forall (x, y) \in H \times H, \quad x * \iota(y) \in H.$$

Then H is a submonoid of M . Moreover, H equipped with the restriction of $*$ forms a group.

Proof. Let e be the neutral element of $(M, *)$. Let $a \in H$. Then $e = a * \iota(a) \in H$.

For any $y \in H$, one has $\iota(y) = e * \iota(y) \in H$. For any $(x, y) \in H \times H$, one has

$$x * y = x * \iota(\iota(y)) \in H.$$

Hence H is closed under $*$, and it contains the neutral element of M . Hence H is a submonoid of M .

For any $y \in H$, one has $\iota(y) \in H$. Hence H equipped with the restriction of $*$ forms a group. $\square$

5.3.5 Corollary. *Let $(M, *)$ be a monoid, G be the set of all invertible elements of M . Then G is a submonoid of M . Moreover, G equipped with the restriction of $*$ forms a group.*

Proof. By definition, any element of G is invertible in M . Moreover, by Proposition 5.2.12,

$$\forall (x, y) \in G \times G, \quad x * \iota(y) \in G.$$

Therefore, Proposition 5.3.4 implies that G is a submonoid of M , and forms a group under the restriction of $*$. $\square$

5.3.6 Notation. Let M be a monoid. If the composition law of M is *not* written additively, we often use $M^\times$ to denote the submonoid of M consisting of all invertible elements of M .

5.3.7 Example. Let X be a set. By Proposition 3.8.2, one has $\mathfrak{S}_X = \mathfrak{M}_X^\times$.

5.3.8 Definition. Let $(G, *)$ be a group, H be a submonoid of G . If, for any $x \in H$, one has $\iota(x) \in H$, then we say that H is a *subgroup* of G .

5.3.9 Proposition. *Let $(M, *)$ be a monoid, H be a non-empty subset of $M^\times$, such that*

$$\forall (x, y) \in H \times H, \quad x * \iota(y) \in H.$$

Then, H is a subgroup of $M^\times$.

Proof. Let e be the neutral element of $(M, *)$. By Proposition 5.3.4, we obtain that H forms a submonoid of $M^\times$. Moreover, for any $y \in H$, one has

$$\iota(y) = e * \iota(y) \in H.$$

Hence H is a subgroup of $M^\times$. $\square$

5.3.10 Proposition. *Let $(G, *)$ be a semigroup (resp., monoid, group), $(H_i)_{i \in I}$ be a family of subsemigroup (resp. submonoid, subgroup) of G , where I is a non-empty set. Then $H = \bigcap_{i \in I} H_i$ is a subsemigroup (resp. submonoid, subgroup) of G .*

Proof. We first consider the case of a semigroup. Let x and y be elements of H . For any $i \in I$, one has $\{x, y\} \subseteq H_i$, and hence $x * y \in H_i$. Hence $x * y \in H$. Therefore H is still a subsemigroup of G .

We then consider the case of monoids. From the previous case we obtain that H is a subsemigroup of G . Moreover, for any $i \in I$, the neutral element of G belongs to H_i . Hence H contains the neutral element of G . Therefore, H is a submonoid of G .

Finally, we consider the case of groups. Note that H is not empty since it contains the neutral element of G . Let x and y be elements of H . For any $i \in I$, one has $\{x, y\} \subseteq H_i$. Hence $x * \iota(y) \in H_i$. Therefore, $x * \iota(y) \in H$. By Proposition 5.3.9, H is a subgroup of G . □

5.4 Homomorphisms

5.4.1 Definition. Let $(M, *)$ and $(N, \star)$ be semigroups. If a mapping $f : M \rightarrow N$ preserves the operation, namely

$$\forall (a, b) \in M \times M, \quad f(a * b) = f(a) \star f(b)$$

then f is said to be a *semigroup homomorphism* from $(M, *)$ to $(N, \star)$.

If $(M, *)$ and $(N, \star)$ are both monoids, $f : M \rightarrow N$ is a semigroup homomorphism, and it maps the neutral element of M to the neutral element of N , then f is said to be a *monoid homomorphism* from $(M, *)$ to $(N, \star)$.

5.4.2 Notation. Let $(M, *)$ and $(N, \star)$ be semigroups (resp. monoids). The notation $f : (M, *) \rightarrow (N, \star)$ signifies that f is a semigroup homomorphism (resp. monoid homomorphism) from $(M, *)$ to $(N, \star)$.

5.4.3 Proposition. Let $f : (S, *) \rightarrow (T, \star)$ be a semigroup homomorphism (resp. monoid homomorphism). If S_1 is a subsemigroup (resp. submonoid) of S , then $f(S_1)$ is a subsemigroup (resp. submonoid) of T .

Proof. Let x and y be elements of $f(S_1)$. By definition, there exist elements a and b in S_1 such that $f(a) = x$ and $f(b) = y$. Since f is a semigroup homomorphism, we have

$$x \star y = f(a) \star f(b) = f(a * b) \in f(S_1).$$

Therefore, $f(S_1)$ is a subsemigroup of T .

If f is a monoid homomorphism and S_1 is a submonoid of S , then the neutral element of S belongs to S_1 , and its image under f is the neutral element of T . Hence, $f(S_1)$ is a submonoid of T . □

5.4.4 Proposition. *Let $f : (M_1, *) \rightarrow (M_2, \star)$ be a monoid homomorphism. For any $a \in M_1^\times$, we have $f(a) \in M_2^\times$, and $\iota(f(a)) = f(\iota(a))$. Furthermore, if G is a subgroup of $M_1^\times$, then $f(G)$ is a subgroup of $M_2^\times$.*

Proof. Let e_1 and e_2 be the neutral elements of M_1 and M_2 , respectively. For any $a \in M_1^\times$, we have

$$\begin{aligned} f(a) \star f(\iota(a)) &= f(a * \iota(a)) = f(e_1) = e_2, \\ f(\iota(a)) \star f(a) &= f(\iota(a) * a) = f(e_1) = e_2. \end{aligned}$$

Hence, $f(a)$ is invertible in M_2 , and $\iota(f(a)) = f(\iota(a))$.

Let G be a subgroup of $M_1^\times$. Let x and y be two elements of $f(G)$, and let a and b be elements of G such that $x = f(a)$ and $y = f(b)$. Then

$$x \star \iota(y) = f(a) \star \iota(f(b)) = f(a) \star f(\iota(b)) = f(a * \iota(b)) \in f(G).$$

Therefore, by Corollary 5.3.9, $f(G)$ is a subgroup of M_2 . □

5.4.5 Proposition. *Let $(M, *)$ be a monoid, $(N, \star)$ be a semigroup, and $f : M \rightarrow N$ be a semigroup homomorphism. If f is surjective, then $(N, \star)$ is a monoid and f is a monoid homomorphism.*

Proof. Let e_M be the neutral element of $(M, *)$. For any $y \in N$, there exists $x \in M$ such that $f(x) = y$. Thus,

$$\begin{aligned} f(e_M) \star y &= f(e_M) \star f(x) = f(e_M * x) = f(x) = y, \\ y \star f(e_M) &= f(x) \star f(e_M) = f(x * e_M) = f(x) = y. \end{aligned}$$

Therefore, $f(e_M)$ is the neutral element of $(N, \star)$. □

5.4.6 Remark. Let $(M, *)$ be a monoid, $(N, \star)$ be a semigroup, and $f : M \rightarrow N$ be a semigroup homomorphism. Proposition 5.4.5 shows that the image of f forms a monoid under the restricted composition law $\star$. However, in general, even if $(N, \star)$ is a monoid, the range of f need not be a submonoid of it.

For example, the constant mapping from the multiplicative monoid of natural numbers $(\mathbb{N}, \times)$ to itself, which maps any $n \in \mathbb{N}$ to 0, is a semigroup homomorphism but not a monoid homomorphism. Its range $\{0\}$ is *not* a submonoid of $(\mathbb{N}, \times)$.

5.4.7 Proposition. *Let $(M, *)$ be a monoid and $(H, \star)$ be a group. Then any semigroup homomorphism from M to H is a monoid homomorphism.*

Proof. Let $f : M \rightarrow H$ be a semigroup homomorphism. Let e_M and e_H be the neutral elements of M and H , respectively. By definition, $e_M = e_M * e_M$, hence

$$e_H \star f(e_M) = f(e_M) = f(e_M * e_M) = f(e_M) \star f(e_M).$$

Thus, by Proposition 5.2.10, we have $f(e_M) = e_H$. □

5.4.8 Definition. Let G and H be groups. A semigroup homomorphism from G to H is also called a *group homomorphism* from G to H . From Propositions 5.4.7 and 5.4.4, it follows that if $f : G \rightarrow H$ is a group homomorphism, then it is a monoid homomorphism, and for any $a \in G$, we have $f(\iota(a)) = \iota(f(a))$.

5.4.9 Proposition. Let $f : G \rightarrow H$ be a group homomorphism. For any subgroup G_0 of G , the set $f(G_0)$ is a subgroup of H .

Proof. This is a direct consequence of Proposition 5.4.4. $\square$

5.4.10 Example. Let M be a semigroup (resp. monoid, group), and let N be a subsemigroup (resp. submonoid, subgroup) of M . Then the inclusion mapping $j_N : N \rightarrow M$ is a semigroup homomorphism (resp. monoid homomorphism, group homomorphism).

5.4.11 Proposition. Let $(X, *)$, $(Y, \star)$, and $(Z, \diamond)$ be semigroups (resp. monoids, groups), and let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be semigroup homomorphisms (resp. monoid homomorphisms, group homomorphisms). Then the composite mapping $g \circ f$ is a semigroup homomorphism (resp. monoid homomorphism, group homomorphism).

Proof. First, consider the case of semigroup homomorphisms (which includes the case of group homomorphisms). Let x_1 and x_2 be elements of X . Then

$$(g \circ f)(x_1 * x_2) = g(f(x_1 * x_2)) = g(f(x_1) \star f(x_2)) = g(f(x_1)) \diamond g(f(x_2)).$$

Therefore, $g \circ f$ is a semigroup homomorphism.

In the case of monoid homomorphisms, if e_X , e_Y , and e_Z denote the neutral elements of X , Y , and Z , respectively, then $g(f(e_X)) = g(e_Y) = e_Z$. Hence, $g \circ f$ is a monoid homomorphism. $\square$

5.4.12 Proposition. Let $(X, *)$ and $(Y, \star)$ be semigroups, and let $f : X \rightarrow Y$ be a semigroup homomorphism. If f is a bijection, then its inverse mapping $f^{-1} : Y \rightarrow X$ is also a semigroup homomorphism.

Proof. Let α and β be elements of Y , and let $a = f^{-1}(\alpha)$, $b = f^{-1}(\beta)$. Then

$$\alpha \star \beta = f(a) \star f(b) = f(a * b).$$

This shows that $f^{-1}(\alpha) * f^{-1}(\beta) = a * b = f^{-1}(\alpha \star \beta)$. $\square$

5.4.13 Definition. Let $f : X \rightarrow Y$ be a semigroup homomorphism. Suppose there exists a semigroup homomorphism $g : Y \rightarrow X$ such that $g \circ f = \text{Id}_X$ and $f \circ g = \text{Id}_Y$, then f is said to be a *semigroup isomorphism*. By Proposition 5.4.14, a semigroup homomorphism $f : X \rightarrow Y$ is a semigroup isomorphism if and only if it is bijective as a mapping.

5.4.14 Proposition. *Let X and Y be monoids, and let $f : X \rightarrow Y$ be a semigroup isomorphism. Then both f and f^{-1} are monoid isomorphisms.*

Proof. This is a direct corollary of Proposition 5.4.5. $\square$

5.4.15 Definition. Let X and Y be monoids (resp. groups). A semigroup isomorphism from X to Y is also called a *monoid isomorphism* (resp. *group isomorphism*) from X to Y .

5.4.16 Proposition. *Let $(S, *)$ be a group. Then the inversion mapping $\iota : S \rightarrow S$ is an isomorphism from $(S, *)$ to the opposite group $(S, \check{*})$.*

Proof. This is a corollary of Proposition 5.2.12. $\square$

5.5 Quotient structure

5.5.1 Definition. Let X be a set and $*$ be a composition law on X . Let $\sim$ be an equivalence relation on X . If, for any elements x, y, x' , and y' in X , the conditions $x \sim x'$ and $y \sim y'$ imply $x * y \sim x' * y'$, then the equivalence relation $\sim$ is said to be *compatible* with $*$.

5.5.2 Proposition. *Let X be a set, $*$ be a composition law on X , and $\sim$ be an equivalence relation on X compatible with $*$. Then*

$$([x], [y]) \longmapsto [x * y]$$

defines an composition law on $X/\sim$ (still denoted as $$). Furthermore, the following statements hold:*

- (1) *If $*$ is commutative, then the composition law $*$ on $X/\sim$ is also commutative.*
- (2) *If $(X, *)$ is a semigroup (resp. monoid, group), then $(X/\sim, *)$ is also a semigroup (resp. monoid, group), and the projection mapping $\pi : X \rightarrow X/\sim$ is a semigroup homomorphism (resp. monoid homomorphism, group homomorphism).*

Proof. Let R be the binary relation on $X \times X$ defined as

$$(x, y) R (x', y') \text{ if and only if } x \sim x' \text{ and } y \sim y'.$$

By Corollary 3.11.11, this is an equivalence relation, and the mapping

$$\Phi : (X \times X)/R \longrightarrow (X/\sim) \times (X/\sim)$$

sending $[(x, y)]$ to $([x], [y])$ is a bijection. Since $\sim$ is compatible with $*$, we obtain that the composed mapping

$$X \times X \xrightarrow{*} X \xrightarrow{\pi} X/\sim$$

is compatible with R and hence induces by passing to quotient a mapping $(X \times X)/R \rightarrow X/\sim$ that sends $[(x, y)]$ to $[x * y]$. The composite of this mapping with the inverse of Φ thus defines a composition law on $X/\sim$ that sends $([x], [y])$ to $[x * y]$.

(1) If the composition law $*$ on X is commutative, then for any $(x, y) \in X \times X$, we have

$$[x] * [y] = [x * y] = [y * x] = [y] * [x],$$

so the composition law on $X/\sim$ is also commutative.

(2) If the composition law $*$ on X is associative, then for any $(x, y, z) \in X \times X \times X$, we have

$$([x] * [y]) * [z] = [x * y] * [z] = [(x * y) * z] = [x * (y * z)] = [x] * [y * z] = [x] * ([y] * [z]).$$

Thus, the composition law on $X/\sim$ is also associative. Moreover, by definition, the projection mapping $\pi : X \rightarrow X/\sim$ is a semigroup homomorphism.

Since the projection mapping is surjective, by Proposition 5.4.5, if $(X, *)$ is a monoid, then $(X/\sim, *)$ is also a monoid, and the projection mapping is a monoid homomorphism. Furthermore, by Proposition 5.4.4, if $(X, *)$ is a group, then $(X/\sim, *)$ is also a group, and the projection mapping is a group homomorphism. $\square$

5.5.3 Definition. Let X be a set, and $*$ be an composition law on X . Let $\sim$ be an equivalence relation on X compatible with $*$. The composition law on the quotient set $X/\sim$ that maps $([x], [y])$ to $[x * y]$ is called the composition law *induced by $*$ by passing to quotient*. Usually, the same symbol as the composition law on X is used to denote the induced composition law on the quotient set.

Let $(X, *)$ be a semigroup (resp. monoid, group), and $\sim$ be an equivalence relation compatible with $*$. Then $(X/\sim, *)$ is called the *quotient semigroup* (resp. *quotient monoid*, *quotient group*) of $(X, *)$ under the equivalence relation $\sim$, and the projection mapping $\pi : X \rightarrow X/\sim$ is called the *quotient homomorphism*.

5.5.4 Theorem. Let $(X, *)$ and $(Y, \star)$ be semigroups (resp. monoids, groups), and let $f : X \rightarrow Y$ be a semigroup homomorphism (resp. monoid homomorphism, group homomorphism).

- (1) Suppose $\sim$ is an equivalence relation compatible with $*$, and the mapping f is compatible with the equivalence relation $\sim$ (see Definition 3.11.9). Then the mapping $\tilde{f} : X/\sim \rightarrow Y$ induced by f (by passing to quotient) is a semigroup homomorphism (resp. monoid homomorphism, group homomorphism).

- (2) The binary relation $\sim_f$ defined as follows is an equivalence relation and is compatible with the composition law $*$:

$$\forall (x, y) \in X \times X, x \sim_f y \text{ if and only if } f(x) = f(y).$$

Furthermore, the restriction of the mapping $\tilde{f} : X/\sim_f \rightarrow Y$ (induced by f by passing to quotient) defines a semigroup isomorphism (resp. monoid isomorphism, group isomorphism) from $(X/\sim_f, *)$ to $(f(X), \star)$.

Proof. (1) For any elements x and y in X , by definition we have

$$\tilde{f}([x] * [y]) = \tilde{f}([x * y]) = f(x * y) = f(x) \star f(y) = \tilde{f}([x]) \star \tilde{f}([y]).$$

Therefore, $\tilde{f}$ is a semigroup homomorphism. If f is a monoid homomorphism, then $f(X) = \tilde{f}(X/\sim)$ is a submonoid of $(Y, \star)$. By Proposition 5.4.5, $\tilde{f}$ is a monoid homomorphism.

(2) By Corollary 3.11.10, $\sim_f$ is an equivalence relation.

Let x, y, x' , and y' be elements of X such that $x \sim_f x'$ and $y \sim_f y'$. Then $f(x) = f(x')$ and $f(y) = f(y')$, hence

$$f(x * y) = f(x) \star f(y) = f(x') \star f(y') = f(x' * y').$$

This shows that $x * y \sim_f x' * y'$. Therefore, $\sim_f$ is compatible with the composition law $*$.

Finally, by Corollary 3.11.10, the restriction of $\tilde{f} : X/\sim_f \rightarrow Y$ defines a bijection from $X/\sim_f$ to $f(X)$, and thus defines a semigroup isomorphism. $\square$

5.5.5 Definition. Let $f : X \rightarrow Y$ be a semigroup homomorphism (resp. monoid homomorphism, group homomorphism), and let $\sim$ be an equivalence relation on X compatible with $*$, such that f is compatible with $\sim$. Then

$$\tilde{f} : X/\sim \longrightarrow Y, \quad [x] \longmapsto f(x)$$

is called the semigroup homomorphism (resp. monoid homomorphism, group homomorphism) *induced by f by passing to quotient*.

5.5.6 Definition. Let $f : G \rightarrow H$ be a group homomorphism, and let e_H be the neutral element of H . The *kernel* of f is defined as the preimage of $\{e_H\}$ under f , denoted as $\text{Ker}(f)$.

5.5.7 Proposition. Let $f : (G, *) \rightarrow (H, \star)$ be a group homomorphism. Then the kernel of f is a subgroup of G . Furthermore, for any $a \in G$ and $x \in \text{Ker}(f)$, we have $a * x * \iota(a) \in \text{Ker}(f)$.

Proof. Let e_H be the neutral element of H . For any $(x, y) \in \text{Ker}(f) \times \text{Ker}(f)$, we have

$$f(x * \iota(y)) = f(x) * \iota(f(y)) = e_H.$$

Hence, $x * \iota(y) \in \text{Ker}(f)$. By Proposition 5.3.9, $\text{Ker}(f)$ is a subgroup of G .

For any $a \in G$ and $x \in \text{Ker}(f)$, we have

$$f(a * x * \iota(a)) = f(a) * f(x) * \iota(f(a)) = f(a) * e_H * \iota(f(a)) = f(a) * \iota(f(a)) = e_H.$$

Therefore, $a * x * \iota(a) \in \text{Ker}(f)$. □

5.5.8 Proposition. *Let $(G, *)$ be a group, and $\sim$ be an equivalence relation compatible with $*$. Let H be the kernel of the quotient homomorphism $\pi : G \rightarrow G/\sim$. Then for any $(x, y) \in G \times G$, $x \sim y$ if and only if $x * \iota(y) \in H$.*

Proof. Let e be the neutral element of G . Let x and y be two elements of G . Then

$$[x * \iota(y)] = [x] * \iota([y]).$$

Therefore, $[x] = [y]$ if and only if $[x * \iota(y)] = [e]$, that is, $x * \iota(y) \in H$. □

5.5.9 Definition. Let $(G, *)$ be a group, and H be a subgroup of G . If for any $(a, x) \in G \times H$, we have $a * x * \iota(a) \in H$, then H is said to be a *normal subgroup* of G . From the definition, it is clear that all subgroups of an abelian group are normal. Moreover, Proposition 5.5.7 shows that the kernel of a group homomorphism is a normal subgroup of the departure group.

5.5.10 Proposition. *Let $(G, *)$ be a group, and H be a normal subgroup of G . Then the binary relation $\sim_H$ on G defined as follows is an equivalence relation compatible with $*$:*

$$x \sim_H y \text{ if and only if } x * \iota(y) \in H.$$

Furthermore, the kernel of the quotient homomorphism $\pi : G \rightarrow G/\sim_H$ is equal to H .

Proof. Let e be the neutral element of $(G, *)$. By definition, for any $x \in G$, we have $x * \iota(x) = e \in H$, so $\sim_H$ is reflexive. Moreover, if $x * \iota(y) \in H$, then by Proposition 5.2.12, we have $y * \iota(x) = \iota(x * \iota(y)) \in H$, so $\sim_H$ is symmetric. Finally, if $x * \iota(y) \in H$ and $y * \iota(z) \in H$, then

$$x * \iota(z) = (x * \iota(y)) * (y * \iota(z)) \in H,$$

hence $\sim_H$ is transitive.

Let x, x', y , and y' be elements of G such that $x' * \iota(x) \in H$ and $y' * \iota(y) \in H$. Then by Proposition 5.2.12, we have

$$(x' * y') * \iota(x * y) = (x' * y') * (\iota(y) * \iota(x)) = x' * (y' * \iota(y)) * \iota(x') * (x' * \iota(x)).$$

Since H is a normal subgroup, from $y' * \iota(y) \in H$, it follows that

$$x' * (y' * \iota(y)) * \iota(x') \in H,$$

and thus $(x' * y') * \iota(x * y) \in H$. This shows that $\sim_H$ is compatible with $*$.

Finally, $x \in \text{Ker}(\pi)$ if and only if $x \sim_H e$, i.e., $x * \iota(e) = x \in H$. $\square$

5.5.11 Notation. Let G be a group, and H be a subgroup of G . The notation G/H denotes the quotient group $G/\sim_H$, where $\sim_H$ is the equivalence relation compatible with the composition law of G defined in Proposition 5.5.10. The group G/H is called the quotient group of G by the normal subgroup H . Proposition 5.5.8 shows that every quotient group of G is obtained by modding out by some normal subgroup.

5.5.12 Proposition. Let $f : X \rightarrow Y$ be a group homomorphism.

- (1) $\text{Ker}(f)$ is a normal subgroup of X .
- (2) $\text{Im}(f)$ is a subgroup of Y .
- (3) The mapping $\tilde{f} : X/\text{Ker}(f) \rightarrow \text{Im}(f)$, $[x] \mapsto f(x)$ is well-defined and is a group isomorphism.

Proof. (1) follows from Proposition 5.5.7.

(2) follows from Proposition 5.4.9.

(3) follows from Theorem 5.5.4 and Proposition 5.5.10. $\square$

5.5.13 Proposition. Let $(X, *)$ and $(Y, \star)$ be groups, and $f : X \rightarrow Y$ be a homomorphism of groups. Then f is injective if and only if $\text{Ker}(f) = \{e_X\}$, where e_X denotes the neutral element of $(X, *)$.

Proof. Let e_Y be the neutral element of $(Y, \star)$. Since f is a homomorphism of groups, one has $f(e_X) = e_Y$. If f is injective, then e_X is the only element of $\text{Ker}(f) := f^{-1}(\{e_Y\})$.

Conversely, suppose that $\text{Ker}(f) = \{e_X\}$. If a and b are two elements of X such that $f(a) = f(b)$, then

$$f(a * \iota(b)) = f(a) * \iota(f(b)) = e_Y,$$

so that $a * \iota(b) = e_X$. This leads to $a = b$. $\square$

5.6 Universal homomorphisms

5.6.1 Proposition. *Let $(M, *)$ be a monoid, and $x \in M$. Then there exists a unique monoid homomorphism f from $(\mathbb{N}, +)$ to M such that $f(1) = x$. In the particular case where x is the neutral element of $(M, *)$, f is the constant mapping sending any element of $\mathbb{N}$ to the neutral element.*

Proof. Define the mapping $f : \mathbb{N} \rightarrow M$ inductively as follows. Let $f(0)$ be the neutral element of M . If $f(0), \dots, f(n)$ have been defined, let

$$f(n+1) := f(n) * x. \quad (5.2)$$

We prove by induction that for any $(n, m) \in \mathbb{N} \times \mathbb{N}$, we have

$$f(n+m) = f(n) * f(m).$$

When $m = 0$, since $f(0)$ is the neutral element of M , the equality holds; the case $m = 1$ follows from (5.2). Assume that the equality

$$f(n+m) = f(n) * f(m).$$

holds. Then

$$f(n+m+1) = f(n+1) * f(m) = f(n) * f(1) * f(m) = f(n) * f(m+1).$$

This shows that f is a monoid homomorphism.

Suppose $g : \mathbb{N} \rightarrow M$ is a monoid homomorphism such that $g(1) = x$. Then by definition, $g(0) = f(0)$ and $g(1) = f(1)$. If for some $n \in \mathbb{N}$, we have $g(n) = f(n)$, then

$$g(n+1) = g(n) * g(1) = f(n) * x = f(n+1).$$

By induction, $g(n) = f(n)$ for all $n \in \mathbb{N}$.

Note that the constant mapping sending any element of $\mathbb{N}$ to the neutral element of $(M, *)$ is a homomorphism of monoids. If x is the neutral element of $(M, *)$, by the uniqueness we obtain that f identifies with this constant mapping. $\square$

5.6.2 Notation. Let $(M, *)$ be a monoid. If $x \in M$, by Proposition 5.6.1, there exists a unique monoid homomorphism from $(\mathbb{N}, +)$ to $(M, *)$ that maps $1 \in \mathbb{N}$ to x . For any $n \in \mathbb{N}$, the image of n under this homomorphism is usually denoted by x^{*n} . However, the following exceptions apply:

- (1) When the monoid M is commutative and its composition law is denoted by $+$, for any $x \in M$ and $n \in \mathbb{N}$, the image of n under the above monoid homomorphism is denoted by nx .

- (2) When the composition law of the monoid is denoted multiplicatively, for any $x \in M$ and $n \in \mathbb{N}$, the image of n under the above monoid homomorphism is denoted by x^n .

5.6.3 Proposition. *Let $(M, *)$ be a monoid.*

- (1) *Let x and y be elements of M . If $x * y = y * x$, then*

$$\forall (n, m) \in \mathbb{N}^2, \quad x^{*m} * y^{*n} = y^{*n} * x^{*m}.$$

- (2) *Let x and y be elements of M . If $x * y = y * x$, then*

$$\forall n \in \mathbb{N}, \quad (x * y)^{*n} = x^{*n} * y^{*n}.$$

- (3) *Let x be an invertible element of M . For any $n \in \mathbb{N}$, x^{*n} is invertible, and the following equality holds:*

$$\iota(x^{*n}) = \iota(x)^{*n}.$$

- (4) *Let x be an invertible element of M . For any natural numbers n and m with $n \geq m$, we have*

$$x^{*(n-m)} = x^{*n} * \iota(x)^{*m}, \quad \iota(x^{*(n-m)}) = \iota(x)^{*n} * x^{*m}.$$

Proof. Let e be the neutral element of $(M, *)$.

(1) First, we prove by induction that $x * y^{*n} = y^{*n} * x$ for all $n \in \mathbb{N}$. If $n = 0$, then y^{*n} is the neutral element of M , so the equality holds. The case $n = 1$ follows from the assumption $x * y = y * x$. Assume that $x * y^{*n} = y^{*n} * x$ holds. Then

$$x * y^{*(n+1)} = x * y^{*n} * y = y^{*n} * x * y = y^{*n} * y * x = y^{*(n+1)} * x.$$

Therefore,

$$\forall n \in \mathbb{N}, \quad x * y^{*n} = y^{*n} * x$$

holds. Applying this equality to y^{*n} and x , we get $y^{*n} * x^{*m} = x^{*m} * y^{*n}$ for all $m \in \mathbb{N}$.

(2) We proceed by induction on n . The case $n = 0$ follows from the definition, and the case $n = 1$ follows from the assumption $x * y = y * x$. Assume that $(x * y)^{*n} = x^{*n} * y^{*n}$. Then

$$(x * y)^{*(n+1)} = (x * y)^{*n} * (x * y) = x^{*n} * (y^{*n} * x) * y.$$

By (1), we have

$$(x * y)^{*(n+1)} = x^{*n} * x * y^{*n} * y = x^{*(n+1)} * y^{*(n+1)}.$$

Hence, the proposition holds.

(3) By (2), we have

$$\iota(x)^{*n} * x^{*n} = (\iota(x) * x)^{*n} = 1^{*n} = e = e^{*n} = (x * \iota(x))^{*n} = x^{*n} * \iota(x)^{*n}.$$

Therefore, $\iota(x)^{*n}$ is the inverse of x^{*n} .

(4) By (3), we have

$$x^{*n} * \iota(x)^{*m} = x^{*(n-m)} * x^{*m} * \iota(x)^{*m} = x^{*(n-m)} * e = x^{*(n-m)}.$$

Applying this proposition to $\iota(x)$, and using (3), we get

$$\iota(x^{*(n-m)}) = \iota(x)^{*(n-m)} = \iota(x)^{*n} * x^{*m}.$$

□

5.6.4 Proposition. *Let $(M, *)$ be a monoid. For any invertible element x of M , there exists a unique monoid homomorphism from $(\mathbb{Z}, +)$ to M that maps $1 \in \mathbb{Z}$ to x .*

Proof. Suppose $f : \mathbb{Z} \rightarrow M$ is a monoid homomorphism such that $f(1) = x$. Since $\mathbb{N}$ is a submonoid of $(\mathbb{Z}, +)$, by Proposition 5.4.11 and Example 5.4.10, the restriction of f to $\mathbb{N}$ is a monoid homomorphism from $(\mathbb{N}, +)$ to M . By Proposition 5.6.1, we have

$$\forall n \in \mathbb{N}, \quad f(n) = x^{*n}.$$

Moreover, by Proposition 5.4.4, we have

$$\forall n \in \mathbb{N}, \quad f(-n) = \iota(x^n).$$

Thus, f is unique. It only remains to verify that the mapping from $\mathbb{Z}$ to M that sends $n \in \mathbb{N}$ to x^{*n} and $-n$ to $\iota(x^{*n})$ is a group homomorphism. By item (3) of Proposition 5.6.3, for any $n \in \mathbb{N}$, we have $\iota(x^{*n}) = \iota(x)^{*n}$.

Let n and m be elements of $\mathbb{N}$. Then

$$x^{*(n+m)} = x^{*n} * x^{*m}, \quad \iota(x)^{*(n+m)} = \iota(x)^{*n} * \iota(x)^{*m}.$$

Furthermore, by item (4) of Proposition 5.6.3, we have

$$\begin{aligned} x^{*n} * \iota(x)^{*m} &= \begin{cases} x^{*(n-m)}, & n \geq m, \\ \iota(x)^{*(m-n)}, & n \leq m. \end{cases} \\ \iota(x)^{*n} * x^{*m} &= \begin{cases} x^{*(m-n)}, & m \geq n, \\ \iota(x)^{*(n-m)}, & m \leq n. \end{cases} \end{aligned}$$

Therefore, the above mapping $\mathbb{Z} \rightarrow M$ is a monoid homomorphism. □

5.6.5 Notation. Let $(M, *)$ be a monoid. If x is an invertible element in M , then by Proposition 5.6.1, there exists a unique monoid homomorphism from $(\mathbb{Z}, +)$ to $(M, *)$ that maps $1 \in \mathbb{Z}$ to x . For any $n \in \mathbb{Z}$, denote by x^{*n} the image of n under this monoid homomorphism. It is often simply written as x^n , except in the following case: if M is commutative and its composition law is denoted additively as $+$, then for any $x \in M$ and $n \in \mathbb{Z}$, denote by nx the image of n under the above monoid homomorphism. Note that $0x = 0$ (the latter 0 denotes the neutral element of M) and $(n + m)x = nx + mx$ for all $(n, m) \in \mathbb{Z}^2$. Additionally, $n0 = 0$ for all $n \in \mathbb{Z}$.

5.7 Product

5.7.1 Proposition. *Let I be a set, and let $((M_i, *_i))_{i \in I}$ be a family of semigroups. Let*

$$M = \prod_{i \in I} M_i.$$

Define a composition law on M as follows: for any elements $x = (x_i)_{i \in I}$ and $y = (y_i)_{i \in I}$ of M , let

$$x * y := (x_i *_i y_i)_{i \in I}.$$

Then M forms a semigroup under this composition law. Moreover, for any $j \in I$, the projection mapping pr_j from M to M_j , which sends $(x_i)_{i \in I}$ to x_j , is a semigroup homomorphism.

If all M_i are monoids (resp. groups), then so is M , and each $\text{pr}_j : M \rightarrow M_j$ is a monoid homomorphism (resp. group homomorphism).

Proof. Let $x = (x_i)_{i \in I}$, $y = (y_i)_{i \in I}$, and $z = (z_i)_{i \in I}$ be elements of M . By definition,

$$(x * y) * z = ((x_i *_i y_i) *_i z_i)_{i \in I} = (x_i *_i (y_i *_i z_i))_{i \in I} = x * (y * z).$$

Thus, the composition law of M is associative. Moreover, for any elements $x = (x_i)_{i \in I}$ and $y = (y_i)_{i \in I}$ in M , we have

$$\forall j \in I, \quad \text{pr}_j(x * y) = \text{pr}_j((x_i *_i y_i)_{i \in I}) = x_j *_j y_j = \text{pr}_j(x) *_j \text{pr}_j(y).$$

Hence, pr_j is a semigroup homomorphism.

Suppose each M_i is a monoid with neutral element e_i . Let $e = (e_i)_{i \in I}$. For any $x = (x_i)_{i \in I} \in M$, we have

$$e * x = (e_i *_i x_i)_{i \in I} = (x_i)_{i \in I} = x, \quad x * e = (x_i *_i e_i)_{i \in I} = (x_i)_{i \in I} = x.$$

Therefore, M is a monoid. Furthermore, $\text{pr}_j(e) = e_j$, so pr_j is a monoid homomorphism.

Suppose for each $i \in I$, M_i is a group. For any $(x_i)_{i \in I} \in M$, we have

$$\begin{aligned}\iota(x_i)_{i \in I} * (x_i)_{i \in I} &= (\iota(x_i) *_i x_i)_{i \in I} = (e_i)_{i \in I} = e, \\ (x_i)_{i \in I} * (\iota(x_i))_{i \in I} &= (x_i *_i \iota(x_i))_{i \in I} = (e_i)_{i \in I} = e.\end{aligned}$$

This shows that $(x_i)_{i \in I}$ is invertible, with inverse $(\iota(x_i))_{i \in I}$. Thus, M forms a group. $\square$

5.7.2 Definition. Let I be a set, and let $(M_i)_{i \in I}$ be a family of semigroups (resp. monoids, groups). The semigroup (resp. monoid, group) $\prod_{i \in I} M_i$ constructed in Proposition 5.7.1 is called the *direct product* of $(M_i)_{i \in I}$. From the definition, we observe that if each M_i is commutative, then the direct product $\prod_{i \in I} M_i$ is also commutative.

5.7.3 Proposition. Let I be a set, $(M_i)_{i \in I}$ be a family of semigroups (resp. monoids, groups), and N be a semigroup (resp. monoid, group). For each $i \in I$, let $f_i : (N, *) \rightarrow (M_i, \star)$ be a semigroup homomorphism (resp. monoid homomorphism, group homomorphism). Then there exists a unique semigroup homomorphism (resp. monoid homomorphism, group homomorphism) $f : N \rightarrow \prod_{i \in I} M_i$ such that $\text{pr}_i \circ f = f_i$ for all $i \in I$.

Proof. By Proposition 3.9.4, there exists a unique mapping $f : N \rightarrow \prod_{i \in I} M_i$ such that $\text{pr}_i \circ f = f_i$ for all $i \in I$. It remains to show that f is a semigroup homomorphism, and in the case where each f_i is a monoid homomorphism, f is also a monoid homomorphism.

First, for any $(x, y) \in N \times N$,

$$\begin{aligned}f(x * y) &= (f_i(x * y))_{i \in I} = (f_i(x) \star_i f_i(y))_{i \in I} \\ &= (f_i(x))_{i \in I} \star (f_i(y))_{i \in I} = f(x) \star f(y).\end{aligned}$$

Thus, f is a semigroup homomorphism.

Now suppose each f_i is a monoid homomorphism. Let e_N be the neutral element of N . Then for each $i \in I$, $f_i(e_N)$ is the neutral element of M_i , so $f(e_N) = (f_i(e_N))_{i \in I}$ is the neutral element of $\prod_{i \in I} M_i$. $\square$

5.7.4 Proposition. Let $(M, *)$ be a commutative monoid, n be a natural number, $(x_i)_{i \in \{1, \dots, n\}} \in M^n$, and $\sigma \in \mathfrak{S}_n$. Then the following equality holds (when $n = 0$, by convention, $x_1 * \dots * x_n$ equals the neutral element of M):

$$x_1 * \dots * x_n = x_{\sigma(1)} * \dots * x_{\sigma(n)}. \quad (5.3)$$

Proof. We proceed by induction on n . When $n = 0$, by convention, both sides of the equality equal the neutral element of M . When $n = 1$, one has $\sigma(1) = 1$, so

the proposition holds. Now assume $n \geq 2$ and that the proposition holds for fewer than n elements in M . If $\sigma(n) = n$, then by the induction hypothesis,

$$x_1 * \cdots * x_{n-1} = x_{\sigma(1)} * \cdots * x_{\sigma(n-1)},$$

which implies (5.3). If there exists $i \in \{1, \dots, n-1\}$ such that $\sigma(i) = n$, then by Proposition 5.1.9 and the commutativity of $*$, we have

$$x_{\sigma(i)} * \cdots * x_{\sigma(n)} = x_{\sigma(i)} * (x_{\sigma(i+1)} * \cdots * x_{\sigma(n)}) = (x_{\sigma(i+1)} * \cdots * x_{\sigma(n)}) * x_{\sigma(i)},$$

which implies

$$x_{\sigma(1)} * \cdots * x_{\sigma(n)} = x_{\sigma(1)} * \cdots * x_{\sigma(i-1)} * x_{\sigma(i+1)} * \cdots * x_{\sigma(n)} * x_n.$$

Since

$$\{\sigma(1), \dots, \sigma(i-1), \sigma(i+1), \dots, \sigma(n)\} = \{1, \dots, n-1\},$$

the induction hypothesis yields (5.3). $\square$

5.7.5 Notation. Let $(M, +)$ be a monoid with neutral element 0, I be a set, and $(x_i)_{i \in I}$ be a family of elements in M . If I is finite, we use the expression

$$\sum_{i \in I} x_i$$

to denote

$$x_{\tau(1)} + \cdots + x_{\tau(n)},$$

where n is a natural number and τ is a bijection from $\{1, \dots, n\}$ to I . Proposition 5.7.4 shows that this element is independent of the choice of the bijection τ . When I is infinite but $\{j \in I \mid x_j \neq 0\}$ is finite, denote

$$\sum_{i \in I} x_i$$

as

$$\sum_{i \in I, x_i \neq 0} x_i.$$

5.7.6 Definition. Let I be a set and $(M_i)_{i \in I}$ be a family of commutative monoids (the composition law of which is written additively). Denote by $\bigoplus_{i \in I} M_i$ the set of elements $(x_i)_{i \in I}$ in $\prod_{i \in I} M_i$ satisfying the condition:

$$\{i \in I \mid x_i \neq 0\} \text{ is finite.}$$

For each $j \in I$, let $\lambda_j : M_j \rightarrow \bigoplus_{i \in I} M_i$ be the mapping that sends $a \in M_j$ to the element $(\lambda_j(a)_i)_{i \in I}$ defined by:

$$\lambda_j(a)_i = \begin{cases} a, & i = j, \\ 0, & i \neq j. \end{cases} \quad (5.4)$$

5.7.7 Proposition. *Let I be a set, $(M_i)_{i \in I}$ be a family of commutative monoids, and N be a commutative monoid. For each $i \in I$, let $g_i : M_i \rightarrow N$ be a monoid homomorphism. Then there exists a unique monoid homomorphism*

$$g : \bigoplus_{i \in I} M_i \longrightarrow N$$

such that $g \circ \lambda_i = g_i$ for all $i \in I$.

Proof. We denote the operations of M_i and N additively. For any $x = (x_i)_{i \in I} \in \bigoplus_{i \in I} M_i$, define

$$g(x) = \sum_{i \in I} g_i(x_i). \quad (5.5)$$

Since $\{i \in I \mid x_i \neq 0\}$ is finite, $\{i \in I \mid g_i(x_i) \neq 0\}$ is also finite since

$$g_i(x_i) \neq 0 \implies x_i \neq 0.$$

Moreover, if $x = (x_i)_{i \in I}$ and $y = (y_i)_{i \in I}$ are elements of $\bigoplus_{i \in I} M_i$, then by Proposition 5.7.4, we have

$$g(x + y) = \sum_{i \in I} g_i(x_i + y_i) = \left(\sum_{i \in I} g_i(x_i) \right) + \left(\sum_{i \in I} g_i(y_i) \right) = g(x) + g(y).$$

Thus, g is a monoid homomorphism. For any $j \in I$, by (5.4) and (5.5), we have

$$\forall a \in M_j, \quad g(\lambda_j(a)) = g_j(a).$$

If $f : \bigoplus_{i \in I} M_i \rightarrow N$ is another monoid homomorphism such that $f \circ \lambda_i = g_i$ for all $i \in I$, then for any $x = (x_i)_{i \in I} \in \bigoplus_{i \in I} M_i$, we have

$$f(x) = f\left(\sum_{\substack{i \in I \\ x_i \neq 0}} \lambda_i(x_i)\right) = \sum_{\substack{i \in I \\ x_i \neq 0}} f(\lambda_i(x_i)) = \sum_{\substack{i \in I \\ x_i \neq 0}} g_i(x_i) = g(x).$$

Hence, $f = g$. □

5.8 Actions of Monoids

5.8.1 Definition. Let $(G, *)$ be a monoid with neutral element e , and let X be a set. Let $\phi : G \times X \rightarrow X$ be a map. If ϕ satisfies the following conditions:

- (1) For all $(a, b) \in G \times G$ and $x \in X$, $\phi(a * b, x) = \phi(a, \phi(b, x))$,

(2) For all $x \in X$, $\phi(e, x) = x$,

then we say that ϕ defines a *left action* of $(G, *)$ on X .

If ϕ satisfies the following conditions:

(1') For all $(a, b) \in G \times G$ and $x \in X$, $\phi(a * b, x) = \phi(b \check{*} a, x) = \phi(a, \phi(b, x))$,

(2') For all $x \in X$, $\phi(e, x) = x$,

that is, ϕ defines a left action of the opposite monoid $(G, \check{*})$ on X , then we say that ϕ defines a *right action* of $(G, *)$ on X .

5.8.2 Notation. Let G be a monoid with its composition law written multiplicatively and its neutral element denoted by 1. Let X be a set. A left action of G on X is usually denoted as

$$((a, x) \in G \times X) \mapsto ax.$$

In this notation, conditions (1) and (2) in Definition 5.8.1 become:

$$\begin{aligned} \forall (a, b) \in G \times G, \forall x \in X, \quad (ab)x &= a(bx), \\ \forall x \in X, \quad 1x &= x. \end{aligned}$$

Similarly, a right action of G on X is usually denoted as

$$((a, x) \in G \times X) \mapsto xa,$$

and conditions (1') and (2') in Definition 5.8.1 become:

$$\begin{aligned} \forall (a, b) \in G \times G, \forall x \in X, \quad x(ab) &= (xa)b, \\ \forall x \in X, \quad x1 &= x. \end{aligned}$$

5.8.3 Example. Let $(G, *)$ be a monoid and H be a submonoid of G . Then the restriction of the composition law

$$* : H \times G \longrightarrow G, \quad (a, b) \mapsto a * b$$

defines a left action of $(H, *)$ on G . The restriction of the opposite composition law

$$\check{*} : H \times G \longrightarrow G, \quad (a, b) \mapsto b * a$$

defines a right action of $(H, *)$ on G . These actions are called the *left canonical action* and the *right canonical action* of $(H, *)$ on G , respectively.

5.8.4 Proposition. Let $(G, *)$ be a monoid, X be a set, and $\phi : G \times X \rightarrow X$ be a left action of $(G, *)$ on X . Define a binary relation R_ϕ on X as follows:

$x R_\phi y$ if and only if there exists $a \in G$ such that $y = \phi(a, x)$.

Then R_ϕ is reflexive and transitive. If $(G, *)$ is a group, then R_ϕ is an equivalence relation on X .

Proof. Let e be the neutral element of $(G, *)$. For any $x \in X$, we have $\phi(e, x) = x$, so $x R_\phi x$.

Let x, y , and z be elements of X such that $x R_\phi y$ and $y R_\phi z$. Then there exist $a, b \in G$ such that $y = \phi(b, x)$ and $z = \phi(a, y)$. Thus,

$$z = \phi(a, \phi(b, x)) = \phi(a * b, x),$$

so $x R_\phi z$.

Now assume $(G, *)$ is a group. Let $(x, y) \in X \times X$ such that $x R_\phi y$. Hence there exists $a \in G$ such that $y = \phi(a, x)$. Then

$$x = \phi(e, x) = \phi(\iota(a) * a, x) = \phi(\iota(a), \phi(a, x)) = \phi(\iota(a), y),$$

so $y R_\phi x$. □

5.8.5 Remark. Note that a right action of a monoid is equivalent to a left action of its opposite monoid. Therefore, the above proposition remains valid when left actions are replaced by right actions.

5.8.6 Definition. Let $(G, *)$ be a monoid, X be a set, and $\phi : G \times X \rightarrow X$ be a left or right action of G on X . For any $x \in X$, denote by $\text{ord}_\phi(x)$ the set

$$\{\phi(a, x) \mid a \in G\},$$

and call it the *orbit* of x under the action ϕ . When there is no ambiguity about ϕ , it is simply denoted as $\text{ord}(x)$.

When $(G, *)$ is a group, by Proposition 3.11.6, the distinct orbits of the group action ϕ are pairwise disjoint, and their union equals X . When ϕ is a left action, the set of all orbits is denoted by ${}_G \backslash X$; when ϕ is a right action, the set of all orbits is denoted by X/G .

5.8.7 Example. Let $(G, *)$ be a group and H be a subgroup of G . The orbits of the canonical left action of the group $(H, *)$ on G are called the *right cosets* of H . For $x \in G$, the right coset containing x is of the form

$$\{h * x \mid h \in H\},$$

denoted by $H * x$. When the group composition law of G is written multiplicatively, the right coset containing x is also denoted by Hx . Similarly, the orbits of the canonical right action of $(H, *)$ on G are called the *left cosets* of H . For $x \in G$, the left coset containing x is denoted by $x * H$. When the group composition law of G is written multiplicatively, the left coset containing x is also denoted by xH .

5.8.8 Example. Let $(G, *)$ be a group with neutral element e . The following mapping defines a left action of $(G, *)$ on G :

$$G \times G \longrightarrow G, \quad (g, x) \longmapsto g * x * \iota(g).$$

Indeed, for any $g, h, x \in G$, we have

$$\begin{aligned} h * (g * x * \iota(g)) * \iota(h) &= (h * g) * x * (\iota(g) * \iota(h)) = (h * g) * x * \iota(h * g), \\ e * x * \iota(e) &= e * x * e = x. \end{aligned}$$

This left action is called the *conjugation action* of $(G, *)$ on itself, and its orbits are called *conjugacy classes*. For any $x \in G$, the conjugacy class containing x is denoted by $\text{cl}(x)$.

Let H be a subgroup of G and $g \in G$. Denote by $g * H * \iota(g)$ the set

$$\{g * h * \iota(g) \mid h \in H\}.$$

For any $(a, b) \in H \times H$, we have

$$(g * a * \iota(g)) * \iota(g * b * \iota(g)) = (g * a * \iota(g)) * (g * \iota(b) * \iota(g)) = g * (a * \iota(b)) * \iota(g) \in H.$$

By Corollary 5.3.9, $g * H * \iota(g)$ forms a subgroup of $(G, *)$. Note that

$$(g, H) \longmapsto g * H * \iota(g)$$

defines a left action of $(G, *)$ on the set of its subgroups, also called the *conjugation action*. Subgroups in the same orbit of this action are said to be *conjugate*. By definition, a subgroup H of G is normal if and only if its conjugacy orbit consists of a single element.

Exercises

1. Consider the composition law $*$ on $\mathbb{Q}$ defined as

$$\forall (x, y) \in \mathbb{Q}^2, \quad x * y = \frac{x + y}{2}.$$

Is this composition law commutative? associative?

2. Consider the composition law $*$ on $\mathbb{R}$ defined as

$$\forall (x, y) \in \mathbb{R}^2, \quad x * y = xy + (x^2 - 1)(y^2 - 1).$$

Is this composition law commutative? associative?

3. For any $(x, y) \in \mathbb{R}^2$, let $x * y = x + y + 3xy$.

(1) Check that, for any $(x, y) \in \mathbb{R}^2$, one has

$$1 + 3(x * y) = (1 + 3x)(1 + 3y).$$

(2) Deduce that the composition law $*$ on $\mathbb{R}$ is commutative and associative.

(3) Prove that $(\mathbb{R}, *)$ forms a monoid.

(4) Prove that an element $x \in \mathbb{R}$ is invertible with respect to $*$ if and only if $x \neq -\frac{1}{3}$.

4. Consider the composition law $*$ on $\mathbb{R}_{\geq 0} := \{x \in \mathbb{R} \mid x \geq 0\}$ defined as

$$\forall (x, y) \in \mathbb{R}_{\geq 0}^2, \quad x * y = \sqrt{x^2 + y^2}.$$

(1) Prove that $(\mathbb{R}_{\geq 0}, *)$ is a commutative monoid. Determine the neutral element of $(\mathbb{R}_{\geq 0}, *)$

(2) Prove that none of the non-zero elements of $\mathbb{R}_{\geq 0}$ is invertible with respect to $*$.

5. Consider the mappings f_1, f_2, f_3, f_4 from $\mathbb{R}^\times := \mathbb{R} \setminus \{0\}$ to itself defined as

$$f_1(x) = x, \quad f_2(x) = -x, \quad f_3(x) = \frac{1}{x}, \quad f_4(x) = -\frac{1}{x}.$$

Check that $\{f_1, f_2, f_3, f_4\}$ equipped with the composition of mappings forms a group.

6. Consider the composition law $*$ on $\mathbb{R}^\times \times \mathbb{R}$ defined as

$$(a, x) * (b, y) = (ab, x + ay).$$

(1) Check that $(\mathbb{R}^\times \times \mathbb{R}, *)$ forms a group. Determine its neutral element.

(2) Is this group commutative?

(3) Let (a, x) be an element of $\mathbb{R}^\times \times \mathbb{R}$ such that $a \neq 1$. Prove that, for any positive integer n , one has

$$(a, x)^n = \left(a^n, \frac{a^n - 1}{a - 1}x\right)$$

7. For each group G in the following questions, determine if the subset H is a subgroup. Justify your answer.

- (1) $G = (\mathbb{Z}, +)$, $H = \{\text{odd numbers}\}$.
- (2) $G = (\mathbb{Q}, +)$, $H = \{x \in \mathbb{Q} \mid x \geq -1\}$.
- (3) $G = (\mathbb{R}^\times, \cdot)$, $H = \{x \in \mathbb{Q} \mid x \neq 0\}$.
- (4) $G = \mathfrak{S}_X$, where X is a non-empty set, $H = \{\sigma \in G \mid \sigma(x) = x\}$, where x is a fixed element of X .

8. We equip $\mathbb{R}$ with the multiplicative law. Prove that the following set is a submonoid of $(\mathbb{R}, \cdot)$:

$$\{a + b\sqrt{2} \mid (a, b) \in \mathbb{Z}^2\}.$$

9. Let n be a positive integer. Prove that

$$\mu_n(\mathbb{C}) := \{z \in \mathbb{C} \mid z^n = 1\}$$

is a subgroup of $(\mathbb{C}^\times, \cdot)$.

10. We consider the equation

$$x^2 - 3y^2 = 1.$$

- (1) Prove that

$$\{x + y\sqrt{3} \mid x \in \mathbb{N}, y \in \mathbb{Z}, x^2 - 3y^2 = 1\}$$

is a subgroup of $(\mathbb{R}_{>0}, \times)$

- (2) Check that $(2, 1)$ is a solution of the equation $x^2 - 3y^2 = 1$.
- (3) Without using the calculator, prove that

$$\left| \sqrt{3} - \frac{97}{56} \right| < 0.0001.$$

11. Prove that the following mappings are morphism of groups. Determine their images.

- (1) $(\mathbb{Z}, +) \rightarrow (\mathbb{R}^\times, \cdot)$, $n \mapsto (-1)^n$.
- (2) $(\mathbb{C}^\times, \cdot) \rightarrow (\mathbb{C}^\times, \cdot)$, $z \mapsto z/|z|$.
- (3) $(\mathbb{C}^\times, \cdot) \rightarrow (\mathbb{R}^\times, \cdot)$, $z \mapsto |z|$.

12. Let G be a group and X be a set equipped with a left action of G .

- (1) For $x \in X$, we define the *stabilizer* of x as

$$\text{Stab}(x) = \{g \in G \mid gx = x\}.$$

Prove that $\text{Stab}(x) = \{g \in G \mid gx = x\}$ is a subgroup of G .

- (2) Prove that the mapping

$$G/\text{Stab}(x) \longrightarrow Gx, \quad g \text{Stab}(x) \longmapsto gx$$

is well defined and is a bijection.

- (3) Prove that, if G is a finite group, then the number of elements of Gx is a divisor of the number of elements of G .

- 13.** Let G be a group and e be its neutral element. For any $a \in G$, let $N(a)$ be the set

$$\{n \in \mathbb{N}_{\geq 1} \mid a^n = e\}.$$

If $N(a)$ is non-empty, we denote by $\text{ord}(a)$ the least element of $N(a)$; otherwise we let $\text{ord}(a) = +\infty$. The element $\text{ord}(a) \in \mathbb{N} \cup \{+\infty\}$ is called the *order* of a .

- (1) Determine the order of the neutral element of G .
 - (2) Let a be an element of G and n be a positive integer. Assume that $a^n = e$. Prove that $\text{ord}(a) \leq n$.
 - (3) Let a be an element of G and n be a positive integer such that $n \leq \text{ord}(a)$. Prove that $a^0, \dots, a^{n-1}$ are distinct.
 - (4) Let a be an element of G . Let $\langle a \rangle$ be the image of the morphism of groups $(\mathbb{Z}, +) \rightarrow G$ that sends $1 \in \mathbb{Z}$ to a . Prove that the order of a is finite if and only if $\langle a \rangle$ is a finite set.
 - (5) Let a be an element of G which is of finite order. Prove that $\text{ord}(a)$ is equal to the number of elements of $\langle a \rangle$.
- 14.** Let G be a group and a and b be two elements of G which are of finite order. Let n and m be respectively the order of a and b . We suppose that $ab = ba$. Let e be the neutral element of G .

- (1) Let N be the least common multiple of m and n . Prove that $(ab)^N = e$. Deduce that ab is of finite order.
- (2) Let r be a positive integer such that $(ab)^r = e$. Prove that

$$e = b^{rn} = a^{rm}.$$

Deduce that r is divisible by $N/\text{gcd}(n, m)$, where $\text{gcd}(n, m)$ denotes the greatest common divisor of n and m .

- (3) Prove that the order of ab is equal to nm if n and m are coprime.

15. Let E be a set. We denote by $\mathfrak{S}_E$ the set of all bijections from E to E . The elements of $\mathfrak{S}_E$ are called *permutations* of E .

- (1) Prove that $\mathfrak{S}_E$ equipped with the composition of mappings forms a group (we write this composition law multiplicatively).
- (2) Let $\sigma \in \mathfrak{S}_E$. Prove that the mapping

$$\varphi_\sigma : \mathbb{Z} \times E \longrightarrow E, \quad (n, x) \longmapsto \sigma^n(x)$$

defines a left action of $\mathbb{Z}$ on E .

- (3) Let $\sigma \in \mathfrak{S}_E$. For any $x \in E$, let $\text{Orb}_\sigma(x) = \{\sigma^n(x) \mid n \in \mathbb{Z}\}$ be the orbite of x under the left action φ_σ . Prove that $\sigma(\text{Orb}_\sigma(x)) \subseteq \text{Orb}_\sigma(x)$ and that the restriction of σ to $\text{Orb}_\sigma(x)$ defines a bijection from $\text{Orb}_\sigma(x)$ to itself.
- (4) Let $\sigma \in \mathfrak{S}_E$. Denote by $_{\langle \sigma \rangle} E$ the set of all orbites of the left action φ_σ . We assume that this set is finite and is of the form $\{O_1, \dots, O_n\}$. For any $i \in \{1, \dots, n\}$, let

$$\sigma_i : E \longrightarrow E, \quad \sigma_i(x) = \begin{cases} \sigma(x), & x \in O_i, \\ x, & x \in E \setminus O_i. \end{cases}$$

Prove that $\sigma = \sigma_1 \cdots \sigma_n$. Determine the orbits of σ_i .

16. Let E be a finite set. We say that a permutation $\sigma \in \mathfrak{S}_E$ is a *cycle* if at most one orbit of φ_σ has more than one element. If $\tau \in \mathfrak{S}_E$ is a cycle and if one of the orbits of φ_σ has exactly two elements, then we say that σ is a *transposition*.

- (1) Let $\tau \in \mathfrak{S}_E$. Prove that τ is a transposition if and only if there exist elements x and y of E such that $x \neq y$ and that

$$\forall z \in E, \quad \tau(z) = \begin{cases} y, & z = x, \\ x, & z = y, \\ z, & z \notin \{x, y\}. \end{cases}$$

We denote by $\tau_{x,y}$ this transposition.

- (2) Let $\sigma \in \mathfrak{S}_E$ be a cycle. Prove that the order of σ is equal to the largest cardinal of its orbits. Is this equality still true for a general permutation?

- (3) Let $\sigma \in \mathfrak{S}_E$ be a cycle. We assume that x is an element of E such that $\text{Orb}_\sigma(x)$ has more than one element. Let p be the order of σ . For any $i \in \{0, \dots, p-1\}$, let $x_i = \sigma^i(x)$. Prove that

$$\sigma = \tau_{x_0, x_1} \tau_{x_1, x_2} \cdots \tau_{x_{p-2}, x_{p-1}}.$$

- (4) Prove that any permutation $\sigma \in \mathfrak{S}_E$ can be written as a composition of transpositions.

- 17.** Let $n \in \mathbb{N}_{\geq 2}$. We denote by $\mathfrak{S}_n$ the permutation group of $\{1, \dots, n\}$. Prove that the mapping

$$\text{sgn} : \mathfrak{S}_n \longrightarrow \mathbb{C}^\times, \quad \text{sgn}(\sigma) := \prod_{\substack{(i,j) \in \{1, \dots, n\}^2 \\ i < j}} \frac{\sigma(j) - \sigma(i)}{j - i}$$

is a morphism of groups. Determine its image.

- 18.** Let E be a finite set of at least two elements. Prove that there are exactly two morphism of groups from $\mathfrak{S}_E$ to $\mathbb{C}^\times$.

Chapter 6

Rings and modules

6.1 Unitary Ring

6.1.1 Definition. Let A be a set, and let $\dagger$ and $*$ be two composition laws on A . If the following conditions hold:

- (1) $(A, \dagger)$ forms a commutative group,
- (2) $(A, *)$ forms a monoid,
- (3) The operation $*$ is distributive over $\dagger$, i.e., for any $(a, b, c) \in A \times A \times A$,

$$a * (b \dagger c) = (a * b) \dagger (a * c), \quad (b \dagger c) * a = (b * a) \dagger (c * a),$$

then $(A, \dagger, *)$ is called a *unitary ring*. If $*$ is also commutative, then $(A, \dagger, *)$ is called a *commutative unitary ring*.

Let $(A, \dagger, *)$ be a unitary ring. The neutral element of the group $(A, \dagger)$ is called the *zero element* of this unitary ring, and the identity element of the monoid $(A, *)$ is called the *unity* of this unitary ring. Additionally, $A^\times$ usually denotes the set of invertible elements in $(A, *)$.

If $(A, \dagger, *)$ is a unitary ring, then $(A, \dagger, \check{*})$ is also a unitary ring. We call it the *opposite ring* of $(A, \dagger, *)$.

6.1.2 Example. The set of integers under the usual addition and multiplication forms a commutative unitary ring. Similarly, the sets of rational numbers, real numbers, and complex numbers under their usual addition and multiplication all form commutative unitary rings.

6.1.3 Notation. When referring to a unitary ring, unless otherwise specified, it is conventional to denote the first operation by $+$ and call it *addition*, denote the second operation by $\cdot$ and call it *multiplication*, denote its zero element by 0 , and denote its unity by 1 .

6.1.4 Proposition. *Let A be a unitary ring.*

(1) *For any $a \in A$, $0a = a0 = 0$.*

(2) *For any $(a, b) \in A \times A$, $(-a)b = a(-b) = -ab$.*

Proof. (1) Since $0 + 0 = 0$, by the distributivity of multiplication with respect to addition, we have

$$0 + 0a = 0a = (0 + 0)a = 0a + 0a.$$

Since $(A, +)$ is a group, any element of A is cancellative by Proposition 5.2.10. Thus we get $0a = 0$. Applying this case to the opposite ring of A , we obtain that $a0 = 0$ also holds.

(2) By the distributivity of multiplication with respect to addition, one has

$$ab + (-a)b = (a + (-a))b = 0b = 0,$$

where the last equality follows from (1). Thus, $(-a)b = -ab$. Applying this result to the opposite ring, we obtain $a(-b) = -ab$. $\square$

6.1.5 Definition. Let $(A, +, \cdot)$ be a unitary ring, 0 be its zero element, and 1 be its unity. If $0 = 1$, by Proposition 6.1.4, we obtain that, for any $a \in A$, one has

$$a = 1a = 0a = 0.$$

Hence $A = \{0\}$. In this case we say that $(A, +, \cdot)$ is the *zero ring*. If $1 \neq 0$, then we say that the unitary ring $(A, +, \cdot)$ is *non-zero*.

6.1.6 Definition. Let $(K, +, \cdot)$ be a unitary ring, 0 its zero element, and 1 its unity. If $0 \neq 1$ and every element in $K \setminus \{0\}$ is invertible in $(K, \cdot)$, then $(K, +, \cdot)$ is called a *division ring*. If $\cdot$ is moreover commutative, then $(K, +, \cdot)$ is called a *field*.

6.1.7 Example. The sets of rational numbers $\mathbb{Q}$, real numbers $\mathbb{R}$, and complex numbers $\mathbb{C}$ under their usual addition and multiplication form fields.

6.1.8 Definition. Let A be a unitary ring, and let A_0 be a subset of A . If A_0 is a subgroup of $(A, +)$ and a submonoid of $(A, \cdot)$, then A_0 is called a *unitary subring* of A . Note that A_0 itself forms a unitary ring under the restricted operations.

6.1.9 Example. $\mathbb{Z}$ is a unitary subring of $\mathbb{Q}$, $\mathbb{Q}$ is a unitary subring of $\mathbb{R}$, and $\mathbb{R}$ is a unitary subring of $\mathbb{C}$.

6.2 Homomorphism

6.2.1 Definition. Let A and B be unitary rings, and let $f : A \rightarrow B$ be a mapping. If f is both a group homomorphism from $(A, +)$ to $(B, +)$ and a monoid homomorphism from $(A, \cdot)$ to $(B, \cdot)$, then f is called a *unitary ring homomorphism*. By definition, if B is a unitary ring and B_0 is a subring of B , then the inclusion mapping $B_0 \rightarrow B$ is a unitary ring homomorphism.

6.2.2 Proposition. *Let R be a unitary ring. The group homomorphism from $(\mathbb{Z}, +)$ to $(R, +)$ that sends $1 \in \mathbb{Z}$ to the unity of R is the unique ring homomorphism from $\mathbb{Z}$ to R .*

Proof. Let 0_R and 1_R denote the zero element and the unity of R , respectively. By Proposition 5.6.4 there exists a unique group homomorphism f from $(\mathbb{Z}, +)$ to $(R, +)$ sending $1 \in \mathbb{Z}$ to 1_R . Thus, it only remains to verify that this mapping is also a homomorphism from $(\mathbb{Z}, \cdot)$ to $(R, \cdot)$, i.e., for any $(n, m) \in \mathbb{Z} \times \mathbb{Z}$, we have

$$f(nm) = f(n)f(m). \quad (6.1)$$

If $n = 0$, then $nm = 0$ and $f(n) = f(nm) = 0_R$. By (1) of Proposition 6.1.4, the equality (6.1) holds in this case. Similarly, if $m = 0$, the equality (6.1) also holds.

Assume n and m are nonzero natural numbers. Under the notation 5.6.2 (which defines $\ell 1_R$), for any nonzero natural number ℓ , we have

$$f(\ell) = \ell 1_R = \underbrace{1_R + \cdots + 1_R}_{\ell \text{ times}}.$$

By distributivity, we get

$$f(n)f(m) = \underbrace{1_R + \cdots + 1_R}_{nm \text{ times}} = f(nm).$$

Finally, by (2) of Proposition 6.1.4, we have

$$\begin{aligned} f((-n)m) &= f(-nm) = -f(nm) = -f(n)f(m) = f(-n)f(m), \\ f(n(-m)) &= f(-nm) = -f(n)f(m) = f(n)(-f(m)) = f(n)f(-m). \end{aligned}$$

Therefore, f is a unitary ring homomorphism. □

6.2.3 Proposition. *Let R be a unitary ring, and let $f : \mathbb{Z} \rightarrow R$ be the unique unitary ring homomorphism. Suppose that for any $n \in \mathbb{Z} \setminus \{0\}$, $f(n) \in R^\times$. Then the map*

$$g : \mathbb{Q} \longrightarrow R, \quad g\left(\frac{a}{b}\right) = f(b)^{-1}f(a), \quad (a, b) \in \mathbb{Z} \times \mathbb{N}_{>0}$$

is well-defined and is the unique unitary ring homomorphism from $\mathbb{Q}$ to R .

Proof. Since f is a unitary ring homomorphism, for any $(a, b) \in \mathbb{Z} \times \mathbb{N}_{>0}$ and $\lambda \in \mathbb{N}_{>0}$, we have

$$f(\lambda b)^{-1}f(\lambda a) = f(b)^{-1}f(\lambda)^{-1}f(\lambda)f(a) = f(b)^{-1}f(a).$$

Thus, g is well-defined. Furthermore, for any $(a, b) \in \mathbb{Z} \times \mathbb{Z}$,

$$f(a)f(b) = f(ab) = f(ba) = f(b)f(a).$$

If $a \neq 0$, then

$$f(b)f(a)^{-1} = f(a)^{-1}f(a)f(b)f(a)^{-1} = f(a)^{-1}f(b)f(a)f(a)^{-1} = f(a)^{-1}f(b).$$

Thus, for any elements (a, b) and (c, d) in $\mathbb{Z} \times \mathbb{N}_{>0}$, we have

$$\begin{aligned} g\left(\frac{a}{b} + \frac{c}{d}\right) &= g\left(\frac{ad + bc}{bd}\right) = f(bd)^{-1}(f(ad + bc)) \\ &= f(b)^{-1}f(d)^{-1}(f(a)f(d) + f(b)f(c)) \\ &= f(b)^{-1}f(a) + f(d)^{-1}f(c) = g\left(\frac{a}{b}\right) + g\left(\frac{c}{d}\right), \end{aligned}$$

$$\begin{aligned} g\left(\frac{a}{b} \cdot \frac{c}{d}\right) &= g\left(\frac{ac}{bd}\right) = f(bd)^{-1}f(ac) \\ &= f(b)^{-1}f(d)^{-1}f(a)f(c) = g\left(\frac{a}{b}\right)g\left(\frac{c}{d}\right). \end{aligned}$$

Therefore, g is a unitary ring homomorphism from $\mathbb{Q}$ to R .

If $h : \mathbb{Q} \rightarrow R$ is a unitary ring homomorphism, then its restriction to $\mathbb{Z}$ coincides with f . For any $(a, b) \in \mathbb{N}_{>0} \times \mathbb{N}_{>0}$ (ensuring $b \neq 0$),

$$f(a) = h(a) = h(b)h(a/b) = f(b)h(a/b).$$

Thus, $h(a/b) = f(b)^{-1}f(a) = g(a/b)$. □

6.3 Modules over unitary rings

In this section, K denotes a unitary ring, 0 denotes the zero element of K , and 1 denotes the unit element of K .

6.3.1 Definition. Let $(V, +)$ be an abelian group. A *left K -module structure* (resp. *right K -module structure*) on $(V, +)$ is a left action (resp. right action)

$$\Phi : K \times V \longrightarrow V$$

of the multiplicative monoid $(K, \cdot)$ on the set V such that

(i) For any $(a, b) \in K \times K$ and $x \in V$,

$$\Phi(a + b, x) = \Phi(a, x) + \Phi(b, x),$$

(ii) For any $a \in K$ and $(x, y) \in V \times V$,

$$\Phi(a, x + y) = \Phi(a, x) + \Phi(a, y).$$

An abelian group equipped with a left K -module structure (resp. right K -module structure) is called a *left K -module* (resp. *right K -module*).

6.3.2 Proposition. *Let $(V, +)$ be an abelian group, $\mathbf{0}$ its neutral element, and $\Phi : K \times V \rightarrow V$ be a mapping. Suppose Φ satisfies condition (i) in Definition 6.3.1, i.e.,*

$$\forall (a, b) \in K \times K, \forall x \in V, \Phi(a + b, x) = \Phi(a, x) + \Phi(b, x).$$

Then for any $x \in V$, $\Phi(0, x) = \mathbf{0}$ and

$$\forall a \in K, \Phi(-a, x) = -\Phi(a, x).$$

Proof. Let $x \in V$. Condition (i) in Definition 6.3.1 shows that the map

$$\Phi(\cdot, x) : K \longrightarrow V, a \longmapsto \Phi(a, x)$$

is a group homomorphism from $(K, +)$ to $(V, +)$. Therefore, it maps the neutral element 0 of $(K, +)$ to $\mathbf{0}$, and the inverse $-a$ of $a \in K$ in $(K, +)$ to the inverse $-\Phi(a, x)$ of $\Phi(a, x)$ in $(V, +)$. $\square$

6.3.3 Remark. Let $(K, +, \cdot)$ be a unitary ring. Let K^{op} denote the opposite ring of K . A right K -module structure on an abelian group V is a left K^{op} -module structure on V , and a left K -module structure on V is a right K^{op} -module structure on V .

When K is a commutative unitary ring, a left K -module structure on an abelian group is also a right K -module structure on the same group. We simply call it a *K -module structure*. We refer to an abelian group equipped with a K -module structure as a *K -module*. If K is a field, then a K -module is also called a *K -vector space*.

6.3.4 Example. Let $(V, +)$ be an abelian group. Then the mapping

$$\mathbb{Z} \times V \longrightarrow V, (n, x) \longmapsto nx$$

defines a $\mathbb{Z}$ -module structure on V , called the *canonical $\mathbb{Z}$ -module structure* on V . Thus, any abelian group can be viewed as a $\mathbb{Z}$ -module.

6.3.5 Example. The multiplicative law of the unitary ring K

$$K \times K \longrightarrow K, (a, x) \longmapsto ax$$

defines a left K -module structure on K . Similarly, the mapping

$$K \times K \longrightarrow K, (a, x) \longmapsto xa$$

defines a right K -module structure on K . Therefore, the unitary ring K itself can be regarded as a left K -module or a right K -module.

6.3.6 Notation. When discussing a left K -module structure on an abelian group $(V, +)$, it is often convenient to denote it in the form $(a, x) \mapsto ax$. Under this notation, conditions in Definition 6.3.1 become

$$\begin{aligned} \forall x \in V, 1x &= x, \\ \forall (a, b) \in K \times K, \forall x \in V, (ab)x &= a(bx), \\ \forall (a, b) \in K \times K, \forall x \in V, (a + b)x &= ax + bx, \\ \forall a \in K, \forall (x, y) \in V \times V, a(x + y) &= ax + ay. \end{aligned}$$

Under this notation, Proposition 6.3.2 implies that, for any $x \in V$, $0x = \mathbf{0}$ and $(-1)x = -x$.

When discussing a right K -module structure on an abelian group $(V, +)$, it is often convenient to denote it in the form $(a, x) \mapsto xa$. Under this notation, conditions in Definition 6.3.1 become

$$\begin{aligned} \forall x \in V, x1 &= x \\ \forall (a, b) \in K \times K, \forall x \in V, x(ab) &= (xa)b, \\ \forall (a, b) \in K \times K, \forall x \in V, x(a + b) &= xa + xb, \\ \forall a \in K, \forall (x, y) \in V \times V, (x + y)a &= xa + ya. \end{aligned}$$

Under this notation, Proposition 6.3.2 implies that for any $x \in V$, $x0 = \mathbf{0}$ and $x(-1) = -x$.

6.3.7 Definition. Let I be a set and $(V_i)_{i \in I}$ be a family of left K -modules (resp. right K -modules). For any $i \in I$, let $\varphi_i : K \times V_i \rightarrow V_i$ be the left K -module structure (resp. right K -module structure) of V_i . It turns out that the direct product group $\prod_{i \in I} V_i$, equipped with the mapping

$$K \times \prod_{i \in I} V_i \longrightarrow \prod_{i \in I} V_i, (a, (x_i)_{i \in I}) \longmapsto (\varphi_i(a, x_i))_{i \in I},$$

forms a left K -module (resp. right K -module). We call it the *direct product* of $(V_i)_{i \in I}$.

In the case where all V_i are equal to the same left K -module (resp. right K -module) V , the direct product of $(V_i)_{i \in I}$ is denoted as V^I . If furthermore $I = \{1, \dots, n\}$, the direct product $V^{\{1, \dots, n\}}$ is also denoted as V^n . An element of V^n can be written in the form $(x_1, \dots, x_n)$, where $x_1, \dots, x_n$ are elements of V .

Let I be a set. Since K is equipped with a structure of left K -module and a structure of right K -module (as explained in Example 6.3.5), the set K^I is naturally equipped with a structure of left K -module and a structure of right K -module. Note that K^I can also be considered the set of mappings from I to K . For any $i \in I$, we denote by $e_i : I \rightarrow K$ that sends i to 1, and sends any $j \in I \setminus \{i\}$ to 0. In the case where $I = \{1, \dots, n\}$, one has

$$(0, \dots, 0, 1, 0, \dots, 0).$$

$\uparrow$
 $i^{\text{th}}\text{-entry}$

A general element $(a_1, \dots, a_n) \in K^n$ can be written as

$$a_1 e_1 + \dots + a_n e_n$$

when we consider K^n as a left K -module. Similarly, it can be written as

$$e_1 a_1 + \dots + e_n a_n$$

when we consider K^n as a right K -module.

6.4 Submodules

In this section, K denotes a unitary ring, 0 denotes the zero element of K , and 1 denotes the unity of K .

6.4.1 Definition. Let V be a left K -module. A subgroup W of the additive group $(V, +)$ is called a *left K -submodule* of V if it satisfies

$$\forall (a, x) \in K \times W, ax \in W.$$

Let V be a right K -module. A subgroup W of the additive group $(V, +)$ is called a *right K -submodule* of V if it satisfies

$$\forall (a, x) \in K \times W, xa \in W.$$

6.4.2 Proposition. Let V be a left K -module, and let W be a non-empty subset of V . Then the following conditions are equivalent:

- (1) W is a left K -submodule of V ,

(2) For any $(a, b) \in K \times K$ and any $(x, y) \in W \times W$, $ax + by \in W$,

(3) For any $a \in K$ and any $(x, y) \in W \times W$, $x + ay \in W$.

Proof. “(1) $\implies$ (2)”: Since W is a left K -submodule of V , for any $(a, b) \in K \times K$ and any $(x, y) \in W \times W$, we have $ax \in W$, $by \in W$, and hence $ax + by \in W$.

(3) is a special case of (2).

“(3) $\implies$ (1)”: By Proposition 6.3.2, for any $(x, y) \in W \times W$,

$$x - y = x + (-1)y \in W.$$

Thus, W is a subgroup of $(V, +)$. Furthermore, for any $a \in K$ and $y \in W$,

$$ay = \mathbf{0} + ay \in W.$$

Therefore, W is a left K -submodule of V . □

6.4.3 Proposition. Let V be a right K -module, and let W be a non-empty subset of V . Then the following conditions are equivalent:

(1) W is a right K -submodule of V ,

(2) For any $(a, b) \in K \times K$ and any $(x, y) \in W \times W$, $xa + yb \in W$,

(3) For any $a \in K$ and any $(x, y) \in W \times W$, $x + ya \in W$.

Proof. Apply Proposition 6.4.2 to the left K^{op} -module structure. □

6.4.4 Example. Let I be a set. Denote by $K^{\oplus I}$ the set of maps $f : I \rightarrow K$ satisfying the condition that

$$I_f := \{i \in I \mid f(i) \neq 0\} \text{ is a finite set.}$$

By definition, for any $j \in I$, the map

$$e_j : I \longrightarrow K, \quad i \longmapsto \begin{cases} 1, & i = j, \\ 0, & i \neq j, \end{cases}$$

belongs to $K^{\oplus I}$.

Note that for any $a \in K$ and any $(f, g) \in K^{\oplus I} \times K^{\oplus I}$,

$$I_{f+ag} \subseteq I_f \cup I_g, \quad I_{f+ga} \subseteq I_f \cup I_g.$$

Thus,

$$\{f + ag, f + ga\} \subseteq K^{\oplus I}.$$

By Proposition 6.4.2, $K^{\oplus I}$ is both a left K -submodule and a right K -submodule of K^I .

6.5 Module homomorphisms

In this section, K denotes a unitary ring, 0 denotes the zero element of K , and 1 denotes the unity of K .

6.5.1 Definition. Let E and F be left K -modules. A *left K -module homomorphism* from E to F is an additive group homomorphism $\varphi : E \rightarrow F$ such that

$$\forall (a, x) \in K \times E, \varphi(ax) = a\varphi(x).$$

If φ is also bijective, it is called a left K -module *isomorphism*.

Denote by $\text{Hom}_{K\text{-Mod}}(E, F)$ the set of all left K -module homomorphisms from E to F . It is a subgroup of F^E .

6.5.2 Proposition. Let E and F be left K -modules, and let $\varphi : E \rightarrow F$ be a map. Then the following statements are equivalent:

- (1) φ is a left K -module homomorphism,
- (2) For any $(a, b) \in K \times K$ and $(x, y) \in E \times E$, $\varphi(ax + by) = a\varphi(x) + b\varphi(y)$,
- (3) For any $a \in K$ and $(x, y) \in E \times E$,

$$\varphi(x + ay) = \varphi(x) + a\varphi(y).$$

Proof. “(1) $\implies$ (2)”: Let $(a, b) \in K \times K$, $(x, y) \in E \times E$. Since φ is a left K -module homomorphism,

$$\varphi(ax) = a\varphi(x), \varphi(by) = b\varphi(y).$$

Moreover, since φ is an additive group homomorphism,

$$\varphi(ax + by) = \varphi(ax) + \varphi(by) = a\varphi(x) + b\varphi(y).$$

(3) is a special case of (2).

“(3) $\implies$ (1)”: Let $\mathbf{0}_E$ and $\mathbf{0}_F$ denote the zero elements of E and F , respectively. For any $(x, y) \in E \times E$,

$$\varphi(x + y) = \varphi(x + 1y) = \varphi(x) + 1\varphi(y) = \varphi(x) + \varphi(y),$$

so φ is a group homomorphism. Furthermore, for any $(a, y) \in K \times E$,

$$\varphi(ay) = \varphi(\mathbf{0}_E + ay) = \varphi(\mathbf{0}_E) + a\varphi(y) = \mathbf{0}_F + a\varphi(y) = a\varphi(y).$$

Therefore, φ is a left K -module homomorphism. □

6.5.3 Proposition. *Let E , F and G be left K -modules. If $\varphi : E \rightarrow F$ and $\psi : F \rightarrow G$ are left K -module homomorphism, then the composite mapping $\psi \circ \varphi : E \rightarrow G$ is a left K -module homomorphism.*

Proof. Let $(x, y) \in E \times E$ and $a \in K$. One has

$$(\psi \circ \varphi)(x + ay) = \psi(\varphi(x + ay)) = \psi(\varphi(x) + a\varphi(y)) = \psi(\varphi(x)) + a\psi(\varphi(y)).$$

By Proposition 6.5.2, we obtain that $\psi \circ \varphi$ is a left K -module homomorphism. $\square$

6.5.4 Proposition. *Let E be a left K -module. For any $x \in E$, there exists a unique homomorphism of left K -modules from K to E that sends $1 \in K$ to x .*

Proof. If $f : K \rightarrow E$ is a homomorphism of left K -modules, then one has $f(a) = f(a \cdot 1) = af(1) = ax$. Hence the uniqueness part of the proposition is true. It remains to check that the mapping $f : K \rightarrow E$, $a \mapsto ax$ is a homomorphism of left K -modules. For $(a, b) \in K \times K$, one has

$$f(a + b) = (a + b)x = ax + bx = f(a) + f(b), \quad f(ab) = (ab)x = a(bx) = af(b).$$

Hence f is effectively a homomorphism of left K -modules. $\square$

6.5.5 Proposition. *Let E and F be left K -modules. Let A be a unitary ring, and suppose F is additionally equipped with a right A -module structure. Then $\text{Hom}_{K\text{-Mod}}(E, F)$ is a right A -submodule of F^E . In particular, when K is a commutative unitary ring, $\text{Hom}_{K\text{-Mod}}(E, F)$ is a K -submodule of F^E .*

Proof. Let φ and ψ be elements of $\text{Hom}_{K\text{-Mod}}(E, F)$. By definition, for any $(a, x, y) \in K \times E \times E$,

$$\begin{aligned} (\varphi - \psi)(x + ay) &= \varphi(x + ay) - \psi(x + ay) \\ &= \varphi(x) + a\varphi(y) - (\psi(x) + a\psi(y)) \\ &= (\varphi(x) - \psi(x)) + a(\varphi(y) - \psi(y)) \\ &= (\varphi - \psi)(x) + a(\varphi - \psi)(y). \end{aligned}$$

Thus, $\text{Hom}_{K\text{-Mod}}(E, F)$ is a subgroup of F^E .

Let $\varphi \in \text{Hom}_{K\text{-Mod}}(E, F)$, $\lambda \in A$. Then for any $(x, y) \in E \times E$ and $a \in K$,

$$\varphi(x + ay)\lambda = (\varphi(x) + a\varphi(y))\lambda = \varphi(x)\lambda + a\varphi(y)\lambda$$

Therefore, $\text{Hom}_{K\text{-Mod}}(E, F)$ is an A -submodule of F^E . $\square$

6.5.6 Theorem. *Let I be a set, V be a left K -module. Let A be a unitary ring, and suppose V is additionally equipped with a right A -module structure. Then the map*

$$\mathrm{Hom}_{K\text{-Mod}}(K^{\oplus I}, V) \longrightarrow V^I, \quad \varphi \longmapsto (\varphi(e_i))_{i \in I} \quad (6.2)$$

is a right A -module isomorphism. In particular, when K is a commutative unitary ring, the above mapping is a K -module isomorphism.

Proof. Let φ and ψ be elements of $\mathrm{Hom}_{K\text{-Mod}}(K^{\oplus I}, V)$. For any $i \in I$,

$$(\varphi + \psi)(e_i) = \varphi(e_i) + \psi(e_i).$$

Moreover, for any $a \in A$,

$$(\varphi(e_i))_{i \in I} a = (\varphi(e_i) a)_{i \in I}.$$

Thus, the mapping described in the theorem is a right A -module homomorphism.

Let $(x_i)_{i \in I}$ be an element of V^I , by Proposition 6.5.4, for any $i \in I$, there exists a unique homomorphism f_i from K to V sending $a \in K$ to ax_i . Moreover, by Proposition 5.7.7, there is a unique homomorphism $f : K^{\oplus I} \rightarrow V$ sending $(a_i)_{i \in I} \in K^{\oplus I}$ to

$$\sum_{i \in I} f_i(a_i) = \sum_{i \in I} a_i x_i.$$

In particular, $f(e_i) = x_i$ for any $i \in I$. Thus $(x_i)_{i \in I}$ is the image of f by the mapping (6.2), so the mapping is surjective.

To show that the mapping (6.2) is injective, by Proposition 5.5.13, it suffices to check that its kernel is the singleton set of the neutral element of $\mathrm{Hom}_{K\text{-Mod}}(K^{\oplus I}, V)$, namely the constant homomorphism that sends any element of $K^{\oplus I}$ to the neutral element 0_V of V . Let $\varphi : K^{\oplus I} \rightarrow V$ be a homomorphism of left K -modules such that $\varphi(e_i) = 0_V$ for any $i \in I$. Then, for any $(a_i)_{i \in I} \in K^{\oplus I}$ one has

$$\varphi((a_i)_{i \in I}) = \sum_{i \in I} \varphi(a_i e_i) = \sum_{i \in I} a_i \varphi(e_i) = 0_V.$$

The theorem is thus proved. □

6.6 Matrices

6.6.1 Definition. Let n be a natural number, and let V be a left K -module. Theorem 6.5.6 implies that for any $(x_1, \dots, x_n) \in V^n$, there exists a unique left

K -module homomorphism that maps $(t_1, \dots, t_n) \in K^n$ to $t_1x_1 + \dots + t_nx_n$. We denote this left K -module homomorphism as a column vector

$$\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

Suppose that $V = K^p$ (viewed as a left K -module), where p is a natural number. Then each $x_i \in K^p$ can be written in the form $(x_{i,1}, \dots, x_{i,p})$, where each $x_{i,j}$ belongs to K . Thus the column vector in Definition 6.6.1 can be written as

$$\begin{pmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,p} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n,1} & x_{n,2} & \cdots & x_{n,p} \end{pmatrix}$$

In general, we call n by p matrix with coefficients in K any homomorphism of left K -modules from K^n to K^p . We denote by $M_{n,p}$ the set of n by p matrices with coefficients in K .

6.6.2 Example. Let n be a natural number and $(b_1, \dots, b_n) \in K^n$. We denote by $\text{diag}(b_1, \dots, b_n)$ the mapping from K^n to K^n that sends $(t_1, \dots, t_n)$ to $(t_1b_1, \dots, t_nb_n)$. This is a homomorphism of left K -modules. Note that

$$(t_1b_1, \dots, t_nb_n) = \sum_{i=1}^n t_i(0, \dots, 0, \underset{\substack{\uparrow \\ i^{\text{th}}\text{-entry}}}{b_i}, 0, \dots, 0).$$

Therefore

$$\text{diag}(b_1, \dots, b_n) = \begin{pmatrix} b_1 & 0 & \cdots & 0 \\ 0 & b_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & b_n \end{pmatrix}.$$

An n by n matrix of the form $\text{diag}(b_1, \dots, b_n)$ is called a *diagonal matrix*.

Note that $\text{diag}(1, \dots, 1)$ is the identity mapping, we denote it as I_n .

6.6.3 Definition. Let n and p be natural numbers, and

$$A = \begin{pmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,p} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,p} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & \cdots & a_{n,p} \end{pmatrix} \in M_{n,p}(K).$$

If

$$\varphi = \begin{pmatrix} x_1 \\ \vdots \\ x_p \end{pmatrix}$$

is a homomorphism of left K -modules from K^p to V , we denote by $A\varphi : K^n \rightarrow V$ the composite left K -module homomorphism $\varphi \circ A$ (see Proposition 6.5.3). By definition, A sends any $(t_1, \dots, t_n) \in K^n$ to

$$\sum_{i=1}^n t_i(a_{i,1}, \dots, a_{i,p}).$$

Therefore, $A\varphi$ sends $(t_1, \dots, t_n)$ to

$$\sum_{i=1}^n t_i(a_{i,1}x_1 + \dots + a_{i,p}x_p),$$

which leads to

$$A\varphi = \begin{pmatrix} a_{1,1}x_1 + \dots + a_{1,p}x_p \\ \vdots \\ a_{n,1}x_1 + \dots + a_{n,p}x_p \end{pmatrix}$$

Let

$$B = \begin{pmatrix} b_{1,1} & b_{1,2} & \cdots & b_{1,p} \\ b_{2,1} & b_{2,2} & \cdots & b_{2,p} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n,1} & b_{n,2} & \cdots & b_{n,p} \end{pmatrix} \in M_{p,r}(K)$$

The above formula shows that

$$AB = \begin{pmatrix} \sum_{j=1}^p a_{1,j}b_{j,1} & \sum_{j=1}^p a_{1,j}b_{j,2} & \cdots & \sum_{j=1}^p a_{1,j}b_{j,r} \\ \sum_{j=1}^p a_{2,j}b_{j,1} & \sum_{j=1}^p a_{2,j}b_{j,2} & \cdots & \sum_{j=1}^p a_{2,j}b_{j,r} \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{j=1}^p a_{n,j}b_{j,1} & \sum_{j=1}^p a_{n,j}b_{j,2} & \cdots & \sum_{j=1}^p a_{n,j}b_{j,r} \end{pmatrix}$$

6.6.4 Proposition. *Let m, n, p and q be natural numbers, $A \in M_{m,n}(K)$, $B \in M_{n,p}(K)$ and $C \in M_{p,q}(K)$. Then one has*

$$(AB)C = A(BC) \in M_{m,q}(K).$$

Moreover,

$$AI_n = I_n A = A.$$

Proof. The first statement is a consequence of Proposition 3.4.3. The second one is a consequence of Proposition 3.4.4. $\square$

6.6.5 Example. Let $n \in \mathbb{N}$, and $(a_1, \dots, a_n)$ and $(b_1, \dots, b_n)$ be elements of K^n . Then $\text{diag}(a_1, \dots, a_n) \text{diag}(b_1, \dots, b_n)$ sends $(t_1, \dots, t_n) \in K^n$ to $(t_1 a_1 b_1, \dots, t_n a_n b_n)$. Hence

$$\text{diag}(a_1, \dots, a_n) \text{diag}(b_1, \dots, b_n) = \text{diag}(a_1 b_1, \dots, a_n b_n).$$

6.7 Linear equations

In this section, we fix a unitary ring K .

6.7.1 Definition. Let p be a natural number. For any $\mathbf{a} = (a_1, \dots, a_p) \in K^p$, we denote by $j(\mathbf{a})$ the least index $i \in \{1, \dots, p\}$ such that $a_i \neq 0$. For example, in the case where $K = \mathbb{R}$, one has

$$j(0, 0, 2, 5) = 3, \quad j(0, 1, 0, 0, 2, 5) = 2.$$

In the case where $\mathbf{a} = (0, \dots, 0)$, by convention $j(\mathbf{a}) = +\infty$.

Let $(n, p) \in \mathbb{N}$ and $A \in M_{n,p}(K)$. We write A in the form of a column

$$\begin{pmatrix} \mathbf{a}^{(1)} \\ \vdots \\ \mathbf{a}^{(n)} \end{pmatrix},$$

where each $\mathbf{a}^{(i)} := (a_{i,1}, \dots, a_{i,p})$ belongs to K^p . If

$$\forall i \in \{1, \dots, n-1\}, \quad j(\mathbf{a}^{(i)}) \leq j(\mathbf{a}^{(i+1)})$$

and the strict inequality $j(\mathbf{a}^{(i)}) < j(\mathbf{a}^{(i+1)})$ holds once $\mathbf{a}^{(i)} \neq (0, \dots, 0)$, then we say that the matrix A is of *row echelon form*. If in addition, for any $i \in \{1, \dots, n\}$ such that $\mathbf{a}^{(i)} \neq (0, \dots, 0)$, one has $a_{i,j(\mathbf{a}^{(i)})} = 1$ and

$$\forall k \in \{1, \dots, n\}, \quad \mathbf{a}_{k,j(\mathbf{a}^{(i)})} = 0,$$

then we say that the matrix A is of the *reduced row echelon form*.

6.7.2 Example. The matrix

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & 0 & 2 & 3 & 4 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

is of row echelon form, while the matrix

$$\begin{pmatrix} 1 & 2 & 0 & 0 & 3 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

is of reduced row echelon form.

6.7.3 Proposition. Let $(n, p) \in \mathbb{N} \times \mathbb{N}$. Let

$$A := \begin{pmatrix} \mathbf{a}^{(1)} \\ \vdots \\ \mathbf{a}^{(n)} \end{pmatrix} \in M_{n,p}(K).$$

If A is of row echelon form, then the set

$$\{i \in \{1, \dots, n\} \mid \mathbf{a}^{(i)} \neq (0, \dots, 0)\}$$

has no more than p elements.

Proof. Assume that $\{i \in \{1, \dots, n\} \mid \mathbf{a}^{(i)} \neq (0, \dots, 0)\}$ has k elements. Then one has

$$1 \leq j(\mathbf{a}^{(1)}) < \dots < j(\mathbf{a}^{(k)}) \leq p,$$

so that $k \leq p$ by Lemma 4.7.2. □

6.7.4 Definition. Let $(n, p) \in \mathbb{N} \times \mathbb{N}$ and

$$A = \begin{pmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,p} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,p} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & \cdots & a_{n,p} \end{pmatrix} \in M_{n,p}(K).$$

Let V be a left K -module and $(b_1, \dots, b_n) \in V^n$. We consider the following system of equations

$$\begin{cases} a_{1,1}x_1 + \cdots + a_{1,p}x_p & = & b_1 \\ & \vdots & \\ a_{n,1}x_1 + \cdots + a_{n,p}x_p & = & b_n. \end{cases} \quad (6.3)$$

An element $(x_1, \dots, x_p)$ of V^p such that all equalities in (6.3) hold is called a *solution* to the system of equations (6.3). The set of all solutions to (6.3) is called the *solution set* of (6.3).

6.7.5 Proposition. *We keep the notation of Definition 6.7.4. Suppose that the matrix A is of reduced row echelon form. Let*

$$I(A) = \{i \in \{1, \dots, n\} \mid (a_{i,1}, \dots, a_{i,p}) \neq (0, \dots, 0)\}, \quad (6.4)$$

$$J_0(A) = \{1, \dots, p\} \setminus \{j(a_{i,1}, \dots, a_{i,p}) \mid i \in I(A)\}. \quad (6.5)$$

(1) *If there exists $i \in \{1, \dots, n\} \setminus I(A)$ such that $b_i \neq 0$, then the system of equations (6.3) does not have any solution.*

(2) *Suppose that*

$$\forall i \in \{1, \dots, n\} \setminus I(A), \quad b_i = 0.$$

Then the system of equations (6.3) has at least one solution. Moreover, the mapping

$$\Phi : V^{J_0(A)} \longrightarrow V^p, \quad (z_\ell)_{\ell \in J_0(A)} \longmapsto (x_1, \dots, x_p)$$

with

$$x_k := \begin{cases} z_k, & \text{if } k \in J_0(A), \\ b_i - \sum_{\ell \in J_0(A)} a_{i,\ell} z_\ell, & \text{if } k = j(a_{i,1}, \dots, a_{i,p}) \end{cases}$$

is injective, and its range identifies with the solutions of the system of equations (6.3).

Proof. (1) The i -th equation of (6.3) can be written as

$$a_{i,1}x_1 + \dots + a_{i,p}x_p = b_i.$$

If $a_{i,1} = \dots = a_{i,p} = 0$ but $b_i \neq 0$, this equation does not have any solution.

(2) We first check that the range of the mapping Φ is contained in the solution set of (6.3). Let $z = (z_\ell)_{\ell \in J_0(A)} \in V^{J_0(A)}$ and $(x_1, \dots, x_p) = \Phi(z)$. For $i \in \{1, \dots, n\} \setminus I(A)$, one has

$$a_{i,1}x_1 + \dots + a_{i,p}x_p = 0 = b_i.$$

For $i \in I(A)$ and $k = j(a_{i,1}, \dots, a_{i,p})$, since $a_{i,k} = 1$ and $a_{i,\ell} = 0$ for any $\ell \in \{1, \dots, p\} \setminus J_0(A)$ such that $\ell \neq k$, one has

$$a_{i,1}x_1 + \dots + a_{i,p}x_p = x_k + \sum_{\ell \in J_0(A)} a_{i,\ell}x_\ell = x_k + \sum_{\ell \in J_0(A)} a_{i,\ell}z_\ell = b_i.$$

Conversely, if $(x_1, \dots, x_p)$ is a solution to (6.3), then for $i \in I(A)$ and $k = j(a_{i,1}, \dots, a_{i,p})$ one has

$$a_{i,1}x_1 + \dots + a_{i,p}x_p = x_k + \sum_{\ell \in J_0(A)} a_{i,\ell}x_\ell$$

and hence

$$x_k = b_i - \sum_{\ell \in J_0(A)} a_{i,\ell}x_\ell.$$

Therefore $(x_1, \dots, x_p) = \Phi((x_\ell)_{\ell \in J_0(A)})$. □

6.7.6 Example. Consider the matrix of reduced row echelon form

$$\begin{pmatrix} 1 & 2 & 0 & 0 & 3 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

The corresponding system of linear equations is

$$\begin{cases} x_1 + 2x_2 & + 3x_5 = b_1 \\ & x_3 & + 2x_5 = b_2 \\ & & x_4 & = b_3 \end{cases}$$

The solution set of the equation is given by

$$\{(b_1 - 2z_2 - 3z_5, z_2, b_2 - 2z_5, b_3, z_5) \mid (z_2, z_5) \in V^2\}.$$

6.7.7 Proposition. Let m, n and p be natural numbers, and $S \in M_{m,n}(K)$, $A \in M_{n,p}(K)$. If $(x_1, \dots, x_p)$ is a solution of the system of linear equations

$$A \begin{pmatrix} x_1 \\ \vdots \\ x_p \end{pmatrix} = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}, \tag{*}$$

then it is also a solution of the system of linear equation

$$SA \begin{pmatrix} x_1 \\ \vdots \\ x_p \end{pmatrix} = S \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}. \tag{*_S}$$

The two systems of linear equations have the same solution set when S is left invertible, namely there exists $R \in M_{n,m}(K)$ such that $RS = I_n$.

Proof. This is a consequence of Proposition 6.6.4. $\square$

6.7.8 Example. Let n be a natural number,

$$\sigma : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$$

be a bijection. Let $P_\sigma : K^n \rightarrow K^n$ be the homomorphism of left K -module defined as

$$P_\sigma(t_1, \dots, t_n) = (t_{\sigma^{-1}(1)}, \dots, t_{\sigma^{-1}(n)}).$$

By definition,

$$P_\sigma P_{\sigma^{-1}} = P_{\sigma^{-1}} P_\sigma = I_n.$$

If V is a left K -module and $(x_1, \dots, x_n) \in V^n$, then

$$P_\sigma \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

sends $(t_1, \dots, t_n) \in K^n$ to

$$\sum_{i=1}^n t_{\sigma^{-1}(i)} x_i = \sum_{j=1}^n t_j x_{\sigma(j)}.$$

Hence

$$P_\sigma \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} x_{\sigma(1)} \\ \vdots \\ x_{\sigma(n)} \end{pmatrix}.$$

One can compute explicitly P_σ as follows:

$$P_\sigma = P_\sigma I_n = P_\sigma \begin{pmatrix} e_1 \\ \vdots \\ e_n \end{pmatrix} = \begin{pmatrix} e_{\sigma(1)} \\ \vdots \\ e_{\sigma(n)} \end{pmatrix}$$

6.7.9 Example. Let n be a natural number and $r = (r_1, \dots, r_n) \in K^n$. We denote by D_r the diagonal matrix

$$\text{diag}(r_1, \dots, r_n).$$

Assume that each r_i is left invertible, namely there exists $s_i \in K$ such that $s_i r_i = 1$. Let $s = (s_1, \dots, s_n)$. One has

$$D_s D_r = I_n.$$

If V is a left K -module and $(x_1, \dots, x_n) \in V^n$, then

$$D_r \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

sends $(t_1, \dots, t_n)$ to

$$\sum_{i=1}^n t_i r_i x_i.$$

Hence

$$D_r \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} r_1 x_1 \\ \vdots \\ r_n x_n \end{pmatrix}.$$

6.7.10 Example. Let $n \in \mathbb{N}$ and $i \in \{1, \dots, n\}$. Let $c = (c_1, \dots, c_n) \in K^n$ such that $c_i = 0$. Let $S_{i,c} \in M_{n,n}(K)$ be the K -linear mapping $K^n \rightarrow K^n$ that sends $(t_1, \dots, t_n)$ to

$$\left(t_1, \dots, t_{i-1}, t_i + \sum_{j=1}^n t_j c_j, t_{i+1}, \dots, t_n \right)$$

Note that

$$S_{i,-c} S_{i,c} = I_n = S_{i,c} S_{i,-c}.$$

If V is a left K -module and $(x_1, \dots, x_n) \in V^n$, then

$$S_{i,c} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

send $(t_1, \dots, t_n)$ to

$$\sum_{j=1}^n t_j x_j + \sum_{j=1}^n t_j c_j x_i = \sum_{j=1}^n t_j (x_j + c_j x_i).$$

Therefore,

$$S_{i,c} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} x_1 + c_1 x_i \\ \vdots \\ x_n + c_n x_i \end{pmatrix}.$$

In particular,

$$S_{i,c} = S_{i,c} I_n = \begin{pmatrix} e_1 + c_1 e_i \\ \vdots \\ e_n + c_n e_i \end{pmatrix}.$$

6.7.11 Definition. Let $G_n(K)$ be the subset of $M_{n,n}(K)$ consisting of matrices S that can be written as $U_1 \dots U_N$ (by convention the product is considered as I_n when $N = 0$), where each U_i is of one of the following forms:

- (1) P_σ with $\sigma : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ being a bijection,
- (2) $\text{diag}(r_1, \dots, r_n)$, with each r_i being left invertible,
- (3) $S_{i,c}$, with $i \in \{1, \dots, n\}$, $c = (c_1, \dots, c_n) \in K^n$, $c_i = 0$.

Let $p \in \mathbb{N}$. We say that a matrix $A \in M_{n,p}(K)$ is *reducible by Gaussian elimination* if there exists $S \in G_n(K)$ such that SA is of reduced row echelon form.

6.7.12 Theorem. Let K be a division ring. For any $(n, p) \in \mathbb{N}^2$, any matrix $A \in M_{n,p}(K)$ is reducible by Gaussian elimination.

Proof. We reason by induction on p . The case where $p = 0$ is trivial since A is the empty matrix. In the following, we suppose that A is of the form

$$\begin{pmatrix} \lambda_1 & & \\ & B & \\ & & \lambda_n \end{pmatrix},$$

where $B \in M_{n,p-1}(K)$.

If $\lambda_1 = \dots = \lambda_n = 0$, we apply the induction hypothesis to B to obtain the existence of $S \in G_n(K)$ such that SB is of reduced row echelon form. Then

$$SA = \begin{pmatrix} 0 & & \\ \vdots & SB & \\ 0 & & \end{pmatrix}$$

is also of reduced row echelon form.

Suppose that $(\lambda_1, \dots, \lambda_n) \neq (0, \dots, 0)$. By permuting the rows, we may assume that $\lambda_1 \neq 0$. Moreover, since K is a division ring, by multiplying the matrix by $\text{diag}(\lambda_1^{-1}, 1, \dots, 1)$, we may assume that $\lambda_1 = 1$. We then add $-\lambda_i$ times the first row of A to the i -th row, where $i \in \{2, \dots, n\}$. Thus we obtain a matrix of the form

$$\begin{pmatrix} 1 & \mu_2 & \cdots & \mu_p \\ 0 & & & \\ \vdots & & C & \\ 0 & & & \end{pmatrix},$$

where $C \in M_{n-1,p-1}(K)$, and $(\mu_2, \dots, \mu_p) \in K^{p-1}$. Applying the induction hypothesis to C , we may assume that

$$C = \begin{pmatrix} C_2 \\ \vdots \\ C_n \end{pmatrix}$$

is of the reduced echelon form. Suppose that $C_2, \dots, C_k$ in $K^p \setminus \{(0, \dots, 0)\}$, $C_{k+1} = \dots = C_n = (0, \dots, 0)$. Note that

$$j(C_2) < \dots < j(C_k),$$

and $C_{i,j(C_\ell)} = 0$ when $i \in \{2, \dots, n\}$, $\ell \in \{2, \dots, k\}$, and $i \neq \ell$. For any $i \in \{2, \dots, k\}$, we add $-\mu_{j(C_i)}$ times the i -th row to the first row. Then we obtain a matrix of reduced row echelon form. The theorem is thus proved. $\square$

6.7.13 Remark. The proof of the above theorem also provides a recursive algorithm to transform a matrix to a matrix of reduced row echelon form. The operations on rows of a matrix described in the proof of the theorem are called *elementary row operations*. We may use this algorithm to resolve a system of linear equation. Let K be a division ring, $A \in M_{n,p}(K)$, V be a left K -module and $(b_1, \dots, b_n) \in V^n$. perform elementary row operations to the matrix

$$\begin{pmatrix} & b_1 \\ A & \vdots \\ & b_n \end{pmatrix}$$

to obtain a new matrix of the form

$$\begin{pmatrix} & b'_1 \\ A' & \vdots \\ & b'_n \end{pmatrix},$$

where $A' \in M_{n,p}(K)$ is of reduced row echelon form. Then, by resolving the equation

$$A' \begin{pmatrix} x_1 \\ \vdots \\ x_p \end{pmatrix} = \begin{pmatrix} b'_1 \\ \vdots \\ b'_n \end{pmatrix},$$

we obtain solutions to the equation

$$A \begin{pmatrix} x_1 \\ \vdots \\ x_p \end{pmatrix} = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}.$$

6.7.14 Example. Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 6 \end{pmatrix}$. Find the left-inverse matrix of A , and solve the linear system of equations

$$AX = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

In fact, we have

$$\begin{aligned} (A | I_3) &= \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 3 & 4 & 0 & 1 & 0 \\ 3 & 4 & 6 & 0 & 0 & 1 \end{array} \right) \\ &\xrightarrow[L_3 \leftarrow L_3 - 3L_1]{L_2 \leftarrow L_2 - 2L_1} \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & -1 & -2 & -2 & 1 & 0 \\ 0 & -2 & -3 & -3 & 0 & 1 \end{array} \right) \xrightarrow[-L_2 \leftarrow -L_2]{L_3 \leftarrow L_3 - 2L_2} \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 2 & 2 & -1 & 0 \\ 0 & 0 & 1 & 1 & -2 & 1 \end{array} \right) \\ &\xrightarrow[L_1 \leftarrow L_1 - 3L_3]{L_2 \leftarrow L_2 - 2L_3} \left(\begin{array}{ccc|ccc} 1 & 2 & 0 & -2 & 6 & -3 \\ 0 & 1 & 0 & 0 & 3 & -2 \\ 0 & 0 & 1 & 1 & -2 & 1 \end{array} \right) \xrightarrow{L_1 \leftarrow L_1 - 2L_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -2 & 0 & 1 \\ 0 & 1 & 0 & 0 & 3 & -2 \\ 0 & 0 & 1 & 1 & -2 & 1 \end{array} \right). \end{aligned}$$

Then we have

$$A^{-1} = \begin{pmatrix} -2 & 0 & 1 \\ 0 & 3 & -2 \\ 1 & -2 & 1 \end{pmatrix}.$$

So the solution to $AX = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ is

$$X = A^{-1} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 & 0 & 1 \\ 0 & 3 & -2 \\ 1 & -2 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}.$$

6.8 Quotient Modules

In this section, K denotes a unitary ring, 0 denotes the zero element of K , and 1 denotes the unity of K .

6.8.1 Proposition. Let E be a left K -module, F a left K -submodule of E , and E/F the quotient group. Then the map

$$K \times (E/F) \longrightarrow E/F, \quad (a, [x]) \longmapsto [ax]$$

is well-defined and defines a left K -module structure on E/F . Furthermore, the projection map

$$\pi : E \longrightarrow E/F, \quad x \longmapsto [x]$$

is a left K -module homomorphism.

Proof. Let x and y be elements of E such that $[x] = [y]$ in E/F , i.e., $x - y \in F$. Then $a(x - y) = ax - ay \in F$, so $[ax] = [ay]$ in E/F .

Now verify that $(a, [x]) \mapsto [ax]$ defines a left K -module structure on E/F . For any $(a, b) \in K \times K$, $x \in E$,

$$\begin{aligned} (ab)[x] &= [(ab)x] = [a(bx)] = a[bx] = a(b[x]), \\ (a + b)[x] &= [(a + b)x] = [ax + bx] = [ax] + [bx] = a[x] + b[x]. \end{aligned}$$

For any $a \in K$, $(x, y) \in E \times E$,

$$a([x] + [y]) = a[x + y] = [a(x + y)] = [ax + ay] = [ax] + [ay] = a[x] + a[y].$$

For any $x \in E$, $1[x] = [1x] = [x]$.

Finally, by definition,

$$\forall (x, y) \in E \times E, \forall a \in K, \quad [x + ay] = [x] + [ay] = [x] + a[y].$$

Thus, the projection mapping $E \rightarrow E/F$ is a left K -module homomorphism. $\square$

6.8.2 Proposition. Let E be a right K -module, F a right K -submodule of E , and E/F the quotient group. Then the map

$$K \times (E/F) \longrightarrow E/F, \quad (a, [x]) \longmapsto [xa]$$

is well-defined and defines a right K -module structure on E/F . Furthermore, the projection map

$$\pi : E \longrightarrow E/F, \quad x \longmapsto [x]$$

is a right K -module homomorphism.

Proof. Apply Proposition 6.8.1 to the opposite ring of K . $\square$

6.8.3 Theorem. Let E and F be left K -modules (resp., right K -modules), and let $f : E \rightarrow F$ be a left K -module homomorphism (resp., right K -module homomorphism).

- (1) The kernel of f is a left K -submodule (resp., right K -submodule) of E .
- (2) The image of f is a left K -submodule (resp., right K -submodule) of F .

- (3) *The induced mapping $\tilde{f} : E/\text{Ker}(f) \rightarrow \text{Im}(f)$, $[x] \mapsto f(x)$ is a left K -module isomorphism (resp., right K -module isomorphism).*

Proof. We only verify the case for left K -modules.

Proposition 5.5.7 shows that $\text{Ker}(f)$ is a subgroup of E ; Proposition 5.4.9 shows that $\text{Im}(f)$ is a subgroup of F . For any $x \in E$ and any $a \in K$, $af(x) = f(ax)$. This shows that $\text{Im}(f)$ is a left K -submodule of F . Moreover, when $x \in \text{Ker}(f)$, for any $a \in K$, $f(ax) = af(x) = 0$, so $ax \in \text{Ker}(f)$. This shows that $\text{Ker}(f)$ is a left K -submodule of E .

Finally, by Theorem 5.5.4 and Proposition 5.5.8, $\tilde{f}$ is a group isomorphism. Furthermore, for any $(a, x) \in K \times E$,

$$\tilde{f}(a[x]) = \tilde{f}([ax]) = f(ax) = af(x) = a\tilde{f}([x]).$$

Therefore, $\tilde{f}$ is a left K -module isomorphism. $\square$

6.9 Quotient ring

6.9.1 Proposition. *Let A be a unitary ring, and let $\sim$ be an equivalence relation on A compatible with both addition and multiplication. Then the quotient set $A/\sim$, endowed with the operations induced by addition and multiplication via the quotient map, forms a unitary ring, and the quotient mapping $A \rightarrow A/\sim$ is a unitary ring homomorphism.*

Proof. Proposition 5.5.2 shows that $A/\sim$ forms an abelian group under the induced addition operation and a monoid under the induced multiplication operation, and that the quotient mapping $A \rightarrow A/\sim$ preserves addition and multiplication. Furthermore, for any elements a, b, c of A ,

$$\begin{aligned} ([a] + [b])[c] &= [a + b][c] = [(a + b)c] = [ac + bc] = [ac] + [bc] = [a][c] + [b][c], \\ [c]([a] + [b]) &= [c][a + b] = [c(a + b)] = [ca + cb] = [ca] + [cb] = [c][a] + [c][b]. \end{aligned}$$

Therefore, $A/\sim$ forms a unitary ring under the induced operations, and the projection mapping is a unitary ring homomorphism. $\square$

6.9.2 Definition. Let A be a unitary ring. The multiplication on A defines a left A -module structure and a right A -module structure on A . A left A -submodule (resp. right A -submodule) of A is called a *left ideal* (resp. *right ideal*) of A . If a subset I of A is both a left ideal and a right ideal, then I is called a *two-sided ideal* of A .

6.9.3 Proposition. *Let A be a unitary ring, and let I be a two-sided ideal of A . Then the binary relation $\sim_I$ on A defined by*

$$x \sim_I y \text{ if and only if } x - y \in I$$

is an equivalence relation compatible with both addition and multiplication on A , and the kernel of the additive group quotient homomorphism $A \rightarrow A/\sim_I$ equals I .

Proof. By Proposition 5.5.10, $\sim_I$ is compatible with the addition operation, and the kernel of the quotient homomorphism $A \rightarrow A/\sim_I$ is I .

Let a, b, x , and y be elements of A such that $x - a \in I$ and $y - b \in I$. Then

$$xy - ab = xy - ay + ay - ab = (x - a)y + a(y - b) \in I.$$

This shows that $\sim_I$ is also compatible with the multiplication operation. $\square$

6.9.4 Notation. Let A be a unitary ring, and let I be a two-sided ideal of A . Denote by A/I the unitary ring $A/\sim_I$, called the *quotient ring of A modulo I* .

6.9.5 Example. For any integer $n \in \mathbb{Z}$, let

$$n\mathbb{Z} := \{nd \mid d \in \mathbb{Z}\}.$$

Then $n\mathbb{Z}$ is an ideal of $\mathbb{Z}$. The quotient ring $\mathbb{Z}/n\mathbb{Z}$ is finite and contains exactly n elements.

6.9.6 Theorem. *Let $f : A \rightarrow B$ be a unitary ring homomorphism.*

- (1) *The image of f is a subring of B .*
- (2) *The kernel of f (viewed as a homomorphism of additive groups) is a two-sided ideal of A .*
- (3) *The induced mapping $\tilde{f} : A/\text{Ker}(f) \rightarrow \text{Im}(f)$ by passing to quotient is a unitary ring isomorphism.*

Proof. (1) is a corollary of Proposition 5.4.3 and 5.4.4.

(2) First, by Proposition 5.5.7, $\text{Ker}(f)$ is an additive subgroup of A . If $x \in \text{Ker}(f)$, then for any $a \in A$,

$$f(ax) = f(a)f(x) = 0, \quad f(xa) = f(x)f(a) = 0.$$

So $\{ax, xa\} \in \text{Ker}(f)$. Therefore, $\text{Ker}(f)$ is a two-sided ideal of A .

(3) By the First Isomorphism Theorem for groups (Proposition 5.5.12), $\tilde{f}$ is an isomorphism of additive groups. Furthermore, for any $(a, b) \in A \times A$,

$$\tilde{f}([a][b]) = \tilde{f}([ab]) = f(ab) = f(a)f(b) = \tilde{f}([a])\tilde{f}([b]).$$

Thus, $\tilde{f}$ is a unitary ring isomorphism. $\square$

6.9.7 Proposition. *Let A be a ring, I be a two-sided ideal, E be a left A -module. If, for any $(a, x) \in I \times E$, one has $ax = 0$, then the mapping*

$$(A/I) \times E \longrightarrow E, ([a], x) \longmapsto ax$$

is well defined, and determine a left A/I -module structure on E .

Proof. If a and b be elements of A such that $a - b \in I$, then, for any $x \in E$, one has

$$ax - bx = (a - b)x = 0,$$

and hence $ax = bx$. Therefore the mapping

$$(A/I) \times E \longrightarrow E, ([a], x) \longmapsto ax$$

is well defined. Moreover, for any $(a, b) \in A \times A$ and $x \in E$, one has

$$\begin{aligned} ([a] \cdot [b])x &= [ab]x = (ab)x = a(bx) = [a]([b]x), \\ ([a] + [b])x &= [a + b]x = (a + b)x = ax + bx = [a]x + [b]x. \end{aligned}$$

For any $a \in A$ and $(x, y) \in E \times E$,

$$[a](x + y) = a(x + y) = ax + ay = [a]x + [a]y.$$

Hence the mapping

$$(A/I) \times E \longrightarrow E, ([a], x) \longmapsto ax$$

defines a structure of A/I -module on E . □

6.9.8 Corollary. *Let A be a ring, I be a two-sided ideal, E be a right A -module. If, for any $(a, x) \in I \times E$, one has $xa = 0$, then the mapping*

$$(A/I) \times E \longrightarrow E, ([a], x) \longmapsto xa$$

is well defined, and determine a right A/I -module structure on E .

6.10 Free modules

In this section, we fix a unitary ring K .

6.10.1 Definition. Let V be a left K -module, and $\mathbf{x} := (x_i)_{i \in I}$ be a family of elements of V . Denote by $\varphi_{\mathbf{x}} : K^{\oplus I} \rightarrow V$ the homomorphism of left K -module that sends $(a_i)_{i \in I}$ to

$$\sum_{i \in I} a_i x_i.$$

The image of this homomorphism is a left sub- K -module of V , called the left sub- K -module *generated by* $(x_i)_{i \in I}$, denoted as

$$\text{Span}_K((x_i)_{i \in I}).$$

An element in the image of this homomorphism is called a K -linear combination of $(x_i)_{i \in I}$.

- (1) If $\varphi_{\mathbf{x}}$ is surjective, that is, for any $y \in V$ there exists $(a_i)_{i \in I} \in K^{\oplus I}$ such that

$$y = \sum_{i \in I} a_i x_i,$$

we say that $(x_i)_{i \in I}$ is a *system of generators* of V . If V has a system of generators parametrized by a finite set, we say that V is *finitely generated*, or is of *finite type*.

- (2) If $\varphi_{\mathbf{x}}$ is injective, that is (by Proposition 5.5.13),

$$\forall (a_i)_{i \in I} \in K^{\oplus I}, \sum_{i \in I} a_i x_i = 0 \implies \forall i \in I, a_i = 0,$$

we say that $(x_i)_{i \in I}$ is K -linearly independent. Note that $(x_i)_{i \in I}$ is *not* K -linearly independent once there exists $i \in I$ such that $x_i = 0$.

- (3) If $\varphi_{\mathbf{x}}$ is an isomorphism, that is any element of V can be written in a unique way as a linear combination of $(x_i)_{i \in I}$, we say that $(x_i)_{i \in I}$ is a *basis* of V . If V has a basis, then we say that V is a free left K -module.

6.10.2 Example. Let I be a set. For any $i \in I$, let e_i be the mapping from I to V sending i to 1 and $j \in I \setminus \{i\}$ to 0. Let $\mathbf{e} = (e_i)_{i \in I}$. Then $\varphi_{\mathbf{e}}$ is the identity mapping of $K^{\oplus I}$. Hence $(e_i)_{i \in I}$ is a basis of $K^{\oplus I}$. We call it the *canonical basis* of $K^{\oplus I}$. In particular, $K^{\oplus I}$ is a free left K -module.

6.10.3 Remark. Let V be a left K -module. If $\mathbf{x} := (x_i)_{i \in I}$ is a system of generators of V , then the homomorphism $\varphi_{\mathbf{x}} : K^{\oplus I} \rightarrow V$ is a surjective homomorphism of left K -modules. By Theorem 6.8.3, the homomorphism $\varphi_{\mathbf{x}}$ induces by passing to quotient an isomorphism from $K^{\oplus I} / \text{Ker}(\varphi_{\mathbf{x}})$ to V . Therefore, any left K -module is a quotient of a free left K -module.

6.10.4 Remark. Let V be a left K -module and $\mathbf{x} = (x_i)_{i \in I}$ be a family of elements of V , that defines a homomorphism of left K -modules $\varphi_{\mathbf{x}} : K^{\oplus I} \rightarrow V$. Let J be a set and $A : K^{\oplus J} \rightarrow K^{\oplus I}$ be a homomorphism of left K -modules. Then the composite mapping $\varphi_{\mathbf{x}} \circ A : K^{\oplus J} \rightarrow V$ corresponds to another family $(y_j)_{j \in J}$ of elements of V .

- (1) If $(x_i)_{i \in I}$ is K -linearly independent and if A is injective, then $\varphi_{\mathbf{x}} \circ A$ is also injective, which means that $(y_j)_{j \in J}$ is also K -linearly independent.
- (2) If A is surjective, then the image of $\varphi_{\mathbf{x}} \circ A$ identifies with that of $\varphi_{\mathbf{x}}$ is also surjective, which means that $(x_i)_{i \in I}$ and $(y_j)_{j \in J}$ generate the same left sub- K -module of V . In particular, if $(x_i)_{i \in I}$ is a system of generators, so is $(y_j)_{j \in J}$.
- (3) If A is a bijection, and if $(x_i)_{i \in I}$ is a basis of V , then $\varphi_{\mathbf{x}} \circ A$ is also a bijection, which means that $(y_j)_{j \in J}$ is also a bijection.

Let n be a natural number, $(x_1, \dots, x_n) \in V^n$ and S be an element of $G_n(K)$. Let

$$\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = S \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

Then $(x_i)_{i=1}^n$ is K -linearly independent if and only if $(y_j)_{j=1}^n$ is K -linearly independent, $(x_i)_{i=1}^n$ is a system of generators if and only if $(y_j)_{j=1}^n$ is a system of generators, $(x_i)_{i=1}^n$ is a basis if and only if $(y_j)_{j=1}^n$ is a basis.

6.10.5 Proposition. *Let n and p be natural numbers and $A \in M_{n,p}(K)$ be a n by p matrix. We write A in the form of*

$$\begin{pmatrix} \mathbf{a}^{(1)} \\ \vdots \\ \mathbf{a}^{(n)} \end{pmatrix}$$

where each $\mathbf{a}^{(i)} = (a_{i,1}, \dots, a_{i,p})$ belongs to K^p . Assume that A is of row echelon form.

- (1) $(\mathbf{a}^{(i)})_{i=1}^n$ is K -linear independent if and only if none of $\mathbf{a}^{(i)}$ is the zero element.
- (2) $(\mathbf{a}^{(i)})_{i=1}^n$ is a system of generators if and only if there are exactly p non-zero elements among $\mathbf{a}^{(1)}, \dots, \mathbf{a}^{(n)}$.

Proof. (1) It suffices to check that, if for any $i \in \{1, \dots, n\}$, $\mathbf{a}^{(i)} \neq (0, \dots, 0)$, then $(\mathbf{a}^{(i)})_{i=1}^n$ is K -linear independent. Let $(\lambda_i)_{i=1}^n$ such that $\square$

6.10.6 Definition. Let V be a left K -module and $(V_i)_{i \in I}$ be a left sub- K -module of V . For any $i \in I$, the inclusion mapping $V_i \rightarrow V$ is a homomorphism of

left K -modules. These homomorphisms induces a unique homomorphism of left K -modules

$$\bigoplus_{i \in I} V_i \longrightarrow V, (x_i)_{i \in I} \longmapsto \sum_{i \in I} x_i.$$

The image of this homomorphism is called the left sub- K -module generated by $(V_i)_{i \in I}$, denoted as $\sum_{i \in I} V_i$. If

$$\sum_{i \in I} V_i = V,$$

we say that V is generated by $(V_i)_{i \in I}$. If the above homomorphism is an isomorphism, we say that V is the *inner direct sum* of $(V_i)_{i \in I}$.

Let $(x_i)_{i \in I}$ be a family of elements of V . If $(x_i)_{i \in I}$ is a basis of V , then V is the inner direct sum of $(Kx_i)_{i \in I}$, where Kx_i denotes $\{\lambda x_i \mid \lambda \in K\}$.

6.10.7 Theorem. *Let K be a division ring and V be a left K -module of finite type, and $(x_i)_{i \in \{1, \dots, n\}}$ be a system of generators of V . There exists a subset I of $\{1, \dots, n\}$ such that $(x_i)_{i \in I}$ forms a basis of V .*

Proof. We reason by induction on n . In the case where $n = 0$, the family $(x_i)_{i \in \{1, \dots, n\}}$ is empty and V is the zero left K -module. In this case the empty family is a basis of V .

In the rest of the proof, we assume that $n \geq 1$ and that the statement is true for left K -modules generated by at most $n - 1$ elements. By definition, if $(x_i)_{i \in \{1, \dots, n\}}$ is linearly independent, then it is a basis of V . Otherwise there exists $(b_1, \dots, b_n) \in K^n \setminus \{(0, \dots, 0)\}$ such that

$$b_1 x_1 + \dots + b_n x_n = 0.$$

By permuting the elements $x_1, \dots, x_n$, we may assume that $b_n \neq 0$. Since b_n is invertible, we deduce that

$$x_n = (-b_n^{-1} b_1) x_1 + \dots + (-b_n^{-1} b_{n-1}) x_{n-1}.$$

For any $x \in V$, which is written as $a_1 x_1 + \dots + a_n x_n$, one has

$$x = (a_1 - a_n b_n^{-1} b_1) x_1 + \dots + (a_{n-1} - a_n b_n^{-1} b_{n-1}) x_{n-1}.$$

Hence $(x_i)_{i \in \{1, \dots, n-1\}}$ is also a system of generators of V . By the induction hypothesis, we obtain the desired result. $\square$

6.10.8 Theorem. *Let K be a unitary ring, V be a left K -module, and W be a left sub- K -module of V . Let $(x_i)_{i=1}^n$ be a family of elements of W and $(\alpha_j)_{j=1}^\ell$ be a family of elements of V/W , where n and ℓ are natural numbers. For any $j \in \{1, \dots, \ell\}$, let x_{n+j} be an element of the equivalence class α_j .*

- (1) If both $(x_i)_{i=1}^n$ and $(\alpha_j)_{j=1}^\ell$ are K -linearly independent, then $(x_i)_{i=1}^{n+\ell}$ is K -linearly independent.
- (2) If both $(x_i)_{i=1}^n$ and $(\alpha_j)_{j=1}^\ell$ are system of generators, then $(x_i)_{i=1}^{n+\ell}$ is a system of generators.

Proof. (1) Let $(b_i)_{i=1}^{n+\ell} \in K^{n+\ell}$ such that

$$\sum_{i=1}^{n+\ell} b_i x_i = 0.$$

Taking the equivalence classes of both sides in V/W , we get

$$\sum_{j=1}^{\ell} b_{n+j} \alpha_j = 0,$$

which leads to

$$b_{n+1} = \cdots = b_{n+\ell} = 0$$

since $(\alpha_j)_{j=1}^\ell$ is K -linearly independent. Thus

$$b_1 x_1 + \cdots + b_n x_n = 0,$$

which leads to

$$b_1 = \cdots = b_n = 0$$

since $(x_i)_{i=1}^n$ is K -linearly independent.

(2) Let $x \in V$. Since $(\alpha_j)_{j=1}^\ell$ is a system of generators, there exists $(c_{n+1}, \dots, c_{n+\ell}) \in K^\ell$ such that

$$c_{n+1} \alpha_1 + \cdots + c_{n+\ell} \alpha_\ell = [x].$$

Let

$$y = x - (c_{n+1} x_{n+1} + \cdots + c_{n+\ell} x_{n+\ell}).$$

One has $y \in W$. Hence there exists $(c_1, \dots, c_n) \in K^n$ such that $y = c_1 x_1 + \cdots + c_n x_n$. Thus

$$x = \sum_{i=1}^{n+\ell} c_i x_i.$$

□

6.11 Rank

6.11.1 Definition. Let K be a division ring and V be a left K -module of finite type. We call *rank* of V the least cardinality of its bases, denoted as $\text{rk}_K(V)$ or simply as $\text{rk}(V)$. Theorem 6.10.7 shows that $\text{rk}_K(V)$ is also the least cardinality of the system of generators of V .

In the case where K is a field, $\text{rk}_K(V)$ is also called the *dimension* of V over K and is denoted as $\dim_K(V)$, or simply as $\dim(V)$.

6.11.2 Theorem. Let K be a division ring and V be a left K -module of finite type, and W be a left sub- K -module of V .

(1) W and V/W are of finite type, and $\text{rk}(W) + \text{rk}(V/W) = \text{rk}(V)$.

(2) Any basis of V has $\text{rk}(V)$ as its cardinality.

Proof. (1) Let $(x_i)_{i=1}^n$ be a basis of V . Then $([x_i])_{i=1}^n$ forms a system of generators of V/W . By Theorem 6.10.7, one can extract a subset I of $\{1, \dots, n\}$ such that $([x_i])_{i \in I}$ forms a basis of V/W . By permuting the elements $x_1, \dots, x_n$, we may assume without loss of generality that $I = \{1, \dots, \ell\}$.

For any $j \in \{\ell + 1, \dots, n\}$, there exists $(b_{j,1}, \dots, b_{j,\ell}) \in K^\ell$ such that

$$[x_j] = \sum_{i=1}^{\ell} b_{j,i} [x_i].$$

Let

$$y_j = x_j - \sum_{i=1}^{\ell} b_{j,i} x_i.$$

For any $x \in W$, there exists $(a_i)_{i=1}^n \in K^n$ such that

$$\begin{aligned} x &= \sum_{i=1}^n a_i x_i = \sum_{i=1}^{\ell} a_i x_i + \sum_{j=\ell+1}^n a_j \left(y_j + \sum_{i=1}^{\ell} b_{j,i} x_i \right) \\ &= \sum_{j=\ell+1}^n a_j y_j + \sum_{i=1}^{\ell} \left(a_i + \sum_{j=\ell+1}^n a_j b_{j,i} \right) x_i. \end{aligned}$$

Taking the equivalence class of x in V/W , we obtain

$$\forall i \in \{1, \dots, \ell\}, \quad a_i + \sum_{j=\ell+1}^n a_j b_{j,i} = 0.$$

Hence

$$x = \sum_{j=\ell+1}^n a_j y_j.$$

Therefore, W is of finite type, and

$$\text{rk}(W) + \text{rk}(V/W) \leq \text{rk}(V).$$

Moreover, Theorem 6.10.8 shows that

$$\text{rk}(V) \leq \text{rk}(W) + \text{rk}(V/W).$$

Hence the equality

$$\text{rk}(V) = \text{rk}(W) + \text{rk}(V/W)$$

holds.

(2) We reason by induction on the rank of V . If $\text{rk}(V) = 0$, then $\emptyset$ is a basis of V . Hence $V = \{0\}$ and the only basis of V is $\emptyset$. If V is of the form Ke , where e is a non-zero element of V , then any basis of V is of the form $\{ae\}$ with $a \in K^\times$. In this case the rank of V is equal to 1.

Suppose that $\text{rk}(V) = n \geq 1$ and that the statement has been proved for left modules of rank $< n$. Let $(e_i)_{i=1}^m$ be a basis of V and $W = Ke_1$. Then $([e_i])_{i=2}^m$ forms a system of generators of V/W . Moreover, if $(a_i)_{i=2}^m$ is an element of K^{m-1} such that

$$\sum_{i=2}^m a_i [e_i] = 0,$$

then one has

$$\sum_{i=2}^m a_i e_i \in W$$

and hence there exists $a_1 \in K$ such that

$$\sum_{i=1}^m a_i e_i = 0,$$

which leads to $a_2 = \cdots = a_m = 0$. Hence $([e_i])_{i=2}^m$ is a basis of V/W . Note that the rank of V/W is equal to $n-1$. By the induction hypothesis, one has $m-1 = n-1$, that is, $m = n$.

□

6.12 Algebra

In this section, we fix a commutative unitary ring K .

6.12.1 Definition. By K -algebra, we mean a K -module A equipped with a composition law

$$A \times A \longrightarrow A, (a, b) \longmapsto ab$$

such that A equipped with the addition and with this composition law forms a unitary ring, and that

$$\forall (\lambda, a, b) \in K \times A \times A, \lambda(ab) = (\lambda a)b = a(\lambda b).$$

Let A and B be K -algebras. We call homomorphism of K -algebras from A to B any mapping $f : A \rightarrow B$ which is in the same time a homomorphism of K -modules and a homomorphism of unitary rings.

6.12.2 Example. The commutative unitary K itself is a K -module. Its multiplicative law defines a structure of K -algebra on K .

6.12.3 Proposition. Let A be a K -algebra and 1_A be its unity. The mapping $K \rightarrow A, \lambda \mapsto \lambda 1_A$ forms a homomorphism of K -algebras. It is actually the unique homomorphism of K -algebras from K to A .

Proof. The mapping described in the proposition is a homomorphism of K -modules. It remains to check that the mapping preserves multiplications. Let λ and μ be two elements of K . One has

$$(\lambda 1_A)(\mu 1_A) = \lambda(1_A(\mu 1_A)) = \lambda(\mu(1_A 1_A)) = \lambda(\mu 1_A) = (\lambda \mu) 1_A.$$

If $f : K \rightarrow A$ is a homomorphism of K -algebras, then it sends $1 \in K$ to 1_A . Since f is a homomorphism of K -modules, it sends $\lambda \in K$ to $\lambda 1_A$. $\square$

6.12.4 Remark. Let A be a K -algebra. By abuse of notation, for any $\lambda \in K$, we use the expression λ to denote $\lambda 1_A$ (as an element of A). Be aware that the mapping $\lambda \mapsto \lambda 1_A$ is in general not injective.

6.12.5 Definition. Let A be a K -algebra and A_0 be a subset of A . If A_0 is a sub- K -module of A and a subring of A , then we say that A_0 is a sub- K -algebra of A .

6.12.6 Proposition. Let A be a K -algebra. If I is an ideal of A , then it is also a sub- K -module of A . Moreover, the multiplicative law on A/I defines a structure of K -algebra on A/I , and the projection mapping $A \rightarrow A/I$ is a homomorphism of K -algebras.

Proof. Let 1_A be the unity of A . By definition, I is a subgroup of $(A, +)$. Let $a \in I$ and $\lambda \in K$. One has $\lambda a = \lambda(1_A a) = (\lambda 1_A) a \in I$. Hence I is a sub- K -module.

For any $(a, b) \in A \times A$ and $\lambda \in K$, one has

$$\lambda([a] \cdot [b]) = \lambda[ab] = [\lambda(ab)] = [(\lambda a)b] = [\lambda a] \cdot [b] = (\lambda[a])[b].$$

Moreover,

$$[\lambda(ab)] = [a(\lambda b)] = [a] \cdot [\lambda b] = [a](\lambda[b]).$$

Hence A/I is a K -algebra.

Since the projection mapping is a homomorphism of K -modules and a homomorphism of unitary rings, it is actually a homomorphism of K -algebras. $\square$

6.12.7 Theorem. *Let $f : A \rightarrow B$ be a homomorphism of K -algebras. Then the image of f is a sub- K -algebra of B . Moreover, f induces by passing to quotient an isomorphism of K -algebras from $A/\text{Ker}(f)$ to $\text{Im}(f)$.*

Proof. Since $\text{Im}(f)$ is a unitary subring of B and a sub- K -module of B , it is actually a sub- K -algebra of B . Moreover, f induces by passing to limit a mapping

$$\tilde{f} : A/\text{Ker}(f) \longrightarrow \text{Im}(f),$$

which is an isomorphism of rings and an isomorphism of K -modules. Hence it is an isomorphism of K -algebras. $\square$

6.13 Formal series

In this section, we fix a commutative unitary ring K .

6.13.1 Definition. Let T be a formal variable. We denote by $K[[T]]$ the set $K^{\mathbb{N}}$ of mappings from $\mathbb{N}$ to K . The elements of $K[[T]]$ are called *formal power series* of coefficients in K .

If $(a_n)_{n \in \mathbb{N}}$ is a mapping from $\mathbb{N}$ to K , viewed as an element of $K[[T]]$, we write it formally as

$$\sum_{n \in \mathbb{N}} a_n T^n$$

or as

$$a_0 + a_1 T + \cdots + a_n T^n + \cdots .$$

In the latter expression, we call a_n the *coefficient* of T^n and often omit the terms with coefficient 0.

We have seen in the previous chapter that $K[[T]]$ forms a K -module. We define a composition law (written multiplicatively) on $K[[T]]$ as follows. If

$$F = \sum_{n \in \mathbb{N}} a_n T^n \text{ and } G = \sum_{n \in \mathbb{N}} b_n T^n$$

are two elements of $K[[T]]$, then we define

$$FG := \sum_{n \in \mathbb{N}} \left(\sum_{\substack{(i,j) \in \mathbb{N} \times \mathbb{N} \\ i+j=n}} a_i b_j \right) T^n.$$

We observe from the definition that this composition law is commutative. This composition law is called *Cauchy product*. By definition, one has

$$T^n T^m = T^{n+m}$$

for any $(n, m) \in \mathbb{N} \times \mathbb{N}$.

6.13.2 Notation. If F is an element of $K[[T]]$, sometimes we use the expression $F(T)$ to denote the formal power series in order to emphasise the formal variable T .

6.13.3 Proposition. *The K -module $K[[T]]$ equipped with the Cauchy product forms a commutative K -algebra.*

Proof. We first check that the Cauchy product is associative. Let

$$F := \sum_{n \in \mathbb{N}} a_n T^n, \quad G := \sum_{n \in \mathbb{N}} b_n T^n, \quad H := \sum_{n \in \mathbb{N}} c_n T^n$$

be elements of $K[[T]]$. One has

$$(FG)H = F(GH) = \sum_{n \in \mathbb{N}} \left(\sum_{\substack{(i,j,k) \in \mathbb{N}^3 \\ i+j+k=n}} a_i b_j c_k \right) T^n.$$

Moreover, by definition one has $1F = F$. Finally,

$$\begin{aligned} (F + G)H &= \left(\sum_{n \in \mathbb{N}} (a_n + b_n) T^n \right) \sum_{n \in \mathbb{N}} c_n T^n = \sum_{n \in \mathbb{N}} \left(\sum_{\substack{(i,j) \in \mathbb{N} \times \mathbb{N} \\ i+j=n}} (a_i + b_i) c_j \right) T^n \\ &= \sum_{n \in \mathbb{N}} \left(\sum_{\substack{(i,j) \in \mathbb{N} \times \mathbb{N} \\ i+j=n}} (a_i c_j + b_i c_j) \right) T^n = FH + GH. \end{aligned}$$

Hence $K[[T]]$ equipped with the addition and the Cauchy product, forms a commutative unitary ring.

Let $\lambda \in K$. For any elements

$$F := \sum_{n \in \mathbb{N}} a_n T^n, \quad G := \sum_{n \in \mathbb{N}} b_n T^n$$

of $K[[T]]$, one has

$$\lambda(FG) = \sum_{n \in \mathbb{N}} \left(\sum_{\substack{(i,j) \in \mathbb{N} \times \mathbb{N} \\ i+j=n}} \lambda(a_i b_j) \right) T^n = \sum_{n \in \mathbb{N}} \left(\sum_{\substack{(i,j) \in \mathbb{N} \times \mathbb{N} \\ i+j=n}} (\lambda a_i) b_j \right) T^n = (\lambda F)G.$$

By the commutativity of Cauchy product, one deduces

$$\lambda(FG) = \lambda(GF) = (\lambda G)F = F(\lambda G).$$

Therefore, $K[[T]]$ is a commutative K -algebra. □

6.13.4 Proposition. *If K is an integral domain, then $K[[T]]$ is also an integral domain.*

Proof. Let

$$F = \sum_{n \in \mathbb{N}} a_n T^n \quad \text{and} \quad G = \sum_{n \in \mathbb{N}} b_n T^n$$

be non-zero elements of $K[[T]]$. Let k be the least natural number such that $a_k \neq 0$ and ℓ be the least natural number such that $b_\ell \neq 0$. Then

$$\sum_{\substack{(i,j) \in \mathbb{N} \times \mathbb{N} \\ i+j=k+\ell}} a_i b_j = a_k b_\ell \neq 0.$$

□

6.13.5 Proposition. *Let*

$$F = \sum_{n \in \mathbb{N}} a_n T^n$$

be an element of $K[[T]]$. It belongs to $K[[T]]^\times$ if and only if $a_0 \in K^\times$.

Proof. Suppose that

$$G = \sum_{n \in \mathbb{N}} b_n T^n$$

is an element of $K[[T]]$ such that

$$FG = \sum_{n \in \mathbb{N}} \left(\sum_{\substack{(i,j) \in \mathbb{N} \times \mathbb{N} \\ i+j=n}} a_i b_j \right) T^n = 1.$$

Then one has $a_0 b_0 = 1$, so that $a_0 \in K^\times$.

In the following, we assume that a_0 is invertible. We construct in a recursive way a family $(b_n)_{n \in \mathbb{N}}$ of elements of K as follows. Let $b_0 = a_0^{-1}$, and recursively, let

$$b_n = -a_0^{-1} \sum_{i=0}^{n-1} b_i a_{n-i}.$$

One then has

$$\sum_{i=0}^n b_i a_{n-i} = 0,$$

that means

$$\sum_{n \in \mathbb{N}} b_n T^n$$

is the inverse of F . □

6.13.6 Example. For $a \in K$, one has

$$(1 + aT)^{-1} = \sum_{n \in \mathbb{N}} (-a)^n T^n.$$

6.13.7 Example. Suppose that, for any non-zero integer n , the canonical image of n in K is invertible. For any $a \in K$, we denote by $\exp(aT)$ the formal power series

$$\sum_{n \in \mathbb{N}} \frac{a^n}{n!} T^n,$$

where by convention $a^0 = 1$.

6.13.8 Proposition. Suppose that, for any non-zero integer n , the canonical image of n in K is invertible. For any $a \in K$, one has

$$\exp(aT) \exp(bT) = \exp((a+b)T).$$

In particular,

$$(\lambda \in K) \mapsto \exp(\lambda T)$$

defines a homomorphism of groups from $(K, +)$ to $K[[T]]^\times$.

Proof. By definition, one has

$$\begin{aligned} \exp(aT) \exp(bT) &= \sum_{n \in \mathbb{N}} \left(\sum_{i=0}^n \frac{a^i b^{n-i}}{i!(n-i)!} \right) T^n = \sum_{n \in \mathbb{N}} \left(\sum_{i=0}^n \binom{n}{i} \frac{a^i b^{n-i}}{n!} \right) T^n \\ &= \sum_{n \in \mathbb{N}} \frac{(a+b)^n}{n!} T^n = \exp((a+b)T). \end{aligned}$$

□

6.13.9 Definition. If we interpret $K^{\mathbb{N}}$ as the set of formal power series $K[[T]]$, we denote by $K[T]$ the subset $K^{\oplus \mathbb{N}}$. It is thus a sub- K -module of $K[[T]]$. The elements of $K[T]$ are called *polynomials* (of one variable) with coefficients in K .

6.13.10 Remark. Let

$$F = \sum_{n \in \mathbb{N}} a_n T^n$$

be an element of $K[[T]]$ and

$$P = \sum_{i=0}^d b_i T^i$$

be a polynomial. By definition, one has

$$PF = \sum_{n \in \mathbb{N}} \left(\sum_{\substack{i \in \{0, \dots, d\} \\ i \leq n}} b_i a_{n-i} \right) T^n = \sum_{i=0}^d \sum_{\substack{n \in \mathbb{N} \\ n \geq i}} b_i a_{n-i} T^n = \sum_{i=0}^d \sum_{n \in \mathbb{N}} b_i a_n T^{n+i}.$$

6.13.11 Proposition. *The set $K[T]$ forms a sub- K -algebra of $K[[T]]$.*

Proof. By definition $K[T]$ is a sub- K -module of $K[[T]]$. It suffices to check that it is a multiplicative submonoid of $K[[T]]$. Let

$$P(T) = a_0 + a_1 T + \dots + a_n T^n \text{ and } Q(T) = b_0 + b_1 T + \dots + b_m T^m$$

be two elements of $K[T]$. By Remark 6.13.10, one has

$$PQ = \sum_{k=0}^{n+m} \sum_{\substack{i \in \{0, \dots, n\} \\ j \in \{0, \dots, m\} \\ i+j=k}} a_i b_j T^k \in K[T].$$

□

6.13.12 Definition. Let

$$P = \sum_{n \in \mathbb{N}} a_n T^n$$

be an element of $K[T]$. If P is not the zero element of $K[T]$, we define the *degree* of P as the largest natural number n such that $a_n \neq 0$. If d is the degree of P , the coefficient a_d of T^d in P is called the *leading coefficient* of P . If P is the zero polynomial, by convention its degree is defined as $-\infty$. We denote by $\deg(P)$ the degree of P .

By definition, we observe that, if P and Q are polynomials in $K[T]$, then

$$\deg(P + Q) \leq \max\{\deg(P), \deg(Q)\}, \quad \deg(PQ) \leq \deg(P) + \deg(Q).$$

Moreover, the equality $\deg(PQ) = \deg(P) + \deg(Q)$ holds if one of the polynomials P and Q is zero, or if the product of the leading coefficients of P and Q is non-zero. In particular, if P and Q are non-zero and if one of the leading coefficients of P and Q is not a zero divisor, then $\deg(PQ) = \deg(P) + \deg(Q)$.

6.13.13 Theorem. *Let P be a non-zero element of $K[T]$ and d be the degree of P . Let c be the leading coefficient of P . Assume that c belongs to $K^\times$. For any polynomial $F \in K[T]$, there exists a unique pair $(Q, R) \in K[T] \times K[T]$ such that $\deg(R) < d$ and that $F = PQ + R$.*

Proof. We first prove the uniqueness. Assume that

$$F = PQ_1 + R_1 = PQ_2 + R_2,$$

where Q_1, Q_2, R_1 and R_2 are polynomials with $\deg(R_1) < d$ and $\deg(R_2) < d$. One has

$$R_1 - R_2 = P(Q_2 - Q_1).$$

Since the leading coefficient of P is non-zero, one has

$$\deg(R_1 - R_2) = \deg(P(Q_2 - Q_1)) = \deg(P) + \deg(Q_2 - Q_1).$$

If $Q_2 - Q_1$ is non-zero, then $\deg(R_1 - R_2) \geq \deg(P) = d$. However,

$$\deg(R_1 - R_2) \leq \max\{\deg(R_1), \deg(R_2)\} < d,$$

which leads to a contradiction. Therefore, one has $Q_1 = Q_2$ and furthermore $R_1 = R_2$.

In the following, we prove the existence part by induction on the degree n of the polynomial F . In the case where $\deg(F) < d$, we just take $Q = 0$ and $R = F$. In what follows, we assume that $n \geq d$. Suppose that F is of the form

$$a_0 + a_1 T + \cdots + a_n T^n,$$

where $a_0, \dots, a_n$ are elements of K . Let

$$G(T) = F(T) - a_n c^{-1} T^{n-d} P(T).$$

Then $\deg(G) < n$. Applying the induction hypothesis to G , we obtain the existence of two polynomials Q_1 and R with $\deg(R) < d$ and $G = PQ_1 + R$. Thus

$$F(T) = P(T)(Q_1(T) + a_n c^{-1} T^{n-d}) + R(T),$$

as required. $\square$

6.13.14 Corollary. *If K is a field, then the polynomial ring $K[T]$ is a principal ideal domain.*

Proof. We have seen that $K[T]$ is an integral domain. It suffices to check that any ideal I of $K[T]$ is a principal ideal. If I is the zero ideal, then it is generated by 0. Otherwise there exists a non-zero $P \in I$ such that

$$\deg(P) = \min\{\deg(F) \mid F \in I, F \neq 0\}.$$

We claim that $I = K[T]P$. Clearly one has $K[T]P \subseteq I$. For any $F \in I$, by Theorem 6.13.13, there exists polynomials Q and R such that $\deg(R) < \deg(P)$ and $F = PQ + R$. Since $P \in I$ and $F \in I$, we obtain that $R \in I$. Since $\deg(R) < \deg(P)$, we obtain that $R = 0$. Hence $I \subseteq K[T]P$. $\square$

Exercises

1. Let $r \in \mathbb{Z}^\times$. We equip the Abelian group $\mathbb{Z}^2$ with the multiplication $*$ defined by, $\forall (x_1, x_2) \in \mathbb{Z}^2$ and $(y_1, y_2) \in \mathbb{Z}^2$ by

$$(x_1, x_2) * (y_1, y_2) = (x_1 y_1 + r x_2 y_2, x_1 y_2 + x_2 y_1).$$

Verify that $(\mathbb{Z}^2, +, *)$ is a commutative unitary ring.

2. Let A be a unitary ring. For all $x, y \in A$, we put $[x, y] = xy - yx$.
 - (1) The Abelian group of A , equipped with the "product" $(x, y) \mapsto [x, y]$, is it a ring?
 - (2) For all $x, y, z \in A$, verify

$$[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0.$$

3. For an $n \in \mathbb{N}_{\geq 1}$, find all the homomorphisms of rings from $\mathbb{Z}^n$ to $\mathbb{Z}$.

4. In a unitary ring A , we suppose that, $\forall x \in A$, we have $x^2 = x$. We call this kind of ring a **Boolean ring**.
- (1) Prove that A is commutative.
 - (2) If A contains at least 3 distinct elements, prove that A has a zero divisor.
 - (3) Prove that A cannot have exactly 3 elements.
 - (4) Construct an example of Boolean ring which has exactly 4 elements.
5. Prove that the only automorphism of $\mathbb{Q}$ is $\text{Id}_{\mathbb{Q}}$, and this is an isomorphism from $\mathbb{Q}$ to $\mathbb{Q}$.
6. Prove that the only sub-field $\mathbb{Q}$ is $\mathbb{Q}$.
7. Let M be a subgroup of the additive group $(\mathbb{Z}, +)$. We assume that $M \neq \{0\}$.
- (1) Show that $M \cap \mathbb{N}_{>0}$ is not empty.
 - (2) Let d be the least element of $M \cap \mathbb{N}_{>0}$. Show that any element of M is divisible by d .
 - (3) Deduce that $M = d\mathbb{Z}$, where $d\mathbb{Z}$ is defined as $\{dn \mid n \in \mathbb{Z}\}$.
8. Let $\mathbb{N}_{\geq 1}$ be the set of positive integers and R the set of functions defined on $\mathbb{N}_{\geq 1}$ with values in a commutative ring K . Define the sum in R to be the ordinary addition of functions, and define the **convolution product** by the formula

$$(f * g)(m) = \sum_{xy=m} f(x)g(y),$$

where the sum is taken over all pairs (x, y) of positive integers such that $xy = m$.

- (1) Show that R is a commutative ring, whose unit element is the function δ such that

$$\delta(x) = \begin{cases} 1, & x = 1 \\ 0, & x \neq 1. \end{cases}$$

- (2) A function f is said to be **multiplicative** if $f(mn) = f(m)f(n)$ whenever m, n are relatively prime. If f, g are multiplicative, show that $f * g$ is multiplicative.

- (3) Let μ be the **Möbius function** such that

$$\mu(n) = \begin{cases} 1, & n = 1, \\ (-1)^r, & n = p_1 \cdots p_r, \text{ } p_1, \dots, p_r \text{ are distinct primes,} \\ 0, & \text{others.} \end{cases}$$

Show that $\mu * \phi_1 = \delta$, where ϕ_1 denotes the constant function having value 1. [**Hint:** Show first that μ is multiplicative, and then prove the assertion for prime powers.] The Möbius inversion formula of elementary number theory is then nothing else but the relation $\mu * \phi_1 * f = f$.

9. Let A be a commutative unitary ring.

- (1) Show that the multiplicative law

$$A \times A \longrightarrow A, \quad (a, b) \longmapsto ab$$

defines a structure of A -module on the additive group $(A, +)$.

- (2) We call *ideal* of A any sub- A -module of A . Let I be a sub- A -module of A . Show that the mapping

$$(A/I) \times (A/I) \longrightarrow A/I, \quad ([a], [b]) \longmapsto [ab]$$

is well defined and determines a structure of monoid on A/I .

- (3) Let I be an ideal of A . Show that the set A/I equipped with the additive law (from the quotient module structure on A/I) and the above multiplicative law forms a commutative unitary ring. Show that the projection mapping $A \rightarrow A/I$ is a morphism of unitary rings.
- (4) Let $f : A \rightarrow B$ be a morphism of unitary rings.
- Show that the kernel of f is an ideal of A .
 - Show that the image of f is a unitary subring of B .
 - Show that the mapping

$$\tilde{f} : A/\text{Ker}(f) \longrightarrow B, \quad [a] \longmapsto f(a)$$

is an isomorphism of unitary rings.

- (5) Show that there exists a unique morphism of unitary rings from $\mathbb{Z}$ to A .

- 10.** Let k be a commutative unitary ring. In this exercise, we write the set of sequences in k parametrized by $\mathbb{N}$ (usually denoted as $k^{\mathbb{N}}$) in the form of $k[[T]]$. If $(a_n)_{n \in \mathbb{N}}$ is a sequence in k , viewed as an element of $k[[T]]$, we express it as

$$a_0T^0 + a_1T + a_2T^2 + \cdots + a_nT^n + \cdots$$

or in the form of

$$\sum_{n \in \mathbb{N}} a_n T^n.$$

We call it a *formal power series* with coefficients in k . In the above formal sum, we often omit the summand with coefficient 0. Moreover, the summand a_0T^0 is often written as a_0 for simplicity. For example, if $a_0 = a_2 = 1$ and $a_n = 0$ for $n \in \mathbb{N} \setminus \{0, 2\}$, then the formal series

$$\sum_{n \in \mathbb{N}} a_n T^n$$

is written in the abbreviated form as

$$1 + T^2.$$

- (1) Show that the set $k[[T]]$ equipped with the following composition law (written additively)

$$\begin{aligned} k[[T]] \times k[[T]] &\longrightarrow k[[T]], \\ \left(\sum_{n \in \mathbb{N}} a_n T^n, \sum_{n \in \mathbb{N}} b_n T^n \right) &\longmapsto \sum_{n \in \mathbb{N}} (a_n + b_n) T^n \end{aligned}$$

forms a commutative group.

- (2) Show that the set $k[[T]]$ equipped with the following composition law (written multiplicatively)

$$\begin{aligned} k[[T]] \times k[[T]] &\longrightarrow k[[T]], \\ \left(\sum_{n \in \mathbb{N}} a_n T^n, \sum_{n \in \mathbb{N}} b_n T^n \right) &\longmapsto \sum_{n \in \mathbb{N}} \left(\sum_{i=0}^n a_i b_{n-i} \right) T^n \end{aligned}$$

forms a commutative monoid.

- (3) Show that, for any $(a, b) \in k \times k$ and any $(n, m) \in \mathbb{N} \times \mathbb{N}$, one has

$$(aT^n)(bT^m) = (ab)T^{n+m}.$$

- (4) Show that $k[[T]]$ equipped with the above (additive and multiplicative) compositions laws forms a commutative unitary ring.

- (5) Show that an element

$$f = \sum_{n \in \mathbb{N}} a_n T^n$$

of $k[[T]]$ is invertible if and only if a_0 is an invertible element of k . Write an algorithm to determine the inverse of f when it is invertible.

- (6) Determine $(1 - aT)^{-1}$, where a is an element of k .

In the rest of the exercise, we suppose that the image of any $n \in \mathbb{N}_{\geq 1}$ by the unique morphism of unitary rings $\mathbb{Z} \rightarrow k$ is invertible.

- (7) Show that there exists an element $f \in k[[T]]$ such that $f^2 = 1 + T$. Determine this element.

- (8) Let $D : k[[T]] \rightarrow k[[T]]$ be the mapping sending

$$f = \sum_{n \in \mathbb{N}} a_n T^n \in k[[T]]$$

to

$$f' := \sum_{n \in \mathbb{N}} (n+1) a_{n+1} T^n.$$

Show that D is a surjective k -linear mapping.

- (9) Determine the kernel of D .
- (10) Determine the set of all formal series $f \in k[[T]]$ which satisfy the equation

$$f' = f.$$

- 11.** Let k be a commutative unitary ring. We denote by $k[T]$ the subring of $k[[T]]$ that is composed of formal power series

$$P = \sum_{n \in \mathbb{N}} a_n T^n$$

such that

$$\deg(P) := \sup\{n \in \mathbb{N} \mid a_n \neq 0\}$$

is not $+\infty$. The element $\deg(P) \in \mathbb{N} \cup \{-\infty\}$ is called the *degree* of P . We say that P is *monic* if its degree is ≥ 0 and if

$$a_{\deg(P)} = 1.$$

The formal power series that belong to $k[T]$ are called *polynomials*.

- (1) Let F and G be polynomials. We assume that F is monic. Show that $\deg(FG) = \deg(F) + \deg(G)$.
- (2) Let P be a monic polynomial. Show that, for any $F \in k[T]$, there exists a unique pair $(Q, R) \in k[T] \times k[T]$ such that $\deg(R) < \deg(P)$ and $F = PQ + R$. We could reason by induction on the degree of F .
- (3) Let I be an ideal of $k[T]$. Suppose that there exists a monic polynomial $P \in I$ such that

$$\forall F \in I \setminus \{0\}, \quad \deg(P) \leq \deg(F).$$

Show that

$$I = Pk[T] = \{PQ \mid Q \in k[T]\}.$$

Prove that such monic polynomial of minimal degree P is unique.

- (4) Suppose that k is a field. Show that any ideal I of $k[T]$ which is different from $\{0\}$ contains a monic polynomial P such that

$$\forall F \in I \setminus \{0\}, \quad \deg(P) \leq \deg(F).$$

- (5) For any $F = a_0 + a_1T + \cdots + a_dT^d \in k[T]$ and any $x \in k$, let

$$F(x) = a_0 + a_1x + \cdots + a_dx^d \in k.$$

Show that, for any $x \in k$, the mapping

$$\text{ev}_x : k[T] \longrightarrow k, \quad F \longmapsto F(x)$$

is a morphism of unitary rings.

- 12.** Let k be a field, P be a non-zero element of $k[T]$ and d be the degree of P . Suppose that P is of the form

$$a_0 + a_1T + \cdots + a_dT^d.$$

If x is an element of k such that $P(x) = 0$, then we say that x is a *root* of P .

- (1) Let $x \in k$. Show that, if $P(x) = 0$, then there exists a unique element $Q \in k[T]$ such that $P = (T - x)Q$.
- (2) Show that P can be written in the form

$$P = (T - x_1)^{d_1} \cdots (T - x_n)^{d_n} \tilde{P},$$

where $x_1, \dots, x_n$ are distinct roots of P in k and $\tilde{P}$ does not have any root in k . Show that $d_1 + \cdots + d_n \leq d$.

- (3) Deduce that the number of roots of P does not exceed d .
- (4) Let $f(T), g(T) \in k[T]$, $\deg(f), \deg(g) \leq n$. Suppose there exist $c_0, \dots, c_n \in k$, such that $f(c_i) = g(c_i)$. $i = 0, \dots, n$. Prove that $f(T) = g(T)$.
- (5) Let $f(T) \in k[T]$ satisfying $\deg(f) \leq n$, and $a_0, \dots, a_n \in k$ be different. Prove

$$f(T) = \sum_{i=0}^n \frac{(T - a_0) \cdots (T - a_{i-1})(T - a_{i+1}) \cdots (T - a_n)}{(a_i - a_0) \cdots (a_i - a_{i-1})(a_i - a_{i+1}) \cdots (a_i - a_n)} f(a_i).$$

This is called **Lagrange interpolation formula**.

- 13.** In this exercise, we let P be the element $T^2 + 1 \in \mathbb{R}[T]$ and we denote by I the ideal

$$P\mathbb{R}[T] = \{PQ \mid Q \in \mathbb{R}[T]\}.$$

We denote by $\mathbb{C}$ the quotient ring $\mathbb{R}[T]/I$ and denote by i the class of $T \in \mathbb{R}[T]$ in the quotient ring $\mathbb{C} = \mathbb{R}[T]/I$.

- (1) Show that $i^2 = -1$ in $\mathbb{C}$.
- (2) Show that any element of $\mathbb{C}$ can be written as $a + bi$, where a and b are real numbers. We could use the fact that any polynomial $F \in \mathbb{R}[T]$ can be written as $PQ + R$ where $\deg(R) \leq 1$.
- (3) Show that the mapping $\mathbb{R}^2 \rightarrow \mathbb{C}$ sending $(a, b) \in \mathbb{R}^2$ to $a + bi$ is an $\mathbb{R}$ -linear bijection.
- (4) Let a, b, x and y be real numbers. Express $(a + bi)(x + yi)$ in terms of $u + vi$ with $(u, v) \in \mathbb{R}^2$. Check that the mapping $\mathbb{R} \rightarrow \mathbb{C}$ sending $a \in \mathbb{R}$ to $a + 0i$ is an injective mapping and also a morphism of unitary rings. Thus we can consider $\mathbb{R}$ as a unitary subring of $\mathbb{C}$.
- (5) For any

$$F = \sum_{n \in \mathbb{N}} a_n T^n \in k[[T]],$$

let

$$\iota(F) = \sum_{n \in \mathbb{N}} (-1)^n a_n T^n.$$

Show that $\iota : k[[T]] \rightarrow k[[T]]$ is a morphism of unitary rings and $\iota \circ \iota$ is the identity mapping. Deduce that ι is an isomorphism of unitary rings.

- (6) Check that $\iota(k[T]) = k[T]$ and $\iota(P) = P$.

- (7) For any $z = a + bi \in \mathbb{C}$, we define $\bar{z} = a - bi$. Show that the mapping $\mathbb{C} \rightarrow \mathbb{C}$ sending z to $\bar{z}$ is an isomorphism of unitary rings. We could use the result of the previous question.
 - (8) For any $z \in \mathbb{C}$, let $N(z) = z\bar{z}$. Show that $N(z)$ is a non-negative real number (in considering $\mathbb{R}$ as a unitary subring of $\mathbb{C}$), and it is positive whenever $z \neq 0$.
 - (9) Let z and w be elements of $\mathbb{C}$. Show that $N(zw) = N(z)N(w)$.
 - (10) Show that $\mathbb{C}$ is a field.
 - (11) Denote by $\mathbb{Z}[i]$ the subset of $\mathbb{C}$ that is composed of elements $a + bi$ with $(a, b) \in \mathbb{Z}^2$. Show that $\mathbb{Z}[i]$ is a unitary subring of $\mathbb{C}$.
 - (12) Show that, for any $z \in \mathbb{Z}[i]$, $N(z)$ is a natural number.
 - (13) Let z be an element of $\mathbb{Z}[i]^\times$. Show that $N(z) = 1$.
 - (14) Determine the set $\mathbb{Z}[i]^\times$.
- 14.** We consider the morphism of unitary rings from $\mathbb{Z}[T]$ to $\mathbb{R}$ which sends $F \in \mathbb{Z}[T]$ to $F(\sqrt{2})$

- (1) Show that

$$\mathbb{Z}[\sqrt{2}] = \{a + b\sqrt{2} \mid (a, b) \in \mathbb{Z}^2\}$$

is a unitary subring of $\mathbb{R}$.

- (2) Show that the mapping $N : \mathbb{Z}[\sqrt{2}] \rightarrow \mathbb{Z}$ which sends $a + b\sqrt{2}$ to $a^2 - 2b^2$ is a morphism of multiplicative monoid. We could inspire from the method of the previous exercise.
- (3) Show that the invertible elements of $\mathbb{Z}[\sqrt{2}]$ are precisely those of the form $a + b\sqrt{2}$ with $a^2 - 2b^2 = 1$ or $a^2 - 2b^2 = -1$.
- (4) Check that $(3, 2)$ is a solution of the equation $a^2 - 2b^2 = 1$ in $\mathbb{N}_{\geq 1}^2$.
- (5) Let $\alpha = 3 + 2\sqrt{2}$. For any $n \in \mathbb{N}_{\geq 1}$, let $(a_n, b_n) \in \mathbb{N}^2$ such that

$$a_n + b_n\sqrt{2} = (3 + 2\sqrt{2})^n.$$

Show that the set of solutions of the equation $a^2 - 2b^2 = 1$ in $\mathbb{N}_{\geq 1}^2$ is precisely $\{(a_n, b_n) \mid n \in \mathbb{N}_{\geq 1}\}$.

- 15.** For a fixed $n \in \mathbb{N}_{\geq 2}$, we denote by

$$\begin{array}{ccc} \gamma_n : \mathbb{Z} & \longrightarrow & \mathbb{Z}/n\mathbb{Z} \\ x & \longmapsto & \bar{x} \end{array}$$

the canonical quotient homomorphism.

- (1) Let $a \in \mathbb{Z}$. Prove that the following statements are equivalent.
- i. $\bar{a}$ is an invertible element in $\mathbb{Z}/n\mathbb{Z}$.
 - ii. $\bar{a}$ is not a zero divisor in $\mathbb{Z}/n\mathbb{Z}$.
 - iii. $\gcd(a, n) = 1$.
- (2) Let $\phi(n) = \#(\mathbb{Z}/n\mathbb{Z})^\times$. Prove

$$\phi(n) = \#\{k \mid k \in \llbracket 0, n-1 \rrbracket, \gcd(n, k) = 1\}.$$

In convention, we define $\phi(1) = 1$.

- (3) Prove that the invertible elements of the ring $\mathbb{Z}/n\mathbb{Z}$ are those $\alpha \in \mathbb{Z}/n\mathbb{Z}$ which generate the additive group $\mathbb{Z}/n\mathbb{Z}$, which means $\mathbb{Z}/n\mathbb{Z} = \{k\alpha\}_{k \in \mathbb{Z}}$.
- (4) Let $m, n \in \mathbb{N}_{\geq 2}$, and $\gcd(m, n) = 1$. Prove

$$\phi(mn) = \phi(m)\phi(n).$$

- (5) Let $a \in \mathbb{Z}$, $\gcd(a, n) = 1$. Prove

$$a^{\phi(n)} \equiv 1 \pmod{n}.$$

- (6) Let $n = p_1^{\alpha_1} \cdots p_k^{\alpha_k}$, where $p_1, \dots, p_k$ are distinct primes, and $\alpha_1, \dots, \alpha_k \geq 1$. Prove

$$\phi(n) = n \left(1 - \frac{1}{p_1}\right) \cdots \left(1 - \frac{1}{p_k}\right).$$

- (7) For a prime p , prove that $\mathbb{Z}/p\mathbb{Z}$ is a field, and $\phi(p) = p - 1$.

In this case, $\mathbb{Z}/p\mathbb{Z}$ is called the **finite field** of order p , usually denoted by $\mathbb{F}_p$.

16. In $\mathbb{R}^3$, let

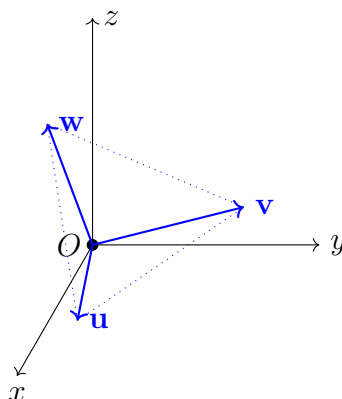
$$\mathbf{u} = (1, -2, 1), \mathbf{v} = (-1, 1, 0).$$

- (1) For $\mathbf{w} = (-6, 3, 3)$, find $a, b \in \mathbb{R}$, such that $\mathbf{w} = a\mathbf{u} + b\mathbf{v}$.
- (2) For $\mathbf{w}' = (6, 3, 3)$, prove $\mathbf{w}' \notin \text{Span}(\mathbf{u}, \mathbf{v})$.
- (3) Give the equation of coordinate of $\text{Span}(\mathbf{u}, \mathbf{v})$.

17. Let $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be three non-zero vectors in $\mathbb{R}^3$, which do not lie in a same plane.

- (1) Prove $\text{Span}(\mathbf{u}, \mathbf{v}, \mathbf{w}) = \mathbb{R}^3$.
- (2) Prove that if $a\mathbf{u} + b\mathbf{v} + c\mathbf{w} = 0$, then $a = b = c = 0$.

- (3) Consider the pyramid below and study the following problems.



- (a) Is $\frac{1}{2}(\mathbf{u} + \mathbf{v} + \mathbf{w})$ inside or outside the pyramid?
 (b) Prove: for an arbitrary point P in the pyramid, if we consider it as the vector $\overrightarrow{OP} = \mathbf{p}$, then there exist $a \geq 0$, $b \geq 0$, $c \geq 0$ and $a + b + c \leq 1$, such that

$$\mathbf{p} = a\mathbf{u} + b\mathbf{v} + c\mathbf{w}.$$

- (4) Write the surface $\Delta(O\mathbf{u}\mathbf{v})$ of the pyramid as a set of the linear combinations of $\{\mathbf{u}, \mathbf{v}\}$.
 (5) Write the surface $\Delta(\mathbf{u}\mathbf{v}\mathbf{w})$ of the pyramid as a set of the linear combinations of $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$.

18. Let

$$\mathbf{u} = (1, 2, -5, 3), \quad \mathbf{v} = (2, -1, 4, 7).$$

For $a, b \in \mathbb{R}$, when $(a, b, -37, -3) \in \text{Span}(\mathbf{u}, \mathbf{v})$?

19. We consider the matrices

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & -1 \\ 2 & 0 \end{pmatrix}, \quad C = (1 \quad -1), \quad D = \begin{pmatrix} -1 \\ 1 \end{pmatrix},$$

$$E = \begin{pmatrix} 1 & -1 & 2 \\ 1 & 0 & 3 \end{pmatrix}, \quad F = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 0 & 3 \end{pmatrix}.$$

- (1) In A, B, C, D, E, F , for which pairs the addition is well-defined?
 (2) In A, B, C, D, E, F , for which pairs the multiplication is well-defined?
 If so, give the number of rows and columns of the product.
 (3) Give AB, BA, CE, EF, FE .

(4) Give the transport of A, B, C, D, E, F .

20. (1) For $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, find a non-zero $B \in M_2(\mathbb{R})$, such that

$$AB = 0_{2,2},$$

and deduce that A is not invertible.

(2) For the matrix $A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}$, give a non-zero matrix $B \in M_{3,2}$, such that

$$AB = 0_{2,2}.$$

(3) Let A be the same as in the previous exercise, give a non-zero matrix $C \in M_{3,2}(\mathbb{R})$, such that

$$AC = I_2.$$

21. Find $A \in M_2(\mathbb{R})$, such that:

(a) $A \neq \pm I_2$, but $A^2 = I_2$.

(b) $A^2 = -I_2$.

(c) $A \neq I_2$, but $A^3 = I_2$.

22. We consider the rotation matrix

$$R(\theta) = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}.$$

(1) Verify

$$(1, 0)R(\theta) = (\cos \theta, \sin \theta), \quad (0, 1)R(\theta) = (-\sin \theta, \cos \theta). \quad (6.6)$$

Explain $R(\theta)$ is the linear transformation with the angle θ .

(2) Verify $\forall \theta_1, \theta_2 \in \mathbb{R}$,

$$R(\theta_1)R(\theta_2) = R(\theta_1 + \theta_2).$$

(3) Find $R(\theta)^k$, $k \in \mathbb{N}$.

(4) Find $R(\theta)^{-1}$.

23. Given $a, b, c, d \in \mathbb{R}$, and we consider the matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Let

$$B = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

(1) Verify $AB = BA = (ad - bc) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

(2) Verify: when $\Delta := (ad - bc) \neq 0$, A is invertible, and

$$A^{-1} = \frac{1}{\Delta} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

(3) When $\Delta = 0$, prove that A is not invertible. (**Hint:** verify $AX = 0$ has a non-zero solution.)

24. For $(a, b) \in \mathbb{R}^2$, we consider

$$M(a, b) = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}.$$

(1) By the fact

$$M(a, b) = aI_2 + bJ, \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

prove:

$$M(a, b)M(c, d) = M(ac - bd, bc + ad).$$

(2) Give the explicit representation of $M(a, b)^k$ ($k \in \mathbb{N}$).

(3) Prove: $M(a, b)$ is invertible if and only if $(a, b) \neq (0, 0)$, and in this time we have

$$M(a, b)^{-1} = M\left(\frac{a}{a^2 + b^2}, -\frac{b}{a^2 + b^2}\right).$$

25. Suppose

$$A = \begin{pmatrix} x^2 & xy & xz \\ xy & y^2 & yz \\ xz & yz & z^2 \end{pmatrix}.$$

Give the explicit representation of A^n , where $n \in \mathbb{N}$.

26. Suppose

$$A = \begin{pmatrix} 0 & 1 & -1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix}.$$

(1) Verify $A^2 - 3A + 2I_3 = 0$.

(2) Determine $a_n, b_n \in \mathbb{R}$, such that

$$x^n = (x^2 - 3x + 2)Q(x) + (a_nx + b_n),$$

where $Q(x)$ is a polynomial.

(3) Give the explicit representation of A^n .

27. It is easy to see that λI_n ($\lambda \in \mathbb{R}$) is commutative with any matrix B of order n . Prove its converse: Suppose $A \in M_n(\mathbb{R})$ is commutative with any $B \in M_n(\mathbb{R})$, then $A = \lambda I_n$, where $\lambda \in \mathbb{R}$.

28. Solve the following linear systems of equations.

$$(1) \begin{cases} x - 2y + 3z = 5, \\ 2x - 4y + z = 5, \\ 3x - 5y + 2z = 8. \end{cases}$$

$$(2) \begin{cases} -5x + y - 3z - 2t = 2, \\ x + 2y - 4z - t = 3, \\ -3x + 5y - 11z - 4t = 8, \\ -9x + 4y - 10z - 5t = 7, \end{cases} \quad \text{where } t \text{ is a parameter.}$$

$$(3) \begin{cases} 2x_1 + x_2 + x_3 + x_4 = a, \\ x_1 + 2x_2 + x_3 + x_4 = b, \\ x_1 + x_2 + 2x_3 + x_4 = c, \\ x_1 + x_2 + x_3 + 2x_4 = d, \end{cases} \quad \text{where } a, b, c, d \in \mathbb{R} \text{ are fixed.}$$

$$(4) \begin{cases} x_1 + x_2 = 0 \\ x_1 + x_2 + x_3 = 0 \\ \quad x_2 + x_3 + x_4 = 0 \\ \quad \quad \ddots \quad \quad \ddots \quad \quad \ddots \quad \vdots \\ \quad \quad \quad x_{n-2} + x_{n-1} + x_n = 0 \\ \quad \quad \quad \quad x_{n-1} + x_n = 0 \end{cases}, \text{ where } n \geq 3.$$

29. For different choices of the parameter m , solve the following linear systems of equations.

$$(1) \begin{cases} x + y + (m-1)z = m, \\ mz = m, \\ -x + (m^2 - m - 1)y + (m-1)z = m. \end{cases}.$$

$$(2) \begin{cases} x - my + m^2z = m, \\ mx - m^2y + mz = 1, \\ mx + y - m^3z = -1. \end{cases}$$

$$(3) \begin{cases} x + y + z = m + 1, \\ mx + y + (m + 1)z = m - 1, \\ x + my + mz = m + 2. \end{cases}$$

30. Find the elementary matrices P_k , such that:

- (a) $(x, y)P_1 = (y, x), \quad \forall (x, y) \in \mathbb{R}^2.$
- (b) $(x, y)P_2 = (x, x + y), \quad \forall (x, y) \in \mathbb{R}^2.$
- (c) $(x, y)P_3 = (-x, y), \quad \forall (x, y) \in \mathbb{R}^2.$
- (d) $(x, y, z)P_4 = (x, z, y), \quad \forall (x, y, z) \in \mathbb{R}^3.$
- (e) $(x, y, z)P_5 = (z, y, x), \quad \forall (x, y, z) \in \mathbb{R}^3.$
- (f) $(x, y, z)P_6 = (x, y, z + x), \quad \forall (x, y, z) \in \mathbb{R}^3.$

31. Give the products of the following matrices.

- (1) $\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}.$
- (2) $\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}.$
- (3) $\begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}.$
- (4) $\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}.$

32. Find the inverses of the matrices below.

$$A = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \\ 2 & 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & 1 & 3 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{pmatrix}.$$

33. If $A \in M_n(\mathbb{R})$ satisfies

$$A^2 = I_n,$$

then we say that A is a *reflection*.

- (1) Give A^{-1} .

- (2) Prove: $\forall \lambda \in \mathbb{R}, |\lambda| \neq 1, A_\lambda = A - \lambda I_n$ is invertible, and give the inverse of A_λ . (**Hint:** calculate A_λ^2 , and write it as the form of $bA_\lambda + cI_n$.)

34. Suppose $A \in M_n(\mathbb{R})$. If $A^2 = A$, then we say that A is a *projection*.

- (1) If A is a projection, prove that so is $B = I_n - A$, and

$$AB = BA = 0;$$

and give the inverse of $I_n + A$.

- (2) Prove that for a projection A , it is invertible if and only if $A = I_n$.

35. For $\mathbf{u} = (x_1, \dots, x_n)^\top, \mathbf{v} = (x_1, \dots, x_n)^\top \in \mathbb{R}^n$, suppose $m_{ij} = x_i y_j - x_j y_i$ ($i, j \in \llbracket 1, n \rrbracket$), $M = (m_{ij})_{1 \leq i, j \leq n}$. Consider the matrix $A = I_n + aM$, where $a \in \mathbb{R}$ is a parameter.

- (a) Prove that if $AX = 0$, then there exist $\lambda, \mu \in \mathbb{R}$ such that $X = \lambda \mathbf{u} + \mu \mathbf{v}$.
 (b) Prove that A is invertible.

36. Let $n \geq 2$, and we consider the matrix

$$J = \begin{pmatrix} 0 & 1 & & & \\ & 0 & 1 & & \\ & & \ddots & \ddots & \\ & & & 0 & 1 \\ & & & & 0 \end{pmatrix}$$

of order n .

- (1) Calculate J^k , $k = 2, \dots, n$, and deduce $J^n = 0_{n,n}$.
 (2) Prove that $I_n + J$ is invertible, and give its inverse matrix.
 (3) For a polynomial $P(x) = \sum_{k=0}^{n-1} a_k x^k$, where $a_0 = 1$, give the representation of $P(J)$. Prove that $P(J)$ is invertible.
 (4) Suppose

$$A = \begin{pmatrix} 1 & \binom{n}{1} & \cdots & \binom{n}{n-1} \\ & \ddots & \ddots & \vdots \\ & & 1 & \binom{n}{1} \\ & & & 1 \end{pmatrix}.$$

Give A^{-1} .

37. (1) Let $A \in M_n(\mathbb{R})$ whose elements in the diagonal are all odd numbers, and elements outside the diagonal are all even numbers. Prove that A is invertible.

(2) Suppose a goatherd has 2025 goats. For every goat, we can always divide the other 2024 ones into two part, each part have 1012 goats and they have the same weight. Prove that all the 2025 goats have the same weight.

38. All the vectors considered are in $\mathbb{R}^n$. Determine the assertions below are true or false.

(1) If $\mathbf{v}$ is not a linear combination of $\mathbf{u}$, then they are linearly independent.

(2) If $\mathbf{v}$ is a linear combination of $\mathbf{u}$, then they are linearly dependent.

(3) If a family of vectors contains the zero vector, then they are linearly dependent.

(4) If a family of vectors does not contain the zero vector, then they are linearly independent.

(5) If $p > n$, then $\mathbf{u}_1, \dots, \mathbf{u}_p$ are linearly dependent.

(6) If $p < n$, then $\mathbf{u}_1, \dots, \mathbf{u}_p$ cannot generate $\mathbb{R}^n$.

(7) If $(\mathbf{u}_1, \dots, \mathbf{u}_n)$ are linearly independent, then $\text{Span}(\mathbf{u}_1, \dots, \mathbf{u}_n) = \mathbb{R}^n$.

(8) If $\text{Span}(\mathbf{u}_1, \dots, \mathbf{u}_n) = \mathbb{R}^n$, then $(\mathbf{u}_1, \dots, \mathbf{u}_n)$ are linearly independent.

(9) If $(\mathbf{u}_1, \dots, \mathbf{u}_p)$ are linearly dependent, $A \in M_{m,n}(\mathbb{R})$ is an arbitrary matrix, then $(A\mathbf{u}_1, \dots, A\mathbf{u}_p)$ are linearly dependent.

(10) If $(\mathbf{u}_1, \dots, \mathbf{u}_p)$ are linearly independent, $A \in M_{m,n}(\mathbb{R})$ is an arbitrary matrix, then $(A\mathbf{u}_1, \dots, A\mathbf{u}_p)$ are linearly independent.

39. Determine whether the families of vectors below are linearly independent.

(1) $\mathbf{u} = (2, 1), \quad \mathbf{v} = (1, 2)$.

(2) $\mathbf{u} = (1, 2, 3), \quad \mathbf{v} = (2, 1, 3), \quad \mathbf{w} = (3, 2, 1)$.

(3) $\mathbf{u} = (1, 2, 1), \quad \mathbf{v} = (1, 1, -2), \quad \mathbf{w} = (1, a, -3)$, where $a \neq \frac{2}{3}$.

(4) In (3), choose $a = \frac{2}{3}$.

(5) $\mathbf{u} = (3, 1, 2, -1), \quad \mathbf{v} = (1, -5, 6, -11), \quad \mathbf{w} = (-1, 2, -3, 5)$.

(6) $\mathbf{u}_1 = (2, 1, 3, 9), \quad \mathbf{u}_2 = (1, 5, 4, 7), \quad \mathbf{u}_3 = (5, 3, 7, 15), \quad \mathbf{u}_4 = (-3, 2, -1, 4)$.

40. Let

$$\mathbf{u} = (1, 2, 1), \quad \mathbf{v} = (2, 3, 3), \quad \mathbf{w} = (3, 7, 1).$$

- (1) Prove that $\mathcal{B} = (\mathbf{u}, \mathbf{v}, \mathbf{w})$ is a basis of $\mathbb{R}^3$.
- (2) Give the coordinate of (x, y, z) with respect to $\mathcal{B}$.

41. Extend the following families of vectors as a basis of the total space.

- (1) $\mathbf{u}_1 = (1, 1, 2), \quad \mathbf{u}_2 = (2, 1, 2).$
- (2) $\mathbf{u}_1 = (1, 0, 1, 0), \quad \mathbf{u}_2 = (0, 1, 0, 1).$
- (3) $\mathbf{u}_1 = (1, 1, 1, 1), \quad \mathbf{u}_2 = (1, 2, 3, 4), \quad \mathbf{u}_3 = (1, 3, 5, 10).$

42. Let K be a field and V be a finite dimensional vector space over K . Prove that, for any $n \in \mathbb{N}$,

$$\dim(V^n) = n \dim(V).$$

43. Let K be a field and V and W be finite dimensional vector spaces over K . Prove that $\text{Hom}_{K\text{-Mod}}(V, W)$ is naturally equipped with a structure of vector space over K and

$$\dim(\text{Hom}_{K\text{-Mod}}(V, W)) = \dim(V) \dim(W).$$

44. Let $u : \mathbb{R}^3 \rightarrow \mathbb{R}^4$ be the linear mapping defined as

$$u(x, y, z) = (-x + y, x - y, -x + z, -y + z).$$

- (1) Let $\{e_1, e_2, e_3\}$ be the canonical basis of $\mathbb{R}^3$ and $\{f_1, f_2, f_3, f_4\}$ be the canonical basis of $\mathbb{R}^4$. Express $u(e_1), u(e_2), u(e_3)$ in terms of f_1, f_2, f_3, f_4 .
 - (2) Write the matrix of u with respect to the canonical basis of $\mathbb{R}^3$ and $\mathbb{R}^4$.
 - (3) Prove that $\{f_1, f_2, u(e_1), u(e_2)\}$ forms a basis of $\mathbb{R}^4$.
 - (4) Write the matrix of u with respect to $\{e_1, e_2, e_3\}$ to $\{f_1, f_2, u(e_1), u(e_2)\}$.
- 45.** Let F, G be two vector subspaces of $\mathbb{R}^5$, $\dim(F) = \dim(G) = 3$. Prove that $F \cap G \neq \{0\}$.
- 46.** Suppose that F is a vector subspace of $\mathbb{R}^n$ of dimension p , $\mathbf{v}_1, \dots, \mathbf{v}_p \in F$. Prove that the following statements are equivalent.

- (a) $(\mathbf{v}_1, \dots, \mathbf{v}_p)$ is a basis of F ;
- (b) $(\mathbf{v}_1, \dots, \mathbf{v}_p)$ are linearly independent;
- (c) $(\mathbf{v}_1, \dots, \mathbf{v}_p)$ generate F : $\text{Span}(\mathbf{v}_1, \dots, \mathbf{v}_p) = F$.

47. Determine whether the following f_k are linear transformation. If so, find the matrix A_k such that $f_k(X) = f_{A_k}(X) = XA_k$.

(a)

$$\begin{aligned} f_1 : \quad \mathbb{R}^2 &\longrightarrow \mathbb{R}^3 \\ (x, y) &\longmapsto (x + y, x - y, x + 2y). \end{aligned}$$

(b)

$$\begin{aligned} f_2 : \quad \mathbb{R}^2 &\longrightarrow \mathbb{R}^3 \\ (x, y) &\longmapsto (x + y, x - y, 1). \end{aligned}$$

(c)

$$\begin{aligned} f_3 : \quad \mathbb{R}^3 &\longrightarrow \mathbb{R}^2 \\ (x, y, z) &\longmapsto (x + y + z + a, x - y - z + b). \end{aligned}$$

(Discuss for different $(a, b) \in \mathbb{R}^2$.)

(d)

$$\begin{aligned} f_4 : \quad \mathbb{R}^3 &\longrightarrow \mathbb{R} \\ (x, y, z) &\longmapsto ax + by + cz. \end{aligned}$$

(e)

$$\begin{aligned} f_5 : \quad \mathbb{R}^3 &\longrightarrow \mathbb{R} \\ (x, y, z) &\longmapsto x^2 + y^2 + z^2. \end{aligned}$$

(f)

$$\begin{aligned} f_6 : \quad \mathbb{R}^3 &\longrightarrow \mathbb{R}^3 \\ (x, y, z) &\longmapsto (x + 2y + 3z, x + 3y + 5z, x + 5y + 9z). \end{aligned}$$

48. In $\mathbb{R}^2$, consider

$$\mathbf{u} = (1, 1), \quad \mathbf{v} = (2, -1).$$

(1) Prove that $(\mathbf{u}, \mathbf{v})$ forms a basis of $\mathbb{R}^2$.

(2) Suppose $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a linear transformation, such that

$$f(\mathbf{u}) = (2, -1), f(\mathbf{v}) = (-4, 2).$$

Find the representation matrix of f .

(3) Find the conditions of the parameter $a \in \mathbb{R}$, such that

$$(5, a) \in \text{Im}(f).$$

49. Consider

$$\begin{aligned} f : \quad \mathbb{R}^3 &\longrightarrow \mathbb{R}^3 \\ (x, y, z) &\longmapsto (-3x - y + z, 8x + 3y - 2z, -4x - y + z). \end{aligned}$$

- (1) Determine the representation matrix of f .
- (2) Determine a basis and the dimension of $\ker(f)$.
- (3) Determine a basis and the dimension of $\text{Im}(f)$.

50. Determine the rank of the following matrices.

$$A = \begin{pmatrix} 0 & -7 & 0 & 2 \\ -1 & -7 & 4 & -2 \\ 2 & 2 & -6 & 0 \\ -2 & -8 & 6 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & -2 & -1 & 1 \\ 7 & 1 & 2 & 4 \\ 2 & 1 & 1 & 1 \\ 5 & 5 & 4 & 2 \end{pmatrix}.$$

51. According to $\lambda \in \mathbb{R}$, determine a basis and the dimension of the row vector spaces of the matrices below, and find their ranks.

$$(1) \begin{pmatrix} -1-\lambda & 2 & 1 \\ 4 & 1-\lambda & -2 \\ 0 & 0 & 3-\lambda \end{pmatrix}.$$

$$(2) \begin{pmatrix} 12-\lambda & -6 & 3 \\ -9 & -5-\lambda & 3 \\ -12 & -8 & 9-\lambda \end{pmatrix}.$$

52. Let $A, B \in M_n(\mathbb{R})$. By the fact

$$(A - I_n)(B - I_n) = (AB - I_n) - (A - I_n) - (B - I_n),$$

prove: $\text{rk}(AB - I_n) \leq \text{rk}(A - I_n) + \text{rk}(B - I_n)$.

53. Suppose $A \in M_{m,n}(\mathbb{R})$. Prove $\text{rk}(AA^T) = \text{rk}(A)$, and $\text{Im}(AA^T) = \text{Im}(A)$.

54. Let $A \in M_{m,n}(\mathbb{R})$.

- (1) If $\text{rk}(A) = n$, $B \in M_{n,p}(\mathbb{R})$. Prove $\text{rk}(AB) = \text{rk}(B)$.
- (2) If $\text{rk}(A) = m$, $C \in M_{p,m}(\mathbb{R})$. Prove $\text{rk}(CA) = \text{rk}(C)$.

55. Suppose $A \in M_{m,n}(\mathbb{R})$, whose rank is $r \geq 1$. Prove that there exist r matrices of rank 1 denoted by $A_1, \dots, A_r$, such that

$$A_1 + \dots + A_r = A.$$

56. Prove that there exists a matrix $A \in M_n(\mathbb{R})$, such that $\ker(A) = \text{Im}(A)$ if and only if n is an even number. (**Hint:** consider the matrix $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$.)

57. Suppose $A \in M_{3,2}(\mathbb{R})$, $B \in M_{2,3}(\mathbb{R})$, such that

$$AB = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Prove: $BA = I_2$.

58. Let $A, B \in M_n(\mathbb{R})$.

(1) Prove

$$\ker(A) \subset \ker(BA), \quad \text{Im}(BA) \subset \text{Im}(B).$$

(2) Prove that $\mathbb{R}^n = \text{Im}(A) + \ker(B)$ is valid if and only if $\text{Im}(BA) = \text{Im}(B)$.

59. Suppose $J \in M_n(\mathbb{R})$ is a nilpotent matrix, which means $\exists p \in \mathbb{N}^*$, such that $J^p = 0_{n,n}$. Prove $J^n = 0_{n,n}$.

60. Suppose $\mathbb{V}(\mathbb{R}^n)$ is the set of all vector subspaces of $\mathbb{R}^n$, $d : \mathbb{V}(\mathbb{R}^n) \rightarrow \mathbb{N}$ is a function which satisfies the following two properties:

- (a) IF $F, G \in \mathbb{V}(\mathbb{R}^n)$, $F \cap G = \{0\} \implies d(F + G) = d(F) + d(G)$.
- (b) $d(\mathbb{R}^n) = n$.

Prove:

(1) If $\dim(F) = 1$, then $d(F) = 1$.

(2) d is the dimension: $\forall F \in \mathbb{V}(\mathbb{R}^n)$, $d(F) = \dim(F)$.

61. Find the quotient and remainder of $f(T)$ dividing $g(T)$.

- (1) $f(T) = T^5 + 4T^4 + T^2 + 2T + 3$, $g(T) = T - 2$;
- (2) $f(T) = T^n - 1$, $g(T) = T - a$, where a is a scalar.

62. Find the greatest common divisor of $f(T)$ and $g(T)$.

- (1) $f(T) = T^5 + T^4 - 5T^3 - 5T^2 + 4T + 4$, $g(T) = T^4 + T^3 - 17T^2 - 21T + 36$;
- (2) $f(T) = 3T^4 - 8T^2 - 3$, $g(T) = T^3 - 2T^2 - 3T + 6$.

63. Find a polynomial $P \in \mathbb{R}[T]$ divisible by $T^2 + 1$ such that $P(T) + 1$ is divisible by $T^3 + T^2 + 1$.

64. Suppose $f \in k[T]$, $f(f(T)) = (f(T))^k$, where $k \in \mathbb{N}$. Find all such f .

- 65.** Let a, b be two strictly positive real numbers. Find all positive integer n , such that the polynomial $T^2 - (a^2 + b^2)$ can divide $T^{2n} - (a^n + b^n)^2$ in $\mathbb{R}[T]$.
- 66.** Find the polynomial $f(T)$ with the smallest degree, such that the remainder of $f(T)$ dividing $(T - 1)^2$ is $2T$, and the remainder of $f(T)$ dividing $(T - 2)^2$ is $3T$.
- 67.** Let k be a field. In the k -algebra $k[T]$, we consider the sub- k -algebra $\mathcal{A}$ generated by T^2 and T^3 , noted $\mathcal{A} = k[T^2, T^3]$. Prove that $k[T]$ and $\mathcal{A}$ are not isomorphic as k -algebras.
- 68.** Let K and L be two fields. If the rings $K[T]$ and $L[T]$ are isomorphic, prove that K and L are isomorphic as fields.
- 69.** Let k be a field, $P \in k[T]$, and $b, c \in \mathbb{N}$ satisfying $\gcd(b, c) = 1$. Prove that $(P^b - 1)(P^c - 1)$ divides $(P - 1)(P^{bc} - 1)$.
- 70.** Let $m_1, \dots, m_k \in \mathbb{N}^+$. Prove that the greatest common divisor of the polynomials $(T^{m_i} - 1)_{1 \leq i \leq k}$ is $T^d - 1$, where $d = \gcd(m_1, \dots, m_k)$.
- 71.** Let k be a field, and $r_1, \dots, r_k \in \mathbb{N}$, such that $0 \leq r_1 < \dots < r_k \leq n - 1$, where $n \in \mathbb{N}_{\geq 2}$. We give $m_1, \dots, m_k \in \mathbb{N}$, such that $m_i \equiv r_i \pmod{n}$ for $i = 1, \dots, k$. Prove that the remainder in the Euclidean division of $T^{m_1} + \dots + T^{m_k}$ by $T^n - 1$ in $k[T]$ is $T^{r_1} + \dots + T^{r_k}$, and calculate the quotient.
- 72.** Let $q, m \in \mathbb{N}^+$. Find a sufficient and necessary condition such that $1 + T^m + \dots + T^{qm}$ is divisible by $1 + T + \dots + T^q$ in $k[T]$, where k is a field.
- 73.** Let $g(X), p(X) \in K[X]$ with $\deg(p) > 0$. Prove that there exist polynomials $\{a_i(X)\}_{i \geq 0}$, such that $\deg(a_i) < \deg(p)$,

$$g(X) = a_0(X) + a_1(X)p(X) + \dots + a_{e-1}(X)p(X)^{e-1},$$

and

$$\frac{g(X)}{p(X)^e} = \frac{a_0(X)}{p(X)^e} + \frac{a_1(X)}{p(X)^{e-1}} + \dots + \frac{a_{e-1}(X)}{p(X)}.$$

Find an analog of this property over $\mathbb{Z}$.

- 74.** Let k be a field, and $f(T) \in k[T]$. We denote by $f'(T)$ the derivative of $f(T)$. Let $c \in k$, and prove the following results:
- (1) c is a multiple root of $f(T)$ if and only if $f(c) = f'(c) = 0$.
 - (2) c is a multiple root of $f(T)$ if and only if c is a root of $\gcd(f(T), f'(T))$.

- (3) If $\gcd(f(T), f'(T)) = 1$, then $f(T)$ has no multiple root (even in the extension of k).
- (4) If $f(T)$ is irreducible over k and $f'(T) \neq 0$, then $f(T)$ has no multiple root (even in the extension of k).
- (5) Let $p(T) \in k[T]$ be irreducible. Then $p(T)$ is a multiple factor of $f(T)$ if and only if $p(T)$ is a common factor of $f(T)$ and $f'(T)$.
- (6) $f(T)$ has no multiple factor if and only if $\gcd(f(T), f'(T)) = 1$.

75. Let k be a field, k' is an extension of k . Let $a \in k'$, and we denote

$$J(a) = \{f \in k[T] \mid f(a) = 0\}.$$

Suppose $J(a) \neq \{0\}$.

- (1) Prove that there exists the unique polynomial $m(T)$ whose leading coefficient is 1, such that $J(a) = \{h(T)m(T) \mid h(T) \in k[T]\}$.
- (2) Prove that the previous $m(T)$ is irreducible.

76. Expand the following rational functions as formal power series.

- (1) $\frac{1}{1+T+\dots+T^{n-1}}$, where $n \in \mathbb{N}_{\geq 2}$.
- (2) $\frac{1}{(1-aT)^p(1-bT)^q}$, where $a \neq b$, $p, q \in \mathbb{N}^+$.
- (3) $\frac{1}{(1-T^p)(1-T^q)}$ with p and q are coprime.

77. For an $n \in \mathbb{N}$, let u_n be the number of permutations $\sigma \in \mathfrak{S}_n$ such that $\sigma^2 = \text{Id}$. We put $u_0 = u_1 = 1$.

- (1) Prove that for all $n \in \mathbb{N}_{\geq 2}$, we have $u_n = u_{n-1} + (n-1)u_{n-2}$.
- (2) We consider the formal power series $S = \sum_{n \geq 0} \frac{u_n}{n!} T^n \in \mathbb{C}[[T]]$. Prove that $S' - (1+T)S = 0$, and deduce that $S = \exp\left(T + \frac{T^2}{2}\right)$. Deduce the expression of u_n .

Chapter 7

Limits

7.1 Filters

7.1.1 Definition. Let X be a set. We call a *filter* on X any non-empty subset $\mathcal{F}$ of $\mathcal{P}(X)$ that satisfies the following conditions:

- (1) for any $(U, V) \in \mathcal{F} \times \mathcal{F}$, $U \cap V \in \mathcal{F}$;
- (2) for any $V \in \mathcal{F}$ and any $U \in \mathcal{P}(X)$ such that $V \subseteq U$, one has $U \in \mathcal{F}$.

Note that, if a filter $\mathcal{F}$ contains $\emptyset$, then it is equal to $\mathcal{P}(X)$. This filter is said to be *degenerate*. If a non-degenerate filter $\mathcal{F}$ is not contained in any non-degenerate filter, then we say that it is an *ultrafilter*.

7.1.2 Proposition. Let X be a set. Let I be a non-empty set and $(\mathcal{F}_i)_{i \in I}$ be a family of filters on X . Then $\mathcal{F} = \bigcap_{i \in I} \mathcal{F}_i$ is a filter on X .

Proof. Let U and V be elements of $\mathcal{F}$. For any $i \in I$ one has $\{U, V\} \subseteq \mathcal{F}_i$. Hence $U \cap V \in \mathcal{F}_i$. Therefore, $U \cap V \in \mathcal{F}$.

Let V be an element of $\mathcal{F}$ and $U \in \mathcal{P}(X)$ such that $V \subseteq U$. For any $i \in I$ one has $V \in \mathcal{F}_i$ and hence $U \in \mathcal{F}_i$. Therefore, $U \in \mathcal{F}$. $\square$

7.1.3 Definition. Let X be a set and S be a subset of $\mathcal{P}(X)$. By Proposition 7.1.2, the intersection of all filters containing S is still a filter containing S . We denote by $\mathcal{F}_S$ this filter and call it the *filter generated by S* .

7.1.4 Definition. Let X be a set. We call a *filter basis* on X any non-empty subset $\mathcal{B}$ of $\mathcal{P}(X)$ such that

$$\forall (A, B) \in \mathcal{B} \times \mathcal{B}, \exists C \in \mathcal{B}, C \subseteq A \cap B.$$

7.1.5 Proposition. *Let X be a set and S be a non-empty subset of $\mathcal{P}(X)$. Then*

$$\mathcal{F}_S = \{U \in \mathcal{P}(X) \mid \exists n \in \mathbb{N}_{\geq 1}, \exists (A_1, \dots, A_n) \in S^n, A_1 \cap \dots \cap A_n \subseteq U\}. \quad (7.1)$$

If S is a filter basis, then

$$\mathcal{F}_S = \{U \in \mathcal{P}(X) \mid \exists A \in S, A \subseteq U\}.$$

Proof. Denote by $\mathcal{F}'_S$ the expression on the right hand side of the equality (7.1). Since S is not empty, one has $\mathcal{F}'_S \neq \emptyset$. For any $U \in \mathcal{F}'_S$, there exists elements $A_1, \dots, A_n$ of S such that $A_1 \cap \dots \cap A_n \subseteq U$. Since $S \subseteq \mathcal{F}_S$, we obtain that $A_1 \cap \dots \cap A_n \in \mathcal{F}_S$ and hence $U \in \mathcal{F}_S$. Therefore, $\mathcal{F}'_S \subseteq \mathcal{F}_S$.

We prove that $\mathcal{F}'_S$ is a filter. By definition, if $V \in \mathcal{F}'_S$ and $U \in \mathcal{P}(X)$ such that $V \subseteq U$, then $U \in \mathcal{F}'_S$. If U and V are two elements of $\mathcal{F}'_S$, and $A_1, \dots, A_n, B_1, \dots, B_m$ be elements of S such that

$$A_1 \cap \dots \cap A_n \subseteq U, B_1 \cap \dots \cap B_m \subseteq V.$$

Hence

$$A_1 \cap \dots \cap A_n \cap B_1 \cap \dots \cap B_m \subseteq U \cap V.$$

By definition, one has $S \subseteq \mathcal{F}'_S$. Since $\mathcal{F}_S$ is the smallest filter containing S , we obtain that $\mathcal{F}_S \subseteq \mathcal{F}'_S$.

Suppose that S is a filter basis. For any $U \in \mathcal{F}_S$, there exist $n \in \mathbb{N}_{\geq 1}$ and $(A_1, \dots, A_n) \in S^n$ such that $A_1 \cap \dots \cap A_n \subseteq U$. Since S is a filter basis, there exists $A \in S$ such that $A \subseteq A_1 \cap \dots \cap A_n \subseteq U$. Hence

$$\mathcal{F}_S \subseteq \{U \in \mathcal{P}(X) \mid \exists A \in S, A \subseteq U\}.$$

Moreover, by (7.1), one has

$$\{U \in \mathcal{P}(X) \mid \exists A \in S, A \subseteq U\}.$$

Hence the equality

$$\mathcal{F}_S = \{U \in \mathcal{P}(X) \mid \exists A \in S, A \subseteq U\}$$

holds. □

7.1.6 Remark. Let $\mathcal{B}$ be a filter basis and $\mathcal{F}$ be a filter containing $\mathcal{B}$. If $\mathcal{B} \subseteq \mathcal{F}$ and

$$\forall U \in \mathcal{F}, \exists A \in \mathcal{B}, A \subseteq U,$$

then $\mathcal{F}$ identifies with the filter generated by $\mathcal{B}$. In fact, the relation $\mathcal{B} \subseteq \mathcal{F}$ implies that $\mathcal{F}_{\mathcal{B}} \subseteq \mathcal{F}$. Conversely, by the second equality in Proposition 7.1.5, one has $\mathcal{F} \subseteq \mathcal{F}_{\mathcal{B}}$. Hence the equality $\mathcal{F} = \mathcal{F}_{\mathcal{B}}$ holds.

7.1.7 Example. Let X be a set and Y be a subset of X . The filter generated by $\{Y\}$ identifies with

$$\{U \in \mathcal{P}(X) \mid Y \subseteq U\}.$$

We call it a *principal filter* and denote it as $\mathcal{F}_Y$.

7.1.8 Proposition. Let X be a set and $\mathcal{F}$ be the set of $U \in \mathcal{P}(X)$ such that $X \setminus U$ is finite. Then $\mathcal{F}$ is a filter on X . We call it the *Fréchet filter* on X .

Proof. Let U and V be elements of $\mathcal{F}$. The sets $X \setminus U$ and $X \setminus V$ are finite. Hence

$$X \setminus (U \cap V) = (X \setminus U) \cup (X \setminus V)$$

is also finite.

Let V be an element of $\mathcal{F}$. If U is a subset of X such that $V \subseteq U$, then $X \setminus U \subseteq X \setminus V$. Hence $X \setminus U$ is a finite set. $\square$

7.2 Order limits

In this section, we fix a partially ordered set $(G, \leq)$, which is assumed to be order complete.

7.2.1 Definition. Let X be a set and $f : X \rightarrow G$ be a mapping. If U is a subset of X , we denote by $f^s(U)$ the element $\sup_{x \in U} f(x)$ and $f^i(U)$ the element $\inf_{x \in U} f(x)$. Thus we obtain two mappings f^s and f^i from $\mathcal{P}(X)$ to G . If we equip $\mathcal{P}(X)$ with the partial order of inclusion $\subseteq$, then f^s is increasing and f^i is decreasing. Moreover, if U is a non-empty subset of X , then one has

$$f^i(U) \leq f^s(U).$$

Let $\mathcal{F}$ be a filter on X . We define

$$\limsup_{\mathcal{F}} f := \inf_{U \in \mathcal{F}} f^s(U), \quad \liminf_{\mathcal{F}} f := \sup_{U \in \mathcal{F}} f^i(U).$$

We call them the *superior limit* and the *inferior limit* of the mapping f along the filter $\mathcal{F}$. If

$$\limsup_{\mathcal{F}} f = \liminf_{\mathcal{F}} f,$$

then we say that f has a *limit* along $\mathcal{F}$ and we denote by

$$\lim_{\mathcal{F}} f$$

the value $\limsup_{\mathcal{F}} f$ (which is equal to $\liminf_{\mathcal{F}} f$).

7.2.2 Example. Let $f : X \rightarrow G$ be a constant mapping and let ℓ be the value of f . For any non-empty subset U of X , one has

$$f^s(U) = f^i(U) = \ell.$$

Therefore, for any non-degenerate filter $\mathcal{F}$ on X , one has

$$\limsup_{\mathcal{F}} f = \liminf_{\mathcal{F}} f = \ell.$$

7.2.3 Notation. Let I be an infinite subset of $\mathbb{N}$ and $\mathcal{F}$ be the Fréchet filter on I . We call a *sequence* in G parametrized by I any element $\mathbf{a} = (a_n)_{n \in I}$ in G^I . Then

$$\limsup_{\mathcal{F}} \mathbf{a}, \quad \liminf_{\mathcal{F}} \mathbf{a}, \quad \lim_{\mathcal{F}} \mathbf{a}$$

are often denoted as

$$\limsup_{n \in I, n \rightarrow +\infty} a_n, \quad \liminf_{n \in I, n \rightarrow +\infty} a_n, \quad \lim_{n \in I, n \rightarrow +\infty} a_n,$$

respectively. If there is no ambiguity on I , we can also denote them as

$$\limsup_{n \rightarrow +\infty} a_n, \quad \liminf_{n \rightarrow +\infty} a_n, \quad \lim_{n \rightarrow +\infty} a_n.$$

7.2.4 Remark. Let $f : X \rightarrow G$ be a mapping, $\mathcal{F}$ be a non-degenerate filter on X , and ℓ be an element of G .

(1) Since

$$\limsup_{\mathcal{F}} f = \inf_{U \in \mathcal{F}} \sup_{x \in U} f(x),$$

we obtain that

$$\limsup_{\mathcal{F}} f \geq \ell$$

if and only if

$$\forall U \in \mathcal{F}, \sup_{x \in U} f(x) \geq \ell.$$

Moreover, if

$$\limsup_{\mathcal{F}} f > \ell,$$

then, for any $U \in \mathcal{F}$, one has

$$\sup_{x \in U} f(x) > \ell.$$

In particular, in the case where $(G, \leq)$ is totally ordered, one has

$$\forall U \in \mathcal{F}, \exists x \in U, f(x) > \ell.$$

(2) Since

$$\liminf_{\mathcal{F}} f = \sup_{U \in \mathcal{F}} \inf_{x \in U} f(x),$$

we obtain that

$$\liminf_{\mathcal{F}} f \leq \ell$$

if and only if

$$\forall U \in \mathcal{F}, \inf_{x \in U} f(x) \leq \ell.$$

Moreover, if

$$\liminf_{\mathcal{F}} f < \ell,$$

then, for any $U \in \mathcal{F}$, one has

$$\inf_{x \in U} f(x) < \ell.$$

In particular, in the case where $(G, \leq)$ is totally ordered, one has

$$\forall U \in \mathcal{F}, \exists x \in U, f(x) < \ell.$$

(3) Assume that $(G, \leq)$ is totally ordered. If

$$\limsup_{\mathcal{F}} f < \ell,$$

then there exists $U \in \mathcal{F}$ such that $f^s(U) < \ell$ (in particular, for any $x \in U$, $f(x) < \ell$).

(4) Assume that $(G, \leq)$ is totally ordered. If

$$\liminf_{\mathcal{F}} f > \ell,$$

then there exists $U \in \mathcal{F}$ such that $f^i(U) > \ell$ (in particular, for any $x \in U$, $f(x) > \ell$).

7.2.5 Proposition. *Let $f : X \rightarrow G$ be a mapping, $\mathcal{F}$ be a non-degenerate filter on X . For any $(U, V) \in \mathcal{F} \times \mathcal{F}$, one has $f^s(U) \geq f^i(V)$. In particular,*

$$\limsup_{\mathcal{F}} f \geq \liminf_{\mathcal{F}} f.$$

Proof. Since f^s is increasing and f^i is decreasing, one has

$$f^s(U) \geq f^s(U \cap V), \quad f^i(U \cap V) \geq f^i(V).$$

Since $U \cap V \in \mathcal{F}$ and $\mathcal{F}$ is non-degenerate, one has $U \cap V \neq \emptyset$. Hence

$$f^s(U \cap V) \geq f^i(U \cap V).$$

Therefore, $f^s(U) \geq f^i(V)$ holds.

By taking the infimum with respect to $U \in \mathcal{F}$, we get

$$\forall V \in \mathcal{F}, \limsup_{\mathcal{F}} f \geq f^i(V).$$

By taking the supremum with respect to $V \in \mathcal{F}$, we get

$$\limsup_{\mathcal{F}} f \geq \liminf_{\mathcal{F}} f.$$

□

7.2.6 Proposition. *Let $f : X \rightarrow G$ be a mapping and $\mathcal{F}$ be a filter on X . Let $\mathcal{B}$ be a filter basis on X that generates $\mathcal{F}$. Then the following equalities hold:*

$$\limsup_{\mathcal{F}} f = \inf_{U \in \mathcal{B}} f^s(U), \quad \liminf_{\mathcal{F}} f = \sup_{U \in \mathcal{B}} f^i(U).$$

Proof. Since $\mathcal{F}$ contains $\mathcal{B}$, one has

$$\begin{aligned} \limsup_{\mathcal{F}} f &= \inf_{U \in \mathcal{F}} f^s(U) \leq \inf_{U \in \mathcal{B}} f^s(U), \\ \liminf_{\mathcal{F}} f &= \sup_{U \in \mathcal{F}} f^i(U) \leq \sup_{U \in \mathcal{B}} f^i(U). \end{aligned}$$

Moreover, by Proposition 7.1.5, for any $W \in \mathcal{F}$, there exists $A \in \mathcal{B}$ such that $A \subseteq W$. Since f^s is increasing, one has

$$f^s(W) \geq f^s(A) \geq \inf_{U \in \mathcal{B}} f^s(U), \quad f^i(W) \leq f^i(A) \leq \sup_{U \in \mathcal{B}} f^i(U).$$

Taking the infimum and the supremum with respect to W in the two formulas, respectively, we get

$$\limsup_{\mathcal{F}} f \geq \inf_{U \in \mathcal{B}} f^s(U), \quad \liminf_{\mathcal{F}} f \leq \sup_{U \in \mathcal{B}} f^i(U).$$

□

7.2.7 Remark. Let I and J be infinite subsets of $\mathbb{N}$. Since

$$\{I_{\geq j} \mid j \in J\}$$

is a filter basis of the Fréchet filter on I , we obtain that, for any sequence $(a_n)_{n \in I} \in G^I$, one has

$$\begin{aligned} \limsup_{n \in I, n \rightarrow +\infty} a_n &= \inf_{j \in J} \sup_{n \in I, n \geq j} a_n, \\ \liminf_{n \in I, n \rightarrow +\infty} a_n &= \sup_{j \in J} \inf_{n \in I, n \geq j} a_n. \end{aligned}$$

7.2.8 Proposition. *Let X be a set and $\mathcal{F}$ be a filter on X . Let $f : X \rightarrow G$ and $g : X \rightarrow G$ be two mappings. Suppose that there exists $A \in \mathcal{F}$ such that*

$$\forall x \in A, f(x) \leq g(x),$$

then one has

$$\limsup_{\mathcal{F}} f \leq \limsup_{\mathcal{F}} g, \quad \liminf_{\mathcal{F}} f \leq \liminf_{\mathcal{F}} g.$$

Proof. Let $\mathcal{B}$ be the set of $B \in \mathcal{F}$ such that $B \subseteq A$. Then $\mathcal{B}$ is a filter basis contained in $\mathcal{F}$. For any $U \in \mathcal{F}$, one has $U \cap A \in \mathcal{B}$ and $U \cap A \subseteq U$. Hence $U \in \mathcal{F}_{\mathcal{B}}$. Therefore, the filter $\mathcal{F}$ is generated by $\mathcal{B}$.

Since $f(x) \leq g(x)$ for any $x \in A$, we obtain that, for any $B \in \mathcal{B}$, one has $f^s(B) \leq g^s(B)$ and $f^i(B) \leq g^i(B)$. By Proposition 7.2.6, we deduce that

$$\begin{aligned} \limsup_{\mathcal{F}} f &= \inf_{B \in \mathcal{B}} f^s(B) \leq \inf_{B \in \mathcal{B}} g^s(B) = \limsup_{\mathcal{F}} g, \\ \liminf_{\mathcal{F}} f &= \sup_{B \in \mathcal{B}} f^i(B) \leq \sup_{B \in \mathcal{B}} g^i(B) = \liminf_{\mathcal{F}} g. \end{aligned}$$

□

7.2.9 Theorem (Squeeze theorem). *Let X be a set and $\mathcal{F}$ be a filter on X . Let f, g and h be mappings from X to G . Assume that there exists $A \in \mathcal{F}$ such that*

$$\forall x \in A, f(x) \leq g(x) \leq h(x).$$

If f and h have a limit along $\mathcal{F}$ and

$$\lim_{\mathcal{F}} f = \lim_{\mathcal{F}} h,$$

then g also has a limit along $\mathcal{F}$, and the following equality holds

$$\lim_{\mathcal{F}} f = \lim_{\mathcal{F}} g = \lim_{\mathcal{F}} h.$$

Proof. By Proposition 7.2.8, one has

$$\liminf_{\mathcal{F}} f \leq \liminf_{\mathcal{F}} g \leq \liminf_{\mathcal{F}} h, \quad \limsup_{\mathcal{F}} f \leq \limsup_{\mathcal{F}} g \leq \limsup_{\mathcal{F}} h.$$

If ℓ is an element of G such that

$$\liminf_{\mathcal{F}} f = \limsup_{\mathcal{F}} f = \liminf_{\mathcal{F}} h = \limsup_{\mathcal{F}} h = \ell,$$

then one has

$$\liminf_{\mathcal{F}} g = \limsup_{\mathcal{F}} g = \ell.$$

□

7.2.10 Proposition. *Let I be an infinite subset of $\mathbb{N}$ and $(a_n)_{n \in I}$ be a sequence in G .*

- (1) *If $(a_n)_{n \in I}$ is increasing, then $\sup_{n \in I} a_n$ is the limit of the sequence $(a_n)_{n \in I}$.*
- (2) *If $(a_n)_{n \in I}$ is decreasing, then $\inf_{n \in I} a_n$ is the limit of the sequence $(a_n)_{n \in I}$.*

Proof. By reversing the direction of the partial order of G , one could deduce (2) from (1). It then suffices to prove (1). Let $\ell := \sup_{n \in I} a_n$.

Let n be an element of I . Since $(a_n)_{n \in I}$ is increasing, one has

$$\inf_{k \in I_{\geq n}} a_k = a_n.$$

Hence

$$\liminf_{n \rightarrow +\infty} a_n = \sup_{n \in I} a_n = \ell.$$

Since $a_n \leq \ell$ for any $n \in I$, by Proposition 7.2.8 one has

$$\limsup_{n \rightarrow +\infty} a_n \leq \ell.$$

By Proposition 7.2.5 we obtain

$$\limsup_{n \rightarrow +\infty} a_n = \liminf_{n \rightarrow +\infty} a_n = \ell.$$

□

7.2.11 Example. Let $\varepsilon > 0$ and $c > 0$. Note that the sequence

$$\left(\frac{c}{n^\varepsilon}\right)_{n \in \mathbb{N}_{\geq 1}}$$

is decreasing. Therefore,

$$\lim_{n \rightarrow +\infty} \frac{c}{n^\varepsilon} = \inf \left\{ \frac{c}{n^\varepsilon} \mid n \in \mathbb{N}_{\geq 1} \right\} = 0.$$

7.2.12 Example. Let a be a real number such that $a > 1$. Let $b = a - 1$. By the binomial equality, one has

$$\forall n \in \mathbb{N}_{\geq 1}, a^n = (1 + b)^n = \sum_{i=0}^n \binom{n}{i} b^i \geq 1 + nb.$$

Therefore, by Proposition 7.2.9, one has

$$\lim_{n \rightarrow +\infty} \frac{1}{a^n} = 0.$$

7.2.13 Proposition. *Let X be a set and Y be a subset of X .*

(1) *If $\mathcal{F}$ is a filter on X , then the set*

$$\mathcal{F}|_Y := \{U \cap Y \mid U \in \mathcal{F}\}$$

is a filter on Y .

(2) *If $\mathcal{B}$ is a filter basis on X , then*

$$\mathcal{B}|_Y := \{B \cap Y \mid B \in \mathcal{B}\}$$

is a filter basis on Y . Moreover, the filter generated by $\mathcal{B}|_Y$ identifies with $\mathcal{F}_{\mathcal{B}}|_Y$.

Proof. (1) Let V_1 and V_2 be elements of $\mathcal{F}|_Y$. There exists elements U_1 and U_2 of $\mathcal{F}$ such that $V_1 = U_1 \cap Y$ and $V_2 = U_2 \cap Y$. Then one has

$$V_1 \cap V_2 = (U_1 \cap Y) \cap (U_2 \cap Y) = (U_1 \cap U_2) \cap Y \in \mathcal{F}|_Y.$$

Let $V \in \mathcal{F}|_Y$ and W be a subset of Y that contains V . By definition there exists $U \in \mathcal{F}$ such that $V = U \cap Y$. Then one has

$$W = (U \cup W) \cap Y.$$

Hence $W \in \mathcal{F}|_Y$.

(2) Since $\mathcal{B}$ is non-empty, $\mathcal{B}|_Y$ is also non-empty. If A_1 and A_2 are elements of $\mathcal{B}|_Y$, then there exist elements B_1 and B_2 of $\mathcal{B}$, such that

$$A_1 = B_1 \cap Y, \quad A_2 = B_2 \cap Y.$$

Since $\mathcal{B}$ is a filter basis, there exists $C \in \mathcal{B}$ such that $C \subseteq B_1 \cap B_2$. Therefore, one has $C \cap Y \in \mathcal{B}|_Y$ and

$$C \cap Y \subset A_1 \cap A_2.$$

This shows that $\mathcal{B}|_Y$ is a filter basis. By definition, one has $\mathcal{B}|_Y \subseteq \mathcal{F}_{\mathcal{B}}|_Y$.

Let U be an element of $\mathcal{F}_{\mathcal{B}}$. There exists $B \in \mathcal{B}$ such that $B \subseteq U$. Then one has $B \cap Y \subseteq U \cap Y$. Hence $U \cap Y$ contains an element of $\mathcal{B}|_Y$. By Remark 7.1.6, we obtain that $\mathcal{F}_{\mathcal{B}}|_Y$ identifies with the filter on Y generated by $\mathcal{B}|_Y$. $\square$

7.2.14 Example. Let I be an infinite subset of $\mathbb{N}$ and J be an infinite subset of I . Let $\mathcal{F}$ be the Fréchet filter on I . Then $\mathcal{F}|_J$ is the Fréchet filter on J . For any sequence $(a_n)_{n \in I}$ in G , $(a_n)_{n \in J}$ is called a *subsequence* of $(a_n)_{n \in I}$.

7.2.15 Proposition. *Let $f : X \rightarrow G$ be a mapping and $\mathcal{F}$ be a filter on X . Let Y be a subset of X .*

(1) One has

$$\limsup_{\mathcal{F}|_Y} f|_Y \leq \limsup_{\mathcal{F}} f, \quad \liminf_{\mathcal{F}|_Y} f|_Y \geq \liminf_{\mathcal{F}} f. \quad (7.2)$$

(2) If Y is an element of $\mathcal{F}$, then

$$\limsup_{\mathcal{F}|_Y} f|_Y = \limsup_{\mathcal{F}} f, \quad \liminf_{\mathcal{F}|_Y} f|_Y = \liminf_{\mathcal{F}} f. \quad (7.3)$$

(3) If $\mathcal{F}|_Y$ is non-degenerate and f admits a limit along $\mathcal{F}$, then $f|_Y$ also admits a limit along $\mathcal{F}|_Y$, and one has

$$\lim_{\mathcal{F}|_Y} f|_Y = \lim_{\mathcal{F}} f.$$

Proof. (1) Let U be an element of $\mathcal{F}$. One has $f^s(U \cap Y) \subseteq f^s(U)$. Therefore,

$$\limsup_{\mathcal{F}|_Y} f|_Y = \inf_{U \in \mathcal{F}} f^s(U \cap Y) \leq \inf_{U \in \mathcal{F}} f^s(U) = \limsup_{\mathcal{F}} f.$$

By reversing the partial order G , we deduce the second inequality of (7.2) from the first one.

(2) If Y belongs to $\mathcal{F}$, then $\mathcal{F}|_Y$ is a filter basis on X that generated $\mathcal{F}$. Hence the equalities (7.3) follow from Proposition 7.2.6

(3) Suppose that $\mathcal{F}|_Y$ is non-degenerate. Then by (7.2) we obtain

$$\liminf_{\mathcal{F}} f \leq \liminf_{\mathcal{F}|_Y} f|_Y \leq \limsup_{\mathcal{F}|_Y} f|_Y \leq \limsup_{\mathcal{F}} f.$$

If ℓ is an element of G such that

$$\liminf_{\mathcal{F}} f = \limsup_{\mathcal{F}} f = \ell,$$

then

$$\liminf_{\mathcal{F}|_Y} f|_Y = \limsup_{\mathcal{F}|_Y} f|_Y = \ell.$$

□

7.2.16 Theorem (Bolzano-Weierstrass). *Suppose that G is totally ordered. Let I be an infinite subset of $\mathbb{N}$ and $(a_n)_{n \in I}$ be a sequence in G .*

(1) *There is an infinite subset J_1 of I such that $(a_n)_{n \in J_1}$ is monotone and admits $\limsup_{n \in I, n \rightarrow +\infty} a_n$ as its limit.*

- (2) *There is an infinite subset J_2 such that $(a_n)_{n \in J_2}$ is monotone and admits $\liminf_{n \in I, n \rightarrow +\infty} a_n$ as its limit.*

Proof. It suffices to prove the first statement since we could deduce the second statement from the first one by reversing the partial order of G .

Let

$$J = \{n \in I \mid \forall m \in I_{\geq n}, a_m \leq a_n\}.$$

If J is infinite, then $(x_n)_{n \in J}$ is a decreasing sequence. Hence it has

$$\inf_{n \in J} x_n$$

as its limit. Moreover, for any $n \in J$, one has

$$a_n = \sup_{m \in I_{\geq n}} a_m.$$

Hence

$$\limsup_{n \in I, n \rightarrow +\infty} a_n = \inf_{n \in J} \sup_{m \in I_{\geq n}} a_m = \inf_{n \in J} a_n,$$

as required.

In what follows, we assume that J is finite. Let $n_0 \in I$ such that

$$\forall n \in J, n < n_0,$$

and let

$$\ell = \sup_{n \in I_{\geq n_0}} a_n.$$

We construct recursively a strictly increasing sequence $(n_k)_{k \in \mathbb{N}}$ in I as follows. Suppose that $n_0, \dots, n_k$ have been chosen. We pick $i \in I$ such that $n_0 \leq i \leq n_k$ and that

$$a_i = \max\{a_j \mid j \in I, n_0 \leq j \leq n_k\}.$$

Since $i \notin J$, there exists $n_{k+1} \in I_{>i}$ such that $a_{n_{k+1}} > a_i$. By definition one has $a_{n_{k+1}} > a_{n_k}$ and $n_{k+1} > n_k$. Let $J_1 = \{n_k \mid k \in \mathbb{N}\}$. The restricted mapping $f|_{J_1}$ is increasing and hence it admits

$$\ell := \sup_{k \in \mathbb{N}} a_{n_k}$$

as its limit. Moreover, since

$$\forall k \in \mathbb{N}, \ell \geq a_{n_{k+1}} > \max\{a_j \mid j \in I, n_0 \leq j \leq n_k\},$$

we obtain that

$$\ell \geq \sup_{i \in I_{\geq n_0}} a_i.$$

Therefore

$$\ell = \sup_{i \in I_{\geq n_0}} a_i.$$

For any $n \in I_{\geq n_0}$ one has $a_n \leq \ell$. Hence, by Proposition 7.2.8 we obtain that

$$\limsup_{n \in I, n \rightarrow +\infty} a_n \leq \ell.$$

Moreover, by Proposition 7.2.15, one has

$$\ell = \limsup_{n \in J_1, n \rightarrow +\infty} a_n \leq \limsup_{n \in I, n \rightarrow +\infty} a_n.$$

Therefore

$$\ell = \limsup_{n \in I, n \rightarrow +\infty} a_n.$$

□

7.2.17 Proposition. *Suppose that $(G, \leq)$ is a totally ordered set that is order complete. Let X be a set and f and g be mappings from X to G . Let $\mathcal{F}$ be a non-degenerate filter on X . Let $f \wedge g$ be the mapping*

$$X \longrightarrow G, \quad x \longmapsto \min\{f(x), g(x)\}$$

and $f \vee g$ be the mapping

$$X \longrightarrow G, \quad x \longmapsto \max\{f(x), g(x)\}.$$

Then the following equalities hold:

$$\limsup_{\mathcal{F}}(f \vee g) = \max \left\{ \limsup_{\mathcal{F}} f, \limsup_{\mathcal{F}} g \right\}, \quad (7.4)$$

$$\liminf_{\mathcal{F}}(f \wedge g) = \min \left\{ \liminf_{\mathcal{F}} f, \liminf_{\mathcal{F}} g \right\} \quad (7.5)$$

Proof. We first prove (7.4). Since $f \vee g \geq f$ and $f \vee g \geq g$, one has

$$\limsup_{\mathcal{F}}(f \vee g) \geq \limsup_{\mathcal{F}} f, \quad \limsup_{\mathcal{F}}(f \vee g) \geq \limsup_{\mathcal{F}} g$$

Note that, for any $U \in \mathcal{F}$, one has

$$(f \vee g)^s(U) = \max\{f^s(U), g^s(U)\}.$$

In fact, since $f \vee g \geq f$ and $f \vee g \geq g$, one has

$$(f \vee g)^s(U) \geq f^s(U), \quad (f \vee g)^s(U) \geq g^s(U).$$

Conversely, for any $x \in U$, one has

$$\max\{f(x), g(x)\} \leq \max\{f^s(U), g^s(U)\}.$$

Hence

$$(f \vee g)^s(U) \leq \max\{f^s(U), g^s(U)\}.$$

Let

$$\mathcal{B}_1 := \{A \in \mathcal{F} \mid f^s(A) \geq g^s(A)\}, \quad \mathcal{B}_2 := \{A \in \mathcal{F} \mid f^s(A) < g^s(A)\}.$$

Assume that, for any $U \in \mathcal{F}$, there exists $A \in \mathcal{B}_1$ such that $A \subseteq U$. Then one has

$$\begin{aligned} \limsup_{\mathcal{F}}(f \vee g) &= \inf_{U \in \mathcal{F}} (f \vee g)^s(U) = \inf_{A \in \mathcal{B}_1} (f \vee g)^s(U) \\ &= \inf_{A \in \mathcal{B}_1} f^s(A) = \inf_{U \in \mathcal{F}} f^s(U) = \limsup_{\mathcal{F}} f. \end{aligned}$$

Suppose that there exists $W \in \mathcal{F}$ that

$$\forall A \in \mathcal{B}_1, A \not\subseteq W.$$

If $\mathcal{B}_1 = \emptyset$, then $\mathcal{B}_2 = \mathcal{F}$. Otherwise there exists $A \in \mathcal{B}_1$. For any $U \in \mathcal{F}$ such that $U \subseteq W$, one has $U \cap A \notin \mathcal{B}_1$ since $U \cap A \subseteq U \subseteq W$. Thus $U \cap A \in \mathcal{B}_2$ and $U \cap A \subseteq W$. Therefore, in both cases, for any $U \in \mathcal{F}$ such that $U \subseteq W$, U contains an element of $\mathcal{B}_2$. Hence

$$\begin{aligned} \limsup_{\mathcal{F}}(f \vee g) &= \inf_{U \in \mathcal{F}|_W} (f \vee g)^s(U) = \inf_{A \in \mathcal{B}_2} (f \vee g)^s(U) \\ &= \inf_{A \in \mathcal{B}_2} g^s(A) = \inf_{U \in \mathcal{F}|_W} g^s(U) = \limsup_{\mathcal{F}} g. \end{aligned}$$

Thus we conclude that

$$\limsup_{\mathcal{F}}(f \vee g) \leq \max \left\{ \limsup_{\mathcal{F}} f, \limsup_{\mathcal{F}} g \right\}.$$

If we apply (7.4) to $(G, \geq)$, we obtain (7.5). □

7.2.18 Corollary. *We keep the notation and the hypothesis of Proposition 7.2.17. If f and g have limits along $\mathcal{F}$, then $f \vee g$ and $f \wedge g$ also have limits along $\mathcal{F}$. Moreover, one has*

$$\lim_{\mathcal{F}}(f \vee g) = \max \left\{ \lim_{\mathcal{F}} f, \lim_{\mathcal{F}} g \right\}, \tag{7.6}$$

$$\lim_{\mathcal{F}}(f \wedge g) = \min \left\{ \lim_{\mathcal{F}} f, \lim_{\mathcal{F}} g \right\} \tag{7.7}$$

Proof. Since $f \vee g \geq f$ and $f \vee g \geq g$, one has

$$\liminf_{\mathcal{F}}(f \vee g) \geq \liminf_{\mathcal{F}} f = \lim_{\mathcal{F}} f, \quad \liminf_{\mathcal{F}}(f \vee g) \geq \liminf_{\mathcal{F}} g = \lim_{\mathcal{F}} g.$$

Therefore,

$$\liminf_{\mathcal{F}}(f \vee g) \geq \max \left\{ \lim_{\mathcal{F}} f, \lim_{\mathcal{F}} g \right\}.$$

Moreover, by (7.4) one has

$$\limsup_{\mathcal{F}}(f \vee g) = \max \left\{ \limsup_{\mathcal{F}} f, \limsup_{\mathcal{F}} g \right\} = \max \left\{ \lim_{\mathcal{F}} f, \lim_{\mathcal{F}} g \right\}.$$

Therefore, the equalities

$$\liminf_{\mathcal{F}}(f \vee g) = \limsup_{\mathcal{F}}(f \vee g) = \max \left\{ \lim_{\mathcal{F}} f, \lim_{\mathcal{F}} g \right\}$$

hold, as required. $\square$

7.3 Partially ordered semigroups

7.3.1 Definition. Let $(G, *)$ be a semigroup (resp. monoid, group), and let $\leq$ be a partial order on G . If for any elements a, b , and c in G , the inequality $a < b$ implies $a * c < b * c$ and $c * a < c * b$ (note that in this case, $a \leq b$ implies $a * c \leq b * c$ and $c * a \leq c * b$), then we say that $(G, *, \leq)$ is a *partially ordered semigroup* (resp. *partially ordered monoid*, *partially ordered group*); if $\leq$ is a total order, then we say that $(G, *, \leq)$ is a *totally ordered semigroup* (resp. *totally ordered monoid*, *totally ordered group*).

7.3.2 Remark. Let $(G, *, \leq)$ be a partially ordered semigroup (resp. partially ordered monoid, partially ordered group, totally ordered semigroup, totally ordered monoid, totally ordered group). Then $(G, *, \check{\leq})$ and $(G, *, \geq)$ are also partially ordered semigroups (resp. partially ordered monoids, partially ordered groups, totally ordered semigroups, totally ordered monoids, totally ordered groups).

7.3.3 Proposition. Let $(G, *, \leq)$ be a partially ordered semigroup, and let a_1, a_2, b_1 , and b_2 be elements of G . If $a_1 \leq b_1$ and $a_2 \leq b_2$, then $a_1 * a_2 \leq b_1 * b_2$. Moreover, if one of the inequalities $a_1 \leq b_1$ and $a_2 \leq b_2$ is strict, then the strict inequality $a_1 * a_2 < b_1 * b_2$ holds.

Proof. By the condition of the partially ordered semigroup, we have

$$a_1 * a_2 \leq a_1 * b_2 \leq b_1 * b_2.$$

If $a_1 < b_1$ or $a_2 < b_2$ holds, then at least one of the inequalities above is strict. By Corollary 4.6.9, it follows that $a_1 * a_2 < b_1 * b_2$. $\square$

7.3.4 Proposition. *Let $(G, *, \leq)$ be a partially ordered semigroup, and let x, y , and a be elements of G . Suppose that $\leq$ is a total order, or that $(G, *)$ is a monoid and $a \in G^\times$. Then the following conditions are equivalent:*

- (1) $x < y$,
- (2) $x * a < y * a$,
- (3) $a * x < a * y$.

Proof. By the definition of a partially ordered monoid, (1) implies (2) and (3).

Assume (2) holds. If $(G, *)$ is a monoid and a is invertible, then

$$x = (x * a) * \iota(a) < (y * a) * \iota(a) = y.$$

If $\leq$ is a total order, and if (1) does not hold, then $y \leq x$, which implies $y * a \leq x * a$. This contradicts (2). Therefore, (2) implies (1). Similarly, it can be shown that (3) implies (1). $\square$

7.3.5 Proposition. *Let $(G, *, \leq)$ be a partially ordered semigroup, and let x, y , and a be elements of G . Suppose that $\leq$ is a total order, or that $(G, *)$ is a monoid and $a \in G^\times$. Then the following conditions are equivalent:*

- (1) $x \leq y$,
- (2) $x * a \leq y * a$,
- (3) $a * x \leq a * y$.

Proof. By the definition of a partially ordered monoid, (1) implies (2) and (3).

Assume (2) holds. If a is invertible, then

$$x = (x * a) * \iota(a) \leq (y * a) * \iota(a) = y.$$

If $\leq$ is a total order, and if (1) does not hold, then $y < x$, which implies $y * a < x * a$. This contradicts (2). Therefore, (2) implies (1). Similarly, it can be shown that (3) implies (1). $\square$

7.3.6 Corollary. *Let $(G, *, \leq)$ be a totally ordered semigroup. Then the elements of G satisfy the left and right cancellation laws.*

Proof. Let x, y , and a be elements of G . Suppose $x * a = y * a$. Then $x * a \leq y * a$ and $y * a \leq x * a$. By Proposition 7.3.5, we have $x \leq y$ and $y \leq x$, hence $x = y$. Similarly, $a * x = a * y$ implies $x = y$. This shows that a satisfies both the left and right cancellation laws. $\square$

7.3.7 Proposition. *Let $(G, *, \leq)$ be a partially ordered monoid. Then the mapping $\iota : G^\times \rightarrow G^\times$ is strictly decreasing.*

Proof. Let e be the neutral element of $(G, *)$. Let x and y be two elements of $G^\times$ such that $x < y$. By Proposition 7.3.4, we have

$$e = \iota(x) * x < \iota(x) * y,$$

and hence

$$\iota(y) = e * \iota(y) < \iota(x) * y * \iota(y) = \iota(x).$$

□

7.3.8 Example. The set of integers $\mathbb{Z}$ forms a totally ordered group under addition and the usual total order. The set of positive rational numbers $\mathbb{Q}_{>0}$ forms a totally ordered group under multiplication and the usual total order.

7.3.9 Proposition. *Let $(G, *, \leq)$ be a partially ordered group, and let e be the neutral element of $(G, *)$. Suppose $G \neq \{e\}$. Then G has neither a greatest element nor a least element.*

Proof. Suppose $(G, \leq)$ has a greatest element β . We first show that $\beta \neq e$. Let $x \in G \setminus \{e\}$. If $\beta = e$, then one has $x < e$. By Proposition 7.3.7 we obtain $e < \iota(x)$, which contradicts the maximality of e .

One then has $e < \beta$. By Proposition 7.3.3, we have $\beta = e * \beta < \beta * \beta$. This contradicts the assumption that β is the greatest element. Therefore, $(G, \leq)$ does not have a greatest element. If we apply this result to $(G, *, \geq)$, we obtain that $(G, \leq)$ does not have a least element.

□

7.4 Enhancement

7.4.1 Definition. Let $(S, *, \leq)$ be a partially ordered semigroup. Suppose $(S, \leq)$ has neither a greatest element nor a least element. Let $\perp$ and $\top$ be two distinct formal elements not belonging to S . Let $\bar{S} = S \cup \{\top, \perp\}$ and extend the partial order $\leq$ to $\bar{S}$ such that

$$\forall s \in S, \quad \perp < s < \top.$$

We extend the composition law $*$ as a mapping to the set

$$(\bar{S} \times \bar{S}) \setminus \{(\top, \perp), (\perp, \top)\}$$

such that

$$\begin{aligned} \forall x \in S \cup \{\top\}, \quad \top * x &= x * \top = \top, \\ \forall x \in S \cup \{\perp\}, \quad \perp * x &= x * \perp = \perp. \end{aligned}$$

Note that $\perp * \top$ and $\top * \perp$ are undefined. Additionally, if $x * y$ is defined and $\top \in \{x, y\}$, then $x * y = \top$; if $x * y$ is defined and $\perp \in \{x, y\}$, then $x * y = \perp$.

The set $\overline{S}$ together with the extended partial order $\leq$ and the extended mapping $*$ is called the *enhancement* of the partially ordered semigroup $(S, *, \leq)$.

If A and B are subsets of $\overline{S}$, denote by $A * B$ the set

$$\{a * b \mid a \in A, b \in B, a * b \text{ is defined}\}.$$

7.4.2 Remark. Let $(S, *, \leq)$ be a partially ordered semigroup. Let $(a, b) \in \overline{S} \times \overline{S}$. Then $a * b$ is defined if and only if $b * a$ is defined. Moreover, if $(S, *)$ is commutative, then $a * b = b * a$ whenever both are defined.

7.4.3 Definition. Let $(S, *, \leq)$ be a partially ordered semigroup, and let $a_0, \dots, a_n$ be elements of $\overline{S}$, where $n \in \mathbb{N}_{\geq 1}$. If $a_0 * \dots * a_{n-1}$ is defined and $(a_0 * \dots * a_{n-1}) * a_n$ is also defined, then $a_0 \dots a_n$ is defined recursively as

$$(a_0 * \dots * a_{n-1}) * a_n.$$

7.4.4 Proposition. Let $(S, *, \leq)$ be a partially ordered semigroup, $n \in \mathbb{N}_{\geq 1}$, and $a_0, \dots, a_n$ be elements of $\overline{S}$. For any $i \in \{0, \dots, n-1\}$, $a_0 * \dots * a_n$ is defined if and only if

$$a_0 * \dots * a_{i-1} * (a_i * a_{i+1}) * a_{i+2} * \dots * a_n$$

is defined. Moreover, when both are defined, the equality

$$a_0 * \dots * a_n = a_0 * \dots * a_{i-1} * (a_i * a_{i+1}) * a_{i+2} * \dots * a_n$$

holds.

Proof. Both expressions are defined if and only if

$$\{\perp, \top\} \not\subseteq \{a_0, \dots, a_n\}.$$

Thus, the first conclusion holds. Furthermore, if $\perp \in \{a_0, \dots, a_n\}$, then both expressions equal $\perp$; if $\top \in \{a_0, \dots, a_n\}$, then both expressions equal $\top$. Finally, if $\{\perp, \top\} \cap \{a_0, \dots, a_n\} = \emptyset$, the equality

$$a_0 * \dots * a_n = a_0 * \dots * a_{i-1} * (a_i * a_{i+1}) * a_{i+2} * \dots * a_n$$

follows from Proposition 5.1.9. □

7.4.5 Proposition. Let $(S, *, \leq)$ be a partially ordered semigroup, and let a and b be elements of $\overline{S}$. If $a < b$, then for any $c \in S$, we have $a * c < b * c$ and $c * a < c * b$.

Proof. When $\{a, b\} \subseteq S$, the inequalities $a * c < b * c$ and $c * a < c * b$ hold by definition. When $a = \perp$, we have $a * c = c * a = \perp$, and $b * c \neq \perp$, $c * b \neq \perp$, so $a * c < b * c$ and $c * a < c * b$. When $b = \top$ and $a \neq \top$, we have $b * c = c * b = \top$, and $a * c \neq \top$, $c * a \neq \top$. Thus, $a * c < b * c$ and $c * a < c * b$. $\square$

7.4.6 Proposition. *Let $(S, *, \leq)$ be a partially ordered semigroup, and let a, b, x, y be elements of $\overline{S}$.*

- (1) *If $x < a$ and $y < b$, then both $x * y$ and $a * b$ are defined, and $x * y < a * b$.*
- (2) *If $x \leq a$ and $y \leq b$, and both $x * y$ and $a * b$ are defined, then $x * y \leq a * b$.*

Proof. (1) Since $x < a$ and $y < b$, we have $\perp \notin \{a, b\}$ and $\top \notin \{x, y\}$, so both $x * y$ and $a * b$ are defined.

If $\top \in \{a, b\}$, then $a * b = \top$. Since $\top \notin \{x, y\}$, by definition, $x * y < \top$. Similarly, if $\perp \in \{x, y\}$, then $x * y = \perp$, and $\perp < a * b$ holds. Finally, if $\perp \notin \{x, y\}$ and $\top \notin \{a, b\}$, then $\{x, y, a, b\} \subseteq S$, and by Proposition 7.3.3, we have $x * y < a * b$.

- (2) If $\top \in \{x, y\}$, then $\top \in \{a, b\}$, so

$$x * y = a * b = \top.$$

Similarly, if $\perp \in \{a, b\}$, then $\perp \in \{x, y\}$, so

$$x * y = a * b = \perp.$$

If $\perp \in \{x, y\}$, then $x * y = \perp \leq a * b$. If $\top \in \{a, b\}$, then $x * y \leq \top = a * b$.

Finally, if $\{x, y, a, b\} \subseteq S$, then by Proposition 7.3.3, we have $x * y \leq a * b$. $\square$

7.4.7 Proposition. *Let $(S, *, \leq)$ be a partially ordered monoid with neutral element e , and let $(a, b) \in \overline{S} \times \overline{S}$. Suppose $a \in S^\times \cup \{\top, \perp\}$. Then the following conditions are equivalent:*

- (1) $a < b$,
- (2) $\iota(a) * b$ is well defined and $e < \iota(a) * b$,
- (3) $b * \iota(a)$ is well defined and $e < b * \iota(a)$.

Proof. Assume $a < b$. Then a and b cannot both be $\top$ or both be $\perp$. Therefore, $\iota(a) * b$ and $b * \iota(a)$ are defined. When $\{a, b\} \subseteq S$, by Proposition 7.3.4, we have $e < \iota(a) * b$ and $e < b * \iota(a)$. When $a = \perp$ and $b = \top$,

$$\iota(a) * b = b * \iota(a) = \top * \top = \top > e.$$

Assume $\iota(a) * b$ is defined and $e < \iota(a) * b$. When $\{a, b\} \subset S$, by Proposition 7.3.4, we have $a < b$. Otherwise, either $a = \perp$ and $b \neq \perp$, or $a \neq \top$ and $b = \top$. In both cases, $a < b$. Similarly, if $b * \iota(a)$ is defined and $e < b * \iota(a)$, then $a < b$. $\square$

7.4.8 Corollary. *Let $(S, *, \leq)$ be a partially ordered monoid, and let a and b be two elements of $S^\times \cup \{\top, \perp\}$. Then $a < b$ if and only if $\iota(b) < \iota(a)$.*

Proof. We only need to prove the necessity. By Proposition 7.4.7, if $a < b$, then

$$\iota(a) * b = \iota(a) * \iota(\iota(b))$$

is defined and $e < \iota(a) * \iota(\iota(b))$. Again by Proposition 7.4.7, we obtain $\iota(b) < \iota(a)$. $\square$

7.4.9 Definition. Let $(S, *, \leq)$ be a partially ordered monoid. We extend $\iota : S^\times \rightarrow S^\times$ to $S^\times \cup \{\top, \perp\}$ by defining

$$\iota(\top) = \perp, \quad \iota(\perp) = \top.$$

7.5 Extrema in a partially ordered monoid

7.5.1 Lemma. *Let $(S, *, \leq)$ be a partially ordered monoid. Let A be a subset of $\overline{S}$, and let $b \in S^\times \cup \{\perp, \top\}$. Then the following statements hold:*

- (1) *If $\sup(A) * b$ is well defined, then $A * \{b\}$ has a supremum in $\overline{S}$, and*

$$\sup(A * \{b\}) = \sup(A) * b.$$

- (2) *If $\inf(A) * b$ is well defined, then $A * \{b\}$ has an infimum in $\overline{S}$, and*

$$\inf(A * \{b\}) = \inf(A) * b.$$

- (3) *If $b * \sup(A)$ is well defined, then $\{b\} * A$ has a supremum in $\overline{S}$, and*

$$\sup(\{b\} * A) = b * \sup(A).$$

- (4) *If $b * \inf(A)$ is well defined, then $\{b\} * A$ has an infimum in $\overline{S}$, and*

$$\inf(\{b\} * A) = b * \inf(A).$$

Proof. (1) Suppose $b = \perp$. Then $\sup(A) * b = \perp$, and $A * \{b\} \subseteq \{\perp\}$, so

$$\sup(A * \{b\}) = \perp = \sup(A) * b.$$

Suppose $b = \top$. If $A * \{b\}$ is nonempty, then $\top \in A * \{b\}$, so

$$\sup(A * \{b\}) = \top = \sup(A) * b.$$

If $A * \{b\} = \emptyset$, then $A \subseteq \{\perp\}$, so $\sup(A) = \perp$. In this case, $\sup(A) * b$ is undefined.

Now assume $b \in S$. For any $a \in A$, we have $a \leq \sup(A)$, so by Proposition 7.4.6,

$$a * b \leq \sup(A) * b.$$

This shows that $\sup(A) * b$ is an upper bound of $A * \{b\}$.

Let M be an upper bound of $A * \{b\}$. When $b \in S^\times$, by Proposition 7.4.6, for any $a \in A$,

$$a = (a * b) * \iota(b) \leq M * \iota(b).$$

Thus, $M * \iota(b)$ is an upper bound of A , so $\sup(A) \leq M * \iota(b)$. Again by Proposition 7.4.6,

$$\sup(A) * b \leq (M * \iota(b)) * b = M.$$

Hence, $\sup(A) * b$ is the supremum of $A * \{b\}$.

Applying (1) to $(S, *, \geq)$ yields (2). Applying (1) and (2) to $(S, \check{*}, \leq)$ yields (3) and (4), respectively. $\square$

7.5.2 Theorem. *Let $(S, *, \leq)$ be a partially ordered group. Let A and B be subsets of $\overline{S}$.*

(1) *If $\sup(A) * \sup(B)$ is defined, then $A * B$ has a supremum in $\overline{S}$, and*

$$\sup(A) * \sup(B) = \sup(A * B).$$

(2) *If $\inf(A) * \inf(B)$ is defined, then $A * B$ has an infimum in $\overline{S}$, and*

$$\inf(A) * \inf(B) = \inf(A * B).$$

Proof. By Proposition 7.4.6, for any $(a, b) \in A \times B$, if $a * b$ is defined, then

$$a * b \leq \sup(A) * \sup(B).$$

This shows that $\sup(A) * \sup(B)$ is an upper bound of $A * B$. In particular, if $\perp \in \{\sup(A), \sup(B)\}$, then $\sup(A) * \sup(B) = \perp$, which must be the supremum of $A * B$.

Below we assume $\perp \notin \{\sup(A), \sup(B)\}$. In particular $A \setminus \{\perp\}$ and $B \setminus \{\perp\}$ are nonempty.

Suppose $\sup(A) = \top$. Let $b \in A \setminus \{\perp\}$. By Lemma 7.5.1, we have

$$\sup(A * \{b\}) = \sup(A) * b = \top.$$

Thus, $\top = \sup(A) * \sup(B)$ is the only upper bound of $A * B$. Similarly, if $\sup(B) = \top$, then $\top = \sup(A) * \sup(B)$ is the supremum of $A * B$.

Now assume $\top \notin \{\sup(A), \sup(B)\}$. For any $b \in B$, $\sup(A) * b$ is well defined since $\top \notin \{\sup(A), b\}$. By Lemma 7.5.1, $A * \{b\}$ has $\sup(A) * b$ as its supremum. Moreover, by Lemma 7.5.1 the set

$$\{\sup(A)\} * B = \{\sup(A) * b \mid b \in B\} = \{\sup(A * \{b\}) \mid b \in B\}$$

has $\sup(A) * \sup(B)$ as its supremum. Since

$$A * B = \bigcup_{b \in B} A * \{b\},$$

by Proposition 4.3.9, $A * B$ also has $\sup(A) * \sup(B)$ as its supremum.

Applying (1) to $(S, *, \geq)$ yields (2). $\square$

7.5.3 Corollary. *Let $(S, *, \leq)$ be a partially ordered group. Let X be a set and $f : X \rightarrow \bar{S}$ and $g : X \rightarrow \bar{S}$ be two mappings. Let Y be the set of $x \in X$ such that $f(x) * g(x)$ is defined. We denote by $f * g$ the mapping from Y to $\bar{S}$ that sends $x \in Y$ to $f(x) * g(x)$.*

(1) *Suppose that $(\sup f) * (\sup g)$ is defined. Then one has*

$$\sup(f * g) \leq (\sup f) * (\sup g).$$

(2) *Suppose that $(\inf f) * (\inf g)$ is defined. Then one has*

$$\inf(f * g) \geq (\inf f) * (\inf g).$$

Proof. (1) Let A and B be respectively the range of f and g . Let C be the range of $f * g$. By definition, one has $C \subseteq A * B$. By Theorem 7.5.2, one has

$$\sup(f * g) = \sup(C) \subseteq \sup(A * B) = \sup(A) * \sup(B) = (\sup f) * (\sup g).$$

If we apply (1) to $(S, *, \geq)$, we obtain (2). $\square$

7.5.4 Theorem. *Let $(S, *, \leq)$ be a partially ordered group. We suppose that $\bar{S}$ is order complete. Let X be a set and $\mathcal{F}$ be a non-degenerate filter on X . Let $f : X \rightarrow \bar{S}$ and $g : X \rightarrow \bar{S}$ be two mappings. We assume that, there exists $W \in \mathcal{F}$ such that, for any $x \in W$, $f(x) * g(x)$ is defined and we denote by $f * g : W \rightarrow \bar{S}$ the mapping sending $x \in W$ to $f(x) * g(x)$. Each of the following inequalities holds, provided that the term on the right hand side of the inequality is defined:*

$$\limsup_{\mathcal{F}}(f * g) \leq \limsup_{\mathcal{F}} f * \limsup_{\mathcal{F}} g \quad (7.8)$$

$$\limsup_{\mathcal{F}}(f * g) \geq \limsup_{\mathcal{F}} f * \liminf_{\mathcal{F}} g \quad (7.9)$$

$$\liminf_{\mathcal{F}}(f * g) \geq \liminf_{\mathcal{F}} f * \liminf_{\mathcal{F}} g \quad (7.10)$$

$$\liminf_{\mathcal{F}}(f * g) \leq \liminf_{\mathcal{F}} f * \limsup_{\mathcal{F}} g. \quad (7.11)$$

Proof. By considering the restrictions of f and g to W , we may assume without loss of generality that $f * g$ is defined on X .

For any $U \in \mathcal{F}$, by Corollary 7.5.3 one has

$$(f * g)^s(U) \leq f^s(U) * g^s(U)$$

provided that $f^s(U) * g^s(U)$ is defined. By Theorem 7.5.2, if

$$\limsup_{\mathcal{F}} f * \limsup_{\mathcal{F}} g$$

is defined, then

$$\limsup_{\mathcal{F}} (f * g) = \inf_{U \in \mathcal{F}} (f * g)^s(U) \leq \inf_{\substack{U \in \mathcal{F} \\ f^s(U) * g^s(U) \text{ is defined}}} f^s(U) * g^s(U).$$

Let U and V be elements of $\mathcal{F}$ such that $f^s(U) * f^s(V)$ is defined. Since f^s is increasing, if $f^s(U \cap V) * g^s(U \cap V)$ is defined, then the inequalities

$$f^s(U) * f^s(V) \geq f^s(U \cap V) * g^s(U \cap V) \geq \limsup_{\mathcal{F}} (f * g)$$

hold. If $f^s(U \cap V) * g^s(U \cap V)$ is not defined, then $f^s(U) * f^s(V)$ should be the greatest element of $\overline{S}$, and the inequality

$$f^s(U) * f^s(V) \geq \limsup_{\mathcal{F}} (f * g)$$

still holds. By Theorem 7.5.2, we get (7.8).

Let U be an element of $\mathcal{F}$. Suppose that $(f * g)^s(U)$ is not the greatest element of $\overline{S}$. For any $V \in \mathcal{F}$, one has

$$\forall x \in U \cap V, (f * g)^s(U) \geq f(x) * g(x) \geq f(x) * g^i(U \cap V) \geq f(x) * g^i(V).$$

By Lemma 7.5.1, we get

$$(f * g)^s(U) \geq f^s(U \cap V) * g^i(V)$$

provided that $f^s(U \cap V) * g^i(V)$ is defined. This inequality is still true when $(f * g)^s(U)$ is equal to the greatest element of $\overline{S}$.

Taking the infimum with respect to $U \in \mathcal{F}$, by Lemma 7.5.1 and Proposition 7.2.15 we obtain that

$$\limsup_{\mathcal{F}} f * g \geq (\limsup_{\mathcal{F}} f) * g^i(V),$$

provided that the term on the right hand side of the inequality is defined. Now, taking the supremum with respect to $V \in \mathcal{F}$, we obtain (7.9).

If we apply (7.8) and (7.9) to $(S, *, \geq)$, we obtain the last two inequalities. $\square$

7.5.5 Corollary. *Let $(S, *, \leq)$ be a partially ordered group. We suppose that $\bar{S}$ is order complete. Let X be a set and $\mathcal{F}$ be a non-degenerate filter on X . Let $f : X \rightarrow \bar{S}$ and $g : X \rightarrow \bar{S}$ be two mappings. We assume that, there exists $W \in \mathcal{F}$ such that, for any $x \in W$, $f(x) * g(x)$ is defined and we denote by $f * g : W \rightarrow \bar{S}$ the mapping sending $x \in W$ to $f(x) * g(x)$. Suppose in addition that g has a limit along $\mathcal{F}$.*

(1) *One has*

$$\limsup_{\mathcal{F}}(f * g) = \limsup_{\mathcal{F}} f * \lim_{\mathcal{F}} g,$$

provided that the term on the right hand side of the equality is defined.

(2) *One has*

$$\liminf_{\mathcal{F}}(f * g) = \liminf_{\mathcal{F}} f * \lim_{\mathcal{F}} g,$$

provided that the term on the right hand side of the equality is defined.

(3) *If f has a limit along $\mathcal{F}$ and*

$$\lim_{\mathcal{F}} f * \lim_{\mathcal{F}} g$$

*is defined, then $f * g$ has a limit along $\mathcal{F}$, and*

$$\lim_{\mathcal{F}}(f * g) = \lim_{\mathcal{F}} f * \lim_{\mathcal{F}} g.$$

Proof. (1) By Theorem 7.5.4, one has

$$\limsup_{\mathcal{F}}(f * g) \leq \limsup_{\mathcal{F}} f * \limsup_{\mathcal{F}} g = \limsup_{\mathcal{F}} f * \lim_{\mathcal{F}} g$$

and

$$\limsup_{\mathcal{F}}(f * g) \geq \limsup_{\mathcal{F}} f * \liminf_{\mathcal{F}} g = \limsup_{\mathcal{F}} f * \lim_{\mathcal{F}} g.$$

Hence the equality holds.

(2) follows from (1) by reversing the partial order $\leq$.

(3) By (1) and (2), one has

$$\limsup_{\mathcal{F}}(f * g) = \liminf_{\mathcal{F}}(f * g) = \lim_{\mathcal{F}} f * \lim_{\mathcal{F}} g.$$

□

7.5.6 Proposition. *Let $(S, *, \leq)$ be a partially ordered group and A be a subset of $\bar{S}$. Let*

$$\iota(A) := \{\iota(a) \mid a \in A\}.$$

(1) If A admits a supremum, then $\iota(A)$ admits an infimum. Moreover, one has

$$\inf(\iota(A)) = \iota(\sup(A)).$$

(2) If A admits an infimum, then $\iota(A)$ admits a supremum. Moreover, one has

$$\sup(\iota(A)) = \iota(\inf(A)).$$

Proof. (1) For any $a \in A$ one has $a \leq \sup(A)$ and hence, by Proposition 7.3.7, one has

$$\iota(\sup(A)) \leq \iota(a),$$

which means that $\iota(\sup(A))$ is a lower bound of $\iota(A)$.

Let m be a lower bound of $\iota(A)$. For any $a \in A$ one has $m \leq \iota(a)$ and hence, by Proposition 7.3.7, one has

$$a = \iota(\iota(a)) \leq \iota(m).$$

Therefore, $\iota(m)$ is an upper bound of A , which leads to $\sup(A) \leq \iota(m)$. By Proposition 7.3.7, we obtain $m \leq \iota(\sup(A))$. Thus $\iota(\sup(A))$ is the infimum of $\iota(A)$.

Applying (1) to $(S, *, \geq)$, we obtain (2). $\square$

7.5.7 Corollary. Let $(S, *, \leq)$ be a partially ordered group, $f : X \rightarrow \overline{S}$ be a mapping and $\mathcal{F}$ be a non-degenerate filter on X . One has

$$\limsup_{\mathcal{F}} \iota(f) = \iota\left(\liminf_{\mathcal{F}} f\right), \quad \liminf_{\mathcal{F}} \iota(f) = \iota\left(\limsup_{\mathcal{F}} f\right),$$

where $\iota(f)$ denotes the mapping from X to $\overline{S}$ that sends $x \in X$ to $\iota(f(x))$.

Proof. For any $U \in \mathcal{F}$, one has

$$\sup_{x \in U} \iota(f(x)) = \iota\left(\inf_{x \in U} f(x)\right),$$

namely $\iota(f)^s(U) = \iota(f^i(U))$. By Proposition 7.5.6, one has

$$\limsup_{\mathcal{F}} \iota(f) = \inf_{U \in \mathcal{F}} \iota(f)^s(U) = \inf_{U \in \mathcal{F}} \iota(f^i(U)) = \iota\left(\sup_{U \in \mathcal{F}} f^i(U)\right) = \iota\left(\liminf_{\mathcal{F}} f\right).$$

Applying the first equality to $(S, *, \geq)$, we obtain the second equality. $\square$

7.6 Absolute value

In this section, we fix a totally ordered Abelian group $(R, +, \leq)$. Note that, for any $a \in R$, either $a = 0$, or $a > 0 > -a$, or $-a > 0 > a$.

We add two formal element $-\infty$ and $+\infty$ to R to form the enhancement of the totally ordered Abelian group $(R, +, \leq)$. More precisely, we let $\overline{R} = R \cup \{-\infty, +\infty\}$ and extend the total order $\leq$ to $\overline{R}$ by setting

$$\forall x \in R, -\infty < x < +\infty.$$

As in the previous section, we extend partially the composition law $+$ to $\overline{R}$ so that

$$\begin{aligned} \forall x \in R, (+\infty) + x = x + (+\infty) = +\infty, (-\infty) + x = x + (-\infty) = -\infty, \\ (+\infty) + (+\infty) = +\infty, (-\infty) + (-\infty) = -\infty. \end{aligned}$$

We also define by convention that

$$-(+\infty) := -\infty, -(-\infty) := +\infty.$$

7.6.1 Definition. For any $a \in \overline{R}$, we define the *absolute value* of a as

$$|a| := \max\{a, -a\}.$$

By definition, one has $|-a| = |a|$.

7.6.2 Proposition. For any $(a, b) \in \overline{R} \times \overline{R}$, if $a + b$ is defined, then one has

$$|a + b| \leq |a| + |b|.$$

Proof. If $\{+\infty, -\infty\} \cap \{a, b\} \neq \emptyset$, then one has $|a| + |b| = +\infty$ and the inequality is true.

Suppose that $\{a, b\} \subseteq R$. By definition, one has $a \leq |a|$ and $b \leq |b|$ and hence

$$a + b \leq |a| + |b|.$$

Similarly, one has $-a \leq |a|$ and $-b \leq |b|$ and hence

$$-(a + b) = (-a) + (-b) \leq |a| + |b|.$$

Therefore,

$$|a + b| \leq |a| + |b|.$$

□

7.6.3 Corollary. *For any $(a, b) \in \overline{R} \times \overline{R}$, one has*

$$||a| - |b|| \leq |a - b|$$

provided that the terms on both sides are defined.

Proof. If $\{-\infty, +\infty\} \cap \{a, b\} \neq \emptyset$, then $|a - b| = +\infty$ and hence the inequality is true.

In the following, we suppose that $\{a, b\} \subseteq R$. One has

$$|a| = |b + (a - b)| \leq |b| + |a - b|,$$

so that

$$|a| - |b| \leq |a - b|.$$

Similarly,

$$|b| = |-b| = |(a - b) + (-a)| \leq |a - b| + |-a| = |a - b| + |a|,$$

so that

$$|b| - |a| \leq |a - b|.$$

Therefore,

$$||a| - |b|| \leq |a - b|.$$

□

7.6.4 Theorem. *Let X be a set, $\mathcal{F}$ be a non-degenerate filter on X , $f : X \rightarrow R$ be a mapping and $\ell \in R$. The following statements are equivalent:*

(1) *The mapping admits ℓ as its limit along $\mathcal{F}$.*

(2)

$$\limsup_{\mathcal{F}} |f - \ell| = 0,$$

where $|f - \ell|$ denotes the mapping from X to R that sends $x \in X$ to $|f(x) - \ell|$.

Moreover, they imply the following condition

(3) *For any $\varepsilon \in R_{>0}$, there exists $U \in \mathcal{F}$ such that*

$$\forall x \in U, |f(x) - \ell| < \varepsilon.$$

The converse is true when $\inf R_{>0} = 0$.

Proof. By Corollaries 7.5.7 and 7.5.5, one has

$$\begin{aligned}\limsup_{\mathcal{F}}(f - \ell) &= \limsup_{\mathcal{F}} f - \ell, \\ \limsup_{\mathcal{F}}(\ell - f) &= \ell + \limsup_{\mathcal{F}}(-f) = \ell - \liminf_{\mathcal{F}} f.\end{aligned}$$

“(1) $\implies$ (2)”: If

$$\limsup_{\mathcal{F}} f = \liminf_{\mathcal{F}} f = \ell,$$

then one has

$$\limsup_{\mathcal{F}}(f - \ell) = \limsup_{\mathcal{F}}(\ell - f) = 0.$$

By Proposition 7.2.17, one has

$$\begin{aligned}\limsup_{\mathcal{F}} |f - \ell| &= \limsup_{\mathcal{F}} \max(f - \ell, \ell - f) \\ &= \max \left\{ \limsup_{\mathcal{F}}(f - \ell), \limsup_{\mathcal{F}}(\ell - f) \right\} = 0.\end{aligned}$$

“(2) $\implies$ (1)”: Since

$$\begin{aligned}\limsup_{\mathcal{F}} |f - \ell| &= \max \left\{ \limsup_{\mathcal{F}}(f - \ell), \limsup_{\mathcal{F}}(\ell - f) \right\}, \\ &= \max \left\{ \limsup_{\mathcal{F}} f - \ell, \ell - \liminf_{\mathcal{F}} f \right\},\end{aligned}$$

if

$$\limsup_{\mathcal{F}} |f - \ell| = 0,$$

then

$$\limsup_{\mathcal{F}} f - \ell \leq 0, \quad \ell - \liminf_{\mathcal{F}} f \leq 0,$$

which leads to

$$\ell \leq \liminf_{\mathcal{F}} f \leq \limsup_{\mathcal{F}} f \leq \ell.$$

Hence

$$\lim_{\mathcal{F}} f = \ell.$$

“(2) $\implies$ (3)”: One can rewrite

$$\limsup_{\mathcal{F}} |f - \ell| = 0$$

as

$$\inf_{U \in \mathcal{F}} \sup_{x \in U} |f(x) - \ell| = 0.$$

Therefore, for any $\varepsilon \in R_{>0}$, ε is not a lower bound of

$$\left\{ \sup_{x \in U} |f(x) - \ell| \mid U \in \mathcal{F} \right\}.$$

Thus there exists $U \in \mathcal{F}$ such that

$$\sup_{x \in U} |f(x) - \ell| < \varepsilon.$$

In particular, for any $x \in U$, one has

$$|f(x) - \ell| < \varepsilon.$$

“(3) $\implies$ (2)”: By Proposition 7.2.8, for any $\varepsilon \in R_{>0}$, one has

$$0 \leq \limsup_{\mathcal{F}} |f - \ell| \leq \varepsilon.$$

Since $\inf R_{>0} = 0$, we obtain that

$$0 \leq \limsup_{\mathcal{F}} |f - \ell| \leq 0,$$

namely

$$\limsup_{\mathcal{F}} |f - \ell| = 0.$$

□

7.6.5 Remark. With the notation and the assumptions of the above theorem, one has

$$0 \leq \liminf_{\mathcal{F}} |f - \ell| \leq \limsup_{\mathcal{F}} |f - \ell|.$$

Therefore, the condition

$$\limsup_{\mathcal{F}} |f - \ell| = 0$$

is also equivalent to

$$\lim_{\mathcal{F}} |f - \ell| = 0.$$

7.6.6 Corollary. *Let X be a set, $\mathcal{F}$ be a non-degenerate filter on X , $f : X \rightarrow R$ be a mapping. Assume that f has a limit along $\mathcal{F}$ and that $\lim_{\mathcal{F}} f \in R$. There exists $M \in \mathbb{R}_{\geq 0}$ and $U \in \mathcal{F}$ such that*

$$\forall x \in U, |f(x)| \leq M.$$

Proof. Let $\ell = \lim_{\mathcal{F}} f \in R$. By Theorem 7.6.4, one has

$$\limsup_{\mathcal{F}} |f - \ell| = \inf_{U \in \mathcal{F}} \sup_{x \in U} |f(x) - \ell| = 0.$$

In particular, there exists $U \in \mathcal{F}$ such that

$$\sup_{x \in U} |f(x) - \ell| \neq +\infty.$$

Let

$$M = \sup_{x \in U} |f(x) - \ell| + |\ell|.$$

For any $x \in U$, one has

$$|f(x)| = |f(x) - \ell + \ell| \leq |f(x) - \ell| + |\ell| = M.$$

□

7.7 Ordered rings

7.7.1 Definition. Let $(R, +, \cdot)$ be a unitary ring with zero element 0, and let $\leq$ be a partial order on R . If $(R, +, \cdot)$ is a partially ordered group, and

$$\forall (a, b) \in R_{>0} \times R_{>0}, ab > 0$$

then $(R, +, \cdot, \leq)$ is called a *partially ordered unitary ring*. If there is no ambiguity regarding the composition laws of R , we may simply say that $(R, \leq)$ is a partially ordered unitary ring.

7.7.2 Proposition. Let $(R, \leq)$ be a partially ordered unitary ring, and let a and b be elements of R .

(1) If $a \geq 0$ and $b \geq 0$, then $ab \geq 0$.

(2) $a < b$ if and only if $-b < -a$.

(3) $a \leq b$ if and only if $-b \leq -a$.

Proof. (1) If either a or b is 0, then by (1) of Proposition 6.1.4, $ab = 0$. Otherwise, by definition, $ab > 0$.

(2) and (3) follow from Proposition 7.3.4 and Proposition 7.3.5, respectively, applied to the additive group $(R, +)$. □

7.7.3 Proposition. Let $(R, \leq)$ be a partially ordered unitary ring, and let λ , a , and b be elements of R .

(1) If $\lambda > 0$ and $a > b$, then

$$\lambda a > \lambda b, \quad a\lambda > b\lambda, \quad (-\lambda)a < (-\lambda)b, \quad a(-\lambda) < b(-\lambda).$$

(2) If $\lambda \geq 0$ and $a \geq b$, then

$$\lambda a \geq \lambda b, \quad a\lambda \geq b\lambda, \quad (-\lambda)a \leq (-\lambda)b, \quad a(-\lambda) \leq b(-\lambda).$$

Proof. (1) and (2) follow from Proposition 7.3.4 and Proposition 7.3.5 (applied to the additive group), respectively, along with the identities

$$\begin{aligned} \lambda a - \lambda b &= (-\lambda)b - (-\lambda)a = \lambda(a - b), \\ a\lambda - b\lambda &= b(-\lambda) - a(-\lambda) = (a - b)\lambda. \end{aligned}$$

These identities rely on distributivity and Proposition 6.1.4. $\square$

7.7.4 Proposition. Let $(R, \leq)$ be a totally ordered unitary ring, and $a \in R$. If $a \neq 0$, then $a^2 > 0$.

Proof. Since R is totally ordered, either $a > 0$ or $a < 0$. If $a > 0$, then by definition $a^2 > 0$. If $a < 0$, then by Proposition 7.3.7 one has $-a > 0$. Therefore, by Proposition 6.1.4 we get

$$a^2 = (-a)(-a) > 0.$$

$\square$

7.7.5 Corollary. Let $(R, \leq)$ be a totally ordered unitary ring, and let $a \in R^\times$. If $a > 0$ (resp. $a < 0$), then $a^{-1} > 0$ (resp. $a^{-1} < 0$).

Proof. We prove the contraposition of the statement. By Proposition 7.7.4, one has $a^2 > 0$. If $a^{-1} \geq 0$, by Proposition 7.7.2, we get $a = a^{-1} \cdot a^2 \geq 0$. If $a^{-1} \leq 0$, then by Proposition 7.7.2 one has $-a^{-1} \geq 0$. Hence, by Proposition 6.1.4, $-a = (-a^{-1}) \cdot a^2 \geq 0$, which leads to $a \leq 0$. $\square$

7.7.6 Proposition. Let $(R, \leq)$ be a nonzero totally ordered unitary ring. Consider the usual order on the ring of integers $\mathbb{Z}$. Then the unique unitary ring homomorphism from $\mathbb{Z}$ to R is strictly increasing.

Proof. Let 0_R and 1_R denote the zero and unit elements of R , respectively. Since R is not the zero ring, $0_R \neq 1_R$. Because $\leq$ is a total order, either $1_R > 0_R$ or $1_R < 0_R$. If $1_R < 0_R$, then by Proposition 7.7.2, we get $-1_R > 0_R$. Consequently,

$$1_R = (-1_R)(-1_R) > 0_R,$$

which leads to a contradiction. Thus, $1_R > 0_R$.

Let $f : \mathbb{Z} \rightarrow R$ be the unitary ring homomorphism sending $1 \in \mathbb{Z}$ to 1_R . We prove by induction that for any $n \in \mathbb{Z}_{>0}$, $f(n) > 0_R$. First, $f(1) = 1_R > 0_R$. If $f(n) > 0_R$, then we have

$$f(n+1) = f(n) + 1_R > 0_R.$$

By induction, $f(n) > 0_R$ holds for all $n \in \mathbb{N}_{\geq 1}$.

Let a and b be integers such that $a < b$. Then $b - a \in \mathbb{N}_{\geq 1}$, so

$$f(b) - f(a) = f(b - a) > 0_R.$$

By Proposition 7.3.4 (for the additive group), we get $f(a) < f(b)$. □

7.7.7 Proposition. *Let $(R, \leq)$ be a non-zero totally ordered unitary ring, and let $f : \mathbb{Q} \rightarrow R$ be a unitary ring homomorphism. Then f is strictly increasing.*

Proof. If (a, b) and (c, d) are elements of $\mathbb{Z} \times \mathbb{N}_{>0}$ such that $a/b > c/d$, then $ad > bc$. By Proposition 7.7.6, we have

$$f(a)f(d) > f(b)f(c)$$

since the restriction of f to $\mathbb{Z}$ preserves the orders and multiplicative laws, and $f(n) > 0$ for $n > 0$. Then, by Proposition 7.3.4, we get

$$g(a/b) = f(b)^{-1}f(a) > f(d)^{-1}f(c) = g(c/d).$$

Therefore, g is strictly increasing. □

7.7.8 Proposition. *Let $(R, \leq)$ be a totally ordered unitary ring. For any $(a, b) \in R \times R$, one has*

$$|ab| = |a| \cdot |b|.$$

Proof. Since $|a| \in \{a, -a\}$ and $|b| \in \{b, -b\}$, one has $|a| \cdot |b| \in \{ab, -ab\}$. By (1) of Proposition 7.7.2, one has $|a| \cdot |b| \geq 0$. Hence $|a| \cdot |b| = |ab|$. □

7.8 Enhancement of a totally ordered ring

7.8.1 Definition. Let $(R, \leq)$ be a totally ordered non-zero unitary ring. We adjoin to two distinct elements $-\infty$ and $+\infty$, not being elements of R , to form a new set $\overline{R}$. We extend the total order $\leq$ to $\overline{R}$ such that

$$\forall x \in R, \quad -\infty < x < +\infty.$$

We also extend the additional law of R to a mapping

$$\overline{R} \times \overline{R} \setminus \{(-\infty, +\infty), (+\infty, -\infty)\} \longrightarrow \overline{R}$$

such that

$$\begin{aligned} \forall x \in R \cup \{+\infty\}, \quad x + (+\infty) &= (+\infty) + x = +\infty, \\ \forall x \in R \cup \{-\infty\}, \quad x + (-\infty) &= (-\infty) + x = -\infty, \end{aligned}$$

Note that $\overline{R}$, equipped with the total order $\leq$ and the extended mapping of the additive law, becomes an enhancement of the totally ordered group $(R, +)$.

We also extend the multiplicative law of R to

$$(\overline{R} \times \overline{R}) \setminus \{(0, +\infty), (0, -\infty), (+\infty, 0), (-\infty, 0)\} \longrightarrow \overline{R}$$

such that

$$\begin{aligned} \forall a \in \overline{R}_{>0}, \quad a(+\infty) &= (+\infty)a = +\infty, \quad a(-\infty) = (-\infty)a = -\infty, \\ \forall a \in \overline{R}_{<0}, \quad a(+\infty) &= (+\infty)a = -\infty, \quad a(-\infty) = (-\infty)a = +\infty. \end{aligned}$$

The set $\overline{R}$, together with the total order $\leq$ and the extensions of the additive and multiplicative laws, is called an *enhancement of the totally ordered unitary ring* $(R, \leq)$.

7.8.2 Remark. The set $R_{>0}$ forms a subsemigroup of $(R, \cdot)$, and $(R_{>0}, \leq)$ forms a totally ordered semigroup. Note that $R_{>0} \cup \{0, +\infty\}$, equipped with $\leq$ and the extension of the multiplicative law, forms an enhancement of the totally ordered semigroup $(R_{>0}, \leq)$.

7.8.3 Proposition. *Let R be a totally ordered unitary ring, and let a and b be elements of $\overline{R}$.*

- (1) *If $a < b$, then for any $c \in R$, $a + c < b + c$.*
- (2) *$a < b$ if and only if $b - a$ is defined and $b - a > 0$.*
- (3) *$a < b$ if and only if $-b < -a$.*

Proof. (1) follows from Proposition 7.4.5, (2) follows from Proposition 7.4.7, (3) follows from Corollary 7.4.8. $\square$

7.8.4 Proposition. *Let $(R, \leq)$ be a totally ordered unitary ring, and let a, b, x, y be elements of $\overline{A}$.*

- (1) *If $x < a$ and $y < b$, then both $x+y$ and $a+b$ are well defined, and $x+y < a+b$.*
- (2) *If $x \leq a$ and $y \leq b$, and both $x+y$ and $a+b$ are well defined, then $x+y \leq a+b$.*

Proof. This is a corollary of Proposition 7.4.6. $\square$

7.8.5 Proposition. *Let R be a totally ordered unitary ring, and let a, b, x, y be elements of $R_{>0} \cup \{0, +\infty\}$.*

- (1) *If $x < a$ and $y < b$, then both xy and ab are well defined, and $xy < ab$.*
- (2) *If $x \leq a$ and $y \leq b$, and both xy and ab are well defined, then $xy \leq ab$.*

Proof. This is a corollary of Proposition 7.4.6. $\square$

7.8.6 Proposition. *Let $(R, \leq)$ be a totally ordered unitary ring, n be a positive integer, and $a, b_1, \dots, b_n$ be elements of $\overline{R}$. If*

$$ab_1 + \dots + ab_n$$

is well defined, then $a(b_1 + \dots + b_n)$ is also well defined, and the equality

$$a(b_1 + \dots + b_n) = ab_1 + \dots + ab_n \quad (7.12)$$

holds. Similarly, if

$$b_1a + \dots + b_na$$

is defined, then $(b_1 + \dots + b_n)a$ is also defined, and the equality

$$(b_1 + \dots + b_n)a = b_1a + \dots + b_na \quad (7.13)$$

holds.

Proof. We only prove the first statement since the second follows from the first one by passing to the opposite ring of R .

When $a = 0$, $b_1, \dots, b_n$ are all elements of R , so $a(b_1 + \dots + b_n)$ is defined and (7.12) follows from the distributivity of the multiplication with respect to the addition. When $a \in \{+\infty, -\infty\}$, either all $b_1, \dots, b_n$ are > 0 or all of them are < 0 since otherwise $ab_1 + \dots + ab_n$ is undefined. If $b_1 > 0, \dots, b_n > 0$, then by Proposition 7.8.4, $b_1 + \dots + b_n > 0$, so

$$a(b_1 + \dots + b_n) = a(b_1 + \dots + b_n) = +\infty.$$

Similarly, if $b_1 < 0, \dots, b_n < 0$, then $b_1 + \dots + b_n < 0$, and (7.12) still holds.

Assume $a \in R \setminus \{0\}$. Then

$$\{+\infty, -\infty\} \not\subseteq \{b_1, \dots, b_n\},$$

otherwise $ab_1 + \cdots + ab_n$ is undefined. If $\{b_1, \dots, b_n\} \subseteq A$, then (7.12) follows from the distributivity of the multiplication with respect to the addition. If $+\infty \in \{b_1, \dots, b_n\}$, then $b_1 + \cdots + b_n = +\infty$, and thus

$$a(b_1 + \cdots + b_n) = ab_1 + \cdots + ab_n = a(+\infty).$$

Similarly, if $-\infty \in \{b_1, \dots, b_n\}$, then

$$a(b_1 + \cdots + b_n) = ab_1 + \cdots + ab_n = a(-\infty).$$

Thus, the proposition holds. □

7.8.7 Proposition. *Let R be a totally ordered unitary ring, and let a and b be elements of $\overline{R}$.*

(1) *Suppose $a < b$. Then for any $\lambda \in R_{>0}$,*

$$\lambda a < \lambda b, \quad (-\lambda)a > (-\lambda)b, \quad a\lambda < b\lambda, \quad a(-\lambda) > b(-\lambda).$$

(2) *Suppose $a \leq b$. Then for any $\lambda \in R_{>0} \cup \{0, +\infty\}$, the following inequalities hold provided the expressions on both sides are well defined:*

$$\lambda a \leq \lambda b, \quad (-\lambda)a \geq (-\lambda)b, \quad a\lambda \leq b\lambda, \quad a(-\lambda) \geq b(-\lambda).$$

Proof. (1) Since $a < b$, Proposition 7.8.3 implies that $b - a$ is defined and $0 < b - a$. Then, by Proposition 7.8.5, $0 < \lambda(b - a)$ and $0 < (b - a)\lambda$. If $\{\lambda b, -\lambda a\} = \{+\infty, -\infty\}$, then $\{b, a\} = \{+\infty, -\infty\}$, which contradicts the assumption that $b - a$ is defined. Thus, $\lambda b - \lambda a$ is defined, and Proposition 7.8.6 implies

$$\lambda b - \lambda a = \lambda(b - a) > 0,$$

so by Proposition 7.8.5, $\lambda a < \lambda b$. By passing to the opposite unitary ring, we obtain $a\lambda < b\lambda$.

(2) When $\lambda = 0$, if λa and λb are well defined, then

$$\lambda a = \lambda b = 0, \quad (-\lambda)a = (-\lambda)b = 0.$$

Similarly, if $a\lambda$ and $b\lambda$ are well defined, then

$$a\lambda = b\lambda = 0, \quad a(-\lambda) = b(-\lambda) = 0.$$

When $\lambda = +\infty$, a and b cannot be 0. Then

$$\lambda a = \lambda b = a\lambda = b\lambda = +\infty, \quad (-\lambda)a = (-\lambda)b = a(-\lambda) = b(-\lambda) = -\infty.$$

Finally, the case where $\lambda \in R_{>0}$ follows from (1). □

7.8.8 Proposition. *Let R be a nonzero totally ordered unitary ring, and let 1_R be its unity. Suppose there exists $n \in \mathbb{N}_{\geq 1}$ such that $n1_R \in R^\times$. Then $\overline{R}$ is a dense totally ordered set.*

Proof. Let 0_R denote the zero element of R . By Proposition 7.7.6, $0_R < 1_R$.

Let $(x, y) \in \overline{R} \times \overline{R}$ such that $x < y$. If $x = -\infty$ and $y = +\infty$, then $x < 0_R < y$. If $x = -\infty$ and $y \in R$, then $x < y - 1_R < y$. If $x \in R$ and $y = +\infty$, then $x < x + 1_R < y$.

Now assume $\{x, y\} \subseteq R$. Since $x < y$, we have $y - x > 0$. Let $b = (n1_R)^{-1}$. Note that

$$nb = \underbrace{b + \cdots + b}_{n \text{ copies}} = \underbrace{1_R b + \cdots + 1_R b}_{n \text{ copies}} = (n1_R)b = 1_R.$$

By Corollary 7.7.5, $b > 0$. Thus, $b(y - x) > 0$. Therefore,

$$x < x + b(y - x) = y - (1_R - b)(y - x).$$

Note that

$$(1_R - b) = nb - b = (n - 1)b > 0.$$

Thus, $y - (1_R - b)(y - x) < y$. The proposition is thus proved. $\square$

7.8.9 Theorem. *Let X be a set, $\mathcal{F}$ be a non-degenerate filter on X , $f : X \rightarrow R$ and $g : X \rightarrow R$ be mappings. Assume that f and g have limits along $\mathcal{F}$ and that*

$$a := \lim_{\mathcal{F}} f \quad \text{and} \quad b := \lim_{\mathcal{F}} g$$

belong to R . Let $fg : X \rightarrow R$ be the mapping that sends $x \in X$ to $f(x)g(x)$. Then fg admits ab as its limit along $\mathcal{F}$.

Proof. By Corollary 7.6.6, there exists $U \in \mathcal{F}$ and $M \in R_{\geq 0}$ such that

$$\forall x \in U, |f(x)| \leq M.$$

Then, for any $x \in U$ one has

$$\begin{aligned} |f(x)g(x) - ab| &= |f(x)g(x) - f(x)b + f(x)b - ab| \\ &\leq |f(x)(g(x) - b)| + |(f(x) - a)b| \\ &= |f(x)| \cdot |g(x) - b| + |f(x) - a| \cdot |b| \\ &\leq M \cdot |g(x) - b| + |f(x) - a| \cdot |b|. \end{aligned}$$

By Theorem 7.5.4 and Corollary 7.5.5 (applied to the multiplication), we obtain

$$\begin{aligned} \limsup_{\mathcal{F}} |fg - ab| &\leq \limsup_{\mathcal{F}} M \cdot |g(x) - b| + \limsup_{\mathcal{F}} |f(x) - a| \cdot |b| \\ &= M \cdot \limsup_{\mathcal{F}} |g(x) - b| + |b| \cdot \limsup_{\mathcal{F}} |f(x) - a| = 0. \end{aligned}$$

$\square$

Exercises

1. State whether the following sequences $(x_n)_{n \in \mathbb{N}}$ converge or not. If yes, please find the sequential limit and if not, please explain why.

- (1) $x_n = (-1)^n$;
- (2) $x_n = (-1)^n \frac{1}{n}$;
- (3) $x_n = \sin n$;
- (4) $x_n = r^n$, where $|r| < 1$;
- (5) $x_n = \sum_{k=1}^n r^k$, where $|r| < 1$.

2. Let $a > 0$, $b > 1$ be two real numbers.

- (1) Prove $\lim_{n \rightarrow \infty} \frac{1}{b^n} = 0$.
- (2) Prove that there exists an $n_0 \in \mathbb{N}$, such that for all $n > n_0$, $\frac{a}{n} \leq \frac{1}{2}$.
- (3) Prove $\lim_{n \rightarrow \infty} \frac{a^n}{n!} = 0$ by the facts above.

3. Prove the following statements about limits.

- (1) $\lim_{n \rightarrow +\infty} \frac{n!}{n^n} = 0$;
- (2) $\lim_{n \rightarrow +\infty} \sqrt[n]{a} = 1$ ($a > 0$);
- (3) $\lim_{n \rightarrow +\infty} \frac{n^k}{a^n} = 0$ ($a > 0$, $k \in \mathbb{N}$);
- (4) $\lim_{n \rightarrow +\infty} \sqrt[n]{n^k} = 1$ ($k \in \mathbb{N}$);
- (5) $\lim_{n \rightarrow +\infty} \left(\frac{1}{n^2} + \frac{1}{(n+1)^2} + \cdots + \frac{1}{(2n)^2} \right) = 0$;
- (6) $\lim_{n \rightarrow +\infty} \left(\frac{1}{n^2} + \frac{2}{n^2} + \cdots + \frac{n}{n^2} \right) = \frac{1}{2}$;
- (7) $\lim_{n \rightarrow +\infty} \left(\frac{1}{2} + \frac{2}{2^2} + \cdots + \frac{n}{2^n} \right) = 2$;
- (8) $\lim_{n \rightarrow +\infty} \frac{3}{2} \cdot \frac{5}{4} \cdots \frac{2^{2^n} + 1}{2^{2^n}} = 2$;
- (9) $\lim_{n \rightarrow +\infty} \left(\frac{1}{2} + \frac{3}{2^2} + \cdots + \frac{2n-1}{2^n} \right) = 3$.
- (10) $\lim_{n \rightarrow \infty} \frac{1}{\sqrt[n]{n!}} = 0$.

$$(11) \lim_{n \rightarrow \infty} ((n+1)^a - n^a) = 0, \text{ where } a < 1.$$

4. Prove the limit by following steps.

$$(1) \text{ For } n \in \mathbb{N}^*, \text{ prove } \frac{1}{2} \cdot \frac{3}{4} \cdots \frac{2n-1}{2n} < \frac{1}{\sqrt{2n+1}}.$$

$$(2) \text{ Prove } \lim_{n \rightarrow \infty} \frac{1}{2} \cdot \frac{3}{4} \cdots \frac{2n-1}{2n} = 0.$$

5. Find or prove the following limits.

$$(1) \text{ Prove } \lim_{n \rightarrow \infty} \sqrt[n]{n} = 1.$$

Hint: Prove $\sqrt[n]{n} \leq 1 + \frac{2}{\sqrt{n}}$.

$$(2) \text{ Find } \lim_{n \rightarrow \infty} \sqrt[n]{1 + \frac{1}{2} + \cdots + \frac{1}{n}}.$$

6. Calculate the following limits.

$$(1) \lim_{n \rightarrow \infty} \frac{a_m n^m + \cdots + a_0}{b_l n^l + \cdots + b_0}, \text{ where } a_0, \dots, a_m, b_0, \dots, b_l \in \mathbb{R}, \text{ and } a_m, b_n \text{ are non-zero. Discuss for the cases of } m > l, m = l \text{ and } m < l.$$

Hint: Begin this subject from the study of $\lim_{n \rightarrow \infty} \frac{1}{n}$.

$$(2) \lim_{n \rightarrow +\infty} (\sqrt{n+1} - \sqrt{n})\sqrt{n};$$

$$(3) \lim_{n \rightarrow +\infty} \frac{\sqrt[3]{n^2+1}}{n+2};$$

$$(4) \lim_{n \rightarrow +\infty} \left(\frac{1}{1 \cdot 3} + \frac{1}{2 \cdot 4} + \cdots + \frac{1}{n(n+2)} \right);$$

$$(5) \lim_{n \rightarrow +\infty} \sum_{k=1}^n \frac{1}{k(k+1)(k+2)}.$$

7. Prove the following statements about limits using the squeeze theorem.

$$(1) \lim_{n \rightarrow +\infty} \left[\frac{1}{(n+1)^2} + \frac{1}{(n+2)^2} + \cdots + \frac{1}{(2n)^2} \right] = 0;$$

$$(2) \lim_{n \rightarrow +\infty} \sqrt[n]{a_1^n + a_2^n + \cdots + a_k^n} = \max_{1 \leq i \leq k} \{a_i\}, (a_i \geq 0, i = 1, 2, \dots, k);$$

$$(3) \lim_{n \rightarrow +\infty} \sqrt[n]{(\sqrt[n]{n} - 1)} = 1;$$

$$(4) \lim_{n \rightarrow +\infty} \sqrt[n]{a_1 a_2 \cdots a_n} = a, (a_n > 0, \lim_{n \rightarrow \infty} a_n = a).$$

8. Suppose that (x_n) is a sequence of real numbers and $\lim_{n \rightarrow +\infty} x_n = a$. Prove that

- (1) $\lim_{n \rightarrow +\infty} \frac{x_1 + 2x_2 + \cdots + nx_n}{n^2} = \frac{a}{2}.$
- (2) Suppose that $0 < r < 1$. Use the conclusion in (1) to calculate the following limits:
- $\lim_{n \rightarrow +\infty} \frac{1}{n^2} (r + 2^2 r^2 + \cdots + n^2 r^n);$
 - $\lim_{n \rightarrow +\infty} \frac{1}{n^2} (1 + 2^{\frac{3}{2}} + 3^{\frac{4}{3}} + \cdots + n^{\frac{n+1}{n}}).$

9. Suppose that $\varphi : \mathbb{N} \rightarrow \mathbb{N}$ such that φ is strictly monotonically increasing, and $\varphi \neq \text{id}_{\mathbb{N}}$.

- Prove that there exists $N \in \mathbb{N}$, such that for any $n \in \mathbb{N}$ and $n \geq N$, $\varphi(n) > n$.
- Let $\varphi^0 := \text{id}_{\mathbb{N}}$, and inductively define

$$\varphi^{n+1} := \varphi \circ \varphi^n, \quad \forall n \in \mathbb{N} \cup \{0\}.$$

Prove that for any $k \in \mathbb{N}$ and $n \in \mathbb{N}$ with $n \geq N$, $\varphi^k(n) \geq n + k$.

- Suppose that (x_n) is a sequence of real numbers satisfying $x_{\varphi(n)} = x_n$ for any $n \in \mathbb{N}$. Prove that if (x_n) is convergent, then there exists $M \in \mathbb{N}$, such that for any $n \in \mathbb{N}$ with $n \geq M$, $x_n = x_M$.

10. Suppose that (x_n) is a sequence of real numbers satisfying $x_n \leq x_{n+1}$ for all $n \in \mathbb{N}$. Prove that if

$$\lim_{n \rightarrow +\infty} \frac{x_1 + x_2 + \cdots + x_n}{n} = a,$$

then $\lim_{n \rightarrow +\infty} x_n = a$.

- Suppose that (a_n) is a sequence of real numbers satisfying

$$\lim_{n \rightarrow +\infty} \frac{a_{n+1}}{a_n} = \lambda,$$

where $|\lambda| < 1$. Prove that $\lim_{n \rightarrow +\infty} a_n = 0$.

- Use the conclusion in (a) to prove the following statements about limits:

- $\lim_{n \rightarrow +\infty} \frac{a^n}{n!} = 0 \quad (a \in \mathbb{R});$
- $\lim_{n \rightarrow +\infty} \frac{1^2 \cdot 3^2 \cdot (2n-1)^2 a^n}{2^2 \cdot 4^2 \cdots (2n)^2} = 0 \quad (|a| < 1).$

11. (1) Suppose that (a_n) is a sequence of real numbers satisfying

$$\lim_{n \rightarrow +\infty} \frac{a_{n+1}}{a_n} = \lambda,$$

where $\lambda > 1$. Prove that $\lim_{n \rightarrow +\infty} a_n = +\infty$.

- (2) Use the conclusion in (a) to prove the following statements about limits:

$$\begin{aligned} \text{i. } & \lim_{n \rightarrow +\infty} \frac{3 \cdot 6 \cdot 9 \cdots 3n}{7 \cdot 10 \cdot 13 \cdots (3n+4)} a^n = +\infty \quad (a > 1); \\ \text{ii. } & \lim_{n \rightarrow +\infty} \frac{1 \cdot 3 \cdot 5 \cdots (2n-1) a^{2n}}{2 \cdot 4 \cdot 6 \cdots (2n-2) \cdot (2n)} = +\infty \quad (|a| > 1). \end{aligned}$$

12. Prove the following statements about superior and inferior limits.

$$(1) \text{ Prove that } \overline{\lim}_{n \rightarrow +\infty} (-1)^n \left(1 + \frac{1}{n}\right) = 1 \text{ and } \underline{\lim}_{n \rightarrow +\infty} (-1)^n \left(1 + \frac{1}{n}\right) = -1.$$

$$(2) \text{ Prove that } \underline{\lim}_{n \rightarrow +\infty} \left[(1 + (-1)^n)n + \frac{1}{n} \right] = 0 \text{ but it has no superior limit.}$$

$$(3) \text{ Prove that } \overline{\lim}_{n \rightarrow +\infty} \left[((-1)^n - 1)n + \frac{1}{n} \right] = 0 \text{ but it has no inferior limit.}$$

$$(4) \text{ Prove that } (n^2) \text{ and } (-n) \text{ have neither superior limit nor inferior limit.}$$

13. (1) Let (u_n) be a sequence that has both superior limit and inferior limit. Prove that $\underline{\lim}_{n \rightarrow +\infty} u_n \leq \overline{\lim}_{n \rightarrow +\infty} u_n$.

$$(2) \text{ Prove that } u_n \text{ converges to a real number } l \text{ if and only if } \underline{\lim}_{n \rightarrow +\infty} u_n = \overline{\lim}_{n \rightarrow +\infty} u_n = l.$$

14. We consider the sequence (x_n) and (y_n) defined by:

$$\begin{cases} x_{2n} = 1 + \frac{1}{2n+1} \\ x_{2n+1} = 0 \end{cases} \quad \begin{cases} y_{2n} = 0 \\ y_{2n+1} = \frac{1}{2n+2} + 1. \end{cases}$$

$$(1) \text{ Calculate } \overline{\lim} x_n, \underline{\lim} x_n \text{ and } \overline{\lim}(x_n + y_n).$$

$$(2) \text{ Calculate } \underline{\lim} x_n, \overline{\lim}(-x_n) \text{ and } \underline{\lim}(-x_n).$$

$$(3) \text{ Prove that if } (u_n) \text{ is a bounded sequence and if } \alpha > 0, \text{ then } \overline{\lim} \alpha x_n = \alpha \overline{\lim} u_n \text{ and } \underline{\lim} \alpha x_n = \alpha \underline{\lim} u_n.$$

15. Let $(r_n)_{n \in \mathbb{N}}$ be a sequence of real numbers defined by $r_n := \sum_{k=1}^n a_k$, where $a_k \in \mathbb{R}$ for all $1 \leq k \leq n$. We suppose that the sequence $(\sqrt[k]{|a_k|})_{k \in \mathbb{N}}$ has a superior limit λ .

- (1) Suppose that $\lambda < 1$ and μ is a real number with $\lambda < \mu < 1$. If there exists an integer K such that $\forall k \in \mathbb{N}$ with $k \geq K$, $\sqrt[k]{|a_k|} < \mu$, prove that the sequence (r_n) is convergent.
- (2) Suppose that $\lambda > 1$ and μ is a real number with $\lambda > \mu > 1$. If for any $k \in \mathbb{N}$, there exists an $p \in \mathbb{N}$ such that $p \geq k$ and $\sqrt[p]{|a_k|} > \mu$, prove that the sequence (r_n) is divergent.
- (3) State the convergence or divergence of the sequence (r_n) , where $r_n := \sum_{k=1}^n a_k$ with a_k defined below:
 - i. $a_{2j} = \left(\frac{j}{2j+1}\right)^{2j}, a_{2j+1} = \left(\frac{1}{2j+1}\right)^{2j+1};$
 - ii. $a_{2j} = \left(\frac{3j-1}{2j+1}\right)^{2j}, a_{2j+1} = \left(\frac{1}{2j+1}\right)^{2j+1}.$

16. Let (r_n) be a sequence of real numbers defined by $r_n := \sum_{k=1}^n a_k$, where $a_k \in \mathbb{R}$ and $a_k > 0$ for all $1 \leq k \leq n$.

- (1) Prove that if the sequence $\left(\frac{a_{k+1}}{a_k}\right)$ has an superior limit strictly smaller than 1, then the sequence (r_n) is convergent.
- (2) Prove that if $\left(\frac{a_{k+1}}{a_k}\right)$ has a inferior limit strictly larger than 1, then the sequence (r_n) is divergent.
- (3) State the convergence or divergence of the sequence (r_n) , where $r_n := \sum_{k=1}^n a_k$ with a_k defined below:
 - i. $a_{2j} = \frac{j!}{1 \cdot 9 \cdot 25 \cdots (2j+1)^2}, a_{2j+1} = \frac{j!}{1 \cdot 9 \cdot 25 \cdots (2j+1)^2(2j+3)};$
 - ii. $a_{2j+1} = \frac{\cos 1 \cdots \cos \frac{1}{2j+1}}{2j+1}, a_{2j+2} = \frac{\cos 1 \cdots \cos \frac{1}{2j+1}}{2j};$
 - iii. $a_{2j+1} = \frac{\exp(-1 + \frac{1}{2} + \cdots + \frac{1}{j})}{2j+1}, a_{2j+2} = \frac{\exp(1 + \frac{1}{2} + \cdots + \frac{1}{j+1})}{2j+2}.$

17. The aim of this exercise is to study the so-called **Stolz–Cesàro theorem**. Let $(a_n)_{n \in \mathbb{N}}$ and $(b_n)_{n \in \mathbb{N}}$ be two series, which satisfy $\lim_{n \rightarrow \infty} a_n = +\infty$ or 0 and $\lim_{n \rightarrow \infty} b_n = +\infty$ or 0 at the same time. In this exercise, we will propose a method to compute $\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$ for some particular cases.

- (1) Suppose $(b_n)_{n \in \mathbb{N}}$ is strictly increasing, $\lim_{n \rightarrow \infty} b_n = +\infty$ and $-\infty \leq l < +\infty$. If

$$\lim_{n \rightarrow \infty} \frac{a_{n+1} - a_n}{b_{n+1} - b_n} = l \in \overline{\mathbb{R}} = \mathbb{R} \cup \{+\infty, -\infty\},$$

prove that there exists a $p \in]l, +\infty[$ and an $n_0 \in \mathbb{N}$, such that for all $n > n_0$, we have

$$\frac{a_{n+1}}{b_{n+1}} < p \left(1 - \frac{b_{n_0}}{b_{n+1}} \right) + \frac{a_{n_0}}{b_{n+1}}.$$

Hint: We can prove $a_{n+1} = (a_{n+1} - a_n) + (a_n - a_{n-1}) + \cdots + (a_{n_0+1} - a_{n_0}) + a_{n_0} < p(b_{n+1} - b_{n_0}) + a_{n_0}$ for some particular $p, n_0 \in \mathbb{R}$.

(2) We keep all the notations and conditions in (1). Let

$$c_n = p \left(1 - \frac{b_{n_0}}{b_n} \right) + \frac{a_{n_0}}{b_n}.$$

Prove that for all $q > p$, there exist $n_1 \in \mathbb{N}$, such that for all $n > n_1$, we have $c_n < q$. Then deduce that there exists an $n_2 \in \mathbb{N}$, such that for all $n > n_2$, we have $\frac{a_{n+1}}{b_{n+1}} < c_{n+1} < q$.

(3) We keep all the notations and conditions in (1). For all real number $q' < l$, there exists an $n_3 \in \mathbb{N}$, such that for all $n > n_3$, we have $q' < \frac{a_{n+1}}{b_{n+1}}$.

Hint: By the same strategy as that in (1) and (2).

(4) Prove that $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = l$ under the assumption in (1).

(5) Suppose $(b_n)_{n \in \mathbb{N}}$ is strictly decreasing, $\lim_{n \rightarrow \infty} a_n = 0$, $\lim_{n \rightarrow \infty} b_n = 0$, and $-\infty \leq l < +\infty$. Prove $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = l$ by a similar strategy as above.

(6) Use Stolz–Casàro theorem to study the following limits.

i. Let $(a_n)_{n \in \mathbb{N}}$ be a series satisfying $\lim_{n \rightarrow \infty} a_n = b \in \mathbb{R}$. Prove

$$\lim_{n \rightarrow \infty} \frac{a_1 + \cdots + a_n}{n} = b.$$

ii. Find $\lim_{n \rightarrow \infty} \frac{1! + 2! + \cdots + n!}{n!}$.

iii. $\lim_{n \rightarrow +\infty} \frac{n^2}{a^{2n}} = 0$ ($|a| > 1$);

iv. $\lim_{n \rightarrow +\infty} \frac{1^k + 2^k + \cdots + n^k}{n^{k+1}} = \frac{1}{k+1}$ ($\forall k \in \mathbb{N}$);

v. $\lim_{n \rightarrow +\infty} \frac{1^k + 2^k + \cdots + n^k}{n^k} - \frac{n}{k+1} = \frac{1}{2}$ ($\forall k \in \mathbb{N}$);

vi. $\lim_{n \rightarrow +\infty} \frac{a_1 + 2a_2 + \cdots + na_n}{n(n+1)} = \pm\infty$ if $\lim_{n \rightarrow +\infty} a_n = \pm\infty$.

18. In this exercise, we will study the limit $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$.

- (1) Prove that the series $\left(\left(1 + \frac{1}{n}\right)^n\right)_{n \in \mathbb{N}}$ is increasing.
- (2) Prove that the series $\left(\left(1 + \frac{1}{n}\right)^{n+1}\right)_{n \in \mathbb{N}}$ is decreasing.
- (3) Deduce that the existences of $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$ and $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^{n+1}$, and prove

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^{n+1}.$$

Note: We denote $e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$, which is an irrational number. We call e the base (or bottom number) of the natural logarithm, and usually we denote $\ln(\cdot) = \log_e(\cdot)$. In fact, we have $e = 2.718281828459 \dots$.

19. For any $a \in \mathbb{R}$, we denote by $|a|$ the element $\max\{a, -a\}$ in $\mathbb{R}$.

- (1) Show that, for any $(a, b) \in \mathbb{R} \times \mathbb{R}$, one has

$$|a + b| \leq |a| + |b|, \quad |a - b| \leq |a| + |b|.$$

- (2) Show that, for any $(a, b) \in \mathbb{R} \times \mathbb{R}$,

$$||a| - |b|| \leq |a - b|$$

- (3) Let I be an infinite subset of $\mathbb{N}$ and $(x_n)_{n \in I}$ be a sequence of real numbers and $\ell \in \mathbb{R}$.

- (a) Show that $(x_n)_{n \in I}$ converges to ℓ if and only if

$$\limsup_{n \in I, n \rightarrow +\infty} |x_n - \ell| = 0.$$

- (b) Show that $(x_n)_{n \in I}$ has a subsequence that converges to ℓ if and only if

$$\liminf_{n \in I, n \rightarrow +\infty} |x_n - \ell| = 0.$$

- (4) Let a and b be two real numbers. Show that

$$\max\{|a|, |b|\} \leq \max\{|a + b|, |a - b|\} \leq 2 \max\{|a|, |b|\}.$$

- (5) Let a and b be real numbers. Show that

$$\frac{1}{2} \leq \max\{|a + b|, |a - b|, |1 - a|\}.$$

(6) Let a and b be real numbers. Show that

$$\max\{a, b\} = \frac{a + b + |a - b|}{2}, \quad \min\{a, b\} = \frac{a + b - |a - b|}{2}$$

20. Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in $\overline{\mathbb{R}} \setminus \{0\}$. Assume that $(|x_n|)_{n \in \mathbb{N}}$ tends to $+\infty$.

(1) Let ε be a positive real number. Show that there exists $N \in \mathbb{N}$ such that

$$\forall n \in \mathbb{N}, n \geq N, \text{ one has } |x_n| > \varepsilon^{-1}.$$

(2) Show that the sequence $(|x_n^{-1}|)_{n \in \mathbb{N}}$ tends to 0.

21. Let ε be a positive real number.

(1) Show that, for any $n \in \mathbb{N}$, $(1 + \varepsilon)^n \geq 1 + n\varepsilon$.

(2) Prove that the sequence $((1 + \varepsilon)^n)_{n \in \mathbb{N}}$ tends to $+\infty$.

(3) Let a be a non-zero real number such that $|a| < 1$. Show that

$$\lim_{n \rightarrow +\infty} a^n = 0.$$

(4) Let a be a non-zero real number such that $|a| < 1$. Determine the limit of the series

$$\sum_{n \in \mathbb{N}} a^n.$$

22. In this exercise, we consider the following two sequences $\alpha = (a_n)_{n \in \mathbb{N}, n \geq 1}$ and $\beta = (b_n)_{n \in \mathbb{N}, n \geq 1}$ defined as

$$a_n = \left(1 + \frac{1}{n}\right)^n, \quad b_n = \left(1 + \frac{1}{n}\right)^{n+1}.$$

(1) Prove the inequality $(1 + t)^n \geq 1 + nt$ for $t \geq -1$ and $n \in \mathbb{N}$. Show that the sequence α is increasing and the sequence β is decreasing.

(2) Show that the series

$$\sum_{n \in \mathbb{N}} \frac{1}{(n+1)(n+2)}$$

is convergent and its limit is 1.

(3) Show that

$$a_n \leq \sum_{k=0}^n \frac{1}{k!} \leq 3.$$

- (4) Prove that the sequences α and β converge to the same limit, which belongs to $[2, 3]$. We denote by e this limit.
- (5) Show that, for any $(n, m) \in \mathbb{N} \times \mathbb{N}$ such that $\min\{n, m\} \geq 1$, one has

$$a_n \leq b_m.$$

23. Consider the sequence $(a_n)_{n \in \mathbb{N}, n \geq 1}$ defined as

$$a_n = \frac{\sin(n)}{n}.$$

- (1) Show that

$$|a_n| \leq \frac{1}{n}.$$

- (2) Show that

$$\limsup_{n \rightarrow +\infty} |a_n| \leq 0.$$

- (3) Does the sequence $(a_n)_{n \in \mathbb{N}, n \geq 1}$ converge?

24. In this exercise, we study the convergence of the series

$$H(\alpha) = \sum_{n \in \mathbb{N}, n \geq 1} \frac{1}{n^\alpha},$$

where α is a real number.

- (1) Show that, if $\alpha \leq 0$, then the series $H(\alpha)$ diverges.
- (2) Let N be a positive integer, show that

$$\sum_{k=N+1}^{2N} \frac{1}{k} \geq \frac{1}{2}.$$

- (3) Show that the series $H(1)$ diverges.
- (4) Let n and ℓ be positive integers. Show that

$$\frac{1}{(n+1)^{1+\frac{1}{\ell}}} \leq \ell \left(\frac{1}{n^{\frac{1}{\ell}}} - \frac{1}{(n+1)^{\frac{1}{\ell}}} \right).$$

One can use a result of Exercice 4.

- (5) Show that, for any $\alpha > 1$, the series $H(\alpha)$ converges.

(6) Show that, for any positive integer n , one has

$$\frac{1}{n+1} \leq \ln(n+1) - \ln(n) \leq \frac{1}{n},$$

where for any $t > 0$, $\ln(t)$ is the unique real number such that

$$e^{\ln(t)} = t.$$

(7) Show that, for any positive integer n , one has

$$\ln(n+1) \leq \sum_{k=1}^n \frac{1}{k} \leq \ln(n) + 1.$$

(8) Show that the sequence

$$\sum_{k=1}^n \frac{1}{k} - \ln(n), \quad n \in \mathbb{N}, n \geq 1$$

is decreasing and is bounded from below by 0.

(9) Show that the sequence in the previous question converges in $\mathbb{R}$.

Note: Usually we denote

$$\gamma = \lim_{n \rightarrow \infty} \left(\sum_{k=1}^n \frac{1}{k} - \ln(n) \right),$$

which is an irrational number. We call it Euler constant. In fact, we have $\gamma = 0.577216 \dots$.

25. Consider the sequence $(u_n)_{n \in \mathbb{N}}$ of real numbers defined in a recursive way as

$$u_0 = 1, \quad u_{n+1} = \frac{u_n}{\sqrt{u_n^2 + 1}}.$$

(1) Show that the mapping $(n \in \mathbb{N}) \mapsto u_n$ is decreasing.

(2) Show that $0 \leq u_n \leq 1$ for any $n \in \mathbb{N}$.

(3) Compute u_1 and u_2 .

(4) Conjecture an expression of u_n . Prove your conjecture by induction on n .

(5) Show that the sequence $(u_n)_{n \in \mathbb{N}}$ has a limite and determine its limit.

- 26.** Consider the sequence $(u_n)_{n \in \mathbb{N}}$ of real numbers parametrized by $\mathbb{N}$ defined in a recursive way as

$$u_0 = 3, \quad u_{n+1} = \frac{u_n - 2}{2u_n + 5}.$$

Show that

$$\forall n \in \mathbb{N} \quad u_n = \frac{9 - 8n}{3 + 8n}.$$

Show that the sequence $(u_n)_{n \in \mathbb{N}}$ converges and determine its limit.

- 27.** Consider Fibonacci's sequence $(F_n)_{n \in \mathbb{N}}$ defined as

$$F_0 = 1, \quad F_1 = 1, \quad F_{n+2} = F_{n+1} + F_n.$$

- (1) Show that, for any $n \in \mathbb{N}$,

$$F_n F_{n+2} - F_{n+1}^2 = (-1)^n.$$

- (2) Show that the function

$$(n \in \mathbb{N}) \mapsto \frac{F_{2n+1}}{F_{2n}}$$

is increasing.

- (3) Show that the function

$$(n \in \mathbb{N}) \mapsto \frac{F_{2n+2}}{F_{2n+1}}$$

is decreasing.

- (4) Let

$$\alpha = \frac{1 + \sqrt{5}}{2}, \quad \beta = \frac{1 - \sqrt{5}}{2}.$$

Show that, for any $n \in \mathbb{N}$,

$$\alpha^{n+2} = \alpha^{n+1} + \alpha^n, \quad \beta^{n+2} = \beta^{n+1} + \beta^n.$$

- (5) Find real numbers λ and μ such that

$$\lambda + \mu = 1, \quad \lambda\alpha + \mu\beta = 1.$$

- (6) Prove that

$$\forall n \in \mathbb{N}, \quad F_n = \lambda\alpha^n + \mu\beta^n.$$

(7) Show that the sequence $(F_n/\alpha^n)_{n \in \mathbb{N}}$ converges. Determine its limit.

28. Consider the sequence $(u_n)_{n \in \mathbb{N}}$ of real numbers defined as

$$u_n = \sqrt{n+1} - \sqrt{n}.$$

(1) Show that, for any $n \in \mathbb{N}$, one has

$$u_n = \frac{1}{\sqrt{n+1} + \sqrt{n}}.$$

(2) Prove that the sequence $(u_n)_{n \in \mathbb{N}}$ is convergent. Determine its limit.

29. Let I be an infinite subset of $\mathbb{N}$, $(a_n)_{n \in I}$ and $(b_n)_{n \in I}$ be bounded sequences in $\mathbb{R}$.

(1) Show that

$$\limsup_{n \in I, n \rightarrow +\infty} (a_n + b_n) \leq \left(\limsup_{n \in I, n \rightarrow +\infty} a_n \right) + \left(\limsup_{n \in I, n \rightarrow +\infty} b_n \right).$$

(2) Show that

$$\liminf_{n \in I, n \rightarrow +\infty} (a_n + b_n) \geq \left(\liminf_{n \in I, n \rightarrow +\infty} a_n \right) + \left(\liminf_{n \in I, n \rightarrow +\infty} b_n \right).$$

(3) Assume that $(b_n)_{n \in I}$ converges to a real number ℓ . Show that

$$\begin{aligned} \limsup_{n \in I, n \rightarrow +\infty} (a_n + b_n) &= \ell + \limsup_{n \in I, n \rightarrow +\infty} a_n, \\ \liminf_{n \in I, n \rightarrow +\infty} (a_n + b_n) &= \ell + \liminf_{n \in I, n \rightarrow +\infty} a_n. \end{aligned}$$

(4) Determine the limit superior and the limit inferior of the sequence

$$u_n = (-1)^n + \frac{1}{n}, \quad n \in \mathbb{N}, \quad n \geq 1.$$

Does this sequence converge?

30. Let $(u_n)_{n \in \mathbb{N}}$ be a sequence in $\mathbb{R}$. We assume that

$$\lim_{n \rightarrow +\infty} |u_{n+1} - u_n| = 0.$$

Let λ be an element of $\mathbb{R}$ such that

$$\liminf_{n \rightarrow +\infty} u_n < \lambda < \limsup_{n \rightarrow +\infty} u_n.$$

- (1) Let n be a natural number. Show that there exist $p \in \mathbb{N}_{\geq n}$ such that $u_p > \lambda$ and $q \in \mathbb{N}_{\geq n}$ such that $u_q < \lambda$.
- (2) For any n , let $\psi(n)$ be the smallest natural number in $\mathbb{N}_{\geq n}$ such that $u_{\psi(n)} > \lambda$ and $\varphi(n)$ be the smallest natural number in $\mathbb{N}_{\geq \psi(n)}$ such that $u_{\varphi(n)} < \lambda$. Show that the sequences $(\varphi(n))_{n \in \mathbb{N}}$ and $(\psi(n))_{n \in \mathbb{N}}$ are increasing and tend to $+\infty$.
- (3) Show that

$$\forall n \in \mathbb{N}, \quad u_{\varphi(n)} < \lambda \leq u_{\varphi(n)-1}.$$

- (4) Show that the sequence $(u_{\varphi(n)})_{n \in \mathbb{N}}$ converges to λ .
- (5) Show that there exists a subsequence of $(u_n)_{n \in \mathbb{N}}$ that converges to λ .
- (6) Is the conclusion of the previous question necessarily true without the condition

$$\lim_{n \rightarrow +\infty} |u_{n+1} - u_n| = 0?$$

Justify your answer.

31. Let a be a positive real number and let N be an integer such that $N \geq 2a$.

- (1) Show that, for $n \geq N$, one has

$$\frac{a^n}{n!} \leq \frac{a^N}{N!} \frac{1}{2^{n-N}}.$$

- (2) Deduce that $a^n = o(n!)$, $n \rightarrow +\infty$.
- (3) Show that the series

$$\sum_{n \in \mathbb{N}} \frac{a^n}{n!}$$

is convergent.

32. The purpose of this exercise is to compare the sequences $(n!)_{n \in \mathbb{N}_{\geq 1}}$ and $(n^n)_{n \in \mathbb{N}_{\geq 1}}$.

- (1) Let a be a real number such that $0 < a < 1$. Show that, for any positive integer n , one has

$$\frac{n!}{n^n} \leq \frac{a^{an}}{a}.$$

- (2) Show that $n! = o(n^n)$, $n \rightarrow +\infty$.

- (3) Show that the series

$$\sum_{n \in \mathbb{N}_{\geq 1}} \frac{n!}{n^n}$$

is convergent.

- 33.** In this exercise, we study the convergence of the sequence

$$u_n = \left(\frac{1}{n!}\right)^{\frac{1}{n}}, \quad n \in \mathbb{N}_{\geq 1}.$$

- (1) Let a be a real number such $0 < a < 1$. Show that, for any $n \in \mathbb{N}_{\geq a-1}$, one has

$$n! \geq (an)^{n-an}.$$

- (2) Deduce that

$$\lim_{n \rightarrow +\infty} \frac{1}{(n!)^{1/n}} = 0.$$

- (3) Show that the sequence $(u_n)_{n \in \mathbb{N}}$ is decreasing. Deduce that the series

$$\sum_{n \in \mathbb{N}_{\geq 1}} (-1)^n u_n$$

converges.

- (4) Does the series

$$\sum_{n \in \mathbb{N}_{\geq 1}} u_n$$

converge?

- 34.** Study the convergence and the absolute convergence of the following series.

$$\sum_{n \in \mathbb{N}} \frac{\ln(n)}{2^n}, \quad \sum_{n \in \mathbb{N}_{\geq 2}} \frac{(-1)^n}{\ln(n)}, \quad \sum_{n \in \mathbb{N}_{\geq 1}} \frac{\sin(n)}{n^2}, \quad \sum_{n \in \mathbb{N}} \frac{n^2 + 1}{3n^4 + 2}, \quad \sum_{n \in \mathbb{N}} \binom{2n}{n}^{-1}.$$

- 35.** (1) Let a and b be positive real numbers. Show that there exists a positive integer N such that

$$\forall n \in \mathbb{N}_{\geq N}, \quad \ln(n) \leq an^b.$$

- (2) Show that, for any $\alpha \in \mathbb{R}_{>1}$ and $b \in \mathbb{R}_{>0}$, there exists a positive integer N' such that

$$\forall n \in \mathbb{N}_{\geq N'}, \quad n^2 \leq \alpha^{n^b}$$

- (3) Let $\alpha \in \mathbb{R}_{>1}$ and $b \in \mathbb{R}_{>0}$. Show that, the series

$$\sum_{n \in \mathbb{N}_{\geq 1}} \frac{1}{\alpha^{n^b}}$$

converges.

- 36.** For $n \in \mathbb{N}_{\geq 2}$, let

$$u_n = \frac{(-1)^n}{n + (-1)^n}.$$

- (1) For any $n \in \mathbb{N}_{\geq 1}$, let $v_n = u_{2n} + u_{2n+1}$. Show that $v_n = O(n^{-2})$.
 (2) Prove that the series

$$\sum_{n \in \mathbb{N}_{\geq 2}} u_n$$

converges and determine its limit. We can use the fact that the sequence

$$\sum_{k=1}^n \frac{1}{k} - \ln(n), \quad n \in \mathbb{N}_{\geq 1}$$

converges in $\mathbb{R}$.

- (3) Does the series

$$\sum_{n \in \mathbb{N}_{\geq 2}} u_n$$

converge absolutely?

- 37.** Let $(u_n)_{n \in \mathbb{N}}$ be a decreasing sequence in $\mathbb{R}_{>0}$ which converges to 0. For any $n \in \mathbb{N}$, let

$$S_n = \sum_{k=0}^n (-1)^k u_k.$$

Recall that we have proved in the course that the sequence $(S_n)_{n \in \mathbb{N}}$ converges in $\mathbb{R}$ to a limit, which we denote as ℓ . For any $n \in \mathbb{N}$, let $R_n = \ell - S_n$. We assume in addition that the following conditions are satisfied:

$$\forall n \in \mathbb{N}, \quad u_{n+2} + u_n \geq 2u_{n+1}, \quad \lim_{n \rightarrow +\infty} \frac{u_{n+1}}{u_n} = 1.$$

- (1) Show that, for any $n \in \mathbb{N}$, $(-1)^{n+1} R_n > 0$.
 (2) Show that, for any $n \in \mathbb{N}$, $|R_n| + |R_{n+1}| = u_{n+1}$.
 (3) For any $n \in \mathbb{N}$, let $v_n = u_n - u_{n+1}$. Show that the sequence $(v_n)_{n \in \mathbb{N}}$ is decreasing and converges to 0.

(4) Show that

$$\forall n \in \mathbb{N}, \quad |R_n| - |R_{n+1}| = (-1)^{n+1} \sum_{k \in \mathbb{N}_{>n}} (-1)^k v_k.$$

Deduce that $|R_n| - |R_{n+1}| \geq 0$ for any $n \in \mathbb{N}$.

(5) Show that, for any $n \in \mathbb{N}$,

$$\frac{u_{n+1}}{2} \leq |R_n| \leq \frac{u_n}{2}.$$

(6) Show that

$$\lim_{n \rightarrow +\infty} (-1)^{n+1} \frac{2R_n}{u_n} = 1.$$

38. Let $(u_n)_{n \in \mathbb{N}}$ be a sequence in $\mathbb{R}$ which converges to some $x \in \mathbb{R}$. Suppose that, for any $n \in \mathbb{N}$, $u_n \neq x$, and that the sequence

$$\frac{u_{n+1} - x}{u_n - x}, \quad n \in \mathbb{N}$$

converges in $\mathbb{R}$ to some $\lambda \in [-1, 1[$.

(1) For any $n \in \mathbb{N}$, let

$$v_n = \frac{u_{n+1} - \lambda u_n}{1 - \lambda}.$$

Show that $v_n - x = o(u_n - x)$, $n \rightarrow +\infty$.

(2) Suppose that $u_n \neq u_{n+1}$ for any $n \in \mathbb{N}$. Show that

$$\lim_{n \rightarrow +\infty} \frac{u_{n+2} - u_{n+1}}{u_{n+1} - u_n} = \lambda.$$

(3) Suppose that $u_n \neq u_{n+1}$ and $u_n + u_{n+2} \neq 2u_{n+1}$ for any $n \in \mathbb{N}$. For any $n \in \mathbb{N}$, let

$$w_n = \frac{u_n u_{n+2} - u_{n+1}^2}{u_n + u_{n+2} - 2u_{n+1}}.$$

Show that $w_n - x = o(u_n - x)$, $n \rightarrow +\infty$.

39. For any real number x , we denote by $\lfloor x \rfloor$ the largest integer that is bounded from above by x . Note that

$$\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1.$$

In this exercise, we fix an integer d which is ≥ 2 .

(1) Let n be an integer. Show that

$$0 \leq n - d \left\lfloor \frac{n}{d} \right\rfloor \leq d - 1.$$

(2) For any $n \in \mathbb{N}_{\geq 1}$, let

$$u_n = \frac{n - d \left\lfloor \frac{n}{d} \right\rfloor}{n(n+1)}.$$

Show that the series

$$\sum_{n \in \mathbb{N}_{\geq 1}} u_n$$

converges.

(3) For any $n \in \mathbb{N}_{\geq 1}$, let

$$S_n = \sum_{k=1}^n u_k, \quad H_n = \sum_{k=1}^n \frac{1}{k}.$$

Show that

$$S_{nd} = H_{nd} - H_n.$$

(4) Show that

$$\lim_{n \rightarrow +\infty} S_{nd} = \ln(d).$$

(5) Prove that

$$\lim_{n \rightarrow +\infty} S_n = \ln(d).$$

(6) Deduce that $\ln(d) \leq d - 1$.

40. Let $(u_n)_{n \in \mathbb{N}}$ be a sequence in $\mathbb{R}_{>0}$. We assume that the series

$$\sum_{n \in \mathbb{N}} u_n$$

converges and we denote by S its sum.

(1) Show that the series

$$\sum_{n \in \mathbb{N}} u_n^2$$

converges to a limit T .

(2) Show that T belongs to $]0, S^2[$.

- (3) Suppose that the series $(u_n)_{n \in \mathbb{N}}$ is of the form $u_n = q^n$, where $0 < q < 1$. Express S and T in terms of q . Show that, for any $\delta \in]0, 1[$ there exists $q \in]0, 1[$ such that $T/S^2 = \delta$.

- 41.** Let $(u_n)_{n \in \mathbb{N}_{\geq 1}}$ be a sequence of real numbers. We suppose that $(u_n)_{n \in \mathbb{N}_{\geq 1}}$ converges to a limit $\ell \in \mathbb{R}$. For any $n \in \mathbb{N}_{\geq 1}$, let $S_n = u_1 + \cdots + u_n$.

- (1) For any $N \in \mathbb{N}_{\geq 1}$, let

$$\varepsilon_N = \sup_{n \in \mathbb{N}_{\geq N}} |u_n - \ell|.$$

Show that, for any natural number n such that $n \geq N$, one has

$$|S_n - n\ell| \leq \sum_{k=1}^N |u_k - \ell| + (n - N)\varepsilon_N.$$

- (2) Show that, for any $N \in \mathbb{N}_{\geq 1}$, one has

$$\limsup_{n \rightarrow +\infty} \left| \frac{S_n}{n} - \ell \right| \leq \varepsilon_N.$$

- (3) Show that the sequence $(S_n/n)_{n \in \mathbb{N}_{\geq 1}}$ converges to ℓ .

- 42.** Let $(u_n)_{n \in \mathbb{N}}$ be a sequence in $\mathbb{R}_{>0}$. For any $n \in \mathbb{N}$, let

$$S_n = \sum_{k=0}^n u_k.$$

- (1) Show that the series

$$\sum_{n \in \mathbb{N}} \frac{u_n}{S_n^2}$$

converges. We can use the inequality $S_n \geq S_{n-1}$.

- (2) Suppose that the sequence $(S_n)_{n \in \mathbb{N}}$ converges to a limit S . Show that the series

$$\sum_{n \in \mathbb{N}} \frac{u_n}{S_n}$$

converges.

- 43.** Let $(u_n)_{n \in \mathbb{N}_{\geq 1}}$ be a sequence in $\mathbb{R}_{>0}$. For any $n \in \mathbb{N}_{\geq 1}$, let

$$P_n = \prod_{k=1}^n (1 + u_k), \quad v_n = \frac{u_n}{P_n}.$$

- (1) Show that the series

$$\sum_{n \geq 1} v_n$$

converges.

- (2) Suppose that the series

$$\sum_{n \in \mathbb{N}_{\geq 1}} u_n$$

diverges.

- (a) Show that $(P_n)_{n \in \mathbb{N}_{\geq 1}}$ tends to $+\infty$.

- (b) Show that

$$\sum_{n=1}^N v_n = 1 - \frac{1}{P_N}.$$

- (c) Deduce that

$$\sum_{n \in \mathbb{N}_{\geq 1}} v_n = 1.$$

Chapter 8

Topological spaces

8.1 Topology

8.1.1 Proposition. *Let X be a set. For any $x \in X$, let $\mathcal{G}_x$ be a filter on X that is contained in the principal filter of $\{x\}$ (this is equivalent to require that $x \in V$ for any $V \in \mathcal{G}_x$). Denote by $\mathcal{T}$ the set of $U \in \mathcal{P}(X)$ such that*

$$\forall x \in U, U \in \mathcal{G}_x.$$

Then the following conditions are satisfied:

- (1) $\{\emptyset, X\} \subseteq \mathcal{T}$,
- (2) for any $(U_1, U_2) \in \mathcal{T} \times \mathcal{T}$, one has $U_1 \cap U_2 \in \mathcal{T}$,
- (3) for any family $(U_i)_{i \in I}$ of elements of $\mathcal{T}$, one has $\bigcup_{i \in I} U_i \in \mathcal{T}$.

Moreover, for any $x \in X$, the set

$$\mathcal{B}_x := \{U \in \mathcal{T} \mid x \in U\}$$

is equal to $\mathcal{T} \cap \mathcal{G}_x$, which is a filter basis contained in $\mathcal{G}_x$. It generates $\mathcal{G}_x$ if and only if, for any $U \in \mathcal{G}_x$, there exists $V \in \mathcal{T}$ such that $x \in V \subseteq U$.

Proof. (1) follows directly from the definition of $\mathcal{T}$.

(2) Let $(U_1, U_2) \in \mathcal{T} \times \mathcal{T}$. For any $x \in U_1 \cap U_2$, One has $\{U_1, U_2\} \in \mathcal{G}_x$ and hence $U_1 \cap U_2 \in \mathcal{G}_x$ since $\mathcal{G}_x$ is a filter. Therefore, $U_1 \cap U_2 \in \mathcal{T}$.

(3) Let $(U_i)_{i \in I}$ be a family of elements of $\mathcal{T}$ and $U = \bigcup_{i \in I} U_i$. For any $x \in U$, there exists $i \in I$ such that $x \in U_i$ and hence $U_i \in \mathcal{G}_x$. Since $\mathcal{G}_x$ is a filter and $U_i \subseteq U$, we obtain $U \in \mathcal{G}_x$. Hence $U \in \mathcal{T}$.

Let $U \in \mathcal{T}$. If $x \in U$, then by definition $U \in \mathcal{G}_x$. Conversely, if $U \in \mathcal{G}_x$, since $\mathcal{G}_x$ is contained in the principal filter of $\{x\}$, one has $x \in U$. Hence

$$\{U \in \mathcal{T} \mid x \in U\} = \{U \in \mathcal{T} \mid U \in \mathcal{G}_x\} = \mathcal{T} \cap \mathcal{G}_x.$$

Since $\mathcal{T}$ and $\mathcal{G}_x$ are both closed under intersection, so is $\mathcal{B}_x$. Hence $\mathcal{B}_x$ is a filter basis contained in $\mathcal{G}_x$. The last statement comes from Proposition 7.1.5. $\square$

8.1.2 Definition. Let X be a set. We call *topology* on X any subset $\mathcal{T}$ of $\mathcal{P}(X)$, that satisfies the following conditions:

- (1) $\{\emptyset, X\} \subseteq \mathcal{T}$,
- (2) for any family $(U_i)_{i \in I}$ of elements of $\mathcal{T}$ (where I is an arbitrary set), the union $\bigcup_{i \in I} U_i$ belongs to $\mathcal{T}$,
- (3) for any pair (U, V) of elements of $\mathcal{T}$, the intersection $U \cap V$ belongs to $\mathcal{T}$.

The pair $(X, \mathcal{T})$ is called a *topological space*.

If two topologies $\mathcal{T}$ and $\mathcal{T}'$ on the same set X satisfies $\mathcal{T} \subseteq \mathcal{T}'$, we say that $\mathcal{T}'$ is *finer* than $\mathcal{T}$ and that $\mathcal{T}$ is *coarser* than $\mathcal{T}'$.

8.1.3 Example. Let X be a set. For any $x \in X$, we let S_x be a subset of the principal filter of $\{x\}$, and let $\mathcal{G}_x$ be the filter generated by S_x . By Proposition 8.1.1,

$$\mathcal{T} := \{U \in \mathcal{T}(X) \mid \forall x \in U, U \in \mathcal{G}_x\}$$

is a topology on X . We call it the *topology generated by the family $(S_x)_{x \in X}$* .

8.1.4 Remark. Let $(X, \mathcal{T})$ be a topological space. For any finite family $(U_i)_{i=1}^n$ of elements of $\mathcal{T}$ (where n is a non-zero natural number), the intersection $\bigcap_{i=1}^n U_i$ belongs to $\mathcal{T}$. This can be proved by induction on n .

8.1.5 Example. Let X be a set. The family $\{\emptyset, X\}$ is a topology on X , called the *trivial topology* on X . Moreover, $\mathcal{P}(X)$ is also a topology on X , called the *discrete topology*.

8.1.6 Proposition. Let $(X, \mathcal{T})$ be a topological space. For any $x \in X$, let

$$\mathcal{B}_x(\mathcal{T}) := \{U \in \mathcal{T} \mid x \in U\}$$

and let $\mathcal{V}_x(\mathcal{T})$ be the filter generated by $\mathcal{B}_x(\mathcal{T})$.

- (1) For any $x \in X$, $\mathcal{B}_x(\mathcal{T})$ is a filter basis on X that is contained in the principal filter of $\{x\}$.

- (2) The topology on X generated by the family of filters $(\mathcal{V}_x(\mathcal{T}))_{x \in X}$ identifies with $\mathcal{T}$.

Proof. (1) If U and V are elements of $\mathcal{B}_x(\mathcal{T})$, then $U \cap V \in \mathcal{T}$ and $x \in U \cap V$. So $U \cap V \in \mathcal{B}_x(\mathcal{T})$.

(2) Let $\mathcal{T}'$ be the topology generated by $(\mathcal{V}_x(\mathcal{T}))_{x \in X}$. Let $U \in \mathcal{T}$. For any $x \in U$ one has $U \in \mathcal{B}_x(\mathcal{T})$. Hence U belongs to $\mathcal{T}'$. Conversely, if U is an element of $\mathcal{T}'$, for any $x \in U$, there exists $V_x \in \mathcal{T}$ such that $x \in V_x \subseteq U$. Hence

$$U = \bigcup_{x \in U} \{x\} \subseteq \bigcup_{x \in U} V_x \subseteq U.$$

Thus

$$U = \bigcup_{x \in U} V_x$$

belongs to $\mathcal{T}$. □

8.1.7 Definition. Let $(X, \mathcal{T})$ be a topological space. For any $x \in X$, let

$$\mathcal{B}_x(\mathcal{T}) = \{U \in \mathcal{T} \mid x \in U\}$$

and let $\mathcal{V}_x(\mathcal{T})$ be the filter generated by $\mathcal{B}_x(\mathcal{T})$. The elements of $\mathcal{B}_x(\mathcal{T})$ are called *open neighbourhoods* of x and the elements of $\mathcal{V}_x(\mathcal{T})$ are called *neighbourhoods* of x . If $\mathcal{B}$ is a filter basis that generates $\mathcal{V}_x(\mathcal{T})$, we say that it is a *neighbourhood basis* of x .

8.1.8 Proposition. Let X be a set, $(\mathcal{T}_i)_{i \in I}$ be a family of topology on X , and $\mathcal{T} = \bigcap_{i \in I} \mathcal{T}_i$. Then $\mathcal{T}$ is a topology on X .

Proof. For any $i \in I$, one has $\{\emptyset, X\} \subseteq \mathcal{T}_i$, so that

$$\{\emptyset, X\} \subseteq \bigcap_{i \in I} \mathcal{T}_i.$$

Let $(U_j)_{j \in J}$ be a family of elements of $\mathcal{T}$. For any $i \in I$, one has $(U_j)_{j \in J} \in \mathcal{T}_i^J$. Hence $\bigcup_{j \in J} U_j \in \mathcal{T}_i$. Therefore,

$$\bigcup_{j \in J} U_j \in \bigcap_{i \in I} \mathcal{T}_i.$$

Let U and V be elements of $\mathcal{T}$. For any $i \in I$, one has $\{U, V\} \subseteq \mathcal{T}_i$ and hence $U \cap V \in \mathcal{T}_i$. Therefore,

$$U \cap V \in \bigcap_{i \in I} \mathcal{T}_i.$$

□

8.1.9 Definition. Let X be a set and S be a subset of $\mathcal{P}(X)$. We denote by $\mathcal{T}_S$ the intersection of all topologies containing S , called the *topology generated by S* . It is the coarsest topology containing S .

8.1.10 Definition. Let X be a set and $\mathcal{B}$ be a subset of $\mathcal{P}(X)$. We say that $\mathcal{B}$ is a *topological basis* of X if the following conditions are satisfied:

- (1) the set X is covered by $\mathcal{B}$, namely $X = \bigcup_{V \in \mathcal{B}} V$,
- (2) for any $(U, V) \in \mathcal{B} \times \mathcal{B}$ and any $x \in U \cap V$, there exists $W_x \in \mathcal{B}$ such that $x \in W_x \subseteq U \cap V$.

8.1.11 Proposition. Let X be a set, $\mathcal{B}$ be a topological basis of X and $\mathcal{T}_\mathcal{B}$ be topology generated by $\mathcal{B}$. Then $\mathcal{T}_\mathcal{B}$ is composed of all subsets A that can be written as the union of a family of elements of $\mathcal{B}$.

Proof. Let $\mathcal{T}$ be the set of $A \in \mathcal{P}(X)$ that can be written as the union of a family of elements of $\mathcal{B}$. Since $\mathcal{T}_\mathcal{B}$ is a topology containing $\mathcal{B}$, one has $\mathcal{T} \subseteq \mathcal{T}_\mathcal{B}$.

To prove the converse inclusion $\mathcal{T}_\mathcal{B} \subseteq \mathcal{T}$, it suffices to check that $\mathcal{T}$ is a topology. By definition one has $\emptyset \in \mathcal{T}$. Moreover, since $\mathcal{B}$ covers X , one has $X \in \mathcal{T}$. By definition, the union of a family of sets in $\mathcal{T}$ is still an element of $\mathcal{T}$. It remains to verify that the intersection of two elements of $\mathcal{T}$ is still an element of $\mathcal{T}$. Let $(U_i)_{i \in I}$ and $(V_j)_{j \in J}$ be two families of elements of $\mathcal{B}$. Let

$$U := \bigcup_{i \in I} U_i, \quad V := \bigcup_{j \in J} V_j.$$

Note that one has

$$U \cap V = \bigcup_{(i,j) \in I \times J} U_i \cap V_j.$$

For any $x \in U_i \cap V_j$, there exists $W_{i,j,x} \in \mathcal{B}$ such that

$$x \in W_{i,j,x} \subseteq U_i \cap V_j.$$

Therefore, we get

$$U \cap V = \bigcup_{\substack{(i,j) \in I \times J \\ x \in U_i \cap V_j}} W_{i,j,x} \in \mathcal{T}.$$

Therefore, $\mathcal{T}$ is a topology containing $\mathcal{B}$. Hence it contains $\mathcal{T}_\mathcal{B}$. □

8.1.12 Proposition. Let X be a set and S be a subset of $\mathcal{P}(X)$. Let $\mathcal{B}_S$ be the set of $U \in \mathcal{P}(X)$ that is either X or can be written as the intersection of finitely many sets in S . Then $\mathcal{B}_S$ is a topological basis on X . Moreover, S and $\mathcal{B}_S$ generate the same topology on X .

Proof. Since X belongs to $\mathcal{B}_S$, we obtain that $\mathcal{B}_S$ covers X . Moreover, by definition, the intersection of two elements of $\mathcal{B}_S$ still belongs to $\mathcal{B}_S$. Therefore, $\mathcal{B}_S$ is a topological basis.

Let $\mathcal{T}_S$ and $\mathcal{T}'_S$ be topologies generated by S and by $\mathcal{B}_S$ respectively. Since $S \subseteq \mathcal{B}_S$, we obtain that $\mathcal{T}_S \subseteq \mathcal{T}'_S$. Conversely, since $\mathcal{T}_S$ is a topology and $S \subseteq \mathcal{T}_S$, we obtain that $\mathcal{B}_S \subseteq \mathcal{T}_S$. Therefore, $\mathcal{T}'_S \subseteq \mathcal{T}_S$. $\square$

8.2 Convergence in a topological space

8.2.1 Definition. Let X and E be sets, and $f : X \rightarrow E$ be a mapping. If $\mathcal{F}$ is a filter on X , we denote by $f_*(\mathcal{F})$ the subset

$$\{A \in \mathcal{P}(E) \mid f^{-1}(A) \in \mathcal{F}\}.$$

8.2.2 Proposition. Let $f : X \rightarrow E$ be a mapping of sets.

- (1) If $\mathcal{F}$ is a filter on X , then $f_*(\mathcal{F})$ is a filter on E .
- (2) If $\mathcal{B}$ is a filter basis on X and $\mathcal{F}$ is the filter generated by $\mathcal{B}$, then

$$f(\mathcal{B}) := \{f(U) \mid U \in \mathcal{B}\}$$

is a filter basis of E . Moreover, $f_*(\mathcal{F})$ is generated by $f(\mathcal{B})$.

Proof. (1) Let A and B be two elements of $f_*(\mathcal{F})$. One has

$$f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B) \in \mathcal{F}.$$

If A is an element of $f_*(\mathcal{F})$ and B is a subset of E that contains A , then

$$f^{-1}(B) \supseteq f^{-1}(A) \in \mathcal{F},$$

which implies that $f^{-1}(B) \in \mathcal{F}$.

(2) Let U and V be elements of $\mathcal{B}$. One has

$$f(U \cap V) \subseteq f(U) \cap f(V).$$

Hence $f(\mathcal{B})$ is a filter basis on E . For any $U \in \mathcal{B}$, one has $U \subseteq f^{-1}(f(U))$ and hence $f(U) \in f_*(\mathcal{F})$. This implies that $f(\mathcal{B}) \subseteq f_*(\mathcal{F})$.

If A is an element of $f_*(\mathcal{F})$, then one has $f^{-1}(A) \in \mathcal{F}$. Hence there exists $V \in \mathcal{B}$ such that $V \subseteq f^{-1}(A)$. Note that $f(V) \subseteq A$. Hence A contains an element of $f(\mathcal{B})$. $\square$

8.2.3 Definition. Let $(E, \mathcal{T})$ be a topological space, $f : X \rightarrow E$ be a mapping, $\mathcal{F}$ be a filter on X , and $\ell \in E$. If $\mathcal{V}_\ell(\mathcal{T}) \subseteq f_*(\mathcal{F})$, we say that f converges to ℓ along $\mathcal{F}$. We denote by

$$\lim_{\mathcal{F}} f = \ell$$

the statement

$$f \text{ converges to } \ell \text{ along } \mathcal{F}.$$

8.2.4 Remark. Notice, that ℓ does not have to be unique in general. Indeed, if we consider $X = \mathbb{R}$ with the right interval topology i.e. $\mathcal{T}_X = \{(a, +\infty), a \in \mathbb{R}\}$, we have $\mathcal{V}_p(\mathcal{T}_X) = \{(a, +\infty) | p \in (a, +\infty)\}$. Choose f to be the identity mapping. One can check, that

$$\lim_{\mathcal{V}_p(\mathcal{T}_X)} f = l$$

for any $l \geq p$.

8.2.5 Proposition. Let $(E, \mathcal{T})$ be a topological space, $f : X \rightarrow E$ be a mapping, $\mathcal{F}$ be a non-degenerate filter on X and $\ell \in E$. The following statements are equivalent:

- (1) f converges to ℓ along $\mathcal{F}$.
- (2) for any neighbourhood U of ℓ , $f^{-1}(U) \in \mathcal{F}$.
- (3) there exists a subset S of $\mathcal{P}(E)$ such that S generates the filter $\mathcal{V}_\ell(\mathcal{T})$ and that, for any

$$\forall U \in S, f^{-1}(U) \in \mathcal{F}.$$

Proof. “(1) $\implies$ (2)”: By definition, if ℓ is a limit of f along $\mathcal{F}$, then $\mathcal{V}_\ell(\mathcal{T}) \subseteq f_*(\mathcal{F})$. Therefore, for any $U \in \mathcal{V}_\ell(\mathcal{T})$, one has $f^{-1}(U) \in \mathcal{F}$.

“(2) $\implies$ (3)”: Since S is a subset of $\mathcal{V}_\ell(\mathcal{T})$, (3) is a particular case of (2).

“(3) $\implies$ (1)”: If the condition (3) holds, then S is a subset of $f_*(\mathcal{F})$. Since $f_*(\mathcal{F})$ is a filter generated by S and $S \subseteq f_*(\mathcal{F})$, we obtain that $\mathcal{V}_\ell(\mathcal{T}) \subseteq f_*(\mathcal{F})$, namely f converges to ℓ along $\mathcal{F}$. $\square$

8.2.6 Example. Let $(G, \leq)$ be a totally ordered set. We call *order topology* on G the topology generated by the family of the form

$$\{G_{>a} | a \in G\} \cup \{G_{<b} | b \in G\}.$$

If ℓ is an element of G , then the neighbourhood filter of ℓ is generated by

$$\{G_{>a} | a \in G, a < \ell\} \cup \{G_{<b} | b \in G, \ell < b\}.$$

Assume that $(G, \leq)$ is order complete. Let X be a set and $\mathcal{F}$ be a non-degenerate filter on X . Let $f : X \rightarrow G$ be a mapping. If f converges to ℓ along $\mathcal{F}$, then, for any $a \in G_{<\ell}$, one has $f^{-1}(G_{>a}) \in \mathcal{F}$. Note that

$$\forall x \in f^{-1}(G_{>a}), f(x) > a.$$

Therefore

$$\liminf_{\mathcal{F}} f \geq a.$$

If $\sup(G_{<\ell}) = \ell$, then, by taking the supremum for $a \in G_{<\ell}$

$$\liminf_{\mathcal{F}} f \geq \ell.$$

Otherwise, in the particular case where $a = \sup(G_{<\ell})$, one has

$$\forall x \in f^{-1}(G_{>a}), f(x) \geq \ell.$$

Therefore one always have

$$\liminf_{\mathcal{F}} f \geq \ell.$$

Similar reasoning shows that

$$\limsup_{\mathcal{F}} f \leq \ell.$$

Therefore,

$$\liminf_{\mathcal{F}} f = \limsup_{\mathcal{F}} f = \ell.$$

Conversely, if

$$\liminf_{\mathcal{F}} f = \ell,$$

for any $a \in G_{<\ell}$, there exists $U \in \mathcal{F}$ such that $f^i(U) > a$. Hence $f^{-1}(G_{>a}) \in \mathcal{F}$. Similarly, for any $b \in G_{>\ell}$, there exists $V \in \mathcal{F}$ such that $f^s(U) < b$. Hence $f^{-1}(G_{<b}) \in \mathcal{F}$.

8.2.7 Example. Let $(E, \mathcal{T})$ be a topological space, $\ell \in E$ and S_ℓ be a subset of $\mathcal{P}(E)$ that generates the filter $\mathcal{V}_\ell(\mathcal{T})$ of neighbourhoods of ℓ .

- (1) Let I be an infinite subset of $\mathbb{N}$ and $\mathcal{F}$ be the Fréchet filter on I . Let $(x_n)_{n \in I}$ be a sequence in E , namely $(x_n)_{n \in I}$ is an element of E^I , considered as a mapping from I to E . If $(x_n)_{n \in I}$ converges to ℓ along $\mathcal{F}$, we say that the sequence $(x_n)_{n \in I}$ converges to ℓ when $n \in I$ tends to the infinity, denoted as

$$\lim_{n \in I, n \rightarrow +\infty} x_n = \ell.$$

This condition is equivalent to, for any $U \in S_\ell$, there exists

8.3 Metric space

8.3.1 Definition. Let X be a set. We call *pseudometric* on X any mapping

$$d : X \times X \longrightarrow \mathbb{R}_{\geq 0}$$

that satisfies the following conditions:

- (1) for any $x \in X$, $d(x, x) = 0$,
- (2) for any $(x, y) \in X \times X$, $d(x, y) = d(y, x)$,
- (3) (triangle inequality) for any $(x, y, z) \in X^3$,

$$d(x, z) \leq d(x, y) + d(y, z).$$

The pair (X, d) is called a *pseudometric space*. If in addition $d(x, y) > 0$ for any $(x, y) \in X \times X$ such that $x \neq y$, we say that d is a *metric* and (X, d) is a *metric space*.

Let (X, d) be a pseudometric space. For any $x \in X$ and $\varepsilon > 0$, we denote by $B(x, \varepsilon)$ the set

$$\{y \in X \mid d(x, y) < \varepsilon\},$$

and call it the *open ball centered at x and of radius ε* .

8.3.2 Definition. Let K be a field. We call *absolute value* on K any mapping

$$|\cdot| : K \longrightarrow \mathbb{R}_{\geq 0}$$

that satisfies the following conditions:

- (1) $|a| = 0$ if and only if a is the zero element of K ,
- (2) for any $(a, b) \in K \times K$, $|ab| = |a| \cdot |b|$,
- (3) for any $(a, b) \in K \times K$, $|a + b| \leq |a| + |b|$.

The couple $(K, |\cdot|)$ is called a *valued field*.

8.3.3 Proposition. Let $(K, |\cdot|)$ be a valued field. Let 0_K and 1_K be respectively the zero element and the unity of K . One has $|1_K| = |-1_K| = 1$.

Proof. Since $1_K^2 = 1_K$, one has

$$|1_K|^2 = |1_K|.$$

Therefore, $|1_K| \in \{0, 1\}$. Since $|1_K| > 0$, one has $|1_K| = 1$. Similarly $(-1_K)^2 = 1_K^2 = 1_K$, so $|-1_K|^2 = |1_K| = 1$. Since $|-1_K| > 0$, one has $|-1_K| = 1$. $\square$

8.3.4 Example. The mapping

$$|\cdot| : \mathbb{R} \longrightarrow \mathbb{R}_{\geq 0}, \quad a \longmapsto \max\{a, -a\}$$

is an absolute value on $\mathbb{R}$.

The mapping

$$|\cdot| : \mathbb{C} \longrightarrow \mathbb{R}_{\geq 0}, \quad (x + yi) \longmapsto \sqrt{x^2 + y^2}$$

is an absolute value on $\mathbb{C}$.

8.3.5 Example. Let p be a prime number. Any non-zero rational number α can be written as

$$\alpha = p^{\text{ord}_p(\alpha)} \frac{m}{n}$$

where $\text{ord}_p(\alpha) \in \mathbb{Z}$, $(m, n) \in \mathbb{Z}^2$, $p \nmid mn$. If $\alpha = 0$ by convention $\text{ord}_p(\alpha) = +\infty$.

Properties:

1. $\text{ord}_p(\alpha\beta) = \text{ord}_p(\alpha) + \text{ord}_p(\beta)$
2. If $\text{ord}_p(\alpha) > \text{ord}_p(\beta)$, then $\text{ord}_p(\alpha + \beta) = \text{ord}_p(\beta)$

Indeed, suppose that

$$\alpha = p^{\text{ord}_p(\alpha)} \frac{m}{n} \quad \beta = p^{\text{ord}_p(\beta)} \frac{u}{v}$$

then

$$\alpha + \beta = p^{\text{ord}_p(\beta)} \frac{p^{\text{ord}_p(\alpha) - \text{ord}_p(\beta)} mv + nu}{nv}$$

since $p \nmid nuv$, we obtain that $\text{ord}_p(\alpha + \beta) = \text{ord}_p(\beta)$.

3. If $\text{ord}_p(\alpha) = \text{ord}_p(\beta)$, then $\text{ord}_p(\alpha + \beta) \geq \text{ord}_p(\beta) = \text{ord}_p(\alpha)$

Suppose that

$$\alpha = p^{\text{ord}_p(\alpha)} \frac{m}{n} \quad \beta = p^{\text{ord}_p(\beta)} \frac{u}{v}$$

then

$$\alpha + \beta = p^{\text{ord}_p(\beta)} \frac{mv + nu}{nv}$$

so $\text{ord}_p(\alpha + \beta) \geq \text{ord}_p(\beta)$. Notice that $p \nmid nv$ but p might divide $mv + nu$.

8.3.6 Proposition. The mapping $|\cdot|_p : \mathbb{Q} \rightarrow \mathbb{R}_{\geq 0}$ defined as

$$|\alpha|_p = \begin{cases} p^{\text{ord}_p(\alpha)} \frac{m}{n} & \text{if } \alpha \neq 0 \\ 0 & \text{if } \alpha = 0 \end{cases}$$

is an absolute value. We call it p -adic absolute value.

Proof. If $\alpha \neq 0$ then $|\alpha|_p > 0$. If $(\alpha, \beta) \in \mathbb{Q}^2$ and $0 \in \{\alpha, \beta\}$, one has $\alpha\beta = 0$ and hence $0 = |\alpha\beta|_p = |\alpha|_p|\beta|_p$. When $0 \notin \{\alpha, \beta\}$, we have

$$|\alpha\beta|_p = p^{-\text{ord}_p(\alpha\beta)} = p^{-\text{ord}_p(\alpha) - \text{ord}_p(\beta)} = |\alpha|_p|\beta|_p.$$

If $\alpha = 0$, $|\alpha + \beta|_p = |\beta|_p$ and if $\beta = 0$, $|\alpha + \beta|_p = |\alpha|_p$. If $0 \notin \{\alpha, \beta\}$, then

$$|\alpha + \beta|_p = p^{-\text{ord}_p(\alpha+\beta)} \leq p^{-\min\{\text{ord}_p(\alpha), \text{ord}_p(\beta)\}} \leq \max\{|\alpha|_p, |\beta|_p\}.$$

□

8.3.7 Remark. Let $(K, |\cdot|)$ be a valued field. If

$$\forall (\alpha, \beta) \in K \times K \quad |\alpha + \beta| \leq \max\{|\alpha|, |\beta|\}$$

we say that $(K, |\cdot|)$ is *non-archimedean*. Otherwise, we say that $(K, |\cdot|)$ is *archimedean*.

8.3.8 Definition. Let $(K, |\cdot|)$ be a valued field and E be a vector space over K . We call *seminorm* on E any mapping

$$\|\cdot\| : E \longrightarrow \mathbb{R}_{\geq 0}$$

that satisfies the following conditions:

- (1) for any $(a, x) \in K \times E$, $\|ax\| = |a| \cdot \|x\|$,
- (2) for any $(x, y) \in E \times E$, $\|x + y\| \leq \|x\| + \|y\|$.

The pair $(E, \|\cdot\|)$ is called a *seminormed vector space* over $(K, |\cdot|)$.

If in addition $\|x\| > 0$ for any $x \in E \setminus \{0\}$, then we say that $\|\cdot\|$ is a norm and $(E, \|\cdot\|)$ is a *normed vector space* over K .

8.3.9 Example. Let $(K, |\cdot|)$ be a valued field. Then $(K, |\cdot|)$ is a normed vector space over $(K, |\cdot|)$.

8.3.10 Proposition. Let $(K, |\cdot|)$ be a valued field and $(E, \|\cdot\|)$ be a seminormed vector space over K . Then the mapping

$$E \times E \longrightarrow \mathbb{R}_{\geq 0}, (x, y) \longmapsto \|x - y\|$$

is a pseudometric on E . It is a metric when $\|\cdot\|$ is a norm.

Proof. First, for any $x \in E$ one has $\|x - x\| = \|0_E\| = \|0_K 0_E\| = |0_K| \cdot \|0_E\| = 0$.

For any $(x, y) \in E \times E$, one has

$$\|y - x\| = \|(-1_K)(x - y)\| = |-1| \cdot \|x - y\| = \|x - y\|.$$

For any $(x, y, z) \in E^3$, one has

$$\|x - z\| = \|(x - y) + (y - z)\| \leq \|x - y\| + \|y - z\|.$$

Therefore, $(x, y) \mapsto \|x - y\|$ defines a pseudometric on E .

Suppose that $\|\cdot\|$ is a norm. If $(x, y) \in E \times E$ such that $x \neq y$, then $x - y \neq 0_E$, so that $\|x - y\| > 0$. Hence $(x, y) \mapsto \|x - y\|$ is a metric. $\square$

8.3.11 Proposition. *Let n be a positive integer and $(E_1, \|\cdot\|_1), \dots, (E_n, \|\cdot\|_n)$ be seminormed vector spaces. Let $E = E_1 \times \dots \times E_n$. Then the following mappings are seminorms on E :*

$$\begin{aligned} \|\cdot\|_{\ell^1} : E &\longrightarrow \mathbb{R}_{\geq 0}, (x_1, \dots, x_n) \longmapsto \|x_1\|_1 + \dots + \|x_n\|_n, \\ \|\cdot\|_{\ell^\infty} : E &\longrightarrow \mathbb{R}_{\geq 0}, (x_1, \dots, x_n) \longmapsto \max\{\|x_1\|_1, \dots, \|x_n\|_n\}. \end{aligned}$$

They are norms if all seminorms $\|\cdot\|_1, \dots, \|\cdot\|_n$ are norms.

Proof. For any $a \in K$ and any $x = (x_1, \dots, x_n) \in E$, one has

$$\|ax\|_{\ell^1} = \|(ax_1, \dots, ax_n)\|_{\ell^1} = \sum_{i=1}^n \|ax_i\|_i = \sum_{i=1}^n |a| \cdot \|x_i\|_i = |a| \sum_{i=1}^n \|x_i\|_i = |a| \|x\|_{\ell^1}.$$

$$\begin{aligned} \|ax\|_{\ell^\infty} &= \|(ax_1, \dots, ax_n)\|_{\ell^\infty} = \max_{i \in \{1, \dots, n\}} \|ax_i\|_i = \max_{i \in \{1, \dots, n\}} |a| \cdot \|x_i\|_i = \\ &= |a| \max_{i \in \{1, \dots, n\}} \|x_i\|_i = |a| \|(x_1, \dots, x_n)\|_{\ell^\infty} = |a| \|x\|_{\ell^\infty}. \end{aligned}$$

and for all $x = (x_1, \dots, x_n) \in E$ and $y = (y_1, \dots, y_n) \in E$

$$\|x + y\|_{\ell^1} = \|(x_1 + y_1, \dots, x_n + y_n)\|_{\ell^1} = \sum_{i=1}^n \|x_i + y_i\|_i \leq \sum_{i=1}^n (\|x_i\|_i + \|y_i\|_i)$$

$$\sum_{i=1}^n (\|x_i\|_i + \|y_i\|_i) = \|x\|_{\ell^1} + \|y\|_{\ell^1}$$

$$\|x + y\|_{\ell^\infty} = \|(x_1 + y_1, \dots, x_n + y_n)\|_{\ell^\infty} = \max_{i \in \{1, \dots, n\}} \|x_i + y_i\|_i \leq \max_{i \in \{1, \dots, n\}} (\|x_i\|_i + \|y_i\|_i)$$

$$\max_{i \in \{1, \dots, n\}} (\|x_i\|_i + \|y_i\|_i) \leq \|x\|_{\ell^\infty} + \|y\|_{\ell^\infty}$$

$\square$

8.3.12 Example. Let $(V, \|\cdot\|)$ be a seminormed vector space over a valued field $(K, |\cdot|)$, W be a vector space over K and $f : W \rightarrow V$ be a K -linear mapping. We denote by $\|\cdot\|_f$ the mapping $W \rightarrow \mathbb{R}_{\geq 0}$ defined as

$$\forall x \in W \quad \|x\|_f := \|f(x)\|.$$

We check that $\|\cdot\|_f$ is a seminorm on W . For any $a \in K$ any $x \in W$ one has

$$\|ax\|_f = \|f(ax)\| = \|af(x)\| = |a| \cdot \|f(x)\| = |a| \cdot \|x\|_f.$$

For any $(x, y) \in W \times W$ one has

$$\|x + y\|_f = \|f(x + y)\| = \|f(x) + f(y)\| \leq \|f(x)\| + \|f(y)\| = \|x\|_f + \|y\|_f.$$

We call $\|\cdot\|_f$ the *seminorm induced by the K -linear mapping f* .

8.3.13 Example. Let $(V, \|\cdot\|)$ be a seminormed vector space over a valued field $(K, |\cdot|)$, E be a vector space over K and $f : V \rightarrow E$ be a surjective K -linear mapping. We denote by $\|\cdot\|_\pi$ the mapping $E \rightarrow \mathbb{R}_{\geq 0}$ defined as

$$E \ni \alpha \mapsto \inf_{x \in \pi^{-1}(\{\alpha\})} \|x\|.$$

For any $\lambda \in K$, $\alpha \in E$

$$\|\lambda\alpha\|_\pi = \inf_{x \in \pi^{-1}(\{\alpha\})} \|\lambda x\| = \inf_{x \in \pi^{-1}(\{\alpha\})} |\lambda| \cdot \|x\| = |\lambda| \cdot \|\alpha\|_\pi.$$

For any $(\alpha, \beta) \in E \times E$

$$\begin{aligned} \|\beta + \alpha\|_\pi &= \inf_{z \in \pi^{-1}(\{\alpha + \beta\})} \|z\| = \inf_{(x, y) \in \pi^{-1}(\{\alpha\}) \times \pi^{-1}(\{\beta\})} \|x + y\| \leq \\ &\leq \inf_{(x, y) \in \pi^{-1}(\{\alpha\}) \times \pi^{-1}(\{\beta\})} (\|x\| + \|y\|) = \inf_{x \in \pi^{-1}(\{\alpha\})} \|x\| + \inf_{y \in \pi^{-1}(\{\beta\})} \|y\| \end{aligned}$$

and

$$\inf_{x \in \pi^{-1}(\{\alpha\})} \|x\| + \inf_{y \in \pi^{-1}(\{\beta\})} \|y\| = \|\alpha\|_\pi + \|\beta\|_\pi.$$

Hence, $\|\cdot\|_\pi$ is a seminorm on E , called the *quotient seminorm of $\|\cdot\|$ by the surjective homomorphism π* .

8.4 Continuity

8.4.1 Definition. Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces, f be a function from X to Y , and $p \in \text{Dom}(f)$. If for any neighbourhood U of $f(p)$, there exists a neighbourhood V of p , such that $f(V) \subseteq U$, we say that the function f is continuous at p . If f is continuous at any $p \in \text{Dom}(f)$, then we say that f is continuous.

8.4.2 Remark. Let $\mathcal{V}_{f(p)}(\mathcal{T}_Y)$ and $\mathcal{V}_p(\mathcal{T}_X)$ be filters of neighbourhoods of $f(p)$ and p respectively. Let $\mathcal{B}_p$ be a filter basis that generates $\mathcal{V}_p(\mathcal{T}_X)$ and let $S_{f(p)}$ be a subset of $U \in \mathcal{V}_{f(p)}(\mathcal{T}_Y)$ that generates $U \in \mathcal{V}_{f(p)}(\mathcal{T}_Y)$. The continuity of f at p is equivalent to

$$\forall U \in S_{f(p)}, \exists V \in \mathcal{B}_p \quad f(V) \subseteq U.$$

in the case where X and Y are metric spaces this condition becomes

$$\forall \epsilon > 0 \in S_{f(p)} \exists \delta > 0 \forall x \in \text{Dom}(f) \quad d_X(x, p) < \delta \implies d_Y(f(x), f(p)) < \epsilon.$$

8.4.3 Example. Let $(X, \mathcal{T}_X)$ be a topological space. Then $Id_X : X \rightarrow X, x \mapsto x$ is continuous. If $x_0 \in X$, then the constant mapping $x_0 : X \rightarrow X, x \mapsto x_0$ is also continuous.

8.4.4 Remark. Notice that $Id_X : (X, \mathcal{T}_X) \rightarrow (X, \mathcal{T}'_X)$ does not have to be continuous when $\mathcal{T}_X$ and $\mathcal{T}'_X$ are different (for example when $\mathcal{T}'_X$ is discrete and $\mathcal{T}_X$ is arbitrary but non-discrete). The constant mapping is continuous independently of the choice of topologies.

8.4.5 Definition. Assume that $\mathcal{V}_p(\mathcal{T}_X)|_{\text{Dom}(f)}$ is non-degenerate as a filter. We say, that $f(x)$ converges to l when $x \in \text{Dom}(f)$ tends to p and write

$$\lim_{x \in \text{Dom}(f), x \rightarrow p} f(x) = l$$

when

$$\lim_{\mathcal{V}_p(\mathcal{T}_X)|_{\text{Dom}(f)}} f = l.$$

8.4.6 Proposition. f is continuous at p if and only if

$$\lim_{x \in \text{Dom}(f), x \rightarrow p} f(x) = f(p),$$

i.e. $f(x)$ converges to $f(p)$ when $x \in \text{Dom}(f)$ tends to p .

Proof. Notice, that we can replace $(X, \mathcal{T}_X)$ by $(\text{Dom}(f), \mathcal{T}_X|_{\text{Dom}(f)})$, where $\mathcal{T}_X|_{\text{Dom}(f)} = \{U \cap \text{Dom}(f) | U \in \mathcal{T}_X\}$, which is also a topological space. Therefore, one can assume, that f is a mapping. First, assume that

$$\lim_{x \in \text{Dom}(f), x \rightarrow p} f(x) = f(p).$$

Take a neighborhood W of $f(p)$. There exists $U \in \mathcal{V}_{f(p)}(\mathcal{T}_Y)$ such that $U \subset W$. Since $f(x)$ tends to $f(p)$ as x tends to p is equivalent to $\mathcal{V}_{f(p)}(\mathcal{T}_Y) \subseteq f_*(\mathcal{V}_p(\mathcal{T}_X))$, we have, that $f^{-1}(U) \in \mathcal{V}_p(\mathcal{T}_X) \subseteq \mathcal{T}_X$.

Conversely, assume that f is continuous at p . Then, for every $U \in \mathcal{V}_{f(p)}(\mathcal{T}_Y)$, there exists $V \in \mathcal{V}_p(\mathcal{T}_X)$, such that $f(V) \subseteq U$, i.e. $V \subseteq f^{-1}(U)$. Since $\mathcal{V}_p(\mathcal{T}_X)$ is a filter, then $f^{-1}(U) \in \mathcal{V}_p(\mathcal{T}_X)$.

□

8.4.7 Proposition. *Let $(X, \mathcal{T}_X), (Y, \mathcal{T}_Y), (Z, \mathcal{T}_Z)$ be topological spaces, f a function from X to Y , g a function from Y to Z , $p \in \text{Dom}(g \circ f)$. Assume that f is continuous at p and g is continuous at $f(p)$. Then $g \circ f$ is continuous at p .*

Proof. Let U be a neighbourhood of $g(f(p))$. Since g is continuous at $f(p)$, there exists a neighbourhood V of $f(p)$, such that $g(V) \subseteq U$. Since f is continuous at p , there exists a neighbourhood W of p , such that $f(W) \subseteq V$. Hence

$$g(f(W)) \subseteq g(V) \subseteq U.$$

One can therefore deduce that $g \circ f$ is continuous at p . $\square$

8.4.8 Theorem. *Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces, f be a function from X to Y . Consider the following conditions*

(i) *f is continuous at p .*

(ii) $\forall (x_n)_{n \in \mathbb{N}} \in \text{Dom}(f)^{\mathbb{N}}$ if $\lim_{n \rightarrow +\infty} x_n = p$ then $\lim_{n \rightarrow +\infty} f(x_n) = f(p)$

One has (i) $\implies$ (ii). Moreover, if p has a countable neighborhood basis, then (ii) $\implies$ (i).

Proof. (i) $\implies$ (ii) Let $(x_n)_{n \in \mathbb{N}} \in \text{Dom}(f)^{\mathbb{N}}$ be such that $\lim_{n \rightarrow +\infty} x_n = p$. Since f is continuous at p , for any neighborhood U of $f(p)$, there exists a neighborhood V of p , such that $f(V) \subseteq U$. Since $\lim_{n \rightarrow +\infty} x_n = p$, there exists $N \in \mathbb{N}$ such that $\forall n \in \mathbb{N}_{>N}$, $x_n \in V$. Hence, $\forall n \in \mathbb{N}_{>N}$ $f(x_n) \in U$. Therefore, $(f(x_n))_{n \in \mathbb{N}}$ converges to $f(p)$.

$\neg(\text{i}) \implies \neg(\text{ii})$ (under the assumption that p has a countable neighborhood basis) Let $(W_n)_{n \in \mathbb{N}}$ be a sequence formed by a countable neighborhood basis of p . For $n \in \mathbb{N}$ let

$$V_n := W_1 \cap \cdots \cap W_n.$$

Notice, that $\{V_n | n \in \mathbb{N}\}$ is also a basis of neighborhood of p . If f wasn't continuous at p , there would exist a neighborhood U of $f(p)$ such that

$$\forall n \in \mathbb{N} \quad f(V_n) \not\subseteq U.$$

We pick $x_n \in V_n$ in a way that $f(x_n) \notin U$. For any neighborhood V of p , there exists $N \in \mathbb{N}$ such that $V_N \subset V$, and consequently $x_v \in V$ for any $n \in \mathbb{N}$. Hence $(x_n)_{n \in \mathbb{N}}$ converges to p . However, since $f(x_n) \notin U$ for any $n \in \mathbb{N}$, $(f(x_n))_{n \in \mathbb{N}}$ does not converge to $f(p)$. $\square$

8.4.9 Proposition. *Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces, f be a mapping from X to Y , S be a subset of $\mathcal{T}_Y$ that generates $\mathcal{T}_Y$. The following statements are equivalent*

1. f is continuous
2. $\forall U \in \mathcal{T}_Y \quad f^{-1}(U) \in \mathcal{T}_X$
3. $\forall U \in S \quad f^{-1}(U) \in \mathcal{T}_X$

Proof. 1. $\implies$ 2. For any $p \in f^{-1}(U)$, one has $f(p) \in U$. Hence, there is a neighborhood V_p of p such that $f(V_p) \subseteq U$ or, equivalently, $V_p \subseteq f^{-1}(U)$ (since f is a mapping). Therefore, $f^{-1}(U)$ is a neighborhood of all $p \in f^{-1}(U)$. We deduce, that $f^{-1}(U) \in \mathcal{T}_X$ using the following:

8.4.10 Lemma. *Let $(X, \mathcal{T}_X)$ be a topological space, $V \subseteq X$. If $\forall p \in V$, V is a neighborhood of p , then $V \in \mathcal{T}_X$.*

Indeed, $\forall p \in V$, there exists $W_p \in \mathcal{T}_X$ such that $p \in W_p \subseteq V$. Hence

$$V = \bigcup_{p \in V} \{p\} \subseteq \bigcup_{p \in V} W_p \subseteq V$$

which lets us conclude that

$$V = \bigcup_{p \in V} W_p \in \mathcal{T}_X.$$

3. is a particular case of 2.

3. $\implies$ 1. Let $\mathcal{T}'_Y = \{U \in \mathcal{P}(Y) | f^{-1}(U) \in \mathcal{T}_X\}$. Then $\{\emptyset, Y\} \subseteq \mathcal{T}'_Y$ by definition. If $(U_1, U_2) \in \mathcal{T}'_Y \times \mathcal{T}'_Y$, then $f^{-1}(U_1 \cap U_2) = f^{-1}(U_1) \cap f^{-1}(U_2) \in \mathcal{T}_X$. If $(U_i)_{i \in I} \in (\mathcal{T}'_Y)^I$, then $f^{-1}(\bigcup_{i \in I} U_i) = \bigcup_{i \in I} f^{-1}(U_i) \in \mathcal{T}_X$. Thus, $\mathcal{T}'_Y$ is a topology. We know, that $S \subseteq \mathcal{T}'_Y$, and therefore also $\mathcal{T}_Y \subseteq \mathcal{T}'_Y$. $\square$

8.5 Initial topology

8.5.1 Definition. Let X be a set, $((Y_i, \mathcal{T}_i))_{i \in I}$ be a family of topological spaces, $(f_i : X \rightarrow Y_i)_{i \in I}$ be a family of mappings. The *initital topology* on X induced by $(f_i)_{i \in I}$ is the topology generated by $\bigcup_{i \in I} \{f^{-1}(U_i) | U_i \in \mathcal{T}_i\}$. It is the coarsest topology on X making all f_i continuous.

8.5.2 Proposition. *Let X be a set, $((Y_i, \mathcal{T}_i))_{i \in I}$ be a family of topological spaces, $(f_i : X \rightarrow Y_i)_{i \in I}$ be a family of mappings and $\mathcal{T}$ be the initial topology on X induced by $(f_i)_{i \in I}$.*

1. The family

$$\mathcal{B} = \left\{ \bigcap_{j \in J} f_j^{-1}(U_j) \mid J \subset I \text{ is finite, } (U_j)_{j \in J} \in \prod_{j \in J} \mathcal{T}_j \right\}$$

is a topological basis generating $\mathcal{T}$.

2. Let $(Z, \mathcal{T}_Z)$ be a topological space, h be a function from Z to X and $p \in \text{Dom}(h)$. The function h is continuous at p if and only if $\forall i \in I$, $f_i \circ h$ is continuous at p .

Proof. 1. Let $S = \bigcup_{i \in I} \{f_i^{-1}(U_i) \mid U_i \in \mathcal{T}_i\}$. It suffices to show that $\mathcal{B}$ identifies with the set $\mathcal{B}'$ of the intersections of finitely many elements of S . Clearly $\mathcal{B} \subseteq \mathcal{B}'$.

To show the opposite inclusion consider the following collection of indices $i_1, \dots, i_n$ from I (there might be repetitions for the same element). Let $J = \{i_1, \dots, i_n\} \subseteq I$ (the repetitions disappear when we consider the collection as a set). For any $j \in J$ put $A_j = \{k \in \{1, \dots, n\} \mid i_k = j\}$ and $W_j = \bigcap_{k \in A_j} U_{i_k}$. Then

$$\bigcap_{k=1}^n f_{i_k}^{-1}(U_{i_k}) = \bigcap_{j \in J} f_j^{-1}(W_j) \in \mathcal{B}.$$

2. Assume, that h is continuous and f_i is continuous. Then $f_i \circ h$ is continuous. Now suppose that $f_i \circ h$ is continuous for any $i \in I$. Then

$$\forall i \in I \forall U_i \in \mathcal{T}_i \quad (f_i \circ h)^{-1}(U_i) = h^{-1}(f_i^{-1}(U_i)).$$

One can therefore see, that for any $V \in S$, $h^{-1}(V) \in \mathcal{T}_Z$. Since $\mathcal{T}$ is generated by S , h is continuous. $\square$

8.5.3 Example. Let $(X_i, \mathcal{T}_i)$ be topological spaces $X := \prod_{i \in I} X_i$, $\pi_i : X \rightarrow X_i$ be the projection, namely $\pi_i((x_j)_{j \in I}) = x_i$. The initial topology on X induced by $(\pi_i)_{i \in I}$ is called the *product topology*.

8.6 Uniform continuity

8.6.1 Definition. Let (X, d_X) and (Y, d_Y) be semimetric spaces, and $f : X \rightarrow Y$ be a function from X to Y , $\alpha \in \mathbb{R}_{\geq 0}$. If

$$\forall (x_1, x_2) \in \text{Dom}(f)^2 \quad d_Y(f(x_1), f(x_2)) \leq \alpha \cdot d_X(x_1, x_2)$$

then we say that f is α -Lipschitzian. If there exists $\alpha \in \mathbb{R}_{\geq 0}$ such that f is α -Lipschitzian, then we say that f is *Lipschitzian*.

8.6.2 Definition. Let (X, d_X) and (Y, d_Y) be semimetric spaces, and $f : X \rightarrow Y$ be a function from X to Y . If

$$\forall \epsilon > 0, \exists \delta > 0, \forall (x_1, x_2) \in \text{Dom}(f)^2 \quad d_x(x_1, x_2) < \delta \implies d_Y(f(x_1), f(x_2)) < \epsilon$$

then we say that f is *uniformly continuous*

8.6.3 Proposition. Let (X, d_X) and (Y, d_Y) be semimetric spaces, and $f : X \rightarrow Y$ be a function from X to Y . If f is uniformly continuous, then f is continuous.

Proof. Let $p \in \text{Dom}(f)$. We know that for any $\epsilon > 0$ there exists $\delta > 0$ such that

$$\forall (x_1, x_2) \in \text{Dom}(f)^2 \quad d_x(x_1, x_2) < \delta \implies d_Y(f(x_1), f(x_2)) < \epsilon.$$

In particular,

$$\forall (x) \in \text{Dom}(f) \quad d_x(x, p) < \delta \implies d_Y(f(x), f(p)) < \epsilon$$

□

8.6.4 Proposition. Let $(V, \|\cdot\|)$ be a seminormed vector space over a valued field $(K, |\cdot|)$.

1. For any $a \in V$, the mapping

$$\tau_a : V \rightarrow V \quad \tau_a(x) := x + a$$

is 1-Lipschitzian.

2. For any $\lambda \in K$, the mapping

$$m_\lambda : V \rightarrow V \quad m_\lambda(x) := \lambda \cdot x$$

is $|\lambda|$ -Lipschitzian.

3. The mapping $\|\cdot\| : V \rightarrow \mathbb{R}_{\geq 0}$ is 1-Lipschitzian.

Proof. 1. $\|\tau_a(x) - \tau_a(y)\| = \|(x + a) - (y + a)\| = \|x - y\|$

2. $\|m_\lambda(x) - m_\lambda(y)\| = \|\lambda \cdot x - \lambda \cdot y\| = |\lambda| \cdot \|x - y\|.$

3. We have

$$\begin{aligned} \|x\| &= \|(x - y) + y\| \leq \|x - y\| + \|y\| \\ \|y\| &= \|(y - x) + x\| \leq \|y - x\| + \|x\| \end{aligned}$$

and therefore

$$|\|x\| - \|y\|| \leq \|x - y\|.$$

□

8.6.5 Definition. Let $(E, \|\cdot\|_E)$ and $(F, \|\cdot\|_F)$ be seminormed vector spaces over a valued field $(K, |\cdot|)$. Let $\varphi : E \rightarrow F$ be a K -linear mapping. We define $\|\varphi\| \in [0, +\infty]$ as

$$\|\varphi\| := \sup_{x \in E, \|x\|_E \neq 0} \frac{\|\varphi(x)\|_F}{\|x\|_E}.$$

In the case where $\|x\|_E = 0$ for any $x \in E$, by convention $\|\varphi\| = 0$.

If $\|\varphi\| < +\infty$, we say that φ is *bounded*. We denote by $\mathcal{L}(E, F)$ the set of all bounded K -linear mappings from E to F .

8.6.6 Proposition.

1. For any $\varphi \in \mathcal{L}(E, F)$ the mapping φ is $\|\varphi\|$ -Lipschitzian. In particular, φ is continuous.
2. Suppose that there exists $\lambda \in K$ such that $|\lambda| > 1$. If $\varphi : E \rightarrow F$ is K -linear mapping that is continuous at 0_E then $\varphi \in \mathcal{L}(E, F)$.

Proof. 1. For any $(x, y) \in E \times E$

$$\|\varphi(x) - \varphi(y)\|_F = \|\varphi(x - y)\|_F \leq \|\varphi\| \cdot \|x - y\|_E$$

2. Since $B(0_F, 1) = \{\alpha \in F \mid \|\alpha\|_F < 1\}$ is a neighborhood of 0_F , there exists $\epsilon > 0$ such that $\varphi(\overline{B}(0_E, \epsilon)) \subseteq B(0_F, 1)$, where $\overline{B}(0_E, \epsilon) := \{x \in E \mid \|x\|_E \leq \epsilon\}$. Let $x \in E \setminus \{0_E\}$. Since $|\lambda| > 1$, there exists $n \in \mathbb{N}$ such that

$$\|\lambda^n x\|_E = |\lambda|^n \|x\|_E < \epsilon \quad \text{and} \quad \|\lambda^{n+1} x\|_E = |\lambda|^{n+1} \|x\|_E \geq \epsilon.$$

Thus

$$\|\varphi(x)\|_F = \|\lambda^{-n} \varphi(\lambda^n x)\|_F \leq |\lambda|^{-n} \leq \frac{|\lambda|}{\epsilon} \|x\|_E.$$

Hence $\|\varphi\| \leq \frac{|\lambda|}{\epsilon}$.

□

8.6.7 Proposition. Let $(E, \|\cdot\|_E)$ and $(F, \|\cdot\|_F)$ be seminormed vector spaces over a valued field $(K, |\cdot|)$. Then $\mathcal{L}(E, F)$ is a vector subspace of F^E , and $\|\cdot\|$ is a seminorm on $\mathcal{L}(E, F)$ called the operator seminorm.

Proof. Let φ and ψ be two K -linear mappings from E to F . For any $x \in E$ such that $\|x\|_E \neq 0$, one has

$$\begin{aligned} \|(\varphi + \psi)(x)\|_F &= \|\varphi(x) + \psi(x)\|_F \leq \|\varphi(x)\|_F + \|\psi(x)\|_F \leq \\ &\leq \|\varphi\| \|x\|_E + \|\psi\| \|x\|_E \leq (\|\varphi\| + \|\psi\|) \|x\|_E \end{aligned}$$

Hence

$$\|\varphi + \psi\| \leq \|\varphi\| + \|\psi\|.$$

In particular, if $\varphi \in \mathcal{L}(E, F)$ and $\psi \in \mathcal{L}(E, F)$, then $\varphi + \psi \in \mathcal{L}(E, F)$.

Let $\lambda \in K^\times$ and $\varphi \in \mathcal{L}(E, F)$. For any $x \in E$ such that $\|x\|_E \neq 0$, one has

$$\|(\lambda\varphi)(x)\|_F = \|\lambda\varphi(x)\|_F = |\lambda| \cdot \|\varphi(x)\|_F \leq |\lambda| \cdot \|\varphi\| \cdot \|x\|_E$$

so that

$$\|\lambda\varphi\| \leq |\lambda| \cdot \|\varphi\|. \quad (8.1)$$

This inequality shows that $\lambda\varphi \in \mathcal{L}(E, F)$. We can conclude that $\mathcal{L}(E, F)$ is a vector subspace of F^E . Moreover, if we apply 8.1 to λ^{-1} and $\lambda\varphi$, we get

$$\|\varphi\| \leq |\lambda^{-1}| \cdot \|\lambda\varphi\| = |\lambda|^{-1} \cdot \|\lambda\varphi\| \text{ so that } |\lambda| \cdot \|\varphi\| \leq \|\lambda\varphi\|$$

Therefore $|\lambda| \cdot \|\varphi\| = \|\lambda\varphi\|$. □

8.6.8 Definition. Let E be a vector space over a valued field $(K, |\cdot|)$, and $\|\cdot\|_1$ and $\|\cdot\|_2$ be seminorms on E . We say that $\|\cdot\|_1$ and $\|\cdot\|_2$ are *equivalent* if there exist positive real numbers c_1 and c_2 such that

$$\forall x \in E \quad c_1 \|\cdot\|_1 \leq \|\cdot\|_2 \leq c_2 \|\cdot\|_1$$

8.6.9 Proposition. Let E be a vector space over a valued field $(K, |\cdot|)$, and $\|\cdot\|_1$ and $\|\cdot\|_2$ be seminorms on E . If $\|\cdot\|_1$ and $\|\cdot\|_2$ are equivalent, then they define the same topology.

Proof. Let $\mathcal{T}_1$ and $\mathcal{T}_2$ be topologies defined by $\|\cdot\|_1$ and $\|\cdot\|_2$ respectively. Then

$$Id_E : (E, \mathcal{T}_1) \rightarrow (E, \mathcal{T}_2)$$

is continuous as it is bounded. We deduce, that $\mathcal{T}_2 \subseteq \mathcal{T}_1$. Similarly

$$Id_E : (E, \mathcal{T}_2) \rightarrow (E, \mathcal{T}_1)$$

is continuous, and hence $\mathcal{T}_1 \subseteq \mathcal{T}_2$. □

8.6.10 Proposition. Let $n \in \mathbb{N}_{\geq 1}$ and (X_i, d_i) , $i \in \{1, \dots, n\}$ be semimetric spaces. Put

$$X = \prod_{i=1}^n X_i \quad \text{and} \quad d : X \times X \rightarrow \mathbb{R}$$

defined as

$$d((x_1, \dots, x_n), (y_1, \dots, y_n)) = \max_{i \in \{1, \dots, n\}} d_i(x_i, y_i).$$

Then d is a semimetric on X , and the topology induced by d is the product topology of those induced by $d_1, \dots, d_n$.

Proof. 1.

$$d((x_1, \dots, x_n), (x_1, \dots, x_n)) = \max_{i \in \{1, \dots, n\}} d_i(x_i, x_i) = 0$$

2.

$$\begin{aligned} d((x_1, \dots, x_n), (y_1, \dots, y_n)) &= \max_{i \in \{1, \dots, n\}} d_i(x_i, y_i) = \\ &= \max_{i \in \{1, \dots, n\}} d_i(y_i, x_i) = d((x_1, \dots, x_n), (y_1, \dots, y_n)) \end{aligned}$$

3.

$$\begin{aligned} d((x_1, \dots, x_n), (z_1, \dots, z_n)) &= \max_{i \in \{1, \dots, n\}} d_i(x_i, z_i) \leq \max_{i \in \{1, \dots, n\}} d_i(x_i, y_i) + \\ &\max_{i \in \{1, \dots, n\}} d_i(x_i, z_i) = d((x_1, \dots, x_n), (y_1, \dots, y_n)) + d((y_1, \dots, y_n), (z_1, \dots, z_n)) \end{aligned}$$

Let $\mathcal{T}$ be the product topology of $\mathcal{T}_1, \dots, \mathcal{T}_n$, and $\mathcal{T}_d$ be the topology induced by d . For any $i \in \{1, \dots, n\}$, let $\pi_i : X \rightarrow X_i$ be the projection mapping. For all $x = (x_1, \dots, x_n), y = (y_1, \dots, y_n)$ in X one has

$$d(x, y) = \max_{i \in \{1, \dots, n\}} d_i(x_i, y_i) = \max_{i \in \{1, \dots, n\}} d_i(\pi_i(x), \pi_i(y))$$

One easily sees, that $d_i(\pi_i(x), \pi_i(y)) \leq d(x, y)$ and hence deduces, that π_i is 1-Lipschitzian. We deduce that $\pi_i : (X, \mathcal{T}_d) \rightarrow (X_i, \mathcal{T}_i)$ is continuous, which implies, using 8.5.2 and the definition of the product topology, that $Id_X : (X, \mathcal{T}_d) \rightarrow (X, \mathcal{T})$ is continuous and that $\mathcal{T} \subseteq \mathcal{T}_d$.

For the opposite inclusion note that $\mathcal{T}_d$ is generated by open balls, namely subsets of the form

$$\begin{aligned} B((p_1, \dots, p_n), \epsilon) &:= \{(x_1, \dots, x_n) \in X \mid d((p_1, \dots, p_n), (x_1, \dots, x_n)) < \epsilon\} = \\ &= \prod_{i=1}^n \{x_i \in X_i \mid d(p_i, x_i) < \epsilon\} = \prod_{i=1}^n B(p_i, \epsilon) \in \mathcal{T} \end{aligned}$$

Therefore $\mathcal{T}_d \subseteq \mathcal{T}$. □

8.6.11 Remark. Let $(E_i, \|\cdot\|_i), i \in \{1, \dots, n\}$ be seminormed vector spaces over a valued field $(K, |\cdot|)$, $E := \bigoplus_{i=1}^n E_i$. Then $\|\cdot\|_{\ell^\infty}$ induces the product topology. Since

$$\|\cdot\|_{\ell^\infty} \leq \|\cdot\|_{\ell^1} \leq n\|\cdot\|_{\ell^\infty}$$

we can conclude that $\|\cdot\|_{\ell^1}$ also induces the product topology.

8.7 Closed subsets

8.7.1 Definition. Let $(X, \mathcal{T})$ be a topological space. If a subset U of X belongs to $\mathcal{T}$, we say that U is *open*. If F is a subset of X such that $X \setminus F$ is open, we say that F is *closed*.

8.7.2 Remark. Let $(X, \mathcal{T})$ be a topological space. We deduce from the axiom of topology the following properties of closed subsets.

- (1) $\emptyset$ and X are closed subsets of X .
- (2) If F_1 and F_2 are two closed subsets of X , then $F_1 \cup F_2$ is a closed subset of X .
- (3) If $(F_i)_{i \in I}$ is a non-empty family of subsets of X , then $\bigcap_{i \in I} F_i$ is a closed subset of X .

8.7.3 Example. In the set $\mathbb{R}$ of real numbers (equipped with the usual topology), intervals of the form $[a, b]$ with $a \leq b$ are closed. In fact, the complement of $[a, b]$ is $]-\infty, a[\cup]b, +\infty[$, which is an open subset of $\mathbb{R}$. Similarly, intervals of the form $]-\infty, a]$ and $[a, +\infty[$ are also closed subsets of $\mathbb{R}$.

8.7.4 Definition. Let $(X, \mathcal{T})$ be a topological space and Y be a subset of X . The adherent points of the principal filter $\mathcal{F}_Y$ are called the *adherent points* of Y . Note that an element p of X is an adherent point of Y if, for any neighbourhood V of x , the intersection $Y \cap V$ is not empty. We denote by $\overline{Y}$ the set of all adherent points of X , called the *closure* of Y .

8.7.5 Proposition. Let $(X, \mathcal{T})$ be a topological space and Y be a subset of X . Then the closure of Y is equal to the intersection of all closed subsets of X containing Y .

Proof. Let $p \in \overline{Y}$. Let F be a closed subset of X . If $p \notin F$, then $p \in X \setminus F$. Since $X \setminus F$ is open, it is a neighbourhood of p . However, its intersection with F is empty, which leads to a contradiction. Hence $p \in F$.

Conversely, suppose that x is not an adherent point of Y , then there exists an open neighbourhood V of x such that $V \cap Y = \emptyset$. Let $F = X \setminus V$. One has $Y \subseteq F$ but $x \notin F$. □

8.7.6 Definition. Let $(X, \mathcal{T})$ be a topological space and Y be a subset of X . We call the *interior* of Y the set of all element $y \in Y$ such that Y is a neighbourhood of Y . We denote the interior of Y as Y° .

8.7.7 Proposition. *Let $(X, \mathcal{T})$ be a topological space and Y be a subset of X . Then Y° is the largest open subset of X contained in Y . Moreover, the following equalities hold:*

$$X \setminus Y^\circ = \overline{X \setminus Y}.$$

Proof. If U is an open subset of X that is contained in Y . For any $x \in U$, U is a neighbourhood of x . Hence $x \in Y^\circ$. Hence Y° contains any open subset of X contained in Y .

Let p be an element of Y° . There exists an open neighbourhood V of p that is contained in Y . By the above argument, one has $V \subseteq Y^\circ$, namely Y° is a neighbourhood of p . Therefore, Y° is an open subset of X .

By the first statement, we have

$$Y^\circ = \bigcup_{U \in \mathcal{T}, U \subseteq Y} U.$$

Hence

$$X \setminus Y = \bigcap_{U \in \mathcal{T}, U \subseteq Y} (X \setminus U) = \bigcap_{\substack{F \text{ closed} \\ X \setminus Y \subseteq F}} F = \overline{X \setminus Y}$$

□

8.7.8 Proposition. *Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces and $f : X \rightarrow Y$ be a mapping. Then f is continuous if and only if, for any closed subset F of Y , the inverse image $f^{-1}(F)$ is closed in X .*

Proof. “ $\implies$ ”: Suppose that f is continuous. Let F be a closed subset of Y and let $U = X \setminus f^{-1}(F)$. Then U is an open subset of X . Hence $f^{-1}(U)$ is an open subset of X . Therefore, $f^{-1}(F) = X \setminus f^{-1}(U)$ is a closed subset of X .

“ $\impliedby$ ”: Suppose that the inverse image of any closed subset of Y is a closed subset of X . Let U be an open subset of Y . Then $F := Y \setminus U$ is a closed subset of Y . Hence

$$f^{-1}(U) = X \setminus f^{-1}(F)$$

is an open subset of X . Therefore, f is a continuous mapping. □

8.7.9 Example. Let (X, d) be a metric space and Y be a non-empty subset of X . For any $x \in X$, we define

$$d(x, Y) := \inf_{y \in Y} d(x, y) \in \mathbb{R}_{\geq 0}.$$

Note that, if x and x' are elements of X , for any $y \in Y$, by the triangle inequality one has

$$d(x, Y) - d(x', y) \leq d(x, y) - d(x', y) \leq d(x, x').$$

Taking the supremum with respect to y , we get

$$d(x, Y) - d(x', Y) \leq d(x, x').$$

By symmetry between x and x' , we obtain

$$d(x', Y) - d(x, Y) \leq d(x, x').$$

Therefore

$$|d(x, Y) - d(x', Y)| \leq d(x, x').$$

This inequality shows that the mapping

$$d(\cdot, Y) : X \longrightarrow \mathbb{R}, \quad (x \in X) \longmapsto d(x, Y)$$

is 1-Lipschitzian, hence is continuous.

For any $\varepsilon > 0$, we let

$$B(Y, \varepsilon) := \{x \in X \mid d(x, Y) < \varepsilon\}, \quad \overline{B}(Y, \varepsilon) := \{x \in X \mid d(x, Y) \leq \varepsilon\}.$$

Note that they are respectively the inverse image of the intervals $] -\varepsilon, +\infty[$ and $[-\varepsilon, \varepsilon]$. Therefore, $B(Y, \varepsilon)$ is an open subset of X , and $\overline{B}(Y, \varepsilon)$ is a closed subset of X . In the case where $Y := \{y\}$ is a one point set, the sets $B(Y, \varepsilon)$ and $\overline{B}(Y, \varepsilon)$ are denoted as $B(y, \varepsilon)$ and $\overline{B}(y, \varepsilon)$ respectively.

8.7.10 Definition. Let $(X, \mathcal{T})$ be a topological space. If the diagonal subset

$$\Delta_X := \{(x, x) \mid x \in X\}$$

of the product space $X \times X$ equipped with the product topology is closed, we say that the topological space $(X, \mathcal{T})$ is a *Hausdorff space*, or is *separated*.

8.7.11 Proposition. *A topological space $(X, \mathcal{T})$ is separated if and only if, for any $(x, y) \in X \times X$ such that $x \neq y$, there exists open neighbourhoods U of x and V of y such that $U \cap V = \emptyset$.*

Proof. By definition, the diagonal Δ_X is closed if and only if, for any $(x, y) \in X \times X$ such that $x \neq y$, there exists a neighbourhood of (x, y) that has the empty intersection with Δ_X . Since

$$\{U \times V \mid (U, V) \in \mathcal{T} \times \mathcal{T}\}$$

forms a topological basis of $X \times X$, we obtain that Δ_X is closed if and only if, for any $(x, y) \in X \times X$ such that $x \neq y$, there exists open neighbourhoods U of x and V of y such that $U \cap V = \emptyset$.

□

8.7.12 Proposition. *Let $(X, \mathcal{F})$ be a Hausdorff topological space. Let $\mathcal{F}$ be a non-degenerate filter on X . If $\mathcal{F}$ has a limit point, then its limit point is unique.*

Proof. We reason by contradiction. Let x and y are two limit points of X . Assume that $x \neq y$. Let U and V be respectively neighbourhoods of x and y such that $U \cap V = \emptyset$. Since both x and y are limit points of $\mathcal{F}$, one has $\{U, V\} \subseteq \mathcal{F}$, this contradicts the assumption that $\mathcal{F}$ is non-degenerate. $\square$

8.7.13 Proposition. *A metric space (X, d) is Hausdorff.*

Proof. Let x and y be elements of X such that $x \neq y$. Then $d(x, y) > 0$. Let $\varepsilon = d(x, y)/2$. Then $B(x, \varepsilon) \cap B(y, \varepsilon)$ should be empty. In fact, if there exists $z \in B(x, \varepsilon) \cap B(y, \varepsilon)$, then

$$d(x, y) \leq d(x, z) + d(z, y) < 2\varepsilon = d(x, y),$$

which leads to a contradiction. Hence (X, d) is a Hausdorff space. $\square$

8.7.14 Proposition. *Let $(X, \mathcal{F})$ be a topological space, Y be a subset of X and $p \in X$. Let Z be a set, $\mathcal{F}$ be a non-degenerate filter on Z and $f : Z \rightarrow X$ be a mapping such that $f(Z) \subseteq Y$. If p is a limit of f along $\mathcal{F}$, then $p \in \overline{Y}$.*

Proof. By definition, for any neighbourhood V of p , one has $f^{-1}(V) \in \mathcal{F}$. In particular, $f^{-1}(V) \neq \emptyset$. Therefore, $f(f^{-1}(V)) \neq \emptyset$.

By definition, $f(f^{-1}(V)) \subseteq V$. Moreover, since $f(Z) \subseteq Y$, one has $f(f^{-1}(V)) \subseteq Y$. Therefore, $f(f^{-1}(V)) \subseteq V \cap Y$. This shows that $V \cap Y \neq \emptyset$. $\square$

8.7.15 Proposition. *Let $(X, \mathcal{F})$ be a topological space, Y be a subset of X and $p \in \overline{Y}$. If p has a countable neighbourhood basis, then there exists a sequence $(y_n)_{n \in \mathbb{N}}$ in Y that converges to p .*

Proof. If p has a countable neighbourhood basis, there then exists a decreasing sequence $(V_n)_{n \in \mathbb{N}}$ of neighbourhoods of p such that $\{V_n \mid n \in \mathbb{N}\}$ forms a neighbourhood basis of p . If p belongs to $\overline{Y}$, then $V_n \cap Y \neq \emptyset$ for any $n \in \mathbb{N}$. We could then take, for any $n \in \mathbb{N}$, an element $x_n \in Y \cap V_n$. Note that, for any $n \in \mathbb{N}$,

$$\{x_k \mid k \in \mathbb{N}, k \geq n\} \subseteq V_n.$$

Therefore, the sequence $(x_n)_{n \in \mathbb{N}}$ converges to p . $\square$

8.7.16 Proposition. *Let (X, d) be a semimetric space and Y be a closed subset of X . For any $x \in X \setminus Y$, one has $d(x, Y) > 0$.*

Proof. Since Y is closed, $X \setminus Y$ is open. There exists then $\varepsilon > 0$ such that $B(x, \varepsilon) \cap Y = \emptyset$. Therefore, for any $y \in Y$, one has $d(x, y) \geq \varepsilon$. $\square$

8.7.17 Corollary. Let $(V, \|\cdot\|)$ be a seminormed vector space over a valued field $(K, |\cdot|)$, and W be a closed vector subspace of V . Then the quotient seminorm $\|\cdot\|_Q$ on $Q := V/W$, defined as

$$\forall \alpha \in Q, \|\alpha\|_Q := \inf_{x \in V, [x] = \alpha} \|x\|.$$

is a norm.

Proof. Let α be a non-zero element of Q and x be an element of V such that $[x] = \alpha$. Since $\alpha \neq 0$, one has $x \notin W$. Since W is closed, one has

$$d(x, W) := \inf_{w \in W} \|x - w\| > 0.$$

Note that

$$\inf_{w \in W} \|x - w\| = \inf_{y \in V, [y] = \alpha} \|y\| = \|\alpha\|_Q.$$

Hence $\|\alpha\|_Q > 0$. □

8.8 Completeness

8.8.1 Definition. Let (E, d) be a semimetric space. For any subset A of E , we define the *diameter* of A as

$$\text{diam}(A) := \sup_{(x, y) \in A \times A} d(x, y) \in [0, +\infty].$$

In the case where $A = \emptyset$, by convention $\text{diam}(\emptyset)$ is set to be 0.

If $\text{diam}(A) < +\infty$, we say that A is *bounded*.

8.8.2 Remark. Let (E, d) be a semimetric space. If A and B are two subsets of E such that $A \subseteq B$, then

$$\text{diam}(A) \leq \text{diam}(B).$$

In particular, if B is bounded, then A is also bounded.

8.8.3 Example. Let (E, d) be a semimetric space and p be an element of E . Let r be a positive real number. If x and y are two elements of $\overline{B}(p, r)$, then one has

$$d(x, y) \leq d(x, p) + d(p, y) \leq r + r = 2r.$$

Hence

$$\text{diam}(\overline{B}(p, r)) \leq 2r.$$

In particular, $\overline{B}(p, r)$ is bounded.

8.8.4 Proposition. *Let (E, d) be a semimetric space, and A be a subset of E . The following conditions are equivalent:*

- (1) A is bounded;
- (2) for any $p \in A$, there exists $r > 0$ such that $A \subseteq \overline{B}(p, r)$;
- (3) A is empty, or there exists $p \in A$ and $r > 0$ such that $A \subseteq \overline{B}(p, r)$.

Proof. “(1) $\implies$ (2)”: Let r be positive real number such that $r \geq \text{diam}(A)$. By definition, for any $p \in A$, one has

$$\forall x \in A, d(p, x) \leq r.$$

Therefore, $A \subseteq \overline{B}(p, r)$.

“(2) $\implies$ (3)” is trivial.

“(3) $\implies$ (1)”: By definition, the empty set is bounded. Suppose that there exists $p \in A$ and $r > 0$ such that $A \subseteq \overline{B}(p, r)$. We have seen in Example 8.8.3 that $\overline{B}(p, r)$ is bounded. Hence A is also bounded. $\square$

8.8.5 Proposition. *Let (E, d) be a semimetric space. If A and B are subsets of E and $(a, b) \in A \times B$, then*

$$\text{diam}(A \cup B) \leq d(x_0, y_0) + \text{diam}(A) + \text{diam}(B).$$

In particular, if both A and B are bounded, so is $A \cup B$. Moreover, if $A \cap B \neq \emptyset$, then

$$\text{diam}(A \cup B) \leq \text{diam}(A) + \text{diam}(B).$$

Proof. Let x and y be two elements of $A \cup B$. If $\{x, y\} \subseteq A$, then $d(x, y) \leq \text{diam}(A)$. Similarly, if $\{x, y\} \subseteq B$, then $d(x, y) \leq \text{diam}(B)$. For the remaining cases, by symmetry we may assume that $x \in A$ and $y \in B$. One has

$$d(x, y) \leq d(x, x_0) + d(x_0, y_0) + d(y_0, y) \leq d(x_0, y_0) + \text{diam}(A) + \text{diam}(B).$$

If $A \cap B \neq \emptyset$, we could take $x_0 = y_0 \in A \cap B$ to obtain the inequality

$$\text{diam}(A \cup B) \leq \text{diam}(A) + \text{diam}(B).$$

$\square$

8.8.6 Definition. Let (E, d) be a semimetric space and $\mathcal{G}$ be a non-degenerate filter. We say that $\mathcal{G}$ is a *Cauchy filter* if

$$\inf_{A \in \mathcal{G}} \text{diam}(A) = 0.$$

Let I be an infinite subset of $\mathbb{N}$ and $\mathcal{F}$ be the Fréchet filter on I . Let $x = (x_n)_{n \in I}$ be a sequence in E . If $x_*(\mathcal{F})$ is a Cauchy filter, then we say that $(x_n)_{n \in I}$ is a Cauchy sequence.

8.8.7 Remark. Let (E, d) be a semimetric space and $\mathcal{G}$ be a non-degenerate filter. The following conditions are equivalent.

- (1) $\mathcal{G}$ is a Cauchy filter;
- (2) for any $\varepsilon > 0$, there exists $A \in \mathcal{G}$ such that $\text{diam}(A) < \varepsilon$.
- (3) for any $\varepsilon > 0$, there exists $A \in \mathcal{G}$ such that $\text{diam}(A) \leq \varepsilon$.

Similarly, if I is an infinite subset of $\mathbb{N}$ and $(x_n)_{n \in I}$ is a sequence in E . The following conditions are equivalent:

- (1) $(x_n)_{n \in I}$ is a Cauchy sequence;
- (2) for any $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that,

$$\forall (n, m) \in I_{\geq N} \times I_{\geq N}, d(x_n, x_m) < \varepsilon;$$

- (3) for any $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that,

$$\forall (n, m) \in I_{\geq N} \times I_{\geq N}, d(x_n, x_m) \leq \varepsilon.$$

8.8.8 Proposition. Let (E, d) be a semimetric space.

- (1) Any finite subsets of E is bounded.
- (2) If I is an infinite subset of $\mathbb{N}$ and $(x_n)_{n \in I}$ is a Cauchy sequence in E , then the set $\{x_n \mid n \in I\}$ is bounded.

Proof. (1) Note that one point subset in E is bounded. By Proposition 8.8.5, by induction we obtain that any finite subset of E is bounded.

(2) By definition, there exists $N \in \mathbb{N}$ such that

$$\text{diam}(\{x_n \mid n \in I, n \geq N\}) \leq 1.$$

Since $\{x_n \mid n \in I, n < N\}$ is a finite set, it is bounded. Therefore, by Proposition 8.8.5 and the first statement of the current proposition, we obtain that $\{x_n \mid n \in I\}$ is bounded. $\square$

8.8.9 Proposition. Let (E, d) be a semimetric space and $\mathcal{G}$ be non-degenerate filter on E . If $\mathcal{G}$ has a limit point, then it is a Cauchy filter. In particular, any convergent sequence in E is a Cauchy sequence.

Proof. Let p be a limit point of $\mathcal{G}$. For any $r > 0$, one has $B(p, r) \in \mathcal{G}$ since $B(p, r)$ is a neighbourhood of p . Note that $\text{diam}(B(p, r)) \leq 2r$. Hence

$$\inf_{A \in \mathcal{G}} \text{diam}(A) = 0.$$

Let I be an infinite subset of $\mathbb{N}$ and $x = (x_n)_{n \in I}$ be a sequence. Let $\mathcal{F}$ be the Fréchet filter on I . If $(x_n)_{n \in I}$ converges, then $x_*(\mathcal{F})$ has a limit point and hence it is a Cauchy filter. Therefore, $(x_n)_{n \in I}$ is a Cauchy sequence. $\square$

8.8.10 Proposition. *Let (E, d) be a semimetric space and $\mathcal{G}$ be a Cauchy filter on E . Any non-degenerate filter containing $\mathcal{G}$ is a Cauchy filter. In particular, if I is an infinite subset of $\mathbb{N}$ and $(x_n)_{n \in I}$ is a Cauchy sequence in E , then any subsequence of $(x_n)_{n \in I}$ is a Cauchy sequence.*

Proof. If $\mathcal{G}'$ is a non-degenerate filter containing $\mathcal{G}$, then

$$0 \leq \inf_{A' \in \mathcal{G}'} \text{diam}(A') \leq \inf_{A \in \mathcal{G}} \text{diam}(A) = 0,$$

so that $\mathcal{G}'$ is a Cauchy filter.

Let $x = (x_n)_{n \in I}$ be a Cauchy sequence. Let J be an infinite subset of I and $\mathcal{F}$ be the Fréchet filter on J . Let $j : J \rightarrow I$ be the inclusion mapping. Then $j_*(\mathcal{F})$ contains the Fréchet filter on I . Therefore,

$$(x \circ j)_*(\mathcal{F}) = x_*(j_*(\mathcal{F}))$$

is a Cauchy filter. $\square$

8.8.11 Proposition. *Let (E, d) be a semimetric space and $\mathcal{F}$ be a non-degenerate filter on E . If $\mathcal{F}$ has a limit point, then it is a Cauchy filter.*

Proof. Let ℓ be a limit point of $\mathcal{F}$. By definition, the neighbourhood filter of ℓ is contained in $\mathcal{F}$. In particular, for any $\varepsilon > 0$, one has $B(\ell, \varepsilon) \in \mathcal{F}$. Note that the diameter of $B(\ell, \varepsilon)$ is bounded from above by 2ε . Therefore,

$$\inf_{A \in \mathcal{F}} \text{diam}(A) = 0.$$

$\square$

8.8.12 Proposition. *Let (E, d) be a semimetric space, and $\mathcal{G}$ be a non-degenerate filter on E . If $\mathcal{G}$ is a Cauchy filter, then any adherent point of $\mathcal{G}$ is necessarily a limit point of $\mathcal{G}$.*

Proof. Let p be an adherent point of $\mathcal{G}$. Let $r > 0$. We pick $A \in \mathcal{G}$ such that $\text{diam}(A) < r/2$. Note that $B(p, r/4) \cap A \neq \emptyset$. Therefore

$$\text{diam}(B(p, r/4) \cup A) \leq \text{diam}(B(p, r/4)) + \text{diam}(A) < r.$$

Hence $A \subseteq B(p, r)$. Therefore $\mathcal{G}$ contains the neighbourhood filter of p . $\square$

8.8.13 Corollary. *Let (E, d) be a semimetric space, I be an infinite subset of $\mathbb{N}$ and $x = (x_n)_{n \in I}$ be a Cauchy sequence in E . If there exists a subsequence of $(x_n)_{n \in I}$ that converges to some $p \in E$, then the sequence $(x_n)_{n \in I}$ converges to p .*

Proof. Let J be an infinite subset of I such that $(x_n)_{n \in J}$ converges to some $p \in E$. Let $\mathcal{F}_J$ and $\mathcal{F}_I$ be the Fréchet filters on J and on I respectively. Let $j : J \rightarrow I$ be the inclusion mapping. By definition, p is a limit point of $(x \circ j)_*(\mathcal{F}_J)$. Since $(x \circ j)_*(\mathcal{F}_J)$ contains $x_*(\mathcal{F}_I)$, we obtain that p is an adherent point of $x_*(\mathcal{F}_I)$. Since $x_*(\mathcal{F}_I)$ is a Cauchy filter, p is also a limit point of $x_*(\mathcal{F}_I)$. $\square$

8.8.14 Proposition. *Let (X, d) be a metric space. Let Y be a subset of X . If Y equipped with the induced metric is complete, then Y is a closed subset of X .*

Proof. Let $(y_n)_{n \in \mathbb{N}}$ be a sequence in Y that converges to some $p \in X$. Then $(y_n)_{n \in \mathbb{N}}$ is a Cauchy sequence in Y . Since Y is complete, the sequence $(y_n)_{n \in \mathbb{N}}$ converges in Y to some $y \in Y$. Since (X, d) is a Hausdorff space, we obtain that $p = y \in Y$. Hence $Y = \bar{Y}$, namely Y is a closed subset of X . $\square$

8.8.15 Proposition. *Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces, and $f : X \rightarrow Y$ be a mapping. Let $\mathcal{F}$ is a non-degenerate filter on X and p be a limit point of $\mathcal{F}$. If f is continuous at p , then $f(p)$ is a limit point $f_*(\mathcal{F})$.*

Proof. Let V be a neighbourhood of $f(p)$. Since f is continuous at p , $f^{-1}(V)$ is a neighbourhood of p . Therefore, $f^{-1}(V) \in \mathcal{F}$ and hence $V \in f_*(\mathcal{F})$. Thus $f(p)$ is a limit point of $f_*(\mathcal{F})$. $\square$

8.8.16 Definition. Let (X, d) be a semimetric space. If any Cauchy filter on X has a limit point, we say that the semimetric space (X, d) is *complete*.

8.8.17 Proposition. *A semimetric (X, d) is complete if and only any Cauchy sequence in X converges.*

Proof. “ $\implies$ ” Let I be an infinite subset of $\mathbb{N}$ and $x := (x_n)_{n \in I}$ is a Cauchy sequence. Let $\mathcal{F}$ be the Fréchet filter on I . The $x_*(\mathcal{F})$ is a Cauchy filter on X , which has a limit point. Hence the sequence $(x_n)_{n \in I}$ converges.

“ $\Leftarrow$ ” Let $\mathcal{G}$ be a Cauchy filter on X . For any $n \in \mathbb{N}$, let $A_n \in \mathcal{G}$ such that $\text{diam}(A_n) \leq 1/(n+1)$. For any $n \in \mathbb{N}$, let

$$B_n = A_0 \cap \cdots \cap A_n$$

and we pick an arbitrary element $x_n \in B_n$. Then the sequence $(x_n)_{n \in \mathbb{N}}$ is a Cauchy sequence in X and hence it converges to some $p \in X$.

We prove that p is an adherent point of $\mathcal{G}$. Let $A \in \mathcal{G}$. For any $n \in \mathbb{N}$, we pick $y_n \in A \cap B_n$. For any $n \in \mathbb{N}$, one has $d(y_n, x_n) \leq 1/(n+1)$ and hence

$$\limsup_{n \rightarrow +\infty} d(y_n, p) \leq \limsup_{n \rightarrow +\infty} (d(y_n, x_n) + d(x_n, p)) = \lim_{n \rightarrow +\infty} d(y_n, x_n) + \limsup_{n \rightarrow +\infty} d(x_n, p) = 0.$$

Hence p is a limit point of $(y_n)_{n \in \mathbb{N}}$ and hence any neighbourhood of p has a non-empty intersection with A .

Since $\mathcal{G}$ is a Cauchy filter, by Proposition 8.8.12 we obtain that p is a limit point of $\mathcal{G}$. The proposition is thus proved. $\square$

8.8.18 Proposition. *Let (X, d_X) and (Y, d_Y) be semimetric spaces, and $f : X \rightarrow Y$ be a mapping. Assume that f is uniformly continuous (this condition is satisfied notammet when f is Lipschitzian). For any Cauchy filter $\mathcal{F}$ on X , $f_*(\mathcal{F})$ is a Cauchy filter on Y .*

Proof. For any $\varepsilon > 0$, there exists $\delta > 0$ such that

$$\forall (x, y) \in X \times X, d_X(x, y) \leq \delta \implies d_Y(f(x), f(y)) \leq \varepsilon.$$

Let $A \in \mathcal{F}$ such that $\text{diam}(A) \leq \delta$. Then $f(A) \in f_*(\mathcal{F})$ and $\text{diam}(f(A)) \leq \varepsilon$. $\square$

8.8.19 Proposition. *Let n be a positive integer, and $(X_1, d_1), \dots, (X_n, d_n)$ be semimetric spaces, $X = X_1 \times \cdots \times X_n$ and d be the semimetric on X defined as*

$$d((x_1, \dots, x_n), (y_1, \dots, y_n)) := \max_{i \in \{1, \dots, n\}} d_i(x_i, y_i).$$

For any $i \in I$, let $\pi_i : X \rightarrow X_i$ be the mapping of projection to the i -th coordinate. Let $\mathcal{F}$ be a non-degenerate filter on X . For any $i \in \{1, \dots, n\}$, let $\mathcal{F}_i = \pi_{i,}(\mathcal{F})$.*

- (1) *The filter $\mathcal{F}$ is a Cauchy filter if and only if $\mathcal{F}_i$ is a Cauchy filter for any $i \in \{1, \dots, n\}$.*
- (2) *The filter $\mathcal{F}$ has a limit point if and only if each filter $\mathcal{F}_i$ has a limit point for any $i \in \{1, \dots, n\}$. Moreover, if p_i is a limit point of $\mathcal{F}_i$, where $i \in \{1, \dots, n\}$, then $(p_1, \dots, p_n)$ is a limit point of $\mathcal{F}$.*
- (3) *If $(X_1, d_1), \dots, (X_n, d_n)$ are complete, then (X, d) is complete.*

Proof. (1) Since each mapping π_i is Lipschitzian, if $\mathcal{F}$ is a Cauchy filter, then each $\mathcal{F}_i$ is a Cauchy filter.

Conversely, let $\mathcal{F}$ be a non-degenerate filter such that each $\mathcal{F}_i$ is a Cauchy filter. For any $\varepsilon > 0$ and any $i \in \{1, \dots, n\}$, there exists $A_i \in \mathcal{F}_i$ such that $\text{diam}(A_i) \leq \varepsilon$. Let

$$A = A_1 \times \cdots \times A_n = \bigcap_{i=1}^n \pi_i^{-1}(A_i) \in \mathcal{F}.$$

For $x := (x_1, \dots, x_n)$ and $y := (y_1, \dots, y_n)$ in A , one has

$$d(x, y) = \max_{i \in \{1, \dots, n\}} d_i(x_i, y_i) \leq \varepsilon.$$

So $\mathcal{F}$ is a Cauchy filter.

(2) Since each $\pi_i : X \rightarrow X_i$ is Lipschitzian, it is continuous. Therefore, if $p = (p_1, \dots, p_n)$ is a limit point of $\mathcal{F}$, then, by Proposition 8.8.15, we obtain that $\pi_i(p) = p_i$ is a limit point of $\mathcal{F}_i$.

Conversely, suppose that, for any $i \in \{1, \dots, n\}$, p_i is a limit point of $\mathcal{F}_i$. If U is a neighbourhood of $p := (p_1, \dots, p_n)$, then there exists open neighbourhoods U_i of p_i such that $U_1 \times \cdots \times U_n \subseteq U$. Note that each U_i belongs to $\mathcal{F}_i$ and hence

$$U = \bigcap_{i=1}^n \pi_i^{-1}(U_i) \in \mathcal{F}.$$

Hence p is a limit point of $\mathcal{F}$. □

8.8.20 Proposition. *Let (X, d) and (X', d') be two semimetric spaces, and $f : X \rightarrow X'$ be a bijection. Assume that there exists positive constant C_1 and C_2 such that*

$$\forall (x, y) \in X \times X, C_1 d(x, y) \leq d'(f(x), f(y)) \leq C_2 d(x, y).$$

- (1) *for any sequence $(x_n)_{n \in I}$ in X , the sequence $(x_n)_{n \in I}$ is a Cauchy sequence if and only if $(f(x_n))_{n \in I}$ is a Cauchy sequence;*
- (2) *for any sequence $(x_n)_{n \in I}$ in X , the sequence $(x_n)_{n \in I}$ is a convergent sequence if and only if $(f(x_n))_{n \in I}$ is a convergent sequence;*
- (3) *if (X, d) is complete, then $(f(X), d')$ is complete and $f(X)$ is closed in X' once d' is a metric.*

Proof. Note that $f : X \rightarrow f(X)$ and $f^{-1} : f(X) \rightarrow X$ are both Lipschitzian. Hence they send Cauchy sequences to Cauchy sequences, and send convergent sequences to convergent sequences. So the first two statements hold. The last statement results from the first two statements and Proposition 8.8.14. □

8.8.21 Theorem. *Let $(K, |\cdot|)$ be a complete valued field and V be a finite-dimensional vector space over K .*

- (1) *All norms on V are equivalent.*
- (2) *For any norm $\|\cdot\|$ on V , the normed vector space $(V, \|\cdot\|)$ is complete.*
- (3) *If we equip V with the topology induced by an arbitrary norm on V , for any normed vector space $(V', \|\cdot\|')$, any K -linear mapping $f : V \rightarrow V'$ is bounded and $f(V)$ is closed in V' .*

Proof. Let $e := (e_i)_{i=1}^n$ be a basis of V . Then the mapping

$$\|\cdot\|_e : V \longrightarrow \mathbb{R}_{\geq 0}, \quad \|a_1e_1 + \cdots + a_ne_n\|_e := \max_{i \in \{1, \dots, n\}} |a_i|$$

is a norm on V . Moreover, if $(V', \|\cdot\|')$ is a normed vector space, for any $(a_1, \dots, a_n) \in K^n$, one has

$$\begin{aligned} \|f(a_1e_1 + \cdots + a_ne_n)\|' &= \|a_1f(e_1) + \cdots + a_nf(e_n)\|' \leq \sum_{i=1}^n |a_i| \cdot \|f(e_i)\|' \\ &\leq \left(\max_{i \in \{1, \dots, n\}} |a_i| \right) \sum_{i=1}^n \|f(e_i)\|' \end{aligned}$$

Therefore, the K -linear mapping $f : (V, \|\cdot\|_e) \rightarrow (V', \|\cdot\|')$ is bounded. In particular, for any norm $\|\cdot\|$ on V , since the identity mapping $(V, \|\cdot\|_e) \rightarrow (V, \|\cdot\|)$ is bounded, there exists a constant $C' > 0$ such that $\|\cdot\| \leq C' \|\cdot\|_e$.

To prove the theorem, we reason by induction on n . In the case where $n = 0$, there is only one norm $\|\cdot\|_0$ on the zero vector space, which is the constant mapping taking 0 as its value. Any sequence in the zero vector space is constant and hence convergent. So the zero normed vector space is complete. Moreover, if f is a K -linear mapping from the zero vector space to a normed vector space $(V', \|\cdot\|')$, it is bounded and the image of f is a one point set, which is closed in V' since V' is Hausdorff.

Consider now the case where $n = 1$. Let $\|\cdot\|$ be a norm on V . Let e_1 be a non-zero element of V , which forms a basis of V . For any $a \in K$ one has

$$\|a_1e_1\| = |a| \cdot \|e_1\| = \|e_1\| \cdot \|a_1e_1\|_{e_1}.$$

Therefore, the norms $\|\cdot\|$ and $\|\cdot\|_{e_1}$ are equivalent. Since $(K, |\cdot|)$ is complete, by Proposition 8.8.20 we obtain that $(V, \|\cdot\|)$ is also complete. If $f : (V, \|\cdot\|) \rightarrow (V', \|\cdot\|')$ is a K -linear mapping, then, either $f(V) = \{0\}$ or $f(V)$ is a one-dimensional vector subspace of V' . In both cases $f(V)$ equipped with the induced metric is complete and hence is closed in V' .

In what follows, we assume that $n \geq 2$ and that the statement is true for any vector space of dimension $\leq n-1$ over K . Let $(V, \|\cdot\|)$ be a vector space of dimension n over K and $\mathbf{e} = (e_1, \dots, e_n)$ be a basis of V over K . Let

$$W = \{a_1e_1 + \dots + a_{n-1}e_{n-1} \mid (a_1, \dots, a_{n-1}) \in K^{n-1}\}.$$

We equip W with the restriction of the norm $\|\cdot\|$. The induction hypothesis shows that $(W, \|\cdot\|)$ is complete. Hence W is a closed vector subspace of V . Let $Q = V/W$. We equip Q with the quotient norm of $\|\cdot\|$, denoted as $\|\cdot\|_Q$.

Applying the induction hypothesis to W , we obtain the existence of $A > 0$ such that, for any $(a_1, \dots, a_{n-1}) \in K^{n-1}$,

$$\max\{|a_1|, \dots, |a_{n-1}|\} \leq A\|a_1e_1 + \dots + a_{n-1}e_{n-1}\|.$$

Let $(b_1, \dots, b_n) \in K^n$, $s := b_1e_1 + \dots + b_ne_n \in V$, $t = b_1e_1 + \dots + b_{n-1}e_{n-1}$, and $\alpha := [s] = b_ne_n \in V/W$. One has

$$\|t\| = \|s - b_ne_n\| \leq \|s\| + |b_n| \cdot \|e_n\|.$$

Let $B = \|e_n\|/\|[e_n]\|_Q$. One has

$$\|s\| \geq \|\alpha\|_Q = |b_n| \cdot \|[e_n]\|_Q = B^{-1}|b_n| \cdot \|e_n\|.$$

Hence

$$(B^{-1} + 1)\|s\| \geq B^{-1}(\|t\| - |b_n| \cdot \|e_n\|) + B^{-1}|b_n| \cdot \|e_n\| = B^{-1}\|t\|,$$

which leads to

$$\|s\| \geq \frac{1}{B+1}\|t\| \geq \frac{A}{B+1} \max\{|b_1|, \dots, |b_{n-1}|\}.$$

Taking

$$C = \min \left\{ \frac{A}{B+1}, \frac{\|e_n\|}{B} \right\},$$

one has

$$\|s\| \geq C \max\{|b_1|, \dots, |b_n|\} = C\|s\|_{\mathbf{e}}.$$

Therefore, the norms $\|\cdot\|$ and $\|\cdot\|_{\mathbf{e}}$ are equivalent. Thus we obtain the equivalence of all norms on V . By Proposition 8.8.19, we obtain that $(V, \|\cdot\|_{\mathbf{e}})$ is complete. By Proposition 8.8.20, we deduce that V equipped with any norm is complete.

Finally, let $(V', \|\cdot\|')$ be a normed vector space and $f : V \rightarrow V'$ be a K -linear mapping. Since $f : (V, \|\cdot\|_{\mathbf{e}}) \rightarrow (V', \|\cdot\|')$ is bounded and $\|\cdot\|$ is equivalent to $\|\cdot\|_{\mathbf{e}}$, we obtain that $f : (V, \|\cdot\|) \rightarrow (V', \|\cdot\|')$ is also bounded.

If f is not injective, then $f(V)$ is of dimension $\leq n-1$ and hence $(f(V), \|\cdot\|')$ is complete. Hence $f(V)$ is closed in V' . If f is injective, since the norms $\|f(\cdot)\|'$ and $\|\cdot\|$ are equivalent, by Proposition 8.8.20 we obtain that $f(V)$ is closed in V' . $\square$

8.8.22 Definition. Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces and $f : X \rightarrow Y$ be a bijection. If both f and f^{-1} are continuous, we say that f is a *homeomorphism*. If there exists a homeomorphism from $(X, \mathcal{T}_X)$ to $(Y, \mathcal{T}_Y)$, then we say that $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ are homeomorphic. Theorem 8.8.21 shows that, all normed vector spaces of the same dimension over a complete valued field are homeomorphic.

8.9 Compactness

8.9.1 Definition. Let X be a set and $\mathcal{F}$ be a non-degenerate filter on X . If there does not exist any non-degenerate filter $\mathcal{G}$ on X such that $\mathcal{F} \subsetneq \mathcal{G}$, then we say that $\mathcal{F}$ is an *ultrafilter*.

8.9.2 Proposition. Let X be a set and $\mathcal{F}$ be a non-degenerate filter on X . The following conditions are equivalent:

- (1) $\mathcal{F}$ is an ultrafilter,
- (2) for any $A \subseteq X$, either $A \in \mathcal{F}$, or $X \setminus A \in \mathcal{F}$,
- (3) for any $(A, B) \in \mathcal{P}(X)^2$, if $A \cup B \in \mathcal{F}$, then $A \in \mathcal{F}$ or $B \in \mathcal{F}$.

Proof. “(1) $\implies$ (2)”: Suppose that A is a subset of X such that $A \notin \mathcal{F}$ and $(X \setminus A) \notin \mathcal{F}$. For any $B \in \mathcal{F}$ one has $B \cap A \neq \emptyset$ since otherwise $B \subseteq X \setminus A$ and hence $X \setminus A \in \mathcal{F}$, which leads to a contradiction. Hence $\mathcal{F} \cup \{A\}$ generates a non-degenerate filter which is strictly larger than $\mathcal{F}$. Hence $\mathcal{F}$ is not an ultrafilter.

“(2) $\implies$ (3)”: Suppose that $B \notin \mathcal{F}$, then $X \setminus B \in \mathcal{F}$. Hence

$$(A \cup B) \cap (X \setminus B) = A \setminus B \in \mathcal{F}.$$

So $A \in \mathcal{F}$.

“(3) $\implies$ (1)”: Suppose that $\mathcal{F}'$ is a non-degenerate filter such that $\mathcal{F} \subsetneq \mathcal{F}'$. Take $A \in \mathcal{F}' \setminus \mathcal{F}$. One has $X = A \cup (X \setminus A) \in \mathcal{F}$. Hence $X \setminus A \in \mathcal{F} \subseteq \mathcal{F}'$. This contradicts the hypothesis that $\mathcal{F}'$ is non-degenerate. $\square$

8.9.3 Corollary. Let $f : X \rightarrow Y$ be a mapping of sets. If $\mathcal{F}$ is an ultrafilter on X , then $f_*(\mathcal{F})$ is an ultrafilter on Y .

Proof. Let A and B be two subsets of Y such that $A \cup B \in f_*(\mathcal{F})$. Then

$$f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B) \in \mathcal{F}.$$

Therefore, either $f^{-1}(A) \in \mathcal{F}$, or $f^{-1}(B) \in \mathcal{F}$. Thus, either $A \in f_*(\mathcal{F})$, or $B \in f_*(\mathcal{F})$. $\square$

8.9.4 Proposition. *Let $(X, \mathcal{T})$ be a topological space and $\mathcal{F}$ be a non-degenerate filter on X . Then there exists an ultrafilter $\mathcal{G}$ on X such that $\mathcal{F} \subseteq \mathcal{G}$.*

Proof. Let Θ be the set of all non-degenerate filter containing $\mathcal{F}$. We equip Θ with the order of inclusion.

If Θ_0 is a non-empty totally ordered subset of Θ , then

$$\mathcal{F}' := \bigcup_{\mathcal{H} \in \Theta_0} \mathcal{H}$$

is an upper bound of Θ_0 . In fact, if V is an element of $\mathcal{F}'$, then there exists $\mathcal{H} \in \Theta_0$ such that $V \in \mathcal{H}$. Hence, for any subset U containing V , one has $U \in \mathcal{H} \subseteq \mathcal{F}'$. If U_1 and U_2 are two elements of $\mathcal{F}'$, then there exists $\mathcal{H}_1$ and $\mathcal{H}_2$ in Θ_0 such that $U_1 \in \mathcal{H}_1$ and $U_2 \in \mathcal{H}_2$. Since Θ_0 is totally ordered, either $\mathcal{H}_1 \subseteq \mathcal{H}_2$, or $\mathcal{H}_2 \subseteq \mathcal{H}_1$. Therefore $U_1 \cap U_2$ belongs to $\mathcal{H}_1$ or belongs to $\mathcal{H}_2$. In any case we have $U_1 \cap U_2 \in \mathcal{F}'$. Hence $\mathcal{F}'$ is a non-degenerate filter. It is thus an upper bound of Θ_0 .

By Zorn's lemma we obtain that Θ has a maximal element $\mathcal{G}$. It is thus an ultrafilter containing $\mathcal{F}$. $\square$

8.9.5 Definition. Let $(X, \mathcal{T})$ be a topological space, Y be a subset of X and $(U_i)_{i \in I}$ be a family of subsets of X . We say that $(U_i)_{i \in I}$ is a *cover* of Y if

$$Y \subseteq \bigcup_{i \in I} U_i.$$

If furthermore U_i is open for any $i \in I$, we say that $(U_i)_{i \in I}$ is an *open cover* of Y . If I is a finite set, we say that $(U_i)_{i \in I}$ is a *finite cover* of Y .

Let Y be a subset of X and $(U_i)_{i \in I}$ be a cover of Y . If J is a subset of I such that

$$Y \subseteq \bigcup_{j \in J} U_j,$$

then we say that $(U_j)_{j \in J}$ is a *subcover* of $(U_i)_{i \in I}$.

8.9.6 Proposition. *Let $(X, \mathcal{T})$ be a topological space and Y be a subset of X . The following conditions are equivalent.*

- (1) *For any ultrafilter $\mathcal{G}$ on X such that $Y \in \mathcal{G}$, the filter $\mathcal{G}$ has a limit point in Y .*
- (2) *For any non-degenerate filter $\mathcal{F}$ on X such that $Y \in \mathcal{F}$, the filter $\mathcal{F}$ has an adherent point in Y .*
- (3) *Any open cover of Y has a finite subcover.*

Proof. “(1) $\implies$ (2)”: Let $\mathcal{F}$ be a filter on X such that $Y \in \mathcal{F}$ and $\mathcal{G}$ be an ultrafilter containing $\mathcal{F}$. By the condition (1), $\mathcal{G}$ has a limit point x . Since $\mathcal{G} \supseteq \mathcal{F}$, we obtain that x is an adherent point of $\mathcal{F}$.

“(2) $\implies$ (3)”: Let $(U_i)_{i \in I}$ be an open cover of Y . For any $i \in I$, let $F_i := X \setminus U_i$. Suppose that $(U_i)_{i \in I}$ does not have any finite subcover of Y . Then, for any finite subset J of I , one has

$$\left(\bigcap_{j \in J} F_j \right) \cap Y \neq \emptyset.$$

Thus

$$\{F_i \mid i \in I\} \cup \{Y\}$$

generates a non-degenerate filter $\mathcal{F}$, which has an adherent point $x \in Y$ by the condition (2). There exists $i_0 \in I$ such that $x \in U_{i_0}$. One has $U_{i_0} \cap F_{i_0} = \emptyset$. This contradicts the hypothesis that x is an adherent point of $\mathcal{F}$.

“(3) $\implies$ (2)”: Let $\mathcal{F}$ be a non-degenerate filter on X such that $Y \in \mathcal{F}$. We suppose that $\mathcal{F}$ does not have any adherent point in Y . For any $x \in Y$, there exists an open neighbourhood U_x of x and an element $A_x \in \mathcal{F}$ such that $U_x \cap A_x = \emptyset$. Let $x_1, \dots, x_n$ be elements of X such that

$$Y \subseteq \bigcup_{i=1}^n U_{x_i}.$$

Let

$$A = \left(\bigcap_{i=1}^n A_{x_i} \right) \cap Y.$$

One has $A \neq \emptyset$ and

$$A = A \cap Y \subseteq \bigcup_{i=1}^n A \cap U_{x_i} = \emptyset,$$

which leads to a contradiction.

“(2) $\implies$ (1)”: Let $\mathcal{G}$ be an ultrafilter on X such that $Y \in \mathcal{G}$, and x be an adherent point of $\mathcal{G}$. Then $\mathcal{G} \cup \mathcal{V}_x(\mathcal{T})$ generates a non-degenerate filter containing $\mathcal{G}$. Since $\mathcal{G}$ is an ultrafilter, one has $\mathcal{V}_x(\mathcal{T}) \subseteq \mathcal{G}$. $\square$

8.9.7 Definition. Let $(X, \mathcal{T})$ be a topological space and F be a subset of X . If the equivalent conditions in Proposition 8.9.6 are satisfied, we say that F is *compact*.

8.9.8 Proposition. Let $f : (X, \mathcal{T}_X) \rightarrow (Y, \mathcal{T}_Y)$ be a continuous mapping between topological spaces. If F is a compact subset of X , then $f(F)$ is a compact subset of Y .

Proof. Let $(V_i)_{i \in I}$ be an open cover of $f(F)$. Then $(f^{-1}(V_i))_{i \in I}$ is an open cover of F . Since F is compact, there exists a finite subset J of I such that $(f^{-1}(V_j))_{j \in J}$ forms an open cover of F . Thus $f(F)$ is covered by $(V_j)_{j \in J}$. $\square$

8.9.9 Proposition. *Let $(X, \mathcal{T})$ be a topological space, F be a compact subset of X and A be a closed subset of X . Then $F \cap A$ is a compact subset of X .*

Proof. Let $(U_i)_{i \in I}$ be an open cover of $F \cap A$. Let $U := X \setminus A$. Then one has

$$F \subseteq U \cup \bigcup_{i \in I} U_i.$$

There then exists a finite subset J of I such that

$$F \subseteq U \cup \bigcup_{j \in J} U_j,$$

so that

$$F \cap A \subseteq \bigcup_{j \in J} U_j.$$

Hence $F \cap A$ is compact. $\square$

8.9.10 Proposition. *Let $(X, \mathcal{T})$ be a Hausdorff topological space and A be a compact subset of X .*

- (1) *For any $x \in X \setminus A$, there exists open subset U and V of X such that $A \subseteq U$, $x \in V$ and $U \cap V = \emptyset$.*
- (2) *The subset A of X is closed.*

Proof. (1) Since $(X, \mathcal{T})$ is Hausdorff, for any $y \in A$, there exists open subsets U_y and V_y of X such that $y \in U_y$, $x \in V_y$ and $U_y \cap V_y = \emptyset$. Since $A \subseteq \bigcup_{y \in A} U_y$, there exists a finite subset $\{y_1, \dots, y_n\}$ of A such that

$$A \subseteq U := \bigcup_{i=1}^n U_{y_i}.$$

Let

$$V := \bigcap_{i=1}^n V_{y_i}.$$

One has

$$U \cap V = \bigcup_{i=1}^n (U_{y_i} \cap V) = \emptyset.$$

(2) The statement in (1) shows that $VX \setminus A$ is a neighbourhood of all its elements. Thus $x \in X \setminus A$ is open and hence A is closed. $\square$

8.9.11 Proposition. *Let $(X, \mathcal{T})$ be a Hausdorff topological space. If A and B are compact subsets of X such that $A \cap B = \emptyset$, then there exists open subsets U and V of X such that $A \subseteq U$, $B \subseteq V$ and $U \cap V = \emptyset$.*

Proof. By the previous proposition, for any $x \in B$, there exists open subsets U_x and V_x of X such that $A \subset U_x$, $x \in V_x$ and $U_x \cap V_x = \emptyset$. Since $(V_x)_{x \in B}$ forms an open cover of B , there exists a finite subset $\{x_1, \dots, x_n\}$ of B such that $(V_{x_i})_{i \in \{1, \dots, n\}}$ forms a cover of B . Let $V = \bigcup_{i=1}^n V_{x_i}$, $U = \bigcap_{i=1}^n U_{x_i}$. One has $A \subseteq U$, $B \subseteq V$ and $U \cap V = \emptyset$. $\square$

8.9.12 Theorem. *Let $(X, \mathcal{T})$ be a topological space and A be a compact subset of X . Let $(I, \leq)$ be a partially ordered set such that any finite subset of I has an upper bound. Let $(F_i)_{i \in I}$ be a decreasing family of closed subsets of X . If*

$$\left(\bigcap_{i \in I} F_i \right) \cap A = \emptyset.$$

Then there exists $i_0 \in I$ such that $F_{i_0} \cap A = \emptyset$.

Proof. For any $i \in I$, let $U_i = X \setminus F_i$. Then $(U_i)_{i \in I}$ is an increasing family of open subsets of X . Since

$$\left(\bigcap_{i \in I} F_i \right) \cap A = \emptyset,$$

one has

$$A \subseteq \bigcup_{i \in I} U_i.$$

Since A is compact, there exists a finite subset J such that

$$A \subseteq \bigcup_{j \in J} U_j.$$

Let i_0 be an upper bound of J . One has $A \subseteq U_{i_0}$, or equivalently $F_{i_0} \cap A = \emptyset$. $\square$

8.9.13 Theorem (Tychonoff). *Let I be a non-empty set and $((X_i, \mathcal{T}_i))_{i \in I}$ be a family of topological spaces. Let X be the product set $\prod_{i \in I} X_i$ equipped with the product topology $\mathcal{T}$. If, for any $i \in I$, A_i is a compact subset of X_i , then the product set $A = \prod_{i \in I} A_i$ is a compact subset of X .*

Proof. Let $\mathcal{F}$ be an ultrafilter on X such that $A \in \mathcal{F}$. For any $i \in I$, let $\mathcal{F}_i = \pi_{i,*}(\mathcal{F})$, where $\pi_i : X \rightarrow X_i$ is the mapping of projection to the i -th coordinate. One has $A_i \in \mathcal{F}_i$ since $\pi_i^{-1}(A_i) \supseteq A$. By Corollary 8.9.3, the filter $\mathcal{F}_i$ is an ultrafilter, and hence has a limit point x_i in A_i . Let $x = (x_i)_{i \in I}$. We will prove that x is a limit point of $\mathcal{F}$. Let U be a neighbourhood of x . By definition of the

product topology, there exists a finite subset J of I , and open neighbourhoods U_j of x_j for $j \in J$ such that

$$\bigcap_{j \in J} \pi_j^{-1}(U_j) \subseteq U.$$

Since $U_j \in \mathcal{F}_j$ for $j \in J$, we obtain $\pi_j^{-1}(U_j) \in \mathcal{F}$. Hence $U \in \mathcal{F}$. $\square$

8.9.14 Definition. Let $(X, \mathcal{T})$ be a topological space. If any $x \in X$ has a compact neighbourhood, we say that the topological space $(X, \mathcal{T})$ is *locally compact*.

8.9.15 Proposition. Let $((X_i, \mathcal{T}_i))_{i \in \{1, \dots, n\}}$ be a finite family of topological spaces. Let $X := X_1 \times \dots \times X_n$, equipped with the product topology $\mathcal{T}$. If all topological spaces $(X_i, \mathcal{T}_i)$ are locally compact, then so is $(X, \mathcal{T})$.

Proof. Let $x := (x_1, \dots, x_n)$ be an element of x . For any $i \in \{1, \dots, n\}$, let F_i be a compact neighbourhood of x_i . Let $F = F_1 \times \dots \times F_n$. Since

$$F = \bigcap_{i=1}^n \pi_i^{-1}(F_i),$$

where $\pi_i : X \rightarrow X_i$ is the i -th projection mapping, the set F is a neighbourhood of x . Moreover, by Theorem 8.9.13, F is compact. Hence $(X, \mathcal{T})$ is locally compact. $\square$

8.10 Compact metric spaces

8.10.1 Definition. Let $(X, \mathcal{T})$ be a topological space and Y be a subset of X . We say that Y is *sequentially compact* if any sequence in Y has a subsequence which converges in Y .

8.10.2 Theorem. Let $(X, \mathcal{T})$ be a topological space and Y be a subset of X . If Y is compact and if any element $y \in Y$ has a countable neighbourhood basis, then Y is sequentially compact.

Proof. Let $x = (x_n)_{n \in \mathbb{N}}$ be a sequence in Y and $\mathcal{F}$ be the Fréchet filter on I . Then $x_*(\mathcal{F})$ is a non-degenerate filter on X such that $Y \in x_*(\mathcal{F})$. Hence $x_*(\mathcal{F})$ has an adherent point $p \in Y$. Let $(U_k)_{k \in \mathbb{N}}$ be a decreasing sequence of neighbourhoods of p that forms a neighbourhood basis of p . We construct a strictly increasing sequence $(n_k)_{k \in \mathbb{N}}$ in I such that $x_{n_k} \in U_k$ for any $k \in \mathbb{N}$. Thus $(x_{n_k})_{k \in \mathbb{N}}$ converges to p when k tends to the infinity. $\square$

8.10.3 Theorem. Let (X, d) be a metric space and Y be a subset of X . The following conditions are equivalent.

(1) Y is compact.

(2) (Y, d) is complete and, for any $\varepsilon > 0$, there is a finite subset A_ε of Y such that

$$Y \subseteq \bigcup_{x \in A_\varepsilon} B(x, \varepsilon).$$

(3) Y is sequentially compact.

Proof. “(1) $\implies$ (2)”: Let $\mathcal{F}$ be a Cauchy filter on Y . Let $f : Y \rightarrow X$ be the inclusion mapping. Then $f_*(\mathcal{F})$ is a Cauchy filter on X containing Y . Since Y is compact, the filter $f_*(\mathcal{F})$ has a adherent point y in Y , which is also a limit point of $f_*(\mathcal{F})$ since $f_*(\mathcal{F})$ is a Cauchy filter. Note that y is also a limit point of $\mathcal{F}$. Hence (Y, d) is complete.

For any $\varepsilon > 0$, one has

$$Y \subseteq \bigcup_{x \in Y} B(x, \varepsilon).$$

Since Y is compact, there exists a finite subset A_ε of Y such that

$$Y \subseteq \bigcup_{x \in A_\varepsilon} B(x, \varepsilon).$$

“(2) $\implies$ (1)”: Let $\mathcal{F}$ be an ultrafilter on X such that $Y \in \mathcal{F}$. For any $\varepsilon > 0$, one has

$$\bigcup_{x \in A_\varepsilon} B(x, \varepsilon) \in \mathcal{F}$$

since the union contains Y . As $\mathcal{F}$ is an ultrafilter, there exists $x \in A_\varepsilon$ such that $B(x, \varepsilon) \in \mathcal{F}$. Therefore, $\mathcal{F}$ is a Cauchy filter. Since (Y, d) is complete, the restriction of $\mathcal{F}$ to Y is non-degenerate and is also a Cauchy filter. It has a limit point x in Y . Note that x is also a limit point of $\mathcal{F}$. Hence Y is compact.

“(1) $\implies$ (3)”: Let $(x_n)_{n \in I}$ be a sequence in Y . Assume that any of the subsequences of $(x_n)_{n \in I}$ does not converge in Y . For any $p \in Y$, there exists $\varepsilon_p > 0$ such that

$$\{n \in I \mid d(p, x_n) < \varepsilon_p\}$$

is finite since otherwise we can construct recursively an increasing sequence $(n_k)_{k \in \mathbb{N}}$ in I such that

$$d(p, x_{n_k}) < \frac{1}{k+1},$$

which produces a subsequence of $(x_n)_{n \in I}$ that converges to p . Since

$$Y \subseteq \bigcup_{p \in Y} B(p, \varepsilon_p),$$

there exists a finite subset $\{p_1, \dots, p_d\}$ such that

$$Y \subseteq \bigcup_{i=1}^d B(p_i, \varepsilon_{p_i}).$$

This is impossible since I is infinite.

“(3) $\implies$ (2)”: We first show that (Y, d) is complete. Let $(x_n)_{n \in I}$ be a Cauchy sequence in Y . Since Y is sequentially compact, $(x_n)_{n \in I}$ has a subsequence that converges to some $\ell \in Y$. Since the sequence $(x_n)_{n \in I}$ is a Cauchy sequence, it converges to ℓ .

Suppose that Y is not covered by a finite family of balls of radius $\varepsilon > 0$ centered at points in Y , then we can construct a sequence $(x_n)_{n \in \mathbb{N}}$ such that

$$x_{n+1} \in Y \setminus \bigcup_{k=0}^n B(x_k, \varepsilon).$$

Thus $(x_n)_{n \in \mathbb{N}}$ does not have any subsequence that is a Cauchy sequence. It cannot have any convergent subsequence. $\square$

8.10.4 Example. Let $[a, b]$ be a closed interval in $\mathbb{R}$, where a and b are real numbers and $a < b$. For any $\varepsilon > 0$, there exists $N \in \mathbb{N}_{\geq 1}$ such that $(b - a)/N < \varepsilon$. For $i \in \{0, \dots, N\}$, let

$$x_i := a + \frac{i}{N}(b - a)$$

Then

$$[a, b] \subseteq \bigcup_{i=0}^n]x_i - \varepsilon, x_i + \varepsilon[.$$

Therefore, $[a, b]$ is compact. In particular, $(\mathbb{R}, |\cdot|)$ is locally compact.

8.10.5 Proposition. *Let (X, d) be a metric space and Y be a subset of X . If Y is compact, then it is bounded and closed.*

Proof. We have seen that compact subsets of X are closed. It suffices to prove that Y is bounded. By the previous theorem, Y is covered by finitely many open balls of radius 1, so it is bounded. $\square$

8.10.6 Proposition. *Let $(K, |\cdot|)$ be a locally compact field and $(V, \|\cdot\|)$ be a finite-dimensional normed vector space over K .*

- (1) *The valued field $(K, |\cdot|)$ is complete.*
- (2) *The normed vector space $(V, \|\cdot\|)$ is locally compact.*

- (3) Assume that there exists $a \in K$ such that $|a| > 1$. A subset of V is compact if and only if it is bounded and closed.

Proof. (1) Let F be a compact neighbourhood of 0. There exists $\varepsilon > 0$ such that $\overline{B}(0_K, \varepsilon)$ is contained in F . This also shows that $\overline{B}(0_K, \varepsilon)$ is compact.

Let $(a_n)_{n \in I}$ be a Cauchy sequence in K . There exists $N \in I$ such that

$$\forall n \in I_{\geq N}, d(a_n, a_N) < \varepsilon.$$

Therefore, $(a_n - a_N)_{n \in I_{\geq N}}$ is a Cauchy sequence contained in F . Since F is compact, this sequence has a convergent subsequence. Hence the sequence converges since it is a Cauchy sequence. We then deduce that the sequence $(a_n)_{n \in I}$ converges.

For the proof of (2) and (3), by Theorem 8.8.21, we may assume without loss of generality that $V = K^n$ and $\|\cdot\| : V \rightarrow \mathbb{R}_{\geq 0}$ is given by

$$\|(a_1, \dots, a_n)\| := \max\{|a_1|, \dots, |a_n|\}.$$

Since

$$\overline{B}(0_V, \varepsilon) = \overline{B}(0_K, \varepsilon)^n,$$

by Tychonoff's theorem we obtain that $\overline{B}(0_V, \varepsilon)$ is compact. Moreover, for any $p \in V$, the mapping

$$\tau_p : V \longrightarrow V, x \longmapsto x + p$$

is Lipschitzian, hence is continuous, hence

$$\overline{B}(p, \varepsilon) = \tau_p(\overline{B}(0_V, \varepsilon))$$

is compact. Therefore, V is locally compact. This proves (2).

In the following, we prove (3). We have seen that any compact subsets of V is bounded and closed. Let A be a bounded closed subset of V . We pick $R > 0$ such that $A \subseteq \overline{B}(0_V, R)$. Since $|a| > 1$, there exists $n \in \mathbb{N}_{\geq 1}$ such that $|a^n|\varepsilon = |a|^n\varepsilon > R$. Let

$$\lambda : V \longrightarrow V, x \longmapsto a^n x.$$

This is a bounded linear mapping, hence is continuous. Therefore, $\lambda(\overline{B}(0_V, \varepsilon)) = a^n \overline{B}(0_V, \varepsilon)$ is compact. Since $|a|^n\varepsilon > R$, one has

$$A \subseteq \overline{B}(0_V, R) \subseteq a^n \overline{B}(0_V, \varepsilon).$$

Hence A is compact. □

8.10.7 Remark. A locally compact metric space is not necessarily complete. One can consider for example the set $]0, 1[$ equipped with the usual metric. This set is locally compact but not complete.

8.10.8 Proposition. *Let $(K, |\cdot|)$ be a locally compact complete valued field, and $(V, \|\cdot\|)$ be a finite-dimensional vector space over K . Assume that there exists $a \in K$ such that $|a| > 1$. Then any bounded closed subset of V is compact.*

Proof. Let A be a bounded closed subset of V . Let $R > 0$ such that $A \subseteq \overline{B}(0, R)$.

Since $(V, \|\cdot\|)$ is locally compact, there exists a compact neighbourhood F of 0 in V . Let $r > 0$ such that $\overline{B}(0, r) \subseteq F$. Then $\overline{B}(0, r)$ is also compact. Take $n \in \mathbb{N}_{\geq 1}$ such that $r|a|^n > R$. Then $\overline{B}(0, R) \subseteq a^n \overline{B}(0, r)$. Note that the mapping

$$f : V \longrightarrow V, \quad s \longmapsto a^n s$$

is bounded and hence is continuous. Therefore,

$$f(\overline{B}(0, r)) = a^n \overline{B}(0, r)$$

is compact. We then deduce that $A = A \cap a^n \overline{B}(0, r)$ is also compact. $\square$

8.10.9 Proposition. *Let A be a non-empty compact subset of $\mathbb{R}$. Then A has a greatest element and a least element.*

Proof. Let M be the supremum of A . Since A is bounded, $M \in \mathbb{R}$. For any $\varepsilon > 0$, $M - \varepsilon$ is not an upper bound of A and hence there exists $t \in A$ such that $M - \varepsilon < t \leq M$. Therefore M belongs to the closure of A in $\mathbb{R}$, which means that $M \in A$ since A is closed. A similar argument shows that the infimum of A also belongs to A . $\square$

8.10.10 Definition. Let X be a set and $f : X \rightarrow \mathbb{R}$ be a mapping.

We say that f attains its maximum at $x_0 \in X$ if for any $x \in X$ one has $f(x) \leq f(x_0)$. The value $f(x_0)$ is called the maximal value of f .

We say that f attains its minimum at $x_1 \in X$ if for any $x \in X$ one has $f(x) \geq f(x_1)$. The value $f(x_1)$ is called the minimal value of f .

8.10.11 Theorem. *Let $(X, \mathcal{T})$ be a topological space and $f : X \rightarrow \mathbb{R}$ be a continuous mapping. For any non-empty compact subset Y of X , the restriction of f at Y attains its maximum and its minimum.*

Proof. Since f is continuous and Y is compact, we obtain that $f(Y)$ is a compact subset of $\mathbb{R}$, hence has a greatest element and a least element. Therefore $f|_Y$ attains its maximum and its minimum. $\square$

8.10.12 Theorem (Heine-Cantor). *Let (X, d_X) and (Y, d_Y) be two metric spaces and $f : X \rightarrow Y$ be a continuous mapping. If X is compact, then f is uniformly continuous.*

Proof. We reason by contradiction. Assume that f is not uniformly continuous, then, there exists $\varepsilon > 0$ and two sequences $(x_n)_{n \in \mathbb{N}}$ and $(y_n)_{n \in \mathbb{N}}$ such that $d_X(x_n, y_n) \leq 1/(n+1)$ and $d_Y(f(x_n), f(y_n)) \geq \varepsilon$. Since X is a compact metric space, it is sequentially compact. Hence there exists an infinite subset I of $\mathbb{N}$ such that $(x_n)_{n \in I}$ and $(y_n)_{n \in I}$ converge in X . Since

$$\lim_{n \in I, n \rightarrow +\infty} d_X(x_n, y_n) = 0,$$

the sequences $(x_n)_{n \in I}$ and $(y_n)_{n \in I}$ converges to the same element p . By the continuity of the mapping f , one has

$$\lim_{n \in I, n \rightarrow +\infty} f(x_n) = \lim_{n \in I, n \rightarrow +\infty} f(y_n) = f(p).$$

This contradicts the condition $d_Y(f(x_n), f(y_n)) \geq \varepsilon$ for any $n \in \mathbb{N}$. $\square$

8.11 Path connectedness

8.11.1 Definition. Let $(X, \mathcal{T})$ be a topological space and C be a subset of X . If, for any $(x, y) \in C \times C$, there exists a continuous mapping $\varphi : [0, 1] \rightarrow X$ such that $\varphi([0, 1]) \subseteq C$ and $\varphi(0) = x$, $\varphi(1) = y$, then we say that C is *path connected*.

8.11.2 Proposition. Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces, and $f : X \rightarrow Y$ be a continuous mapping. If C is a path connected subset of X , then $f(C)$ is a path connected subset of Y .

Proof. Let a and b be elements of $f(C)$. There exists $(x, y) \in C \times C$ such that $f(x) = a$ and $f(y) = b$. Since C is path connected, there exists a continuous mapping $\varphi : [0, 1] \rightarrow X$ such that $\varphi([0, 1]) \subseteq C$, $\varphi(0) = x$ and $\varphi(1) = y$. Then $f \circ \varphi$ is a continuous mapping from $[0, 1]$ to Y such that $f(\varphi([0, 1])) \subseteq f(C)$, and $(f \circ \varphi)(0) = f(x) = a$, $(f \circ \varphi)(1) = f(y) = b$. $\square$

8.11.3 Theorem. A subset I of $\mathbb{R}$ is path connected if and only if it is an interval.

Proof. Assume that I is an interval. For any $(a, b) \in I \times I$ such that $a \leq b$, one has $[a, b] \subseteq I$. Consider the mapping $f : [0, 1] \rightarrow \mathbb{R}$ defined as

$$f(t) = a + t(b - a).$$

This is the sum of a constant mapping and a linear mapping. Hence it is continuous. Moreover, $f(0) = a$ and $f(1) = b$. Hence I is path connected.

Assume that I is parth connected. We show that I is an interval. Let a and b be elements of I such that $a \leq b$. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous mapping such that $f(0) = a$, $f(1) = b$ and $f([0, 1]) \subseteq I$.

Let c be an element of $[a, b]$. Assume that c does not belong to the image of f . Then one has $a < c < b$. We now construct in a recursive way two sequences $(x_n)_{n \in \mathbb{N}}$ and $(y_n)_{n \in \mathbb{N}}$ in $[0, 1]$ as follows: let $x_0 = 0$ and $y_0 = 1$. If x_n and y_n are chosen so that $x_n < y_n$, $f(x_n) < c < f(y_n)$. Let $z_n = (x_n + y_n)/2$. If $f(z_n) > c$, then we take $x_{n+1} = x_n$ and $y_{n+1} = z_n$; if $f(z_n) < c$, then we take $x_{n+1} = z_n$ and $y_{n+1} = y_n$. The sequence $(x_n)_{n \in \mathbb{N}}$ is increasing, the sequence $(y_n)_{n \in \mathbb{N}}$ is decreasing, and

$$0 \leq y_n - x_n \leq \frac{1}{2^n}.$$

Hence both sequence converges to the same point $w \in [0, 1]$. Note that

$$\forall n \in \mathbb{N}, (c - f(x_n))(c - f(y_n)) < 0.$$

Taking the limit when n tends to the infinity, by the continuity of f we get

$$0 \leq (c - f(w))^2 = \lim_{n \rightarrow +\infty} (c - f(x_n))(c - f(y_n)) \leq 0.$$

Therefore $f(w) = c$, which leads to a contradiction. $\square$

8.11.4 Corollary. *Let I be an interval in $\mathbb{R}$. If $f : I \rightarrow \mathbb{R}$ is a continuous mapping, then $f(I)$ is an interval. In particular, if f has a positive value and a negative value, then it vanishes at some point of I .*

Proof. The topological space I is path connected since it is an interval in $\mathbb{R}$, so $f(I)$ is path connected, and hence is an interval in $\mathbb{R}$. $\square$

Exercises

1. We equip $\mathbb{R}$ with the topology defined by the usual metric $(a, b) \mapsto |a - b|$.
 - (1) Let a and b be two real numbers such that $a < b$. Express the interval $]a, b[$ as an open ball. Determine its center and its radius.
 - (2) Let a be a real number. Show that the intervals $]a, +\infty[$ and $] -\infty, a[$ are open subsets of $\mathbb{R}$.
 - (3) Let a be a real number. Show that the intervals $[a, +\infty[$ and $] -\infty, a]$ are closed subsets of $\mathbb{R}$.
 - (4) Let a and b be real numbers such that $a \leq b$. Show that $[a, b]$ is a closed subset of $\mathbb{R}$.
2. (1) If $\{\mathcal{T}_i\}_{i \in I}$ is a family of topologies on X , show that $\bigcap_{i \in I} \mathcal{T}_i$ is a topology on X . Is $\bigcup_{i \in I} \mathcal{T}_i$ a topology on X ?

- (2) Let $\{\mathcal{T}_i\}_{i \in I}$ be a family of topologies on X . Show that there is a unique smallest topology on X containing all the collections $\mathcal{T}_i$, and a unique largest topology contained in all $\mathcal{T}_i$.
- (3) If $X = \{a, b, c\}$, let

$$\mathcal{T}_1 = \{\emptyset, X, \{a\}, \{a, b\}\} \text{ and } \mathcal{T}_2 = \{\emptyset, X, \{a\}, \{b, c\}\}.$$

Find the smallest topology containing $\mathcal{T}_1$ and $\mathcal{T}_2$, and the largest topology contained in $\mathcal{T}_1$ and $\mathcal{T}_2$.

3. Let X and Y be topological spaces. The **product topology** on $X \times Y$ is the topology having as basis the collection $\mathcal{B}$ of all sets of the form $U \times V$, where U is an open subset of X and V is an open subset of Y .

- (1) Prove that such a collection $\mathcal{B}$ is a topological basis.
- (2) Give an example to show that the collection $\mathcal{B}$ is not a topology on $X \times Y$.
- (3) A subbasis $\mathcal{S}$ for a topology on X is a collection of subsets of X whose union equals X . The topology generated by the subbasis $\mathcal{S}$ is defined to be the collection $\mathcal{T}$ of all unions of finite intersections of elements of $\mathcal{S}$. In other words, we have

$$\mathcal{T} = \left\{ \bigcup_{B \in \mathcal{B}'} B \mid \mathcal{B}' \subset \mathcal{B} \right\},$$

where $\mathcal{B} = \{S_1 \cap \cdots \cap S_n \mid S_i \in \mathcal{S}, i = 1, \dots, n, n \in \mathbb{N}^+\}$.

Prove that the above definition is well-defined, which means you need to prove that $\mathcal{T}$ is a topology. **Hint:** you only need to prove that all the finite intersections of elements in $\mathcal{S}$ forms a topological basis.

- (4) Let $\mathcal{B}$ be a basis for the topology of X , and $\mathcal{C}$ be a basis for the topology of Y . Prove that the collection

$$\mathcal{D} = \{B \times C \mid B \in \mathcal{B}, C \in \mathcal{C}\}$$

is a basis for the topology of $X \times Y$.

- (5) Let X, Y be two topological spaces,

$$\begin{array}{ccc} \pi_1 : & X \times Y & \longrightarrow X \\ & (x, y) & \mapsto x \end{array},$$

and

$$\begin{array}{ccc} \pi_2 : & X \times Y & \longrightarrow Y \\ & (x, y) & \mapsto y \end{array}$$

be the projections. Prove that

$$\mathcal{S} = \{\pi_1^{-1}(U) \mid U \text{ open in } X\} \cup \{\pi_2^{-1}(V) \mid V \text{ open in } Y\}$$

is a subbasis for the product topology on $X \times Y$.

- (6) Describe the product topology of $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$, where each $\mathbb{R}$ is equipped with the standard topology.

4. Let X be a topological space with topology $\mathcal{T}$. If Y is a subset of X , we define the collection

$$\mathcal{T}_Y = \{Y \cap U \mid U \in \mathcal{T}\}.$$

- (1) Prove that $\mathcal{T}_Y$ is a topology on Y , which is called the **subspace topology**. With this topology, Y is called a subspace of X , where its open sets consist of all intersections of open sets of X with Y .
- (2) Let $\mathcal{B}$ be a basis for the topology of X . Prove that the collection

$$\mathcal{B}_Y = \{B \cap Y \mid B \in \mathcal{B}\}$$

is a basis for the subspace topology on Y .

- (3) With all the above notations, give an example to show that an open subset of Y with respect to $\mathcal{T}_Y$ do not necessarily to be an open subset of X with respect to $\mathcal{T}$.
- (4) Let Y be a subspace of X . If U is open in Y and Y is open in X , then U is open in X .
- (5) Let A be a subspace of X and B be a subspace of Y . Prove that the product topology on $A \times B$ is the same as the topology $A \times B$ inherits as a subspace of $X \times Y$.
5. We equip $\mathbb{R}$ with the topology defined by the usual metric. Let I be an open subset of $\mathbb{R}$. We assume that I is not empty. We define a binary relation $\sim$ on I as follows: for any $(a, b) \in I \times I$, $a \sim b$ if and only if the closed interval with extremities a and b is contained in I , that is, $[a, b] \subseteq I$ when $a \leq b$, and $[b, a] \subseteq I$ when $b \leq a$.
- (1) Show that $\sim$ is an equivalence relation on I .
- (2) Let J be an equivalence class of I under the equivalence relation $\sim$. Show that J is an interval.
- (3) Let J be an interval in $\mathbb{R}$ which is contained in I . Show that J is contained in some equivalence class of I under the equivalence relation $\sim$.

- (4) Show that any equivalence class of I under the equivalence relation $\sim$ is an interval of the form $]a, b[$ with $(a, b) \in [-\infty, +\infty]^2$, $a < b$.
- (5) Show that I is a disjoint union of countably many open intervals.
6. Let n be a positive integer and (X_i, d_i) , $i \in \{1, \dots, n\}$ be a family of metric spaces. Let X be the product set $X_1 \times \dots \times X_n$. Recall that the elements of X are n -tuples $x = (x_1, \dots, x_n)$ with $x_1 \in X_1, \dots, x_n \in X_n$. We define a function $d : X \times X \rightarrow \mathbb{R}_{\geq 0}$ as follows: for any elements $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$ of X ,

$$d(x, y) := \max_{i \in \{1, \dots, n\}} d_i(x_i, y_i).$$

- (1) Show that d is a metric on X . We call it the *product metric* of $d_1, \dots, d_n$.
- (2) Let $x = (x_1, \dots, x_n)$ be an element of X and $r > 0$. Show that

$$B_d(x, r) = B_{d_1}(x_1, r) \times \dots \times B_{d_n}(x_n, r).$$

- (3) Let $(x^{(k)})_{k \in \mathbb{N}}$ be an element of $X^{\mathbb{N}}$, and $x = (x_1, \dots, x_n) \in X$. We write each $x^{(k)} \in X$ in the form of $(x_1^{(k)}, \dots, x_n^{(k)})$, where $x_i^{(k)} \in X_i$ for any $i \in \{1, \dots, n\}$.
- (3.a) Show that $(x^{(k)})_{k \in \mathbb{N}}$ is a Cauchy sequence if and only if each $(x_i^{(k)})_{k \in \mathbb{N}}$ is a Cauchy sequence, where $i \in \{1, \dots, n\}$.
- (3.b) Show that the sequence $(x^{(k)})_{k \in \mathbb{N}}$ converges in X to x if and only if, for each $i \in \{1, \dots, n\}$, the sequence $(x_i^{(k)})_{k \in \mathbb{N}}$ in X converges to x_i .
- (4) Show that, if all metric spaces $(X_1, d_1), \dots, (X_n, d_n)$ are complete, then (X, d) is also complete.
- (5) Let Y be a topological space. For any $i \in \{1, \dots, n\}$, let $f_i : Y \rightarrow X_i$ be a mapping. Let $f : Y \rightarrow X_1 \times \dots \times X_n$ be the mapping sending $y \in Y$ to $(f_1(y), \dots, f_n(y))$. Show that the mapping f is continuous at some $y_0 \in Y$ if and only if each mapping f_i is continuous at y_0 , where $i \in \{1, \dots, n\}$.

7. Let V be a vector space over $\mathbb{R}$. We call *norm on V* any mapping $\|\cdot\| : V \rightarrow \mathbb{R}_{\geq 0}$ that satisfies the following conditions:

- (i) for any $x \in V \setminus \{\mathbf{0}_V\}$, $\|x\| > 0$,
- (ii) for any $(a, x) \in \mathbb{R} \times V$, $\|ax\| = |a| \cdot \|x\|$,
- (iii) for any $(x, y) \in V \times V$, $\|x + y\| \leq \|x\| + \|y\|$.

The couple $(V, \|\cdot\|)$ is called a normed vector space over $\mathbb{R}$.

- (1) Show that the function

$$d : V \times V \longrightarrow \mathbb{R}_{\geq 0}, \quad (x, y) \in V \times V \longmapsto \|x - y\|$$

is a metric on V . We call this metric the *metric associated with the norm* $\|\cdot\|$.

- (2) Let n be a positive integer. Show that the mapping

$$\|\cdot\|_{\ell^\infty} : \mathbb{R}^n \longrightarrow \mathbb{R}_{\geq 0}, \quad \|(x_1, \dots, x_n)\|_{\ell^\infty} := \max_{i \in \{1, \dots, n\}} |x_i|.$$

Show that $\|\cdot\|_{\ell^\infty}$ is a norm on $\mathbb{R}^n$.

- (3) Let $d_\infty : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}_{\geq 0}$ be the metric associated with the norm $\|\cdot\|_{\ell^\infty}$. Show that the metric space $(\mathbb{R}^n, d_\infty)$ is complete. One can express d_∞ as a product metric.

- (4) Show that the mapping

$$\|\cdot\|_{\ell^1} : \mathbb{R}^n \longrightarrow \mathbb{R}_{\geq 0}, \quad \|(x_1, \dots, x_n)\|_{\ell^1} := \sum_{i=1}^n |x_i|$$

is a norm on $\mathbb{R}^n$.

- (5) Show that, for any $x \in \mathbb{R}^n$, one has

$$\|x\|_{\ell^\infty} \leq \|x\|_{\ell^1} \leq n\|x\|_{\ell^\infty}.$$

- (6) Let $d_1 : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}_{\geq 0}$ the metric associated with the norm $\|\cdot\|_{\ell^1}$. Let $(x^{(k)})_{k \in \mathbb{N}}$ be a sequence in $\mathbb{R}^n$ and x be an element of $\mathbb{R}^n$.

(6.a) Show that $(x^{(k)})_{k \in \mathbb{N}}$ is a Cauchy sequence in (X, d_∞) if and only if it is a Cauchy sequence in (X, d_1) .

(6.b) Show that $(x^{(k)})_{k \in \mathbb{N}}$ converges to x in (X, d_∞) if and only if it converges to x in (X, d_1) .

- (7) Deduce that the metric space (X, d_1) is complete.

8. Let X and Y be metric spaces and α be a positive constant. We say that a mapping $f : X \rightarrow Y$ is α -Lipschitzian if

$$\forall (a, b) \in X \times X, \quad d(f(a), f(b)) \leq \alpha d(a, b).$$

- (1) Let X and Y be metric spaces and $f : X \rightarrow Y$ be a mapping which is α -Lipschitzian for some positive constant α . Show that f is uniformly continuous.

- (2) Let (X, d) be a metric space and x_0 be an element of X . Show that the mapping $f : X \rightarrow \mathbb{R}$ sending $x \in X$ to $d(x_0, x)$ is 1-Lipschitzian.
- (3) Let (X, d) be a metric space and x_0 be an element of X . Show that, for any $r > 0$, the set

$$\overline{B}(x_0, r) := \{x \in X \mid d(x_0, x) \leq r\}$$

is a closed subset of X .

- (4) Let $(V, \|\cdot\|)$ be a normed vector space over $\mathbb{R}$. Show that the mapping $\|\cdot\| : V \rightarrow \mathbb{R}$ is 1-Lipschitzian.
9. Let X and Y be metric spaces, and $f : X \rightarrow Y$ be a mapping, and x_0 be an element of X , we say that f is locally Lipschitzian at x_0 if there exists $r > 0$ and $\alpha > 0$ such that

$$\forall (a, b) \in B(x_0, r) \times B(x_0, r), \quad d(f(a), f(b)) \leq \alpha d(a, b).$$

Prove that, if f is locally Lipschitzian at x_0 , then it is continuous at x_0 .

10. We equip $\mathbb{R}$ with the usual metric and $\mathbb{R}^2$ with the product metric.

- (1) Show that the mapping

$$+ : \mathbb{R} \times \mathbb{R} \longrightarrow \mathbb{R}, \quad (a, b) \longmapsto a + b$$

is 2-Lipschitzian. Deduce that this mapping is continuous.

- (2) Let Y be a topological space and $f : Y \rightarrow \mathbb{R}$ and $g : Y \rightarrow \mathbb{R}$ be mappings. Assume that both f and g are continuous at some $y_0 \in Y$. Show that the mapping

$$f + g : Y \longrightarrow \mathbb{R}, \quad (y \in Y) \longmapsto f(y) + g(y)$$

is continuous at y_0 .

- (3) Let a be a real number. Show that the mapping

$$\mathbb{R} \longrightarrow \mathbb{R}, \quad t \longmapsto at$$

is $|a|$ -Lipschitzian. Deduce that the mapping is continuous.

- (4) Let Y be a topological space and $f : Y \rightarrow \mathbb{R}$ be a mapping which is continuous at some $y_0 \in Y$. Show that, for any $a \in \mathbb{R}$, the mapping

$$af : Y \longrightarrow \mathbb{R}, \quad (y \in Y) \longmapsto af(y)$$

is continuous at y_0 .

- (5) Show that the mapping

$$\mathbb{R} \times \mathbb{R}, \quad (a, b) \longmapsto ab$$

is locally Lipschitzian at any point $(a_0, b_0) \in \mathbb{R} \times \mathbb{R}$.

- (6) Let Y be a topological space and $f : Y \rightarrow \mathbb{R}$ and $g : Y \rightarrow \mathbb{R}$ be mappings. Assume that both f and g are continuous at some $y_0 \in Y$. Show that the mapping

$$fg : Y \longrightarrow \mathbb{R}, \quad (y \in Y) \longmapsto f(y)g(y)$$

is continuous at y_0 .

- (7) Let P be an element of the polynomial ring $\mathbb{R}[T]$. Show that the mapping

$$\mathbb{R} \longrightarrow \mathbb{R}, \quad x \longmapsto P(x)$$

is continuous.

- (8) Let n be a positive integer. We equip $\mathbb{R}^n$ with the product metric of the usual metric on $\mathbb{R}$.

- (8.a) Show that, for any $i \in \{1, \dots, n\}$, the mapping

$$p_i : \mathbb{R}^n \longrightarrow \mathbb{R}, \quad (x_1, \dots, x_n) \longmapsto x_i$$

is 1-Lipschitzian.

- (8.b) Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be an $\mathbb{R}$ -linear mapping. Show that there exists $(a_1, \dots, a_n) \in \mathbb{R}^n$ such that

$$\forall (x_1, \dots, x_n) \in \mathbb{R}^n, \quad f(x_1, \dots, x_n) = a_1x_1 + \dots + a_nx_n.$$

- (8.c) Deduce that any $\mathbb{R}$ -linear mapping from $\mathbb{R}^n$ to $\mathbb{R}$ is continuous.

- (8.d) Let m be another positive integer. We equip $\mathbb{R}^m$ with the product metric of the usual metric on $\mathbb{R}$. Show that any $\mathbb{R}$ -linear mapping from $\mathbb{R}^n$ to $\mathbb{R}^m$ is continuous.

- 11.** Prove that the following subsets of $\mathbb{R}^2$ (equipped with the product metric of the usual metric) are open. Draw the picture of each of these subsets.

- (1) $\{(x, y) \in \mathbb{R}^2 \mid x < y\}$.
- (2) $\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 < 1\}$.
- (3) $\{(x, y) \in \mathbb{R}^2 \mid xy > 1\}$.
- (4) $\{(x, y) \in \mathbb{R}^2 \mid x \neq 0\}$

$$(5) \{(x, y) \in \mathbb{R}^2 \mid |x| < |y|\}.$$

- 12.** Let $(a_n)_{n \in \mathbb{N}}$ be a sequence in $\mathbb{R}$. Let R be a positive real number such that the sequence $(a_n R^n)_{n \in \mathbb{N}}$ is bounded.

- (1) Let t be an element of $\mathbb{R}$ such that $|t| < R$. Show that the series

$$\sum_{n \in \mathbb{N}} a_n t^n$$

converges absolutely. We denote by $f(t)$ the limit of this series.

- (2) For any $n \in \mathbb{N}$, let $f_n : \mathbb{R} \rightarrow \mathbb{R}$ be the mapping sending $t \in \mathbb{R}$ to

$$\sum_{k=0}^n a_k t^k.$$

Let r be a positive real number such that $r < R$. Show that the sequence of functions $(f_n)_{n \in \mathbb{N}}$ converges uniformly to f on $B(0, r)$.

- (3) Deduce that the mapping $f : B(0, R) \rightarrow \mathbb{R}$ is continuous.
 (4) Prove that the function

$$\exp : \mathbb{R} \longrightarrow \mathbb{R}, \quad t \longmapsto \sum_{n \in \mathbb{N}} \frac{t^n}{n!}$$

is well-defined and is continuous.

- (5) Prove that the function

$$B(0, 1) \longrightarrow \mathbb{R}, \quad t \longmapsto \sum_{n \in \mathbb{N}} \frac{(-1)^n}{n} t^n$$

is well-defined and is continuous.

- 13.** Let $P \in \mathbb{R}[T]$ be a polynomial of odd degree. Show that there exists $t \in \mathbb{R}$ such that $P(t) = 0$.

- 14.** In this exercise, we consider the mapping $f : \mathbb{R} \rightarrow \mathbb{R}$ given by

$$f(x) = x^3 - 3x - 3.$$

- (1) Show that the mapping f is strictly increasing on $[1, +\infty[$.
 (2) Show that the equation $x^3 - 3x - 3 = 0$ has a unique solution in the interval $[2, 2 + \frac{1}{9}]$.

- (3) Use the algorithm of dichotomy to give an upper and a lower bound of the solution with an error $< 100^{-1}$.

15. Let I be an interval in $\mathbb{R}$ and $f : I \rightarrow \mathbb{R}$ be a monotone mapping.

- (1) Show that, for any interval J in $\mathbb{R}$, $f^{-1}(J)$ is an interval.
- (2) Assume that $f(I)$ is an interval. Show that f is continuous.
- (3) Prove that, if f is strictly monotone and continuous, then $f^{-1} : f(I) \rightarrow \mathbb{R}$ is a continuous mapping.
- (4) Denote by $\ln :]0, +\infty[\rightarrow \mathbb{R}$ the inverse mapping of $\exp$. Show that $\ln$ is a continuous mapping.

16. We now define three topologies on the real line $\mathbb{R}$, all of which are of interest. If $\mathcal{B}$ is the collection of all open intervals in the real line,

$$]a, b[= \{x \mid a < x < b\},$$

the topology generated by $\mathcal{B}$ is called the **standard topology** (or **Euclidean topology**) on the real line. Whenever we consider $\mathbb{R}$, we shall suppose it is given this topology unless we specifically state otherwise.

If $\mathcal{B}'$ is the collection of all half-open intervals of the form

$$[a, b[= \{x \mid a \leq x < b\},$$

where $a < b$, the topology generated by $\mathcal{B}'$ is called the **lower limit topology** on $\mathbb{R}$. When $\mathbb{R}$ is given the lower limit topology, we denote it by $\mathbb{R}_\ell$.

Finally let K denote the set of all numbers of the form $\frac{1}{n}$, for $n \in \mathbb{N}^+$, and let $\mathcal{B}''$ be the collection of all open intervals (a, b) , along with all sets of the form $(a, b) \setminus K$. The topology generated by $\mathcal{B}''$ will be called the **K -topology** on $\mathbb{R}$. When $\mathbb{R}$ is given this topology, we denote it by $\mathbb{R}_K$.

- (1) Prove that all three of these collections are bases.
- (2) Prove that the topologies of $\mathbb{R}_\ell$ and $\mathbb{R}_K$ are strictly finer than the standard topology on $\mathbb{R}$, but are not comparable with one another.

17. Check that $\mathcal{T} = \{(a, +\infty) \mid a \in \mathbb{R}\}$ is a topology on $\mathbb{R}$, find $\mathcal{V}_p(\mathcal{T})$ for $p \in \mathbb{R}$.

18. Show that

$$\lim_{\mathcal{F}} f$$

does not need to be unique.

- 19.** What happens with $\lim_{\mathcal{F}} f$ for the degenerate $\mathcal{F}$? Do we really need it?

Assume that $\mathcal{V}_p(\mathcal{T}_x)/\text{Dom}(f)$ is non-degenerate. Let $(X, \mathcal{T}_X), (Y, \mathcal{T}_Y)$ be two topological spaces, $f : X \rightarrow Y$ be a continuous map. We say that $f(x)$ converges to ℓ when $x \in \text{Dom}(f)$ tends to p and write

$$\lim_{x \in \text{Dom}(f), x \rightarrow p} f(x) = \ell$$

when

$$\lim_{\mathcal{V}_p(\mathcal{T}_X)/\text{Dom}(f)} f = \ell.$$

- 20.** (a) Prove that f is continuous at p if and only if

$$\lim_{x \in \text{Dom}(f), x \rightarrow p} f(x) = f(p),$$

i.e., $f(x)$ converges to $f(p)$ when $x \in \text{Dom}(f)$ tends to p .

- (b) Let $\mathcal{T}_1$ and $\mathcal{T}_2$ be two topologies on X . Prove that

$$\text{id} : (X, \mathcal{T}_1) \rightarrow (X, \mathcal{T}_2)$$

is continuous if and only if $\mathcal{T}_2 \subseteq \mathcal{T}_1$.

- (c) Prove that

$$\begin{array}{ccc} x_0 : (X, \mathcal{T}_1) & \longrightarrow & (X, \mathcal{T}_2) \\ x & \mapsto & x_0 \end{array}$$

is locally continuous.

- 21.** Let $(L, |\cdot|)$ be a valued field. Show that $|1_K| = |-1_K| = 1$.

- 22.** Let $(E, \|\cdot\|)$ be a semi-normed vector space. Show that

$$\begin{array}{ccc} E \times E & \longrightarrow & \mathbb{R} \\ (x, y) & \mapsto & \|x - y\| \end{array}$$

is a semi-norm.

- 23.** Let (X, d) be a pseudo-metric space.

- (1) For $x, y \in X$, check that $x \sim y \Leftrightarrow d(x, y) = 0$ is an equivalent relation.
 (2) Let $X^* = X/\sim$ be the quotient space. Show that

$$\begin{array}{ccc} d^* : X^* \times X^* & \longrightarrow & \mathbb{R}_{\geq 0} \\ ([x], [y]) & \mapsto & d(x, y) \end{array}$$

is a metric.

- 24.** Let $(K, |\cdot|)$ be a non-trivial valued field, where $K = \mathbb{C}$ or $\mathbb{R}$, $(E, \|\cdot\|)$ and $(F, \|\cdot\|)$ be two seminormed spaces over K , and $f : E \rightarrow F$ be a morphism. Prove that the following statements are equivalent.

- (1) f is continuous.
- (2) $\exists p \in E$, such that f is continuous at p .
- (3) f is continuous at $0 \in E$.
- (4) f is bounded in some neighborhood of 0.
- (5) $\exists M \geq 0$, such that $\forall x \in E, \|x\| \leq 1 \Rightarrow \|f(x)\| \leq M$.
- (6) $\exists M \geq 0$, such that $\forall x \in E, \|f(x)\| \leq M\|x\|$.
- (7) f is uniformly continuous.

- 25.** Let $\mathcal{L}(E, F)$ be the set of linear maps from E to F , where $(E, \|\cdot\|_E)$ and $(F, \|\cdot\|_F)$ are two normed vector space over a valued field $(K, |\cdot|)$. For a $f \in \mathcal{L}(E, F)$, we define the **operator norm** of f as

$$\|f\| := \sup_{x \in E, \|x\| \neq 0} \frac{\|f(x)\|_F}{\|x\|_E}.$$

Prove the following statements.

- (1) $\|f\| = \sup\{\|f(x)\|_F \mid x \in E, \|x\|_E \leq 1\}$.
 - (2) $\|f\| = \inf\{M \geq 0 \mid \forall x \in E, \|f(x)\|_F \leq M\|x\|_E\}$.
 - (3) If $E \neq \{0\}$, then $\|f\| = \sup\{\|f(x)\|_F \mid x \in E, \|x\|_E = 1\}$.
- 26.** Let X be a topological space. We say that X is **irreducible** if $X = X_1 \cup X_2$ with the closed subsets X_1 and X_2 , then we have $X = X_1$ or $X = X_2$.
- (1) Let $\mathbb{R}$ be the topological space equipped with the standard topology. Prove that if I is an irreducible subspace of $\mathbb{R}$, then I consists of a single point.
 - (2) Let X be an irreducible topological space. Prove that the intersection of two non-empty open subsets of X is non-empty.
 - (3) Let X be a topological space. Prove that the set of irreducible subspaces of X admits maximal elements for the inclusion relation, and such a maximal element is a closed subset.

Let X be a topological space. We call these maximal elements determined in (3) the **irreducible components** of X .

- (4) Prove that the union of all irreducible components of a topological space X is just X .
 - (5) Let X be irreducible. Prove that any non-empty open subset of X is dense in X and is irreducible.
 - (6) Let U be an open subset of X . Prove that the irreducible components of U are the $\{X_i \cap U\}_i$, where the X_i are the irreducible components of X which meet U .
 - (7) Suppose that X is a finite union of irreducible closed subsets $(Z_j)_{j=1}^n$. Prove that every irreducible component Z of X is equal to one of the Z_j , $j = 1, \dots, n$. If, moreover, there is no inclusion relation between the $(Z_j)_{j=1}^n$, then prove that the $(Z_j)_{j=1}^n$ are exactly the irreducible components of X .
- 27.** Let $\mathbb{A}_{\mathbb{C}}^n = \{(x_1, \dots, x_n) \mid x_1, \dots, x_n \in \mathbb{C}\}$, and $A = \mathbb{C}[T_1, \dots, T_n]$. For a $P = (a_1, \dots, a_n) \in \mathbb{A}_{\mathbb{C}}^n$ and an $f \in A$, we define $f(P) = f(a_1, \dots, a_n)$ for simplicity. We define

$$Z(f) = \{P \in \mathbb{A}_{\mathbb{C}}^n \mid f(P) = 0\}$$

as the set of zeros of f . Let $T \subseteq A$, we define

$$Z(T) = \{P \in \mathbb{A}_{\mathbb{C}}^n \mid f(P) = 0, \forall f \in T\}$$

as the common zeros of the elements in T .

Let $Y \subseteq \mathbb{A}_{\mathbb{C}}^n$. We say that Y is an **algebraic set** if there exists a $T \subseteq A$, such that $Y = Z(T)$.

- (1) Prove that the union of two algebraic sets is an algebraic set, the intersection of any family of algebraic sets is an algebraic set, and the empty set and the whole space are algebraic sets.
- (2) We take the open subsets of $\mathbb{A}_{\mathbb{C}}^n$ as the complements of the algebraic sets. Prove that this operation defines a topology on $\mathbb{A}_{\mathbb{C}}^n$. We call it the **Zariski topology** on $\mathbb{A}_{\mathbb{C}}^n$.
- (3) We consider $\mathbb{A}_{\mathbb{C}}^1$. Describe the algebraic set of $\mathbb{A}_{\mathbb{C}}^1$. Is the Zariski topology on $\mathbb{A}_{\mathbb{C}}^1$ Hausdorff or not?
- (4) Prove that the Zariski topology on $\mathbb{A}_{\mathbb{C}}^1$ is irreducible (see the previous exercise for the definition of irreducible topology).
- (5) Let $Y \subseteq \mathbb{A}_{\mathbb{C}}^n$. We define $I(Y) = \{f \in A \mid f(P) = 0, \forall P \in Y\}$. Prove the following results.

- i. If $T_1 \subseteq T_2$ are subsets of A , then $Z(T_1) \supseteq Z(T_2)$.
- ii. If $Y_1 \subseteq Y_2$ are subsets of $\mathbb{A}_{\mathbb{C}}^n$, then $I(Y_1) \supseteq I(Y_2)$.
- iii. For any two subsets Y_1, Y_2 of $\mathbb{A}_{\mathbb{C}}^n$, we have $I(Y_1 \cup Y_2) = I(Y_1) \cap I(Y_2)$.

28. Prove the following statements.

- (1) $\lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0$;
- (2) $\lim_{x \rightarrow a} \sqrt[n]{x} = \sqrt[n]{a}$ ($a > 0$, $n \in \mathbb{N}$);
- (3) $\lim_{x \rightarrow 1} (3x^3 - 2) = 1$;
- (4) $\lim_{x \rightarrow 3} (2x^2 - x) = 15$;
- (5) $\lim_{x \rightarrow \pm\infty} \frac{[x]}{x} = 1$, where $[x]$ is the largest integer smaller than $x \in \mathbb{R}$;
- (6) $\lim_{x \rightarrow \pm\infty} \frac{2x^2 - 1}{3x^2 - x} = \frac{2}{3}$;
- (7) $\lim_{x \rightarrow \pm\infty} \frac{3x^2 - 1}{x^2 + x + 1} = 3$;
- (8) $\lim_{x \rightarrow +\infty} (\sqrt{x^2 + x} - x) = \frac{1}{2}$.

29. Prove that $\lim_{x \rightarrow 0} \frac{1}{x}$ does not exist.

30. Prove that if $\lim_{x \rightarrow a} f(x) = A$, then $\lim_{x \rightarrow a} |f(x)| = |A|$. Here we assume that $a \in \mathbb{R}$ or $a = \pm\infty$.

31. Let the function f be defined as

$$f : x \mapsto x - [x], \quad x \in \mathbb{R}.$$

Prove that

- (1) for any $n \in \mathbb{Z}$, $\lim_{x \rightarrow n} f(x)$ does not exist;
- (2) for any $a \notin \mathbb{Z}$, $\lim_{x \rightarrow a} f(x)$ exists.

32. Calculate the following limits.

- (1) $\lim_{x \rightarrow a} \frac{\sin x - \sin a}{x - a}$;
- (2) $\lim_{x \rightarrow 0} \frac{\sin nx}{\sin mx}$ ($n, m \in \mathbb{N}$);

- (3) $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2};$
 (4) $\lim_{x \rightarrow 0} \frac{\sqrt{1 + \sin x} - \sqrt{1 - \sin x}}{x};$
 (5) $\lim_{x \rightarrow 0} \frac{(1+mx)^n - (1+nx)^m}{x^2} \quad (m, n \in \mathbb{N});$
 (6) $\lim_{x \rightarrow 0} \frac{(1+x)^{\frac{1}{n}} - 1}{x} \quad (n \in \mathbb{N});$
 (7) $\lim_{x \rightarrow a} \frac{x^{\frac{1}{m}} - a^{\frac{1}{m}}}{x^{\frac{1}{n}} - a^{\frac{1}{n}}} \quad (a > 0, m, n \in \mathbb{N}).$

33. Determine the parameters a, b and c such that the following limit

$$\lim_{x \rightarrow +\infty} \left(\frac{x^3 + 2}{x + 1} - ax^2 - bx - c \right)$$

equals to 0 or 1.

34. Suppose that $f : X \rightarrow \mathbb{R}$ and $g : Y \rightarrow \mathbb{R}$ are two functions such that $f(X) \subseteq Y$, $a \in \mathbb{R}$ is a limit point of X and $\lim_{x \rightarrow a} f(x) = A$. Moreover, $A \in \mathbb{R}$ is a limit point of Y and $\lim_{y \rightarrow A} g(y) = B$.

- (1) Is it always true that $\lim_{x \rightarrow a} g \circ f(x) = B$?

Hint: Consider the example where $f, g : \mathbb{R} \rightarrow \mathbb{R}$ are two functions defined by

$$f(x) = \begin{cases} 0 & x = 0, \\ x \sin(\frac{1}{x}) & x \neq 0, \end{cases} \quad \text{and} \quad g(y) = \begin{cases} 1 & y = 0, \\ 0 & y \neq 0. \end{cases}$$

- (2) Prove that if there exists $\delta > 0$ such that $A \notin f(X \cap (]a - \delta, a + \delta[\setminus \{a\}))$, then $\lim_{x \rightarrow a} g \circ f(x) = B$.

35. Suppose that $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ is an increasing function. Prove that the following three statements are all equivalent:

- (1) $\lim_{x \rightarrow +\infty} f(x)$ exists;
 (2) the sequence $(f(n))$ converges;
 (3) f is bounded above on $\mathbb{R}^+$.

36. Let $f : [0, 1] \rightarrow [0, 1]$ be an arbitrary increasing function.

- (1) Prove that there exists $x \in [0, 1]$ such that $f(x) = x$.

- (2) Suppose $f(0) = 0$ and for any $x \in]0, 1]$, $f(x) < x$. Define the sequence (x_n) by $x_{n+1} = f(x_n)$ for any $n \in \mathbb{N}$ and $x_0 \in]0, 1]$. Does the sequence (x_n) converge?

37. Find the sets of the points that the following functions are continuous at. For the points of discontinuity, specify their kind of discontinuity.

- (1) $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ defined by $f(x) = \sqrt{x}$;
 (2) $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} x^2 & x \in \mathbb{Q} \\ 0 & x \notin \mathbb{Q}; \end{cases}$$

- (3) $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} \frac{\sin x}{|x|} & x \neq 0 \\ 1 & x = 0; \end{cases}$$

- (4) $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} \frac{x}{1-|x|} & |x| \neq 1 \\ 1 & |x| = 1; \end{cases}$$

- (5) $f : [0, a^2] \rightarrow \mathbb{R}$ defined by $f(x) = \sin(\cos x + \sqrt{1 + a^2})$.

38. Prove the following properties about continuous functions.

- (1) Prove that if f is continuous at a , then so is $|f|$. Is the converse true?
 (2) Prove that every continuous function f can be written as $f = E + O$, where E is even and continuous, and O is odd and continuous.
 (3) Prove that if f and g are continuous, then so are $\max(f, g)$ and $\min(f, g)$.
 (4) Prove that every continuous function f can be written as $f = g - h$, where g and h are nonnegative and continuous.

39. Let $I \subseteq \mathbb{R}$ denote an interval, $\mathbb{Q}^c$ be the complement of $\mathbb{Q}$ in $\mathbb{R}$, and $f : I \rightarrow \mathbb{R}$ a function.

- (1) Prove that if f is continuous on I , and $f(I) \subseteq \mathbb{Q}$, then f is a constant function.
 (2) Prove that if f is continuous on I , and $f(I) \subseteq \mathbb{Q}^c$, then f is a constant function.

- (3) Prove that if I is not a point set and $f(I \cap \mathbb{Q}) \subseteq \mathbb{Q}^c, f(I \cap \mathbb{Q}^c) \subseteq \mathbb{Q}$, then f is not continuous on I .

40. Suppose that $f, g : [a, b] \rightarrow \mathbb{R}$ are two continuous functions satisfying

$$f(x) \neq 0, \quad f^2(x) = g^2(x), \quad \forall x \in [a, b].$$

Prove that only one of the following cases hold:

- 1) $f(x) = g(x), \forall x \in [a, b];$
- 2) $f(x) = -g(x), \forall x \in [a, b].$

41. Let $P(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0$. Prove that

- (1) if $a_0 < 0$, then $P_n(x) = 0$ has at least one positive root;
- (2) if $a_0 < 0$ and n is even, then $P_n(x) = 0$ has at least one negative root;
- (3) if n is even, then there exists $\tilde{x} \in \mathbb{R}$ such that $P_n(x) \geq P_n(\tilde{x})$ for any $x \in \mathbb{R}$;
- (4) conclude from (c) that there exists $\beta \in \mathbb{R}$ such that for any $b \geq \beta$, $P_n(x) = b$ has a real root, and for any $b < \beta$, the equation $P_n(x) = b$ has no real roots.

42. Let $f : [a, b] \rightarrow \mathbb{R}$ be a function such that all points of discontinuity are removable discontinuities.

- (1) Prove that for any $\epsilon > 0$, the set

$$E_\epsilon := \{x \in [a, b] : |\lim_{y \rightarrow x} f(y) - f(x)| > \epsilon\}$$

is a finite set.

- (2) Prove that the set of discontinuities of f is countable. Conclude from this that there does not exist a function $f : [a, b] \rightarrow \mathbb{R}$ such that every point in $[a, b]$ is a removable discontinuity.

43. State whether the following functions are uniformly continuous, and provide a proof of your statement.

- (1) $f : [0, \infty) \rightarrow \mathbb{R}$ defined by $f(x) = \sqrt{x}$;
- (2) $f : [0, a] \rightarrow \mathbb{R}$ defined by $f(x) = x^2$;
- (3) $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^2$;
- (4) $f :]a, 1[\rightarrow \mathbb{R}$ defined by $f(x) = \frac{1}{x^6}$, where $0 \leq a < 1$;
- (5) $f :]0, +\infty[\rightarrow \mathbb{R}$ defined by $f(x) = \sin(\frac{1}{x})$.

- 44.** Suppose that $a \in \mathbb{R}$ and $f : [a, +\infty[\rightarrow \mathbb{R}$ is a continuous function such that $\lim_{x \rightarrow \infty} f(x) = A$, where A is a finite real number or $+\infty$.
- (1) Prove that f takes all values between $f(a)$ and A .
 - (2) Prove that if $A = +\infty$, then f is bounded from below and f attains its greatest lower bound.
 - (3) Prove that if A is a finite real number, then f is uniformly continuous on $[a, \infty)$.
- 45.** Let X be a set and $f : X \rightarrow \mathbb{R}$ be a function. Determine whether the following statements are true or false. If true, please provide a proof; if false, give a counterexample.
- (1) If X is a bounded set and f is continuous on X , then f is bounded on X .
 - (2) If X is a bounded set and f is uniformly continuous on X , then f is bounded on X .
 - (3) If X is closed, and f is bounded and continuous on X , then f is uniformly continuous.
 - (4) If X is bounded, and f is bounded and continuous on X , then f is uniformly continuous.
 - (5) If X is closed, and f is monotonic, bounded and continuous on X , then f is uniformly continuous.
 - (6) If X is bounded, and f is uniformly continuous on X , then f is bounded and continuous on X .
- 46.** Let $E \subseteq \mathbb{R}$. The *modulus of continuity* of a function $f : E \rightarrow \mathbb{R}$ is the function $\omega(\delta)$ defined for $\delta > 0$ as follows:

$$\omega(\delta) = \sup_{\substack{x_1, x_2 \in E \\ |x_1 - x_2| < \delta}} |f(x_1) - f(x_2)|.$$

Show that

- (1) the modulus of continuity is a nondecreasing nonnegative function having the limit

$$\omega(0^+) := \lim_{\delta \rightarrow +0} \omega(\delta);$$

- (2) for every $\epsilon > 0$, there exists $\delta > 0$ such that for any points $x_1, x_2 \in E$, the relation $|x_1 - x_2| < \delta$ implies $|f(x_1) - f(x_2)| < \omega(0^+) + \epsilon$;

- (3) if E is a closed interval, an open interval, or a half-open interval, the relation

$$\omega(\delta_1 + \delta_2) \leq \omega(\delta_1) + \omega(\delta_2)$$

holds for the modulus of continuity of a function $f : E \rightarrow \mathbb{R}$;

- (4) the moduli of continuity of the functions x and $\sin(x^2)$ on the whole real axis are respectively $\omega(\delta) = \delta$ and the constant $\omega(\delta) = 2$ in the domain $\delta > 0$;

- (5) a function f is uniformly continuous on E if and only if $\omega(0^+) = 0$.

47. Let I denote a non-empty interval of $\mathbb{R}$, and f a function from I into I . A sequence (x_n) is defined by choosing an element x_0 of I and setting $x_{n+1} = f(x_n)$ for $n \in \mathbb{N}$. Such a sequence is called a **recursive sequence**.

- (1) Prove that if the sequence (x_n) converges to an element l of I and if f is continuous at the point l , then $f(l) = l$.

- (2) Assume that f is increasing on I .

- i. Show that the sequence (x_n) is monotonic.
- ii. Let l be an element of I such that $f(l) = l$. Prove that if $x_0 \leq l$ (respectively $x_0 \geq l$), then $(\forall n \in \mathbb{N})(x_n \leq l)$ (respectively $(\forall n \in \mathbb{N})(x_n \geq l)$).
- iii. Additionally, suppose that I is a closed and bounded interval of $\mathbb{R}$, and f is continuous on I . Prove that the sequence (x_n) converges to an element l of I such that $f(l) = l$.

- (3) Assume that f is decreasing on I .

- i. Prove that the sequences (x_{2n}) and (x_{2n+1}) are monotonic and have opposite directions of variation.
- ii. Let I_0 be the closed interval with endpoints x_0 and x_1 , and let l be an element of I such that $f(l) = l$. Prove that l belongs to I_0 .
 - A. Suppose that x_2 belongs to I_0 . Prove that the sequences (x_{2n}) and (x_{2n+1}) are convergent. Let $l = \lim_{n \rightarrow +\infty} x_{2n}$ and $l' = \lim_{n \rightarrow +\infty} x_{2n+1}$. Prove that l and l' are elements of I , and if f is continuous at the points l and l' , then $l = f(l')$ and $l' = f(l)$.
 - B. Prove that if $x_2 \notin I_0$, the sequence (x_n) is divergent.

- (4) i. Prove that the sequence (x_n) , defined by $x_0 > -1$ and $x_{n+1} = \frac{x_n^2}{x_n+1}$ for $n \in \mathbb{N}$, has the limit 0.
- ii. Prove that the sequence (x_n) , defined by $x_0 \in \mathbb{R}_+$ and $x_{n+1} = \frac{2x_n^2+3x_n}{x_n+1}$ for $n \in \mathbb{N}$, tends to 0 if $x_0 = 0$ and to $+\infty$ if $x_0 > 0$.

- iii. Prove that the sequence (x_n) , defined by $x_0 \in [0, 2]$ and $x_{n+1} = -\frac{x_n^2}{2} + 2$ for $n \in \mathbb{N}$, diverges if $x_0 \neq -1 + \sqrt{5}$ and converges if $x_0 = -1 + \sqrt{5}$.
 - iv. Prove that the sequence (x_n) , defined by $x_0 \in [0, 2]$ and $x_{n+1} = 1 - \frac{x_n^2}{4}$ for $n \in \mathbb{N}$, converges to $2(\sqrt{2} - 1)$.
 - v. Let $t \in [0, 1]$. Prove that the sequence (x_n) , defined by $x_0 \in [0, 1]$ and $x_{n+1} = x_n + \frac{1}{2}(t - x_n^2)$ for $n \in \mathbb{N}$, converges to $\sqrt{t}$.
- 48.** The goal of this problem is to determine the maximal ideals of the ring $A = C(I, \mathbb{R})$ of real-valued functions that are defined and continuous on a closed bounded interval I of $\mathbb{R}$. The finite covering theorem will be used.

For a subset $\mathcal{M} \subset A$, if it satisfies the following three conditions:

- 1) $\mathcal{M} \neq \emptyset$ and $\mathcal{M} \neq A$;
- 2) for any $p, q \in \mathcal{M}$, we have $p + q \in \mathcal{M}$;
- 3) for any $p \in \mathcal{M}$ and $f \in A$, we have $p \cdot f \in \mathcal{M}$;

then we call I an ideal of A . Suppose m is an ideal of A and there does not exist another ideal I such that $m \subsetneq I \subsetneq A$ (i.e., if I is an ideal and $m \subset I$, then $I = m$). Then we call m a maximal ideal of A .

- (1)
 - i. Let $x_0 \in I$. Prove that the set $\mathcal{M}_{x_0}$ of functions in A that vanish at x_0 is an ideal of A .
 - ii. Let J be an ideal of A such that $\mathcal{M}_{x_0} \subset J$ and $\mathcal{M}_{x_0} \neq J$. Let $f \in J \setminus \mathcal{M}_{x_0}$. Prove that there exists an element g of $\mathcal{M}_{x_0}$ such that $f^2 + g$ does not vanish on I . Deduce that $\mathcal{M}_{x_0}$ is a maximal ideal of A .
- (2) Let $\mathcal{M}$ be a maximal ideal of A .
 - i. Let $f \in \mathcal{M}$. Prove that the equation $f(x) = 0$ has a solution in I .
 - ii. For each element $f \in A$, define

$$Z(f) = \{x \in I \mid f(x) = 0\}.$$

Suppose that $\bigcap_{f \in \mathcal{M}} Z(f) = \emptyset$.

Prove that for every element x of I , there exists an element $f_x \in \mathcal{M}$ such that $x \notin Z(f_x)$. Deduce that there exists a finite number $f_1, \dots, f_n$ of functions in $\mathcal{M}$ such that

$$Z(f_1) \cap \dots \cap Z(f_n) = \emptyset.$$

- iii. Using the finite covering theorem, prove that there exists a finite number of functions $f_1, \dots, f_n$ in A such that:

$$Z(f_1) \cap \dots \cap Z(f_n) = \emptyset.$$

- iv. Prove that $f_1^2 + \dots + f_n^2 \notin \mathcal{M}$. Deduce a contradiction.
 v. Prove that $\bigcap_{f \in \mathcal{M}} Z(f)$ consists of a single point x_0 in I , and that $\mathcal{M} = \mathcal{M}_{x_0}$.

A side remark: In this exercise the set I can be notably replaced by any metric space X that we introduced in the end of the chapter. We can imagine the maximal ideals in the ring $C(X)$ as points in the space, thereby studying the geometry on X through the algebraic objects in $C(X)$. This has served as the initial observation in the field of algebraic geometry, as well as the mathematical foundation theory of quantum mechanics.

49. Let $f : [0, +\infty[\rightarrow \mathbb{R}$ be a function.

- (1) Suppose that f is bounded on any bounded and closed interval $[a, b]$ and

$$\lim_{x \rightarrow +\infty} f(x+1) - f(x) = l,$$

where l is a finite real number. Prove that $\lim_{x \rightarrow +\infty} \frac{f(x)}{x} = l$.

- (2) Suppose that f is bounded from below on any bounded and closed interval $[a, b]$ and

$$\lim_{x \rightarrow +\infty} f(x+1) - f(x) = +\infty.$$

Prove that $\lim_{x \rightarrow +\infty} \frac{f(x)}{x} = +\infty$.

50. Prove the following identities about inverse functions of trigonometric functions.

- (1) $\arccos(-x) = \arccos x$;
- (2) $\arctan(-x) = -\arctan x$;
- (3) $\arcsin x + \arccos x = \frac{\pi}{2}$;
- (4) $\arctan x + \operatorname{arccot} x = \frac{\pi}{2}$.

51. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(y) = y - \lambda \sin y$, where $0 \leq \lambda < 1$.

- (1) Prove that f is a continuous and strictly increasing function;

- (2) Prove that there exists a unique continuous inverse function $g : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$g(x) - \lambda \sin(g(x)) = x.$$

- 52.** Suppose that $I \rightarrow \mathbb{R}$ is a nonempty interval, and $f : I \rightarrow \mathbb{R}$ is an increasing function. Prove that the following statements are equivalent:

- (1) f is continuous on I ;
- (2) $f(I)$ is an interval.

- 53.** We consider the mapping $f : [0, +\infty[\rightarrow [0, +\infty[$ sending $x \in [0, +\infty[$ to x^2 .

- (1) Show that f is locally Lipschitzian but not Lipschitzian.
- (2) Show that f is continuous but not uniformly continuous.
- (3) Show that f is a bijection and its inverse mapping is continuous.
- (4) Show that f^{-1} is not locally Lipschitzian at 0.
- (5) Show that the restriction of f^{-1} to $[1, +\infty[$ is Lipschitzian.
- (6) Show that the restriction of f^{-1} to $[0, 2]$ is uniformly continuous.
- (7) Deduce that f^{-1} is uniformly continuous.

- 54.** Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the mapping defined as

$$f(x) = \frac{1}{1 + |x|}.$$

Show that the mapping f is 1-Lipschitzian.

- 55.** Let n be a positive integer, $(X_1, \mathcal{T}_1), \dots, (X_n, \mathcal{T}_n)$ be topological spaces, and X be the product space $X_1 \times \dots \times X_n$. We equip X with the product topology. For any $i \in \{1, \dots, n\}$, let $p_i : X \rightarrow X_i$ be the mapping of projection to the i -th coordinate that sends $(x_1, \dots, x_n) \in X_1 \times \dots \times X_n$ to x_i .

- (1) For any $i \in \{1, \dots, n\}$, let U_i be an open subset of X_i . Show that $U_1 \times \dots \times U_n$ is an open subset of X . One could write it as the intersection of $p_i^{-1}(U_i)$.

- (2) Let

$$\mathcal{B} := \{U_1 \times \dots \times U_n \mid \forall i \in \{1, \dots, n\}, U_i \in \mathcal{T}_i\}.$$

Show that the intersection of two sets in $\mathcal{B}$ still belongs to $\mathcal{B}$.

- (3) Let $\mathcal{T}$ be the set of elements $U \in \mathcal{P}(X)$ that can be written as a union of some sets in $\mathcal{B}$. Show that $\mathcal{T}$ is a topology and deduce that it identifies with the product topology of X .

- (4) Let Y be a topological space and $p \in Y$. For any $i \in \{1, \dots, n\}$, let $f_i : Y \rightarrow X_i$ be a mapping which is continuous at p . Let $f : Y \rightarrow X$ be the mapping sending $y \in Y$ to $(f_1(y), \dots, f_n(y))$.
- (a) For any $i \in \{1, \dots, n\}$, let V_i be a neighbourhood of $f_i(p)$. Show that $f^{-1}(V_1 \times \dots \times V_n)$ is a neighbourhood of p .
- (b) Deduce that f is continuous at p .

56. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the mapping which is defined as

$$f(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q}, \\ 0, & \text{if } x \notin \mathbb{Q}. \end{cases}$$

- (1) Show that $\sqrt{2}$ is an irrational number.
- (2) Deduce that, for any $n \in \mathbb{N}_{\geq 1}$, $\frac{\sqrt{2}}{n}$ is an irrational number.
- (3) Let $p \in \mathbb{R}$.
- (a) Show that there exists a sequence $(u_n)_{n \in \mathbb{N}}$ in $\mathbb{R} \setminus \mathbb{Q}$ that converges to p .
- (b) Suppose that $p \in \mathbb{Q}$. Show that there exists a sequence in $\mathbb{Q}$ that converges to p .
- (c) Suppose that $p \in \mathbb{R} \setminus \mathbb{Q}$. Show that the sequence of rational numbers

$$\frac{\lfloor np \rfloor}{n}, \quad n \in \mathbb{N}_{\geq 1}$$

converges to p .

- (d) Deduce that f is not continuous at p .
- (4) Let g_0 and g_1 be continuous mappings from $\mathbb{R}$ to $\mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be the mapping defined as

$$g(x) = \begin{cases} g_0(x), & \text{if } x \in \mathbb{Q}, \\ g_1(x), & \text{if } x \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$$

Show that g is continuous at some $p \in \mathbb{R}$ if and only if $g_0(p) = g_1(p)$.

- 57.** (1) Show that the mapping from $\mathbb{R}$ to $\mathbb{R}$ sending $x \in \mathbb{R}$ to $|x|$ is 1-Lipschitzian.
- (2) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a mapping and $p \in \mathbb{R}$. We assume that f is continuous at p . Show that the function $|f|$ sending $x \in \mathbb{R}$ to $|f(x)|$ is continuous at p .
- (3) Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a mapping such that $|g|$ is continuous at some $p \in \mathbb{R}$. Is the mapping g necessarily continuous at p ?

- (4) Let f and g be continuous mappings from $\mathbb{R}$ to $\mathbb{R}$. Show that the mappings

$$(x \in \mathbb{R}) \longmapsto \max\{f(x), g(x)\}$$

and

$$(x \in \mathbb{R}) \longmapsto \min\{f(x), g(x)\}$$

are both continuous.

- 58.** We consider the mapping $g : \mathbb{R} \rightarrow \mathbb{R}$ defined as

$$g(x) = \begin{cases} \frac{1}{\ln|x|}, & \text{if } x \notin \{-1, -0, 1\}, \\ 0, & \text{if } x \in \{-1, -0, 1\}. \end{cases}$$

- (1) Show that the mapping $x \mapsto x^{-1}$ from $\mathbb{R} \setminus \{0\}$ to $\mathbb{R} \setminus \{0\}$ is locally Lipschitzian. Deduce that it is a continuous mapping.
- (2) Show that the mapping g is continuous at $\mathbb{R} \setminus \{-1, 0, 1\}$.
- (3) Show that g is continuous at 0.
- (4) Determine the left limit and the right limit of g at 1. Is the mapping g continuous at 1?
- (5) Determine the left limit and the right limit of g at -1 . Is the mapping g continuous at -1 ?

- 59.** Let I be a non-empty open interval of $\mathbb{R}$ and $f : I \rightarrow \mathbb{R}$ be an increasing mapping.

- (1) Let $p \in I$, show that the mapping f has a left limit $f(p-)$ and a right limit $f(p+)$ at p .
- (2) Show that, for any $p \in I$, $f(p-) \leq f(p+)$.
- (3) Show that f is continuous at $p \in I$ if and only if $f(p-) = f(p+)$.
- (4) Let n be a positive integer and

$$I_n = \{p \in I \mid f(p+) - f(p-) \geq 1/n\}.$$

Show that I_n is countable.

- (5) Show that the set of $p \in I$ at which f is discontinuous is countable.

- 60.** Let $\mathcal{F}$ be the Fréchet filter on $\mathbb{N}$ and $(X, \mathcal{T})$ be a topological space. Let $u = (u_n)_{n \in \mathbb{N}}$ be a sequence in X and x be an element of X . Show that the sequence $(u_n)_{n \in \mathbb{N}}$ converges to x if and only if the filter $u_*(\mathcal{F})$ contains $\mathcal{V}_x$.

61. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous mapping such that

$$\forall (x, y) \in \mathbb{R}^2, \quad f(x + y) = f(x) + f(y).$$

- (1) Show that $f(0) = 0$.
- (2) Show that, for any $x \in \mathbb{R}$, $f(-x) = -f(x)$.
- (3) Show that, for any $n \in \mathbb{Z}$ and $x \in \mathbb{R}$, one has $f(nx) = nf(x)$.
- (4) Show that, for any $r \in \mathbb{Q}$ and $x \in \mathbb{R}$, one has

$$f(rx) = rf(x).$$

- (5) Show that, for any $x \in \mathbb{R}$ one has

$$f(x) = f(1)x.$$

62. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous mapping.

- (1) Show that the mapping $F : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined as $F(x, y) = y - f(x)$ is continuous.
- (2) Show that the graph of f , defined as

$$\{(x, y) \in \mathbb{R} \times \mathbb{R} \mid y = f(x)\},$$

is a closed subset of $\mathbb{R}^2$.

63. Let (X, d) be a metric space, $p \in X$ and $r > 0$. Let K be a compact subset of X such that $K \subseteq B(p, r)$. Show that there exists $r' \in]0, r[$ such that

$$\forall x \in K, \quad d(p, x) < r'.$$

64. Let (X, d) be a metric space and K_1, K_2 be two compact subsets of X that are disjoint. Show that

$$d(K_1, K_2) := \inf_{(x, y) \in K_1 \times K_2} d(x, y) > 0.$$

We could use the compactness of the product $K_1 \times K_2$.

65. Let V and W be vector spaces over $\mathbb{R}$, and $f : V \rightarrow W$ be an $\mathbb{R}$ -linear mapping. We equip V and W with norms $\|\cdot\|_V$ and $\|\cdot\|_W$ respectively. Recall that these norms define metrics on V and W , which also determine topologies on these vector spaces.

- (1) Assume that f is continuous at $\mathbf{0}_V$. Show that it is continuous.

- (2) We denote by $\|f\|$ the element

$$\sup_{x \in V \setminus \{0_V\}} \frac{\|f(x)\|_W}{\|x\|_V}$$

in $[0, +\infty]$. Show that

$$\|f\| = \sup_{x \in V, \|x\|_V \leq 1} \|f(x)\|_W = \sup_{x \in V, \|x\|_V = 1} \|f(x)\|_W.$$

- (3) Show that f is continuous if and only if $\|f\| < +\infty$.
 (4) Let n be a positive integer. Show that the product topology on $\mathbb{R}^n$ identifies with the topology induced by the norm $\|\cdot\|_{\ell^\infty}$ defined as

$$\|(a_1, \dots, a_n)\|_{\ell^\infty} := \max\{|a_1|, \dots, |a_n|\}.$$

- (5) Let $\|\cdot\|$ be an arbitrary norm on $\mathbb{R}^n$. Show that $\|\cdot\|$ is continuous, where we consider the product topology on $\mathbb{R}^n$. We could prove that there exists a constant $C_1 > 0$ such that $\|x\| \leq C_1 \|x\|_{\ell^\infty}$ for any $x \in \mathbb{R}^n$.
 (6) Show that $\{x \in \mathbb{R}^n \mid \|x\|_{\ell^\infty} \leq 1\}$ is compact.
 (7) Prove that there exists $C_2 > 0$ such that $\|x\| \geq C_2 \|x\|_{\ell^\infty}$.
 (8) Show that the topology induced by $\|\cdot\|$ identifies with the product topology.
 (9) Show that $\mathbb{R}^n$ is complete.

66. Let I be an interval in $\mathbb{R}$ and $f : I \rightarrow \mathbb{R}$ be a continuous mapping. We assume that f is injective.

- (1) Let a, b and c be elements of I such that $a < b < c$. Show that

$$(f(b) - f(a))(f(c) - f(b)) > 0.$$

- (2) Show that the mapping f is strictly monotone.

67. Show that the mapping

$$f : \mathbb{R} \longrightarrow \mathbb{R}, \quad f(x) = \lfloor x \rfloor + \sqrt{x - \lfloor x \rfloor}$$

is well defined and is continuous.

68. We consider the mapping $|\cdot| : \mathbb{C} \rightarrow \mathbb{R}_{\geq 0}$ defined as

$$\forall (x, y) \in \mathbb{R}^2, \quad |x + iy| = \sqrt{x^2 + y^2}.$$

For any $z = x + iy \in \mathbb{C}$ with $(x, y) \in \mathbb{R}^2$, we denote by $\bar{z}$ the complex number $x - iy$.

- (1) We consider $\mathbb{C}$ is a vector space over $\mathbb{R}$. Show that $|\cdot| : \mathbb{C} \rightarrow \mathbb{R}$ is a norm and $|z|^2 = z\bar{z}$ for any $z \in \mathbb{C}$.
- (2) For any $z \in \mathbb{C}$ and $r > 0$, let

$$D(z, r) := \{w \in \mathbb{C} \mid |w - z| \leq r\}.$$

Show that $D(z, r)$ is compact.

- (3) Let $z \in \mathbb{C}$. Show that the sequence

$$\sum_{n \in \mathbb{N}} \frac{z^n}{n!}$$

converges in $\mathbb{C}$ to a limit which we denoted as $\exp(z)$.

- (4) Show that $\exp(\cdot)$ is a morphism of groups from the additive group $(\mathbb{C}, +)$ to the multiplicative group $(\mathbb{C}^\times, \times)$, where $\mathbb{C}^\times := \mathbb{C} \setminus \{0\}$.
- (5) Show that $\exp(\bar{z}) = \overline{\exp(z)}$.
- (6) Show that, for any $\theta \in \mathbb{R}$ one has $|\exp(i\theta)| = 1$.
- (7) Let G be a subgroup of $(\mathbb{R}, +)$. Show that, either $\overline{G} = \mathbb{R}$, or G is discrete, namely for any $x \in G$, x has a neighbourhood V such that $V \cap G = \{x\}$.
- (8) Show that the kernel K of the group homomorphism

$$(\mathbb{R}, +) \longrightarrow (\mathbb{C}^\times, \times), \quad (\theta \in \mathbb{R}) \longmapsto \exp(i\theta)$$

is discrete.

- (9) For any $\theta \in \mathbb{R}$, let $\cos(\theta)$ and $\sin(\theta)$ be real numbers such that

$$\exp(i\theta) = \cos(\theta) + i \sin(\theta).$$

Show that $\cos(2) < -\frac{1}{3}$.

- (10) Deduce that $K \cap \mathbb{R}_{>0}$ is non-empty and there exists a unique real number $\pi > 0$ such that $K = 2\pi\mathbb{Z}$.
- (11) Show that $\exp(i\pi) = -1$.
- (12) Show that the image of $\cos(\cdot)$ is $[-1, 1]$.
- (13) Show that $\cos(\cdot)$ is strictly decreasing on $[0, \pi/2]$ and $\sin(\cdot)$ is strictly increasing on $[0, \pi/2]$. Deduce that $\sin(\pi/2) = 1$.
- (14) Show that the image of the mapping

$$\exp(i\cdot) : \mathbb{R} \longrightarrow \mathbb{C}, \quad (\theta \in \mathbb{R}) \longmapsto \exp(i\theta)$$

is equal to $S^1 := \{z \in \mathbb{C} \mid |z| = 1\}$. Deduce that S^1 is a subgroup of $\mathbb{C}^\times$.

(15) Show that the morphism of groups $\exp(\cdot) : \mathbb{C} \rightarrow \mathbb{C}^\times$ is surjective. What is its kernel?

69. Let $D = \{z \in \mathbb{C} \mid |z| \leq 1\}$ be the unit disc of the complex plane. For $u, v \in D$, we define $d(u, v) = |u - v|$ if $v = ru$ for some $r \in \mathbb{R}$, and $d(u, v) = |u| + |v|$ otherwise.

(1) Show that d is a metric on D .

(2) What is the topology induced by d on the unit circle $\mathbb{T}$?

70. Recall that $C^n([0, 1])$ is the space of real-valued functions whose n -th derivatives are continuous. For $f, g \in C^n([0, 1])$, we define

$$d_n(f, g) = \sup_{x \in [0, 1]} |f^{(n)}(x) - g^{(n)}(x)|.$$

(1) Is $d_1(f, g)$ a metric on $C^1([0, 1])$?

(2) Show that

$$d(f, g) = \sum_{n=0}^{\infty} \frac{1}{2^n} \cdot \frac{d_n(f, g)}{1 + d_n(f, g)}$$

is a metric on $C^\infty([0, 1])$.

71. Show that the following subspaces of $\mathbb{C}$ are homeomorphic:

$$A = \mathbb{C} \setminus \{0\}, \quad B = \{z \in \mathbb{C} \mid |z| > 1\}, \quad C = \{z \in \mathbb{C} \mid 0 < |z| < 1\}, \quad D = \mathbb{C} \setminus [0, 1]$$

72. The following examples show that completeness is not a topological notion.

(1) Let ϕ be the function on $\mathbb{R}$ given by $\phi(x) = \frac{x}{1+|x|}$. We define

$$d(x, y) = |\phi(x) - \phi(y)|, \quad x, y \in \mathbb{R}.$$

Show that d is a metric that induces the same topology as the usual one on $\mathbb{R}$. However, $\mathbb{R}$ is not complete under the metric d .

(2) More generally, let O be an open subset of a complete metric space (E, d) . Consider the map $\phi : O \rightarrow E \times \mathbb{R}$ defined by

$$\phi(x) = \left(x, \frac{1}{d(x, O^c)} \right) \stackrel{\text{def}}{=} (x, \rho(x)).$$

Show that ϕ is a homeomorphism from O onto a closed subset of $E \times \mathbb{R}$. Deduce that there exists a complete metric on O that induces the topology given by d on O . (**Remark:** $(O, d|_O)$ is in general not complete.)

- 73.** Show that a metric space (E, d) is complete if and only if every sequence $\{x_n\}_{n=1}^{\infty} \subset E$ satisfying $d(x_n, x_{n+1}) \leq 2^{-n}$ for all $n \geq 1$ converges.
- 74.** Let (E, d) be a metric space and let $\{x_n\}_{n=1}^{\infty} \subseteq E$ be a Cauchy sequence. Let $A \subset E$ be a subset such that the closure $\bar{A}$ is complete and that $\lim_{n \rightarrow \infty} d(x_n, A) = 0$. Prove that $\{x_n\}_{n=1}^{\infty}$ converges in E .
- 75.** Let (E, d) be a metric space and $\alpha > 0$. Assume that $A \subset E$ such that $d(x, y) \geq \alpha$ for any $x, y \in A$ with $x \neq y$. Show that A is complete.
- 76.** Let (E, d) be a metric space and $A \subset E$ be a subset. Assume that every Cauchy sequence of A converges in E . Show that the closure $\bar{A}$ of A is complete.
- 77.** Let (E, d) be a metric space and let $\{x_n\}_{n=1}^{\infty} \subseteq E$ be a divergent Cauchy sequence.
- (1) Show that for any $x \in E$, the sequence $\{d(x, x_n)\}_{n=1}^{\infty}$ converges to a positive number $g(x)$.
 - (2) Show that the function $x \mapsto \frac{1}{g(x)}$ is continuous from E to $\mathbb{R}$.
 - (3) Show that the function above is unbounded.
- 78.** Let $f : (E, d) \rightarrow (F, \delta)$ be a uniformly continuous bijection such that the inverse f^{-1} is uniformly continuous too. Show that for any $A \subset E$, $f(A)$ is complete if and only if A is complete.
- 79.** Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a uniformly continuous function. Show that there exist two nonnegative constants a and b such that

$$|f(x)| \leq a\|x\| + b,$$

where $\|x\|$ is the Euclidean norm of x .

- 80.** Let $f : E \rightarrow F$ be a continuous map between two metric spaces. Assume that f is uniformly continuous on every bounded subset of E .
- (1) Show that $\{f(x_n)\}_{n=1}^{\infty}$ is a Cauchy sequence for every Cauchy sequence $\{x_n\}_{n=1}^{\infty} \subseteq E$.
 - (2) Let $E \subset E'$ with E dense in E' and assume that F is complete. Show that f admits a unique continuous extension from E' to F .

- 81.** Show, by a counter-example, that the assumption that f is a contraction in the Fixed Point Theorem cannot be relaxed to the following: $d(f(x), f(y)) < d(x, y)$ for all $x, y \in E$ with $x \neq y$.

(Hint: Consider the function $f(x) = (x^2 + 1)^{1/2}$ on $[0, +\infty[$).

- 82.** Let (E, d) be a complete metric space and let f be a map such that $f^n = f \circ \cdots \circ f$ (n times) is a contraction. Show that f has a unique fixed point. Show, by an example, that such an f is not necessarily continuous.

- 83.** Let $I = (0, \infty)$ be equipped with its usual topology τ .

- (1) Show that τ is defined by the following complete distance:

$$d(x, y) = |\log x - \log y|.$$

- (2) Let $f \in C^1(I)$ such that for some $\lambda < 1$, $x|f'(x)| \leq \lambda f(x)$ for all $x \in I$. Show that f has a unique fixed point in I .

- 84.** Let $E = \{a_n\}_{n=1}^\infty$ be a countable set. We define

$$d(a_m, a_n) = \begin{cases} 0 & m = n \\ 10 + 1/m + 1/n & m \neq n \end{cases}$$

- (1) Show that d is a metric and that E is complete under this metric.

- (2) Let $f : E \rightarrow E$ be defined by $f(a_n) = a_{n+1}$. Show that $d(f(a_m), f(a_n)) < d(a_m, a_n)$ for $m \neq n$ but f does not have any fixed point.

- 85.** The purpose here is to give an alternate proof of the Fixed Point Theorem. Let (E, d) be a nonempty complete metric space and let $f : E \rightarrow E$ be a Lipschitzian function of constant $\lambda < 1$. For any $R \geq 0$, set $A_R = \{x \in E : d(x, f(x)) \leq R\}$.

- (1) Show that $f(A_R) \subset A_{\lambda R}$.

- (2) Show that A_R is a nonempty closed subset of E for any $R > 0$.

- (3) Show that for any $x, y \in A_R$, it holds that $d(x, y) \leq 2R + d(f(x), f(y))$. Then deduce that $\text{diam}(A_R) \leq 2R/(1 - \lambda)$.

- (4) Show that A_0 is nonempty.

- 86.** Let (E, d) be a complete metric space. Let f and g be two contractions on E that commute (i.e., $f \circ g = g \circ f$). Show that f and g have a unique common fixed point.

Show, by a counter-example, that the conclusion is not true if the commuting assumption is removed.

- 87.** The aim of this exercise is to show that the conclusion of the preceding one can be extended, in a certain sense, to non-commuting contractions.

Let (E, d) be a complete metric space. Given $A \subset E$, let d_A be the distance function associated to A : $d_A(x) = \inf\{d(x, a) \mid a \in A\}$. Let $\mathcal{C}$ denote the family of all compact subsets of E . For $A, B \in \mathcal{C}$, we define

$$h(A, B) = \sup_{x \in E} |d_A(x) - d_B(x)|.$$

- (1) Show that h is a distance on $\mathcal{C}$.
 (2) For $F \subset E$, we write $F_\varepsilon = \{x \mid d_F(x) \leq \varepsilon\}$. Show that

$$h(A, B) = \inf\{\varepsilon \geq 0 \mid A \subset A_\varepsilon, B \subset B_\varepsilon\}.$$

- (3) Show that $(\mathcal{C}, h)$ is complete.
 (4) Now let $f_1, \dots, f_n$ be n contractions on E . Define T on $(\mathcal{C}, h)$ by

$$T(A) = \bigcup_{k=1}^n f_k(A).$$

Show that T is a contraction and that there exists a unique compact set K such that $T(K) = K$.

- 88.** Let K and H be two compact subsets of a topological space. Is their intersection $K \cap H$ also compact?
- 89.** Let A, B be two disjoint closed subsets of a compact Hausdorff E . Show that there exist two disjoint open sets U, V such that $A \subset U$ and $B \subset V$. Show, by a counterexample, that the compactness condition cannot be removed.
- 90. (Separable metric space).** A metric space X is said to be separable provided that there exists a countable subset of X that is dense in X . Show that
- (1) A compact metric space is separable.
 (2) A metric space X is separable if and only if there is a countable collection of open subsets $\{O_n\}_{n=1}^\infty$ such that each open subset of X is the union of a subcollection of $\{O_n\}_{n=1}^\infty$.
- 91.** Let E, F be two metric spaces, and let A be a compact subset of E and B a compact subset of F . Assume that Ω is an open set in $E \times F$ such that $A \times B \subset \Omega$. Show that there exist open sets $U \subset E$ and $V \subset F$ such that $A \times B \subset U \times V \subset \Omega$.

- 92. (Lebesgue's covering lemma).** Let $\{O_\lambda\}_{\lambda \in \Lambda}$ be an open cover of a compact metric space X . Show the existence of $\varepsilon > 0$ such that for each $x \in X$ the open ball $B(x, \varepsilon)$ is contained in some member of the open cover.
- 93.** Let (E, d) be a compact metric space. Suppose that $f : E \rightarrow E$ satisfy $d(f(x), f(y)) < d(x, y)$ for all $x, y \in E$ with $x \neq y$. Show that f has a unique fixed point.
- 94.** For $f \in C^0([0, 1])$, we define $\|f\|_\infty = \sup_{x \in [0, 1]} |f(x)|$. Is the set

$$S = \{f \in C^0([0, 1]) \mid \|f\|_\infty \leq 1\}$$

compact?

- 95.** Consider the set of functions

$$S = \left\{ \sum_{n \in \mathbb{Z}} c_n e^{inx} \mid |c_n| \leq (1 + |n|)^{-2} \right\}.$$

- (1) Show that every function in S lies in $C^0([0, 2\pi])$.
- (2) Show that S is compact in $C^0([0, 2\pi])$ (under the supremum norm $\|\cdot\|_\infty$).
- 96.** Let $\mathbb{R}^n$ be equipped with its euclidean topology. Show that a connected open subset of $\mathbb{R}^n$ with $n \geq 2$ is not homeomorphic to any subset of $\mathbb{R}$.
- 97.** Let E be a topological space and $A \subset E$. Show that if a connected subset $C \subset E$ meets A and A^c , it must meet ∂A .
- 98.** (1) Show that the subset of $\mathbb{R}^2$ consisting of points that have at least one coordinate is irrational and connected.
- (2) Give an example of two homeomorphic subsets $A, B \subset \mathbb{R}^2$ such that A^c is connected but B^c is disconnected.
- 99.** Let (E, d) be a metric space. Assume that $\overline{B(x, r)} = \overline{B}(x, r)$ for all $x \in E$ and $r > 0$. Show that E is connected.
- 100.** Show that the following set $\Gamma \subseteq \mathbb{R}^2$ is connected

$$\Gamma = \left(\bigcup_{r \in [0, 1] \cap \mathbb{Q}} \{r\} \times [0, 1] \right) \cup \left(\bigcup_{r \in [0, 1] \setminus \mathbb{Q}} \{r\} \times [-1, 0] \right).$$

Is Γ path-connected?

- 101.** Let A, B be two subsets of a topological space E .

- (1) Show that if A and B are closed and both $A \cup B$, $A \cap B$ are connected, then A and B are connected.
- (2) Show that if A and B are connected and $\overline{A} \cap B \neq \emptyset$, then $A \cup B$ is connected.

102. (Furstenberg's topological proof of infinitely many primes). Let $\mathbb{Z}$ and $\mathbb{Z}_+$ be the sets of integers and positive integers, respectively. We define arithmetic sequences

$$V(a, b) = \{an + b : n \in \mathbb{Z}\},$$

where $a \in \mathbb{Z}_+$, $b \in \mathbb{Z}$. Let τ be a family of subsets of $\mathbb{Z}$, whose members are $\emptyset$, $\mathbb{Z}$, and the sets which can be represented as the union of arithmetic sequences. Show that

- (1) τ is a topology on $\mathbb{Z}$.
- (2) $V(a, b)$ is both open and closed.
- (3) Apply (a) and (b) to show that there are infinitely many primes.

103. (Vitali's covering lemma). We denote by $B(x, r_x)$ the closed ball of radius $r_x > 0$ and centered at x . Let $\mathcal{F} = \{B(x, r_x) : x \in S\}$ be a collection of closed balls in $\mathbb{R}^d$ with uniformly bounded diameters, i.e.,

$$\sup\{r_x : B(x, r_x) \in \mathcal{F}\} < \infty.$$

Show that there exists a countable family of disjoint balls $\mathcal{G} = \{B(y, r_y) : y \in T\} \subseteq \mathcal{F}$ such that

$$\cup_{x \in S} B(x, r_x) \subset \cup_{y \in T} B(y, 5r_y).$$

104. Suppose that X is a vector space.

- (1) If $A, B \subset X$ are convex, show that $A + B$ and $A \cap B$ are convex. What about $A \cup B$?
- (2) Show that $2A \subset A + A$. When is it true that $2A = A + A$?

105. Let $\mathbb{R}^d$ be endowed with the Euclidean metric. Let $K \subset \mathbb{R}^d$ be a symmetric, convex, compact set with $0 \in K^\circ$, that is, the origin is an interior point of K . Show that

$$\rho_K(x) = \inf\{r > 0 : r^{-1}x \in K\}$$

defines a norm on $\mathbb{R}^d$.

- 106.** Let X be an infinite dimensional normed vector space. A sequence $\{e_n\}_{n=1}^{\infty} \subset X$ is called a **Schauder basis** for X if for every $x \in X$ there is a unique sequence of scalars $\{\alpha_n\}_{n=1}^{\infty}$ such that

$$\|x - (\alpha_1 e_1 + \cdots + \alpha_n e_n)\| \rightarrow 0 \text{ as } n \rightarrow \infty,$$

and we have the expansion

$$x = \sum_{n=1}^{\infty} \alpha_n e_n.$$

- (1) Find a Schauder basis for ℓ^p for $1 \leq p < \infty$.
- (2) Show that if a normed vector space has a Schauder basis, then it is separable. Is the converse true?
- 107.** Let X be a normed vector space and let M be a nonempty subset. The annihilator of M is defined as $M^a = \{f \in X^* : f(x) = 0 \text{ for all } x \in M\}$. Show that M^a is a closed subspace of X^* . What are X^a and $\{0\}^a$?
- 108.** Let X be a normed vector space over $\mathbb{C}$ and let $X_{\mathbb{R}}$ be the same space treated as a normed space over $\mathbb{R}$. Show that, for each $\mathbb{R}$ -linear functional $\varphi : X_{\mathbb{R}} \rightarrow \mathbb{R}$, there exists a unique $\mathbb{C}$ -linear functional $\tilde{\varphi} : X \rightarrow \mathbb{C}$ such that $\varphi(x) = \operatorname{Re} \tilde{\varphi}(x)$ for each $x \in X$ and it is give by

$$\tilde{\varphi}(x) = \varphi(x) - i\varphi(ix).$$

- 109.** Let $\Lambda^{\alpha}(\mathbb{R})$ be the space of all bounded functions on $\mathbb{R}$ which satisfy a Hölder condition of exponent α with $0 < \alpha \leq 1$, that is

$$\sup_{x \neq y} \frac{|f(x) - f(y)|}{|x - y|^{\alpha}} < \infty.$$

- (1) Show that a function which satisfies a Hölder condition of exponent α with $\alpha > 1$ is a constant.
- (2) Show that

$$\|f\|_{\Lambda^{\alpha}(\mathbb{R})} = \sup_{x \in \mathbb{R}} |f(x)| + \sup_{x \neq y} \frac{|f(x) - f(y)|}{|x - y|^{\alpha}}$$

defines a norm on $\Lambda^{\alpha}(\mathbb{R})$, and with this norm, $\Lambda^{\alpha}(\mathbb{R})$ is a Banach space.

- (3) We define $C^{k,\alpha}(\mathbb{R})$ to be the class of functions f on $\mathbb{R}$ whose derivatives of order less than or equal to k belong to $\Lambda^\alpha(\mathbb{R})$. Show that

$$\|f\|_{C^{k,\alpha}(\mathbb{R})} = \sum_{\ell=0}^k \|f^{(\ell)}\|_{\Lambda^\alpha(\mathbb{R})}$$

defines a norm on $C^{k,\alpha}(\mathbb{R})$, and with the norm, $C^{k,\alpha}(\mathbb{R})$ is a Banach space.

110. A normed vector space X is said to be **uniformly convex** if for every $0 < \varepsilon \leq 2$, there exists $\delta > 0$ such that for any two $x, y \in X$ with $\|x\| = 1$, $\|y\| = 1$ and $\|x - y\| \geq \varepsilon$, it holds that

$$\left\| \frac{x + y}{2} \right\| \leq 1 - \delta.$$

Let X be a uniformly convex Banach space. Let $K \subset X$ be a closed convex set. Show that for any $x \in X$, there exists a unique $x^* \in K$ such that

$$\|x - x^*\| = \inf_{y \in K} \|x - y\|.$$

111. Let X be a normed vector space and suppose S and T are in the dual space X^* . If $S(x) = 0$ implies $T(x) = 0$, show that there is a constant c such that $S(x) = cT(x)$ for all $x \in X$.
112. Let (X, ρ) be a metric space. We denote by $\text{Lip}_0(X)$ the set of real-valued Lipschitz functions f on X that vanish at $x_0 \in X$.

- (1) Show that $\text{Lip}_0(X)$ is a Banach space endowed with the following norm

$$\|f\| = \sup_{x \neq y} \frac{|f(x) - f(y)|}{\rho(x, y)}.$$

- (2) For each $x \in X$, we define the linear functional $T_x : f \rightarrow f(x)$ for $f \in \text{Lip}_0(X)$. Show that $\|T_x - T_y\| = \rho(x, y)$. Thus X is isometric to a subset of the Banach space $\text{Lip}_0(X)^*$. (Since any closed subset of a complete metric space is complete, this provides another proof of the existence of a completion for any metric space X . It also shows that any metric space is isometric to a subset of a normed linear space.)

113. Let $\{\Lambda_n\}_{n=1}^\infty$ be a sequence of bounded linear operators from a normed vector space X to a Banach space Y with $\|\Lambda_n\| \leq M < \infty$ for all $n \geq 1$. Suppose that there is a dense set $E \subset X$ such that $\{\Lambda_n x\}_{n=1}^\infty$ converges for each $x \in E$. Show that $\{\Lambda_n x\}_{n=1}^\infty$ converges for each $x \in X$.

- 114.** Let X and Y be normed vector spaces. Let T be a bounded linear operator from X to Y with kernel $\ker(T) = Z$. Show that there is a unique bounded linear operator S from X/Z to Y such that $T = S \circ \varphi$, where $\varphi : X \rightarrow X/Z$ is the natural map. Moreover, show that $\|T\| = \|S\|$.
- 115.** (1) Let X be a complex normed vector space and let $T : X \rightarrow \mathbb{C}$ be a linear map. Show that T is continuous if and only if the kernel $\ker(T)$ is closed.
- (2) More generally, let X and Y be normed vector spaces and let $T : X \rightarrow Y$ be a linear operator. If X is finite-dimensional, show that T is continuous. If Y is finite dimensional, show that T is continuous if and only if $\ker(T)$ is closed.
- 116.** Let X be the normed space of sequences of complex numbers $x = \{x_i\}_{i=1}^{\infty}$ with only finitely many nonzero terms and normed by $\|x\| = \sup_i |x_i|$. Let $T : X \rightarrow X$ be defined by

$$Tx = \left\{ x_1, \frac{x_2}{2}, \frac{x_3}{3}, \dots \right\}.$$

Show that T is a bounded linear operator, but T^{-1} is unbounded.

- 117.** Show that a subset of a finite-dimensional normed vector space is compact if and only if it is closed and bounded.
- 118.** (1) Give an open cover of the closed unit ball of $C([0, 1])$ (with the norm $\|f\| = \max_{x \in [0, 1]} |f(x)|$) that has no finite subcover.
- (2) For $1 \leq p \leq \infty$, give an open cover of the closed unit ball of the Banach space ℓ^p that has no finite subcover.
- 119. (Another proof of Riesz's Theorem).** Let X be an infinite-dimensional normed vector space. Let B be the closed unit ball in X , and let B_0 be the unit open ball in X . If B is compact, then the open cover $\{x + B_0/3\}_{x \in B}$ of B has a finite subcover $\{x_i + B_0/3\}_{i=1}^n$. Use Riesz's Lemma with $Y = \text{span}\{x_1, x_2, \dots, x_n\}$ to derive a contradiction.

Riesz's Lemma: Let Y be a closed proper subspace of a normed vector space X . Then for each $\varepsilon > 0$, there exists a unit vector $x_0 \in X$ such that

$$\|x_0 - y\| > 1 - \varepsilon \quad \text{for all } y \in Y.$$

- 120.** Define $X = \{f \in C([0, 1]) : f(0) = 0\}$ and $V = \{f \in X : \int_0^1 f(x) dx = 0\}$. Show that V is a closed subspace of X , yet there is no $x \in X$ with $\|x\| = 1$ such that $d(x, V) = 1$. (This assertion means that, in general, we can not avoid the $\varepsilon > 0$ in Riesz's lemma).

- 121.** Let X be a normed vector space and let B be the open unit ball. Show that X is infinite-dimensional if and only if B contains an infinite collection of non-overlapping open balls of radius $1/4$.
- 122.** Show that, for every infinite-dimensional normed vector space X , there exists an infinite sequence $\{x_n\}_{n=1}^{\infty}$ of norm one such that $\|x_m - x_n\| > 1$ for all $m, n \in \mathbb{N}$ and $m \neq n$.
- 123.** (1) Suppose X is a vector space. The algebraic dual of X is the set of all linear functionals on X , and is also a vector space. Suppose also that X is a normed vector space. Show that X has finite dimension if and only if the algebraic dual and the dual space X^* coincide.
- (2) Let X be an infinite dimensional normed vector space over $\mathbb{R}$ or $\mathbb{C}$. Prove that there exists a non-continuous linear functional.
- (3) For any discontinuous linear functional Λ on a normed vector space X , show that $\overline{\ker(\Lambda)} = X$, that is, the kernel of Λ is a dense subspace of X .
- 124.** (1) Show that a Banach space $\mathcal{B}$ is separable if its dual space $\mathcal{B}^*$ is separable.
- (2) Show that a normed vector space X is separable if and only if there is a compact subset $K \subset X$ such that $\overline{\text{span}(K)} = X$.
- 125.** Let X be a normed vector space and let X^* be its dual space with the norm

$$\|f\| = \sup_{\|x\| \leq 1} |f(x)|.$$

- (1) For each $x \in X$, show that the mapping $T_x : f \rightarrow f(x)$ is a bounded linear functional on X^* . Moreover, show that $\|T_x\| = \|x\|$. (This gives a natural embedding of X into X^{**} , the dual space of X^*).
- (2) Let $\{x_n\}_{n=1}^{\infty} \subset X$ be a sequence such that $\{f(x_n)\}_{n=1}^{\infty}$ is bounded for every $f \in X^*$. Show that $\{\|x_n\|\}_{n=1}^{\infty}$ is bounded.
- 126.** Let $C([a, b])$ be the family of continuous functions on $[a, b]$ with the maximum norm. We say that a sequence $\{f_n\}_{n=1}^{\infty}$ weakly converges to f , denoted by $f_n \xrightarrow{w} f$ if, for all $\Lambda \in C([a, b])^*$, we have $\Lambda(f_n) \rightarrow \Lambda(f)$.
- (1) Suppose that $f_n \xrightarrow{w} f$. Show that $\{f_n\}_{n=1}^{\infty}$ is also pointwise convergent, that is, $f_n(x) \rightarrow f(x)$ for all $x \in [a, b]$.
- (2) Show that a weakly convergent sequence in $C^1([a, b])$ is convergent in $C([a, b])$. Is this still true when $[a, b]$ is replaced by $\mathbb{R}$.

127. For a complex sequence $x = \{x_i\}_{i=1}^\infty$, we define its ℓ^1 and ℓ^∞ norms as

$$\|x\|_1 = \sum_{i=1}^{\infty} |x_i|, \quad \|x\|_\infty = \sup_{i \geq 1} |x_i|.$$

The Banach spaces ℓ^1 and ℓ^∞ are defined as

$$\ell^1 = \{x : \|x\|_1 < \infty\}, \quad \ell^\infty = \{x : \|x\|_\infty < \infty\}$$

and c_0 is the subspace of ℓ^∞ consisting of all x such that $\lim_{i \rightarrow \infty} x_i = 0$.

- (1) For each $y \in \ell^1$, show that the mapping $\Lambda_y(x) = \sum_{i=1}^{\infty} x_i y_i$ is a bounded linear functional on c_0 with norm $\|\Lambda_y\| = \|y\|_1$. Moreover, every member in c_0^* can be obtained in this way and hence $c_0^* = \ell^1$.
 - (2) Similarly, show that $(\ell^1)^* = \ell^\infty$.
 - (3) Show that every $y \in \ell^1$ induces a bounded linear functional on ℓ^∞ as in (a). However, this does not give all mappings of $(\ell^\infty)^*$.
 - (4) Show that c_0 and ℓ^1 are separable but ℓ^∞ is not.
- 128.** Suppose that X is a Banach space, M and N are linear subspaces, and that $X = M \oplus N$, that is

$$X = M + N = \{m + n : m \in M, n \in N\}$$

and $M \cap N = \{0\}$. Let P be the projection of X on to M , that is, if $x = m + n$, then $P(x) = m$. Show that P is well defined and linear. Prove that P is bounded if and only if both M and N are closed.

129. Let X be a normed vector space. Given $A \subset X$ and $B \subset X^*$, we define

$$A^\perp = \{f \in X^* : f(x) = 0 \text{ for all } x \in A\}$$

$${}^\perp B = \{x \in X : f(x) = 0 \text{ for all } f \in B\}$$

- (1) Show that $A^\perp$ and ${}^\perp B$ are closed subspaces of X^* and X , respectively.
 - (2) Let $V \subset X$ be a subspace. Show that $\overline{V} = {}^\perp (V^\perp)$.
- 130.** An operator T on a Banach space X is called compact if, for every bounded subset B , the image $T(B)$ has compact closure. Alternatively, T is compact if and only if for every sequence $\{x_n\}_{n=1}^\infty \subset X$, $\{T(x_n)\}_{n=1}^\infty \subset X$ has a convergent subsequence.
- (Fredholm alternative).** Suppose that T is a compact operator on a Banach space X . Show that exactly one of the following holds

- (1) $I - T$ is invertible with a bounded inverse.
 - (2) There exists a nonzero $x \in X$ such that $Tx = x$ (i.e., 1 is an eigenvalue of T).
- 131.** Prove that a subset A of a metric space (X, d) is bounded if and only if every countable subset of A is bounded.
- 132.** Let X be a normed vector space and $M \subset X$. For $\varepsilon > 0$, we say that a set $A \subset M$ is an ε -net in M provided that, for every $x \in M$ there is $y \in A$ such that $\|x - y\| < \varepsilon$. We say that A is ε -separated if $\|x - y\| \geq \varepsilon$ for all $x, y \in A$ and $x \neq y$.
- Suppose that X has dimension n . Show that every ε -net in the unit ball of X contains at least ε^{-n} points. On the other hand, show that there exists an ε -net in the unit ball with at most $(1 + 2/\varepsilon)^n$ points.

Chapter 9

Differential calculus

9.1 Landau symbol

In this section, we fix a complete valued field $(K, |\cdot|)$ and a normed vector space $(V, \|\cdot\|)$ over K .

9.1.1 Definition. Let X be a set, $f : X \rightarrow V$, $g : X \rightarrow \mathbb{R}_{\geq 0}$ be mappings.

Let Y be a subset of X . We use the expression

$$f(x) = O(g(x)), x \in Y$$

to denote the statement

$$\exists C > 0, \forall x \in Y, \|f(x)\| \leq Cg(x).$$

Let $\mathcal{F}$ be a filter on X . We use the expression

$$f(x) = O(g(x)) \text{ along } \mathcal{F}$$

to denote

$$\exists A \in \mathcal{F}, f(x) = O(g(x)), x \in A.$$

We use the expression

$$f(x) = o(g(x)) \text{ along } \mathcal{F}$$

to denote the following statement: there exists a function $\varepsilon : X \rightarrow \mathbb{R}_{\geq 0}$ and an element $A \in \mathcal{F}$ such that

$$\lim_{\mathcal{F}} \varepsilon = 0 \text{ and } \forall x \in A, \|f(x)\| \leq \varepsilon(x)g(x).$$

9.1.2 Example. Let I be an infinite subset of $\mathbb{N}$, $(x_n)_{n \in I}$ be a sequence in V and $(b_n)_{n \in I}$ be a sequence in $\mathbb{R}_{\geq n}$. Let $\mathcal{F}$ be the Fréchet filter on I . Then

$$x_n = o(b_n) \text{ along } \mathcal{F} \quad (9.1)$$

if and only if there exists a sequence $(\varepsilon_n)_{n \in I}$ and $N \in \mathbb{N}$ such that

$$\lim_{n \rightarrow +\infty} \varepsilon_n = 0 \text{ and } \forall n \in I_{\geq N}, \|x_n\| \leq \varepsilon_n b_n.$$

Usually, we use the notation

$$x_n = o(b_n), n \rightarrow +\infty$$

to denote the statement (9.1).

9.1.3 Example. Let $(X, \mathcal{T})$ be a topological space, Y be a subset of X , and y_0 be an adherent point of Y . Let $f : Y \rightarrow V$ and $g : Y \rightarrow \mathbb{R}_{\geq 0}$ be two mappings. Let $\mathcal{F}$ be a filter of neighbourhoods of y_0 . Then

$$f(x) = o(g(x)) \text{ along } \mathcal{F}|_Y \quad (9.2)$$

if and only if there exists a mapping $\varepsilon : Y \rightarrow \mathbb{R}_{\geq 0}$ and a neighbourhood V of y_0 such that

$$\lim_{x \in Y, x \rightarrow y_0} \varepsilon(x) = 0 \text{ and } \forall x \in Y \cap V, \|f(x)\| \leq \varepsilon(x)g(x).$$

Usually we use the notation

$$f(x) = o(g(x)), x \in Y, x \rightarrow y_0$$

to denote the statement (9.2)

9.1.4 Example. Let $\mathcal{F}$ be the filter on $\mathbb{R}$ generated by subsets of the form $[a, +\infty[$. Let Y be an element of $\mathcal{F}$ (namely Y contains a subset of the form $[a, +\infty[$), and $f : Y \rightarrow V$ and $g : Y \rightarrow \mathbb{R}_{\geq 0}$ be two mappings. Then

$$f(x) = o(g(x)) \text{ along } \mathcal{F}|_Y \quad (9.3)$$

if and only if there exists a mapping $\varepsilon : Y \rightarrow \mathbb{R}_{\geq 0}$ and $a \in \mathbb{R}$ such that

$$\lim_{x \in Y, x \rightarrow +\infty} \varepsilon(x) := \lim_{\mathcal{F}|_Y} \varepsilon = 0 \text{ and } \forall x \in Y \cap [a, +\infty[, \|f(x)\| \leq \varepsilon(x)g(x).$$

Usually, we use the notation

$$f(x) = o(g(x)), x \in Y, x \rightarrow +\infty$$

to denote the statement (9.2)

9.1.5 Example. Let $\mathcal{F}$ be the filter on $\mathbb{R}$ generated by subsets of the form $]-\infty, a]$. Let Y be an element of $\mathcal{F}$ (namely Y contains a subset of the form $]-\infty, a]$), and $f : Y \rightarrow V$ and $g : Y \rightarrow \mathbb{R}_{\geq 0}$ be two mappings. Then

$$f(x) = o(g(x)) \text{ along } \mathcal{F}|_Y \quad (9.4)$$

if and only if there exists a mapping $\varepsilon : Y \rightarrow \mathbb{R}_{\geq 0}$ and $a \in \mathbb{R}$ such that

$$\lim_{x \in Y, x \rightarrow -\infty} \varepsilon(x) := \lim_{\mathcal{F}|_Y} \varepsilon = 0 \text{ and } \forall x \in Y \cap]-\infty, a], \|f(x)\| \leq \varepsilon(x)g(x).$$

Usually we use the notation

$$f(x) = o(g(x)), x \in Y, x \rightarrow -\infty$$

to denote the statement (9.2)

9.1.6 Remark. Let $(X, \mathcal{T})$ be a topological space, (M, d) be a metric space $f : X \rightarrow M$ be a mapping, and $x_0 \in X$. Then f has $\ell \in M$ as limit at x_0 if and only if

$$d(f(x), \ell) = o(1), x \rightarrow x_0.$$

9.1.7 Proposition. Let X be a set and $\mathcal{F}$ be a filter on X .

(1) Let $f : X \rightarrow V$ and $g : X \rightarrow \mathbb{R}_{\geq 0}$ be mappings. If

$$f(x) = o(g(x)) \text{ along } \mathcal{F},$$

then

$$f(x) = O(g(x)) \text{ along } \mathcal{F}.$$

(2) Let $f_1 : X \rightarrow V$, $f_2 : X \rightarrow V$ and $g : X \rightarrow \mathbb{R}_{\geq 0}$ be mappings.

(2.a) If

$$f_1(x) = O(g(x)) \text{ along } \mathcal{F}, \quad f_2(x) = O(g(x)) \text{ along } \mathcal{F},$$

then

$$f_1(x) + f_2(x) = O(g(x)) \text{ along } \mathcal{F}.$$

(2.b) If

$$f_1(x) = o(g(x)) \text{ along } \mathcal{F}, \quad f_2(x) = o(g(x)) \text{ along } \mathcal{F},$$

then

$$f_1(x) + f_2(x) = o(g(x)) \text{ along } \mathcal{F}.$$

(3) Let $\lambda : X \rightarrow K$, $f : X \rightarrow V$, $g : X \rightarrow \mathbb{R}_{\geq 0}$ and $h : X \rightarrow \mathbb{R}_{\geq 0}$ be mappings.

(3.a) *If*

$$\lambda(x) = O(g(x)) \text{ along } \mathcal{F}, \quad f(x) = O(h(x)) \text{ along } \mathcal{F},$$

then

$$\lambda(x)f(x) = O(g(x)h(x)) \text{ along } \mathcal{F}.$$

(3.b) *If*

$$\lambda(x) = O(g(x)) \text{ along } \mathcal{F}, \quad f(x) = o(h(x)) \text{ along } \mathcal{F},$$

or if

$$\lambda(x) = o(g(x)) \text{ along } \mathcal{F}, \quad f(x) = O(h(x)) \text{ along } \mathcal{F},$$

then

$$\lambda(x)f(x) = o(g(x)h(x)) \text{ along } \mathcal{F}.$$

Proof. (1) By definition, there exists $\varepsilon : X \rightarrow \mathbb{R}_{\geq 0}$ and $A \in \mathcal{F}$ such that

$$\lim_{\mathcal{F}} \varepsilon = 0 \text{ and } \forall x \in A, \|f(x)\| \leq \varepsilon(x)g(x).$$

Since

$$\lim_{\mathcal{F}} \varepsilon = 0,$$

there exists $B \in \mathcal{F}$ such that

$$\forall x \in B, |\varepsilon(x)| \leq 1.$$

Hence

$$\forall x \in A \cap B, \|f(x)\| \leq g(x),$$

as required.

(2.a) There exist $(A_1, A_2) \in \mathcal{F} \times \mathcal{F}$ and $(C_1, C_2) \in \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0}$ such that

$$\forall x \in A_1, \|f_1(x)\| \leq C_1g(x), \quad \forall x \in A_2, \|f_2(x)\| \leq C_2g(x).$$

Thus,

$$\forall x \in A_1 \cap A_2, \|f_1(x) + f_2(x)\| \leq (C_1 + C_2)g(x),$$

as required.

(2.b) There exist mappings $\varepsilon_1 : X \rightarrow \mathbb{R}_{\geq 0}$, $\varepsilon_2 : X \rightarrow \mathbb{R}_{\geq 0}$ and $(A_1, A_2) \in \mathcal{F} \times \mathcal{F}$ such that

$$\lim_{\mathcal{F}} \varepsilon_1 = \lim_{\mathcal{F}} \varepsilon_2$$

and

$$\forall x \in A_1, \|f_1(x)\| \leq \varepsilon_1(x)g(x), \quad \forall x \in A_2, \|f_2(x)\| \leq \varepsilon_2(x)g(x).$$

Then, one has

$$\lim_{\mathcal{F}} (\varepsilon_1 + \varepsilon_2)(x) = 0,$$

and

$$\forall x \in A_1 \cap A_2, \|f_1(x) + f_2(x)\| \leq (\varepsilon_1(x) + \varepsilon_2(x))g(x).$$

(3.a) There exists $(C_1, C_2) \in \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0}$ and $(A_1, A_2) \in \mathcal{F} \times \mathcal{F}$ such that

$$\forall x \in A_1, |\lambda(x)| \leq C_1 g(x), \quad \forall x \in A_2, \|f(x)\| \leq C_2 h(x).$$

Hence

$$\forall x \in A_1 \cap A_2, \|\lambda(x)f(x)\| = |\lambda(x)| \cdot \|f(x)\| \leq C_1 C_2 g(x)h(x).$$

(3.b) We assume that

$$\lambda(x) = O(g(x)) \text{ along } \mathcal{F}, \quad f(x) = o(h(x)) \text{ along } \mathcal{F}.$$

There exists $(A_1, A_2) \in \mathcal{F} \times \mathcal{F}$, $C \in \mathbb{R}_{>0}$ and a mapping $\varepsilon : X \rightarrow \mathbb{R}_{\geq 0}$ such that

$$\forall x \in A_1, |\lambda(x)| \leq Cg(x), \quad \forall x \in A_2, \|f(x)\| \leq \varepsilon(x)h(x).$$

Then one has

$$\lim_{\mathcal{F}} C\varepsilon(x) = 0$$

and

$$\forall x \in A_1 \cap A_2, \|\lambda(x)f(x)\| = |\lambda(x)| \cdot \|f(x)\| \leq C\varepsilon(x)g(x)h(x),$$

as required. The proof under the second hypothesis is very similar, and we omit the details. $\square$

9.2 Differentiability

In this section, we fix a complete non-trivially valued field $(K, |\cdot|)$ and two normed vector space $(E, \|\cdot\|_E)$ and $(F, \|\cdot\|_F)$ over K . We denote by $\mathcal{L}(E, F)$ be vector space of bounded linear mappings from $(E, \|\cdot\|_E)$ to $(F, \|\cdot\|_F)$. We denote by $\|\cdot\|$ the operator norm of $\mathcal{L}(E, F)$.

9.2.1 Definition. Let U be an open subset of E , $f : U \rightarrow F$ be a mapping and $p \in U$. If there is an element $Df(p) \in \mathcal{L}(E, F)$, such that

$$f(p+h) - f(p) - Df(p)(h) = o(\|h\|_E), \quad h \rightarrow 0_E,$$

then we say that the mapping f is *differentiable* at p and that the bounded linear mapping $Df(p)$ is the *differential* of f at the point p . If f is differentiable at any point of U , we say that f is *differentiable* on U .

9.2.2 Proposition. *Let U be an open subset of E , $f : U \rightarrow F$ be a mapping, and $p \in U$. If f is differentiable at p , then it is continuous at p .*

Proof. For $x \in U$, one has

$$f(x) - f(p) - Df(p)(x - p) = o(\|x - p\|_E), \quad x \rightarrow p,$$

so that

$$f(x) - f(p) = O(\|x - p\|_E) = o(1), \quad x \rightarrow p.$$

□

9.2.3 Lemma. *Let $\varphi : E \rightarrow F$ be a K -linear mapping. For any $r > 0$, one has*

$$\|\varphi\| = \sup_{x \in E, 0 < \|x\| \leq r} \frac{\|\varphi(x)\|_F}{\|x\|_E}.$$

Proof. By definition,

$$\|\varphi\| := \sup_{x \in E \setminus \{0\}} \frac{\|\varphi(x)\|_F}{\|x\|_E} \geq \sup_{x \in E, 0 < \|x\| \leq r} \frac{\|\varphi(x)\|_F}{\|x\|_E}.$$

Since the absolute value $|\cdot|$ is not trivial, there exists $a \in K$ such that $0 < |a| < 1$. Let y be an arbitrary non-zero element of E . There exists $n \in \mathbb{N}_{\geq 1}$ such that $|a|^n \cdot \|y\|_E \leq r$. Hence

$$\frac{\|\varphi(y)\|_F}{\|y\|_E} = \frac{\|\varphi(a^n y)\|_F}{\|a^n y\|_E} \leq \sup_{x \in E, 0 < \|x\| \leq r} \frac{\|\varphi(x)\|_F}{\|x\|_E}.$$

□

9.2.4 Proposition. *Let $\varphi : E \rightarrow F$ be a bounded linear mapping. If*

$$\varphi(x) = o(\|x\|), \quad x \rightarrow 0_E,$$

then φ is the zero linear mapping.

Proof. By definition, there exists a mapping $\varepsilon : E \rightarrow \mathbb{R}_{\geq 0}$ and a neighbourhood U of 0_E such that

$$\lim_{x \rightarrow 0_E} \varepsilon(x) = 0$$

and

$$\|\varphi(x)\|_F \leq \varepsilon(x) \|x\|_E.$$

By Lemma 9.2.3, one has

$$\forall r \in \mathbb{R}_{\geq 0}, \quad \|\varphi\| = \sup_{x \in E, 0 < \|x\| \leq r} \frac{\|\varphi(x)\|_F}{\|x\|_E} \leq \sup_{x \in E, 0 < \|x\| \leq r} \varepsilon(x).$$

Taking the infimum with respect to r we obtain

$$\|\varphi\| \leq \limsup_{x \rightarrow 0_E} \varepsilon(x) = 0,$$

namely φ is the zero linear mapping. $\square$

9.2.5 Corollary. *Let U be an open subset of E , $f : U \rightarrow F$ be a mapping and $p \in U$. If f is differentiable at p , then its differential at p is unique.*

Proof. Let φ and ψ be bounded K -linear mappings from E to F such that

$$f(p+h) - f(p) - \varphi(h) = o(\|h\|_E), \quad h \rightarrow 0_E,$$

$$f(p+h) - f(p) - \psi(h) = o(\|h\|_E), \quad h \rightarrow 0_E.$$

Then one has

$$\|\varphi(h) - \psi(h)\|_F = o(\|h\|_E), \quad h \rightarrow 0_E.$$

By Proposition 9.2.4, we obtain that $\varphi = \psi$. $\square$

9.2.6 Example. Let $f : E \rightarrow F$ be a bounded linear mapping. For any $p \in E$, one has

$$f(p+h) - f(p) - f(h) = 0_F.$$

Therefore, f is differentiable at p and

$$Df(p) = f.$$

9.2.7 Proposition. *Let $(E, \|\cdot\|_E)$ be a normed vector space over $(K, |\cdot|)$, U be an open subset of E and p be an element of U . Let $n \in \mathbb{N}_{\geq 1}$. For any $i \in \{1, \dots, n\}$, let $(F_i, \|\cdot\|_i)$ be a normed vector space over $(K, |\cdot|)$ and $f_i : U \rightarrow F_i$ be a mapping. Let $F := F_1 \times \dots \times F_n$, equipped with the norm $\|\cdot\|_F$ defined as*

$$\|(x_1, \dots, x_n)\|_F := \max_{i \in \{1, \dots, n\}} \|x_i\|_i.$$

Let

$$f : U \longrightarrow F$$

be the mapping that sends $x \in U$ to $(f_1(x), \dots, f_n(x))$. If each f_i is differentiable at $p \in U$, then f is also differentiable at p . Moreover,

$$Df(p) : E \longrightarrow F$$

sends $h \in E$ to

$$(Df_1(p)(h), \dots, Df_n(p)(h)).$$

Proof. One has

$$\begin{aligned} & \|f(p+h) - f(p) - (Df_1(p)(h), \dots, Df_n(p)(h))\|_F \\ &= \|(f_1(p+h) - f_1(p) - Df_1(p)(h), \dots, f_n(p+h) - f_n(p) - Df_n(p)(h))\|_F \\ &\leq \sum_{i=1}^n \|f_i(p+h) - f_i(p) - Df_i(p)(h)\|_i = o(\|h\|_E), \quad h \rightarrow 0_E. \end{aligned}$$

□

9.2.8 Proposition. *Let $(E, \|\cdot\|_E)$, $(F, \|\cdot\|_F)$ and $(G, \|\cdot\|_G)$ be normed vector spaces over K , U be an open subset of E , V be an open subset of F , $f : U \rightarrow F$ and $g : V \rightarrow G$ be mappings. We assume that $f(U) \subseteq V$. Let p be an element of U . If f is differentiable at p and g is differentiable at $f(p)$, then $g \circ f$ is differentiable at p . Moreover, one has*

$$\forall h \in E, \quad D(g \circ f)(p)(h) = Dg(f(p))(Df(p)(h)).$$

Proof. By definition, one has

$$f(p+h) - f(p) - Df(p)(h) = o(\|h\|_E), \quad h \rightarrow 0_E.$$

Since

$$Df(p)(h) = O(\|h\|_E), \quad h \rightarrow 0_E,$$

we obtain that

$$\|f(p+h) - f(p)\|_F = O(\|h\|_E), \quad h \rightarrow 0_E.$$

Thus

$$\begin{aligned} & g(f(p+h)) - g(f(p)) - Dg(f(p))(f(p+h) - f(p)) \\ &= o(\|f(p+h) - f(p)\|_F) = o(\|h\|_E), \quad h \rightarrow 0_E. \end{aligned}$$

Moreover,

$$\begin{aligned} & \|Dg(f(p))(f(p+h) - f(p) - Df(p)(h))\|_G \\ &\leq \|Dg(f(p))\| \cdot \|f(p+h) - f(p) - Df(p)(h)\|_F = o(\|h\|_E), \quad h \rightarrow 0_E. \end{aligned}$$

Hence

$$g(f(p+h)) - g(f(p)) - Dg(f(p))(Df(p)(h)) = o(\|h\|_E), \quad h \rightarrow 0_E.$$

□

9.2.9 Definition. Let U be an open subset of K , $(F, \|\cdot\|)$ be a normed vector space over K , and $f : U \rightarrow F$ be a mapping. If f is differentiable at $p \in U$, we denote by $f'(p)$ the element

$$Df(p)(1) \in F.$$

Note that

$$f(p+h) - f(p) - hf'(p) = o(h), \quad h \rightarrow 0.$$

Hence

$$f'(p) = \lim_{h \rightarrow 0} \frac{f(p+h) - f(p)}{h}.$$

9.3 Convexity

9.3.1 Definition. Let E be a vector space over $\mathbb{R}$ and S be a non-empty subset of E . We call *affine combination* of elements of S any element of E of the form

$$a_1x_1 + \cdots + a_nx_n,$$

where n is a positive integer, $(x_1, \dots, x_n) \in E^n$, and $(a_1, \dots, a_n) \in \mathbb{R}^n$ such that $a_1 + \cdots + a_n = 1$. We denote by $\text{Aff}(S)$ the set of all affine combinations of elements of S , called the *affine hull* of S . If S is equal to its affine hull (namely any affine combination of elements of S belongs to S), then we say that S is an *affine subspace* of E .

9.3.2 Proposition. Let E be a vector space over $\mathbb{R}$.

- (1) If F is a vector subspace of E and $p \in E$, then

$$p + F := \{p + x \mid x \in F\}$$

is an affine subspace of E .

- (2) If A is an affine subspace of E . For any $p \in A$, the set

$$A - p := \{x - p \mid x \in A\}$$

is a vector subspace of E , which does not depend on the choice of $p \in A$. We call $A - p$ the vector subspace associate with the affine subspace A .

- (3) Let S be a non-empty subset of E . Then the affine hull of S is the smallest affine subspace of E containing S .

Proof. (1) Let n be a positive integer, $(x_1, \dots, x_n) \in F^n$ and $(a_1, \dots, a_n) \in \mathbb{R}^n$ such that $a_1 + \dots + a_n = 1$. One has

$$\sum_{i=1}^n a_i(p + x_i) = \sum_{i=1}^n a_i p + \sum_{i=1}^n a_i x_i = p + \sum_{i=1}^n a_i x_i \in p + F.$$

(2) Let n be a positive integer, $(x_1, \dots, x_n) \in A^n$ and $(b_1, \dots, b_n) \in \mathbb{R}^n$. One has

$$\sum_{i=1}^n b_i(x_i - p) = b_1 x_1 + \dots + b_n x_n + (1 - b_1 - \dots - b_n)p - p \in A - p.$$

Therefore $A - p$ is a vector subspace of E .

If q is a another element of A . For any $x \in A$, one has

$$x - p = (x - q) - (p - q) \in A - q$$

since $A - q$ is a vector subspace of E . Hence $A - p \subseteq A - q$. By the symmetry between p and q , we obtain that $A - q \subseteq A - p$. Hence $A - q = A - p$.

(3) By definition, if $(A_i)_{i \in I}$ is a non-empty family of affine subspaces of E , then $\bigcap_{i \in I} A_i$ is an affine subspace of E . If A is an affine subspace of E that contains S , then A contains the affine hull of S . Thus $\text{Aff}(S)$ is contained in the intersection of all affine subspaces containing S (which is hence the smallest affine subspace containing S). It remains to check that $\text{Aff}(S)$ is itself an affine subspace. Let $p \in S$, the set $\text{Aff}(S) - p$ is composed of elements of the form

$$\sum_{i=1}^n a_i x_i - p = \sum_{i=1}^n a_i (x_i - p),$$

where n is a positive integer, $(x_1, \dots, x_n) \in S^n$, and $(a_1, \dots, a_n) \in \mathbb{R}^n$ such that $a_1 + \dots + a_n = 1$. Hence

$$\text{Aff}(S) - p \subseteq \text{Span}_{\mathbb{R}}(S - p),$$

where $S - p := \{x - p \mid x \in S\}$. Conversely, for any positive integer n , $(x_1, \dots, x_n) \in S^n$ and $(b_1, \dots, b_n) \in \mathbb{R}^n$, one has

$$\sum_{i=1}^n b_i(x_i - p) = \left(\sum_{i=1}^n b_i x_i + (1 - b_1 - \dots - b_n)p \right) - p \in \text{Aff}(S) - p.$$

Therefore,

$$\text{Aff}(S) - p = \text{Span}_{\mathbb{R}}(S - p)$$

is a vector subspace of E and hence $\text{Aff}(S)$ is an affine subspace. $\square$

9.3.3 Definition. Let E be a vector space over $\mathbb{R}$ and S be a non-empty subset of E . We call *convex combination* of elements of S any element of E of the form

$$a_1x_1 + \cdots + a_nx_n,$$

where n is a positive integer, $(x_1, \dots, x_n) \in E^n$, and $(a_1, \dots, a_n) \in [0, 1]^n$ such that $a_1 + \cdots + a_n = 1$. We denote by $\text{Conv}(S)$ the set of all convex combinations of elements of S , called the *convex hull* of S . If S is equal to its convex hull, we say that it is a *convex subset* of E . Note that the convex hull of S is contained in the affine hull of S .

9.3.4 Proposition. Let E be a vector space over $\mathbb{R}$ and C be a non-empty subset of E . Then C is convex if and only if, for any $(x, y) \in C \times C$ and any $t \in [0, 1]$, one has

$$tx + (1 - t)y \in C.$$

Proof. Suppose that C is convex. For any $(x, y) \in C \times C$ and $t \in [0, 1]$, $tx + (1 - t)y$ is a convex combination of x and y . Hence it belongs to C .

Conversely, suppose that the convex combination of two element elements of C belongs to C . We will prove by induction that the convex combination of n elements of C belongs to C , where n is an arbitrary positive integer. The case where $n = 1$ and $n = 2$ are either trivial or from the hypothesis. Suppose that $n \geq 3$ and the convex combination of $< n$ elements of C belongs to C . Let $(x_1, \dots, x_n) \in C^n$ and $(a_1, \dots, a_n) \in [0, 1]^n$ such that $a_1 + \cdots + a_n = 1$. If $a_n = 1$, then $a_1 = \cdots = a_{n-1} = 0$ and hence $a_1x_1 + \cdots + a_nx_n = x_n \in C$. Otherwise, by the induction hypothesis one has

$$\sum_{i=1}^{n-1} \frac{a_i}{1 - a_n} x_i \in C.$$

Still by the induction hypothesis we obtain that

$$a_1x_1 + \cdots + a_nx_n = (1 - a_n) \sum_{i=1}^{n-1} \frac{a_i}{1 - a_n} x_i + a_nx_n \in C.$$

□

9.3.5 Example. Let E be a vector space over $\mathbb{R}$ and $(x, y) \in E \times E$ such that $x \neq y$. Then the convex hull of $\{x, y\}$ is of the form

$$\{tx + (1 - t)y \mid t \in \mathbb{R}\} = \{y + t(x - y) \mid t \in \mathbb{R}\}.$$

It is called the *line segment* linking x to y , denoted as $[x, y]$.

9.4 Mean value property

9.4.1 Proposition (Gronwall inequality). *Let $(F, \|\cdot\|_F)$ be a normed vector space over $\mathbb{R}$, $(a, b) \in \mathbb{R}^2$ such that $a < b$. Let $f : [a, b] \rightarrow F$ and $g : [a, b] \rightarrow \mathbb{R}$ be two continuous mappings that are differentiable on $]a, b[$. Assume that, for any $t \in]a, b[$ one has $\|f'(t)\| \leq g(t)$. Then*

$$\|f(b) - f(a)\|_F \leq g(b) - g(a)$$

Proof. Let $m \in]a, b[$. Let $\varepsilon > 0$ and

$$J = \{t \in [m, b] \mid \forall s \in [m, t], \|f(s) - f(m)\| \leq g(s) - g(m) + \varepsilon(s - m)\}.$$

By definition, J is an interval. Since f and g are continuous, J is actually a closed interval. Hence is of the form $[m, c]$. If $c < b$, then, for $h > 0$ sufficiently small, one has

$$f(c + h) - f(c) = hf'(c) + o(h), \quad g(c + h) - g(c) = hg'(c) + o(h).$$

Hence there exists $\delta > 0$ such that $c + \delta \leq b$ and $\forall h \in [0, \delta]$,

$$\|f(c + h) - f(c)\|_F \leq \|f'(c)\|_F h + \frac{\varepsilon}{2}h, \quad g(c + h) - g(c) \geq g'(c)h - \frac{\varepsilon}{2}h.$$

Hence

$$\|f(c + h) - f(c)\| \leq g(c + h) - g(c) + \varepsilon h.$$

Moreover, $\|f(c) - f(m)\| \leq g(c) - g(m) + \varepsilon(c - m)$. Thus

$$\|f(c + h) - f(m)\| \leq g(c + h) - g(m) + \varepsilon(c + h - m).$$

This means $J \supseteq [m, c + \delta]$, contradiction. Hence we obtain

$$\|f(b) - f(c)\|_F \leq g(b) - g(c) + \varepsilon(b - c).$$

For the same reason,

$$\|f(c) - f(a)\|_F \leq g(c) - g(a) + \varepsilon(c - a).$$

Hence

$$\|f(b) - f(a)\|_F \leq g(b) - g(a) + \varepsilon(b - a).$$

Since $\varepsilon > 0$ is arbitrary, we obtain the desired result. $\square$

9.4.2 Theorem (Mean value inequality). *Let $(F, \|\cdot\|)$ be a normed vector space over $\mathbb{R}$. Let $(a, b) \in \mathbb{R}^2$ such that $a < b$, and $f : [a, b] \rightarrow F$ be a continuous mapping that is differentiable on $]a, b[$. Then one has*

$$\|f(b) - f(a)\|_F \leq \sup_{t \in]a, b[} \|f'(t)\|_F \cdot (b - a).$$

Proof. Without loss of generality, we assume that $a < b$ and

$$\sup_{t \in]a, b[} \|f'(t)\|_F < +\infty.$$

Let M be a real number such that

$$M > \sup_{c \in]a, b[} \|f'(t)\|_F.$$

Let $m = (a + b)/2$ and

$$J = \{x \in [m, b] \mid \forall t \in [m, x], \|f(t) - f(m)\|_F \leq M(t - m)\}.$$

By definition, J is an interval containing m . It is then of the form $[m, c]$ or $[m, c[$. For any $t \in [m, c[$, one has

$$\|f(t) - f(m)\|_F \leq M(t - m).$$

Taking the limit when $t \rightarrow c$, by the continuity of f , we obtain

$$\|f(c) - f(m)\|_F = \lim_{t \in [m, c[, t \rightarrow c} \|f(t) - f(m)\|_F \leq \lim_{t \in [m, c[, t \rightarrow c} M(t - m) = M(c - m).$$

Hence $J = [m, c]$. If $c < b$, then the mapping is differentiable at c , so that, for any $h \in]0, b - c[$, one has

$$\|f(c + h) - f(c)\|_F = \|hf'(c) + o(h)\| \leq \|f'(c)\|_F h + o(h), \quad h \rightarrow 0.$$

Since $M > \|f'(c)\|_F$, there exists $h_0 > 0$ such that

$$\forall h \in]0, h_0], \|f(c + h) - f(c)\|_F \leq Mh,$$

and hence, for any $h \in]0, h_0]$,

$$\begin{aligned} \|f(c + h) - f(m)\|_F &\leq \|f(c + h) - f(c)\|_F + \|f(c) - f(m)\|_F \\ &\leq Mh + M(c - m) = M(c + h - m). \end{aligned}$$

This contradicts the definition of J . Therefore one has $c = b$ and hence

$$\|f(b) - f(m)\|_F \leq M(b - m).$$

A similar argument in considering

$$J' = \{x \in [a, m] \mid \forall t \in [x, m], \|f(m) - f(t)\|_F \leq M(m - t)\}$$

shows that

$$\|f(m) - f(a)\|_F \leq M(m - a).$$

Therefore,

$$\begin{aligned} \|f(b) - f(a)\|_F &\leq \|f(b) - f(m)\|_F + \|f(m) - f(a)\|_F \\ &\leq M(b - m) + M(m - a) = M(b - a). \end{aligned}$$

□

9.4.3 Corollary. Let $(E, \|\cdot\|)$ and $(F, \|\cdot\|)$ be normed vector spaces over $\mathbb{R}$, and U be an open subset of E . Let $(x, y) \in U \times U$ such that the line segment $[x, y]$ is contained in U . Let $f : U \rightarrow F$ be a differentiable mapping. Then one has

$$\|f(x) - f(y)\|_F \leq \left(\sup_{t \in]0, 1[} \|Df(tx + (1-t)y)\| \right) \|x - y\|_E,$$

where $\|Df(tx + (1-t)y)\|$ denotes the operator norm of $Df(tx + (1-t)y)$.

Proof. Let $g : [0, 1] \rightarrow F$ be the mapping that sends $t \in [0, 1]$ to $f(tx + (1-t)y)$. This mapping is continuous and differentiable on $]0, 1[$. Moreover, one has

$$g'(t) = Df(tx + (1-t)y)(x - y).$$

In particular,

$$\|g'(t)\|_F \leq \|Df(tx + (1-t)y)\| \cdot \|x - y\|_E.$$

By Theorem 9.4.2, one has

$$\|f(x) - f(y)\|_F = \|g(0) - g(1)\|_F \leq \sup_{t \in]0, 1[} \|g'(t)\|_F.$$

□

9.4.4 Definition. Let $(E, \|\cdot\|)$ be a normed vector space over $\mathbb{R}$, U be an open subset of E , $f : U \rightarrow \mathbb{R}$ be a mapping. Let $p \in U$. We say that p is a *local maximum point* of f if there exists a neighbourhood V of p such that $V \subseteq U$ and

$$\forall x \in V, f(x) \leq f(p).$$

Similarly, we say that p is a *local minimum point* of f if there exists a neighbourhood V of p such that $V \subseteq U$ and

$$\forall x \in V, f(x) \geq f(p).$$

If p is a local maximum point or a local minimum point of f , we say that p is a *local extremum point*.

If f is differentiable at p and if $Df(p)$ is the zero linear mapping, we say that p is a *critical point*.

9.4.5 Theorem. Let $(E, \|\cdot\|)$ be a normed vector space over $\mathbb{R}$, U be an open subset of E , $f : U \rightarrow \mathbb{R}$ be a mapping. Any extremum point of f is a critical point.

Proof. Let p be an extremum point of f . There exists $r > 0$ such that $p + B(0, r) \subseteq U$ and $f(p + h) \geq f(p)$ for any $h \in B(0, r)$. Note that

$$h \longmapsto f(p + h) - f(p)$$

does not change sign on U . For any $h \in B(0, r)$, one has

$$(f(p + h) - f(p))(f(p - h) - f(p)) \geq 0.$$

Let $h \in B(0, r)$. For any $t \in]0, 1]$, one has

$$\begin{aligned} f(p + th) - f(p) &= tDf(p)(h) + o(t), \\ f(p - th) - f(p) &= tDf(p)(-h) + o(t), \end{aligned}$$

when $t \rightarrow 0$. Hence one has

$$0 \leq (f(p + h) - f(p))(f(p - h) - f(p)) = -Df(p)(h)^2 t^2 + o(t^2).$$

Dividing the two sides by t^2 , taking the limit when $t \rightarrow 0$, we obtain that

$$0 \leq -Df(p)(h)^2,$$

which leads to $Df(p)(h) = 0$. □

9.4.6 Theorem (Mean value theorem of Rolle). *Let $(a, b) \in \mathbb{R}^2$ such that $a < b$. Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous mapping that is differentiable on $]a, b[$. If $f(a) = f(b)$, then there exists $t \in]a, b[$ such that $f'(t) = 0$.*

Proof. Since $[a, b]$ is compact and f is continuous, we obtain that f attains its maximum and its minimum on $[a, b]$. Moreover, since $f(a) = f(b)$, the mapping f has an extremum point in $]a, b[$ since otherwise the maximal value and the minimal value of f are both $f(a)$, which means that f is a constant mapping. By Theorem 9.4.5, we obtain that f has a critical point in $]a, b[$, namely there exists $t \in]a, b[$ such that $f'(t) = 0$. □

9.4.7 Theorem (Mean value theorem of Lagrange). *Let $(a, b) \in \mathbb{R}^2$ such that $a < b$. Let $f : [a, b] \rightarrow \mathbb{R}$ and $g : [a, b] \rightarrow \mathbb{R}$ be a continuous mapping that are differentiable on $]a, b[$. Then there exists $\xi \in]a, b[$ such that*

$$g'(\xi)(f(b) - f(a)) = f'(\xi)(g(b) - g(a)).$$

In particular, there exists $c \in]a, b[$ such that

$$f(b) - f(a) = f'(c)(b - a).$$

Proof. If $g(b) = g(a)$, by Rolle's theorem there exists $\xi \in]a, b[$ such that $g'(\xi) = 0$. Hence the desired result holds.

In the rest of the proof, we suppose that $g(b) \neq g(a)$. Let $h : [a, b] \rightarrow \mathbb{R}$ be the mapping defined as

$$h(t) = f(t) - Cg(t).$$

where C is a constant such that

$$f(a) - Cg(a) = f(b) - Cg(b),$$

or equivalently

$$C = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

Then h is a continuous mapping, which is differentiable on $]a, b[$, and

$$h'(t) = f'(t) - Cg'(t).$$

Since $h(a) = h(b) = 0$, by Theorem 9.4.6, there exists $\xi \in]a, b[$ such that $h'(\xi) = 0$, or equivalently $f'(\xi) = Cg'(\xi)$. Thus

$$f'(\xi)(g(b) - g(a)) = g'(\xi)(f(b) - f(a)).$$

□

9.4.8 Theorem (Darboux). *Let I be an open interval in $\mathbb{R}$ and $f : I \rightarrow \mathbb{R}$ be a differentiable mapping. Then $f'(I)$ is an interval in $\mathbb{R}$.*

Proof. Let $(a, b) \in I \times I$ such that $a < b$. Consider the mappings

$$g : [a, b] \longrightarrow \mathbb{R}, \quad g(x) := \begin{cases} \frac{f(x) - f(a)}{x - a}, & x \neq a \\ f'(a), & x = a, \end{cases}$$

and

$$h : [a, b] \longrightarrow \mathbb{R}, \quad h(x) := \begin{cases} \frac{f(b) - f(x)}{b - x}, & x \neq b \\ f'(b), & x = b. \end{cases}$$

Note that the mappings g and h are continuous. Moreover, by the mean value theorem, one has

$$g(I) \cup h(I) \subseteq f'(I).$$

Since $g(I)$ and $h(I)$ are intervals and

$$\frac{f(b) - f(a)}{b - a} \in g(I) \cap h(I),$$

we obtain that $g(I) \cup h(I)$ is an interval. Hence $f'(I)$ contains the interval linking $f'(a)$ and $f'(b)$. Thus $f'(I)$ is itself an interval. □

9.5 Multilinear mappings

In this section, we fix a commutative unitary ring K .

9.5.1 Definition. Let n be a positive integer. Let $E_1, \dots, E_n, F$ be K -modules. We say that a mapping

$$f : E_1 \times \cdots \times E_n \longrightarrow F$$

is n -linear (or simply *multilinear* if there is no ambiguity on n) if, for any $i \in \{1, \dots, n\}$ and any

$$(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) \in E_1 \times \cdots \times E_{i-1} \times E_{i+1} \times \cdots \times E_n,$$

the mapping from E_i to F sending $x_i \in E_i$ to

$$f(x_1, \dots, x_{i-1}, x_i, x_{i+1}, \dots, x_n),$$

is a morphism of K -modules. We denote by

$$\text{Hom}_K^{(n)}(E_1, \dots, E_n, F)$$

The set of all n -linear mappings from $E_1 \times \cdots \times E_n$ to F . It is actually a sub- K -module of

$$F^{E_1 \times \cdots \times E_n}.$$

In the particular case where $n = 2$, a 2-linear mapping from $E_1 \times E_2$ to F is also called a *bilinear* mapping.

9.5.2 Proposition. Let n be an integer that is ≥ 1 , $E_1, \dots, E_n, F$ be K -modules, and $i \in \{1, \dots, n-1\}$. Then the mapping

$$f : \text{Hom}_K^{(n)}(E_1, \dots, E_n, F) \longrightarrow \text{Hom}_K^{(i)}(E_1, \dots, E_i, \text{Hom}_K^{(i)}(E_{i+1}, \dots, E_n, F))$$

sending $\varphi \in \text{Hom}_K^{(n)}(E_1, \dots, E_n, F)$ to the mapping

$$(x_1, \dots, x_i) \longmapsto ((x_{i+1}, \dots, x_n) \longmapsto \varphi(x_1, \dots, x_i, x_{i+1}, \dots, x_n))$$

is an isomorphism of K -modules.

Proof. The inverse of the mapping f is given by

$$\psi \longmapsto ((x_1, \dots, x_i, x_{i+1}, \dots, x_n) \longmapsto \psi(x_1, \dots, x_i)(x_{i+1}, \dots, x_n)).$$

□

9.5.3 Definition. Let K be a commutative unitary ring. Let n be a positive integer, E and F be K -modules. We say that an n -linear mapping

$$\varphi : E^n = \underbrace{E \times \cdots \times E}_{n \text{ copies}} \longrightarrow F$$

is symmetric if, for any bijection $\sigma : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$, one has

$$\forall (x_1, \dots, x_n) \in E^n, \varphi(x_{\sigma(1)}, \dots, x_{\sigma(n)}) = \varphi(x_1, \dots, x_n).$$

Let $P : E \rightarrow F$ be a mapping. Assume that there exists a symmetric n -linear mapping from

$$\varphi : E^n = \underbrace{E \times \cdots \times E}_{n \text{ copies}} \longrightarrow F$$

such that

$$\forall x \in E, P(x) = \varphi(x, \dots, x),$$

then we say that P is a *homogeneous polynomial mapping of degree n* from E to F and that φ is a *polarisation* of P . In the case where $F = K$, a homogeneous polynomial mapping of degree n from E to K is also called a *homogeneous polynomial* of degree n on E .

9.6 Bounded multilinear mappings

In this section, we fix a complete valued field $(K, |\cdot|)$.

9.6.1 Definition. Let n be a positive integer, and $(E_1, \|\cdot\|_1), \dots, (E_n, \|\cdot\|_n), (F, \|\cdot\|_F)$ be normed vector spaces over $(K, |\cdot|)$. For any $\varphi \in \text{Hom}_K^{(n)}(E_1, \dots, E_n, F)$, we denote by $\|\varphi\|$ the element

$$\sup_{\substack{(x_1, \dots, x_n) \in E_1 \times \cdots \times E_n \\ x_1 \neq 0, \dots, x_n \neq 0}} \frac{\|\varphi(x_1, \dots, x_n)\|_F}{\|x_1\|_1 \cdots \|x_n\|_n} \in [0, +\infty].$$

If $\|\varphi\| < +\infty$, we say that φ is *bounded*. We denote by

$$\mathcal{L}^{(n)}(E_1, \dots, E_n, F)$$

the set of all bounded multilinear mappings from $E_1 \times \cdots \times E_n$ to F . It is a K -vector subspace of $F^{E_1 \times \cdots \times E_n}$. Moreover, $\|\cdot\|$ is a norm on this vector space, called the *operator norm*.

9.6.2 Proposition. Let n be a positive integer, and $(E_1, \|\cdot\|_1), \dots, (E_n, \|\cdot\|_n), (F, \|\cdot\|_F)$ be normed vector spaces over $(K, |\cdot|)$. Assume that the normed vector space $(F, \|\cdot\|_F)$ is complete, then $(\mathcal{L}^{(n)}(E_1, \dots, E_n, F), \|\cdot\|)$ is also complete.

Proof. Let $(\varphi_i)_{i \in \mathbb{N}}$ be a Cauchy sequence in $\mathcal{L}^{(n)}(E_1, \dots, E_n, F)$. By definition, for any $\mathbf{x} = (x_1, \dots, x_n) \in E_1 \times \dots \times E_n$, the mapping

$$f_{\mathbf{x}} : \mathcal{L}^{(n)}(E_1, \dots, E_n, F) \longrightarrow F, \varphi \longmapsto \varphi(x_1, \dots, x_n)$$

is a bounded linear mapping since

$$\|\varphi(x_1, \dots, x_n)\|_F \leq \|\varphi\| \cdot \|x_1\|_1 \cdots \|x_n\|_n.$$

Hence it sends the Cauchy sequence $(\varphi_i)_{i \in \mathbb{N}}$ to a Cauchy sequence $(\varphi_i(x_1, \dots, x_n))_{i \in \mathbb{N}}$, that converges to some element $\psi(x_1, \dots, x_n) \in F$ since F is assumed to be complete. Thus we obtain a mapping $\psi : E_1 \times \dots \times E_n \rightarrow F$ which is the pointwise limit of $(\varphi_i)_{i \in \mathbb{N}}$. In particular, ψ is a multilinear mapping.

Since $\|\cdot\|$ is a norm on $\mathcal{L}^{(n)}(E_1, \dots, E_n, F)$, it is Lipschitzian. Hence $(\|\varphi_i\|)_{i \in \mathbb{N}}$ is a Cauchy sequence in $\mathbb{R}$, hence has a limit, which we denote as C . Taking the limit on both sides of the inequality

$$\|\varphi_i(x_1, \dots, x_n)\|_F \leq \|\varphi_i\| \cdot \|x_1\|_1 \cdots \|x_n\|_n$$

when $i \rightarrow +\infty$, we obtain

$$\|\psi(x_1, \dots, x_n)\|_F \leq C \|x_1\|_1 \cdots \|x_n\|_n.$$

Therefore, ψ is a bounded multilinear mapping. Finally, for any $\varepsilon > 0$, there exists $a \in \mathbb{N}$ such that, for any $(i, j) \in \mathbb{N}_{\geq a} \times \mathbb{N}_{\geq a}$, one has $\|\varphi_i - \varphi_j\| \leq \varepsilon$ and hence, for any $(x_1, \dots, x_n) \in E_1 \times \dots \times E_n$,

$$\|\varphi_i(x_1, \dots, x_n) - \varphi_j(x_1, \dots, x_n)\|_F \leq \varepsilon \|x_1\|_1 \cdots \|x_n\|_n.$$

Taking the limit when $j \rightarrow +\infty$, we obtain

$$\forall i \in \mathbb{N}_{\geq a}, \|\varphi_i - \psi\| \leq \varepsilon.$$

Hence

$$\lim_{i \rightarrow +\infty} \|\varphi_i - \psi\| = 0,$$

as required. □

9.6.3 Proposition. Let n be an integer that is ≥ 2 , $(E_1, \|\cdot\|_1), \dots, (E_n, \|\cdot\|_n)$ and $(F, \|\cdot\|_F)$ be normed vector spaces, and $i \in \{1, \dots, n-1\}$, the mapping

$$\mathcal{L}^{(n)}(E_1, \dots, E_n, F) \longrightarrow \mathcal{L}^{(i)}(E_1, \dots, E_i, \mathcal{L}^{(n-i)}(E_{i+1}, \dots, E_n, F))$$

that sends $\varphi \in \mathcal{L}^{(n)}(E_1, \dots, E_n, F)$ to

$$(x_1, \dots, x_i) \longmapsto \left((x_{i+1}, \dots, x_n) \longmapsto \varphi(x_1, \dots, x_i, x_{i+1}, \dots, x_n) \right)$$

is an isomorphism of normed vector spaces (namely an isomorphism of vector spaces and an isometry).

Proof. For any $(x_1, \dots, x_i) \in E_1 \times \dots \times E_i$, one has

$$\|\varphi(x_1, \dots, x_i, \cdot)\| \leq \|\varphi\| \cdot \|x_1\|_1 \cdots \|x_i\|_i.$$

and hence $\varphi(x_1, \dots, x_i, \cdot) \in \mathcal{L}^{(n-i)}(E_{i+1}, \dots, E_n, F)$. Moreover, one has

$$\begin{aligned} \|f(\varphi)\| &= \sup_{\substack{(x_1, \dots, x_i) \in E_1 \times \dots \times E_i \\ x_1 \neq 0, \dots, x_i \neq 0}} \frac{\|\varphi(x_1, \dots, x_i, \cdot)\|}{\|x_1\|_1 \cdots \|x_i\|_i} \\ &= \sup_{\substack{(x_1, \dots, x_i) \in E_1 \times \dots \times E_i \\ x_1 \neq 0, \dots, x_i \neq 0}} \sup_{\substack{(x_{i+1}, \dots, x_n) \in E_{i+1} \times \dots \times E_n \\ x_{i+1} \neq 0, \dots, x_n \neq 0}} \frac{\|\varphi(x_1, \dots, x_i, x_{i+1}, \dots, x_n)\|_F}{\|x_1\|_1 \cdots \|x_n\|_n} \\ &= \|\varphi\|. \end{aligned}$$

Hence f is an isometry. In particular, f is injective.

To show that f is surjective, we observe that, for any

$$\psi \in \mathcal{L}^{(i)}(E_1, \dots, E_i, \mathcal{L}^{(n-i)}(E_{i+1}, \dots, E_n, F)),$$

the mapping

$$\varphi : (x_1, \dots, x_n) \mapsto \psi(x_1, \dots, x_i)(x_{i+1}, \dots, x_n)$$

belongs to $\mathcal{L}^{(n)}(E_1, \dots, E_n, F)$, and one has $f(\varphi) = \psi$. $\square$

9.6.4 Corollary. Let n be an integer that is ≥ 1 , $(E_1, \|\cdot\|_1), \dots, (E_n, \|\cdot\|_n)$ and $(F, \|\cdot\|_F)$ be normed vector spaces. Suppose that $E_1, \dots, E_n$ are of finite dimension over K . Then any n -linear mapping from $E_1 \times \dots \times E_n$ to F is bounded.

9.6.5 Theorem. Let n be a positive integer, $(E_1, \|\cdot\|_1), \dots, (E_n, \|\cdot\|_n)$, $(F, \|\cdot\|_F)$ be normed vector spaces over $(K, |\cdot|)$. Let φ be an element of $\mathcal{L}^{(n)}(E_1, \dots, E_n, F)$. We equip $E_1 \times \dots \times E_n$ with the norm $\|\cdot\|$ defined as

$$\|(x_1, \dots, x_n)\| := \max_{i \in \{1, \dots, n\}} \|x_i\|_i.$$

For any $p = (p_1, \dots, p_n) \in E_1 \times \dots \times E_n$, the mapping φ is differentiable at p . Moreover, for any $h = (h_1, \dots, h_n) \in E_1 \times \dots \times E_n$,

$$D\varphi(p)(h) = \sum_{i=1}^n \varphi(p_1, \dots, p_{i-1}, h_i, p_{i+1}, \dots, p_n).$$

Proof. For $p = (p_1, \dots, p_n)$ and $h = (h_1, \dots, h_n)$ in $E_1 \times \dots \times E_n$, one has

$$\begin{aligned} &\varphi(p+h) - \varphi(p) \\ &= \sum_{i=1}^n (\varphi(p_1 + h_1, \dots, p_{i-1} + h_{i-1}, p_i + h_i, p_{i+1}, \dots, p_n) \\ &\quad - \varphi(p_1 + h_1, \dots, p_{i-1} + h_{i-1}, p_i, p_{i+1}, \dots, p_n)) \\ &= \sum_{i=1}^n \varphi(p_1 + h_1, \dots, p_{i-1} + h_{i-1}, h_i, p_{i+1}, \dots, p_n) \end{aligned}$$

Note that, for any $i \in \{2, \dots, n\}$,

$$\begin{aligned} & \varphi(p_1 + h_1, \dots, p_{i-1} + h_{i-1}, h_i, p_{i+1}, \dots, p_n) - \varphi(p_1, \dots, p_{i-1}, h_i, p_{i+1}, \dots, p_n) \\ &= \sum_{j=1}^{i-1} \varphi(p_1 + h_1, \dots, p_{j-1} + h_{j-1}, h_j, p_{j+1}, \dots, p_{i-1}, h_i, p_{i+1}, \dots, p_n) \\ &= O(\|h\|^2) = o(\|h\|). \end{aligned}$$

Hence,

$$\varphi(p + h) - \varphi(p) = \sum_{i=1}^n \varphi(p_1, \dots, p_{i-1}, h_i, p_{i+1}, \dots, p_n) + o(\|h\|).$$

□

9.6.6 Corollary. *Let $(E, \|\cdot\|_E)$ and $(F, \|\cdot\|_F)$ be normed vector spaces over K . Let U be an open subset of E , $f : U \rightarrow K$ and $g : U \rightarrow F$ be two mappings, and $p \in U$. Suppose that f and g are both differentiable at p , then fg is also differentiable at p . Moreover, for any $h \in E$ one has*

$$D(fg)(p)(h) = f(p)Dg(p)(h) + Df(p)(h)g(p).$$

In the particular case where $(E, \|\cdot\|_E) = (K, |\cdot|)$, one has

$$(fg)'(p) = f(p)g'(p) + f'(p)g(p).$$

Proof. Let $\varphi : K \times F \rightarrow F$ be the mapping that sends (a, s) to as . It is a bounded bilinear mapping since

$$\|as\|_F = |a| \cdot \|s\|_F.$$

Let $\Gamma : U \rightarrow F \times F$ be the mapping that sends $x \in U$ to $(f(x), g(x))$. Then fg identifies with the composite $\varphi \circ \Gamma$.

Note that

$$D\Gamma(p)(h) = (Df(p)(h), Dg(p)(h))$$

and

$$D\varphi(a, s)(b, t) = \varphi(a, t) + \varphi(b, s) = at + bs.$$

Hence

$$\begin{aligned} D(fg)(p)(h) &= D\varphi(\Gamma(p))(D\Gamma(p)(h)) \\ &= D\varphi(f(p), g(p))(Df(p)(h), Dg(p)(h)) \\ &= f(p)Dg(p)(h) + Df(p)(h)g(p). \end{aligned}$$

□

9.6.7 Proposition. *Let $(K, |\cdot|)$ be a valued field, n be a positive integer, $(E, \|\cdot\|_E)$ and $(F, \|\cdot\|_F)$ be normed vector spaces over K . Let $P : E \rightarrow F$ be a homogeneous polynomial mapping of degree n . Assume that P admits a polarisation φ which is a bounded symmetric n -linear mapping. Then P is differentiable on E , and, for any $x \in E$ one has*

$$\forall h \in E, DP(x)(h) = n\varphi(x, \dots, x, h).$$

Proof. Let $\Delta : E \rightarrow E^n$ that sends $x \in E$ to $(x, \dots, x)$. Then $P = \varphi \circ \Delta$, so that

$$DP(x)(h) = D\varphi(\Delta(x))(D\Delta(x)(h)).$$

Since Δ is a bounded linear mapping, one has

$$D\Delta(x)(h) = \Delta(h) = (h, \dots, h).$$

Moreover, since φ is a bounded n -linear mapping, one has

$$D\varphi(\Delta(x))(z_1, \dots, z_n) = \sum_{i=1}^n \varphi(x, \dots, x, z_i, x, \dots, x).$$

Hence, by the assumption that φ is symmetric, one has

$$DP(x)(h) = n\varphi(x, \dots, x, h).$$

□

9.6.8 Remark. In the particular case where $(E, \|\cdot\|_E) = (K, |\cdot|)$, a homogeneous polynomial mapping of degree n is of the form

$$P(x) = x^n s,$$

where s is an element of F . A polarization of P is given by

$$\varphi : K^n \longrightarrow F, \varphi(a_1, \dots, a_n) = a_1 \dots a_n s.$$

Therefore,

$$P'(x) = DP(x)(1) = nx^{n-1}s.$$

9.7 Higher differential

In this section, we fix a complete non-trivial valued field $(K, |\cdot|)$ and a normed K -vector spaces $(E, \|\cdot\|_E)$ and $(F, \|\cdot\|_F)$.

9.7.1 Definition. Let U be an open subset of E and $f : U \rightarrow F$ be a mapping.

Let $p \in U$. If f is continuous at p , we say that it is 0-times differentiable at p and we denote by $D^0 f(p)$ the value $f(p)$. If f is differentiable on an open neighbourhood V of p and if $Df : V \rightarrow \mathcal{L}(E, F)$ is $(n-1)$ -times differentiable at p , we say that f is n times differentiable at p and we denote by $D^n f(p) \in \mathcal{L}^{(n)}(E, \dots, E, F)$ which sends $(h_1, \dots, h_n) \in E^n$ to

$$D^{n-1}(Df)(p)(h_1, \dots, h_{n-1})(h_n).$$

In the case where $E = K$, we denote by $f^{(n)}(p)$ the element

$$D^n f(1, \dots, 1).$$

Note that, for any $i \in \{1, \dots, n-1\}$, one has

$$D^n f(p)(h_1, \dots, h_n) = D^i(D^{n-i} f)(p)(h_1, \dots, h_i)(h_{i+1}, \dots, h_n).$$

In the case where $E = K$, one has

$$(f^{(n-i)})^{(i)}(p) = f^{(n)}(p).$$

Sometimes, f'' denotes $f^{(2)}$ and f''' denotes $f^{(3)}$.

If f is n -times differentiable on U and if the mapping

$$D^n f : U \longrightarrow \mathcal{L}^{(n)}(E, \dots, E, F)$$

is continuous, then we say that f is of class C^n .

9.7.2 Definition. Let X be a set. We denote by $\mathfrak{S}_X$ the set of all bijections from X to itself. The elements of $\mathfrak{S}_X$ are called *permutations* on X . The set $\mathfrak{S}_X$ equipped with the composition of mappings forms a group, called the *symmetric group* of X .

Let n be an integer ≥ 2 and $x_1, \dots, x_n$ be distinct elements in $\mathfrak{S}_X$. We denote by

$$(x_1 \ x_2 \ \cdots \ x_n)$$

the mapping that sends x_i to x_{i+1} for $i \in \{1, \dots, n-1\}$, x_n to x_1 and any $y \in X \setminus \{x_1, \dots, x_n\}$ to y itself. Such a permutation is called an *n -cycle*. A 2-cycle is also called a *transposition*.

Note that the group $\mathfrak{S}_X$ acts on X as follows:

$$\mathfrak{S}_X \times X \longrightarrow X, (\sigma, x) \longmapsto \sigma(x).$$

For any $\sigma \in \mathfrak{S}_x$ and $x \in X$, we denote by $\text{orb}_\sigma(x)$ the set

$$\{\sigma^n(x) \mid n \in \mathbb{Z}\}.$$

9.7.3 Proposition. *Let X be a set, $\sigma \in \mathfrak{S}_X$ and $x \in X$. If $\text{orb}_\sigma(x)$ is a finite set of d elements, where $d \in \mathbb{N}_{\geq 1}$, then one has $\sigma^d(x) = x$ and*

$$\text{orb}_\sigma(x) = \{x, \sigma(x), \dots, \sigma^{d-1}(x)\}.$$

The restriction of σ to $\text{orb}_\sigma(x)$ coincides to the d -cycle

$$(x \ \sigma(x) \ \dots \ \sigma^{d-1}(x)).$$

Proof. By the hypothesis of the proposition, the set

$$\{(n, m) \in \mathbb{N} \times \mathbb{N} \mid n < m, \sigma^n(x) = \sigma^m(x)\}$$

is not empty. Let

$$\ell := \inf\{m - n \mid (n, m) \in \mathbb{N}^2, n < m, \sigma^n(x) = \sigma^m(x)\}.$$

Then $x, \sigma(x), \dots, \sigma^{\ell-1}(x)$ are distinct elements of $\text{orb}_\sigma(x)$. Moreover, for any $n \in \mathbb{N}$, we can write n into the form $n = \ell q + r$, where $q \in \mathbb{N}$, $r \in \{0, \dots, \ell - 1\}$, then one has

$$\sigma^n(x) = \sigma^r(\sigma^{\ell q}(x)) = \sigma^r(x).$$

which implies that $\text{orb}_\sigma(x) = \{x, \sigma(x), \dots, \sigma^{\ell-1}(x)\}$. Hence $d = \ell$. $\square$

9.7.4 Theorem. *Let X be a finite set and $\sigma \in \mathfrak{S}_X$. There exists $d \in \mathbb{N}$, $(n_1, \dots, n_d) \in \mathbb{N}_{\geq 2}^d$ and pairwise disjoint subsets $X_1, \dots, X_d$ of cardinalities $n_1, \dots, n_d$ of X respectively, together with n_i -cycles τ_i of elements of X_i ($i \in \{1, \dots, d\}$), such that*

$$\sigma = \tau_1 \circ \dots \circ \tau_d.$$

In other words, any permutation can be decomposed into the composition of finitely many cycles on disjoint subsets.

Proof. We reason by induction on the cardinality N of X . The case where $\sigma = \text{Id}_X$ is trivial (this also includes the case where $N = 0$ and $N = 1$). In the following, we assume that $N \geq 2$ and $\sigma \neq \text{Id}_X$. We pick $x \in X$ such that $\sigma(x) \neq x$ and let $X_1 = \text{orb}_\sigma(x)$. Let n_1 be the cardinality of $\text{orb}_\sigma(x)$. The restriction of σ to X_1 corresponds to the n_1 -cycle given by

$$\tau_1 := (x \ \sigma(x) \ \dots \ \sigma^{n_1-1}(x)).$$

Let $Y := X \setminus X_1$. For any $y \in Y$ one has $\sigma(y) \in Y$ since otherwise $y = \sigma^{-1}(\sigma(y)) \in X_1$. Let $\tau := \sigma|_Y \in \mathfrak{S}_Y$. Applying the induction hypothesis on Y , we obtain disjoint subsets $X_2, \dots, X_d$ of Y of cardinalities $n_2, \dots, n_d$, with n_i -cycles τ_i of elements of X_i ($i \in \{2, \dots, d\}$), such that $\tau = \tau_2 \circ \dots \circ \tau_d$. Hence $\sigma = \tau_1 \circ \dots \circ \tau_d$, as required. $\square$

9.7.5 Corollary. *Let X be a finite set. Any element $\sigma \in \mathfrak{S}_X$ can be written as a composition of transpositions.*

Proof. By the previous theorem, it suffices to check that each cycle is a composition of transpositions. Let n be an integer such that $n \geq 2$. One has

$$(x_1 x_2 \cdots x_n) = (x_1 x_2) \circ (x_2 x_3 \cdots x_n).$$

Therefore, by induction we obtain

$$(x_1 x_2 \cdots x_n) = (x_1 x_2) \circ (x_2 x_3) \circ \cdots \circ (x_{n-1} x_n).$$

□

9.7.6 Corollary. *Let $\mathfrak{S}_n$ be the symmetric group of $\{1, \dots, n\}$. Then any element $\sigma \in \mathfrak{S}_n$ can be written as a composition of transpositions of the form $(i \ i+1)$, where $i \in \{1, \dots, n-1\}$ (such a transposition is called an adjacent transposition).*

Proof. By the previous corollary, it suffices to check that any transposition τ can be written as a composition of the special transpositions of the form $(i \ i+1)$, where $i \in \{1, \dots, n-1\}$. Let j and k be elements of $\{1, \dots, n\}$ such that $1 < j < k$. One has

$$(j-1 \ j) \circ (j \ k) \circ (j-1 \ j) = (j-1 \ k).$$

Hence by induction we obtain that, for any $(j, k) \in \{1, \dots, n\}^2$ such that $j < k$, one has

$$(j \ k) = (j \ j+1) \circ (j+1 \ j+2) \circ \cdots \circ (k-1 \ k) \circ (k-2 \ k-1) \circ \cdots \circ (j \ j+1).$$

□

9.7.7 Theorem (Schwarz). *Let U be an open subset of E , $f : U \rightarrow F$ be a mapping, $p \in U$ and n be a positive integer. Assume that f is n -times differentiable at p . Then $D^n f(p)$ is a symmetric n -linear mapping from E^n to F .*

Proof. We first treat the case where $n = 2$. Since f is two times differentiable at p , we obtain that f is of class C^1 on an open neighbourhood V of p and Df is differentiable at p .

Let $\varepsilon > 0$. We suppose that δ is a positive real number such that, for any $h \in E$ with $\|h\|_E \leq 2\delta$, one has $p+h \in V$ and

$$\|Df(p+h)(\cdot) - Df(p)(\cdot) - D^2f(p)(h, \cdot)\| \leq \varepsilon \|h\|_E.$$

Let h such that $\|h\|_E < \delta$. Let $g : B(0, \delta) \rightarrow F$ be the mapping defined as

$$g_h(k) = f(p+h+k) - f(p+h) - f(p+k) + f(p) - D^2f(p)(h, k).$$

One has

$$\begin{aligned} Dg_h(k)(\cdot) &= Df(p+h+k)(\cdot) - Df(p+k)(\cdot) - D^2f(p)(h, \cdot) \\ &= (Df(p+h+k)(\cdot) - Df(p)(\cdot) - D^2f(p)(h+k, \cdot)) \\ &\quad - (Df(p+k)(\cdot) - Df(p)(\cdot) - D^2f(p)(k, \cdot)). \end{aligned}$$

Therefore, for $(h, k) \in B(0, \delta) \times B(0, \delta)$, one has

$$\|Dg_h(k)(\cdot)\| \leq \varepsilon(\|h+k\|_E + \|k\|_E) \leq 3\varepsilon \max\{\|h\|_E, \|k\|_E\}.$$

By the mean value inequality, since $g_h(0) = 0$, one has

$$\|g_h(k)\|_F \leq 3\varepsilon \max\{\|h\|_E, \|k\|_E\}^2$$

Therefore,

$$f(p+h+k) - f(p+h) - f(p+k) + f(p) - D^2f(p)(h, k) = o(\max\{\|h\|_E, \|k\|_E\}^2).$$

By the symmetry between k and h , we obtain

$$f(p+h+k) - f(p+h) - f(p+k) + f(p) - D^2f(p)(k, h) = o(\max\{\|h\|_E, \|k\|_E\}^2).$$

Therefore

$$D^2f(p)(k, h) - D^2f(p)(h, k) = o(\max\{\|h\|_E, \|k\|_E\}^2).$$

In particular, for fixed $(h, k) \in E \times E$, one has

$$D^2f(p)(tk, th) - D^2f(p)(th, tk) = o(|t|^2).$$

Dividing the two sides by t^2 and let $t \rightarrow 0$, we obtain

$$D^2f(p)(k, h) = D^2f(p)(h, k).$$

To prove the general case, we reason by induction on n . Assume that $n \geq 3$. One has

$$D^n f(p)(h_1, \dots, h_n) = D^{n-1}(Df)(p)(h_1, \dots, h_{n-1}, h_n).$$

Therefore, if σ is an adjacent transposition of the for $(j \ j+1)$ with $j < n-1$, one has

$$D^n f(p)(h_1, \dots, h_n) = D^n f(p)(h_{\sigma(1)}, \dots, h_{\sigma(n)}).$$

It remains the case where σ is of the form $(n-1 \ n)$. In this case we use the fact that

$$\begin{aligned} D^n f(p)(h_1, \dots, h_n) &= D^2(D^{n-2}f)(h_1, \dots, h_{n-2})(h_{n-1}, h_n) \\ &= D^2(D^{n-2}f)(h_1, \dots, h_{n-2})(h_n, h_{n-1}) \end{aligned}$$

to conclude. □

9.8 Taylor formulae

In this section, we let $(E, \|\cdot\|_E)$ and $(F, \|\cdot\|_F)$ be normed vector spaces over $\mathbb{R}$.

9.8.1 Theorem (Taylor-Young formula). *Let U be an open subset of E , $f : U \rightarrow F$ be a mapping, $p \in U$, and n be a positive integer. Suppose that the mapping f is n -times differentiable at p . Then one has*

$$f(p+h) = \sum_{k=0}^n \frac{1}{k!} D^k f(p)(h, \dots, h) + o(\|h\|_E^n), \quad h \rightarrow 0.$$

Proof. We reason by induction on n . The case where $n = 0$ comes from the definition of continuity of f at p .

In the following, we suppose that $n \geq 1$ and that the theorem holds for mappings that are $(n-1)$ -times differentiable. We define a mapping $g : U \rightarrow F$ as

$$g(x) = f(x) - \sum_{k=0}^n \frac{1}{k!} D^k f(p)(x-p, \dots, x-p)$$

Note that

$$Dg(x)(h) = Df(x)(h) - \sum_{k=1}^n \frac{1}{(k-1)!} D^k f(p)(x-p, \dots, x-p, h).$$

Thus

$$Dg(x) = Df(x) - \sum_{k=0}^{n-1} \frac{1}{k!} D^k (Df)(p)(x-p, \dots, x-p)$$

The induction hypothesis applied to Df shows that

$$Dg(x) = o(\|x-p\|^{n-1}), \quad x \rightarrow p.$$

For any $\varepsilon > 0$, there exists $r > 0$ such that $p + B(0, r) \subseteq U$, g is differentiable on $p + B(0, r)$ and

$$\forall x \in p + B(0, r), \quad \|Dg(x)\| \leq \varepsilon \|x-p\|_E^{n-1}.$$

By the mean value theorem, one has

$$\forall x \in p + B(0, r), \quad \|g(x)\|_F = \|g(x) - g(p)\|_F \leq \varepsilon \|x-p\|_E^n.$$

Hence

$$g(x) = o(\|x-p\|_E^n), \quad x \rightarrow p.$$

□

9.8.2 Theorem (Taylor-Lagrange formula). *Let a and b be real numbers such that $a < b$ and $f : [a, b] \rightarrow \mathbb{R}$ be a mapping. Suppose that f is $n+1$ -times differentiable on $]a, b[$, and, for any $k \in \{0, \dots, n\}$, $f^{(k)}$ extends to a continuous mapping from $[a, b]$ to $\mathbb{R}$. Then there exists $\xi \in]a, b[$ such that*

$$f(b) - \sum_{k=0}^n \frac{(b-a)^k}{k!} f^{(k)}(a) = \frac{f^{(n+1)}(\xi)(b-a)^{n+1}}{(n+1)!}.$$

Proof. Consider the mapping $g : [0, 1] \rightarrow \mathbb{R}$ defined as

$$g(t) = \sum_{k=0}^n \frac{(b-t)^k}{k!} f^{(k)}(t) + \frac{C}{(n+1)!} (b-t)^{n+1},$$

where C is a constant such that $g(a)$ and $g(b)$ are equal, namely

$$\sum_{k=0}^n \frac{(b-a)^k}{k!} f^{(k)}(a) + \frac{C}{(n+1)!} (b-a)^{n+1} = f(b).$$

Note that

$$\begin{aligned} g'(t) &= \sum_{k=0}^n \frac{(b-t)^k}{k!} f^{(k+1)}(t) - \sum_{k=1}^n \frac{(b-t)^{k-1}}{(k-1)!} f^{(k)}(t) - \frac{C}{n!} (b-t)^n \\ &= \frac{(b-t)^n}{n!} f^{(n+1)}(t) - \frac{C(b-t)^n}{n!}. \end{aligned}$$

By Rolle's mean value theorem, there exists $\xi \in]a, b[$ such that $g'(\xi) = 0$, namely $C = f^{(n+1)}(\xi)$, as required. $\square$

9.8.3 Theorem. *Let $n \in \mathbb{N}$, U be an open subset of E , $f : U \rightarrow F$ be a mapping that is $(n+1)$ -times differentiable. Let $p \in U$ and $h \in E$ such that $p+th \in U$ for any $t \in [0, 1]$. Let*

$$M = \sup_{t \in]0, 1[} \|D^{(n+1)}f(p+th)\|.$$

Then

$$\left\| f(p+th) - \sum_{k=0}^n \frac{1}{k!} D^k f(p)(h, \dots, h) \right\| \leq \frac{M}{(n+1)!} \|h\|^{n+1}.$$

Proof. Let $\phi : [0, 1] \rightarrow F$ be defined as

$$\phi(t) = f(p+th) + \sum_{k=1}^n \frac{(1-t)^k}{k!} D^k f(p+th)(h, \dots, h).$$

Then $\phi(1) = f(p + h)$,

$$\phi(0) = \sum_{k=0}^n \frac{1}{k!} D^k f(p)(h, \dots, h).$$

Moreover,

$$\begin{aligned} \phi'(t) &= \sum_{k=0}^n \frac{(1-t)^k}{k!} D^{k+1} f(p + th)(h, \dots, h) - \frac{(1-t)^{k-1}}{(k-1)!} \sum_{k=1}^n D^k f(p + th)(h, \dots, h) \\ &= \sum_{k=0}^n \frac{(1-t)^k}{k!} D^{k+1} f(p + th)(h, \dots, h) - \sum_{k=0}^{n-1} \frac{(1-t)^k}{k!} D^{k+1} f(p + th)(h, \dots, h) \\ &= \frac{(1-t)^n}{n!} D^{n+1} f(p + th)(h, \dots, h). \end{aligned}$$

Hence

$$\|\phi'(t)\| \leq M \frac{(1-t)^n}{n!} \|h\|^{n+1} = \left(-M \frac{(1-t)^{n+1}}{(n+1)!} \|h\|^{n+1} \right)'.$$

By Gronwall's inequality,

$$\|\phi(1) - \phi(0)\| \leq \frac{M}{(n+1)!} \|h\|^{n+1}.$$

□

9.9 Series in a Banach space

In this section, we fix a complete non-trivially valued field $(K, |\cdot|)$. In particular, there is an element $a \in K$ such that $|a| < 1/2$.

9.9.1 Proposition. *Let (X, d) be a metric space and $(x_n)_{n \in \mathbb{N}}$ be a sequence in X . If*

$$\sum_{n \in \mathbb{N}} d(x_n, x_{n+1}) < +\infty,$$

then the sequence $(x_n)_{n \in \mathbb{N}}$ is a Cauchy sequence.

Proof. For any $N \in \mathbb{N}$, and any $(n, m) \in \mathbb{N}_{\geq N} \times \mathbb{N}_{\geq N}$, by the triangle inequality one has

$$d(x_n, x_m) \leq \sum_{k \in \mathbb{N}_{\geq N}} d(x_k, x_{k+1}).$$

Hence

$$\lim_{N \rightarrow +\infty} \sup_{(n, m) \in \mathbb{N}_{\geq N}^2} d(x_n, x_m) \leq \lim_{N \rightarrow +\infty} \sum_{k \in \mathbb{N}_{\geq N}} d(x_k, x_{k+1}) = 0.$$

□

9.9.2 Definition. Let $(E, \|\cdot\|)$ be a normed vector space. If E equipped with the metric

$$E \times E \longrightarrow \mathbb{R}_{\geq 0}, (x, y) \longmapsto \|x - y\|$$

is complete, we say that $(E, \|\cdot\|)$ is a *Banach space*.

Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in a Banach space E . If

$$\sum_{n \in \mathbb{N}} \|x_n\| < +\infty,$$

then we say that the series $\sum_{n \in \mathbb{N}} x_n$ *converges absolutely*. Proposition 9.9.1 shows that an absolutely convergent series in a Banach space always converges.

9.9.3 Remark. Let $(E, \|\cdot\|)$ be a Banach space. If $(x_n)_{n \in \mathbb{N}}$ is a sequence in E such that $\sum_{n \in \mathbb{N}} x_n$ converges, then

$$\sum_{n \in \mathbb{N}} x_n := \left(\sum_{k=0}^n x_k \right)_{n \in \mathbb{N}}$$

is a Cauchy sequence. Hence

$$\lim_{n \rightarrow +\infty} \|x_n\| = \lim_{n \rightarrow +\infty} \left\| \sum_{k=0}^n x_k - \sum_{k=0}^{n-1} x_k \right\| = 0.$$

In particular, the sequence $(x_n)_{n \in \mathbb{N}}$ is bounded.

9.9.4 Theorem (Root test of Cauchy). *Let $(E, \|\cdot\|)$ be a Banach space and $(x_n)_{n \in \mathbb{N}}$ be a sequence in E . Let*

$$r = \limsup_{n \rightarrow +\infty} \|x_n\|^{1/n}.$$

If $r < 1$, then the series $\sum_{n \in \mathbb{N}} x_n$ converges absolutely. If $r > 1$, then the series diverges.

Proof. Suppose that $r < 1$. Let $\varepsilon \in \mathbb{R}$ such that $r < \varepsilon < 1$. There exists $N \in \mathbb{N}$ such that, for any $n \in \mathbb{N}_{\geq N}$, one has

$$\|x_n\| \leq \varepsilon^n.$$

Hence

$$\sum_{n \in \mathbb{N}_{\geq N}} \|x_n\| \leq \sum_{n \in \mathbb{N}_{\geq N}} \varepsilon^n = \frac{\varepsilon^N}{1 - \varepsilon} < +\infty.$$

Therefore the series $\sum_{n \in \mathbb{N}} x_n$ converges absolutely.

If $r > 1$, we let $\beta > 1$ such that $r > \beta$. By Bolzano-Weierstrass theorem, there is an infinite subset I of $\mathbb{N}$ such that $(\|x_n\|^{1/n})_{n \in I}$ converges to r . Hence there exists $N \in \mathbb{N}$ such that

$$\forall n \in \mathbb{N}_{\geq N}, \|x_n\| \geq \beta^n.$$

Thus the sequence $(x_n)_{n \in \mathbb{N}}$ is not bounded. $\square$

9.9.5 Theorem (Ratio test of D'Alembert). *Let $(E, \|\cdot\|)$ be a Banach space and $(x_n)_{n \in \mathbb{N}}$ be a sequence in E .*

(1) *If*

$$\limsup_{n \rightarrow +\infty} \frac{\|x_{n+1}\|}{\|x_n\|} < 1,$$

then the series $\sum_{n \in \mathbb{N}} x_n$ converges absolutely.

(2) *If*

$$\liminf_{n \rightarrow +\infty} \frac{\|x_{n+1}\|}{\|x_n\|} > 1,$$

then the series $\sum_{n \in \mathbb{N}} x_n$ converges

Proof. (1) Let $\alpha \in]0, 1[$ such that

$$\limsup_{n \rightarrow +\infty} \frac{\|x_{n+1}\|}{\|x_n\|} < \alpha.$$

There exists $N \in \mathbb{N}$ such that $\|x_{n+1}\| \leq \alpha \|x_n\|$ for $n \in \mathbb{N}_{\geq N}$. Hence

$$\sum_{n \geq N} \|x_n\| \leq \|x_N\| \sum_{n \geq N} \alpha^{n-N} = \frac{\|x_N\|}{1 - \alpha} < +\infty.$$

(2) Let $\beta > 1$ such that

$$\liminf_{n \rightarrow +\infty} \frac{\|x_{n+1}\|}{\|x_n\|} > \beta.$$

There exists $N \in \mathbb{N}$ such that $\|x_N\| > 0$ and

$$\forall n \geq N, \|x_{n+1}\| \geq \beta \|x_n\| \geq \dots \geq \beta^{n-N} \|x_N\|.$$

Hence the sequence $(x_n)_{n \in \mathbb{N}}$ is unbounded and the series $\sum_{n \in \mathbb{N}} x_n$ diverges. $\square$

9.10 Local inversion

9.10.1 Theorem (Baire). *Let (X, d) be a complete metric space. Let $(\Omega_n)_{n \in \mathbb{N}}$ be a family of open and dense subsets of X , and Ω be the intersection $\bigcap_{n \in \mathbb{N}} \Omega_n$. Then the subset Ω is dense in X .*

Proof. Let x_0 be an arbitrary element of X and $\varepsilon > 0$. We construct in a recursive way a sequence as follows. Let $r_0 = \varepsilon$. Suppose that $x_n \in X$ and r_n is chosen. Then $B(x_n, r_n) \cap \Omega_n$ is a non-empty open subset. We pick $x_{n+1} \in X$ and $r_{n+1} \in]0, r_n/2]$ such that

$$\overline{B}(x_{n+1}, r_{n+1}) \subseteq B(x_n, r_n) \cap \Omega_n.$$

By Proposition 9.9.1, the sequence $(x_n)_{n \in \mathbb{N}}$ is a Cauchy sequence, which converges in X since (X, d) is complete. Let $x \in X$ be the limit of $(x_n)_{n \in \mathbb{N}}$. For any $n \in \mathbb{N}$ and any $k \in \mathbb{N}_{\geq n+1}$, one has $x_k \in \overline{B}(x_{n+1}, r_{n+1})$ and hence

$$x \in \overline{B}(x_{n+1}, r_{n+1}) \subseteq B(x_n, r_n) \cap \Omega_n.$$

Thus we obtain that $x \in \Omega \cap B(x_0, \varepsilon)$. Therefore, Ω is dense in X . $\square$

9.10.2 Remark. The same statement holds for a locally compact topological space.

9.10.3 Corollary. *Let (X, d) be a complete non-empty metric space. If $(Y_n)_{n \in \mathbb{N}}$ is a family of closed subsets of X such that $X = \bigcup_{n \in \mathbb{N}} Y_n$, then there exists $n \in \mathbb{N}$ such that $Y_n^\circ \neq \emptyset$.*

Proof. We reason by contradiction. Assume that $Y_n^\circ = \emptyset$ for any $n \in \mathbb{N}$. Then $\Omega_n := X \setminus Y_n$ is dense for any $n \in \mathbb{N}$. Hence $\bigcap_{n \in \mathbb{N}} \Omega_n$ is dense, which implies that $\bigcup_{n \in \mathbb{N}} Y_n$ has an empty interior. This is absurd. $\square$

9.10.4 Theorem (Banach). *Let E be a vector space over K . Let $\|\cdot\|_1$ and $\|\cdot\|_2$ be norms on E such that $(E, \|\cdot\|_1)$ and $(E, \|\cdot\|_2)$ are both complete. Assume that there exists $C > 0$ such that $\|\cdot\|_2 \leq C\|\cdot\|_1$, then the norms $\|\cdot\|_1$ and $\|\cdot\|_2$ are equivalent.*

Proof. For any $x \in E$ and $r > 0$, we let

$$B_i(x, r) := \{y \in E \mid \|y - x\|_i < r\}, \quad i \in \{1, 2\}.$$

Moreover, for any $Y \subseteq E$, we let $\overline{Y}^{\|\cdot\|_2}$ be the closure of Y in $(E, \|\cdot\|_2)$. Since

$$X = \bigcup_{n \in \mathbb{N}_{\geq 1}} \overline{B_1(0, n)}^{\|\cdot\|_2},$$

there exists $n_0 \in \mathbb{N}_{\geq 1}$, $p \in E$ and $r_0 > 0$ such that

$$B_2(p, r_0) \subseteq \overline{B_1(0, n_0)}^{\|\cdot\|_2},$$

or equivalently

$$B_2(0, r_0) \subseteq \overline{B_1(-p, n_0)}^{\|\cdot\|_2}.$$

By the triangle inequality, one has

$$B_1(-p, n_0) \subseteq B_1(0, n_0 + \|p\|_1),$$

which implies that

$$\overline{B_1(-p, n_0)}^{\|\cdot\|_2} \subseteq \overline{B_1(0, n_0 + \|p\|_1)}^{\|\cdot\|_2}$$

Therefore, by letting $r_1 := n_0 + \|p\|_1$, one has

$$B_2(0, r_0) \subseteq \overline{B_1(0, r_1)}^{\|\cdot\|_2} \subseteq B_1(0, r_1) + B_2(0, r_0|a|).$$

Pick an element $a \in K$ such that $0 < |a| < 1/2$. By rescaling by powers of a , we obtain that

$$B_2(0, r_0|a|^n) \subseteq B_1(0, r_1|a|^n) + B_2(0, r_0|a|^{n+1}).$$

Let y be an element of $B_2(0, r_0)$. We choose

$$(x_0, y_0) \in B_1(0, r_0) \times B_2(r_0/2)$$

such that $y = x_0 + y_0$. If

$$(x_n, y_n) \in B_1(0, r_0|a|^n) \times B_2(0, r_0|a|^{n+1})$$

is chosen, we pick

$$(x_{n+1}, y_{n+1}) \in B_1(0, r_0|a|^{n+1}) \times B_2(0, r_0|a|^{n+2})$$

such that $y_n = x_{n+1} + y_{n+1}$. Thus we construct in a recursive way two sequences $(x_n)_{n \in \mathbb{N}}$ and $(y_n)_{n \in \mathbb{N}}$. Note that the series $\sum_{n \in \mathbb{N}} x_n$ converges in $(E, \|\cdot\|_1)$ to an element $x \in E$ and hence converges in $(E, \|\cdot\|_2)$ to the same element x . Moreover, the relation

$$y = \sum_{k=0}^{\infty} x_k + y_n$$

shows that the series $\sum_{n \in \mathbb{N}} x_n$ converges in $(E, \|\cdot\|_2)$ to y . Hence $x = y$ and

$$\|y\|_1 = \|x\|_1 \leq \sum_{n \in \mathbb{N}} \|x_n\|_1 \leq \frac{r_1}{1 - |a|}.$$

Therefore,

$$B_2(0, r_0) \subseteq B_1(0, r_1/(1 - |a|)),$$

which shows that $\|\cdot\|_1$ is bounded from above by a constant times $\|\cdot\|_2$. $\square$

9.10.5 Corollary. *Let $(E, \|\cdot\|_E)$ and $(F, \|\cdot\|_F)$ be Banach spaces, and $f : E \rightarrow F$ be bounded linear mapping.*

- (1) *If f is a bijection, then f^{-1} is also bounded.*
- (2) *If f is a surjective mapping, then it is an open mapping, namely, for any open subset U of E , $f(U)$ is an open subset of F .*

Proof. (1) We let $\|\cdot\|'_E$ be the norm on E defined as

$$\forall x \in E, \|x\|'_E := \|f(x)\|_F.$$

Note that $\|\cdot\|'_E \leq \|f\| \cdot \|\cdot\|_E$. By Theorem 9.10.4, there exists $C > 0$ such that $\|\cdot\|_E \leq C\|\cdot\|_{E'}$. In particular, for any $y \in F$, one has

$$C^{-1}\|y\|_F = C^{-1}\|f(f^{-1}(y))\|_F = C^{-1}\|f^{-1}(y)\|_{E'} \geq \|f^{-1}(y)\|_E.$$

Hence f^{-1} is bounded.

(2) Let E_0 be the kernel of f . Since f is bounded, it is continuous and hence E_0 is closed in E . Let $\pi : E \rightarrow E/E_0$ be the quotient mapping and $g : E/E_0 \rightarrow F$ be the mapping that sends $[x]$ to $f(x)$. We equip $Q := E/E_0$ with the quotient norm $\|\cdot\|_Q$.

Let $x \in E$ and $r > 0$. For any $\alpha \in B([x], r)$, one has

$$\|\alpha - [x]\|_Q = \inf_{y \in \alpha} \|y - x\|_E < r.$$

Hence there exists $y \in \alpha$ such that $\|y - x\|_E < r$, which means that $B([x], r) \subseteq \pi(B(x, r))$. Therefore, π is an open mapping. Since g is a bounded bijection, by (1) we obtain that g^{-1} is also bounded. Therefore, g is an open mapping. We then deduce that $f = g \circ \pi$ is an open mapping. $\square$

9.10.6 Proposition. *Let $(E, \|\cdot\|_E)$, $(F, \|\cdot\|_F)$ and $(G, \|\cdot\|_G)$ be normed vector spaces over $(K, |\cdot|)$, U be an open subset of E , V be an open subset of F . Let $f : U \rightarrow V$ and $g : V \rightarrow G$ be mappings. Let $n \in \mathbb{N}$.*

- (1) *Let $p \in U$. If f is n -times differentiable at p and g is n -times differentiable at $f(p)$, then $g \circ f$ is n -times differentiable at p .*
- (2) *If f is of class C^n on U and g is of class C^n on V , then $g \circ f$ is of class C^n on U .*

Proof. We reason by induction on n . The case where $n = 0$ comes from topology part of the lecture. Suppose that $n \geq 1$ and the proposition is true for natural numbers $< n$. For the statement (1), in the case where $n = 1$, we just use the chain rule for the differentiability. If $n > 1$, then Df and Dg are respectively defined in a neighbourhood U' and V' of p and $f(p)$, such that $f(U') \subset V'$. $\square$

9.10.7 Definition. Let $(E, \|\cdot\|_E)$ and $(F, \|\cdot\|_F)$ be normed vector spaces over $(K, |\cdot|)$, U be an open subset of E and $f : U \rightarrow F$ be a mapping, and $n \in \mathbb{N}$. If f is n -times differentiable at each point of U and if $D^n f$ is continuous, we say that f is of class C^n . Mappings of class C^1 are also said to be *continuously differentiable*. If f is of class C^n for any $n \in \mathbb{N}$, we say that f is of class C^∞ . Note that, if $n \in \mathbb{N}_{\geq 1}$ and if f is n -times differentiable, then $D^{n-1}f$ is differentiable and hence continuous. Thus f is of class C^{n-1} . This observation also shows that f is of class C^∞ if and only if it is n -times differentiable for any $n \in \mathbb{N}$.

9.10.8 Theorem. Let $(E, \|\cdot\|)$ be a Banach space. We denote by $\text{GL}(E)$ the set of invertible bounded K -linear mappings from E to E .

- (1) For any $\varphi \in \text{GL}(E)$, φ^{-1} is a bounded K -linear mapping.
- (2) For any $\varphi \in \mathcal{L}(E, E)$ such that $\|\varphi\| < 1$, one has $\text{Id}_E + \varphi \in \text{GL}(E)$.
- (3) If we equip $\mathcal{L}(E, E)$ with the topology induced by the operator norm, then $\text{GL}(E)$ is an open subset of $\mathcal{L}(E, E)$.
- (4) The mapping

$$\iota : \text{GL}(E) \longrightarrow \text{GL}(E),$$

$$\varphi \longmapsto \varphi^{-1}, \text{ is of class } C^\infty.$$

Proof. (1) Let $\|\cdot\|'$ be the norm on E defined as

$$\forall x \in E, \|x\|' := \|\varphi(x)\|.$$

Since φ is bounded, one has $\|\cdot\|' \leq \|\varphi\| \cdot \|\cdot\|$. By Theorem 9.10.4, there exists $C > 0$ such that $\|\cdot\| \leq C\|\cdot\|'$, namely, for any $y \in E$,

$$\|\varphi^{-1}(y)\| \leq C\|\varphi(\varphi^{-1}(y))\| = C\|y\|.$$

- (2) Note that the series

$$\sum_{n \in \mathbb{N}} (-1)^n \varphi^n.$$

Note that, for any $n \in \mathbb{N}$, one has

$$\|\varphi^n\| \leq \|\varphi\|^n.$$

Since $\|\varphi\| < 1$, this series converges absolutely. Therefore, it converges to some $\eta \in \mathcal{L}(E, E)$. Moreover, for $n \in \mathbb{N}$, one has

$$(\text{Id}_E + \varphi) \sum_{i=0}^n (-1)^i \varphi^i = \left(\sum_{i=0}^n (-1)^i \varphi^i \right) (\text{Id}_E + \varphi) = \text{Id}_E + (-1)^n \varphi^{n+1}.$$

Taking the limit when $n \rightarrow +\infty$, we obtain that

$$\psi \circ \eta = \eta \circ \psi = \text{Id}_E.$$

(3) If f is an element of $\text{GL}(E)$, for any $\varphi \in \mathcal{L}(E, E)$ such that $\|\varphi\| < 1/\|f^{-1}\|$, one has

$$f + \varphi = f \circ (\text{Id}_E + f^{-1} \circ \varphi) \in \text{GL}(E).$$

Hence $\text{GL}(E)$ is an open subset of $\mathcal{L}(E, E)$.

(3) Let $f \in \text{GL}(E)$. For any $h \in \mathcal{L}(E, E)$ such that $\|h\| < 1/\|f^{-1}\|$, one has

$$\begin{aligned} \iota(f + h) - \iota(f) &= (f + h)^{-1} - f^{-1} = (f \circ (\text{Id}_E + f^{-1} \circ h))^{-1} - f^{-1} \\ &= (\text{Id}_E + f^{-1} \circ h)^{-1} \circ f^{-1} - f^{-1} \\ &= \left(\sum_{n \in \mathbb{N}} (-1)^n (f^{-1} \circ h)^n \right) \circ f^{-1} - f^{-1} \\ &= -f^{-1} \circ h \circ f^{-1} + o(\|h\|). \end{aligned}$$

Therefore, the mapping ι is differentiable, and

$$D\iota(f)(h) = -\varphi(\iota(f), h, \iota(f)),$$

where $\varphi : \mathcal{L}(E, E)^3 \rightarrow \mathcal{L}(E, E)$ is the bounded 3-linear mapping that sends (u, v, w) to $u \circ v \circ w$. Since bounded multilinear mappings are differentiable, by induction we obtain that ι is n -times differentiable for any n . Hence it is of class C^n . $\square$

9.10.9 Definition. Let (X, d) be a metric space, and $f : X \rightarrow X$ be a mapping. If there exists $\alpha \in [0, 1[$ such that f is α -Lipschitzian, then we say that f is a *contraction*.

9.10.10 Definition. Let X be a set and $f : X \rightarrow X$ be a mapping. If x is an element of X such that $f(x) = x$, then we say that f is a *fixed point* of f .

9.10.11 Theorem (Banach fixed point theorem). *Let (X, d) be a complete metric space and $f : X \rightarrow X$ be a contraction. Then f has a unique fixed point.*

Proof. Let x_0 be an arbitrary element of X . We define a sequence $(x_n)_{n \in \mathbb{N}}$ in a recursive way as

$$x_n = f(x_{n-1}), \quad n \in \mathbb{N}_{\geq 1}.$$

One has

$$d(x_n, x_{n+1}) \leq \alpha d(x_{n-1}, x_n) \leq \dots \leq \alpha^n d(x_0, x_1).$$

Therefore

$$\sum_{n \in \mathbb{N}} d(x_n, x_{n+1}) \leq \sum_{n \in \mathbb{N}} \alpha^n d(x_0, x_1) = d(x_0, x_1) \sum_{n \in \mathbb{N}} \alpha^n = \frac{d(x_0, x_1)}{1 - \alpha} < +\infty.$$

So $(x_n)_{n \in \mathbb{N}}$ is a Cauchy sequence, that converges in X to some point x . Since f is Lipschitzian, it is continuous. Hence

$$f(x) = \lim_{n \rightarrow +\infty} f(x_n) = \lim_{n \rightarrow +\infty} x_{n+1} = x.$$

Therefore, f has a fixed point.

If y is another fixed point of X , one has

$$0 \leq d(x, y) = d(f(x), f(y)) \leq \alpha d(x, y).$$

Since $\alpha < 1$, one has $d(x, y) = 0$ and hence $x = y$. □

9.10.12 Definition. Let $(E, \|\cdot\|_E)$ and $(F, \|\cdot\|_F)$ be normed vector spaces, U and V be respectively open subsets of E and F , and $f : U \rightarrow V$ be a mapping. Let $n \in \mathbb{N} \cup \{\infty\}$. If f is invertible and both f and f^{-1} are of class C^n , then we say that f is a C^n -diffeomorphism.

9.10.13 Theorem (Local inversion). Let $(E, \|\cdot\|_E)$ and $(F, \|\cdot\|_F)$ be Banach spaces over $\mathbb{R}$, U be an open subset of E and $f : U \rightarrow F$ be a mapping of class C^n , where $n \in \mathbb{N}_{\geq 1} \cup \{\infty\}$. Suppose that p is an element of U such that $Df(p) \in \mathcal{L}(E, F)$ is invertible, then there exists an open neighbourhood V of p contained in U such that $f|_V : V \rightarrow f(V)$ is a C^n -homeomorphism. Moreover, one has

$$\forall y \in f(V), Df^{-1}(y) = Df(f^{-1}(y))^{-1}.$$

Proof. By replacing f by

$$\tilde{f} : (x \in E) \mapsto (Df(p))^{-1}(f(p+x) - f(p)),$$

we may assume that $E = F$, $p = f(p) = 0$ and $Df(p) = \text{Id}_E$. In fact, one has

$$D\tilde{f}(0)(h) = D((Df(p))^{-1})(0)(Df(p)(h)) = (Df(p))^{-1}(Df(p)(h)) = h.$$

In the following, we let $u : U \rightarrow E$ be the mapping defined as $u(x) = f(x) - x$. Then one has $Du(0) = 0$. Since f is of class C^1 , so is u . There then exists $r > 0$ such that

$$\forall x \in \overline{B}(0, r), \|Du(x)\| \leq \frac{1}{2}.$$

By the mean value inequality, the restriction of u to $\overline{B}(0, r)$ is $\frac{1}{2}$ -Lipschitzian. Moreover, for any $(x, y) \in \overline{B}(0, r)^2$, one has

$$\|f(x) - f(y)\| = \|x + u(x) - y - u(y)\| \geq \|x - y\| - \|u(x) - u(y)\| \geq \frac{1}{2}\|x - y\|.$$

Hence f is injective.

Let $a \in \overline{B}(0, r/2)$. For any $x \in \overline{B}(0, r)$, one has

$$\|a - u(x)\| \leq \|a\| + \|u(x)\| \leq r$$

Therefore, the mapping

$$v : \overline{B}(0, r) \longrightarrow E, \quad v(x) := a - u(x)$$

sends $\overline{B}(0, r)$ to $\overline{B}(0, r)$, and hence defines a contraction on $\overline{B}(0, r)$. By the fixed point theorem, we obtain the existence of an element $g(a) \in \overline{B}(0, r)$ such that

$$g(a) = v(g(a)) = a - u(g(a)) = a - f(g(a)) + g(a),$$

or equivalently

$$f(g(a)) = a.$$

Let $W := B(0, r/2)$ and $V = f^{-1}(W) \cap B(0, r)$. Then $f|_V$ is surjective onto W . Let f^{-1} be the inverse of this mapping. It remains to check that f^{-1} is of class C^1 .

For $z \in B(0, r)$, one has $\|Du(z)\| \leq 1/2$. Hence $Df(z) = \text{Id}_E + Du(z)$ is invertible on $B(0, r)$.

Let $(x, x_0) \in V \times V$ and $(y, y_0) \in W \times W$ such that $f(x) = y$, $f(x_0) = y_0$. Then

$$y - y_0 = f(x) - f(x_0) = Df(x_0)(x - x_0) + o(\|x - x_0\|).$$

Moreover,

$$\|x - x_0\| = \|(f(x) - f(x_0)) - (u(x) - u(x_0))\| \leq \|y - y_0\| + \frac{1}{2}\|x - x_0\|.$$

Hence

$$\|f^{-1}(y) - f^{-1}(y_0)\| = \|x - x_0\| \leq 2\|y - y_0\|,$$

which shows that h is Lipschitzian, which is also continuous. This also implies that

$$o(\|x - x_0\|) = o(\|y - y_0\|).$$

Therefore,

$$Df(x_0)(x - x_0) = y - y_0 + o(\|y - y_0\|),$$

or equivalently

$$\begin{aligned} f^{-1}(y) - f^{-1}(y_0) &= x - x_0 = Df(x_0)^{-1}(y - y_0) + o(\|y - y_0\|) \\ &= Df(f^{-1}(y_0))^{-1}(y - y_0) + o(\|y - y_0\|). \end{aligned}$$

Therefore, f^{-1} is differentiable at y_0 and one has

$$Df^{-1}(y_0) = Df(f^{-1}(y_0))^{-1}.$$

Finally, one has $Df^{-1} = \iota \circ Df \circ f^{-1}$ is continuous. Therefore, Df^{-1} is of class C^{n-1} . The theorem is thus proved. $\square$

9.11 Uniform convergence

9.11.1 Definition. Let X be a set and $(Y, \mathcal{T})$ be a topological space, and $(f_n)_{n \in \mathbb{N}}$ be a sequence of mappings from X to Y . Let $f : X \rightarrow Y$ be a mapping. We say that the sequence $(f_n)_{n \in \mathbb{N}}$ converges pointwise to f if, for any $x \in X$, the sequence $(f_n(x))_{n \in \mathbb{N}}$ converges to $f(x)$.

Suppose that (Y, d) is a metric space. We say that the sequence $(f_n)_{n \in \mathbb{N}}$ converges uniformly to f if

$$\lim_{n \rightarrow +\infty} \sup_{x \in X} d(f_n(x), f(x)) = 0.$$

9.11.2 Theorem. Let $(X, \mathcal{T}_X)$ be a topological space and (Y, d_Y) be a metric space. Let $(f_n)_{n \in \mathbb{N}}$ be a sequence of mappings from X to Y , that converges uniformly to a mapping $f : X \rightarrow Y$. Let $p \in X$. If each mapping f_n is continuous at p , then f is continuous at p .

Proof. Let $\varepsilon > 0$. We will prove that $f^{-1}(B(f(p), \varepsilon))$ is a neighbourhood of p . Let $n \in \mathbb{N}$ such that

$$\sup_{x \in X} d(f_n(x), f(x)) < \varepsilon/3.$$

We claim that

$$f_n^{-1}(B(f_n(p), \varepsilon/3)) \subseteq f^{-1}(B(f(p), \varepsilon))$$

Let x be an element of X such that

$$d(f_n(x), f_n(p)) < \varepsilon/3.$$

One has

$$d(f(x), f(p)) \leq d(f(x), f_n(x)) + d(f_n(x), f_n(p)) + d(f_n(p), f(p)) < \varepsilon.$$

Since f_n is continuous, $f_n^{-1}(B(f_n(p), \varepsilon/3))$ is a neighbourhood of p . We then deduce that $f^{-1}(B(f(p), \varepsilon))$ is also a neighbourhood of p . $\square$

9.11.3 Theorem. Let (X, d_X) and (Y, d_Y) be metric spaces, $(f_n)_{n \in \mathbb{N}}$ be a sequence of uniformly continuous mappings from X to Y . Suppose that the sequence $(f_n)_{n \in \mathbb{N}}$ converges uniformly to a mapping $f : X \rightarrow Y$, then f is uniformly continuous.

Proof. Let $\varepsilon > 0$. There exists $n \in \mathbb{N}$ such that

$$\sup_{x \in X} d_Y(f_n(x), f(x)) < \varepsilon/3.$$

Since f_n is uniformly continuous, there exists $\delta > 0$ such that

$$\forall (x, y) \in X \times X, d_X(x, y) < \delta \implies d_Y(f_n(x), f_n(y)) < \varepsilon/3.$$

Therefore, for any $(x, y) \in X \times X$ such that $d_X(x, y) < \delta$, one has

$$d_Y(f(x), f(y)) \leq d_Y(f(x), f_n(x)) + d_Y(f_n(x), f_n(y)) + d_Y(f_n(y), f(y)) < \varepsilon.$$

Therefore, f is uniformly continuous. $\square$

9.11.4 Theorem. Let $(E, \|\cdot\|_E)$ and $(F, \|\cdot\|_F)$ be normed vector space over a complete non-trivially valued field $(K, |\cdot|)$, U be an open subset of E , and $(f_n)_{n \in \mathbb{N}}$ be a sequence of differentiable mappings from U to F . Let $f : U \rightarrow F$ and $g : U \rightarrow \mathcal{L}(E, F)$ be mappings, and $p \in U$. Suppose that

- (1) the sequence $(Df_n)_{n \in \mathbb{N}}$ converges uniformly to g ,
- (2) $(f_n)_{n \in \mathbb{N}}$ converges uniformly to f .
- (3) there exists $N \in \mathbb{N}$ and a mapping $\delta : U \rightarrow \mathbb{R}_{\geq 0}$ such that

$$\lim_{x \rightarrow p} \delta(x) = 0$$

and that, for any $n \in \mathbb{N}_{\geq N}$ and any $x \in U$,

$$\|f_n(x) - f_n(p) - Df_n(p)(x - p)\|_F \leq \delta(x)\|x - p\|_E.$$

Then the mapping f is differentiable and $Df = g$.

Proof. For any $n \in \mathbb{N}$, let

$$\varepsilon_n := \sup_{x \in U} \|Df_n(x) - g(x)\|, \quad d_n := \sup_{x \in U} \|f_n(x) - f(x)\|.$$

For sufficiently large n one has

$$\begin{aligned} \|f(x) - f(p) - g(p)(x - p)\| &\leq \|(f(x) - f_n(x)) - (f(p) - f_n(p))\| \\ &\quad + \|f_n(x) - f_n(p) - Df_n(p)(x - p)\| + \|Df_n(p)(x - p) - g(p)(x - p)\| \\ &\leq 2d_n + \delta(x)\|x - p\|_E + \varepsilon_n\|x - p\|_E. \end{aligned}$$

Taking the limit when $n \rightarrow +\infty$, we obtain

$$\|f(x) - f(p) - g(p)(x - p)\| \leq \delta(x)\|x - p\|_E,$$

namely

$$f(x) - f(p) - g(p)(x - p) = o(\|x - p\|_E), \quad x \rightarrow p.$$

□

9.11.5 Theorem. *Let $(E, \|\cdot\|_E)$ and $(F, \|\cdot\|_F)$ be normed vector space over $\mathbb{R}$, U be an open subset of E , $(f_n)_{n \in \mathbb{N}}$ be a sequence of differentiable mappings from U to F and $g : U \rightarrow \mathcal{L}(E, F)$ be a mapping. Suppose that*

- (1) *the sequence $(Df_n)_{n \in \mathbb{N}}$ converges uniformly to g ,*
- (2) *$(f_n)_{n \in \mathbb{N}}$ converges pointwise to a mapping $f : U \rightarrow F$.*

Then the mapping f is differentiable and $Df = g$.

Proof. Let $p \in U$. For any $(n, m) \in \mathbb{N}^2$, let

$$c_{m,n} := \sup_{x \in U} \|Df_m(x) - Df_n(x)\|.$$

For any $n \in \mathbb{N}$, let

$$\varepsilon_n := \sup_{x \in U} \|Df_n(x) - g(x)\|.$$

By the mean value inequality, for $r > 0$ such that $B(0, r) \subseteq U$, one has

$$\|(f_n(x) - f_m(x)) - (f_n(p) - f_m(p))\|_F \leq c_{m,n}\|x - p\|_E$$

Taking the limit when $m \rightarrow +\infty$, one obtains

$$\|(f_n(x) - f(x)) - (f_n(p) - f(p))\|_F \leq \varepsilon_n\|x - p\|_E.$$

Therefore,

$$\begin{aligned} \|f(x) - f(p) - g(p)(x - p)\| &\leq \|(f(x) - f_n(x)) - (f(p) - f_n(p))\| \\ &\quad + \|f_n(x) - f_n(p) - Df_n(p)(x - p)\| + \|Df_n(p)(x - p) - g(p)(x - p)\| \\ &\leq \varepsilon_n\|x - p\|_E + \|f_n(x) - f_n(p) - Df_n(p)(x - p)\|_F + \varepsilon_n\|x - p\|_E. \end{aligned}$$

Since f_n is differentiable at p , one has

$$\lim_{x \rightarrow p} \frac{\|f_n(x) - f_n(p) - Df_n(p)(x - p)\|_F}{\|x - p\|_E} = 0.$$

Therefore,

$$\limsup_{x \rightarrow p} \frac{\|f(x) - f(p) - g(p)(x - p)\|}{\|x - p\|_E} \leq 2\varepsilon_n.$$

Taking the limit when $n \rightarrow +\infty$, we obtain

$$\limsup_{x \rightarrow p} \frac{\|f(x) - f(p) - g(p)(x - p)\|}{\|x - p\|_E} = 0,$$

namely

$$f(x) - f(p) - g(p)(x - p) = o(\|x - p\|_E), \quad x \rightarrow p.$$

□

9.11.6 Proposition. *Let $(E, \|\cdot\|_E)$ and $(F, \|\cdot\|_F)$ be normed vector spaces over $\mathbb{R}$. We assume that $(F, \|\cdot\|_F)$ is a Banach space. Let U be a path connected subset of E and $(f_n)_{n \in \mathbb{N}}$ be a sequence of differentiable mappings from U to F . Suppose that*

- (1) *the sequence $(Df_n)_{n \in \mathbb{N}}$ converges uniformly to a mapping $g : U \rightarrow \mathcal{L}(E, F)$,*
- (2) *there exists $p \in U$ such that $(f_n(p))_{n \in \mathbb{N}}$ converges.*

Then the sequence $(f_n)_{n \in \mathbb{N}}$ converges pointwise on U to a differentiable mapping $f : U \rightarrow F$ such that $Df = g$.

Proof. We first treat the case where U is convex. For any $(n, m) \in \mathbb{N}^2$, let

$$c_{m,n} := \sup_{x \in U} \|Df_m(x) - Df_n(x)\|.$$

Let $x \in U$. By the mean value inequality, for $(n, m) \in \mathbb{N}^2$, one has

$$\|(f_n(x) - f_m(x)) - (f_n(p) - f_m(p))\|_F \leq c_{m,n} \|x - p\|_E,$$

which leads to

$$\|f_n(x) - f_m(x)\|_F \leq \|f_n(p) - f_m(p)\|_F + c_{m,n} \|x - p\|_E.$$

Therefore, $(f_n(x))_{n \in \mathbb{N}}$ is a Cauchy sequence, which converges in $(F, \|\cdot\|_F)$. Thus the statement of the proposition follows from Theorem 9.11.5.

We now treat the general case. Let $x \in U$ and let $\gamma : [0, 1] \rightarrow U$ be a continuous mapping such that $\gamma(0) = p$ and $\gamma(1) = x$. Let I be the set of $t \in [0, 1]$ such that $(f_n(\gamma(s)))_{n \in \mathbb{N}}$ converges for all $s \in [0, t]$. By definition, I is an interval containing 0, hence is of the form $[0, c]$ or $[0, c[$. Let $r > 0$ such that $B(\gamma(c), r) \subseteq U$. Note that $\gamma^{-1}(B(\gamma(c), r))$ is an open subset of $[0, 1]$ that contains c . Since c belongs to the closure of I , the intersection of $\gamma^{-1}(B(\gamma(c), r))$ with I is not empty. Therefore,

there exists $q \in \gamma(I) \cap B(\gamma(c), r)$ such that $(f_n(q))_{n \in \mathbb{N}}$ converges. Since $B(\gamma(c), r)$ is convex, by the particular case proved above, we obtain that $(f_n(x))_{n \in \mathbb{N}}$ converges for $x \in B(\gamma(c), r)$, which implies that $c \in I$. If $c \neq b$, then c is an adherent point of $]c, 1]$. Since $\gamma^{-1}(B(\gamma(c), r))$ is open in $[0, 1]$, there exists $r' > 0$ such that $B(c, r') \subseteq \gamma^{-1}(B(\gamma(c), r))$. In particular, $B(c, r') \cap]c, 1]$ is a non-empty and therefore must be an open interval in $[0, 1]$, which leads to a contradiction. Hence $c = 1$ and $(f_n(x))_{n \in \mathbb{N}}$ converges. $\square$

9.11.7 Definition. Let X be a set and $(F, \|\cdot\|_F)$ be a Banach space on a complete valued field $(K, |\cdot|)$. Let $(f_n)_{n \in \mathbb{N}}$ be a sequence of mappings from X to F . We say that the series $\sum_{n \in \mathbb{N}} f_n$ converges normally if

$$\sum_{n \in \mathbb{N}} \sup_{x \in X} \|f_n(x)\|_F$$

converges.

9.11.8 Proposition. Let X be a set and $(F, \|\cdot\|_F)$ be a Banach space on a complete valued field $(K, |\cdot|)$. Let $(f_n)_{n \in \mathbb{N}}$ be a sequence of mappings from X to F . If the series $\sum_{n \in \mathbb{N}} f_n$ converges normally, then it converges uniformly.

Proof. For any $n \in \mathbb{N}$, let

$$c_n = \sup_{x \in X} \|f_n(x)\|_F.$$

By the assumption of the theorem, one has

$$\forall x \in X, \quad \sum_{n \in \mathbb{N}} \|f_n(x)\|_F \leq \sum_{n \in \mathbb{N}} c_n < +\infty.$$

Therefore, the sequence

$$\left(\sum_{k=0}^n f_k(x) \right)_{n \in \mathbb{N}}$$

converges in F to some element that we denote as $f(x)$.

For any $n \in \mathbb{N}$ and $x \in X$, let

$$g_n(x) = \sum_{k=0}^n f_k(x).$$

Then

$$\begin{aligned} \|g_n(x) - f(x)\|_F &= \lim_{m \rightarrow +\infty} \|g_n(x) - g_m(x)\|_F = \lim_{m \rightarrow +\infty} \left\| \sum_{k=n+1}^m f_k(x) \right\|_F \\ &\leq \sum_{k=n+1}^{+\infty} \|f_k(x)\|_F \leq \sum_{k=n+1}^{+\infty} c_k. \end{aligned}$$

Therefore,

$$\lim_{n \rightarrow +\infty} \sup_{x \in X} \|g_n(x) - f(x)\|_F = 0.$$

□

9.11.9 Corollary. *Let $(E, \|\cdot\|_E)$ be and $(F, \|\cdot\|_F)$ be normed vector spaces over a complete valued field $(K, |\cdot|)$. We suppose that $(F, \|\cdot\|_F)$ is a Banach space. Let U be an open subset of E and $(f_n)_{n \in \mathbb{N}}$ be a sequence of differentiable mappings from U to F . Let $p \in U$. Suppose that*

- (1) *the series $\sum_{n \in \mathbb{N}} f_n$ converges normally,*
- (2) *the series $\sum_{n \in \mathbb{N}} Df_n$ converges normally,*
- (3) *there exists $N \in \mathbb{N}$ a family of mappings $(\delta_n : U \rightarrow \mathbb{R}_{\geq 0})_{n \in \mathbb{N}_{\geq N}}$ such that*

$$\forall n \in \mathbb{N}_{\geq N}, \lim_{x \rightarrow p} \delta_n(x) = \delta_n(p) = 0,$$

that the series $\sum_{n \in \mathbb{N}} \delta_n$ converges normally, and that, for any $n \in \mathbb{N}_{\geq N}$ and any $x \in U$,

$$\|f_n(x) - f_n(p) - Df_n(p)(x - p)\|_F \leq \delta_n(x) \|x - p\|_E.$$

Let f and g be respectively the limit of the series $\sum_{n \in \mathbb{N}} f_n$ and $\sum_{n \in \mathbb{N}} Df_n$. Then the mapping f is differentiable at p and $Df(p) = g(p)$.

Proof. Without loss of generality, we assume that $N = 0$. Let $S_n = \sum_{k=0}^n f_k$. Thus $DS_n = \sum_{k=0}^n Df_k$. By Proposition 9.11.8, the sequence $(S_n)_{n \in \mathbb{N}}$ converges uniformly to f and $(DS_n)_{n \in \mathbb{N}}$ converges uniformly to g . Moreover, the condition (3) implies that

$$\|S_n(x) - S_n(p) - DS_n(p)(x - p)\|_F \leq \|x - p\|_E \sum_{k=0}^n \delta_k(x).$$

Since the series $\sum_{n \in \mathbb{N}} \delta_n$ converges normally, it converges uniformly. Let δ be the limit of this series. Moreover, for any $n \in \mathbb{N}$, one has

$$\|f_n(x) - f_n(p) - Df_n(p)(x - p)\|_F \leq \|x - p\|_E \delta(x)$$

By Theorem 9.11.2 we obtain that δ is continuous at p , namely

$$\lim_{x \rightarrow p} \delta(x) = 0.$$

Therefore, by Theorem 9.11.4 we obtain the desired result. □

9.11.10 Corollary. Let $(E, \|\cdot\|_E)$ and $(F, \|\cdot\|_F)$ be normed vector spaces over $\mathbb{R}$. We suppose that $(F, \|\cdot\|_F)$ is a Banach space. Let U be an open subset of E , $(f_n)_{n \in \mathbb{N}}$ be a sequence of differentiable mappings from U to F . Suppose that

- (1) the series $\sum_{n \in \mathbb{N}} Df_n$ converges normally to a mapping $g : U \rightarrow \mathcal{L}(E, F)$,
- (2) the series $\sum_{n \in \mathbb{N}} f_n$ converges pointwise to a mapping $f : U \rightarrow F$.

Then the mapping f is differentiable and $Df = g$.

9.11.11 Corollary. Let $(E, \|\cdot\|_E)$ and $(F, \|\cdot\|_F)$ be normed vector spaces over $\mathbb{R}$. We assume that $(F, \|\cdot\|_F)$ is a Banach space. Let U be a path connected subset of U and $(f_n)_{n \in \mathbb{N}}$ be a sequence of differentiable mappings from U to F . Suppose that

- (1) the series $\sum_{n \in \mathbb{N}} Df_n$ converges normally to a mapping $g : U \rightarrow \mathcal{L}(E, F)$,
- (2) there exists $p \in U$ such that the series $\sum_{n \in \mathbb{N}} f_n(p)$ converges.

Then the series $\sum_{n \in \mathbb{N}} f_n$ converges pointwise on U to a differentiable mapping $f : U \rightarrow F$ such that $Df = g$.

9.12 Power series

In this section, we fix a complete non-trivially valued field $(K, |\cdot|)$. Let $(E, \|\cdot\|)$ be a Banach space over K .

9.12.1 Definition. Let $(s_n)_{n \in \mathbb{N}}$ be a sequence in E and $a \in K$. We call *power series centered at b* with coefficients $(s_n)_{n \in \mathbb{N}}$ the sequence of polynomial mappings on K as follows

$$\left(z \mapsto \sum_{k=0}^n (z - b)^k s_k \right)_{n \in \mathbb{N}}. \quad (9.5)$$

We use the expression

$$\sum_{n \in \mathbb{N}} (z - b)^n s_n$$

to denote this sequence. If $z \in K$ is an element such that the sequence (9.5) converges, by abuse of notation we still use the expression

$$\sum_{n \in \mathbb{N}} (z - b)^n s_n.$$

The element

$$\left(\limsup_{n \rightarrow +\infty} \|s_n\|^{1/n} \right)^{-1} \in [0, +\infty]$$

is called the *radius of convergence* of the power series $\sum_{n \in \mathbb{N}} (z - b)^n s_n$. If S is a power series, we denote by $R(S)$ the radius of convergence of S .

9.12.2 Proposition. *Let $S = \sum_{n \in \mathbb{N}} (z - b)^n s_n$ be a power series.*

- (1) *For any $a \in K$ such that $|a - b| < R(S)$, the series $\sum_{n \in \mathbb{N}} (a - b)^n s_n$ converges absolutely.*
- (2) *For any $a \in K$ such that $|a - b| > R(S)$, the series $\sum_{n \in \mathbb{N}} (a - b)^n s_n$ diverges.*
- (3) *If $r > 0$ is such that $(r^n \|s_n\|)_{n \in \mathbb{N}}$ is bounded, then $R(S) \geq r$.*

Proof. One has

$$\limsup_{n \rightarrow +\infty} \|(a - b)^n s_n\|^{1/n} = \limsup_{n \rightarrow +\infty} |a - b| \cdot \|s_n\|^{1/n} = |a - b| \cdot \limsup_{n \rightarrow +\infty} \|s_n\|^{1/n}.$$

- (1) If $|a - b| < R(S)$, then

$$|a - b| \cdot \limsup_{n \rightarrow +\infty} \|s_n\|^{1/n} < 1.$$

Hence the series

$$\sum_{n \in \mathbb{N}} (a - b)^n s_n$$

converges uniformly.

- (2) If $|a - b| > R(S)$, then

$$|a - b| \cdot \limsup_{n \rightarrow +\infty} \|s_n\|^{1/n} > 1.$$

Hence the series

$$\sum_{n \in \mathbb{N}} (a - b)^n s_n$$

diverges.

- (3) One has

$$\|s_n\|^{1/n} = (r^n \|s_n\|)^{1/n} \cdot \frac{1}{r}.$$

Since $(r^n \|s_n\|)_{n \in \mathbb{N}}$ is bounded, one has

$$\limsup_{n \rightarrow +\infty} (r^n \|s_n\|)^{1/n} \leq 1.$$

Hence

$$\limsup_{n \rightarrow +\infty} \|s_n\|^{1/n} \leq \frac{1}{r},$$

which leads to $R(S) \geq r$. □

9.12.3 Proposition. Let $S = \sum_{n \in \mathbb{N}} (z - b)^n s_n$ be a power series. For any $r \in \mathbb{R}$ such that $0 \leq r < R(s)$, the series of functions S converges normally on $\overline{B}(b, r)$. In particular,

$$(z \in B(b, R(s))) \longrightarrow S(z) := \sum_{n \in \mathbb{N}} (z - b)^n s_n$$

defines a continuous mapping from $B(b, R(s))$ to E .

Proof. Since $r < R(s)$, one has

$$\sum_{n \in \mathbb{N}} \sup_{z \in \overline{B}(b, r)} \|(z - b)^n s_n\|_F \leq \sum_{n \in \mathbb{N}} r^n \|s_n\|_F < +\infty.$$

Therefore By Proposition 9.11.2, the sequence $\sum_{n \in \mathbb{N}} (z - b)^n s_n$ converges uniformly on $\overline{B}(b, r)$ to a continuous mapping from $\overline{B}(b, r)$ to E . Since $r < R(s)$ is arbitrary, we obtain that

$$(z \in B(b, R(s))) \longrightarrow S(z) := \sum_{n \in \mathbb{N}} (z - b)^n s_n$$

defines a continuous mapping from $B(b, R(s))$ to E . □

9.12.4 Proposition. Let $S = \sum_{n \in \mathbb{N}} (z - b)^n s_n$ be a power series, and P be a non-zero polynomial in $K[T]$. For any integer n , we let $P(n)$ be the element $P(n1_K)$. Let S_P be the power series

$$\sum_{n \in \mathbb{N}} (z - b)^n P(n) s_n.$$

Then $R(S_P) \geq R(S)$.

Proof. Let d be the degree of P . Then P is of the form

$$P(T) = c_d T^d + c_{d-1} T^{d-1} + \cdots + c_0,$$

where $c_d \neq 0$. One has, for any $b > 1$

$$|P(n)| = O(n^d) = o(b^n), \quad n \rightarrow +\infty.$$

Hence there exists $C > 0$ such that

$$|P(n)| \leq b^n, \quad n \in \mathbb{N}_{\geq 1}$$

Hence

$$\limsup_{n \rightarrow +\infty} \|P(n) s_n\|^{1/n} \leq b \limsup_{n \rightarrow +\infty} \|s_n\|^{1/n}.$$

Since $b > 1$ is arbitrary, one has

$$\limsup_{n \rightarrow +\infty} \|P(n) s_n\|^{1/n} \leq \limsup_{n \rightarrow +\infty} \|s_n\|^{1/n}.$$

□

9.12.5 Lemma. *Let z and z_0 be two element of K , and n be a positive integer. The following equality holds:*

$$z^n - z_0^n - nz_0^{n-1}(z - z_0) = (z - z_0)^2 \sum_{j=0}^{n-2} (n - j - 1) z^j z_0^{n-2-j}.$$

Proof. One has

$$z^n - z_0^n = \sum_{i=0}^{n-1} z^i z_0^{n-1-i}.$$

Hence

$$\begin{aligned} z^n - z_0^n - nz_0^{n-1}(z - z_0) &= \sum_{i=1}^{n-1} z_0^{n-1-i} (z^i - z_0^i) = (z - z_0)^2 \sum_{i=1}^{n-1} z_0^{n-1-i} \sum_{j=0}^{i-1} z^j z_0^{i-1-j} \\ &= (z - z_0)^2 \sum_{j=0}^{n-2} z^j \sum_{i=j+1}^{n-1} z_0^{(n-1-i)+(i-1-j)} \\ &= (z - z_0)^2 \sum_{j=0}^{n-2} (n - 1 - j) z^j z_0^{n-2-j}. \end{aligned}$$

□

9.12.6 Theorem. *Let $\sum_{n \in \mathbb{N}} (z - b)^n s_n$ be a power series, and R be its radius of convergence of S . For any $z \in B(b, R)$, let $S(z)$ be the limit of the series $\sum_{n \in \mathbb{N}} (z - b)^n s_n$. Then the mapping $S : B(b, R) \rightarrow E$ is differentiable, and its derivative is given by the limit of the power series*

$$\sum_{n \in \mathbb{N}_{\geq 1}} (z - b)^{n-1} n s_n = \sum_{n \in \mathbb{N}} (z - b)^n (n + 1) s_{n+1}.$$

Proof. We have seen that the power series converges on $B(b, R)$. Let $r < R$ and z and z_0 be two elements of $\overline{B}(b, r)$. By Lemma 9.12.5, one has

$$\begin{aligned} &\left\| (z - b)^n s_n - (z_0 - b)^n s_n - (z - z_0)(z_0 - b)^{n-1} n s_n \right\|_E \\ &= |z - z_0|^2 \cdot \left\| \sum_{j=0}^{n-2} (n - 1 - j) (z - b)^j (z_0 - b)^{n-2-j} s_n \right\|_E. \end{aligned}$$

Note that

$$\begin{aligned} \left\| \sum_{j=0}^{n-2} (n - 1 - j) (z - b)^j (z_0 - b)^{n-2-j} s_n \right\|_E &\leq \sum_{j=0}^{n-2} (n - 1 - j) r^{n-2} \|s_n\|_E \\ &= \frac{n(n-1)}{2} r^{n-2} \|s_n\|_E \end{aligned}$$

By Proposition 9.12.4, we obtain that

$$\sum_{n \in \mathbb{N}} \frac{n(n-1)}{2} r^{n-2} \|s_n\| < +\infty.$$

Therefore, the result follows from Corollary 9.11.9. $\square$

9.12.7 Definition. Let $(a_n)_{n \in \mathbb{N}} \in K^{\mathbb{N}}$ and $(s_n)_{n \in \mathbb{N}} \in E^{\mathbb{N}}$. We call *Cauchy product* of the series $\sum_{n \in \mathbb{N}} a_n$ and $\sum_{n \in \mathbb{N}} s_n$ as the series

$$\sum_{n \in \mathbb{N}} \left(\sum_{k=0}^n a_k s_{n-k} \right).$$

This series is called the *Cauchy product* of $\sum_{n \in \mathbb{N}} a_n$ and $\sum_{n \in \mathbb{N}} s_n$.

9.12.8 Theorem (Mertens). Let $(a_n)_{n \in \mathbb{N}} \in K^{\mathbb{N}}$ and $(s_n)_{n \in \mathbb{N}} \in E^{\mathbb{N}}$. Suppose that the series $\sum_{n \in \mathbb{N}} a_n$ and $\sum_{n \in \mathbb{N}} s_n$ converge to $b \in K$ and $t \in E$ respectively.

- (1) If at least one of the series $\sum_{n \in \mathbb{N}} a_n$ and $\sum_{n \in \mathbb{N}} s_n$ converges absolutely, then the Cauchy product

$$\sum_{n \in \mathbb{N}} \left(\sum_{k=0}^n a_k s_{n-k} \right)$$

converges to bt .

- (2) If both series $\sum_{n \in \mathbb{N}} a_n$ and $\sum_{n \in \mathbb{N}} s_n$ converge absolutely, then the Cauchy product

$$\sum_{n \in \mathbb{N}} \left(\sum_{k=0}^n a_k s_{n-k} \right)$$

converges absolutely

Proof. (1) We suppose that the series $\sum_{n \in \mathbb{N}} a_n$ converges absolutely.

For any $n \in \mathbb{N}$, let

$$A_n = \sum_{n \in \mathbb{N}} a_n, \quad S_n = \sum_{\ell=0}^n s_\ell.$$

For any $N \in \mathbb{N}$, let

$$t_N = \sum_{n=0}^N \left(\sum_{k=0}^n a_k s_{n-k} \right) = \sum_{\substack{(k,\ell) \in \mathbb{N}^2 \\ k+\ell \leq N}} a_k s_\ell = \sum_{k=0}^N a_k S_{N-k} = A_N t + \sum_{k=0}^N a_k (S_{N-k} - t).$$

Then

$$t_N - bt = (A_N - b)t + \sum_{k=0}^N a_k(S_{N-k} - t).$$

Let

$$\alpha := \sum_{n \in \mathbb{N}} |a_n|.$$

For any $n \in \mathbb{N}$, let

$$\varepsilon_n = \sup_{m \in \mathbb{N}, m \geq n} \|S_m - t\|.$$

For any $\ell \in \{0, \dots, N\}$, one has

$$\begin{aligned} \left\| \sum_{k=0}^N a_k(S_{N-k} - t) \right\| &\leq \sum_{k=0}^{N-\ell} |a_k| \cdot \|S_{N-k} - t\| + \sum_{k=N-\ell+1}^N |a_k| \cdot \|S_{N-k} - t\| \\ &\leq \varepsilon_\ell \alpha + \max_{i \in \{0, \dots, \ell-1\}} \|S_i - t\| \sum_{k=N-\ell+1}^N |a_k|. \end{aligned}$$

Taking the superior limit when $N \rightarrow +\infty$, we get

$$\forall \ell \in \mathbb{N}, \limsup_{N \rightarrow +\infty} \left\| \sum_{k=0}^N a_k(S_{N-k} - t) \right\| \leq \varepsilon_\ell \alpha.$$

Taking the infimum with respect to ℓ , we get

$$\limsup_{N \rightarrow +\infty} \left\| \sum_{k=0}^N a_k(S_{N-k} - t) \right\| = 0.$$

Therefore, we deduce

$$\lim_{N \rightarrow +\infty} t_N = bt,$$

as required.

The case where $\sum_{n \in \mathbb{N}} s_n$ converges absolutely is simimilar, we omit the details.

(2) Let

$$\alpha := \sum_{n \in \mathbb{N}} |a_n|, \quad \beta := \sum_{n \in \mathbb{N}} \|s_n\|.$$

For any $N \in \mathbb{N}$ one has

$$\sum_{n=0}^N \left\| \sum_{k=0}^n a_k s_{n-k} \right\| \leq \sum_{n=0}^N \sum_{k=0}^n |a_k| \cdot \|s_{n-k}\| \leq \sum_{\substack{(k,\ell) \in \mathbb{N} \times \mathbb{N} \\ k \leq N, \ell \leq N}} |a_k| \cdot \|s_\ell\| \leq \alpha \beta.$$

Hence the Cauchy product converges absolutely. □

9.12.9 Example. Consider the following power series on $\mathbb{C}$:

$$\sum_{n \in \mathbb{N}} \frac{z^n}{n!}.$$

By the ratio test of D'Alembert, for any $r > 0$, one has

$$\sum_{n \in \mathbb{N}} \frac{r^n}{n!} < +\infty.$$

Hence the convergence radius of the above power series is $+\infty$. For $z \in \mathbb{C}$, we denote by $\exp(z)$ the limit of the series

$$\sum_{n \in \mathbb{N}} \frac{z^n}{n!}.$$

Sometimes we also write it as e^z . More generally, for any $a \in \mathbb{C}$, we let $\exp(az)$ be the limit of the series

$$\sum_{n \in \mathbb{N}} \frac{(az)^n}{n!}.$$

Note that

$$\exp'(az) = a \exp(az).$$

We let

$$\begin{aligned} \cos(z) &:= \frac{e^{iz} + e^{-iz}}{2}, \quad \sin(z) := \frac{e^{iz} - e^{-iz}}{2i}, \\ \cosh(z) &:= \frac{e^z + e^{-z}}{2}, \quad \sinh(z) := \frac{e^z - e^{-z}}{2}. \end{aligned}$$

9.12.10 Proposition. *Let a and b be complex numbers. One has*

$$\exp((a+b)z) = \exp(az) \exp(bz).$$

Proof. The Cauchy product of

$$\sum_{n \in \mathbb{N}} \frac{(az)^n}{n!} \quad \text{and} \quad \sum_{n \in \mathbb{N}} \frac{(bz)^n}{n!}$$

is

$$\sum_{n \in \mathbb{N}} \frac{(az + bz)^n}{n!}.$$

Hence the result follows from Theorem 9.12.8. □

9.13 Directional differential

9.13.1 Definition. Let $(K, |\cdot|)$ be a complete non-trivially valued field, $(E, \|\cdot\|_E)$ and $(F, \|\cdot\|_F)$ be normed vector spaces over K , U be an open subset of E and $f : U \rightarrow F$ be a mapping. Let $p \in U$ and $h \in E$. If the limit

$$\lim_{t \rightarrow 0} \frac{f(p + th) - f(p)}{t}$$

exists, we say that f admits the *directional derivative* at p along h , and we denote by $\partial_h f(p)$ the limit

$$\lim_{t \rightarrow 0} \frac{f(p + th) - f(p)}{t}.$$

In the rest of this section, we fix a complete non-trivially valued field $(K, |\cdot|)$. Let n be a positive integer, $(E_1, \|\cdot\|_1), \dots, (E_n, \|\cdot\|_n)$ and $(F, \|\cdot\|_F)$ be normed vector spaces over $(K, |\cdot|)$. Let $E = E_1 \times \dots \times E_n$ and $\|\cdot\|$ be the norm on E defined as

$$\|(s_1, \dots, s_n)\| = \max_{i \in \{1, \dots, n\}} \|s_i\|_i.$$

9.13.2 Definition. Let U be an open subset of E and $f : U \rightarrow F$. We say that f has the i -th *partial differential* at $p = (p_1, \dots, p_n) \in U$ if the mapping

$$x_i \mapsto f(p_1, \dots, p_{i-1}, x_i, p_{i+1}, \dots, p_n)$$

is differentiable at p_i . We denote by $D_i f(p) \in \mathcal{L}(E_i, F)$ the differential of the mapping

$$x_i \mapsto f(p_1, \dots, p_{i-1}, x_i, p_{i+1}, \dots, p_n)$$

at p_i , and call it the i -th partial differential of f at p . In the case where $E_i = K$, the element $D_i f(p)(1)$ is also written as $\partial_i f(p)$ or as

$$\frac{\partial f}{\partial x_i}(p),$$

where x_i denotes i -th variable of f . Note that $\partial_i f(p)$ also identifies with the directional derivative of f at p along

$$(0, \dots, 0, 1, 0, \dots, 0).$$

$\uparrow$
 $i^{\text{th}}\text{-entry}$

9.13.3 Theorem. Let U be an open subset of E and $f : U \rightarrow F$ be a mapping.

- (1) Let $p = (p_1, \dots, p_n) \in U$. Assume that f is differentiable at p . Then for any $i \in \{1, \dots, n\}$, f has the i -th partial differential at p , and, for any $h = (h_1, \dots, h_n) \in E$ one has

$$D_i f(p)(h_i) = Df(p)(0, \dots, 0, \underset{\substack{\uparrow \\ i^{\text{th}}\text{-entry}}}{h_i}, 0, \dots, 0).$$

- (2) Assume that $K = \mathbb{R}$, F is finite-dimensional, and, for any $i \in \{1, \dots, n\}$, f has the i -th partial differential on U , and $D_i f : U \rightarrow \mathcal{L}(E_i, F)$ is continuous. Then f is continuously differentiable on U , and, for any $h = (h_1, \dots, h_n) \in E$ one has

$$Df(p)(h) = \sum_{i=1}^n D_i f(p)(h_i).$$

Proof. (1) Since f is differentiable at p , one has

$$\begin{aligned} & f(p_1, \dots, p_{i-1}, p_i + h_i, p_{i+1}, \dots, p_n) - f(p_1, \dots, p_n) \\ &= Df(p)(0, \dots, 0, h_i, 0, \dots, 0) + o(\|h_i\|_i). \end{aligned}$$

Therefore, f has the i -th partial differential, and

$$D_i f(p)(h_i) = Df(p)(0, \dots, 0, h_i, 0, \dots, 0).$$

(2) We first treat the case where $F = \mathbb{R}$. Let $p = (p_1, \dots, p_n) \in U$ and $r > 0$ such that $B(p, r) \subseteq U$. Let $h = (h_1, \dots, h_n) \in E$ such that $\|h\| < r$. One can rewrite $f(p+h) - f(p)$ as

$$\sum_{i=1}^n f(p_1 + h_1, \dots, p_i + h_i, p_{i+1}, \dots, p_n) - f(p_1 + h_1, \dots, p_{i-1} + h_{i-1}, p_i, \dots, p_n)$$

By the mean value theorem, there exist $(t_1(h), \dots, t_n(h)) \in [0, 1]^n$ such that

$$\begin{aligned} & f(p_1 + h_1, \dots, p_i + h_i, p_{i+1}, \dots, p_n) - f(p_1 + h_1, \dots, p_{i-1} + h_{i-1}, p_i, \dots, p_n) \\ &= D_i f(p_1 + h_1, \dots, p_{i-1} + h_{i-1}, p_i + t_i(h)h_i, p_{i+1}, \dots, p_n)(h_i). \end{aligned}$$

Since $D_i f$ is assumed to be continuous, one has

$$\|D_i f(p_1 + h_1, \dots, p_{i-1} + h_{i-1}, p_i + t_i(h)h_i, p_{i+1}, \dots, p_n) - D_i f(p_1, \dots, p_n)\| = o(1)$$

when $h \rightarrow 0_E$. Therefore,

$$f(p+h) - f(p) = \sum_{i=1}^n D_i f(p)(h_i) + o(\|h\|),$$

as required. □

9.14 Convexity

9.14.1 Definition. Let E be a vector space over $\mathbb{R}$ and U be a convex subset of E . We say that a mapping $f : U \rightarrow \mathbb{R}$ is *convex* if the *epigraph* of f , which is defined as

$$\Gamma_+(f) := \{(x, a) \mid x \in U, f(x) \leq a\},$$

is a convex subset of $E \times \mathbb{R}$. Similarly, we say that a mapping $f : U \rightarrow \mathbb{R}$ is *concave* if its *hypograph*, which is defined as

$$\Gamma_-(f) := \{(x, a) \mid x \in U, f(x) \geq a\},$$

is a convex subset of $E \times \mathbb{R}$.

9.14.2 Proposition. Let E be a vector space over $\mathbb{R}$, U be a convex subset of E , and $f : U \rightarrow \mathbb{R}$ be a mapping. The following conditions are equivalent:

- (1) f is convex,
- (2) for any $(x, y) \in U \times U$ and $t \in [0, 1]$, one has

$$f(tx + (1 - t)y) \leq tf(x) + (1 - t)f(y).$$

Proof. “ $\implies$ ”: Note that $(x, f(x))$ and $(y, f(y))$ belongs to the epigraph of f . Hence

$$t(x, f(x)) + (1 - t)(y, f(y)) = (tx + (1 - t)y, tf(x) + (1 - t)f(y)) \in \Gamma_+(f).$$

Hence

$$f(tx + (1 - t)y) \leq tf(x) + (1 - t)f(y).$$

“ $\impliedby$ ”: Let (x, a) and (y, b) be elements of $\Gamma_+(f)$. Then $a \geq f(x)$, $b \geq f(y)$. For any $t \in [0, 1]$, one has

$$ta + (1 - t)b \geq tf(x) + (1 - t)f(y) \geq f(tx + (1 - t)y).$$

Hence

$$(tx + (1 - t)y, ta + (1 - t)b) \in \Gamma_+(f).$$

□

9.14.3 Proposition. Let E be a vector space over $\mathbb{R}$, U be a convex subset of E , and $f : U \rightarrow \mathbb{R}$ be a mapping. The following conditions are equivalent:

- (1) f is concave,

(2) for any $(x, y) \in U \times U$ and $t \in [0, 1]$, one has

$$f(tx + (1 - t)y) \geq tf(x) + (1 - t)f(y).$$

In particular, f is concave if and only if $-f$ is convex.

Proof. The proof is quite similar to that of the previous proposition. We omit the detail. $\square$

9.14.4 Proposition. Let E be a vector space over $\mathbb{R}$, U be a convex subset of E and $f : U \rightarrow \mathbb{R}$ be a mapping. Suppose that $(f_i)_{i \in I}$ is a family of linear forms on E and $(c_i)_{i \in I}$ is a family of real numbers such that,

$$\forall p \in U, f(p) = \sup_{i \in I} (f_i(p) + c_i).$$

Then f is convex.

Proof. Let x and y be elements of U and $t \in [0, 1]$. One has

$$\forall i \in I, f_i(tx + (1 - t)y) + c_i = t(f_i(x) + c_i) + (1 - t)(f_i(y) + c_i) \leq tf(x) + (1 - t)f(y).$$

Taking the supremum with respect to i , we obtain

$$f(tx + (1 - t)y) \leq tf(x) + (1 - t)f(y).$$

$\square$

9.14.5 Proposition. Let $(E, \|\cdot\|)$ be a normed vector space over $\mathbb{R}$ and U be an open convex subset of E , and $f : U \rightarrow \mathbb{R}$ be a mapping. Assume that f is differentiable on U . Then, f is convex if and only if

$$\forall (p, x) \in U, f(x) \geq f(p) + Df(p)(x - p).$$

Moreover, when f is convex, for any $x \in U$, one has

$$f(x) = \sup_{p \in U} (f(p) + Df(p)(x - p)).$$

Proof. For $p \in U$, let

$$g_p : U \longrightarrow \mathbb{R}, \quad g_p(x) := f(p) + Df(p)(x - p) = Df(p)(x) + f(p) - Df(p)(p).$$

Then $f(p) = g_p(p)$. If, for any $p \in U$ one has

$$\forall x \in U, f(x) \geq g_p(x),$$

then one has

$$f = \sup_{p \in U} g_p.$$

Therefore, by the previous proposition we obtain that f is convex.

Conversely, we assume that f is convex. For any $(p, x) \in U \times U$ and any $t \in [0, 1]$, one has

$$f(tx + (1-t)p) = f(p + t(x-p)) \leq tf(x) + (1-t)f(p) = f(p) + t(f(x) - f(p)).$$

Since f is differentiable at p , one has

$$f(p + t(x-p)) = f(p) + tDf(p)(x-p) + o(|t|).$$

Hence

$$f(x) - f(p) \geq Df(p)(x-p) + o(1).$$

Taking the limit when $t \rightarrow 0$, we get

$$f(x) - f(p) \geq Df(p)(x-p).$$

□

9.14.6 Definition. Let E be a vector space over $\mathbb{R}$. We call a *bilinear form* on E any bilinear mapping from $E \times E$ to $\mathbb{R}$. Let $\varphi : E \times E \rightarrow \mathbb{R}$ be a symmetric bilinear form. If

$$\forall x \in E, \varphi(x, x) \geq 0$$

we say that φ is *semipositive*. If

$$\forall x \in E \setminus \{0\}, \varphi(x, x) > 0,$$

then we say that φ is *positive definite*.

9.14.7 Example. The following symmetric bilinear form on $\mathbb{R}^n$ is positive definite:

$$((x_1, \dots, x_n), (y_1, \dots, y_n)) \mapsto \sum_{i=1}^n x_i y_i.$$

9.14.8 Definition. Let E be a vector space over $\mathbb{R}$ and $\varphi : E \times E \rightarrow \mathbb{R}$ be a symmetric bilinear form on E . We call *kernel* of φ the set of $x \in E$ such that

$$\forall y \in E, \varphi(x, y) = 0.$$

We denote by $\text{Ker}(\varphi)$ the kernel of φ .

Note that the kernel of φ is the intersection of

$$\text{Ker}(\varphi(\cdot, y)), y \in E.$$

It is a vector subspace of E .

We call the *isotropic cone* of φ the set of $x \in E$ such that $\varphi(x, x) = 0$. Clearly the kernel of φ is contained in the isotropic cone of φ .

9.14.9 Proposition. *Let E be a vector space over $\mathbb{R}$ and $\varphi : E \times E \rightarrow \mathbb{R}$ be a symmetric bilinear form on E . If φ is semipositive, then the kernel of φ is equal to its isotropic cone.*

Proof. It suffices to check that any element y of the isotropic cone is contained in the kernel of φ . Let x be an arbitrary element of E . For any $t \in \mathbb{R}$ one has

$$\varphi(x + ty, x + ty) = \varphi(x, x) + 2t\varphi(x, y) + t^2\varphi(y, y) \geq 0.$$

Since $\varphi(y, y) = 0$, we obtain

$$\forall t \in \mathbb{R}, \varphi(x, x) + 2t\varphi(x, y) \geq 0.$$

Replacing t by $-t$ and then taking the product of the two inequalities, we get

$$\forall t \in \mathbb{R}, (\varphi(x, x) + 2t\varphi(x, y))(\varphi(x, x) - 2t\varphi(x, y)) = \varphi(x, x)^2 - 4t^2\varphi(x, y)^2 \geq 0$$

Dividing the two sides of the inequality by t^2 and then taking the limit when $|t| \rightarrow 0$, we get $-\varphi(x, y)^2 \geq 0$. Therefore, one has $\varphi(x, y) = 0$. □

9.14.10 Theorem (Cauchy-Schwarz). *Let E be a vector space over $\mathbb{R}$ and $\varphi : E \times E \rightarrow \mathbb{R}$ be a semipositive bilinear form. For any $(x, y) \in E \times E$, one has*

$$\varphi(x, y)^2 \leq \varphi(x, x)\varphi(y, y).$$

The equality holds if and only if the vector subspace of $E/\text{Ker}(\varphi)$ generated by the equivalence classes of x and y is of dimension at most 1.

Proof. We first show that, if the classes of x and y are colinear in $E/\text{Ker}(\varphi)$, then the equality

$$\varphi(x, y)^2 = \varphi(x, x)\varphi(y, y)$$

holds. In fact, there exists $h \in E$ and $(a, b) \in \mathbb{R} \times \mathbb{R}$ such that $\{x - ah, y - bh\} \subseteq \text{Ker}(\varphi)$. Note that

$$\varphi(x, y) = \varphi((x - ah) + ah, (y - bh) + bh) = \varphi(ah, bh) = ab\varphi(h, h).$$

Similarly

$$\varphi(x, x) = \varphi(ah, ah) = a^2\varphi(h, h), \quad \varphi(y, y) = \varphi(bh, bh) = b^2\varphi(h, h).$$

Hence

$$\varphi(x, y)^2 = \varphi(x, x)\varphi(y, y)$$

We have seen in the previous proposition that, if $\varphi(y, y) = 0$, then y belongs to the kernel of φ , and hence $\varphi(x, y) = \varphi(y, y) = 0$. Moreover, in this case $[y] = 0$ in $E/\text{Ker}(\varphi)$ and hence the classes of x and y are colinear in $E/\text{Ker}(\varphi)$.

In the following, we treat the case where $\varphi(y, y) \neq 0$. For any $t \in \mathbb{R}$, one has

$$\varphi(x + ty, x + ty) = t^2\varphi(y, y) + 2t\varphi(x, y) + \varphi(x, x) \geq 0.$$

In the particular case where $t = -\varphi(x, y)/\varphi(y, y)$, we get

$$\frac{\varphi(x, y)^2}{\varphi(y, y)} - 2\frac{\varphi(x, y)^2}{\varphi(y, y)} + \varphi(x, x) \geq 0,$$

which leads to

$$\varphi(x, y)^2 \leq \varphi(x, x)\varphi(y, y).$$

Moreover, if the equality holds, then $\varphi(x + ty, x + ty) = 0$ for the choice of $t = -\varphi(x, y)/\varphi(y, y)$ and hence $x + ty \in \text{Ker}(\varphi)$. Therefore, the classes of x and y in $E/\text{Ker}(\varphi)$ are colinear. $\square$

9.14.11 Theorem. *Let $(E, \|\cdot\|)$ be a normed vector space over $\mathbb{R}$, U be an open and convex subset of E , and $f : U \rightarrow \mathbb{R}$ be a mapping. If f is second order differentiable and $D^2f(p)$ is semipositive for any $p \in U$, then f is convex.*

Proof. Let p and x be two elements of U . Consider the mapping

$$g : [0, 1] \longrightarrow \mathbb{R}, \quad g(t) = f(tx + (1 - t)p) = f(p + t(x - p)).$$

One has

$$g'(t) = Df(p + t(x - p))(x - p)$$

and

$$g''(t) = D^2f(p + t(x - p))(x - p, x - p) \geq 0.$$

By Taylor-Lagrange formula, there exists $\xi \in]0, 1[$ such that

$$g(1) - g(0) = g'(0) + \frac{1}{2}g''(\xi) \geq g'(0) = Df(p)(x - p).$$

Hence we obtain

$$f(x) - f(p) \geq Df(p)(x - p).$$

By Proposition 9.14.5, we obtain that f is convex. $\square$

9.14.12 Example. The mapping $\exp : \mathbb{R} \rightarrow \mathbb{R}$ is convex since $\exp'' = \exp$ takes non-negative values. Moreover, the mapping $\ln : \mathbb{R}_{>0} \rightarrow \mathbb{R}$ is also convex since $\ln'(x) = 1/x > 0$ for any $x \in \mathbb{R}_{>0}$

Let $n \in \mathbb{N}$. The mapping $f_n : \mathbb{R} \rightarrow \mathbb{R}$, $x \mapsto x^n$ is convex on $\mathbb{R}$ when n is even or $n \leq 1$. In fact, one has $f_n'' = n(n-1)f_{n-2}$ when $n \geq 3$. Moreover, when $n \leq 2$, f_n'' is a constant mapping, which takes 0 as its value when $n \leq 1$ and takes 2 as its value when $n = 2$.

Let E be a vector space over $\mathbb{R}$ and $\varphi : E \times E \rightarrow \mathbb{R}$ be a symmetric bilinear form on E . Let $q : E \rightarrow \mathbb{R}$ be the associated quadratic form, which is defined as

$$\forall x \in E, q(x) := \varphi(x, x).$$

If the symmetric bilinear form φ is semipositive, then the quadratic form q is convex. In fact, for any $(x, y) \in E \times E$ and any $t \in [0, 1]$, one has

$$\begin{aligned} q(x) - q(p) - 2\varphi(p, x - p) &= \varphi(x, x) - \varphi(p, p) - 2\varphi(p, x) + 2\varphi(p, p) \\ &= \varphi(x, x) - 2\varphi(p, x) + \varphi(p, p) = \varphi(x - p, x - p) \geq 0. \end{aligned}$$

Therefore, for any $x \in E$,

$$q(x) = \sup_{p \in E} (q(p) + 2\varphi(p, x - p)),$$

which shows that q is a convex mapping.

Exercises

1. State whether the following statements are true or false.

- (1) $o(g(x)) \pm o(g(x)) = o(g(x))$, $x \rightarrow x_0$;
- (2) $O(g_1(x)) \cdot o(g_2(x)) = o(g_1(x) \cdot g_2(x))$, $x \rightarrow x_0$;
- (3) $o(O(g(x))) = o(g(x))$, $x \rightarrow x_0$;
- (4) $O(o(g(x))) = o(g(x))$, $x \rightarrow x_0$;
- (5) $o(x) = O(x)$, $o(x^2)/x = o(x)$, $x \rightarrow x_0$;
- (6) $O(g_1(x)) \cdot O(g_2(x)) = O(g_1(x) \cdot g_2(x))$, $x \in I$;
- (7) $o(g_1(x)) \cdot o(g_2(x)) = o(g_1(x) \cdot g_2(x))$, $x \rightarrow x_0$;
- (8) $o(o(g(x))) = o(g(x))$, $x \rightarrow x_0$;
- (9) $O(O(g(x))) = O(g(x))$, $x \rightarrow x_0$.

2. For all $\alpha > 0$, prove that $\exp\left(\frac{\ln B}{\ln \ln B}\right) = O(B^\alpha)$ when $B \geq 3$.

3. In this exercise, we denote by $\mathbb{C}[[T]]$ the set of formal power series with coefficients in $\mathbb{C}$. Remind that, as a set, $\mathbb{C}[[T]]$ identifies with $\mathbb{C}^{\mathbb{N}}$, the set of all sequences in $\mathbb{C}$ parametrized by $\mathbb{N}$. Moreover, a sequence $(a_n)_{n \in \mathbb{N}}$ in $\mathbb{C}^{\mathbb{N}}$, which is viewed as a formal power series, is written formally as

$$\sum_{n \in \mathbb{N}} a_n T^n.$$

We equip $\mathbb{C}[[T]]$ with an additive composition law and a multiplicative composition law as follows:

$$\begin{aligned} \left(\sum_{n \in \mathbb{N}} a_n T^n \right) + \left(\sum_{n \in \mathbb{N}} b_n T^n \right) &= \sum_{n \in \mathbb{N}} (a_n + b_n) T^n, \\ \left(\sum_{n \in \mathbb{N}} a_n T^n \right) \left(\sum_{n \in \mathbb{N}} b_n T^n \right) &= \sum_{n \in \mathbb{N}} \left(\sum_{\substack{(\ell, m) \in \mathbb{N}^2 \\ \ell + m = n}} a_\ell b_m \right) T^n. \end{aligned}$$

Recall that $\mathbb{C}[[T]]$ forms a commutative unitary ring under these composition laws. Moreover, the mapping

$$\mathbb{C} \times \mathbb{C}[[T]] \longrightarrow \mathbb{C}[[T]], \quad \left(\lambda, \sum_{n \in \mathbb{N}} a_n T^n \right) \longmapsto \sum_{n \in \mathbb{N}} (\lambda a_n) T^n$$

equip $\mathbb{C}[[T]]$ with a structure of vector space over $\mathbb{R}$. We denote by $D : \mathbb{C}[[T]] \rightarrow \mathbb{C}[[T]]$ the formal derivative mapping defined as

$$D \left(\sum_{n \in \mathbb{N}} a_n T^n \right) = \sum_{n \in \mathbb{N}} (n+1) a_{n+1} T^n.$$

For any $R \in]0, +\infty]$, let $\mathbb{U}_R$ be the set of formal power series

$$\sum_{n \in \mathbb{N}} a_n T^n$$

such that

$$\forall r \in [0, R[, \quad \sum_{n \in \mathbb{N}} |a_n| r^n < +\infty.$$

In the remaining of the exercise, we fix a positive real number R .

- (1) Show that D is an $\mathbb{C}$ -linear mapping, which satisfy the following Leibniz rule:

$$\forall (\varphi, \psi) \in \mathbb{C}[[T]], \quad D(\varphi\psi) = \varphi D(\psi) + D(\varphi)\psi.$$

- (2) Prove that $\mathbb{U}_R$ is an $\mathbb{C}$ -vector subspace and a subring of $\mathbb{C}[[T]]$.
 (3) Prove that, for any $\varphi \in \mathbb{U}_R$, one has $D(\varphi) \in \mathbb{U}_R$.
 (4) Let

$$\varphi = \sum_{n \in \mathbb{N}} a_n T^n \in \mathbb{U}_R.$$

Prove that, for any $t \in]-R, R[$, the series

$$\sum_{n \in \mathbb{N}} a_n t^n$$

converges absolutely. We denote by $f_\varphi(t)$ the limit of this series.

- (5) Let $\varphi \in \mathbb{U}_R$. Prove that $f_\varphi :]-R, R[\rightarrow \mathbb{C}$ is a continuous mapping.
 (6) Let $\varphi \in \mathbb{U}_R$. Prove that the mapping $f_\varphi(\cdot)$ is differentiable on any point of $]-R, R[$ and the following equality holds

$$f'_\varphi = f_{D(\varphi)}.$$

- (7) For any $t \in \mathbb{R}$, let $\exp(t)$ be the limit of the following series

$$\sum_{n \in \mathbb{N}} \frac{t^n}{n!}.$$

Prove that the mapping $\exp(\cdot)$ is differentiable on $\mathbb{R}$ and that $\exp' = \exp$.

- (8) Recall that we have shown that $\exp(\cdot)$ is a strictly increasing mapping from $\mathbb{R}$ to $\mathbb{R}_{>0}$. Let $\ln : \mathbb{R}_{>0} \rightarrow \mathbb{R}$ be the inverse mapping of $\exp$. Show that $\ln$ is differentiable and that

$$\forall x \in \mathbb{R}_{>0}, \quad \ln'(x) = \frac{1}{x}.$$

- (9) Prove that the function

$$\mathbb{R}^\times \longrightarrow \mathbb{R}, \quad x \longmapsto \ln |x|$$

is differentiable at any point of $\mathbb{R}^\times$ and its derivative is $x \mapsto 1/x$.

- (10) Let $a \in \mathbb{R}$. Denote by $f_a : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ the mapping sending $t \in \mathbb{R}_{>0}$ to $t^a := \exp(a \ln(t))$. Show that f_a is differentiable and

$$f'_a(t) = at^{a-1}.$$

- (11) Let $b > 0$. Denote by $g_b : \mathbb{R} \rightarrow \mathbb{R}_{>0}$ the mapping sending $t \in \mathbb{R}$ to $b^t := \exp(t \ln(b))$. Prove that g_b is differentiable on $\mathbb{R}$ and

$$g'_b(t) = b^t \ln(b).$$

- (12) Let ω be a complex number. Prove that the mapping from $\mathbb{R}$ to $\mathbb{C}$ sending t to $\exp(\omega t)$ is differentiable and determine its derivative. One can show that the mapping from $\mathbb{R}$ to $\mathbb{C}$ sending $t \in \mathbb{R}$ to $\omega t \in \mathbb{C}$ is a bounded $\mathbb{R}$ -linear mapping.

- (13) Prove that, for any $t \in \mathbb{R}$, one has

$$\cos(t) = \frac{\exp(it) + \exp(-it)}{2}, \quad \sin(t) = \frac{\exp(it) - \exp(-it)}{2i}.$$

- (14) Prove that the mappings $\sin : \mathbb{R} \rightarrow \mathbb{R}$ and $\cos : \mathbb{R} \rightarrow \mathbb{R}$ are differentiable. Moreover, the following equality holds:

$$\sin' = \cos, \quad \cos' = -\sin.$$

4. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable mapping. We suppose that there exists $\ell \in \mathbb{R}$ such that

$$\lim_{x \rightarrow -\infty} f(x) = \lim_{x \rightarrow +\infty} f(x) = \ell.$$

- (1) Prove that there exists $(a, b, c) \in \mathbb{R}^3$ such that $a < b < c$ and that

$$(f(b) - f(a))(f(b) - f(c)) \geq 0.$$

- (2) Deduce that there exists $\xi \in \mathbb{R}$ such that $f'(\xi) = 0$.

5. Let $(a, b) \in \mathbb{R}^2$ such that $a < b$. Let $f, g : [a, b] \rightarrow \mathbb{R}$ be continuous mapping that is differentiable on $]a, b[$. We assume that $g'(x) \neq 0$ for any $x \in]a, b[$.

- (1) Denote by α the real number

$$\frac{f(b) - f(a)}{g(b) - g(a)}.$$

Prove that $h = f - \alpha g$ is continuous on $[a, b]$ and differentiable on $]a, b[$. Verify that $h(b) = h(a)$.

- (2) Prove that there exists $\xi \in]a, b[$ such that

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(\xi)}{g'(\xi)}.$$

6. We denote by $\iota : \mathbb{R}^\times \rightarrow \mathbb{R}^\times$ the mapping sending $x \in \mathbb{R}^\times$ to x^{-1} .

(1) Let $p \in \mathbb{R}^\times$. Prove that

$$\iota(x) - \iota(p) = -\frac{\iota(x)}{p}(x - p) = -\frac{1}{p^2}(x - p) + \frac{\iota(x)}{p^2}(x - p)^2.$$

(2) Prove that ι is differentiable on $\mathbb{R}^\times$ and

$$\iota'(x) = -\frac{1}{x^2}.$$

(3) Let I be an open interval and $f : I \rightarrow \mathbb{R}^\times$. Prove that, if f is differentiable at some point $x \in I$, then $1/f$ is also differentiable at x , and its derivative at x is equal to

$$-\frac{f'(x)}{f(x)^2}.$$

(4) Let n be an integer and $f_n : \mathbb{R}^\times \rightarrow \mathbb{R}^\times$ be the mapping sending $x \in \mathbb{R}^\times$ to x^n . Show that

$$f'_n(x) = nx^{n-1}.$$

(5) For any $t \in \mathbb{R} \setminus \{(n + \frac{1}{2})\pi \mid n \in \mathbb{Z}\}$, let

$$\tan(t) := \frac{\sin(t)}{\cos(t)}.$$

Prove that the mapping $\tan(\cdot)$ is differentiable and determine its derivative.

(6) For any $t \in \mathbb{R} \setminus \{n\pi \mid n \in \mathbb{Z}\}$, let

$$\cot(t) = \frac{\cos(t)}{\sin(t)}.$$

Show that the mapping $\cot(\cdot)$ is differentiable and determine its derivative.

7. (1) Prove that the restriction of the mapping $\sin$ to $[-\frac{\pi}{2}, \frac{\pi}{2}]$ is strictly increasing and the image of $[-\frac{\pi}{2}, \frac{\pi}{2}]$ by $\sin$ is $[-1, 1]$.

(2) Let $\arcsin : [-1, 1] \rightarrow [-\frac{\pi}{2}, \frac{\pi}{2}]$ be the inverse mapping of the restriction of $\sin$ to $[-\frac{\pi}{2}, \frac{\pi}{2}]$. Prove that $\arcsin$ is continuous on $[-1, 1]$ and is differentiable on $] -1, 1[$. Determine its derivative.

- (3) Prove that the restriction of the mapping $\cos$ to $[0, \pi]$ is strictly decreasing and the image of $[0, \pi]$ by $\cos$ is $[-1, 1]$.
- (4) Let $\arccos : [-1, 1] \rightarrow [0, \pi]$ be the inverse mapping of the restriction of $\cos$ to $[0, \pi]$. Prove that $\arccos$ is continuous on $[-1, 1]$ and is differentiable on $] -1, 1[$. Determine its derivative.
- (5) Prove that the restriction of $\tan$ to $] -\frac{\pi}{2}, \frac{\pi}{2}[$ is strictly increasing and its image is $\mathbb{R}$.
- (6) Let $\arctan : \mathbb{R} \rightarrow] -\frac{\pi}{2}, \frac{\pi}{2}[$ be the inverse of the restriction of $\tan$ to $] -\frac{\pi}{2}, \frac{\pi}{2}[$. Prove that the mapping $\arctan$ is differentiable and determine its derivative.
8. Let a and b be positive real numbers such that $a < b$. Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous mapping that is differentiable on $]a, b[$.

- (1) Prove that there exists $\xi \in]a, b[$ such that

$$\frac{f(b) - f(a)}{\ln(b) - \ln(a)} = \xi f'(\xi).$$

- (2) Let x be a real number such that $x > 1$. Prove that, for any $n \in \mathbb{N}_{\geq 1}$, there exists $\xi_n \in]1, x[$ such that

$$x^{1/n} - 1 = \frac{1}{n} \xi_n^{1/n} \ln(x).$$

- (3) Prove that

$$\lim_{n \rightarrow +\infty} n(x^{1/n} - 1) = \ln(x).$$

9. For any $n \in \mathbb{N}_{\geq 1}$, let $f_n : [0, 1] \rightarrow [0, 1]$ be the mapping sending $t \in [0, 1]$ to t^n .

- (1) Show that the sequence $(f_n)_{n \in \mathbb{N}_{\geq 1}}$ converges simply to some mapping. Determine the limit mapping.
- (2) Is the above convergence uniform?

10. Let X be a topological space and $(E, \|\cdot\|)$ be a Banach space. Let I be an infinite subset of $\mathbb{N}$ and $(f_n)_{n \in \mathbb{N}}$ be a sequence of mappings from X to E . For any $n \in \mathbb{N}$, let

$$\|f_n\|_{\sup} := \sup_{x \in X} \|f_n(x)\| \in [0, +\infty].$$

We say that the series $\sum_{n \in I} f_n$ converges normally if

$$\sum_{n \in I} \|f_n\|_{\sup} < +\infty.$$

- (1) Suppose that the series $\sum_{n \in I} f_n$ converges normally. Prove that $\sum_{n \in I} f_n$ converges uniformly.
- (2) For any $n \in \mathbb{N}_{\geq 1}$ and any $x \in \mathbb{R}$, let

$$f_n(x) = \frac{nx}{1 + n^5 x^2}.$$

Prove that the series $\sum_{n \in \mathbb{N}_{\geq 1}} f_n$ converges normally.

- 11.** (1) Let a be a positive real number. Prove that

$$\frac{1}{a+1} < \ln(a+1) - \ln(a) < \frac{1}{a}.$$

- (2) Prove that the function $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ defined as

$$f(x) = \left(1 + \frac{1}{x}\right)^x$$

is increasing.

- (3) Prove that the function $g : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ defined as

$$g(x) = \left(1 + \frac{1}{x}\right)^{x+1}$$

is decreasing.

- (4) Determine

$$\lim_{x \rightarrow +\infty} f(x) \quad \text{and} \quad \lim_{x \rightarrow +\infty} g(x)$$

- 12.** Consider the Dirichlet function $f : \mathbb{R} \rightarrow \{0, 1\}$ defined by

$$D(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \in \mathbb{Q}^c. \end{cases}$$

- (a) Prove that D is not differentiable at any $x \in \mathbb{R}$.
- (b) Let $g : \mathbb{R} \rightarrow \mathbb{R}$ defined by $g(x) = x^2 D(x)$. Prove that g is differentiable at 0 and $g'(0) = 0$.

- 13.** Let $a \in \mathbb{R}$ with $a > 0$. Prove that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = |x|^a$ is differentiable at 0 if and only if $a > 1$.

- 14.** Suppose that $f, g, h : X \rightarrow \mathbb{R}$ are functions satisfying the following conditions:

- (a) $f(a) = g(a) = h(a)$, $f'(a) = g'(a)$;
- (b) $f(x) \leq h(x) \leq g(x)$, for any $x \in X$.

Prove that h is differentiable at a and $h'(a) = f'(a) = g'(a)$.

15. Suppose that $I \subseteq \mathbb{R}$ is an open interval and $0 \in I$. Let $f : I \rightarrow \mathbb{R}$ be a function which is continuous at 0 and $f(0) = 0$. Prove that if

$$\lim_{x \rightarrow 0} \frac{f(2x) - f(x)}{x} = A \in \mathbb{R},$$

then f is differentiable at 0 and $f'(0) = A$.

16. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by

$$f(x) = \begin{cases} x^{2n} \sin \frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$$

Prove that at $x = 0$, $f'(0), f''(0), \dots, f^{(n)}(0)$ exist, but $f^{(n+1)}(0)$ does not exist.

17. Let $f : X \rightarrow \mathbb{R}$ be a function. Determine whether the following statements are true or false. If true, please provide a proof; if false, give a counterexample.

- (a) If f is differentiable at $x_0 \in X$, then f is continuous at x_0 .
- (b) If f is differentiable at x_0 , then $|f|$ is also differentiable at x_0 .
- (c) If f is differentiable at x_0 , then f is differentiable on a neighborhood of x_0 .
- (d) If f is differentiable on an interval $I \subseteq X$, then f' is also continuous on I .

18. Let f be a real-valued function defined on a neighborhood of $x_0 \in \mathbb{R}$, and $\alpha \in \mathbb{R}$. We say that f satisfies the *Lipschitz condition* of order α at x_0 if there exist $M, \delta > 0$ such that

$$|f(x) - f(x_0)| < M|x - x_0|^\alpha, \quad \forall x \in (x_0 - \delta, x_0 + \delta).$$

- (a) Prove that if f satisfies the Lipschitz condition of order α at x_0 for $\alpha > 0$, then f is continuous at x_0 .
- (b) Prove that if f satisfies the Lipschitz condition of order α at x_0 for $\alpha > 1$, then f is differentiable at x_0 and $f'(x_0) = 0$.

- (c) Find a function f which satisfies the Lipschitz condition of order α , and is continuous at x_0 but not differentiable at x_0 .

19. Let f be a function defined on a neighborhood of $x_0 \in \mathbb{R}$. We define the *symmetric derivative* of f at x_0 by

$$f'_s(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0 - h)}{2h}$$

provided the limit exists.

- (a) Prove that if f has a right-hand derivative and a left-hand derivative at x_0 , then it has a symmetric derivative at x_0 . Calculate $f'_s(x_0)$ in terms of left-hand and right-hand derivatives at x_0 .
- (b) Prove that the function f defined by

$$f(x) = \begin{cases} x \sin \frac{1}{x} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

has a symmetric derivative at 0, but does not have left-hand or right-hand derivatives at 0.

- (c) If $f : [a, b] \rightarrow \mathbb{R}$ is an increasing function and has a symmetric derivative at every point of this interval, then the symmetric derivative is positive.
- (d) If a function has a local extreme at a point x_0 , can one conclude that $f'_s(x_0) = 0$? If yes, please prove your statement; if not, provide a counterexample.
- 20.** (a) Suppose that $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function which satisfies $f(0) = 0$ and $f'(0)$ exists. Let (x_n) be a sequence defined by

$$x_n = f\left(\frac{1}{n^2}\right) + f\left(\frac{2}{n^2}\right) + \cdots + f\left(\frac{n}{n^2}\right).$$

Calculate $\lim_{n \rightarrow +\infty} x_n$.

- (b) Find the limits of the following sequences.

- i. $\lim_{n \rightarrow +\infty} \sin \frac{1}{n^2} + \sin \frac{2}{n^2} + \cdots + \sin \frac{n}{n^2};$
- ii. $\lim_{n \rightarrow +\infty} \left(1 + \frac{1}{n^2}\right) \left(1 + \frac{2}{n^2}\right) \cdots \left(1 + \frac{n}{n^2}\right).$

- 21.** (a) Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function. Suppose that $x_0, x_1 \in [a, b]$ are two points where local maximums are attained and $x_0 < x_1$. Prove that there exists $\tilde{x} \in (x_0, x_1)$ such that $f(\tilde{x})$ is a local minimum.

- (b) Let $f : [a, b] \rightarrow \mathbb{R}$ be a function that is continuously differentiable and only has one local extreme. Prove that if the local extreme is a local maximum, then it is the global maximum; if the local extreme is a local minimum, then it is the global minimum.
- 22.** Let $f : X \rightarrow \mathbb{R}$ be a differentiable function. Prove that
- (a) if f is an even function, then f' is an odd function;
 - (b) if f is an odd function, then f' is an even function;
 - (c) if f is a periodic function with period T , so is f' .
- 23.** Calculate the derivatives of the following functions.
- (a) $f(x) = \sin^2((x + \sin x)^6)$;
 - (b) $f(x) = \log \frac{\sqrt{1+x^3}-1}{\sqrt{1+x^3}+1}$
 - (c) $f(x) = \arcsin(\cos x^2) + \arccos(\sin x^2)$;
 - (d) $f(x) = x^{\frac{1}{x}}$;
 - (e) $f(x) = (\log x)^x + x^{\alpha \log x}$;
 - (f) $f(x) = \sin \left(\frac{x^3}{\sin(\frac{x^3}{\sin x})} \right)$.
- 24.** Find f' in terms of g' if
- (a) $f(x) = g(x + g(a))$;
 - (b) $f(x) = g(x \cdot g(a))$;
 - (c) $f(x) = g(x + g(x))$;
 - (d) $f(x) = g(x)(x - a)$;
 - (e) $f(x) = g(a)(x - a)$;
 - (f) $f(x + 3) = g(x^2)$.
- 25.** Let $X \subseteq \mathbb{R}$. Suppose that a function $f : X \rightarrow \mathbb{R}$ is differentiable and satisfy the equation
- $$f(x)^x = x^{f(x)},$$
- for any $x \in X$. Calculate $f'(x)$.
- 26.** (a) Prove that the function $f : [-1, 1] \rightarrow \mathbb{R}$ defined by $f(x) = \arcsin x$ satisfies that

$$\arcsin^{(n)}(0) = \begin{cases} 0 & \text{if } n = 2k \\ ((2k-1)(2k-3) \cdots 3 \cdot 1)^2 & \text{if } n = 2k+1. \end{cases}$$

- (b) Prove that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} e^{-\frac{1}{x^2}} & x \neq 0 \\ 0 & x = 0. \end{cases}$$

is infinitely differentiable and

$$f^{(n)}(0) = 0, \quad \forall n \in \mathbb{N}.$$

- 27.** Use Leibniz rule for $(x-a)^n(x-b)^n$ to prove that

$$\binom{n}{0}^2 + \binom{n}{1}^2 + \cdots + \binom{n}{n}^2 = \frac{2n(2n-1) \cdots (n+1)}{n!}.$$

- 28.** The following problem concerns properties about hyperbolic and inverse hyperbolic functions.

- (a) The inverse function of $\sinh x$ is given by $\operatorname{arsinh} y = \ln(y + \sqrt{1+y^2})$, $\forall y \in \mathbb{R}$;
- (b) The functions inverse to $\cosh x$ on $\mathbb{R}_+$ and $\mathbb{R}_-$ are respectively given by $\operatorname{arcosh}_+ y = \ln(y + \sqrt{y^2-1})$ and $\operatorname{arcosh}_- y = \ln(y - \sqrt{y^2-1})$, $\forall y \geq 1$;
- (c) $\operatorname{arsinh}' y = \frac{1}{\sqrt{1+y^2}}$, $\forall y \in \mathbb{R}$;
- (d) $\operatorname{arcosh}' y = -\frac{1}{\sqrt{y^2-1}}$, $\forall y \geq 1$;
- (e) $\operatorname{artanh} y = \frac{1}{2} \ln \frac{1+y}{1-y}$, $\forall |y| < 1$;
- (f) $\operatorname{arcoth} y = \frac{1}{2} \ln \frac{y+1}{y-1}$, $\forall |y| > 1$;
- (g) $\operatorname{artanh}' y = \frac{1}{1-y^2}$, $\forall |y| < 1$;
- (h) $\operatorname{arcoth}' y = -\frac{1}{y^2-1}$, $\forall |y| > 1$;
- (i) $\sinh(x+y) = \sinh x \cdot \cosh y + \cosh x \cdot \sinh y$;
- (j) $\cosh(x+y) = \cosh x \cdot \cosh y + \sinh x \cdot \sinh y$.

- 29.** (Lorentz transform) Fix $c, v \geq 0$. Let $\alpha, \gamma, \delta \in \mathbb{R}$ and $\beta = \alpha v$. Suppose that the equations

$$\begin{pmatrix} \tilde{x} \\ \tilde{t} \end{pmatrix} = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} x \\ t \end{pmatrix}$$

hold for all nonzero functions $x, \tilde{x}, \tilde{t} : [0, \infty) \rightarrow \mathbb{R}$ satisfying

$$x(t)^2 - c^2 t^2 = \tilde{x}(t)^2 - c^2 \tilde{t}(t)^2.$$

Find the the matrix $\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$ in terms of c and v .

30. Let $y(t) = e^t$ and $x(t) = \ln t$. Prove that

$$\frac{dy}{dx} = te^t \quad \text{and} \quad \frac{d^2y}{dx^2} = t(t+1)e^t.$$

31. Assume that f is C^∞ on an open interval $I \subset \mathbb{R}$ and $f'(x_0) \neq 0$ for a point $x_0 \in I$. Then there exists a neighborhood U of x_0 and a neighborhood V of $f(x_0)$ such that f has an inverse f^{-1} on U and $f^{-1} : V \rightarrow U$ is also C^∞ .

32. (a) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable function, and $\lim_{x \rightarrow +\infty} f(x) = \lim_{x \rightarrow -\infty} f(x)$ (equal to either a finite number or $\pm\infty$). Prove that there exists $\xi \in \mathbb{R}$ such that $f'(\xi) = 0$.

(b) Let $f : [a, b] \rightarrow \mathbb{R}$ be a function satisfying

- i. f is continuous on $]a, b[$;
- ii. f is differentiable on (a, b) ;
- iii. $\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow b^-} f(x) = +\infty$.

Prove that for any $A \in \mathbb{R}$, there exists $\xi \in (a, b)$ such that $f'(\xi) = A$.

33. Let $f : [a, b] \rightarrow \mathbb{R}$ be a differentiable function satisfying $f'(a) = f'(b)$. Prove that there exists $\xi \in (a, b)$ such that

$$f'(\xi) = \frac{f(\xi) - f(a)}{\xi - a}.$$

34. Let $f : [a, b] \rightarrow \mathbb{R}$ be a function and $\alpha > 0$.

- (a) If f is continuously differentiable on $[a, b]$, then f is Lipschitz of order 1.
- (b) If f is differentiable on $[a, b]$ and is Lipschitz of order 1, then f' is bounded on $[a, b]$.
- (c) If f is Lipschitz of order α for $\alpha > 1$, show that f is constant on $[a, b]$.

35. (a) Let f be a real-valued function, defined and continuous on the closed interval $[a, b]$, differentiable at every point of the open interval (a, b) , except possibly at a point $x_0 \in (a, b)$. Prove that if the derivative f' has a limit at the point x_0 , then the function f is differentiable at x_0 and

$$f'(x_0) = \lim_{x \rightarrow x_0} f'(x).$$

- (b) Let $n \in \mathbb{N}$, and let $f : I \setminus \{a\} \rightarrow \mathbb{R}$ be a function of class C^n . If, for all $k \in [0, n]$, the derivative $f^{(k)}$ has a finite limit α_k at a , then f can be uniquely extended to $\tilde{f} : I \rightarrow \mathbb{R}$ such that $\tilde{f}$ is of class C^n .

- 36.** Let $f : [a, \infty) \rightarrow \mathbb{R}$ be a differentiable function. If $\lim_{x \rightarrow +\infty} f(x) + f'(x) = l$. Prove that $\lim_{x \rightarrow +\infty} f(x) = l$. (Hint: assume without loss of generality that $l = 0$, and consider $g(x) := e^x f(x)$ for which one invokes Cauchy mean-value theorem.)

- 37.** Find all the functions that satisfy the following conditions:

- (a) $f(1) = 1, f'(0) > 0$;
 (b) $\forall t \in \mathbb{R}, f'(t) \cdot f'(f(t)) = 1$.

(Hint: use Darboux theorem.)

- 38.** (a) Let I be a closed interval, $K \in [0, 1)$, and $f : I \rightarrow I$ be a K -Lipschitz function (i.e. $|f(x) - f(y)| \leq K|x - y|$, $\forall x, y \in I$). Prove that any sequence $(u_n)_{n \in \mathbb{N}}$ defined by

$$u_0 \in I, \quad u_{n+1} = f(u_n) \quad \forall n \in \mathbb{N},$$

converges to a fixed point of f . (Hint: consider the cases where $I = [a, b]$, $I = [a, +\infty)$, $I = (-\infty, a]$, and $I = \mathbb{R}$.)

- (b) Let $(u_n)_{n \in \mathbb{N}}$ be the sequence defined by

$$u_0 \in [0, 1], \quad u_{n+1} = 1 - \frac{u_n^2}{4}.$$

Does the sequence converge? If yes, please provide a proof and find the limit; if not, explain why.

- 39.** Let f be a function of class C^n defined on I , which vanishes at $x_1 < \dots < x_n$, with $n > 1$. Prove that for all $x \in I$, there exists $\xi \in I$ such that

$$f(x) = \frac{(x - x_1) \cdots (x - x_n)}{n!} f^{(n)}(\xi).$$

(Hint: for $x \in I \setminus \{x_1, \dots, x_n\}$, consider $\varphi : t \mapsto f(t) - A(t - x_1) \cdots (t - x_n)$, where $A \in \mathbb{R}$ is chosen so that $\varphi(x) = 0$.)

- 40.** Find the local extrema and global extrema of the following functions:

- (a) $f(x) = x^3 + 2x^2 + x + 1$, $x \in [-2, 2]$;
 (b) $f(x) = 3x^4 - 8x^3 + 6x^2$, $x \in [-1, 1]$;
 (c) $f(x) = \frac{1}{x^5 + x + 1}$, $x \in [-\frac{1}{2}, 1]$;

- (d) $f(x) = \frac{x}{x^2-1}$, $x \in [0, 5]$;
- (e) $f(x) = \begin{cases} 0 & x \notin \mathbb{Q}, \\ x & x \in \mathbb{Q}; \end{cases}$
- (f) $f(x) = \begin{cases} 0 & x \notin \mathbb{Q}, \\ \frac{1}{q} & x = \frac{p}{q} \text{ in lowest terms.} \end{cases}$

41. Prove the following inequalities:

- (a) $\frac{2}{\pi}x < \sin x < x$ for any $x \in (0, \frac{\pi}{2})$;
- (b) $x < \arcsin x < \frac{x}{\sqrt{1-x^2}}$ ($0 < x < 1$);
- (c) $e^x < \frac{1}{1-x}$ ($x < 1$);
- (d) $(x^\alpha + y^\alpha)^{\frac{1}{\alpha}} < (x^\beta + y^\beta)^{\frac{1}{\beta}}$ ($x > 0, y > 0, \beta > \alpha > 0$);
- (e) $|1 + x|^p \geq 1 + px + c_p \varphi_p(x)$, where c_p is a constant depending only on p . If $1 < p \leq 2$,

$$\varphi_p(x) = \begin{cases} |x|^2 & \text{for } |x| \leq 1, \\ |x|^p & \text{for } |x| > 1; \end{cases}$$

and if $p > 2$, $\varphi_p(x) = |x|^p$ on $\mathbb{R}$.

42. (Sufficient conditions for an extremum in terms of higher-order derivatives) Suppose a real-valued function f defined on a neighborhood of $x_0 \in \mathbb{R}$ has derivatives of order up to n inclusive at x_0 ($n \geq 1$). Suppose that $f'(x_0) = \dots = f^{(n-1)}(x_0) = 0$ and $f^{(n)}(x_0) \neq 0$.

- (a) If n is odd, prove that there is no extremum at x_0 .
- (b) If n is even, prove that x_0 is a local extremum. In particular, if $f^{(n)}(x_0) < 0$, x_0 is a local maximum and if $f^{(n)}(x_0) > 0$, x_0 is a local minimum.

43. (a) Let $f : [a, b] \rightarrow \mathbb{R}$ be a function that is differentiable on a neighborhood of a point $x_0 \in (a, b)$. Suppose that f' is continuous at x_0 and $f'(x_0) \neq 0$. Prove that there exists a neighborhood of x_0 such that f is monotonic on this neighborhood.

- (b) If f' is not continuous at x_0 , does the above conclusion still hold? (Hint:

$$\text{consider the function } f(x) = \begin{cases} \frac{x}{2} + x^2 \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0. \end{cases})$$

44. State whether the following functions are convex, concave or neither. If yes, please provide a proof; if not, please explain why.

- (a) $f : [0, \infty) \rightarrow \mathbb{R}$ defined by $f(x) = x^\alpha$ ($\alpha > 0$);
- (b) $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x \sin x^2$;
- (c) $f : (0, \frac{\pi}{2}) \rightarrow \mathbb{R}$ defined by $f(x) = \log \left(\frac{x}{\sin x} \right)$;
- (d) $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} (x-2)^2 & x \in [0, \infty) \\ (x+2)^2 & x \in (-\infty, 0]. \end{cases}$$

45. Let f and g be two convex, positive, and monotonically increasing functions on I .

- (a) Assume f and g are twice differentiable. Show that $f \cdot g$ is convex.
- (b) Show that the result holds in the general case.

46. Let I be an interval and $f : I \rightarrow \mathbb{R}$ a function satisfying

$$f\left(\frac{x+y}{2}\right) \leq \frac{1}{2}(f(x) + f(y)), \quad \forall x, y \in I.$$

- (a) Prove that $\forall k = \frac{m}{2^n} \in [0, 1]$, one has

$$f(kx + (1-k)y) \leq kf(x) + (1-k)f(y), \quad \forall x, y \in I.$$

- (b) If f is continuous on I , then f is convex.

47. Let I be an open interval and $f : I \rightarrow \mathbb{R}$ be a convex function.

- (a) Prove that for any $x \in I$, $\lim_{y \rightarrow x^+} \frac{f(y) - f(x)}{y - x}$ and $\lim_{y \rightarrow x^-} \frac{f(y) - f(x)}{y - x}$ exist, and

$$\lim_{y \rightarrow x^-} \frac{f(y) - f(x)}{y - x} \leq \lim_{y \rightarrow x^+} \frac{f(y) - f(x)}{y - x}.$$

- (b) Let $g, h : I \rightarrow \mathbb{R}$ be functions defined by $g(x) = \lim_{y \rightarrow x^-} \frac{f(y) - f(x)}{y - x}$ and $h(x) = \lim_{y \rightarrow x^+} \frac{f(y) - f(x)}{y - x}$. Prove that g and h are monotonically increasing, and

$$\forall a, b \in I, \quad h(a) \leq \frac{f(b) - f(a)}{b - a} \leq g(b).$$

- 48.** Let I be an open interval and $f : I \rightarrow \mathbb{R}$ a function. Prove that f is convex if and only if f is continuous on I and the left-hand and right-hand derivatives exist and are monotonically increasing on I .
- 49.** (a) Show that a function f is both convex and concave if and only if f is affine.
- (b) Let f be a convex function that is bounded above on $\mathbb{R}$. Show that f is constant. Does the result hold if f is convex and bounded above on $[A, +\infty[$?
- (c) Deduce that if f is a twice-differentiable function on $\mathbb{R}$, bounded and non-constant, then there exist t_1 and t_2 such that $f''(t_1)f''(t_2) < 0$.
- 50.** Find the following limits of functions:
- (a) $\lim_{x \rightarrow 0} \frac{x - \tan x}{x^3}$;
- (b) $\lim_{x \rightarrow 0} \frac{xe^x - \log(1+x)}{x^2}$;
- (c) $\lim_{x \rightarrow 0} \sin x \log x$;
- (d) $\lim_{x \rightarrow 1} \frac{1}{\log x} - \frac{1}{x-1}$;
- (e) $\lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{\frac{1}{x^2}}$;
- (f) $\lim_{x \rightarrow 0} \left(\frac{\cos x}{\cosh x} \right)^{\frac{1}{x^2}}$;
- (g) $\lim_{x \rightarrow \frac{\pi}{2}} (\sin x)^{\tan x}$;
- 51.** (a) Find the shortest distance from the point $(0, h)$ ($h > 0$) to the parabola $y = x^2$.
- (b) For all the isosceles triangles with a fixed perimeter, what is the ratio of the side to the base such that the volume of the solid generated by rotating the triangle about its base is maximized?
- (c) Two rivers intersect perpendicularly, with widths a and b respectively. What is the maximum length the boat can have to transition from one river to the other?
- 52.** (a) Two bodies with masses m_1 and m_2 are moving in space under the action of their mutual gravitation alone. Using Newton's laws (i.e. the general

law of motion and the law of universal gravitation), verify that the quantity

$$E = \left(\frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 \right) + \left(-G\frac{m_1m_2}{r} \right) =: K + U,$$

where v_1 and v_2 are velocities of the bodies and r the distance between them, does not vary during this motion.

(b) Extend this result to the case of the motion of n bodies.

53. (Newton's method) Let α, β be two real numbers such that $\alpha < \beta$, and let $f : [\alpha, \beta] \rightarrow \mathbb{R}$ be a function satisfying the following conditions:

- 1) f is twice continuously differentiable on $[\alpha, \beta]$;
- 2) f' does not vanish on $[\alpha, \beta]$;
- 3) $f(\alpha)f(\beta) < 0$.

(a) Prove that the equation $f(x) = 0$ has a unique root l in $[\alpha, \beta]$.

(b) Let a be an element of the interval (α, β) . Let $\tilde{Q}_1$ be the linear function whose graph is the tangent to the graph of f at a . Show that the root x_1 of the equation $\tilde{Q}_1(x) = 0$ is

$$x_1 = a - \frac{f(a)}{f'(a)}.$$

(c) Let $g : [\alpha, \beta] \rightarrow \mathbb{R}$ be the function defined by

$$g(x) = x - \frac{f(x)}{f'(x)}.$$

i. Show that g is differentiable on (α, β) and that

$$\forall x \in (\alpha, \beta), \quad g'(x) = \frac{f(x)f''(x)}{(f'(x))^2}.$$

ii. Show that there exist two points $a, b \in (\alpha, \beta)$ such that $a < l < b$, $l - a = b - l$, and

$$\forall x \in [a, b], \quad |g'(x)| < 1.$$

Deduce that $g([a, b]) \subseteq [a, b]$.

iii. We construct a sequence defined recursively by

$$x_0 = a, \quad x_{n+1} = g(x_n), \quad \forall n \in \mathbb{N}.$$

Prove that the sequence converges to l .

- (d) Use Newton's method to find the decimal approximation of $\sqrt{2}$, accurate to the third decimal place.

- 54.** (Geometry of trigonometric functions) Let $S, C : \mathbb{R} \rightarrow \mathbb{R}$ be differential functions satisfying

$$\begin{cases} S' = C, & S(0) = 0 \\ C' = -S, & C(0) = 1 \end{cases}$$

Prove the following assertions:

- (a) $S^2 + C^2 = 1$.
 (b) There exists $A > 0$ such that
 i. $S(0) = 0$, $S(A) = 1$ and S is strictly increasing on $[0, A]$;
 ii. $C(0) = 1$, $C(A) = 0$ and C is strictly decreasing on $[0, A]$.

Moreover there exists $B > 0$ such that

- i. $S(A) = 1$, $S(A+B) = 0$ and S is strictly decreasing on $[A, A+B]$;
 ii. $C(0) = 0$, $C(A+B) = -1$ and C is strictly increasing on $[A, A+B]$.
 (c) $\pi := A + B$ is the first zero of S on $\mathbb{R}_{>0}$.
 (d) $S(x) = -S(x + \pi)$, $C(x) = -C(x + \pi)$ and 2π is the smallest period of S and C . Moreover, $A = B = \pi/2$.
 (e) $\sin \theta < \theta < \tan \theta$ for $0 < \theta < \pi/2$.

(We will see in the next chapter that π is the length of the semicircle and θ is the length of the curve from $(0, 1)$ to $(\cos \theta, \sin \theta)$ on the semicircle for $0 \leq \theta \leq \pi$.)

- 55.** Let X be a subset of $\mathbb{R}$ equipped with the subspace topology, and f, g be two bounded functions defined over X and valued in $\mathbb{R}$. For an $x \in X$, define $d(f(x), g(x)) = |f(x) - g(x)|$.

- (a) Show that

$$\sup_{x \in X} d(f(x), g(x))$$

is a metric on the space of bounded functions defined over X .

- (b) Prove that the uniform convergence of a sequence of bounded functions $(f_n)_{n \in \mathbb{N}}$ is equivalent to the convergence of $(f_n)_{n \in \mathbb{N}}$ (as a standard sequence) with respect to the above metric.

56. Find the following limits if they exist.

- | | |
|---|---|
| (1) $\lim_{x \rightarrow \infty, y \rightarrow \infty} \frac{ x + y }{x^2 + y^2}$ | (2) $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x + y}$ |
| (3) $\lim_{(x,y) \rightarrow (0,0)} (x^2 + y^2)^{x^2 y^2}$ | (4) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^6 y^8}{(x^2 + y^4)^5}$ |
| (5) $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}$ | (6) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^3 - y^3}{x^3 + y^3}$ |
| (7) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^4 y^4}{(x^3 + y^6)^2}$ | (8) $\lim_{(x,y) \rightarrow (0,0)} \frac{\arctan(x^3 + y^3)}{x^2 + y^2}$ |
| (9) $\lim_{(x,y) \rightarrow (0,0)} \left(x \sin \frac{1}{y} + y \sin \frac{1}{x} \right)$ | (10) $\lim_{(x,y) \rightarrow (0,0)} \frac{\sin xy}{x}$ |
| (11) $\lim_{(x,y) \rightarrow (0,0)} \frac{\log(x + e^y)}{\sqrt{x^2 + y^2}}$ | (12) $\lim_{(x,y) \rightarrow (0,0)} (x + y) \sin \frac{1}{x} \sin \frac{1}{y}$ |
| (13) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^2 y^2 + (x - y)^2}$ | (14) $\lim_{(x,y) \rightarrow (0,0)} \frac{xy(x^2 - y^2)}{x^2 + y^2}$ |
| (15) $\lim_{(x,y) \rightarrow (0,0)} \frac{\sqrt{xy}}{(x^2 + y^2)^p}, p > 0$ | (16) $\lim_{(x,y) \rightarrow (0,0)} \frac{\sin xy}{\sqrt{x^2 + y^2}}$ |

57. Let $U \subset \mathbb{R}^d$ be a non-empty open convex set. We say $f : U \rightarrow \mathbb{R}$ is convex if and only if for all $x, y \in U$ and all $\lambda \in [0, 1]$ we have

$$f((1 - \lambda)x + \lambda y) \leq (1 - \lambda)f(x) + \lambda f(y).$$

(We say that f is strictly convex if the strictly inequality holds.)

- (1) Suppose f is differentiable. Show that f is convex if and only if for all $x = (x_1, \dots, x_d), y = (y_1, \dots, y_d) \in U$ we have

$$f(y) - f(x) \geq \sum_{i=1}^d \frac{\partial f}{\partial x_i}(x)(y_i - x_i).$$

(Show that f is strictly convex if the strict inequality holds.)

- (2) Suppose that f is convex on $\mathbb{R}^d$ and $f(0) = 0$. Show that there exist α, β such that for all $x \in \mathbb{R}^d$ we have

$$f(x) \geq \alpha \|x\| + \beta.$$

58. Let $f : U \rightarrow \mathbb{R}$ be a convex function, where $U \subset \mathbb{R}^d$ is a non-empty open convex set.

- (1) Show that any local minimal point of f is also a global minimal point.
 - (2) Show that the global minimal points of f form a convex set.
 - (3) Show that if f is strictly convex then it has at most one global minimal point.
 - (4) Suppose that $f \in C^1$ and $x^* \in U$ is a global minimal point. Then we have $\frac{\partial f}{\partial x_i}(x^*) = 0$ for all $i = 1, 2, \dots, d$.
- 59.** The mapping $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is given by $f(x_1, \dots, x_d) = x_1 x_2 \cdots x_d$. Find the extremal points of f on the set

$$\left\{ (x_1, x_2, \dots, x_d) \in \mathbb{R}^d \mid \sum_{i=1}^d \prod_{j \neq i} x_j = 1 \right\}.$$

- 60.** Suppose $f(x, y, z)$ is continuous on the cube $[a, b] \times [a, b] \times [a, b]$. Show that

$$g(x, y) = \max_{a \leq z \leq b} f(x, y, z)$$

is continuous in the square $[a, b] \times [a, b]$.

- 61.** Let $\Omega \subset \mathbb{R}^d$ be an open set. Suppose $f : \Omega \rightarrow \mathbb{R}$ has bounded partial derivatives (that is, there exists $M > 0$ such that $|\frac{\partial f}{\partial x_i}(x)| \leq M$ for all $i = 1, 2, \dots, d$ and all $x \in \Omega$). Show that f is continuous on Ω . Is f differentiable on Ω ?
- 62.** Suppose that $f(x, y)$ is separately continuous. Show that $f(x, y)$ is continuous provided that one of the following conditions hold.
- (1) The continuity of $f(x, y)$ as a function of x is uniform with respect to y , that is, for any x and $\varepsilon > 0$, there exists δ depending on x, ε but not y such that for all x' such that $|x' - x| \leq \delta$ and all y we have

$$|f(x, y) - f(x', y)| \leq \varepsilon.$$

- (2) The continuity of $f(x, y)$ as a function of y is uniform with respect to x ,
- (3) Particularly, $f(x, y)$ is λ -Lipschitz with respect to y , that is, for all y, y' and x , we have

$$|f(x, y) - f(x, y')| \leq \lambda |y - y'|.$$

- (4) $f(x, y)$ is λ -Lipschitz with respect to x .

- 63.** Let $V \subset \mathbb{R}^d$ be a non-empty open set. Let $K \subset V$ be a bounded closed set. Show that there exists a continuous function $f : \mathbb{R}^d \rightarrow [0, 1]$ such that

$$f(x) = \begin{cases} 1, & \text{if } x \in K \\ 0, & \text{if } x \notin V. \end{cases}$$

- 64.** Suppose that $f(x)$ is continuous in a bounded open set $D \subset \mathbb{R}^d$. Show that f is uniformly continuous in D if and only if for any $x_0 \in \partial D$, $\lim_{x \in D, x \rightarrow x_0} f(x)$ exists. Here, ∂D means the boundary of D .

- 65.** Show that

$$f(x, y) = \begin{cases} \frac{xy}{\sqrt{x^2+y^2}}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

is continuous on $\mathbb{R}^2$ but not differentiable at $(0, 0)$.

- 66.** Define

$$f(x, y) = \begin{cases} xy \sin \frac{1}{\sqrt{x^2+y^2}}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0). \end{cases}$$

Prove that

- (1) $\frac{\partial f}{\partial x}(0, 0)$ and $\frac{\partial f}{\partial y}(0, 0)$ exist.
- (2) $\frac{\partial f}{\partial x}(x, y)$ and $\frac{\partial f}{\partial y}(x, y)$ are not continuous at $(0, 0)$.
- (3) $f(x, y)$ is differentiable at $(0, 0)$.

- 67.** Let $\alpha, \beta > 0$. The function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is given by

$$f(x, y) = \begin{cases} 0, & (x, y) = (0, 0) \\ \sin \left(\frac{|x|^\alpha + |y|^\beta}{x^2 + y^2} \right), & (x, y) \neq (0, 0). \end{cases}$$

When is $f \in C^1(\mathbb{R}^2)$?

- 68.** Suppose that $\frac{\partial f}{\partial x}(x, y)$ exists at $(0, 0)$ and that $\frac{\partial f}{\partial y}(x, y)$ is continuous at $(0, 0)$. Show that $f(x, y)$ is differentiable at $(0, 0)$.
- 69.** The function φ satisfies $\varphi(0) = 0$ and $|\varphi(t)| \leq t^2$ in a small neighbourhood of 0. Show that $f(x, y) := \varphi(|xy|)$ is differentiable at $(0, 0)$.
- 70.** Let $\Omega \subset \mathbb{R}^d$ be an open set. Let $f : \Omega \rightarrow \mathbb{R}$ be a differentiable function. Prove the following statements.

- (1) If f achieves the maximal value at x_0 , then $df(x_0) = 0$.
 - (2) Suppose that Ω is open and convex. If $df(x) = 0$ for all $x \in \Omega$, then f is a constant.
 - (3) Suppose that Ω is open and connected. (Connectivity means that $\emptyset$ and Ω are the only subsets of Ω that are both open and closed). If $df(x) = 0$ for all $x \in \Omega$, then f is a constant.
- 71.** Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous mapping, such that $f(x) = 0$ if and only if $x = a$ or $x = b$. Show that $f(x) \geq 0$ for all $x \in [a, b]$ or $f(x) \leq 0$ for all $x \in [a, b]$.
- 72.** Give an example of a continuous mapping $f :]-1, 1[\rightarrow \mathbb{R}$ which attains a global maximum at 0, but is not differentiable at 0.
- 73.** Give an example of a differentiable mapping $f :]-1, 1[\rightarrow \mathbb{R}$ whose derivative at 0 equals 0, but such that 0 is neither a local minimum, nor a local maximum.
- 74.** Let $f :]a, b[\rightarrow \mathbb{R}$ be a differentiable mapping. Show that if f is increasing (decreasing), then $f'(x) \geq 0$ ($f'(x) \leq 0$) for all $x \in]a, b[$.
- 75.** Let $f :]a, b[\rightarrow \mathbb{R}$ be a differentiable mapping such that $f'(x) > 0$ ($f'(x) < 0$) for all $x \in]a, b[$. Show that f is strictly increasing (decreasing).
- 76.** Give an example of a function $f :]-1, 1[\rightarrow \mathbb{R}$ which is continuous and strictly increasing, but not differentiable at 0.
- 77.** Give an example of a function $f : \mathbb{R} \rightarrow \mathbb{R}$ which is strictly increasing and differentiable, but whose derivative at 0 is 0.
- 78.** Give an example of a subset $X \subseteq \mathbb{R}$ and a mapping $f : X \rightarrow \mathbb{R}$, such that $f'(x) > 0$ for all $x \in X$, but f is not increasing.
- 79.** Let $f :]a, b[\rightarrow \mathbb{R}$ be a differentiable mapping such that $f'(x) = 0$ for all $x \in]a, b[$. Show that f is constant on $]a, b[$.
- 80.** Let $f, g :]a, b[\rightarrow \mathbb{R}$ be differentiable mappings such that $f'(x) = g'(x)$ for all $x \in]a, b[$. Show that there exists a constant $c \in \mathbb{R}$ such that $f = g$ on $]a, b[$.
- 81.** Prove that the equation $x^3 + x - 1$ has exactly one solution.
- 82.** Find an equation of a line tangent to the graph of the function $f(x) = x^3 - x$ parallel to the line passing through the points $f(0)$ and $f(2)$.

- 83.** Let $M > 0$ and let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function differentiable on $]a, b[$ such that $|f'(x)| \leq M$ for all $x \in]a, b[$. Show that f is Lipschitzian.
- 84.** Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable mapping, such that f' is bounded. Show that f is uniformly continuous.
- 85.** We say that a function $f :]a, b[\rightarrow \mathbb{R}$ is concave upward on $]a, b[\subset \mathbb{R}$ if

$$f(tx + (1 - t)y) \geq tf(x) + (1 - t)f(y).$$

We say that a function $f :]a, b[\rightarrow \mathbb{R}$ is concave downward on $]a, b[\subset \mathbb{R}$ if

$$f(tx + (1 - t)y) \leq tf(x) + (1 - t)f(y).$$

Assume that $f, Df :]a, b[\rightarrow \mathbb{R}$ are differentiable. Show that f is concave upward, if and only if $f''(x) > 0$ for all $x \in]a, b[$ and that f is concave downward if and only if $f''(x) < 0$ for all $x \in]a, b[$.

- 86.** Formulate the definition of concavity of mappings in the language of convexity of sets.
- 87.** (The First Derivative Test) Let $f :]a, b[\rightarrow \mathbb{R}$ be a differentiable mapping with a continuous derivative such that $f'(c) = 0$ for some $c \in]a, b[$. Prove the following statements:
- (a) if the sign of f' changes from positive to negative at c , then f has a local maximum at c
 - (b) if the sign of f' changes from negative to positive at c , then f has a local minimum at c
 - (c) if the sign of f' does not change at c , then f does not have a local minimum nor a local maximum at c . In this case we say, that f has an inflection point at c .
- 88.** (The Second Derivative Test) Let $f :]a, b[\rightarrow \mathbb{R}$ be a twice differentiable mapping with a continuous second derivative. Prove the following statements:
- (a) if $f'(c) = 0$ and $f''(c) > 0$ for $c \in]a, b[$, then f has a local minimum at c
 - (b) if $f'(c) = 0$ and $f''(c) < 0$ for $c \in]a, b[$, then f has a local maximum at c
- 89.** (Cauchy's Mean Value Theorem) Let $f, g :]a, b[\rightarrow \mathbb{R}$ be continuous mappings differentiable on $]a, b[$. then there exists $c \in]a, b[$ such that

$$(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c).$$

(Hint: Consider the function $h(x) = (g(b) - g(a))f(x) - (f(b) - f(a))g(x)$).

- 90.** Let $(E, \|\cdot\|)$ and $(F, \|\cdot\|)$ be normed vector spaces over $\mathbb{R}$, let $U \subseteq E$ be an open, convex and connected set, and let $f : U \rightarrow F$ be a differentiable mapping such that

$$\forall x \in U \quad \|Df(x)\| \leq K$$

Let $a, b \in U$. Show that

$$\|f(b) - f(a)\|_F \leq K\|b - a\|_E$$

- 91.** We say, that a topological space X is connected if $X = A \cup B$ for $A, B \subseteq X$ open $A \cap B = \emptyset$ implies that $A = \emptyset$ and $B = X$ or $B = \emptyset$ and $A = X$. Show that the following conditions are equivalent:

- (a) X is connected
- (b) X cannot be presented as a disjoint sum of two non-empty open sets
- (c) X cannot be presented as a disjoint sum of two non-empty closed sets
- (d) The only sets that are simultaneously open and closed are X and $\emptyset$
- (e) All continuous functions $f : X \rightarrow \{0, 1\}$ are constant, where $\{0, 1\}$ is endowed with discrete topology.
- (f) All discrete-valued continuous maps $f : X \rightarrow \{1, \dots, n\}$, for any natural n are constant

- 92.** Show that any path connected topological space is connected.

- 93.** Let $(E, \|\cdot\|)$ and $(F, \|\cdot\|)$ be normed vector spaces over $\mathbb{R}$, let $U \subseteq E$ be an open and connected set, and let $f : U \rightarrow F$ be a differentiable mapping such that

$$\forall x \in U \quad Df(x) = 0.$$

Show that f is constant.

- 94.** Let $(E, \|\cdot\|)$ and $(F, \|\cdot\|)$ be normed vector spaces over $\mathbb{R}$, let $U \subseteq E$ be an open and connected set, and let $f, g : U \rightarrow F$ be differentiable mappings such that

$$\forall x \in U \quad Df(x) = Dg(x).$$

Show that $f = g + C$ on U for some constant C .

- 95.** Show that the connectivity assumption in the exercises 23,25 and the convexity assumption in the exercise 20 is necessary.

- 96.** Let $(E, \|\cdot\|)$ and $(F, \|\cdot\|)$ be normed vector spaces over $\mathbb{R}$, let $U \subseteq E$ be an open set, $f : U \rightarrow F$ be a mapping and $p \in U$. Show that f is differentiable at p if and only if, there exists a linear mapping $Df(p) \in \mathcal{L}(E, F)$ such that
- $$\forall \epsilon > 0 \exists r > 0 (\|h\|_E < r, p+h \in U) \Rightarrow (\|f(p+h) - f(p) - Df(p)(h)\|_F \leq \epsilon \|h\|_E)$$
- 97.** Let U be an open subset of a valued field K , $(F, \|\cdot\|)$ be a normed vector space over K , and $f : U \rightarrow F$ be a mapping. If f is differentiable at $p \in U$, we denote by $f'(p)$ the element

$$Df(p)(1) \in F.$$

Show that:

$$f'(p) = \lim_{h \rightarrow 0} \frac{f(p+h) - f(p)}{h}.$$

- 98.** Let $\{a_n\}_{n=1}^\infty$ be a sequence of positive real numbers such that $\sum_{n=1}^\infty a_n$ is convergent. Show that $\sum_{n=1}^\infty \sqrt{a_n}/n^\alpha$ is also convergent when $\alpha > 1/2$. Is the statement valid when $\alpha = 1/2$?
- 99.** Let $\{a_n\}_{n=1}^\infty$ be a decreasing sequence of positive real numbers. Show that $\sum_{n=1}^\infty a_n$ and $\sum_{n=1}^\infty 2^n a_{2^n}$ either converge or diverge at the same time.
- 100.** Let $\{a_n\}_{n=1}^\infty$ be a sequence of positive real numbers such that the series $\sum_{n=1}^\infty a_n$ is convergent.
- If $\{a_n\}_{n=1}^\infty$ is decreasing, show that $\lim_{n \rightarrow \infty} n a_n = 0$.
 - Give a counter example of (a) if the monotonicity assumption is removed.
- 101.** Let $\{a_n\}_{n=1}^\infty$ be a sequence of positive reals. Denote $S_n = \sum_{k=1}^n a_k$ and $R_n = \sum_{k=n}^\infty a_k$.
- Suppose that $\sum_{n=1}^\infty a_n$ diverges. Show that $\sum_{n=1}^\infty a_n/S_n^\alpha$ converges if $\alpha > 1$ and diverges if $\alpha \leq 1$.
 - Suppose that $\sum_{n=1}^\infty a_n$ converges. Show that $\sum_{n=1}^\infty a_n/R_n^\alpha$ diverges if $\alpha \geq 1$ and converges if $\alpha < 1$.
- 102.** Let $\sum_{n=1}^\infty a_n$ be a positive convergent series. Denote $b_n = (\prod_{k=1}^n a_k)^{1/n}$. Show that $\sum_{n=1}^\infty b_n$ is convergent.
- 103.** Let $\{a_n\}_{n=1}^\infty$ be a sequence of positive reals. Suppose that $\sum_{n=1}^\infty 1/a_n$ is convergent. Show the convergence of the following series

$$\sum_{n=1}^\infty \frac{n}{a_1 + \cdots + a_n} \quad \text{and} \quad \sum_{n=1}^\infty \frac{n^2 a_n}{(a_1 + \cdots + a_n)^2}.$$

- 104. Cesàro summation.** Let $\{a_n\}_{n=1}^{\infty}$ be a sequence of real numbers. Define $S_n = \sum_{k=1}^n a_k$. We say that $\{a_n\}_{n=1}^{\infty}$ is Cesàro summable, with Cesàro sum A , if

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n S_k = A.$$

- (a) Suppose that $\sum_{n=1}^{\infty} a_n$ is convergent. Show that $\{a_n\}_{n=1}^{\infty}$ is Cesàro summable and

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n S_k = \sum_{n=1}^{\infty} a_n.$$

- (b) Is the sequence $1, -1, 1, -1, \dots$ Cesàro summable?

- 105.** Let $\sum_{n=1}^{\infty} a_n$ be a convergent positive series. Show that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n k a_k = 0.$$

- 106.** Let $\{a_n\}_{n=1}^{\infty}$ be a real sequence such that $a_n \searrow 0$. Show that $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} n(a_n - a_{n+1})$ either converge or diverge at the same time. If they converge, then

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} n(a_n - a_{n+1}).$$

- 107.** Let $\{b_n\}_{n=1}^{\infty}$ be a decreasing sequence and $\lim_{n \rightarrow \infty} b_n = 0$. Suppose that $\sum_{n=1}^{\infty} a_n b_n$ is convergent. Define $S_n = \sum_{k=1}^n a_k$. Show that

(a) $\lim_{n \rightarrow \infty} S_n \cdot b_n = 0$.

(b) $\sum_{n=1}^{\infty} S_n(b_n - b_{n+1})$ is convergent and

$$\sum_{n=1}^{\infty} S_n(b_n - b_{n+1}) = \sum_{n=1}^{\infty} a_n b_n.$$

- 108.** Suppose that $\sum_{n=1}^{\infty} a_n$ converges and that $\sum_{n=1}^{\infty} (b_{n+1} - b_n)$ is absolutely convergent. Show that $\sum_{n=1}^{\infty} a_n b_n$ converges.

- 109. (Riemann rearrangement theorem)** Let $\sum_{n=1}^{\infty} u_n$ be a conditional convergent series. We denote by v_n (resp. w_n) the n -th nonnegative (resp. negative) term of $\{u_n\}_{n=1}^{\infty}$. We define $V_n = \sum_{k=1}^n v_k$ and $W_n = \sum_{k=1}^n w_k$.

- (a) Show that $\lim_{n \rightarrow \infty} V_n = +\infty$ and $\lim_{n \rightarrow \infty} W_n = -\infty$.

- (b) Let $\ell \in \mathbb{R}$. Show, by induction, that there exist two strictly increasing sequences of integers $\{m_k\}_{k=0}^{\infty}$ and $\{n_k\}_{k=0}^{\infty}$ such that $\ell < V_{m_0}$ and for all $k \geq 0$ that

$$\begin{aligned} W_{n_k} + V_{m_{k+1}} - v_{m_{k+1}} &\leq \ell < W_{n_k} + V_{m_{k+1}}, \\ W_{n_k} + V_{m_k} &< \ell \leq W_{n_k} + V_{m_k} - w_{n_k}. \end{aligned}$$

Deduce that there exists a bijection $\sigma : \mathbb{N} \rightarrow \mathbb{N}$ such that the series $\sum_{n=1}^{\infty} u_{\sigma(n)} = \ell$.

- (c) Show, by induction, that there exist two strictly increasing sequences of integers $\{p_k\}_{k=0}^{\infty}$ and $\{q_k\}_{k=0}^{\infty}$ such that $V_{p_0} > 0$ and for all $k \geq 1$ that

$$\begin{aligned} W_{q_k} + V_{p_{k+1}} - v_{p_{k+1}} &\leq k < W_{q_k} + V_{p_{k+1}}, \\ W_{q_k} + V_{p_k} &< k - 1 \leq W_{q_k} + V_{p_k} - w_{p_k}. \end{aligned}$$

Deduce that there exists a bijection $\nu : \mathbb{N} \rightarrow \mathbb{N}$ such that the series $\sum_{n=1}^{\infty} u_{\nu(n)}$ diverges.

- (d) Let $z_n = i^n/n$ for $n \geq 1$. Show that the complex series $\sum_{n=1}^{\infty} z_n$ is conditionally convergent. Show that for any complex number $z \in \mathbb{C}$ there exists a bijection $\varphi : \mathbb{N} \rightarrow \mathbb{N}$ such that the series $\sum_{n=1}^{\infty} z_{\varphi(n)}$ converges to z .

110. By rearranging the terms of the series

$$S_{1,1} := \frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \frac{1}{5} + \cdots = 1 - \log 2$$

we obtain

$$S_{p,q} := \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2p} - \frac{1}{3} - \frac{1}{5} - \cdots - \frac{1}{2q+1} + \frac{1}{2p+2} + \cdots$$

in which blocks of p positive terms alternate with blocks of q negative terms. Show that

$$S_{p,q} - S_{1,1} = \frac{1}{2} \log \frac{p}{q}.$$

- 111.** Let $\{a_n\}_{n=1}^{\infty}$ be positive sequence such that $\lim_{n \rightarrow \infty} a_n = 0$ and $\sum_{n=1}^{\infty} a_n$ diverges. Denote $S_n = \sum_{k=1}^n a_k$. Show that $\{S_n - \lfloor S_n \rfloor\}_{n=1}^{\infty}$ is dense in $[0, 1]$.

- 112.** Let $\sum_{n=1}^{\infty} u_n$ be a convergent series of positive terms such that

$$u_n \leq \sum_{k=n+1}^{\infty} u_k, \quad \forall n \in \mathbb{N}.$$

Put

$$r_n = \sum_{k=n}^{\infty} u_k.$$

Show that for any $x \in (0, r_0]$ there exists a strictly increasing sequence $\{n_k\}_{k=0}^{\infty} \subset \mathbb{N}$ such that

$$u_{n_0} + u_{n_1} + \cdots + u_{n_{k-1}} + r_{n_k} \geq x > u_{n_0} + u_{n_1} + \cdots + u_{n_{k-1}} + r_{n_k+1}.$$

Deduce that

$$\sum_{k=0}^{\infty} u_{n_k} = x.$$

- 113.** Let $\sum_{n=1}^{\infty} u_n$ be a convergent series of nonnegative terms. Let $R_n = \sum_{k=n}^{\infty} u_k$ be the remainder of order n . Assume that there exists a constant $C > 0$ such that $R_n \leq C u_n$ for all $n \in \mathbb{N}$. Show that there exist $a \in (0, 1)$ and $M > 0$ such that $u_n \leq M a^n$ for all $n \in \mathbb{N}$.

- 114.** The **Betrand series** is defined as follows

$$\sum_{n=2}^{\infty} \frac{1}{n^{\alpha} \log^{\beta} n}.$$

When is this series convergent?

- 115.** For any $k \in \mathbb{Z}_+$ and $x > 0$, we define $\log^{(k)} x = \log(|\log^{(k-1)} x|)$ with $\log^{(1)} x = \log x$. Show that the following series is convergent

$$\sum_{n=2}^{\infty} \frac{1}{n \log^{(1)} n \cdots \log^{(k-1)} n (\log^{(k)} n)^2}.$$

- 116. Kummer's test:** Let $\{a_n\}_{n=1}^{\infty}$ and $\{b_n\}_{n=1}^{\infty}$ be two positive sequences.

- (a) Show that $\sum_{n=1}^{\infty} a_n$ is convergent if

$$\liminf_{n \rightarrow \infty} \left(b_n \cdot \frac{a_n}{a_{n+1}} - b_{n+1} \right) > 0.$$

- (b) Suppose that $\sum_{n=1}^{\infty} 1/b_n$ diverges and

$$\limsup_{n \rightarrow \infty} \left(b_n \cdot \frac{a_n}{a_{n+1}} - b_{n+1} \right) < 0.$$

Show that $\sum_{n=1}^{\infty} a_n$ diverges.

117. Raabe's test: Let $\{u_n\}_{n=1}^{\infty}$ be a positive sequence. Put

$$\alpha_n = n \left(1 - \frac{u_{n+1}}{u_n} \right), \quad n \geq 1.$$

We propose to show the following

(a) If $\ell = \liminf \alpha_n > 1$, then $\sum_{n=1}^{\infty} u_n$ converges.

(b) If $L = \limsup \alpha_n < 1$, then $\sum_{n=1}^{\infty} u_n$ diverges.

(a) Let $\beta \in \mathbb{R}$ and $v_n = n^{-\beta}$. Show that

$$\frac{v_{n+1}}{v_n} = 1 - \frac{\beta}{n} + o\left(\frac{1}{n}\right) \quad \text{as } n \rightarrow \infty.$$

(b) Show that if $\beta < \ell$, there exists a positive constant A such that

$$u_n \leq \frac{A}{n^{\beta}}, \quad n \geq 1;$$

and if $\beta > L$, there exists a positive constant B such that

$$u_n \geq \frac{B}{n^{\beta}}, \quad n \geq 1.$$

(c) Deduce the desired results.

(d) Application: Let $a, b, \lambda \in \mathbb{R}$ with a, b not negative integers. Determine the nature of the series of general term given by

$$u_n = \left[\frac{(a+1)(a+2) \cdots (a+n)}{(a+1)(b+2) \cdots (b+n)} \right]^{\lambda}.$$

118. Bertrand's test: Let $\{u_n\}_{n=1}^{\infty}$ be a positive sequence. Assume that

$$\frac{u_{n+1}}{u_n} = 1 - \frac{1}{n} - \frac{\alpha_n}{n \log n}.$$

Show that

(a) If $\liminf \alpha_n > 1$, the series $\sum_{n=1}^{\infty} u_n$ converges;

(b) If $\limsup \alpha_n < 1$, the series $\sum_{n=1}^{\infty} u_n$ diverges.

State a similar criterion when

$$\frac{u_{n+1}}{u_n} = 1 - \frac{1}{n} - \frac{1}{n \log n} - \frac{\alpha_n}{n \log \log n}.$$

Extend the results.

119. Let $\sum_{n=1}^{\infty} a_n$ be a complex series. Suppose that

$$\frac{a_{n+1}}{a_n} = 1 + \frac{A}{n} + \frac{B_n}{n^2},$$

where $\operatorname{Re}(A) < -1$ and B_n is bounded. Show that $\sum_{n=1}^{\infty} |a_n|$ is convergent.

120. Let $\sum_{n=1}^{\infty} a_n$ be a real series such that $\lim_{n \rightarrow \infty} a_n = 0$ and

$$-1 \leq \frac{a_{n+1}}{a_n} < r, \quad \text{where } r \in (0, 1).$$

Show that $\sum_{n=1}^{\infty} a_n$ is convergent.

121. Let $\{u_n\}_{n=1}^{\infty}$ be a positive sequence. Show that

- (a) $\limsup_{n \rightarrow \infty} \left(\frac{u_{n+1}}{u_n} \right)^n < \frac{1}{e}$, the series $\sum_{n=1}^{\infty} u_n$ converges;
- (b) $\liminf_{n \rightarrow \infty} \left(\frac{u_{n+1}}{u_n} \right)^n > \frac{1}{e}$, the series $\sum_{n=1}^{\infty} u_n$ diverges.

122. Let $\{a_n\}_{n=1}^{\infty}$ be a sequence of positive real numbers.

- (a) Suppose there is $\alpha \in (0, 1)$ such that $a_n^{1/n} \leq 1 - n^{-\alpha}$ for all $n \in \mathbb{Z}_+$. Is $\sum_{n=1}^{\infty} a_n$ convergent?
- (b) Suppose there is $\beta > 1$ such that $n(a_n/a_{n+1} - 1) \geq \beta$ for all $n \in \mathbb{Z}_+$. Is $\sum_{n=1}^{\infty} a_n$ convergent?

123. Let $\{a_n\}_{n=1}^{\infty}$ be a positive sequence such that

$$a_n^{1/n} < 1 - \frac{1}{n} - \frac{\log \log n}{\log n}, \quad n \geq N.$$

Is $\sum_{n=1}^{\infty} a_n$ convergent?

124. Let $\{a_n\}_{n=1}^{\infty}$ be a positive sequence such that $\lim_{n \rightarrow \infty} na_n = 0$ and

$$\frac{a_{n+1}}{a_n} = \frac{n+a}{n+b}, \quad a, b \in \mathbb{R}.$$

- (a) Show that $\sum_{n=1}^{\infty} a_n$ is convergent and determine its value.
- (b) Determine the value of $\sum_{n=1}^{\infty} a_n$, where

$$a_n = \frac{1 \cdot 3 \cdot \dots \cdot (2n-1)}{2 \cdot 4 \cdot \dots \cdot 2n \cdot (2n+2)}$$

125. Let $\sum_{n=1}^{\infty} a_n$ be a complex series. Suppose that

$$\frac{a_{n+1}}{a_n} = e^{i\theta} \left(1 - \frac{b}{n} + O\left(\frac{1}{n^2}\right) \right), \quad \theta, b \in \mathbb{R},$$

Show that $\sum_{n=1}^{\infty} a_n$ diverges if $b < 0$, and converges if $b > 0$ and $\theta \neq 2k\pi, k \in \mathbb{Z}$.

126. Let $\alpha \in (0, 1)$.

(a) Show that the series $\sum_{n \geq 1} \frac{|\sin(n^\alpha)|}{n}$ diverges.

(b) Show that there exists a positive constant A such that

$$\left| \int_0^x \sin(t^\alpha) dt \right| \leq Ax^{1-\alpha}, \quad \left| \int_0^n \sin(t^\alpha) dt - \sum_{k=0}^n \sin(k^\alpha) \right| \leq An^\alpha, \quad x > 0, n \in \mathbb{N}.$$

(c) Deduce that if $\beta > \max(\alpha, 1 - \alpha)$, the series $\sum_{n \geq 1} \frac{\sin(n^\alpha)}{n^\beta}$ converges.

127. Let $\sum_{n=1}^{\infty} a_n$ be a positive series. The sequence $\{s_n\}_{n=1}^{\infty}$ is defined via the following equation

$$2s_{n+1} = s_n + \sqrt{s_n^2 + a_n} \text{ with } s_1 = 1.$$

Show that $\lim_{n \rightarrow \infty} s_n$ exists provided that $\sum_{n=1}^{\infty} a_n$ converges. Is the converse statement true?

128. Let $f : \mathbb{R}_+ \rightarrow (0, \infty)$ be a decreasing function. Suppose that

$$\lim_{x \rightarrow \infty} \frac{e^x f(e^x)}{f(x)} = \lambda.$$

Show that $\sum_{n=1}^{\infty} f(n)$ converges if $\lambda < 1$ and diverges if $\lambda > 1$.

129. Let $f : [1, +\infty) \rightarrow \mathbb{R}_+$ be a decreasing function such that the integral $\int_1^{+\infty} f(x) dx$ diverges. Let $u_n = f(n)$. Show that

$$\sum_{k=1}^n u_k \sim \int_1^n f(x) dx \text{ as } n \rightarrow +\infty.$$

130. Let $f : \mathbb{R}_+ \rightarrow (0, \infty)$ be a continuously differentiable function such that $\lim_{x \rightarrow \infty} \frac{f'(x)}{f(x)} = \lambda \neq 0$.

(a) If $\int_0^\infty f(x)dx$ converges, show that

$$\sum_{k=n+1}^{\infty} f(k) \sim \frac{\lambda}{1 - e^{-\lambda}} \int_n^{\infty} f(x)dx \text{ as } n \rightarrow \infty.$$

(b) If $\int_0^\infty f(x)dx$ diverges, show that

$$\sum_{k=n+1}^{\infty} f(k) \sim \frac{\lambda}{1 - e^{-\lambda}} \int_0^n f(x)dx \text{ as } n \rightarrow \infty.$$

131. Let $\{u_n\}_{n=1}^\infty$ be a positive sequence. Assume that there exist $\alpha > 1$ and $\lambda > 0$ such that $u_{n+1} - u_n \sim -\lambda u_n^\alpha$ as $n \rightarrow \infty$.

(a) Show that starting from a certain rank $\{u_n\}_{n=1}^\infty$ is decreasing and $\lim_{n \rightarrow \infty} u_n = 0$.

(b) Show that

$$\lim_{n \rightarrow \infty} \left(\frac{1}{u_{n+1}^{\alpha-1}} - \frac{1}{u_n^{\alpha-1}} \right) = \lambda(\alpha - 1).$$

(c) Deduce that

$$u_n \sim \left(\frac{1}{\lambda(\alpha - 1)n} \right)^{\frac{1}{\alpha-1}}.$$

(d) Determine the nature of the series $\sum_{n=1}^\infty u_n^\beta$ according to the positive values of β .

(e) Application: the sequence $\{u_n\}_{n=1}^\infty$ is defined by $u_{n+1} = \sin u_n$.

132. We aim to study the harmonic series $\sum_{n=1}^\infty 1/n$ and the Euler constant. Let

$$\sigma_n = \sum_{k=1}^n \frac{1}{k} - \log n \text{ and } u_1 = \sigma_1, \quad u_n = \sigma_n - \sigma_{n-1}, \quad n \geq 2.$$

(a) Show that

$$u_n \sim \frac{-1}{2n^2} \text{ as } n \rightarrow \infty.$$

(b) Deduce that the sequence $\{\sigma_n\}_{n=1}^\infty$ converges; its limit is called the Euler constant γ .

(c) Show that the remainder R_n of order n of $\sum u_n$ is equivalent to $\frac{-1}{2n}$ as $n \rightarrow \infty$.

(d) Deduce that

$$\sigma_n = \gamma + \frac{1}{2n} + o\left(\frac{1}{n}\right).$$

133. Let $\sum_{n=0}^{\infty} z_n$ be a conditional convergent complex series. Let $z_n = x_n + \mathfrak{B}y_n$ with $x_n, y_n \in \mathbb{R}$.

- (a) Show that there exists at most one $\lambda \in \mathbb{R}$ such that the series $\sum_{n=0}^{\infty} (\lambda x_n + y_n)$ absolutely converges.
- (b) Assume that there exists one $\lambda \in \mathbb{R}$ such that $\sum_{n=0}^{\infty} (\lambda x_n + y_n)$ absolutely converges. Show that the set

$$\left\{ \sum_{n=0}^{\infty} z_{\sigma(n)} : \sigma : \mathbb{N} \rightarrow \mathbb{N} \text{ is a bijection s.t. } \sum_{n=0}^{\infty} z_{\sigma(n)} \text{ converges} \right\}$$

is a line in the complex plane.

134. Let $\sum_{n=0}^{\infty} u_n, \sum_{n=0}^{\infty} v_n$ be two convergent series of sums U, V respectively, and let $\sum_{n=0}^{\infty} w_n$ be their product series. Assume that $\sum_{n=0}^{\infty} w_n$ converges to W . Let U_n, V_n, W_n be the respective n -th partial sums.

- (a) Show that $W_0 + W_1 + \cdots + W_n = U_0 V_n + U_1 V_{n-1} + \cdots + U_n V_0$.
- (b) Deduce that $W = UV$.

135. Discuss the convergence of the following series.

- (1) $\sum_{n=1}^{\infty} \frac{\sin(2\pi en!)}{n^\alpha}$
- (2) $\sum_{n=1}^{\infty} \sin(\pi en!)$
- (3) $\sum_{n=2}^{\infty} \cos\left(\pi n^2 \log \frac{n}{n-1}\right)$
- (4) $\sum_{n=1}^{\infty} \frac{\cos^k(n\theta)}{n^\alpha}, (\alpha > 0, \theta \in \mathbb{R}, k \in \mathbb{N})$
- (5) $\sum_{n=1}^{\infty} (-1)^n n^\alpha a^{n^\beta}, (\alpha, \beta, a > 0)$
- (6) $\sum_{n=1}^{\infty} \left(\sum_{k=1}^n \frac{1}{k}\right) \frac{\cos(n\theta)}{n + (-1)^n}, (\theta \in \mathbb{R})$
- (7) $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^\alpha + \sin(n\theta)}, (\alpha > 0, \theta \in \mathbb{R})$
- (8) $\sum_{n=1}^{\infty} (-1)^n \frac{n^\alpha \sin(n^{-\alpha})}{n^\beta + (-1)^n}, (\alpha, \beta > 0)$
- (9) $\sum_{n=1}^{\infty} \left(1 - \frac{\alpha \log n}{n}\right)^n$
- (10) $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^{p+1/n}}$
- (11) $\sum_{n=1}^{\infty} (-1)^n \sqrt{n} \sin \frac{1}{n}$
- (12) $\sum_{n=1}^{\infty} \log n \log \left(1 + \frac{(-1)^n}{n}\right)$
- (13) $\sum_{n=1}^{\infty} \sin(\pi \sqrt{n^2 + 1})$
- (14) $\sum_{n=1}^{\infty} \frac{\cos n}{\sqrt{n} + \cos n}$
- (15) $\sum_{n=1}^{\infty} \left(\cos\left(\frac{1}{n}\right)\right)^{n^\alpha}, (\alpha \in \mathbb{R})$
- (16) $\sum_{n=1}^{\infty} \left|\sin \frac{\pi \sqrt{n}}{n^\alpha}\right|, (\alpha \in \mathbb{R})$
- (17) $\sum_{n=1}^{\infty} \left(\frac{1 \cdot 3 \cdots 2n-1}{2 \cdot 4 \cdots 2n}\right)^\alpha, (\alpha \in \mathbb{R})$
- (18) $\sum_{n=1}^{\infty} \frac{a(a+1) \cdots (a+n-1)b(b+1) \cdots (b+n-1)}{n!c(c+1) \cdots (c+n-1)}, (a, b, c \in \mathbb{R})$

136. Discuss the convergence of the following infinite products.

- (1) $\prod_{n=1}^{\infty} \left(\cos \frac{1}{n^\alpha}\right)^n, \alpha > 0$
- (2) $\prod_{n=0}^{\infty} (1 + z^{2^n}), z \in \mathbb{C}$ (value in case of convergence)
- (3) $\prod_{n=1}^{\infty} \left(1 + \frac{i}{n}\right), \prod_{n=1}^{\infty} \left|1 + \frac{i}{n}\right|$
- (4) $\prod_{n=0}^{\infty} \cos \frac{a}{2^n}, a \in \mathbb{R}, \prod_{n=0}^{\infty} \operatorname{ch} \frac{a}{2^n}$ (value)

137. Let $(u_n) \subset \mathbb{C} \setminus \{-1\}$ such that the series $\sum u_n$ absolutely converges. Show that for any bijection $\sigma : \mathbb{N} \rightarrow \mathbb{N}$

$$\prod_{n=0}^{\infty} (1 + u_n) = \prod_{n=0}^{\infty} (1 + u_{\sigma(n)}).$$

138. (More on Cesàro summation) Let $(s_n)_{n \geq 0}$ be a sequence of complex numbers, and define

$$\sigma_n = \frac{s_0 + \cdots + s_n}{1 + n} \quad (n = 0, 1, 2, \dots).$$

- (a) Can it happen that $\lim_{n \rightarrow \infty} s_n = +\infty$ but $\lim_{n \rightarrow \infty} \sigma_n = 0$?
 (b) For $n \geq 1$, put $a_n = s_n - s_{n-1}$. Show that

$$s_n - \sigma_n = \frac{1}{n+1} \sum_{k=1}^n k a_k.$$

Assume that $\lim_{n \rightarrow \infty} n a_n = 0$ and that (σ_n) converges, show that (s_n) converges to the same limit.

- (c) We aim to show that the last conclusion still holds under the weak assumption that there exists $M > 0$, such that $|n a_n| \leq M$ for all n .

i. For $m < n$, show that

$$s_n - \sigma_n = \frac{m+1}{n-m} (\sigma_n - \sigma_m) + \frac{1}{n-m} \sum_{i=m+1}^n (s_n - s_i).$$

ii. For $i = m+1, \dots, n$, show that

$$|s_n - s_i| \leq \frac{(n-i)M}{i+1} \leq \frac{(n-m-1)M}{m+2}.$$

- iii. Fix an arbitrary $\varepsilon > 0$. For any $i \leq n$, choose m to be the unique integer such that

$$m \leq \frac{n-\varepsilon}{1+\varepsilon} < m+1.$$

Show that

$$\frac{m+1}{n-m} \leq \frac{1}{\varepsilon}$$

and that

$$|s_n - s_i| \leq M\varepsilon.$$

- iv. Conclude, i.e. assume $M > 0$, $|n a_n| \leq M$ for all n , and (σ_n) converges to σ , then (s_n) also converges to σ .

139. (Iterated series.) Consider an infinite array of numbers $(a_{ij})_{i,j \geq 0}$ defined by

$$a_{ij} = \begin{cases} 0 & \text{if } i < j; \\ -1 & \text{if } i = j; \\ 2^{j-i} & \text{if } i > j. \end{cases}$$

- (a) Compute $\sum_{i=0}^{\infty} \left(\sum_{j=0}^{\infty} a_{ij} \right)$ and $\sum_{j=0}^{\infty} \left(\sum_{i=0}^{\infty} a_{ij} \right)$.
- (b) Show that if $(a_{ij})_{i,j \geq 0}$ is an infinite array of *non-negative* numbers, then

$$\sum_{i=0}^{\infty} \left(\sum_{j=0}^{\infty} a_{ij} \right) = \sum_{j=0}^{\infty} \left(\sum_{i=0}^{\infty} a_{ij} \right).$$

Here, we allow $\infty = \infty$.

- 140. (There are many prime numbers.)** It is known that Riemann's ζ -function is defined as

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

- (a) Show that

$$\zeta(s) = \prod_{p \in \mathcal{P}} \frac{1}{1 - p^{-s}},$$

where $\mathcal{P}$ is the set of all primes.

- (b) Deduce that the series

$$\sum_{p \in \mathcal{P}} \frac{1}{p^s}$$

converges if $s > 1$, and diverges if $0 < s \leq 1$. Particularly, it implies that there are infinitely many primes.

Chapter 10

Integral calculus

10.1 Differential 1-forms

10.1.1 Definition. Let $(K, |\cdot|)$ be a complete non-trivially valued field and $(E, \|\cdot\|_E)$ and $(F, \|\cdot\|_F)$ be normed vector spaces over K .

Let U be an open subset of E . We call *1-form* on U with coefficients in F any mapping from U to $\mathcal{L}(E, F)$.

In the particular case where $F = K$, a mapping from U to $\mathcal{L}(E, K)$ is just called a 1-form on U .

10.1.2 Definition. Let $(K, |\cdot|)$ be a complete non-trivially valued field. We call *complete valued extension* of $(K, |\cdot|)$ any complete non-trivially valued field $(K', |\cdot|')$ such that $K' \supseteq K$ and that the restriction of $|\cdot|'$ to K coincides with $|\cdot|$.

10.1.3 Example. $(\mathbb{C}, |\cdot|)$ is a complete valued extension of $(\mathbb{R}, |\cdot|)$.

10.1.4 Notation. Let $(K, |\cdot|)$ be a complete non-trivially valued field and $(K', |\cdot|')$ be a complete valued extension of $(K, |\cdot|)$. Let F be a normed vector space over K' .

If $\alpha : U \rightarrow \mathcal{L}(E, K')$ and $s : U \rightarrow F$ are mappings, we denote by

$$\alpha \otimes s : U \longrightarrow \mathcal{L}(E, F)$$

the mapping that sends $p \in U$ to the mapping

$$(\ell \in E) \longmapsto \alpha(p)(\ell)s(p).$$

Note that the above mapping belongs to $\mathcal{L}(E, F)$ since

$$\|\alpha(p)(\ell)s(p)\|_F = |\alpha(p)(\ell)|' \cdot \|s(p)\|_F \leq \|\alpha(p)\| \cdot |\ell| \cdot \|s(p)\|_F$$

In the particular case where F is a field containing K , the mapping $\alpha \otimes s$ is also denoted as $s\alpha$.

10.1.5 Example. Let φ be an element of $\mathcal{L}(E, F)$. We denote by

$$d\varphi : E \longrightarrow \mathcal{L}(E, F)$$

the constant mapping that sends E to φ . More generally, if U is an open subset of E and $f : U \rightarrow F$ is a differentiable mapping, we denote by df the differential of f on U . It is a 1-form on U with coefficients in F (note that there is no difference between Df and df).

10.1.6 Remark. Consider the case where $E = K^n$, with $n \in \mathbb{N}_{\geq 1}$. For $i \in \{1, \dots, n\}$, let $x_i : K^n \rightarrow K$ be the projection mapping to the i -th coordinate. If U is an open subset of K^n and $f : U \rightarrow F$ is a differentiable mapping, then one has

$$df = \sum_{i=1}^n dx_i \otimes \frac{\partial f}{\partial x_i}.$$

Sometimes, in order to emphasize the coordinate symbols $x_1, \dots, x_n$, we write the above formula as

$$df(x_1, \dots, x_n) = \sum_{i=1}^n dx_i \otimes \frac{\partial f}{\partial x_i}(x_1, \dots, x_n).$$

In the particular case where E is a field containing K , we write

$$df(x_1, \dots, x_n) = \sum_{i=1}^n \frac{\partial f}{\partial x_i}(x_1, \dots, x_n) dx_i$$

10.1.7 Example. Let $c \in \mathbb{C}$. Consider the mapping

$$f : \mathbb{R} \longrightarrow \mathbb{C}, \quad f(t) := e^{ct}.$$

Then one has $f'(t) = ce^{ct}$ and hence

$$de^{ct} = ce^{ct} dt.$$

In particular,

$$\begin{aligned} d \sin(t) &= \frac{1}{2i} d(e^{it} - e^{-it}) = \frac{1}{2i} (ie^{it} + ie^{-it}) dt = \cos(t) dt, \\ d \cos(t) &= \frac{1}{2} d(e^{it} + e^{-it}) = \frac{1}{2} (ie^{it} - ie^{-it}) dt = -\sin(t) dt \end{aligned}$$

10.1.8 Definition. Let U be an open subset of E and α be a 1-form on U . We say that α is *exact* if there exists a differentiable mapping $f : U \rightarrow F$ such that

$$\alpha = df.$$

10.1.9 Proposition. *Let U be an open subset of E and $f : U \rightarrow K$ and $g : U \rightarrow F$ be two mappings. One has*

$$d(fg) = f dg + df \otimes g.$$

Proof. This comes from Leibniz's rule for the differential of products. $\square$

10.1.10 Proposition. *Let U be an open subset of E , and V be an open subset of K . Let $f : U \rightarrow V$ and $g : V \rightarrow F$ be mappings. One has*

$$d(g \circ f) = dg \otimes (g' \circ f).$$

Proof. For $p \in U$ and $h \in E$, one has

$$D(g \circ f)(p)(h) = Dg(f(p))(Df(p)(h)) = Df(p)(h)g'(f(p)).$$

Hence we obtain the required equality. $\square$

10.2 Primitive functions

10.2.1 Proposition. *Let $(E, \|\cdot\|_E)$ and $(F, \|\cdot\|_F)$ be two normed vector spaces over $\mathbb{R}$ and U be a path connected open subset of E . If $f : U \rightarrow F$ is a differentiable mapping such that $df = 0$, then f is a constant mapping.*

Proof. Let p and q be two elements of U . Let $\gamma : [0, 1] \rightarrow U$ be a continuous mapping that is differentiable on $]0, 1[$, such that $\gamma(0) = p$ and $\gamma(1) = q$. By the mean value inequality, one has

$$\|f(q) - f(p)\|_F = \|(f \circ \gamma)'(t)\| \leq \sup_{t \in]0, 1[} \|Df(\gamma(t))(\gamma'(t))\| = 0.$$

Therefore, f is a constant mapping. $\square$

10.2.2 Definition. Let I be an open interval in $\mathbb{R}$ and $f : I \rightarrow F$ be a mapping. If there exists $F : I \rightarrow F$ such that $F' = f$, or equivalently

$$dF(t) = dt \otimes f(t),$$

then we say that F is a *primitive function* of f . Proposition 10.2.1 shows that two primitive functions of f differ by a constant mapping from I to $\mathbb{R}$. We often use a formal symbol

$$\int f(t) dt$$

to present an arbitrary primitive function of f . Note that, if $F : I \rightarrow \mathbb{R}$ is a differentiable function on I , then one has

$$\int dF(t) = F(t) + C,$$

where C denotes a constant in $\mathbb{R}$.

10.2.3 Example. Let $c \in \mathbb{C} \setminus \{0\}$. Consider the mapping $f_c : \mathbb{R} \rightarrow \mathbb{C}$, $t \mapsto e^{ct}$. One has

$$f'_c = cf_c.$$

Therefore

$$\int e^{ct} dt = \frac{e^{ct}}{c} + C,$$

where C denotes a complex constant.

10.2.4 Example. Consider the mapping $\ln : \mathbb{R}_{>0} \rightarrow \mathbb{R}$. It is the inverse of $\exp$. Hence

$$\ln'(t) = \frac{1}{\exp'(\ln(t))} = \frac{1}{\exp(\ln(t))} = \frac{1}{t}.$$

Therefore,

$$\int \frac{1}{t} dt = \ln(t) + C.$$

10.2.5 Example. Consider the polynomial mapping

$$F : \mathbb{R} \rightarrow \mathbb{R}, F(t) = \frac{t^{n+1}}{n+1},$$

where $n \in \mathbb{N}$. One has $F'(t) = t^n$. Therefore,

$$\int t^n dt = \frac{1}{n+1} t^{n+1} + C.$$

10.2.6 Proposition. Let I be an open interval on $\mathbb{R}$. Let $f : I \rightarrow \mathbb{R}$ and $g : I \rightarrow \mathbb{R}$ be mappings which have primitive functions F and G respectively. Then one has

$$\int F(t) dG(t) + \int dF(t) \otimes g(t) = F(t)G(t) + C.$$

Proof. This is because

$$d(FG) = F dG + dF \otimes G.$$

□

10.2.7 Example. One has

$$\int t e^t dt = \int t d e^t = t e^t - \int e^t dt = t e^t - e^t + C$$

10.2.8 Proposition. Let U be an open subset of E and V be an open subset of $\mathbb{R}$. Let $f : U \rightarrow V$ and $g : V \rightarrow F$ be differentiable mappings. One has

$$\int df(t) \otimes g'(f(t)) = g(f(t)) + C,$$

where C is a constant in F .

Proof. This comes from Proposition 10.1.10. □

10.2.9 Example. One has

$$\int \sin(t) \cos(t) dt = \int \sin(t) d \sin(t) = \frac{1}{2} \sin(t)^2 + C.$$

10.3 Daniell integral

In this section, we fix a set Ω . We equip $\mathbb{R}^\Omega$ with the following partial order $\leq$:

$$f \leq g \text{ if and only if } \forall \omega \in \Omega, f(\omega) \leq g(\omega).$$

Note that, any finite subset $\{f_1, \dots, f_n\}$ has a supremum and an infimum in $\mathbb{R}^\Omega$. One has

$$\begin{aligned} \inf\{f_1, \dots, f_n\}(\omega) &= \min\{f_1(\omega), \dots, f_n(\omega)\}, \\ \sup\{f_1, \dots, f_n\}(\omega) &= \max\{f_1(\omega), \dots, f_n(\omega)\}. \end{aligned}$$

10.3.1 Definition. We call *Riesz space* on Ω a real vector subspace S of $\mathbb{R}^\Omega$ which satisfies the following condition: for any $(f, g) \in S \times S$, $\inf\{f, g\}$ belongs to S .

10.3.2 Remark. Let S be a Riesz space on Ω . For any $(f, g) \in S \times S$, one has

$$\sup\{f, g\} = f + g - \inf\{f, g\} \in S,$$

$$|f| = \sup\{f, 0\} - \inf\{f, 0\} \in S.$$

By induction for all $n \in \mathbb{N}_{\geq 1}$ and $(f_1, \dots, f_n) \in S^n$

$$\sup\{f_1, \dots, f_n\} \in S \quad \inf\{f_1, \dots, f_n\} \in S.$$

10.3.3 Definition. Let Ω be a set and S be a Riesz space on Ω . We say that an $\mathbb{R}$ -linear mapping $I : S \rightarrow \mathbb{R}$ is an *integral operator* on S if it satisfies the following conditions:

- (1) if $(f, g) \in S \times S$ is such that $f \leq g$, then $I(f) \leq I(g)$,
- (2) if $(f_n)_{n \in \mathbb{N}}$ is a decreasing sequence in S that converges pointwise to the constant zero mapping, then one has

$$\lim_{n \rightarrow +\infty} I(f_n) = 0.$$

10.3.4 Proposition. *Let Ω be a set and S be a Riesz space on Ω , and $I : S \rightarrow \mathbb{R}$ be an $\mathbb{R}$ -linear mapping such that for any $f, g \in S$, $I(f) \leq I(g)$ whenever $f \leq g$. Then the following conditions are equivalent.*

- (1) I is an integral operator.
- (2) For any increasing sequence $(f_n)_{n \in \mathbb{N}}$ in S that converges pointwise to some $f \in S$, one has

$$I(f) = \lim_{n \rightarrow +\infty} I(f_n).$$

Proof. “(1) $\implies$ (2)”: The sequence $(f - f_n)_{n \in \mathbb{N}}$ is decreasing and converges pointwise to the constant zero mapping. Hence

$$0 = \lim_{n \rightarrow +\infty} I(f - f_n) = \lim_{n \rightarrow +\infty} (I(f) - I(f_n)).$$

Hence the sequence $(I(f_n))_{n \in \mathbb{N}}$ converges to $I(f)$.

“(2) $\implies$ (1)”: The sequence $(-f_n)_{n \in \mathbb{N}}$ is increasing and converges pointwise to the constant zero mapping. Hence

$$\lim_{n \rightarrow +\infty} I(-f_n) = 0,$$

which implies that

$$\lim_{n \rightarrow +\infty} I(f_n) = 0.$$

□

10.3.5 Proposition. *Let X be a set, S be a Riesz space on X , and $I : S \rightarrow \mathbb{R}$ be an integral operator. Let $(f_n)_{n \in \mathbb{N}}$ be an increasing sequence in S and $f \in S$. If f is bounded from above by $\lim_{n \rightarrow +\infty} f_n$, then*

$$I(f) \leq \lim_{n \rightarrow +\infty} I(f_n).$$

Proof. The sequence $(\inf\{f, f_n\})_{n \in \mathbb{N}}$ is increasing and converges to f . Hence, by Proposition 10.3.4,

$$I(f) = \lim_{n \rightarrow +\infty} I(\inf\{f, f_n\}) \leq \lim_{n \rightarrow +\infty} I(f_n).$$

□

10.3.6 Proposition. *Let Ω be a set, S be a Riesz space on Ω , and $I : S \rightarrow \mathbb{R}$ be an integral operator. If*

$$\lim_{n \rightarrow +\infty} f_n \leq \lim_{n \rightarrow +\infty} g_n,$$

then $\lim_{n \rightarrow +\infty} I(f_n) \leq \lim_{n \rightarrow +\infty} I(g_n)$.

Proof. For any $m \in \mathbb{N}$, one has

$$f_m \leq \lim_{n \rightarrow +\infty} f_n \leq \lim_{n \rightarrow +\infty} g_n.$$

By Proposition 10.3.5, we obtain

$$I(f_m) \leq \lim_{n \rightarrow +\infty} I(g_n).$$

Taking the limit when $m \rightarrow +\infty$, we obtain

$$\lim_{m \rightarrow +\infty} I(f_m) \leq \lim_{n \rightarrow +\infty} I(g_n)$$

□

10.3.7 Definition. Let Ω be a set and S be a Riesz space on X . We denote by $S^\uparrow$ the set of all mappings f from Ω to $]-\infty, +\infty]$ that can be written as the pointwise limit of an increasing sequence in S . Note that $S^\uparrow$ is stable by addition and by multiplication by a non-negative scalar. Moreover, for any elements f and g in $S^\uparrow$, the mappings $\inf\{f, g\}$ and $\sup\{f, g\}$ belong to $S^\uparrow$.

10.3.8 Definition. Let Ω be a set, S be a Riesz space on Ω , and $I : S \rightarrow \mathbb{R}$ be an integral operator. We extend I to $S^\uparrow$ as follows: for any $f \in S^\uparrow$ which is written as the pointwise limit of an increasing sequence $(f_n)_{n \in \mathbb{N}}$, we let

$$I(f) = \lim_{n \rightarrow +\infty} I(f_n).$$

Note that Proposition 10.3.6 shows that the extended mapping $I : S^\uparrow \rightarrow]-\infty, +\infty]$ is well defined. Moreover, it satisfies the following properties:

- (1) for any elements f and g of $S^\uparrow$, one has $I(f + g) = I(f) + I(g)$,
- (2) for any $f \in S^\uparrow$ and any $a \geq 0$, one has $I(af) = aI(f)$,
- (3) if f and g are two elements of $S^\uparrow$ such that $f \leq g$, then $I(f) \leq I(g)$.

10.3.9 Proposition. *Let Ω be a set and S be a Riesz space on Ω . Let $(f_n)_{n \in \mathbb{N}}$ be an increasing sequence in $S^\uparrow$ that converges to a mapping $f : \Omega \rightarrow]-\infty, +\infty]$*

- (1) The mapping f belongs to $S^\uparrow$.
 (2) If $I : S \rightarrow \mathbb{R}$ is an integral operator, then

$$I(f) = \lim_{n \rightarrow +\infty} I(f_n).$$

Proof. For any $k \in \mathbb{N}$, let $(g_{k,m})_{m \in \mathbb{N}} \in S^\mathbb{N}$ be an increasing sequence such that

$$f_k = \lim_{m \rightarrow +\infty} g_{k,m}.$$

For $n \in \mathbb{N}$, let

$$h_n = \sup\{g_{0,n}, \dots, g_{n,n}\} \in S.$$

The sequence $(h_n)_{n \in \mathbb{N}}$ is increasing. Moreover, for $(k, n) \in \mathbb{N}^2$ such that $n \geq k$, one has $f_n \geq f_k \geq g_{k,n}$. Hence $f_n \geq h_n$. Note that

$$h_{n+1} = \sup\{g_{0,n+1}, \dots, g_{n,n+1}, g_{n+1,n+1}\} \geq \sup\{g_{0,n}, \dots, g_{n,n}, g_{n+1,n+1}\} \geq h_n.$$

Taking the limit when $n \rightarrow +\infty$, we get

$$f = \lim_{n \rightarrow +\infty} f_n \geq \lim_{n \rightarrow +\infty} h_n \geq \lim_{n \rightarrow +\infty} g_{k,n} = f_k.$$

Taking the limit when $k \rightarrow +\infty$, we get

$$f = \lim_{n \rightarrow +\infty} h_n.$$

Hence $f \in S^\uparrow$ and

$$I(f) = \lim_{n \rightarrow +\infty} I(h_n) \leq \lim_{n \rightarrow +\infty} I(f_n).$$

Conversely, for any $n \in \mathbb{N}$, $f \geq f_n$. Hence

$$I(f) \geq \lim_{n \rightarrow +\infty} I(f_n).$$

□

10.3.10 Definition. Let Ω be a set and S be a Riesz space on Ω . We denote by $S^\downarrow$ the set of all mappings f from Ω to $[-\infty, +\infty[$ that can be written as the pointwise limit of a decreasing sequence in S . Note that $S^\downarrow$ is stable by addition and by multiplication by a non-negative scalar. Note that $f \in S^\downarrow$ if and only if $-f \in S^\uparrow$.

For any elements f and g in $S^\downarrow$, the mappings $\inf\{f, g\}$ and $\sup\{f, g\}$ belong to $S^\downarrow$. The integral operator I can be extended to a mapping $I : S^\downarrow \rightarrow [-\infty, +\infty[$ such that, for any $f \in S^\downarrow$ which is the pointwise limit of a decreasing sequence $(f_n)_{n \in \mathbb{N}}$, one has

$$I(f) = \lim_{n \rightarrow +\infty} I(f_n).$$

Moreover, the extended mapping satisfies the following properties.

- (1) For any elements f and g of $S^\downarrow$, one has $I(f + g) = I(f) + I(g)$.
- (2) For any $f \in S^\downarrow$ and $a \in \mathbb{R}_{\geq 0}$, one has $I(af) = aI(f)$.
- (3) If f and g are two elements of $S^\downarrow$ such that $f \leq g$, then $I(f) \leq I(g)$.
- (4) If $(f_n)_{n \in \mathbb{N}}$ is a decreasing sequence of mappings in $S^\downarrow$, then the pointwise limit f of the sequence $(f_n)_{n \in \mathbb{N}}$ belongs to $S^\downarrow$, and one has

$$I(f) = \lim_{n \rightarrow +\infty} I(f_n),$$

- (5) A mapping $f : \Omega \rightarrow [-\infty, +\infty[$ belongs to $S^\downarrow$ if and only if $-f$ belongs to $S^\uparrow$. Moreover, one has $I(-f) = -I(f)$.

10.3.11 Proposition. Let Ω be a set, S be a Riesz space on Ω , and $I : S \rightarrow \mathbb{R}$ an integral operator. Let $(f, g) \in (S^\uparrow \cup S^\downarrow)^2$. If $f \leq g$, then $I(f) \leq I(g)$.

Proof. It suffices to treat the cases where $(f, g) \in S^\uparrow \times S^\downarrow$ or $(f, g) \in S^\downarrow \times S^\uparrow$.

If $(f, g) \in S^\uparrow \times S^\downarrow$, then $(-f, g) \in S^\downarrow \times S^\downarrow$. Similarly, if $(f, g) \in S^\downarrow \times S^\uparrow$, then $(-f, g) \in S^\uparrow \times S^\uparrow$. In both cases, we obtain

$$I(g) - I(f) = I(g) + I(-f) = I(g - f) \geq 0.$$

□

10.3.12 Definition. Let Ω be a set, S be a Riesz space on Ω , and $I : S \rightarrow \mathbb{R}$ an integral operator. For any mapping $f : \Omega \rightarrow \mathbb{R}$, we let

$$\bar{I}(f) := \inf_{u \in S^\uparrow, u \geq f} I(u), \quad \underline{I}(f) := \sup_{\ell \in S^\downarrow, \ell \leq f} I(\ell).$$

We denote by $\mathcal{L}^1(I)$ the set of mappings $f : \Omega \rightarrow \mathbb{R}$ such that

$$\bar{I}(f) = \underline{I}(f) \in \mathbb{R}.$$

A mapping f that belong to $\mathcal{L}^1(I)$ is said to be Daniell integrable. If f is Daniell integrable, we denote by $I(f)$ the real number $\bar{I}(f)$ (which is equal to $\underline{I}(f)$). Thus we obtain a mapping $I : \mathcal{L}^1(I) \rightarrow \mathbb{R}$.

Note that a mapping $f : \Omega \rightarrow \mathbb{R}$ belongs to $\mathcal{L}^1(I)$ if and only if, for any $\varepsilon > 0$, there exist $u \in S^\uparrow$ and $\ell \in S^\downarrow$ such that $\ell \leq f \leq u$ and that $I(u) - I(\ell) \leq \varepsilon$.

We denote by $\mathcal{L}_R^1(I)$ the set of mappings $f : \Omega \rightarrow \mathbb{R}$ such that

$$\sup_{\ell \in S, \ell \leq f} I(\ell) = \inf_{u \in S, u \geq f} I(u) \in \mathbb{R}.$$

A mapping $f : \Omega \rightarrow \mathbb{R}$ that belongs to $\mathcal{L}_R^1(I)$ is said to be *Riemann integrable* with respect to I . By definition, if a mapping $f : \Omega \rightarrow \mathbb{R}$ is Riemann integrable, then it is Daniell integrable.

10.3.13 Lemma. *Let Ω be a set, S be a Riesz space on Ω , and $I : S \rightarrow \mathbb{R}$ be an integral operator.*

(1) *For any mapping $f : \Omega \rightarrow \mathbb{R}$, one has*

$$\underline{I}(-f) = -\bar{I}(f), \quad \bar{I}(-f) = \underline{I}(f).$$

In particular, if $f \in \mathcal{L}^1(I)$, then $-f \in \mathcal{L}^1(I)$, and $I(-f) = -I(f)$.

(2) *For any mappings $f : \Omega \rightarrow \mathbb{R}$ and $g : \Omega \rightarrow \mathbb{R}$, one has*

$$\bar{I}(f + g) \leq \bar{I}(f) + \bar{I}(g), \quad \underline{I}(f + g) \geq \underline{I}(f) + \underline{I}(g)$$

In particular, if $(f, g) \in \mathcal{L}^1(I) \times \mathcal{L}^1(I)$, then $f + g \in \mathcal{L}^1(I)$, and one has

$$I(f + g) = I(f) + I(g).$$

(3) *For any $f : \Omega \rightarrow \mathbb{R}$ and any $\lambda \in \mathbb{R}_{>0}$, one has*

$$\bar{I}(\lambda f) = \lambda \bar{f}, \quad \underline{I}(\lambda f) = \lambda \underline{I}(f).$$

In particular, if $f \in \mathcal{L}^1(I)$, then $\lambda f \in \mathcal{L}^1(I)$, and $I(\lambda f) = \lambda I(f)$.

(4) *If $f : \Omega \rightarrow \mathbb{R}$ and $g : \Omega \rightarrow \mathbb{R}$ are two mappings such that $f \leq g$, then*

$$\underline{I}(f) \leq \underline{I}(g), \quad \bar{I}(f) \leq \bar{I}(g).$$

(5) *If f is an element of $S^\uparrow \cup S^\downarrow$ such that $I(f) \in \mathbb{R}$, then $f \in \mathcal{L}^1(I)$.*

Proof. (1) For any $u \in S^\uparrow$ such that $u \geq f$, one has $-u \in S^\downarrow$ and $-u \leq -f$. Hence

$$-I(u) = I(-u) \leq \underline{I}(-f).$$

Taking the supremum with respect to u , we get $-\bar{I}(f) \leq \underline{I}(-f)$. Applying this inequality to $-f$, we get $-\bar{I}(-f) \leq \underline{I}(f)$.

Similarly, for any $\ell \in S^\downarrow$ such that $\ell \leq f$, one has $-\ell \in S^\uparrow$ and $-\ell \geq -f$. Hence

$$-I(\ell) = I(-\ell) \geq \bar{I}(-f).$$

Taking the infimum with respect to ℓ , we obtain $-\underline{I}(f) \geq \bar{I}(-f)$. Applying this inequality to $-f$, we obtain $-\underline{I}(-f) \geq \bar{I}(f)$.

Therefore, one has

$$\bar{I}(-f) = -\underline{I}(f), \quad \underline{I}(-f) = -\bar{I}(f).$$

(2) For any $(\ell_1, \ell_2) \in S^\downarrow \times S^\downarrow$ such that $\ell_1 \leq f_1$ and $\ell_2 \leq f_2$, one has

$$\ell_1 + \ell_2 \leq f_1 + f_2.$$

Taking the supremum with respect to (ℓ_1, ℓ_2) , we obtain

$$\underline{I}(f_1) + \underline{I}(f_2) \leq \underline{I}(f_1 + f_2).$$

Similarly, for any $(u_1, u_2) \in S^\uparrow \times S^\uparrow$ such that

$$f_1 \leq u_1, \quad f_2 \leq u_2,$$

one has

$$f_1 + f_2 \leq u_1 + u_2.$$

Taking the infimum with respect to (u_1, u_2) , we obtain

$$\bar{I}(f_1 + f_2) \leq \bar{I}(u_1) + \bar{I}(u_2).$$

(3) One has

$$\underline{I}(\lambda f) = \sup_{\substack{\ell \leq \lambda f \\ \ell \in S^\downarrow}} I(\ell) = \sup_{\substack{\ell \leq f \\ \ell \in S^\downarrow}} I(\lambda \ell) = \lambda \underline{I}(f),$$

and

$$\lambda \bar{I}(f) = \lambda \inf_{\substack{u \geq f \\ u \in S^\uparrow}} I(\lambda u) = \sup_{\substack{u \geq \lambda f \\ u \in S^\uparrow}} I(u) = \bar{I}(\lambda f).$$

(4) For any $\ell \in S^\downarrow$ such that $\ell \leq f$, one has $\ell \leq g$ and hence $I(\ell) \leq \underline{I}(g)$.

Taking the supremum with respect to ℓ , we get $\underline{I}(f) \leq \underline{I}(g)$.

For any $u \in S^\uparrow$ such that $g \leq u$, one has $f \leq u$ and hence

$$\bar{I}(f) \leq I(u).$$

Taking the infimum with respect to u , we get

$$\bar{I}(f) \leq \bar{I}(g).$$

(5) We first consider the case where $f \in S^\uparrow$. By definition

$$\bar{I}(f) = I(f).$$

Moreover, there exists an increasing sequence $(f_n)_{n \in \mathbb{N}} \in S^\mathbb{N} \subseteq (S^\downarrow)^\mathbb{N}$ that converges pointwise to f (hence $f_n \leq f$ for any $n \in \mathbb{N}$) with

$$I(f) = \lim_{n \rightarrow +\infty} I(f_n).$$

Hence $\underline{I}(f) = I(f)$. Thus we deduce that $f \in \mathcal{L}^1(I)$ once $I(f) \in \mathbb{R}$.

The proof of the case where $f \in S^\downarrow$ is quite similar, so we omit the details. $\square$

10.3.14 Theorem (Beppo Levi). *Let $(f_n)_{n \geq 0}$ be a monotone sequence in $\mathcal{L}^1(I)$ that converges pointwisely to a limit mapping $f : \Omega \rightarrow \mathbb{R}$. If the limit*

$$\lim_{n \rightarrow +\infty} I(f_n)$$

is a real number, then $f \in \mathcal{L}^1(I)$ and

$$I(f) = \lim_{n \rightarrow +\infty} I(f_n).$$

Proof. Without loss of generality, we may assume that $(f_n)_{n \in \mathbb{N}}$ is increasing. Moreover, by replacing f_n by $f_n - f_0$, we may assume that f_0 is the constant zero mapping.

Let $\varepsilon > 0$. For any $n \in \mathbb{N}$, let $u_n \in S^\uparrow$ such that $f_n - f_{n-1} \leq u_n$ and that

$$I(f_n - f_{n-1}) \geq I(u_n) - \frac{\varepsilon}{2^n}.$$

Hence

$$f_n = \sum_{k=1}^n (f_k - f_{k-1}) \leq u_1 + \cdots + u_n$$

and

$$I(f_n) \geq \sum_{k=1}^n \left(I(u_k) - \frac{\varepsilon}{2^k} \right) \geq I(u_1) + \cdots + I(u_n) - \varepsilon.$$

Let

$$u = \sum_{n \in \mathbb{N}, n \geq 1} u_n \in S^\uparrow.$$

By definition

$$I(u) = \sum_{n \in \mathbb{N}} I(u_n).$$

One has $u \geq f$.

Let α be the limit of $(I(f_n))_{n \in \mathbb{N}}$. By definition, one has

$$\alpha \geq I(u) - \varepsilon \geq \bar{I}(f) - \varepsilon.$$

For any $n \in \mathbb{N}$, let $\ell_n \in S^\downarrow$ such that $\ell_n \leq f_n$ and $I(\ell_n) \geq I(f_n) - \varepsilon$. Since $\ell_n \leq f_n \leq f$, one has

$$\alpha - \varepsilon \leq \liminf_{n \rightarrow +\infty} I(\ell_n) \leq \underline{I}(f).$$

Thus we obtain

$$\alpha - \varepsilon \leq \underline{I}(f) \leq \bar{I}(f) \leq \alpha + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, we get

$$\bar{I}(f) = \underline{I}(f) = \alpha.$$

□

10.3.15 Theorem (Daniell). *The set $\mathcal{L}^1(I)$ forms a Riesz space on Ω , and $I : \mathcal{L}^1(I) \rightarrow \mathbb{R}$ is an integral operator extending $I : S \rightarrow \mathbb{R}$.*

Proof. By Lemma 10.3.13, $\mathcal{L}^1(I)$ is a vector subspace of $\mathbb{R}^\Omega$ and $I : \mathcal{L}^1(I) \rightarrow \mathbb{R}$ is an $\mathbb{R}$ -linear mapping such that, for any $(f, g) \in \mathcal{L}^1(\mathbb{R}) \times \mathcal{L}^1(\mathbb{R})$,

$$f \leq g \implies I(f) \leq I(g).$$

Let f_1 and f_2 be two elements of $\mathcal{L}^1(I)$. For any $\varepsilon > 0$, there exists $(\ell_1, \ell_2) \in S^\downarrow \times S^\downarrow$ and $(u_1, u_2) \in S^\uparrow \times S^\uparrow$, such that

$$\ell_1 \leq f_1 \leq u_1, \quad \ell_2 \leq f_2 \leq u_2,$$

and

$$I(u_1 - \ell_1) \leq \frac{\varepsilon}{2}, \quad I(u_2 - \ell_2) \leq \frac{\varepsilon}{2}.$$

Note that

$$\inf\{\ell_1, \ell_2\} \leq \inf\{f_1, f_2\} \leq \inf\{u_1, u_2\}.$$

Moreover, one has

$$\inf\{u_1, u_2\} - \inf\{\ell_1, \ell_2\} \leq (u_1 - \ell_1) + (u_2 - \ell_2).$$

Hence

$$\begin{aligned} \bar{I}(\inf\{f_1, f_2\}) - \underline{I}(\inf\{f_1, f_2\}) &\leq I(\inf\{u_1, u_2\}) - I(\inf\{\ell_1, \ell_2\}) \\ &\leq I(u_1 - \ell_1) + I(u_2 - \ell_2) \leq \varepsilon. \end{aligned}$$

Since $\varepsilon > 0$ is arbitrary, we obtain that

$$\underline{I}(\inf\{f_1, f_2\}) = \bar{I}(\inf\{f_1, f_2\}) \in \mathbb{R}.$$

Hence $\inf\{f_1, f_2\} \in \mathcal{L}^1(I)$.

Finally, by Beppo Levi's theorem, if $(f_n)_{n \in \mathbb{N}}$ is an increasing sequence in $\mathcal{L}^1(I)$ that converges pointwise to some $f \in \mathcal{L}^1(I)$, then one has

$$I(f) = \lim_{n \rightarrow +\infty} I(f_n).$$

Hence I is an integral operator. □

10.3.16 Remark. Let Ω be a set and S be a Riesz space on Ω . For any $f \in S$, let $|f|$ be the mapping that sends $\omega \in \Omega$ to $|f(\omega)|$. One has

$$|f| = \sup\{f, 0\} - \inf\{f, 0\} \in S.$$

In particular, if $I : S \rightarrow \mathbb{R}$ is an integral operator, then, for any $f \in \mathcal{L}^1(I)$, one has $|f| \in \mathcal{L}^1(I)$.

10.3.17 Theorem (Fatou's lemma). *Let $(f_n)_{n \in \mathbb{N}}$ be a sequence in $\mathcal{L}^1(I)$. Assume that there exists $g \in \mathcal{L}^1(I)$ such that $f_n \geq g$ for any $n \in \mathbb{N}$. Then one has*

$$\liminf_{n \rightarrow +\infty} f_n \in \mathcal{L}^1(I)^\dagger$$

and

$$I\left(\liminf_{n \rightarrow +\infty} f_n\right) \leq \liminf_{n \rightarrow +\infty} I(f_n).$$

Moreover, if

$$\liminf_{n \rightarrow +\infty} I(f_n) < +\infty,$$

then $\liminf_{n \rightarrow +\infty} f_n \in \mathcal{L}^1(I)$.

Proof. For any $n \in \mathbb{N}$, let g_n be the limit of the following decreasing sequence in $\mathcal{L}^1(I)$:

$$\left(\inf\{f_n, f_{n+1}, \dots, f_{n+k}\}\right)_{k \in \mathbb{N}}.$$

For any $k \in \mathbb{N}$, one has

$$f_n \geq \inf\{f_n, f_{n+1}, \dots, f_{n+k}\} \geq g,$$

and hence

$$I(f_n) \geq \lim_{k \rightarrow +\infty} I(\inf\{f_n, f_{n+1}, \dots, f_{n+k}\}) \geq I(g).$$

By the theorem of Beppo Levi, one has $g_n \in \mathcal{L}^1(I)$ and

$$I(g_n) = \lim_{k \rightarrow +\infty} I(\inf\{f_n, f_{n+1}, \dots, f_{n+k}\}) \leq I(f_n).$$

By definition, the sequence $(g_n)_{n \in \mathbb{N}}$ is increasing, and one has

$$\liminf_{n \rightarrow +\infty} f_n = \lim_{n \rightarrow +\infty} g_n.$$

Therefore,

$$\liminf_{n \rightarrow +\infty} f_n \in \mathcal{L}^1(I)^\dagger$$

and

$$I\left(\liminf_{n \rightarrow +\infty} f_n\right) = \lim_{n \rightarrow +\infty} I(g_n) \leq \liminf_{n \rightarrow +\infty} I(f_n).$$

In the case where

$$\liminf_{n \rightarrow +\infty} I(f_n) < +\infty,$$

one has

$$\liminf_{n \rightarrow +\infty} I(f_n) \in \mathcal{L}^1(I).$$

□

10.3.18 Theorem (Dominated convergence, Lebesgue). *Let $(f_n)_{n \in \mathbb{N}}$ be a sequence in $\mathcal{L}^1(I)$ that converges pointwise to a mapping f . Assume that there exists $g \in \mathcal{L}^1(I)$ such that*

$$\forall n \in \mathbb{N}, \quad |f_n| \leq g.$$

Then, one has $f \in \mathcal{L}^1(I)$ and

$$I(f) = \limsup_{n \rightarrow +\infty} I(f_n).$$

Proof. We apply Fatou's lemma to $(f_n)_{n \in \mathbb{N}}$ and $(-f_n)_{n \in \mathbb{N}}$ to get

$$\lim_{n \rightarrow +\infty} f_n \in \mathcal{L}^1(I)$$

and

$$I(f) \leq \liminf_{n \rightarrow +\infty} I(f_n), \quad I(f) = -I(-f) \geq -\liminf_{n \rightarrow +\infty} I(-f_n) = \limsup_{n \rightarrow +\infty} I(f_n).$$

Hence the sequence $(I(f_n))_{n \in \mathbb{N}}$ converges to $I(f)$. □

10.4 Set semirings

10.4.1 Definition. Let Ω be a non-empty set. We call *semiring* on Ω and subset $\mathcal{C}$ of $\mathcal{P}(\Omega)$ that satisfies the following conditions:

- (1) $\emptyset \in \mathcal{C}$,
- (2) for any $(A, B) \in \mathcal{C}^2$, $A \cap B \in \mathcal{C}$,
- (3) for any $(A, B) \in \mathcal{C}^2$ there exists a finite family $(C_i)_{i=1}^n$ of pairwise disjoint sets in $\mathcal{C}$ such that $B \setminus A = \bigcup_{i=1}^n C_i$.

10.4.2 Proposition. *Let Ω be a set and $\mathcal{C}$ be a semiring on Ω .*

- (1) *Let B be an element of $\mathcal{C}$ and $A_1, \dots, A_n$ be a finite family of elements of $\mathcal{C}$. Then $B \setminus (A_1 \cup \dots \cup A_n)$ can be written as the union of a family of pairwise disjoint elements of $\mathcal{C}$.*
- (2) *Let Θ be a finite subset of $\mathcal{C}$. There exists a finite family Φ of pairwise disjoint elements in $\mathcal{C}$ such that each element of Θ can be written as the union of some elements of Φ .*

(3) Let $\mathcal{A}$ be the following set

$$\{A \in \mathcal{P}(\Omega) \mid \exists n \in \mathbb{N}, \exists (A_1, \dots, A_n) \in \mathcal{C}^n, A = A_1 \cup \dots \cup A_n\}.$$

Then, any element of $\mathcal{A}$ can be written as the union of a finite family of pairwise disjoint elements of $\mathcal{C}$. Moreover, for any $(A, A') \in \mathcal{A} \times \mathcal{A}$, one has

$$\{A \cup A', A \cap A', A \setminus A'\} \subseteq \mathcal{A}.$$

Proof. (1) We reason by induction on n . The case where $n = 0$ is trivial. In what follows, we assume that $n \geq 1$. By the induction hypothesis, there exists a family $(C_i)_{i=1}^m$ of pairwise disjoint elements of $\mathcal{C}$, such that

$$B \setminus (A_1 \cup \dots \cup A_{n-1}) = \bigcup_{i=1}^m C_i.$$

Since $\mathcal{C}$ is a semialgebra, for any $i \in \{1, \dots, m\}$, $C_i \setminus A_n$ can be written as the union of a family of pairwise disjoint elements $(D_{i,j})_{j=1}^{d_i}$ in $\mathcal{C}$. Since $C_1, \dots, C_m$ are pairwise disjoint, we obtain that

$$(D_{i,j})_{i \in \{1, \dots, m\}, j \in \{1, \dots, d_i\}}$$

is a pairwise disjoint family of elements of $\mathcal{C}$, such that

$$B \setminus (A_1 \cup \dots \cup A_n) = \bigcup_{i=1}^m C_i \setminus A_n = \bigcup_{i=1}^m \bigcup_{j=1}^{d_i} D_{i,j},$$

as required.

(2) We reason by induction on the cardinality n of Θ . The case where $n = 0$ is trivial. In the following, we assume that $n \geq 1$ and that the statement is true for any subset of cardinality $< n$. Let Θ' be a subset of cardinality $n - 1$ of Θ . Let Φ' be a family of pairwise disjoint elements of $\mathcal{C}$ such that each element of Θ' can be written as the union of some elements of Φ' .

Let B be the only element of $\Theta \setminus \Theta'$. Let $(C_i)_{i=1}^m$ be a family of pairwise disjoint elements of $\mathcal{C}$ such that

$$B \setminus \bigcup_{A \in \Theta'} A = C_1 \cup \dots \cup C_m.$$

For each $C \in \Phi'$, let U_C be a finite family of pairwise disjoint elements of $\mathcal{C}$ such that

$$C \setminus B = \bigcup_{D \in U_C} D.$$

Let

$$\Phi = \left(\bigcup_{C \in \Phi'} U_C \right) \cup \{C \cap B \mid C \in \Phi'\} \cup \{C_1, \dots, C_m\}.$$

This is a family of pairwise disjoint elements of $\mathcal{C}$. Note that any element $C \in \Phi'$ can be written as the union of a family of elements of Φ as follows:

$$C = (C \cap B) \cup (C \setminus B) = (C \cap B) \cup \bigcup_{D \in U_C} D.$$

Therefore, any element of Θ' can also be written as the union of a family of elements of Φ . Moreover, one has

$$B = \left(\bigcup_{C \in \Phi'} (C \cap B) \right) \cup \left(B \setminus \bigcup_{A \in \Theta'} A \right) = \left(\bigcup_{C \in \Phi'} (C \cap B) \right) \cup C_1 \cup \dots \cup C_m.$$

Hence, any element of Θ can be written as the union of some elements of Φ .

(3) The first statement comes from (2). If $A = A_1 \cup \dots \cup A_n$ and $A' = A'_1 \cup \dots \cup A'_m$ are elements of $\mathcal{A}$, then, by definition one has $A \cup A' \in \mathcal{A}$. Moreover,

$$A \cap A' = \bigcup_{(i,j) \in \{1, \dots, n\} \times \{1, \dots, m\}} A_i \cap A'_j \in \mathcal{A}.$$

Finally, by (1), for any $i \in \{1, \dots, n\}$, one has $A_i \setminus A' \in \mathcal{A}$. Therefore,

$$A \setminus A' = \bigcup_{i=1}^n A_i \setminus A' \in \mathcal{A}.$$

□

10.4.3 Proposition. *Let Ω be a non-empty set, $\mathcal{C}$ be a semialgebra on Ω , and S be the vector subspace of $\mathbb{R}^\Omega$ generated by mappings of the form $\mathbb{1}_A$, where $A \in \mathcal{C}$.*

(1) *Any pair $(f, g) \in S^2$ can be written in the form*

$$f = \sum_{i=1}^n a_i \mathbb{1}_{C_i}, \quad g = \sum_{i=1}^n b_i \mathbb{1}_{C_i},$$

where n is a natural number, $(a_1, \dots, a_n) \in \mathbb{R}^n$, $(b_1, \dots, b_n) \in \mathbb{R}^n$, and $C_1, \dots, C_n$ are pairwise disjoint sets in $\mathcal{C}$.

(2) *S is a Riesz space.*

Proof. (1) Suppose that

$$f = \sum_{A \in \Theta_f} \lambda_A \mathbb{1}_A, \quad g = \sum_{B \in \Theta_g} \mu_B \mathbb{1}_B,$$

where Θ_f and Θ_g . Let $\Theta = \Theta_f \cup \Theta_g$. By (2), there exists a finite family $(C_i)_{i=1}^n$ of pairwise disjoint elements of $\mathcal{C}$ such that any element of Θ can be written as the union of some elements among $C_1, \dots, C_n$. For any $i \in \{1, \dots, n\}$, let

$$a_i = \sum_{A \in \Theta_f, A \cap C_i \neq \emptyset} \lambda_A, \quad b_i = \sum_{B \in \Theta_g, B \cap C_i \neq \emptyset} \mu_B.$$

Then one has

$$f = \sum_{i=1}^n a_i \mathbb{1}_{C_i}, \quad g = \sum_{i=1}^n b_i \mathbb{1}_{C_i}.$$

(4) Let f and g be two elements of S . We write f and g as

$$f = \sum_{i=1}^n a_i \mathbb{1}_{C_i}, \quad g = \sum_{i=1}^n b_i \mathbb{1}_{C_i},$$

where n is a natural number, $(a_1, \dots, a_n) \in \mathbb{R}^n$, $(b_1, \dots, b_n) \in \mathbb{R}^n$, and $C_1, \dots, C_n$ are pairwise disjoint sets in $\mathcal{C}$. Then one has

$$\inf\{f, g\} = \sum_{i=1}^n \min\{a_i, b_i\} \mathbb{1}_{C_i}.$$

□

10.4.4 Example. Let $\mathcal{C}$ be the set of elements of the form $]a, b]$, where $(a, b) \in \mathbb{R} \times \mathbb{R}$, $a \leq b$. Then $\mathcal{C}$ is a semialgebra on $\mathbb{R}$. In fact, if $]a, b]$ and $]c, d]$ are two elements of $\mathcal{C}$, then one has

$$]a, b] \setminus]c, d] = \begin{cases} \emptyset, & \text{if } c \leq a \text{ and } b \leq d, \\]a, c], & \text{if } c > a \text{ and } b \leq d, \\]d, b], & \text{if } c \leq a \text{ and } b > d, \\]a, c] \cup]d, b], & \text{if } c > a \text{ and } b > d. \end{cases}$$

Therefore, $\mathcal{C}$ is a semialgebra.

10.4.5 Example. Let X be a topological space. If $f : X \rightarrow \mathbb{R}$ is a mapping. We call *support* of f the closure of the set

$$\{x \in X \mid f(x) \neq 0\},$$

denoted as $\text{Supp}(f)$. Note that, for any $f \in \mathbb{R}^X$ and any $c \in \mathbb{R} \setminus \{0\}$

$$\text{Supp}(f) = \text{Supp}(|f|) = \text{Supp}(cf).$$

Moreover, for any $(f, g) \in \mathbb{R}^X \times \mathbb{R}^X$,

$$\begin{aligned}\text{Supp}(f + g) &\subseteq \text{Supp}(f) \cup \text{Supp}(g), \\ \text{Supp}(fg) &\subseteq \text{Supp}(f) \cap \text{Supp}(g)\end{aligned}$$

Let X be a topological space and $C_c(X)$ be the set of mappings $f : X \rightarrow \mathbb{R}$ which are of compact support. Then $C_c(X)$ is a vector subspace of $\mathbb{R}^X$. Note that, for any $(f, g) \in C_c(X)$, one has

$$\inf\{f, g\} = \frac{(f + g) - |f - g|}{2} \in C_c(X).$$

Therefore, $C_c(X)$ is a Riesz space over X .

10.5 σ -additive functions

10.5.1 Definition. Let Ω be a non-empty set, $\mathcal{C}$ be a subset of $\mathcal{P}(\Omega)$, and $\mu : \mathcal{C} \rightarrow \mathbb{R}_{\geq 0}$ be a mapping. We say that μ is additive if, for any finite family $(A_i)_{i=1}^n$ of pairwise disjoint sets in $\mathcal{C}$ such that $\bigcup_{i=1}^n A_i \in \mathcal{C}$, one has

$$\mu\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n \mu(A_i).$$

Note that, if $\emptyset \in \mathcal{C}$, then one has

$$\mu(\emptyset) = \mu(\emptyset \cup \emptyset) = \mu(\emptyset) + \mu(\emptyset).$$

Hence $\mu(\emptyset) = 0$.

10.5.2 Example. Let $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ be an increasing right continuous mapping. Let $\mathcal{C}$ be the set of intervals of the form $]a, b]$, where $(a, b) \in \mathbb{R} \times \mathbb{R}$, $a \leq b$. Then the mapping $\mu_\varphi : \mathcal{C} \rightarrow \mathbb{R}_{\geq 0}$ sending $]a, b]$ to $\varphi(b) - \varphi(a)$ is an additive mapping. In fact, if $a_0 < a_1 < \dots < a_n$ are real numbers, so that

$$]a_0, a_n] = \bigcup_{i=1}^n]a_{i-1}, a_i],$$

then

$$\sum_{i=1}^n \mu_\varphi(]a_{i-1}, a_i]) = \sum_{i=1}^n (\varphi(a_i) - \varphi(a_{i-1})) = \varphi(a_n) - \varphi(a_0) = \mu_\varphi(]a_0, a_n]).$$

10.5.3 Proposition. *Let Ω be set, $\mathcal{C}$ be a semiring on Ω , $\mu : \mathcal{C} \rightarrow \mathbb{R}_{\geq 0}$ be an additive mapping, S be the $\mathbb{R}$ -vector subspace of $\mathbb{R}^\Omega$ generated by mappings of the form $\mathbb{1}_A$, where with $A \in \mathcal{C}$.*

- (1) *There is a unique $\mathbb{R}$ -linear mapping $I : S \rightarrow \mathbb{R}$ such that $I(\mathbb{1}_A) = \mu(A)$ for any $A \in \mathcal{C}$.*
- (2) *Let*

$$\mathcal{A} := \{A \in \mathcal{P}(\Omega) \mid \exists n \in \mathbb{N}, \exists (A_1, \dots, A_n) \in \mathcal{C}^n, A = A_1 \cup \dots \cup A_n\}.$$

The the mapping μ extends in a unique way to an additive mapping from $\mathcal{A}$ to $\mathbb{R}_{\geq 0}$

Proof. (1) Since any element of S is a linear combination of the form $\mathbb{1}_A$, $A \in \mathcal{C}$, if the $\mathbb{R}$ -linear mapping I exists, it is unique, and it sends

$$\sum_{i=1}^n a_i \mathbb{1}_{A_i} \in S$$

to

$$\sum_{i=1}^n a_i \mu(A_i),$$

where $n \in \mathbb{N}$, $(\lambda_1, \dots, \lambda_n) \in \mathbb{R}^n$, $(A_1, \dots, A_n) \in \mathcal{C}^n$. It remains to check that the above $\mathbb{R}$ -linear mapping is well defined, namely, if a mapping $f \in S$ is written as

$$f = \sum_{A \in \Theta} \lambda_A \mathbb{1}_A = \sum_{B \in \Theta'} \lambda'_B \mathbb{1}_B,$$

where Θ and Θ' are finite subsets of $\mathcal{C}_0$, and $(\lambda_A)_{A \in \Theta}$ and $(\lambda'_B)_{B \in \Theta'}$ are families of real numbers, then

$$\sum_{A \in \Theta} \lambda_A \mu(A) = \sum_{B \in \Theta'} \lambda'_B \mu(B).$$

By Proposition 10.4.3 (3), there exist a family of pairwise disjoint sets $C_1, \dots, C_n$ in $\mathcal{C}$, such that each $A \in \Theta \cup \Theta'$ can be written as the union of certain sets among $C_1, \dots, C_n$, together with $(b_1, \dots, b_n) \in \mathbb{R}^n$, such that

$$f = \sum_{i=1}^n b_i \mathbb{1}_{C_i}.$$

Thus

$$\forall i \in \{1, \dots, n\}, b_i = \sum_{A \in \Theta, A \cap C_i \neq \emptyset} \lambda_A = \sum_{B \in \Theta', B \cap C_i \neq \emptyset} \lambda'_B.$$

Therefore,

$$\begin{aligned} \sum_{A \in \Theta} \lambda_A \mu(A) &= \sum_{A \in \Theta} \lambda_A \sum_{\substack{i \in \{1, \dots, n\}, \\ A \cap C_i \neq \emptyset}} \mu(C_i) = \sum_{i=1}^n \mu(C_i) \sum_{A \in \Theta, A \cap C_i \neq \emptyset} \lambda_A \\ &= \sum_{i=1}^n \mu(C_i) \sum_{B \in \Theta', B \cap C_i \neq \emptyset} \lambda'_B = \sum_{B \in \Theta'} \sum_{\substack{i \in \{1, \dots, n\} \\ B \cap C_i \neq \emptyset}} \mu(C_i) = \sum_{B \in \Theta'} \lambda'_B \mu(B). \end{aligned}$$

Therefore, the $\mathbb{R}$ -linear mapping $I : S \rightarrow \mathbb{R}$ is well defined.

(2) Note that any element of $\mathcal{A}$ can be written as the union of a finite family of pairwise disjoint elements of $\mathcal{C}$, we obtain that, if the extension of μ exists, it is necessarily unique.

In the following, we prove the existence of the extension. Note that for any $A \in \mathcal{A}$, one has $\mathbb{1}_A \in S$ and $I(\mathbb{1}_A) \in \mathbb{R}_{\geq 0}$. Thus

$$(A \in \mathcal{A}) \longmapsto I(\mathbb{1}_A)$$

is a mapping from $\mathcal{A}$ to $\mathbb{R}_{\geq 0}$ that extends μ . Since the mapping I is $\mathbb{R}$ -linear, we obtain that the extended mapping is additive. $\square$

10.5.4 Definition. Let $\mathcal{C}$ be a subset of $\mathcal{P}(\Omega)$. We say that a mapping $\mu : \mathcal{C} \rightarrow \mathbb{R}_{\geq 0}$ is σ -additive if, for any countable set Θ and any family $(C_i)_{i \in \Theta}$ of pairwise disjoint elements of $\mathcal{C}$ such that $\bigcup_{i \in \Theta} C_i \in \mathcal{C}$, one has

$$\mu\left(\bigcup_{i \in \Theta} C_i\right) = \sum_{i \in \Theta} \mu(C_i),$$

where the infinite sum

$$\sum_{i \in \Theta} \mu(C_i)$$

should be understood as

$$\sup_{\substack{J \subseteq \Theta \\ J \text{ is finite}}} \sum_{j \in J} \mu(C_j).$$

10.5.5 Proposition. Let $\mathcal{C}$ be a semiring on Ω and

$$\mathcal{A} := \{A \in \mathcal{P}(\Omega) \mid \exists n \in \mathbb{N}, \exists (A_1, \dots, A_n) \in \mathcal{C}^n, A = A_1 \cup \dots \cup A_n\}.$$

Let $\mu : \mathcal{C} \rightarrow \mathbb{R}_{\geq 0}$ be an additive mapping, that extends in a unique way to an additive mapping $\mu : \mathcal{A} \rightarrow \mathbb{R}_{\geq 0}$. Let S be the vector subspace of $\mathbb{R}^\Omega$ generated by mappings of the form $\mathbb{1}_A$, $A \in \mathcal{C}$. The following conditions are equivalent.

- (1) The mapping $\mu : \mathcal{C} \rightarrow \mathbb{R}_{\geq 0}$ is σ -additive.
- (2) The extended mapping $\mu : \mathcal{A} \rightarrow \mathbb{R}_{\geq 0}$ is σ -additive.
- (3) For any decreasing sequence $(A_n)_{n \in \mathbb{N}} \in \mathcal{A}^{\mathbb{N}}$ such that

$$\bigcap_{n \in \mathbb{N}} A_n = \emptyset,$$

one has

$$\lim_{n \rightarrow +\infty} \mu(A_n) = 0.$$

- (4) The $\mathbb{R}$ -linear mapping $I : S \rightarrow \mathbb{R}$ that sends, for any $A \in \mathcal{C}$, $\mathbb{1}_A$ to $\mu(A)$, is an integral operator.

Proof. “(1) $\implies$ (2)”: Let $(A_n)_{n \in \mathbb{N}}$ be a family of pairwise disjoint elements in $\mathcal{A}$. For any $n \in \mathbb{N}$, we write A_n in the form

$$A_n = \bigcup_{j=0}^{d_n} C_{n,j},$$

where $(C_{n,j})_{j=0}^{d_n}$ is a family of pairwise disjoint elements in $\mathcal{C}$. Let $A = \bigcup_{n \in \mathbb{N}} A_n$. We also write A as a disjoint union of elements in $\mathcal{C}$:

$$A = B_1 \cup \dots \cup B_m.$$

For any $i \in \{1, \dots, m\}$, one has

$$B_i = B_i \cap A = \bigcup_{n \in \mathbb{N}} \bigcup_{j=0}^{d_n} (B_i \cap C_{n,j}).$$

Since μ is σ -additive on $\mathcal{C}$, we obtain that

$$\mu(B_i) = \sum_{n \in \mathbb{N}} \sum_{j=0}^{d_n} \mu(B_i \cap C_{n,j}) = \lim_{N \rightarrow +\infty} \sum_{n=0}^N \sum_{j=0}^{d_n} \mu(B_i \cap C_{n,j}).$$

Therefore,

$$\begin{aligned} \mu(A) &= \sum_{i=1}^m \mu(B_i) = \lim_{N \rightarrow +\infty} \sum_{i=1}^m \sum_{n=0}^N \sum_{j=0}^{d_n} \mu(B_i \cap C_{n,j}) \\ &= \lim_{N \rightarrow +\infty} \sum_{n=0}^N \sum_{i=1}^m \sum_{j=0}^{d_n} \mu(B_i \cap C_{n,j}) = \lim_{N \rightarrow +\infty} \sum_{n=0}^N \mu(A_n) = \sum_{n \in \mathbb{N}} \mu(A_n). \end{aligned}$$

“(2) $\implies$ (3)”: For any $n \in \mathbb{N}$, let $B_n = A_n \setminus A_{n+1}$. By Proposition 10.4.2, one has $B_n \in \mathcal{A}$. Moreover, since $\mu : \mathcal{A} \rightarrow \mathbb{R}_{\geq 0}$ is additive, one has

$$\mu(A_n) = \mu(B_n) + \mu(A_{n+1}).$$

Therefore, one has

$$\mu(A_0) = \mu(B_0) + \cdots + \mu(B_{N-1}) + \mu(A_N)$$

for any $N \in \mathbb{N}$. Since A_0 is the disjoint union of $(B_n)_{n \in \mathbb{N}}$, one has

$$\mu(A_0) = \lim_{N \rightarrow +\infty} \sum_{n=0}^{N-1} \mu(B_n) = \lim_{N \rightarrow +\infty} (\mu(A_0) - \mu(A_N)),$$

so that

$$\lim_{N \rightarrow +\infty} \mu(A_N) = 0.$$

“(3) $\implies$ (4)”: Let $(f_n)_{n \in \mathbb{N}}$ be a decreasing sequence in S that converges pointwise to 0. Let

$$B := \{\omega \in \Omega \mid f_0(\omega) > 0\} \in \mathcal{A}, \quad M = \max\{f_0(\omega) \mid \omega \in \Omega\}.$$

For any $\varepsilon > 0$ and any $n \in \mathbb{N}$, let

$$A_n^\varepsilon := \{\omega \in \Omega \mid f_n(\omega) \geq \varepsilon\} \in \mathcal{A}.$$

Note that $(A_n^\varepsilon)_{n \in \mathbb{N}}$ is a decreasing family in $\mathcal{A}$. Moreover, since $(f_n)_{n \in \mathbb{N}}$ converges pointwise to the zero mapping, we obtain that $\bigcap_{n \in \mathbb{N}} A_n^\varepsilon = \emptyset$. Note that

$$\forall n \in \mathbb{N}, \quad 0 \leq f_n \leq \varepsilon \mathbb{1}_B + M \mathbb{1}_{A_n^\varepsilon},$$

so that

$$\forall n \in \mathbb{N}, \quad 0 \leq I(f_n) \leq \varepsilon I(\mathbb{1}_B) + MI(\mathbb{1}_{A_n^\varepsilon}) = \varepsilon \mu(B) + M \mu(A_n^\varepsilon).$$

Taking the superior limit when $n \rightarrow +\infty$, we get

$$\limsup_{n \rightarrow +\infty} I(f_n) \leq \varepsilon \mu(B),$$

which leads to

$$\lim_{n \rightarrow +\infty} I(f_n) = 0,$$

as required.

“(4) $\implies$ (1)” : Let $(C_n)_{n \in \mathbb{N}}$ be a family of pairwise disjoint sets in $\mathcal{C}$, such that

$$A := \bigcup_{n \in \mathbb{N}} C_n \in \mathcal{C}.$$

For any $n \in \mathbb{N}$, let

$$f_n := \sum_{k=0}^n \mathbb{1}_{C_k}.$$

Then the sequence $(f_n)_{n \in \mathbb{N}}$ is increasing and converge pointwise to $\mathbb{1}_A$. Therefore

$$\mu(A) := I(\mathbb{1}_A) = \lim_{n \rightarrow +\infty} I(f_n) = \lim_{n \rightarrow +\infty} \sum_{k=0}^n \mu(C_k) = \sum_{k \in \mathbb{N}} \mu(C_k).$$

□

10.5.6 Proposition. Let $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ be an increasing and right continuous mapping. Let $\mathcal{C} \subseteq \mathcal{P}(\mathbb{R})$ be the set consisting of intervals of the form $]a, b]$, where $(a, b) \in \mathbb{R} \times \mathbb{R}$, $a < b$. Let S be the vector subspace of $\mathbb{R}^{\mathbb{R}}$ generated by mappings of the form $\mathbb{1}_A$ with $A \in \mathcal{C}$. Let $I_\varphi : S \rightarrow \mathbb{R}$ be the $\mathbb{R}$ -linear mapping that sends $\mathbb{1}_{]a, b]}$ to $\varphi(b) - \varphi(a)$. Let $\mathcal{A}$ be the subset of $\mathcal{P}(\mathbb{R})$ consisting of sets that could be written as a union of finitely many elements of $\mathcal{C}$.

- (1) Let $\varepsilon > 0$ and $A \in \mathcal{A}$, $A \neq \emptyset$. Then there exists $B \in \mathcal{A}$ such that $\emptyset \neq \overline{B} \subseteq A$ and $I_\varphi(\mathbb{1}_A) - I_\varphi(\mathbb{1}_B) \leq \varepsilon$.
- (2) I_φ is an integral operator.

Proof. (1) We first consider the case where $A \in \mathcal{C}$, namely A is of the form $]a, b]$, where $a < b$. By the right continuity of φ , there exists $a' \in]a, b[$ such that $\varphi(a') - \varphi(a) \leq \varepsilon$. Let $B =]a', b]$. One has

$$\overline{B} = [a', b] \subseteq]a, b].$$

Moreover,

$$I_\varphi(\mathbb{1}_B) = \varphi(b) - \varphi(a'), \quad I_\varphi(\mathbb{1}_A) = \varphi(b) - \varphi(a),$$

and hence

$$I_\varphi(\mathbb{1}_A) - I_\varphi(\mathbb{1}_B) = \varphi(a') - \varphi(a) \leq \varepsilon.$$

In general, the set A can be written as a disjoint union $A_1 \cup \dots \cup A_n$ with $\{A_1, \dots, A_n\} \subseteq \mathcal{C}$. For any $i \in \{1, \dots, n\}$, there exists $B_i \in \mathcal{C}$ with $\emptyset \neq \overline{B_i} \subseteq A_i$ and $I_\varphi(\mathbb{1}_{A_i}) - I_\varphi(\mathbb{1}_{B_i}) \leq \varepsilon/n$. Let $B = \bigcup_{i=1}^n B_i$. Then

$$I_\varphi(\mathbb{1}_A) - I_\varphi(\mathbb{1}_B) = \sum_{i=1}^n I_\varphi(\mathbb{1}_{A_i}) - I_\varphi(\mathbb{1}_{B_i}) \leq \varepsilon.$$

(2) Let $(A_n)_{n \in \mathbb{N}}$ be a decreasing sequence in $\mathcal{A}$ such that

$$\bigcap_{n \in \mathbb{N}} A_n = \emptyset.$$

Let $\varepsilon > 0$. For any $n \in \mathbb{N}$, let $B_n \in \mathcal{A}$ such that $\overline{B_n} \subseteq A_n$ and

$$I_\varphi(\mathbb{1}_{A_n}) - I_\varphi(\mathbb{1}_{B_n}) \leq \frac{\varepsilon}{2^n}.$$

Note that $\overline{B_n}$ is compact. For any $n \in \mathbb{N}$, let $C_n = B_0 \cap \cdots \cap B_n \subseteq \overline{B_0} \cap \cdots \cap \overline{B_n}$. Since $\bigcap_{n \in \mathbb{N}} A_n = \emptyset$, we obtain that $\bigcap_{n \in \mathbb{N}} \overline{B_n} = \emptyset$. Hence there exists $N \in \mathbb{N}$ such that

$$\bigcap_{n=0}^N \overline{B_n} = \emptyset.$$

Moreover,

$$B_n \setminus C_n = B_n \setminus (B_n \cap C_{n-1}) = B_n \setminus C_{n-1} \subseteq A_n \setminus C_{n-1} \subseteq A_{n-1} \setminus C_{n-1}.$$

Hence

$$I_\varphi(\mathbb{1}_{A_n \setminus C_n}) = I_\varphi(\mathbb{1}_{B_n \setminus C_n}) + I_\varphi(\mathbb{1}_{A_n \setminus B_n}) \leq I_\varphi(\mathbb{1}_{A_{n-1} \setminus C_{n-1}}) + \frac{\varepsilon}{2^n}.$$

So

$$\forall n \geq N, \quad I_\varphi(\mathbb{1}_{A_n}) \leq \frac{\varepsilon}{2^n} + \frac{\varepsilon}{2^{n-1}} + \cdots + \frac{\varepsilon}{2} \leq \varepsilon.$$

Thus

$$\lim_{n \rightarrow +\infty} I_\varphi(\mathbb{1}_{A_n}) = 0.$$

□

10.6 Measurable space

10.6.1 Definition. Let Ω be a set. We call σ -algebra on Ω any subset $\mathcal{A}$ of $\mathcal{P}(\Omega)$ that satisfies the following conditions:

(1) for any countable set I and any $(A_i)_{i \in I} \in \mathcal{A}^I$, one has

$$\bigcup_{i \in I} A_i \in \mathcal{A},$$

(2) for any $A \in \mathcal{A}$, one has $\Omega \setminus A \in \mathcal{A}$.

Note that the condition (1) implies that $\emptyset \in \mathcal{A}$, and hence, by condition (2), $\Omega \in \mathcal{A}$. Moreover, if I is a non-empty countable set and $(A_i)_{i \in I} \in \mathcal{A}^I$, one has

$$\bigcap_{i \in I} A_i = \Omega \setminus \left(\bigcup_{i \in I} (\Omega \setminus A_i) \right) \in \mathcal{A}.$$

If Ω is a set and $\mathcal{A}$ is a σ -algebra on Ω , then $(\Omega, \mathcal{A})$ is called a *measurable space*.

10.6.2 Proposition. *Let Ω be a set. Let J be a non-empty set and $(\mathcal{A}_j)_{j \in J}$ be a family of σ -algebras on Ω , then $\mathcal{A} := \bigcap_{j \in J} \mathcal{A}_j$ is a σ -algebra on Ω .*

Proof. Let I be a countable set and $(A_i)_{i \in I} \in \mathcal{A}^I$. For any $j \in J$ one has $(A_i)_{i \in I} \in \mathcal{A}_j$ and hence

$$\bigcup_{i \in I} A_i \in \mathcal{A}_j.$$

Thus

$$\bigcup_{i \in I} A_i \in \bigcap_{j \in J} \mathcal{A}_j = \mathcal{A}.$$

Let $A \in \mathcal{A}$. For any $j \in J$ one has $A \in \mathcal{A}_j$ and hence $\Omega \setminus A \in \mathcal{A}_j$. Thus we obtain

$$\Omega \setminus A \in \bigcap_{j \in J} \mathcal{A}_j.$$

□

10.6.3 Example. Let Ω be a set. Then $\mathcal{P}(\Omega)$ is a σ -algebra on Ω . Moreover, if $\mathcal{C}$ is a subset of $\mathcal{P}(\Omega)$, we denote by $\sigma(\mathcal{C})$ the intersection of all σ -algebras containing $\mathcal{C}$. It is a σ -algebra on Ω , which is the smallest σ -algebra containing $\mathcal{C}$. We call it the σ -algebra generated by $\mathcal{C}$.

10.6.4 Example. Let $(X, \mathcal{T})$ be a topological space, the σ -algebra generated by the topology $\mathcal{T}$ is called the *Borel σ -algebra* on $(X, \mathcal{T})$.

10.6.5 Proposition. *Let A be a subset of $[-\infty, +\infty]$. We define a binary relation $\sim$ on A as follows: for $(x, y) \in A \times A$, $x \sim y$ if and only if there exists an interval J contained in A such that $\{x, y\} \subseteq J$. Then $\sim$ is an equivalence relation on A and each equivalence class of $\sim$ is an interval.*

Proof. By definition the binary relation $\sim$ is reflexive and symmetric. If $x \sim y$ and $y \sim z$, then there exists intervals J_1 and J_2 contained in A , such that $\{x, y\} \subseteq J_1$, $\{y, z\} \subseteq J_2$. Since $J_1 \cap J_2 \neq \emptyset$, it is an interval contained in A . Moreover,

$\{x, z\} \subseteq J_1 \cup J_2$. Hence $x \sim z$. Thus $\sim$ is an equivalence relation. We can then write A as a disjoint union of equivalence classes of $\sim$.

It remains to check that each equivalence class of $\sim$ is an interval. Let x and y be two elements in the same equivalence class α of $\sim$ such that $x < y$. Since $x \sim y$, there is an interval J such that $\{x, y\} \subseteq J \subseteq A$. Hence $[x, y] \subseteq A$. Therefore, all elements of $[x, y]$ are equivalent to x and to y under $\sim$. Hence $[x, y] \subseteq \alpha$. Therefore, α is an interval. $\square$

10.6.6 Remark. We equip the extended real line $[-\infty, +\infty]$ with the order topology. Let U be an open subset of $[-\infty, +\infty]$. Let $\sim$ be the equivalence relation on U as defined in the previous proposition. Let α be an equivalence class of $\sim$. For any $x \in \alpha$, there is a neighbourhood of x contained in U which is an interval. Hence α is a neighbourhood of x . This observation shows that the equivalence class α is an open subset. In particular, any open subset $[-\infty, +\infty]$ can be written as a disjoint union of intervals which are open in $[-\infty, +\infty]$. Note that each interval that is open in $[-\infty, +\infty]$ contains at least a rational number. We deduce that U is a disjoint union of countably many intervals that are open in $[-\infty, +\infty]$.

10.6.7 Example. We equip the extended real line $[-\infty, +\infty]$ with the order topology $\mathcal{T}$. Then its Borel σ -algebra $\sigma(\mathcal{T})$ is generated by intervals of the form $[-\infty, b[$ for $b \in \mathbb{Q}$. In fact, if we denote by $\mathcal{A}$ the σ -algebra

$$\sigma(\{[-\infty, b[\mid b \in \mathbb{Q}\}),$$

then by definition one has $\mathcal{A} \subseteq \sigma(\mathcal{T})$

Conversely, for any $x \in \mathbb{R} \cup \{+\infty\}$, one has

$$[-\infty, x[= \bigcup_{b \in \mathbb{Q}, b < x} [-\infty, b[\in \mathcal{A}.$$

We then deduce that, for any $x \in [-\infty, +\infty]$, one has

$$[x, +\infty] = [-\infty, +\infty] \setminus [-\infty, x[\in \mathcal{A}.$$

Finally, for any $x \in \mathbb{R} \cup \{-\infty\}$, one has

$$]x, +\infty] = \bigcup_{a \in \mathbb{Q}, a > x} [a, +\infty] \in \mathcal{A}.$$

Moreover, for any $(a, b) \in [-\infty, +\infty]^2$ such that $a < b$, one has

$$]a, b[= [-\infty, b[\cap]a, +\infty].$$

Therefore, all intervals that are open in $[-\infty, +\infty]$ belongs to $\mathcal{A}$. Finally, by Remark 10.6.6, we obtain that any open subset of $[-\infty, +\infty]$ belongs to $\mathcal{A}$.

10.6.8 Definition. Let $(\Omega_1, \mathcal{A}_1)$ and $(\Omega_2, \mathcal{A}_2)$ be measurable spaces, and $f : \Omega_1 \rightarrow \Omega_2$ be a mapping. If, for any $A \in \mathcal{A}_2$, one has $f^{-1}(A) \in \mathcal{A}_1$, then we say that the mapping f is $\mathcal{A}_1$ -measurable. In the case where there is no ambiguity on $\mathcal{A}_1$, we just say that f is measurable.

10.6.9 Example. Let $(\Omega, \mathcal{A})$ be a measurable space and A is an element of $\mathcal{A}$. Then $\mathbb{1}_A$ is $\mathcal{A}$ -measurable.

10.6.10 Proposition. Let $(\Omega_1, \mathcal{A}_1)$, $(\Omega_2, \mathcal{A}_2)$ and $(\Omega_3, \mathcal{A}_3)$ be measurable spaces, $f : \Omega_1 \rightarrow \Omega_2$ and $g : \Omega_2 \rightarrow \Omega_3$ be mappings. If f and g are both measurable, so is $g \circ f$.

Proof. If $A \in \mathcal{A}_3$, then $g^{-1}(A) \in \mathcal{A}_2$ and hence

$$(g \circ f)^{-1}(A) = f^{-1}(g^{-1}(A)) \in \mathcal{A}_1.$$

Hence $g \circ f$ is measurable. □

10.6.11 Proposition. Let $(\Omega, \mathcal{A})$ be a measurable space, X be a set, and $\mathcal{C}$ be a subset of $\mathcal{P}(X)$. Let $f : \Omega \rightarrow X$ be a mapping. If, for any $A \in \mathcal{C}$, one has $f^{-1}(A) \in \mathcal{A}$, then the mapping f is measurable when we equip X with the σ -algebra $\sigma(\mathcal{C})$. In particular, a continuous mapping between topological spaces is measurable if one equips the topological spaces with Borel σ -algebras.

Proof. Let $\mathcal{A}'$ be the set of $A \in \mathcal{P}(X)$ such that $f^{-1}(A) \in \mathcal{A}$. By the assumption of the proposition, one has $\mathcal{C} \subseteq \mathcal{A}'$. In the following, we show that $\mathcal{A}'$ is a σ -algebra.

Let I be a countable set and $(A_i)_{i \in I}$ be a family of element of $\mathcal{A}'$ parametrized by I . One has

$$f^{-1}\left(\bigcup_{i \in I} A_i\right) = \bigcup_{i \in I} f^{-1}(A_i) \in \mathcal{A}.$$

Hence

$$\bigcup_{i \in I} A_i \in \mathcal{A}'.$$

For any $A \in \mathcal{A}'$, one has

$$f^{-1}(X \setminus A) = \Omega \setminus f^{-1}(A) \in \mathcal{A}.$$

Hence $X \setminus A \in \mathcal{A}'$. Thus we obtain that $\mathcal{A}'$ is a σ -algebra.

Since $\sigma(\mathcal{C})$ is the smallest σ -algebra containing $\mathcal{C}$, we obtain $\mathcal{C} \subseteq \mathcal{A}'$. □

10.6.12 Definition. Let Ω be a set and $((E_i, \mathcal{E}_i))_{i \in \Theta}$ be a family of measurable spaces. For any $i \in \Theta$, we fix a mapping $f_i : \Omega \rightarrow E_i$. Let f be the family $(f_i)_{i \in \Theta}$, we denote by $\sigma(f)$ the σ -algebra

$$\sigma\left(\bigcup_{i \in \Theta} \{f_i^{-1}(A_i) \mid A_i \in \mathcal{E}_i\}\right).$$

It is the smallest σ -algebra on Ω that makes all f_i measurable. In the particular case where Ω is equal to the product $\prod_{i \in \Theta} E_i$, and $f_i : \Omega \rightarrow E_i$ is the projection to the i -th coordinate, the σ -algebra $\sigma(f)$ is called the product σ -algebra of $(\mathcal{E}_i)_{i \in \Theta}$. It is often denoted as $\bigotimes_{i \in \Theta} \mathcal{E}_i$.

10.6.13 Proposition. *Let X be a set, and $((E_i, \mathcal{E}_i))_{i \in \Theta}$ be a family of measurable spaces. For any $i \in I$, let $f_i : X \rightarrow E_i$ be a mapping. We equip X with the σ -algebra $\sigma((f_i)_{i \in I})$. Let $(\Omega, \mathcal{A})$ be a measurable space and $g : \Omega \rightarrow X$ be a mapping. Then g is measurable if and only if $f_i \circ g$ is measurable for any $i \in \Theta$.*

Proof. By definition each f_i is measurable. If g is measurable, by Proposition 10.6.10 we obtain that $f_i \circ g$ is measurable.

Conversely, if $f_i \circ g$ is measurable for any $i \in \Theta$, then, for any $i \in \Theta$ and $A_i \in \mathcal{E}_i$, one has

$$g^{-1}(f_i^{-1}(A_i)) = (f_i \circ g)^{-1}(A_i) \in \mathcal{A}.$$

By Proposition 10.6.11, we obtain that g is measurable. $\square$

10.6.14 Remark. Let n be a positive integer, and $(X_1, \mathcal{T}_1), \dots, (X_n, \mathcal{T}_n)$ be topological spaces. Recall that the product topology on $X = X_1 \times \dots \times X_n$ is generated by the family of subsets of the form $U_1 \times \dots \times U_n$, with $(U_1, \dots, U_n) \in \mathcal{T}_1 \times \dots \times \mathcal{T}_n$. Since the projections $X \rightarrow X_i$ are continuous, they are measurable. Hence the Borel σ -algebra on X contains the product σ -algebra of $\sigma(\mathcal{T}_1), \dots, \sigma(\mathcal{T}_n)$. They are actually equal if each $\mathcal{T}_i$ admit a countable topological basis. In fact, if $\mathcal{B}_1, \dots, \mathcal{B}_n$ are respectively countable topological bases of $\mathcal{T}_1, \dots, \mathcal{T}_n$, then any open subset of X can be written as a countable union of some sets in

$$\{U_1 \times \dots \times U_n \mid (U_1, \dots, U_n) \in \mathcal{B}_1 \times \dots \times \mathcal{B}_n\}.$$

Hence any open subset of X belongs to the product σ -algebra $\sigma(\mathcal{T}_1) \otimes \dots \otimes \sigma(\mathcal{T}_n)$.

Note that a countable topological basis of the real number line $\mathbb{R}$ is given by open intervals of the form $]p, q[$, where $(p, q) \in \mathbb{Q} \times \mathbb{Q}$, $p < q$. Therefore, the product σ -algebra on $\mathbb{R}^n$ coincides with the Borel σ -algebra on $\mathbb{R}^n$ (equipped with any norm). We then deduce that, for any pair (f, g) of measurable mappings from a measure space $(\Omega, \mathcal{A})$ to $\mathbb{R}^n$, the mappings $f + g$ and fg are both measurable.

10.6.15 Notation. Let Ω be a set and $f : \Omega \rightarrow [-\infty, +\infty]$ be a mapping. If P is a condition on $[-\infty, +\infty]$, we use the expression $\{P(f)\}$ to denote the set

$$f^{-1}(\{t \in [-\infty, +\infty] \mid P(t)\}) = \{\omega \in \Omega \mid P(f(\omega))\}.$$

For example, the expression $\{f > 0\}$ denotes $\{\omega \in \Omega \mid f(\omega) > 0\}$.

10.6.16 Remark. Let Ω be a set and $f : \Omega \rightarrow \mathbb{R}_{\geq 0}$ be a mapping. For any $n \in \mathbb{N}$, let

$$f_n := \sum_{k=1}^{n2^n-1} \frac{k}{2^n} \mathbb{1}_{\{\frac{k}{2^n} \leq f \leq \frac{k+1}{2^n}\}} + n \mathbb{1}_{\{f \geq n\}}.$$

Then the sequence $(f_n)_{n \in \mathbb{N}}$ is increasing and converges pointwise to f .

10.6.17 Theorem. Let $(\Omega, \mathcal{A})$ be a measurable space, $(X, \mathcal{T})$ be a topological space. Assume that there exists a metric d on X such that $\mathcal{T}$ coincides with the topology defined by d . Let $(f_n)_{n \in \mathbb{N}}$ be a sequence of measurable mappings from Ω to X . Assume that the sequence $(f_n)_{n \in \mathbb{N}}$ converges pointwise to a mapping $f : \Omega \rightarrow X$. Then the limit mapping f is measurable.

Proof. Let Y be a closed subset of X . The mapping

$$d(\cdot, Y) : X \rightarrow \mathbb{R}_{\geq 0}$$

is continuous (since it is Lipschitzian). Therefore,

$$\begin{aligned} f^{-1}(Y) &= \left\{ \omega \in \Omega \mid \lim_{n \rightarrow +\infty} d(f_n(\omega), Y) = 0 \right\} \\ &= \left\{ \omega \in \Omega \mid \limsup_{n \rightarrow +\infty} d(f_n(\omega), Y) = 0 \right\} \\ &= \bigcap_{\substack{m \in \mathbb{N} \\ m \geq 1}} \bigcup_{N \in \mathbb{N}} \bigcap_{\substack{n \in \mathbb{N} \\ n \geq N}} \{ \omega \in \Omega \mid d(f_n(\omega), Y) < m^{-1} \} \end{aligned}$$

Since f_n is measurable and $d(\cdot, Y)$ is continuous (and hence measurable), the set

$$\left\{ \omega \in \Omega \mid \limsup_{n \rightarrow +\infty} d(f_n(\omega), Y) = 0 \right\}$$

belongs to $\mathcal{A}$ and hence $f^{-1}(Y)$ belongs to $\mathcal{A}$, too. $\square$

10.7 Monotone class theorems

10.7.1 Definition. Let Ω be a set and $\mathcal{H}$ be a family of bounded mappings from Ω to $\mathbb{R}_{\geq 0}$. We say that $\mathcal{H}$ is a λ -family if the following conditions are satisfied:

- (1) $\mathbb{1}_\Omega \in \mathcal{H}$,
- (2) if $(f, g) \in \mathcal{H} \times \mathcal{H}$, then $f + g \in \mathcal{H}$,
- (3) if $(f, g) \in \mathcal{H} \times \mathcal{H}$ such that $f \leq g$, then $g - f \in \mathcal{H}$,

- (4) if $f \in \mathcal{H}$ and $\lambda \in \mathbb{R}_{\geq 0}$, then $\lambda f \in \mathcal{H}$,
- (5) if $(f_n)_{n \in \mathbb{N}}$ is an increasing family in $\mathcal{H}$ that converges pointwise to a bounded mapping $f : \Omega \rightarrow \mathbb{R}_{\geq 0}$, then $f \in \mathcal{H}$.

10.7.2 Remark. Let Ω be a set and $\mathcal{H}$ be a λ -family on Ω . Assume that, for any $(f, g) \in \mathcal{H} \times \mathcal{H}$, one has $\sup\{f, g\} \in \mathcal{H}$. Then for any countable family $(f_i)_{i \in \Theta}$ of elements of $\mathcal{H}$ which are uniformly bounded, one has

$$f := \sup\{f_i \mid i \in \Theta\} \in \mathcal{H}.$$

In fact, without loss of generality we may assume that $\Theta = \mathbb{N}$. We let $g_n = \sup\{f_0, \dots, f_n\} \in \mathcal{H}$. The sequence $(g_n)_{n \in \mathbb{N}}$ is increasing and converges pointwise to f .

10.7.3 Lemma. Let n be a positive integer. Let $\varphi_n : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ the mapping that sends x to x^n .

- (1) φ_n is convex.
- (2) for any $x \in \mathbb{R}_{> 0}$,

$$x^n = \sup_{a \in \mathbb{Q}_{> -}} \max\{na^{n-1}x - (n-1)x^n, 0\}.$$

10.7.4 Theorem. Let Ω be a set and $\mathcal{H}$ be a λ -family of mappings from Ω to $\mathbb{R}_{\geq 0}$. Suppose that, for any $(f, g) \in \mathcal{H} \times \mathcal{H}$, one has $\inf\{f, g\} \in \mathcal{H}$. Then any bounded $\sigma(\mathcal{H})$ -measurable mappings from Ω to $\mathbb{R}_{\geq 0}$ belongs to $\mathcal{H}$. In particular, $\sigma(\mathcal{H})$ identifies with the set of $A \in \mathcal{P}(\Omega)$ such that $\mathbb{1}_A \in \mathcal{H}$.

Proof. If f and g are two elements of $\mathcal{H}$, then

$$\sup\{f, g\} = f + g - \inf\{f, g\} \in \mathcal{H}.$$

In particular, if $f \in \mathcal{H}$ and $a \in \mathbb{R}_{> 0}$, then

$$\sup\{f - a, 0\} = \sup\{f, a\mathbb{1}_\Omega\} - a\mathbb{1}_\Omega.$$

Thus,

$$f^n = \sup_{a \in \mathbb{Q}_{> 0}} \sup\{na^{n-1}f - (n-1)a^n, 0\} \in \mathcal{H}.$$

Let $\mathcal{A}$ be the set of $A \in \mathcal{P}(\Omega)$ such that $\mathbb{1}_A \in \mathcal{H}$. Then $\mathcal{A}$ is a σ -algebra. Let $f \in \mathcal{H}$ and $t > 0$. One has $\inf\{t^{-1}f, \mathbb{1}_\Omega\} \in \mathcal{H}$. Moreover, the sequence

$$(\mathbb{1}_\Omega - (t^{-1}f, \mathbb{1}_\Omega)^n)_{n \in \mathbb{N}}$$

is increasing and converges to $\mathbb{1}_{\{f < t\}}$. Hence $\{f < t\} \in \mathcal{A}$. In particular, any element of $\mathcal{H}$ is $\mathcal{A}$ -measurable, namely $\sigma(\mathcal{H}) \subseteq \mathcal{A}$. $\square$

10.7.5 Theorem. *Let Ω be a set and $\mathcal{H}$ be a λ -family of bounded mappings from Ω to $\mathbb{R}_{\geq 0}$. Let $\mathcal{H}_0$ be a subset of $\mathcal{H}$. If, for any $(f, g) \in \mathcal{H}_0 \times \mathcal{H}_0$, one has $fg \in \mathcal{H}_0$, then any bounded $\sigma(\mathcal{H}_0)$ -measurable mapping from Ω to $\mathbb{R}_{\geq 0}$ belongs to $\mathcal{H}$.*

Proof. By replacing $\mathcal{H}$ by the intersection of all λ -families containing $\mathcal{H}_0$, we may assume that $\mathcal{H}$ is the smallest λ -family containing $\mathcal{H}_0$.

Let $\mathcal{H}_1$ be the set of $f \in \mathcal{H}$ such that $fg \in \mathcal{H}$ for any $g \in \mathcal{H}_0$. Then $\mathcal{H}_1$ is a λ -family containing $\mathcal{H}_0$. Thus $\mathcal{H}_1 = \mathcal{H}$. Now let $\mathcal{H}_2$ be the set of $f \in \mathcal{H}$ such that $fg \in \mathcal{H}$ for any $g \in \mathcal{H}$. Then $\mathcal{H}_2$ is a λ -family that contains $\mathcal{H}_0$. Hence $\mathcal{H}_2 = \mathcal{H}$. This observation shows that $\mathcal{H}$ is stable by multiplication.

Let $(f, g) \in \mathcal{H} \times \mathcal{H}$. We prove that $|f - g| \in \mathcal{H}$. Without loss of generality, we may assume that $|f - g| \leq 1$. Note that

$$(f - g)^2 = f^2 + g^2 - 2fg \in \mathcal{H}.$$

Let $(f_n)_{n \in \mathbb{N}}$ be a sequence of functions defined as

$$f_0 = 0, \quad f_{n+1} = f_n + \frac{1}{2}((f - g)^2 - f_n^2)$$

Then $f_n \in \mathcal{H}$ and $f_n \leq |f - g|$. In fact, this inequality is clearly satisfied when $n = 0$. If $f_n \in \mathcal{H}$ and $f_n \leq |f - g|$, then $f_{n+1} \in \mathcal{H}$. Moreover, since $|f - g| \leq 1$, one has $f_{n+1} \leq |f - g|$ since the mapping $z \mapsto z - \frac{1}{2}z^2$ is increasing on $[0, 1]$. The sequence $(f_n)_{n \in \mathbb{N}}$ is increasing and converge pointwise to $|f - g|$. Hence $|f - g| \in \mathcal{H}$. We then deduce that

$$\inf\{f, g\} = \frac{1}{2}(f + g - |f - g|) \in \mathcal{H}.$$

Hence any bounded and $\sigma(\mathcal{H}_0)$ -measurable mapping from Ω to $\mathbb{R}_{\geq 0}$ belongs to $\mathcal{H}$. The theorem is thus proved. $\square$

10.7.6 Theorem. *Let Ω be a set, $\mathcal{L}$ be a Riesz space on Ω . We assume that, for any $(f, g) \in \mathcal{L} \times \mathcal{L}^\uparrow$, one has $\inf\{f, g\} \in \mathcal{L}$. Then the following statements hold.*

(1) *If $(f, g) \in \mathcal{L}^\uparrow \times \mathcal{L}^\uparrow$ is such that $f \leq g$ and $g - f$ is a well defined as a mapping from Ω to $]-\infty, +\infty]$, then $g - f \in \mathcal{L}^\uparrow$.*

(2) *Let $\mathcal{A}$ be the set*

$$\{A \in \mathcal{P}(\Omega) \mid \mathbb{1}_A \in \mathcal{L}^\uparrow\}.$$

(2.a) *For any $(A, B) \in \mathcal{A} \times \mathcal{A}$, $B \setminus A \in \mathcal{A}$.*

(2.b) *For any countable family $(A_i)_{i \in \Theta}$ of elements of $\mathcal{A}$, one has*

$$\bigcup_{i \in \Theta} A_i \in \mathcal{A}.$$

In particular, if $\Omega \in \mathcal{A}$, then $\mathcal{A}$ is a σ -algebra.

(3) Let $f \in \mathcal{L}^\uparrow$ such that $f \geq 0$. We suppose that

$$B := \{\omega \in \Omega \mid f(\omega) > 0\} \in \mathcal{A}.$$

For any positive real number t , the set

$$\{\omega \in \Omega \mid 0 < f(\omega) < t\}$$

belongs to $\mathcal{A}$.

(4) Let $f : \Omega \rightarrow [0, +\infty]$ be a mapping such that

$$B := \{\omega \in \Omega \mid f(\omega) > 0\} \in \mathcal{A}$$

If, for any positive rational number t , the set

$$A_t := \{\omega \in \Omega \mid 0 < f(\omega) < t\}$$

belongs to $\mathcal{A}$. Then $f \in \mathcal{L}^\uparrow$.

Proof. (1) Let $(g_n)_{n \in \mathbb{N}}$ be an increasing sequence that converges pointwise to g . For any $n \in \mathbb{N}$, by the assumption of the theorem, one has $\inf\{g_n, f\} \in \mathcal{L}$. Note that

$$(g_n - \inf\{g_n, f\})_{n \in \mathbb{N}} = (\sup\{g_n - f, 0\})_{n \in \mathbb{N}}$$

is an increasing sequence in $\mathcal{L}$, that converges pointwise to $g - f$. Hence $g - f$ belongs to $\mathcal{L}^\uparrow$.

(2.a) First, one has $\mathbb{1}_{A \cap B} = \inf\{\mathbb{1}_A, \mathbb{1}_B\} \in \mathcal{L}^\uparrow$. Hence, by (1) one has

$$\mathbb{1}_{B \setminus A} = \mathbb{1}_B - \mathbb{1}_{A \cap B} \in \mathcal{L}^\uparrow.$$

Therefore, $B \setminus A \in \mathcal{A}$.

(2.b) It suffices to treat the case where $\Theta = \mathbb{N}$. If $(A_n)_{n \in \mathbb{N}}$ is a sequence of elements of $\mathcal{A}$ and $A = \bigcup_{n \in \mathbb{N}} A_n$. Then $\mathbb{1}_A$ identifies with the limit of the following increasing sequence

$$(\sup\{\mathbb{1}_{A_0}, \dots, \mathbb{1}_{A_n}\})_{n \in \mathbb{N}}.$$

Hence $\mathbb{1}_A \in \mathcal{L}^\uparrow$ and $A \in \mathcal{A}$.

(3) For any $a \in \mathbb{R}_{>0}$, one has

$$\sup\{f - a, 0\} = \sup\{f, a\mathbb{1}_B\} - a\mathbb{1}_B \in \mathcal{L}^\uparrow.$$

Let $n \in \mathbb{N}_{\geq 1}$. Note that, for any $x \in \mathbb{R}_{\geq 0}$, one has

$$x^n = \sup_{a \in \mathbb{Q}_{>0}} \max\{na^{n-1}x - (n-1)x^n, 0\}.$$

Therefore,

$$f^n = \sup_{a \in \mathbb{Q}_{>0}} \sup \{na^{n-1}f - (n-1)a^n, 0\} \in \mathcal{L}^\uparrow.$$

If f is an element of $S^\uparrow$, $t \in \mathbb{R}_{>0}$. One has

$$\inf \{t^{-1}f, \mathbb{1}_B\} \in \mathcal{L}^\uparrow.$$

Moreover, the sequence

$$(\mathbb{1}_B - \inf \{t^{-1}f, 1\}^n)_{n \in \mathbb{N}, n \geq 1}$$

is increasing and converges to

$$\mathbb{1}_{\{\omega \in \Omega \mid 0 < f(\omega) < t\}}.$$

Hence the latter belongs to $\mathcal{L}^\uparrow$, which shows that

$$\{\omega \in \Omega \mid 0 < f(\omega) < t\}$$

belongs to $\mathcal{A}$.

(4) For any $n \in \mathbb{N}_{\geq 1}$, let

$$\begin{aligned} f_n &:= \sum_{k=0}^{n2^n-1} \frac{k}{2^n} \mathbb{1}_{\{\omega \in \Omega \mid k/2^n \leq f(\omega) < (k+1)/2^n\}} + n \mathbb{1}_{\{\omega \in \Omega \mid f(\omega) \geq n\}} \\ &= \sum_{k=1}^{n2^n-1} \frac{k}{2^n} \mathbb{1}_{A_{(k+1)/2^n} \setminus A_{k/2^n}} + n(\mathbb{1}_B \setminus A_n). \end{aligned}$$

By (2) and (3), we obtain that f_n belongs to $\mathcal{L}^\uparrow$. Since f is the pointwise limit of the increasing sequence $(f_n)_{n \in \mathbb{N}}$, we obtain that $f \in \mathcal{L}^\uparrow$. $\square$

10.7.7 Remark. Let Ω be a set, S be a Riesz space on Ω , and $I : S \rightarrow \mathbb{R}$ be an integral operator. If $(f, g) \in \mathcal{L}^1(I) \times \mathcal{L}^1(I)^\uparrow$, then $\inf\{f, g\} \in \mathcal{L}^1(I)^\uparrow$ and

$$I(\inf\{f, g\}) \leq I(f) < +\infty.$$

By Beppo Levi's theorem, we obtain that $\inf\{f, g\} \in \mathcal{L}^1(I)$.

10.8 Measure space

10.8.1 Definition. We call measure space any triple $(\Omega, \mathcal{A}, \mu)$, where Ω is a set, $\mathcal{A}$ is a σ -algebra on Ω , and $\mu : \mathcal{A} \rightarrow \mathbb{R}_{\geq 0}$ is a mapping such that $\mu(\emptyset) = 0$ and, for any countable family $(A_i)_{i \in \Theta}$ of pairwise disjoint sets in $\mathcal{A}$, one has

$$\mu\left(\bigcup_{i \in \Theta} A_i\right) = \sum_{i \in \Theta} \mu(A_i).$$

10.8.2 Proposition. Let $(\Omega, \mathcal{A}, \mu)$ be a measure space. If $(A_n)_{n \in \mathbb{N}}$ is an increasing sequence in $\mathcal{A}$ and $A = \bigcup_{n \in \mathbb{N}} A_n$, then

$$\mu(A) = \lim_{n \rightarrow +\infty} \mu(A_n).$$

Proof. Let $B_0 = A_0$. For any $n \in \mathbb{N}$, such that $n \geq 1$, let $B_n = A_n \setminus A_{n-1}$. The $(B_n)_{n \in \mathbb{N}}$ is a family of pairwise disjoint elements in $\mathcal{A}$. Moreover, $A_n = B_0 \cup \dots \cup B_n$, and $A = \bigcup_{n \in \mathbb{N}} B_n$. Therefore,

$$\mu(A) = \sum_{n \in \mathbb{N}} \mu(B_n) = \lim_{n \rightarrow +\infty} \mu(B_0 \cup \dots \cup B_n) = \lim_{n \rightarrow +\infty} \mu(A_n) < +\infty,$$

which leads to a contradiction. $\square$

10.8.3 Proposition. Let $(\Omega, \mathcal{A}, \mu)$ be a measure space. Let $\mathcal{C}$ be the set of $A \in \mathcal{A}$ such that $\mu(A) < +\infty$. Then $\mathcal{C}$ is a semiring on Ω , and the restriction of μ to $\mathcal{C}$ is σ -additive.

Proof. Since $\mu(\emptyset) = 0$, one has $\emptyset \in \mathcal{C}$.

If $(A, B) \in \mathcal{C} \times \mathcal{C}$, then $A \cap B \in \mathcal{A}$ and $\mu(A \cap B) \leq \mu(A) < +\infty$. Hence $A \cap B \in \mathcal{C}$.

If A and B are two elements of $\mathcal{C}$ such that $A \subseteq B$. Then $B \setminus A = B \cap (\Omega \setminus A) \in \mathcal{A}$ and

$$\mu(B \setminus A) \leq \mu(B) < +\infty.$$

Hence $B \setminus A \in \mathcal{C}$.

By definition, the restriction of μ to $\mathcal{C}$ is σ -additive. $\square$

10.8.4 Definition. Let $(\Omega, \mathcal{A}, \mu)$ be a measure space,

$$\mathcal{C} = \{A \in \mathcal{A} \mid \mu(A) < +\infty\},$$

and S be the vector space generated by mappings of the form $\mathbb{1}_A$, where $A \in \mathcal{C}$. Let $I : S \rightarrow \mathbb{R}$ be the unique integral operator such that $I(\mathbb{1}_A) = \mu(A)$ for any $A \in \mathcal{C}$. We denote by $\mathcal{L}^1(\Omega, \mathcal{A}, \mu)$ the set of $\mathcal{A}$ -measurable mappings that belong to $\mathcal{L}^1(I)$. The elements of $\mathcal{L}^1(\Omega, \mathcal{A}, \mu)$ are said to be *integrable* with respect to the measure μ .

If $f : \Omega \rightarrow \mathbb{R}$ is an element of $\mathcal{L}^1(\Omega, \mathcal{A}, \mu)$, we denote by

$$\int_{\Omega} f \, d\mu \quad \text{or} \quad \int_{\Omega} f(x) \mu(dx)$$

the real number $I(f)$. The mappings of $\mathcal{L}^1(\Omega, \mathcal{A}, \mu)$ are said to be *integrable* with respect to μ .

Let $f : \Omega \rightarrow \mathbb{R}$ be an $\mathcal{A}$ -measurable mapping that is bounded from below by some element of $\mathcal{L}^1(\Omega, \mathcal{A}, \mu)$. If f does not belong to $\mathcal{L}^1(\Omega, \mathcal{A}, \mu)$, then, by convention

$$\int_{\Omega} f \, d\mu$$

is defined to be $+\infty$.

Let $f : \Omega \rightarrow \mathbb{R}$ be an $\mathcal{A}$ -measurable mapping that is bounded from below by some element of $\mathcal{L}^1(\Omega, \mathcal{A}, \mu)$. If f does not belong to $\mathcal{L}^1(\Omega, \mathcal{A}, \mu)$, then, by convention

$$\int_{\Omega} f \, d\mu$$

is defined to be $-\infty$.

10.8.5 Proposition. *We keep the notation of Definition Let A be an element of $\mathcal{A}$. Then*

$$\int_{\Omega} \mathbb{1}_A \, d\mu = \mu(A).$$

Proof. By definition, the equality is true when $\mu(A) < +\infty$. It suffices to check the case where $\mu(A) = +\infty$. We reason by contradiction in assuming that $\mathbb{1}_A \in \mathcal{L}^1(\Omega, \mathcal{A}, \mu)$.

We first show that, for any $f \in S$ such that $f \geq 0$, one has $\inf\{\mathbb{1}_A, f\} \in S$. In fact, f is of the form

$$\sum_{i=1}^n b_i \mathbb{1}_{B_i},$$

where $(b_1, \dots, b_n) \in \mathbb{R}_{\geq 0}^n$, $B_1, \dots, B_n$ are pairwise disjoint elements of $\mathcal{C}$. Note that

$$\inf\{\mathbb{1}_A, f\} = \sum_{i=1}^n \min\{b_i, 1\} \mathbb{1}_{A \cap B_i} \in S.$$

Since $\mathbb{1}_A \in \mathcal{L}^1(\Omega, \mathcal{A}, \mu)$, there exists $g \in S^{\uparrow}$ such that $\mathbb{1}_A \leq g$ and $I(g) < +\infty$. Let $(f_n)_{n \in \mathbb{N}}$ be an increasing sequence in S that converges pointwise to g . By replacing f_n by $\sup\{f_n, 0\}$, we may assume that $f_n \geq 0$. Note that $(\inf\{f_n, \mathbb{1}_A\})_{n \in \mathbb{N}}$ is an increasing sequence that converges pointwise to $\mathbb{1}_A$. In particular, there exists an increasing sequence $(A_n)_{n \in \mathbb{N}}$ in $\mathcal{C}$ (with $A_n = \{\omega \in A \mid f_n(\omega) \geq 1\}$) such that

$$\lim_{n \rightarrow +\infty} \mu(A_n) \leq I(g) < +\infty$$

and $A = \bigcup_{n \in \mathbb{N}} A_n$. This contradicts the assumption that $\mu(A) = +\infty$. $\square$

10.8.6 Proposition. *Let $(\Omega, \mathcal{A}, \mu)$ be a measure space.*

- (1) Let f be a measurable mapping from Ω to $\mathbb{R}_{\geq 0}$. If f belongs to $\mathcal{L}^1(\Omega, \mathcal{A}, \mu)$, then, for any $t \in \mathbb{R}_{>0}$, one has

$$\mu(\{\omega \in \Omega \mid f(\omega) \geq t\}) < +\infty.$$

- (2) Let f and g be $\mathcal{A}$ -measurable mappings from Ω to $\mathbb{R}_{\geq 0}$ such that $f \leq g$. If $g \in \mathcal{L}^1(\Omega, \mathcal{A}, \mu)$, then f also belongs to $\mathcal{L}^1(\Omega, \mathcal{A}, \mu)$.
- (3) An $\mathcal{A}$ -measurable mapping $f : \Omega \rightarrow \mathbb{R}$ belongs to $\mathcal{L}^1(\Omega, \mathcal{A}, \mu)$ if and only if $|f| \in \mathcal{L}^1(\Omega, \mathcal{A}, \mu)$.

Proof. (1) By Theorem 10.7.6, for any $x \in \mathbb{R}_{>0}$, one has

$$\mathbb{1}_{\{\omega \in \Omega \mid 0 < f(\omega) < x\}} \in \mathcal{L}^1(I)^\uparrow.$$

Moreover, if $0 < t < x$, then

$$\mathbb{1}_{\{\omega \in \Omega \mid t \leq f(\omega) < x\}} = \mathbb{1}_{\{\omega \in \Omega \mid 0 < f(\omega) < x\}} - \mathbb{1}_{\{\omega \in \Omega \mid 0 < f(\omega) < t\}} \in \mathcal{L}^1(I)^\uparrow.$$

Therefore, we obtain that

$$\mathbb{1}_{\{\omega \in \Omega \mid t \leq f(\omega)\}} \in \mathcal{L}^1(I)^\uparrow$$

since it is the pointwise limit of the increasing sequence

$$\left(\mathbb{1}_{\{\omega \in \Omega \mid t \leq f(\omega) < t+n\}} \right)_{n \in \mathbb{N}}.$$

Since

$$t \mathbb{1}_{\{\omega \in \Omega \mid t \leq f(\omega)\}} \leq f,$$

we obtain that

$$t \mu(\{\omega \in \Omega \mid t \leq f(\omega)\}) \leq \int_{\Omega} f \, d\mu < +\infty.$$

- (2) By (1), one has

$$\forall a > 0, \mu(\{\omega \in \Omega \mid g(\omega) \geq a\}) < +\infty$$

and hence

$$\forall a > 0, \mu(\{\omega \in \Omega \mid f(\omega) \geq a\}) < +\infty.$$

Hence

$$\mathbb{1}_{\{\omega \in \Omega \mid f(\omega) > 0\}} \in \mathcal{L}^1(I)^\uparrow.$$

Moreover, for any $t > 0$, one has

$$\forall a > 0, \mu(\{\omega \in \Omega \mid a \leq f(\omega) < t\}) < +\infty,$$

which leads to

$$\mathbb{1}_{\{\omega \in \Omega \mid 0 < f(\omega) < t\}} \in \mathcal{L}^1(I)^\uparrow.$$

By Theorem 10.7.6, we obtain that $f \in \mathcal{L}^1(I)^\uparrow$. Since $f \leq g$, one has

$$I(f) \leq I(g) < +\infty,$$

and hence $f \in \mathcal{L}^1(\Omega, \mathcal{A}, \mu)$.

(3) By Theorem 10.3.15, $\mathcal{L}^1(I)$ is a Riesz space. In particular, if f belongs to $\mathcal{L}^1(\Omega, \mathcal{A}, \mu)$, then

$$|f| = \sup\{f, 0\} - \inf\{f, 0\}$$

is $\mathcal{A}$ -measurable and hence $|f|$ is also $\mathcal{A}$ -measurable. Conversely, by (2), if $|f|$ is integrable, then

$$\sup\{f, 0\} \text{ and } \sup\{-f, 0\}$$

are both integrable since they are non-negative and bounded from above by $|f|$. Hence

$$f = \sup\{f, 0\} - \sup\{-f, 0\} \in \mathcal{L}^1(\Omega, \mathcal{A}, \mu).$$

□

10.8.7 Notation. Let $(\Omega, \mathcal{A}, \mu)$ be a measure space, and $f : \Omega \rightarrow \mathbb{R}$ be an element of $\mathcal{L}^1(\Omega, \mathcal{A}, \mu)$. For any $A \in \mathcal{A}$, one has

$$|\mathbb{1}_A f| \leq |f|,$$

and hence $\mathbb{1}_A f \in \mathcal{L}^1(\Omega, \mathcal{A}, \mu)$. We denote by

$$\int_A f \, d\mu \text{ or } \int_A f(x) \, \mu(dx)$$

the integral

$$\int_\Omega \mathbb{1}_A f \, d\mu.$$

Consider the particular case where $\Omega = \mathbb{R}$, $\mathcal{A}$ is the Borel σ -algebra on $\mathbb{R}$ and $\mu_\varphi([a, b]) = \varphi(b) - \varphi(a)$, where $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is an increasing right continuous mapping. If $f : \mathbb{R} \rightarrow \mathbb{R}$ is an element of $\mathcal{L}^1(\mathbb{R}, \sigma(\mathcal{T}_\mathbb{R}), \mu_\varphi)$, then

$$\int_{[a, b]} f \, d\mu_\varphi$$

is denoted as

$$\int_a^b f(t) \, d\varphi(t).$$

The measure μ_φ is called the Stieltjes measure associated with φ . In the particular case where $\varphi = \text{Id}_\mathbb{R}$, the measure μ_φ is called Lebesgue measure.

10.8.8 Definition. Let $(\Omega, \mathcal{A})$ be a measurable space. We say that a measure μ on $(\Omega, \mathcal{A})$ is σ -finite if there exists an increasing sequence $(A_n)_{n \in \mathbb{N}}$ in $\mathcal{A}$ such that $\mu(A_n) < +\infty$ for any $n \in \mathbb{N}$ and that $\Omega = \bigcup_{n \in \mathbb{N}} A_n$.

10.8.9 Theorem (Carathéodory). *Let Ω be a set, $\mathcal{C}$ be a semiring on Ω and $\mu : \mathcal{C} \rightarrow \mathbb{R}_{\geq 0}$ be an σ -additive mapping. If there exists an increasing sequence $(A_n)_{n \in \mathbb{N}}$ in $\mathcal{C}$ such that $\mu(A_n) < +\infty$ for any $n \in \mathbb{N}$ and that $\Omega = \bigcup_{n \in \mathbb{N}} A_n$, then μ extends in a unique way to a σ -finite measure on $\sigma(\mathcal{C})$.*

Proof. Let S be the vector subspace of $\mathbb{R}^\Omega$ generated by mappings of the form $\mathbb{1}_A$, where $A \in \mathcal{C}$. Let $I : S \rightarrow \mathbb{R}$ be the $\mathbb{R}$ -linear mapping that sends $\mathbb{1}_A$ to $\mu(A)$. Then I is an integral operator. Let $\mathcal{L}^1(I)$ be the vector space of Daniell integrable functions. It is a Riesz space and $I : \mathcal{L}^1(I) \rightarrow \mathbb{R}$ is an integral operator. Moreover, if $(f, g) \in \mathcal{L}(I) \times \mathcal{L}^1(I)$, then $\inf\{f, g\} \in \mathcal{L}^1(I)$. By Theorem 10.7.6, the set

$$\mathcal{A} = \{A \in \mathcal{P}(\Omega) \mid \mathbb{1}_A \in \mathcal{L}^1(I)^\uparrow\}$$

is a σ -algebra containing $\mathcal{C}$, and

$$(A \in \mathcal{A}) \longmapsto I(\mathbb{1}_A)$$

is a measure on $\mathcal{A}$ extending μ .

Suppose that ν is a measure on $(\Omega, \sigma(\mathcal{C}))$ extending μ . Let S' be the vector subspace of $\mathbb{R}^\Omega$ generated by mappings of the form $\mathbb{1}_A$ with $A \in \sigma(\mathcal{C})$, $\nu(A) < +\infty$. Then $S \subseteq S' \subseteq \mathcal{L}^1(I)$. By Beppo Levi's theorem, the restriction of I_ν to $S^\uparrow$ and $S^\downarrow$ coincides with I and furthermore I_ν coincides with I . Therefore $\nu = \mu$ on $\sigma(\mathcal{C})$. $\square$

10.8.10 Definition. Let $(\Omega, \mathcal{A}, \mu)$ be a measure space. We say that a subset A of Ω is μ -negligeable if there exists $B \in \mathcal{A}$ such that $A \subseteq B$ and $\mu(B) = 0$.

10.8.11 Proposition. *Let $(\Omega, \mathcal{A}, \mu)$ be a measure space. If $f \in \mathcal{L}(\Omega, \mathcal{A}, \mu)$ and if $A \in \mathcal{A}$ is μ -negligeable, then*

$$\int_{\Omega} \mathbb{1}_A(x) f(x) \mu(dx) = 0.$$

Proof. Without loss of generality, assume that f is of the form

$$\sum_{i=1}^n a_i \mathbb{1}_{B_i},$$

where $B_i \in \mathcal{A}$ and $a_i \geq 0$. Then

$$\mathbb{1}_A f = \sum_{i=1}^n a_i \mathbb{1}_{B_i \cap A}.$$

Note that $\mu(B_i \cap A) = 0$, so that

$$\int_{\Omega} \mathbb{1}_A(x) f(x) \mu(dx) = \sum_{i=1}^n a_i \mu(B_i \cap A) = 0.$$

□

10.8.12 Theorem. *Let Y be a metric space, $f : X \times Y \rightarrow \mathbb{R}$ be a mapping, g be an element of $\mathcal{L}^1(X, \mathcal{F}, \mu)$ and $a \in Y$. Assume that*

- (1) *there is a μ -negligible set A such that, for any $x \in X \setminus A$,*

$$f(x, \cdot) : Y \longrightarrow \mathbb{R}, \quad y \longmapsto f(x, y)$$

is continuous at a ,

- (2) *for any $y \in Y$,*

$$f(\cdot, y) : X \longrightarrow \mathbb{R}, \quad x \longmapsto f(x, y)$$

is $\mathcal{F}$ -measurable,

- (3) *for any $y \in Y$, there exists a μ -negligible set A_y such that*

$$\forall x \in X \setminus A_y, \quad |f(x, y)| \leq g(x).$$

Then the mapping

$$(y \in Y) \longmapsto \int_X f(x, y) \mu(dx)$$

is continuous.

Proof. Let $(y_n)_{n \in \mathbb{N}}$ be a sequence in Y that converges to $y \in Y$. For any $n \in \mathbb{N}$, let $f_n : X \rightarrow \mathbb{R}$ be the mapping that sends $x \in X$ to $f(x, y_n)$. Since f_n is $\mathcal{F}$ -measurable and $|f_n| \leq g$, we obtain that $f_n \in \mathcal{L}^1(X, \mathcal{F}, \mu)$. Moreover, by dominated convergence theorem, we obtain that

$$\lim_{n \rightarrow +\infty} \int_X f_n(x) \mu(dx) = \lim_{n \rightarrow +\infty} \int_X f(x, y_n) \mu(dx) = \int_X f(x, y) \mu(dx).$$

□

10.8.13 Corollary. *Let $(a, b) \in \mathbb{R}^2$ such that $a < b$. Let μ be a measure on $[a, b]$ and $f : [a, b] \rightarrow \mathbb{R}$ be a Borel measurable mapping that is integrable with respect to μ . Let $F : [a, b] \rightarrow \mathbb{R}$ be the mapping defined as*

$$F(x) := \int_a^x f(t) dt.$$

If $x_0 \in [a, b]$ is such that $\mu(\{x_0\}) = 0$, then F is continuous at x_0 .

Proof. By definition, one has

$$F(x) = \int_{]a,b]} \mathbb{1}_{]a,x]}(t) f(t) \, dt$$

for any $x \in [a, b]$. If $t \neq x_0$, then the mapping

$$x \longmapsto \mathbb{1}_{]a,x]}(t)$$

is continuous at x_0 . Hence F is continuous at x_0 as well. $\square$

10.8.14 Theorem. *Let $(a, b) \in \mathbb{R}^2$ with $a < b$. Let $f : [a, b] \rightarrow \mathbb{R}$ be a Borel measurable mapping which is integrable with respect to the Lebesgue measure. Let $F : [a, b] \rightarrow \mathbb{R}$ be the mapping defined as*

$$F(x) = \int_0^x f(t) \, dt.$$

- (1) *If f is continuous at $x \in]a, b[$, then F is differentiable at x and $F'(x) = f(x)$. In particular, F is a primitive function of f .*
- (2) *Suppose that f is continuous and that $G : [a, b] \rightarrow \mathbb{R}$ is a continuous mapping which is a primitive function of f on $]a, b[$. Then one has*

$$\int_a^b f(t) \, dt = G(b) - G(a).$$

In particular, there exists $\xi \in]a, b[$ such that

$$\int_a^b f(t) \, dt = f(\xi)(b - a).$$

Proof. (1) For $h > 0$ such that

$$a < x - h < x < x + h < b,$$

let

$$u_h = \sup_{t \in [x-h, x+h]} |f(t) - f(x)|.$$

Then,

$$|F(x+h) - F(x) - f(x)h| = \left| \int_x^{x+h} (f(t) - f(x)) \, dt \right| \leq u_h h.$$

Similarly,

$$|F(x-h) - F(x) + f(x)h| \leq u_h h.$$

Since

$$\lim_{h \rightarrow 0} u_h = 0,$$

we get the differentiability of F at x .

(2) Since f is continuous, F is differentiable on $]a, b[$. Moreover, by Theorem 10.8.12, we obtain that F is continuous on $[a, b]$. Thus $G - F$ is a constant mapping. Hence

$$\int_a^b f(t) dt = F(b) = F(b) - F(a) = G(b) - G(a).$$

The last statement comes from the mean value theorem. $\square$

10.8.15 Theorem. *Let I be an open interval in $\mathbb{R}$. Let $f : X \times I \rightarrow \mathbb{R}$ be a mapping and g be an element of $\mathcal{L}^1(X, \mathcal{F}, \mu)$. Let A be a μ -negligible subset of X . We assume that,*

(1) *for any $(x \in (X \setminus A))$, the mapping*

$$f(x, \cdot) : (t \in I) \mapsto f(x, t)$$

is differentiable, and its derivative $\frac{\partial f}{\partial t}(x, \cdot)$ satisfies

$$\forall t \in I, \quad \left| \frac{\partial f}{\partial t}(x, t) \right| \leq g(x).$$

(2) *for any $t \in I$, the mapping*

$$f(\cdot, t) : (x \in X) \mapsto f(x, t)$$

belongs to $\mathcal{L}^1(X, \mathcal{F}, \mu)$.

Then, the mapping

$$(t \in I) \mapsto \int_X f(x, t) \mu(dx)$$

is differentiable, and its differential is

$$(t \in I) \mapsto \int_X \frac{\partial f}{\partial t}(x, t) \mu(dx).$$

10.9 Product measure

In this section, we fix two measure space $(X, \mathcal{A}_X)$ and $(Y, \mathcal{A}_Y)$. Let μ_X and μ_Y be σ -finite measures on $(X, \mathcal{A}_X)$ and $(Y, \mathcal{A}_Y)$, respectively. We equip $X \times Y$ with the product σ -algebra $\mathcal{A}_X \otimes \mathcal{A}_Y$.

10.9.1 Proposition. *Let $f : X \times Y \rightarrow \mathbb{R}$ be a measurable mapping. For any $x \in X$, the mapping $f(x, \cdot) : Y \rightarrow \mathbb{R}$ is $\mathcal{A}_Y$ -measurable.*

Proof. Since

$$f = \sup\{f, 0\} - \sup\{-f, 0\},$$

without loss of generality, we may assume that f is non-negative. Moreover, since any non-negative measurable mapping can be written as the pointwise limit of an increasing sequence of bounded measurable non-negative mappings, it suffices to treat the case where f is non-negative and bounded.

Let $\mathcal{H}$ be the set of bounded mappings f from $X \times Y$ to $\mathbb{R}_{\geq 0}$ such that $f(x, \cdot) : Y \rightarrow \mathbb{R}_{\geq 0}$ is $\mathcal{A}_Y$ -measurable. By definition, $\mathcal{H}$ is a λ -family. Moreover, $\mathcal{H}$ contains the family $\mathcal{C}$ of linear combinations with non-negative coefficients of mappings of the form $(x, y) \mapsto f_1(x)f_2(y)$, where f_1 and f_2 are non-negative bounded and measurable mappings from $(X, \mathcal{A}_X)$ and $(Y, \mathcal{A}_Y)$ to $\mathbb{R}_{\geq 0}$, respectively. The family $\mathcal{C}$ is stable by multiplication. Hence $\mathcal{H}$ contains all bounded $\sigma(\mathcal{C})$ -measurable mappings from $X \times Y$ to $\mathbb{R}_{\geq 0}$. Since $\sigma(\mathcal{C}) = \mathcal{A}_X \otimes \mathcal{A}_Y$, we obtain the result of the proposition. $\square$

10.9.2 Theorem (Fubini-Tonelli). *Let $(X, \mathcal{A}_X, \mu_X)$ and $(Y, \mathcal{A}_Y, \mu_Y)$ be σ -finite measure spaces. Let $f : X \times Y \rightarrow [0, +\infty]$ be an $\mathcal{A}_X \otimes \mathcal{A}_Y$ -measurable mapping. Then the mapping*

$$(x \in X) \mapsto \int_Y f(x, y) \mu_Y(dy)$$

is $\mathcal{A}_X$ -measurable. Moreover, there is a unique σ -finite measure $\mu_X \otimes \mu_Y$, such that

$$\int_{X \times Y} f(x, y) d(\mu_X \otimes \mu_Y) = \int_X \int_Y f(x, y) \mu_Y(dy) \mu_X(dx).$$

Proof. Let $\mathcal{H}$ be the set of bounded $\mathcal{A}_X \otimes \mathcal{A}_Y$ -measurable mapping $f : X \times Y \rightarrow \mathbb{R}_{\geq 0}$ such that the mapping

$$(x \in X) \mapsto \int_Y f(x, y) \mu_Y(dy)$$

is $\mathcal{A}_X$ -measurable. Then $\mathcal{H}$ is a λ -family. Moreover, $\mathcal{H}$ contains the set $\mathcal{C}$ of positive linear combinations of mappings of the form $f_1(x)f_2(y)$, where $f_1 : X \rightarrow \mathbb{R}_{\geq 0}$ and $f_2 : Y \rightarrow \mathbb{R}_{\geq 0}$ are bounded and measurable mappings. Note that the family $\mathcal{C}$ is stable by multiplication. We then obtain that $\mathcal{H}$ contains all bounded $\mathcal{A}_X \otimes \mathcal{A}_Y$ -measurable mappings from $X \times Y$ to $\mathbb{R}_{\geq 0}$.

Let $\mathcal{A}_0$ be the family of subsets of $X \times Y$ which can be written as $A \times B$, where $(A, B) \in \mathcal{A}_X \times \mathcal{A}_Y$. This is a semi-ring on $X \times Y$ and

$$\nu : A \times B \mapsto \mu_X(A)\mu_Y(B) = \int_X \int_Y \mathbb{1}_{A \times B}(x, y) d(\mu_X \otimes \mu_Y)$$

defines a σ -additive mapping from $\mathcal{G}_0$ to $[0, +\infty]$. By Carathéodory theorem, it extends in a unique way to a σ -finite measure ν on $\sigma(\mathcal{G}_0) = \mathcal{G}$. Moreover, by definition, the set of bounded $\mathcal{A}_X \otimes \mathcal{A}_Y$ -measurable mappings satisfying the equality

$$\int_{X \times Y} f(x, y) \, d\nu = \int_X \int_Y f(x, y) \, \mu_Y(dy) \mu_X(dx)$$

forms a λ -family containing $\mathcal{C}$. Hence it contains all bounded $\mathcal{A}_X \otimes \mathcal{A}_Y$ -measurable mappings from $X \times Y$ to $\mathbb{R}$. $\square$

10.9.3 Theorem. *Let $(X, \mathcal{A}_X, \mu_X)$ and $(Y, \mathcal{A}_Y, \mu_Y)$ be σ -finite measure spaces. A mapping $f : X \times Y \rightarrow \mathbb{R}$ is $\mu_X \otimes \mu_Y$ -integrable if and only if*

$$\int_X \int_Y |f(x, y)| \, \mu_Y(dy) \mu_X(dx) < \infty.$$

Moreover, in the case where f is $\mu_X \otimes \mu_Y$ -integrable, one has

$$\int_{X \times Y} f(x, y) \, d(\mu_X \otimes \mu_Y) = \int_X \int_Y f(x, y) \, \mu_Y(dy) \mu_X(dx).$$

Proof. The first statement is a consequence of the previous theorem. As for the second equality, it suffices to write f as the difference of two integrable and non-negative mappings. $\square$

10.9.4 Definition. Let a and b be real numbers such that $a < b$ and $\varphi : [a, b] \rightarrow \mathbb{R}$ be a right continuous mapping, which can be written as $\varphi_1 - \varphi_2$, where φ_1 and φ_2 are two right continuous increasing mappings. For any $x \in]a, b]$, we denote by $\varphi(x-)$ the limit

$$\lim_{t \leq x, t \rightarrow x} \varphi(t) = \lim_{t \leq x, t \rightarrow x} \varphi_1(t) - \lim_{t \leq x, t \rightarrow x} \varphi_2(t),$$

and by $\Delta\varphi(x)$ the difference $\varphi(x) - \varphi(x-)$.

If $f : [a, b] \rightarrow \mathbb{R}$ is a bounded Borel measurable mapping, we denote by

$$\int_a^b f(x) \, d\varphi(x)$$

the difference

$$\int_a^b f(x) \, d\varphi_1(x) - \int_a^b f(x) \, d\varphi_2(x).$$

Note that the difference does not depend on the decomposition $\varphi = \varphi_1 - \varphi_2$.

10.9.5 Theorem. Let $(a, b) \in \mathbb{R} \times \mathbb{R}$, $a < b$. Let φ and ψ be right continuous mappings from $[a, b]$ to $\mathbb{R}$, which can be written as the difference of two right continuous and increasing mapping from $[a, b]$ to $\mathbb{R}$. Then the following equality holds:

$$\int_a^b \varphi(t) d\psi(t) + \int_a^b \psi(t) d\varphi(t) = \varphi(b)\psi(b) - \varphi(a)\psi(a) + \sum_{t \in]a, b]} \Delta\varphi(t)\Delta\psi(t).$$

Proof. Without loss of generality, we may assume that ψ and φ are both increasing. Let η be the Borel measure on $]a, b] \times]a, b]$ that is the product of $d\varphi$ and $d\psi$. Then

$$\begin{aligned} \int_a^b (\varphi(y) - \varphi(a)) d\psi(y) &= \int_{]a, b]} \int_{]a, y]} d\varphi(x) d\psi(y) \\ &= \int_{]a, b] \times]a, b]} \mathbb{1}_{\{(x, y) \in]a, b] \times]a, b] \mid x \leq y\}} d\eta \end{aligned}$$

Similarly,

$$\begin{aligned} \int_a^b (\psi(x) - \psi(a)) d\varphi(x) &= \int_{]a, b]} \int_{]a, x]} d\psi(y) d\varphi(x) \\ &= \int_{]a, b] \times]a, b]} \mathbb{1}_{\{(x, y) \in]a, b] \times]a, b] \mid y \leq x\}} d\eta \end{aligned}$$

Therefore, we obtain

$$\begin{aligned} &\int_a^b \varphi(y) d\psi(y) - \varphi(a)(\psi(b) - \psi(a)) + \int_a^b \psi(x) d\varphi(x) - \psi(a)(\varphi(b) - \varphi(a)) \\ &= \eta([a, b] \times]a, b]) + \int_{]a, b]} \mathbb{1}_D d\eta = (\psi(b) - \psi(a))(\varphi(b) - \varphi(a)) + \int_{]a, b]} \mathbb{1}_D d\eta, \end{aligned}$$

where $D = \{(t, t) \mid t \in]a, b]\}$ denotes the diagonal subset of $]a, b] \times]a, b]$. By Fubini's theorem, one has

$$\int_{]a, b]} \mathbb{1}_D d\eta = \int_a^b \Delta\psi(t) d\varphi(t) = \sum_{t \in]a, b]} \Delta\varphi(t)\Delta\psi(t).$$

Thus we get the desired result. $\square$

10.9.6 Example. Let $\lfloor \cdot \rfloor : \mathbb{R} \rightarrow \mathbb{R}$ be the mapping that sends $x \in \mathbb{R}$ to the greatest integer bounded from below by x . This mapping is increasing and right continuous. For any $(a, b) \in \mathbb{R} \times \mathbb{R}$ such that $a < b$, one has

$$\lfloor b \rfloor - \lfloor a \rfloor = \sum_{\substack{n \in \mathbb{Z} \\ a < n \leq b}} 1 = \text{card}([a, b] \cap \mathbb{Z}).$$

Note that the mapping

$$\mu : A \mapsto \text{card}(A \cap \mathbb{Z})$$

defines a measure on $(\mathbb{R}, \mathcal{P}(\mathbb{R}))$. For any mapping $f : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$, one has

$$\int_{\mathbb{R}} f \, d\mu = \sum_{n \in \mathbb{Z}} f(n).$$

By Carathéodory's theorem, the Borel measure $\mu_{[\cdot]}$ identifies with the restriction of μ on the Borel σ -algebra. We then deduce that, for any Borel measurable mapping $f : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$, one has

$$\int_{\mathbb{R}} f(x) \, d[x] = \sum_{n \in \mathbb{Z}} f(n).$$

Let $(a, b) \in \mathbb{R} \times \mathbb{R}$ and $f : [a, b] \rightarrow \mathbb{R}$ be a mapping which can be written as the difference of two right continuous increasing mappings from $[a, b]$ to $\mathbb{R}$. One has

$$\int_a^b f(x) \, d[x] + \int_a^b [x] \, df(x) = f(b)[b] - f(a)[a] + \sum_{\substack{n \in \mathbb{Z} \\ a < n \leq b}} \Delta f(n).$$

In particular, if f is continuous, one has

$$\int_a^b f(x) \, d[x] + \int_a^b [x] \, df(x) = f(b)[b] - f(a)[a].$$

This formula is useful in the estimation of some series. For example, for any real number $x > 1$, one has

$$\begin{aligned} \sum_{\substack{n \in \mathbb{N} \\ 1 \leq n \leq x}} \frac{1}{n} &= 1 + \sum_{\substack{n \in \mathbb{N} \\ 1 < n \leq x}} \frac{1}{n} = 1 + \int_1^x \frac{1}{t} d[t] = 1 + \frac{[x]}{x} - 1 - \int_1^x [t] d\left(\frac{1}{t}\right) \\ &= \frac{[x]}{x} + \int_1^x \frac{[t]}{t^2} dt = 1 - \frac{\langle x \rangle}{x} + \int_1^x \frac{1}{t} dt - \int_1^x \frac{\langle t \rangle}{t^2} dt, \end{aligned}$$

where $\langle t \rangle = t - [t] \in [0, 1[$. Note that, for $x \geq 1$,

$$\int_x^{+\infty} \frac{\langle t \rangle}{t^2} dt \leq \int_x^{+\infty} \frac{1}{t^2} dt = \frac{1}{x} \leq 1.$$

Let

$$\gamma = 1 - \int_1^{+\infty} \frac{\langle t \rangle}{t^2} dt.$$

One has

$$\sum_{\substack{n \in \mathbb{N} \\ 1 \leq n \leq x}} \frac{1}{n} = \ln(x) + \gamma + O\left(\frac{1}{x}\right).$$

Exercises

1. In this exercise, let Ω be a non-empty set, S be a vector subspace of $\mathbb{R}^\Omega$ which is stable by $\wedge$ and $I : S \rightarrow \mathbb{R}$ be an integral operator. Let $(f_n)_{n \in \mathbb{N}}$ be an increasing sequence of elements of the vector space $\tilde{L}^1(I)$ and let $f : \Omega \rightarrow]-\infty, +\infty]$ be the pointwise limit of the sequence $(f_n)_{n \in \mathbb{N}}$. We denote by α the limit of the sequence $(I(f_n))_{n \in \mathbb{N}}$.

- (1) Let $\varepsilon \in \mathbb{R}_{>0}$. Prove that, for any $n \in \mathbb{N}_{\geq 1}$ there exists $u_n \in S^\uparrow$ such that $f_n - f_{n-1} \leq u_n$ and that

$$I(f_n - f_{n-1}) \geq I(u_n) - \frac{\varepsilon}{2^n}.$$

- (2) Prove that, for any $n \in \mathbb{N}_{\geq 1}$,

$$f_n \leq u_1 + \cdots + u_n$$

and

$$I(f_n) \geq I(u_1) + \cdots + I(u_n) - \varepsilon.$$

- (3) Let u be the pointwise limit of the series $\sum_{n \in \mathbb{N}_{\geq 1}} u_n$. Prove that $u \in S^\uparrow$ and

$$I(u) = \sum_{n \in \mathbb{N}_{\geq 1}} I(u_n).$$

- (4) Prove that $u \geq f$ and

$$\alpha \geq I(u) - \varepsilon \geq \bar{I}(f) - \varepsilon.$$

- (5) Prove that, for any $n \in \mathbb{N}$, there exists $\ell_n \in S^\downarrow$ such that $\ell_n \leq f_n$ and

$$I(\ell_n) \geq I(f_n) - \varepsilon.$$

Deduce that

$$\underline{I}(f) \geq I(f_n) - \varepsilon.$$

- (6) Prove that $\underline{I}(f) \geq \alpha - \varepsilon$.
 (7) Suppose that $\alpha \in \mathbb{R}$. Prove that f belongs to $\tilde{L}^1(I)$ and $I(f) = \alpha$.
 (8) Deduce that $I : \tilde{L}^1(I) \rightarrow \mathbb{R}$ is an integral operator.
 (9) Suppose that $\alpha = +\infty$. Prove that $I(f) = +\infty$.

(10) Prove that, if $(g_n)_{n \in \mathbb{N}}$ is a decreasing sequence in $\tilde{L}^1(I)$ such that

$$\lim_{n \rightarrow +\infty} I(g_n) \in \mathbb{R},$$

then the sequence $(g_n)_{n \in \mathbb{N}}$ converges pointwisely to an element $g \in \tilde{L}^1(I)$ and

$$I(g) = \lim_{n \rightarrow +\infty} I(g_n).$$

2. In this exercise, let Ω be a non-empty set, S be a vector subspace of $\mathbb{R}^\Omega$ which is stable by $\wedge$ and $I : S \rightarrow \mathbb{R}$ be an integral operator. Let $(f_n)_{n \in \mathbb{N}}$ be a sequence in $\tilde{L}^1(I)$. Assume that there exists $g \in \tilde{L}^1(I)$ such that $f_n \geq g$ for any $n \in \mathbb{N}$.

(1) For any $n \in \mathbb{N}$, let

$$g_n = \inf_{m \in \mathbb{N}_{\geq n}} f_m.$$

Prove that g_n is the pointwise limit of the decreasing sequence

$$(f_n \wedge \cdots \wedge f_{n+k})_{k \in \mathbb{N}}.$$

(2) Prove that, for any $n \in \mathbb{N}$, $g_n \in \tilde{L}^1(I)$ and $I(g_n) \leq I(f_n)$.

(3) Prove that

$$\lim_{n \rightarrow +\infty} I(g_n) \leq \liminf_{n \rightarrow +\infty} I(f_n).$$

(4) Prove that

$$\liminf_{n \rightarrow +\infty} f_n \in \tilde{L}^1(I)^\uparrow$$

and

$$I(\liminf_{n \rightarrow +\infty} f_n) \leq \liminf_{n \rightarrow +\infty} I(f_n).$$

3. In this exercise, let Ω be a non-empty set, S be a vector subspace of $\mathbb{R}^\Omega$ which is stable by $\wedge$ and $I : S \rightarrow \mathbb{R}$ be an integral operator. Let $(f_n)_{n \in \mathbb{N}}$ be a sequence in $\tilde{L}^1(I)$, which converges simply to a mapping $f : \Omega \rightarrow \mathbb{R}$. Assume that there exists $g \in \tilde{L}^1(I)$ such that $|f_n| \leq g$ for any $n \in \mathbb{N}$.

(1) Prove that $f \in \tilde{L}^1(I)$.

(2) Prove that

$$I(f) \leq \liminf_{n \rightarrow +\infty} I(f_n), \quad I(-f) \leq \liminf_{n \rightarrow +\infty} I(-f_n).$$

(3) Deduce that $(I(f_n))_{n \in \mathbb{N}}$ converges to $I(f)$.

4. In this exercise, we let S be the set of mappings in $\mathbb{R}^{\mathbb{R}}$ which can be written in the form of

$$\sum_{i=1}^n \lambda_i \mathbb{1}_{[a_i, b_i]},$$

where $n \in \mathbb{N}$, $(\lambda_1, \dots, \lambda_n) \in \mathbb{R}^n$, and for any $i \in \{1, \dots, n\}$,

$$(a_i, b_i) \in \mathbb{R}^2, \quad a_i < b_i.$$

Denote by $I : S \rightarrow \mathbb{R}$ the integral operator defined as

$$I(\lambda_1 \mathbb{1}_{[a_1, b_1]} + \dots + \lambda_n \mathbb{1}_{[a_n, b_n]}) = \sum_{i=1}^n \lambda_i (b_i - a_i).$$

If A is a subset of $\mathbb{R}$ and f is a function from $\mathbb{R}$ to $\mathbb{R}$ such that the mapping

$$f_A : \mathbb{R} \rightarrow \mathbb{R}, \quad t \mapsto \begin{cases} f(t), & \text{if } t \in A \cap \text{Dom}(f), \\ 0, & \text{else} \end{cases}$$

belongs to $\tilde{L}^1(I)^\uparrow \cup \tilde{L}^1(I)^\downarrow$, then we denote by

$$\int_A f(t) \, dt$$

the value $I(f_A)$. If $f_A \in \tilde{L}^1(I)$, then we say that f is *integrable* on A .

- (1) Let a and b be two real numbers such that $a \leq b$. Prove that, for any $f \in \tilde{L}^1(I)$, the mappings

$$\mathbb{1}_{]a, b]} f, \quad \mathbb{1}_{[a, b[} f, \quad \mathbb{1}_{]a, b[} f, \quad \mathbb{1}_{[a, b]} f$$

are all elements of $\tilde{L}^1(I)$, and

$$\int_{]a, b]} f(t) \, dt = \int_{[a, b[} f(t) \, dt = \int_{]a, b[} f(t) \, dt = \int_{[a, b]} f(t) \, dt.$$

In what follows, we denote by

$$\int_a^b f(t) \, dt$$

this value.

(2) Let a and b be real numbers such that $a < b$ and $f : [a, b] \rightarrow \mathbb{R}$ be a mapping. We assume that f is integrable on $[a, b]$.

(a) Prove that, for any $x \in [a, b]$, the function f is integrable on $[a, x]$, and the mapping

$$(x \in [a, b]) \longrightarrow \int_a^x f(t) \, dt$$

is continuous.

(b) Suppose that f is continuous. Prove that the mapping

$$x \longmapsto \int_a^x f(t) \, dt$$

is differentiable on $]a, b[$ and its derivative coincides with the restriction of f to $]a, b[$.

(c) Let $F : [a, b] \rightarrow \mathbb{R}$ be a continuous mapping that is differentiable on $]a, b[$ and satisfies $F'(x) = f(x)$ for any $x \in]a, b[$. Prove that

$$\int_a^b f(t) \, dt = F(b) - F(a).$$

(3) Let a and b be real numbers such that $a < b$. Determine

$$\int_a^b t^n \, dt, \quad \int_a^b \exp(t) \, dt, \quad \int_a^b \cos(t) \, dt, \quad \int_a^b \sin(t) \, dt.$$

(4) Prove that the mapping

$$\mathbb{R} \longrightarrow \mathbb{R}, \quad t \longmapsto \frac{1}{1+t^2}$$

is integrable on $\mathbb{R}$ and determine its integral.

(5) Prove that the mapping

$$\mathbb{R} \longrightarrow \mathbb{R}, \quad t \longmapsto \exp(-|t|)$$

is integrable on $\mathbb{R}$ and determine its integral.

(6) Prove that the mapping

$$\mathbb{R} \longrightarrow \mathbb{R}, \quad t \longmapsto \exp(-t^2)$$

is integrable on $\mathbb{R}$.

5. In this exercise, we let S be the set of mappings in $\mathbb{R}^{\mathbb{R}}$ which can be written in the form of

$$\sum_{i=1}^n \lambda_i \mathbb{1}_{[a_i, b_i]},$$

where $n \in \mathbb{N}$, $(\lambda_1, \dots, \lambda_n) \in \mathbb{R}^n$, and for any $i \in \{1, \dots, n\}$,

$$(a_i, b_i) \in \mathbb{R}^2, \quad a_i < b_i.$$

Let $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ be an increasing and right continuous mapping. Denote by $I_\varphi : S \rightarrow \mathbb{R}$ the integral operator defined as

$$I_\varphi(\lambda_1 \mathbb{1}_{[a_1, b_1]} + \dots + \lambda_n \mathbb{1}_{[a_n, b_n]}) = \sum_{i=1}^n \lambda_i (\varphi(b_i) - \varphi(a_i)).$$

If A is a subset of $\mathbb{R}$ and f is a function from $\mathbb{R}$ to $\mathbb{R}$ such that the mapping

$$f_A : \mathbb{R} \rightarrow \mathbb{R}, \quad t \mapsto \begin{cases} f(t), & \text{if } t \in A \cap \text{Dom}(f), \\ 0, & \text{else} \end{cases}$$

belongs to $\tilde{L}^1(I)^\uparrow \cup \tilde{L}^1(I)^\downarrow$, then we denote by

$$\int_A f(t) d\varphi(t)$$

the value $I(f_A)$. If $f_A \in \tilde{L}^1(I)$, then we say that f is $d\varphi$ -integrable on A . For any $a \in \mathbb{R}$, we denote by $\Delta\varphi(a)$ the difference $\varphi(a) - \varphi_-(a)$, where

$$\varphi_-(a) := \lim_{t>0, t \rightarrow 0} \varphi(a-t).$$

- (1) Let a be a real number, show that $\mathbb{1}_{\{a\}}$ is $d\varphi$ -integrable on $\mathbb{R}$. Show that

$$\int_{\mathbb{R}} \mathbb{1}_{\{a\}}(t) d\varphi(t) = \Delta\varphi(a).$$

- (2) Let a and b be two real numbers such that $a \leq b$. Prove that, for any $f \in \tilde{L}^1(I_\varphi)$, the mappings

$$\mathbb{1}_{]a, b]} f, \quad \mathbb{1}_{[a, b[} f, \quad \mathbb{1}_{]a, b[} f, \quad \mathbb{1}_{[a, b]} f$$

are all elements of $\tilde{L}^1(I)$, and

$$\begin{aligned} \int_{]a, b]} f(t) d\varphi(t) &= \int_{]a, b[} f(t) d\varphi(t) + f(b) \Delta\varphi(b), \\ \int_{[a, b[} f(t) d\varphi(t) &= \int_{]a, b[} f(t) d\varphi(t) + f(a) \Delta\varphi(a), \\ \int_{[a, b]} f(t) d\varphi(t) &= \int_{]a, b[} f(t) d\varphi(t) + f(a) \Delta\varphi(a) + f(b) \Delta\varphi(b). \end{aligned}$$

- (3) Let a and b be real numbers such that $a < b$. Assume that ψ is an integrable function on $[a, b]$ and that

$$\forall x \in [a, b], \quad \varphi(x) = \varphi(a) + \int_a^x \psi(t) \, dt.$$

Prove that, for any function f which is $d\varphi$ -integrable on $]a, b]$, the function $f\psi$ is integrable on $]a, b]$, and

$$\int_{]a, b]} f(t) \, d\varphi(t) = \int_a^b f(t)\psi(t) \, dt.$$

- (4) We assume that φ is continuous. Let a and b be real numbers such that $a < b$. Prove that, for any integrable function f on $[\varphi(a), \varphi(b)]$, the function $f \circ \varphi$ is $d\varphi$ -integrable on $]a, b]$, and one has

$$\int_{[\varphi(a), \varphi(b)]} f(x) \, dx = \int_{]a, b]} f(\varphi(t)) \, d\varphi(t).$$

- (5) Compute

$$\int_{\mathbb{R}} x \exp(-x^2) \, dx.$$

6. Find the primitive functions of the following mappings.

- (1) $(x \in \mathbb{R}_{>0}) \mapsto x^\alpha$, where $\alpha \in \mathbb{R}$,
- (2) $(x \in \mathbb{R} \setminus \{0\}) \mapsto x^{-1}$,
- (3) $(x \in \mathbb{R}) \mapsto a^x$, where $a \in \mathbb{R}_{>0}$,
- (4) $(x \in \mathbb{R}) \mapsto \cos(x)$,
- (5) $(x \in \mathbb{R}) \mapsto \sin(x)$,
- (6) $(x \in]-\pi/2, \pi/2[) \mapsto 1/\cos(x)^2$,
- (7) $(x \in]-1, 1[) \mapsto 1/\sqrt{1-x^2}$,
- (8) $(x \in]-1, 1[) \mapsto 1/(1+x^2)$.

7. Determine the primitive functions of the following functions.

- (1) $f(x) = x^3 - x + 1$,
- (2) $f(x) = \cos(x) - \sin(x)$,
- (3) $f(x) = 1 - e^x + x$,
- (4) $f(x) = \sqrt{x} + \frac{1}{x} + \frac{2}{x^2}$,

(5) $f(x) = (x + 1)/x^2$.

8. Using the method of change of variables, determine the primitive functions of the following functions.

(1) $f(x) = x/(1 + x^2)$,

(2) $f(x) = \cos(x) \sin(x)^2$,

(3) $f(x) = \ln(x)/x$,

(4) $f(x) = 1/(x \ln(x))$,

(5) $f(x) = e^x/(1 + e^x)$,

(6) $f(x) = x\sqrt{1 + x^2}$.

9. Compute the following integrals.

$$\int_0^\pi (1 - \cos(3x)) \, dx, \quad \int_0^{\sqrt{\pi}} x \cos(x^2) \, dx, \quad \int_1^2 \frac{\ln(x)}{x} \, dx.$$

10. Using the method of integration by parts, compute the following integrals.

$$\int_0^1 x e^x \, dx, \quad \int_1^e x^2 \ln(x) \, dx.$$

11. Using the method of integration by parts, determine the primitive function of the following functions.

(1) $x \mapsto \arctan(x)$,

(2) $x \mapsto \ln(x)^2$,

(3) $x \mapsto \sin(\ln(x))$,

(4) $x \mapsto \ln(x + \sqrt{1 + x^2})$,

(5) $x \mapsto e^{2x}(\tan(x) + 1)^2$.

12. The purpose of this exercise is to determine the primitive function of the mapping $x \mapsto \ln(x)^n$. We denote by F_n a primitive function of $x \mapsto \ln(x)^n$ such that $F_n(1) = (-1)^n n!$.

- (1) Prove that, for any positive integer n , one has

$$F_n(x) + nF_{n-1}(x) = x \ln(x)^n.$$

(2) Prove that

$$F_n(x) = \sum_{k=0}^n (-1)^k \frac{n!}{(n-k)!} x \ln(x)^{n-k}.$$

13. Let $(a, b, m, n) \in \mathbb{R}^2 \times \mathbb{N}^2$ such that $a < b$. We denote by $A_{m,n}$ the integral

$$\int_a^b (t-a)^m (t-b)^n dt.$$

(1) Prove that

$$A_{m,n} = -\frac{m}{n+1} A_{m-1,n+1}.$$

(2) Determine the value of $A_{0,n}$ for $n \in \mathbb{N}$.

(3) Prove that, for any $n \in \mathbb{N}$,

$$A_{n,n} = \frac{(-1)^{n+1} (a-b)^{2n+1}}{(2n+1) \binom{2n}{n}}.$$

14. Using a change of variables, determine the following integrals.

(1) $\int_1^4 \frac{1-\sqrt{t}}{\sqrt{t}} dt,$

(2) $\int_0^\pi \frac{\sin(t)}{1+\cos(t)^2} dt,$

(3) $\int_1^e \frac{1}{(2t \ln(t) + t)} dt.$

(4) $\int_0^1 \frac{1}{1+e^t} dt.$

(5) $\int_1^3 \frac{\sqrt{t}}{t+1} dt,$

(6) $\int_{-1}^1 \sqrt{1-t^2} dt.$

(7) $\int_0^{\pi/6} \frac{1}{\cos(t)} dt$ (by the change of variables $t \mapsto \sin(t)$),

(8) $\int_1^2 \frac{2t}{\sqrt{1+t}} dt$ (by the change of variables $t \mapsto \sqrt{1+t}$),

(9) $\int_0^3 \frac{1}{\sqrt{1+\sqrt{1+t}}} dt$ (by the change of variables $t \mapsto \sqrt{1+t}$).

15. Determine the primitive function of the following functions.

- (1) $x \mapsto \frac{1}{x^3 - 1},$
- (2) $x \mapsto \frac{x^3 + 2x}{x^2 + x + 1},$
- (3) $x \mapsto \frac{1}{x^3 - 7x + 6},$
- (4) $x \mapsto \frac{4x^2}{x^4 - 1}.$

16. For any $n \in \mathbb{N}_{\geq 1}$, let

$$A_n = \int_0^1 \frac{1}{(x^2 + 1)^n} dx.$$

- (1) Determine A_1 .
- (2) Prove that, for any $n \in \mathbb{N}$ such that $n \geq 1$, one has

$$A_{n+1} = \frac{2n-1}{2n} A_n + \frac{1}{n2^{n+1}}.$$

- (3) Determine A_3 .

17. Prove that the function $\ln(\cdot)$ is Lebesgue integrable on $]0, 1]$ and determine its integral.

18. Prove that the function $t \mapsto e^{-|t|}$ is Lebesgue integrable on $\mathbb{R}$ and determine its integral.

19. Prove that the function $t \mapsto e^{-t^2}$ is Lebesgue integrable on $\mathbb{R}$.

20. The purpose of this exercise is to study the Lebesgue integrability of the mapping

$$f_\alpha : \mathbb{R}_{>0} \longrightarrow \mathbb{R}_{>0}, \quad x \longmapsto \frac{1}{x^\alpha}.$$

- (1) Find a primitive function F_α of the mapping f_α .
- (2) Prove that the mapping f_α is Lebesgue integrable on $]0, 1]$ if and only if $\alpha < 1$.
- (3) Prove that the mapping f_α is Lebesgue integrable on $[1, +\infty[$ if and only if $\alpha > 1$.

- 21.** Let α and β be real numbers. The purpose of this exercise is to determine the Lebesgue integrability of the function

$$f_{\alpha,\beta}(x) = \frac{1}{x^\alpha \ln(x)^\beta}$$

on $[e, +\infty[$.

- (1) Suppose that $\alpha > 1$. Prove that $f_{\alpha,\beta}$ is Lebesgue integrable on $[e, +\infty[$.
 - (2) Determine a primitive function of $f_{1,\beta}$.
 - (3) Prove that $f_{1,\beta}$ is Lebesgue integrable on $[e, +\infty[$ if and only if $\beta > 1$.
 - (4) Suppose that $\alpha < 1$. Prove that $f_{\alpha,\beta}$ is not Lebesgue integrable on $[e, +\infty[$.
- 22.** Let $f : [0, +\infty[$ be a continuous mapping which is Lebesgue integrable on $[0, +\infty[$.

- (1) Prove that, if the limit

$$\lim_{x \rightarrow +\infty} f(x)$$

exists, then it is necessarily 0.

- (2) Suppose that f is uniformly continuous. Prove that

$$\lim_{x \rightarrow +\infty} f(x) = 0.$$

- (3) Construct a continuous and integrable function f on $[0, +\infty[$ which does not converge when $x \rightarrow +\infty$.

- 23.** For any $x \in \mathbb{R}$, denote by f_x the mapping

$$(t \in]0, +\infty[) \mapsto \frac{\sin(xt)}{te^t}$$

- (1) Show that, for any $x \in \mathbb{R}$, the mapping f_x is Lebesgue integrable on $]0, +\infty[$.
- (2) For any $x \in \mathbb{R}$, let

$$F(x) = \int_0^{+\infty} \frac{\sin(xt)}{te^t} dt.$$

Prove that the mapping F is of class C^1 and determine F' .

- (3) Prove that $F(x) = \arctan(x)$.

24. In this exercise, we consider the following integral

$$F(x) = \int_0^{+\infty} \frac{e^{-xt}}{1+t^2} dt, \quad x \in \mathbb{R}_{>0}.$$

(1) Prove that, for any $x \in \mathbb{R}_{>0}$, $F(x)$ is well defined and takes value in $\mathbb{R}_{>0}$.

(2) Prove that

$$\lim_{x \rightarrow +\infty} F(x) = 0.$$

(3) Prove that F is of class C^2 .

(4) Prove that F verifies the following equation

$$F''(x) + F(x) = \frac{1}{x}.$$

25. The purpose of this exercise is to compute the value of the following integral

$$A = \int_0^{+\infty} e^{-t^2} dt.$$

For any $x \in \mathbb{R}_{\geq 0}$, let

$$f(x) = \int_0^x e^{-t^2} dt, \quad g(x) = \int_0^1 \frac{e^{-(t^2+1)x^2}}{t^2+1} dt.$$

(1) Prove that, for any $x \in \mathbb{R}_{\geq 0}$, one has

$$g(x) + f(x)^2 = \frac{\pi}{4}.$$

(2) Determine the limits of $f(x)$ and $g(x)$ when $x \rightarrow +\infty$.

(3) Deduce that $A = \sqrt{\pi}/2$.

26. Let $f : [0, 1] \rightarrow \mathbb{R}_{>0}$ be a continuous mapping. For any $\alpha \in \mathbb{R}_{\geq 0}$, let

$$F(\alpha) = \int_0^1 f(t)^\alpha dt.$$

(1) Prove that F is differentiable on $\mathbb{R}_{>0}$ and that $F'(\alpha)$ has a limit when $\alpha \rightarrow 0$.

(2) Prove that

$$\lim_{\alpha \rightarrow 0} \exp(F(\alpha)^{1/\alpha}) = \int_0^1 \ln f(t) dt.$$

27. Prove that the following limits exist and find the limits.

- (a) $\lim_{n \rightarrow +\infty} \frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{n+n};$
- (b) $\lim_{n \rightarrow +\infty} \frac{n}{n^2+1} + \frac{n}{n^2+2} + \cdots + \frac{n}{n^2+n};$
- (c) $\lim_{n \rightarrow +\infty} \frac{1}{n} \left(\sin \frac{\pi}{n} + \sin \frac{2\pi}{n} + \cdots + \sin \frac{n\pi}{n} \right);$
- (d) $\lim_{n \rightarrow +\infty} \left(\frac{1^\alpha}{n^{\alpha+1}} + \frac{2^\alpha}{n^{\alpha+1}} + \cdots + \frac{n^\alpha}{n^{\alpha+1}} \right);$
- (e) $\lim_{n \rightarrow +\infty} \sin \frac{\pi}{n} \cdot \sum_{k=0}^{n-1} \frac{1}{2 + \cos \frac{k\pi}{n}}.$

28. Find the derivatives of the following functions.

- (a) $f(x) = \int_0^{x^3} \sin^3 t dt;$
- (b) $f(x) = \int_1^x \frac{1}{1+t^2+\sin^2 t} dt;$
- (c) $f(x) = \sin \left(\int_0^x \cos \left(\int_0^y \sin^2 t dt \right) dy \right);$
- (d) $f^{-1}(x)$, where $f(x) = \int_1^x \frac{1}{t} dt;$
- (e) $f^{-1}(x)$, where $f(x) = \int_0^x \frac{1}{\sqrt{1-t^2}} dt, \quad x \in (0, 1).$

29. (a) Let $I, K \subseteq \mathbb{R}$ denote intervals. Suppose that $f : I \rightarrow \mathbb{R}$ is a continuous function and $\alpha, \beta : K \rightarrow I$ are differentiable functions. Define a function $F : K \rightarrow \mathbb{R}$ by

$$F(x) = \int_{\alpha(x)}^{\beta(x)} f(t) dt.$$

Prove that F is differentiable on K and in particular,

$$F'(x) = f(\beta(x))\beta'(x) - f(\alpha(x))\alpha'(x), \quad \forall x \in K.$$

(b) Find the derivative for the function $F : (1, \infty) \rightarrow \mathbb{R}$ defined by

$$F(x) = \int_{x^a}^{x^b} \frac{1}{\log t} dt,$$

where $a, b > 0$.

- 30.** (Oscillation and Lebesgue's theorem). We say that a subset $E \subset \mathbb{R}$ has (Lebesgue) *measure zero* if for any $\varepsilon > 0$ there exists a countable family of open intervals (I_n) such that $E \subset \cup_n I_n$ and $\sum_n l(I_n) \leq \varepsilon$, where $l(I_n)$ denotes the length of I_n , i.e., $l(I_n) = b_n - a_n$ if I_n has endpoints a_n and b_n .

(a) Prove that a countable union of sets of measure zero has again measure zero. In particular, any countable set has measure zero.

In the following consider a bounded function $f : [a, b] \rightarrow \mathbb{R}$. The *oscillation* of f on a subset $A \subset [a, b]$ is the quantity

$$\omega(f; A) = \sup_{x_1, x_2 \in A} |f(x_1) - f(x_2)|$$

and the *oscillation* of f at a point $x \in [a, b]$ is defined to be

$$\omega(f; x) = \lim_{\delta \rightarrow 0^+} \omega(f; [x - \delta, x + \delta] \cap [a, b]).$$

Denote by E the set of discontinuities of f .

- (b) Prove that $E = \{x \mid \omega(f; x) > 0\}$.
- (c) Prove that if f is Riemann integrable, then $\{x \mid \omega(f; x) > 1/n\}$ has measure zero for all $n \in \mathbb{N} \setminus \{0\}$. (Hint: use Darboux sums)
- (d) (*Lebesgue's theorem*) Prove that f is Riemann integrable if and only if E has measure zero.
- 31.** (Second mean value theorem for integrals) Let a and b be two real numbers such that $a < b$, f a continuous and monotone function on the closed interval $[a, b]$, and g a continuous function on $[a, b]$. Let G be the function defined by

$$G(x) = \int_a^x g(t) dt$$

for every x in $[a, b]$, and consider the integral:

$$I = \int_a^b f(t)g(t) dt = \int_a^b f(t)G'(t) dt.$$

- (a) Prove that for every real number $\varepsilon > 0$, one can find a finite family of elements a_i (for $i = 0, 1, \dots, n$) in $[a, b]$ such that

$$\left| I - \sum_{i=1}^n \int_{a_{i-1}}^{a_i} f(a_i)G'(t) dt \right| < \varepsilon.$$

(b) Establish the equality:

$$\sum_{i=1}^n \int_{a_{i-1}}^{a_i} f(a_i) G'(t) dt = f(b)G(b) - \sum_{i=2}^n G(a_{i-1})[f(a_i) - f(a_{i-1})].$$

(c) Let

$$\mathcal{I}_\varepsilon = \sum_{i=1}^n \int_{a_{i-1}}^{a_i} f(a_i) G'(t) dt.$$

Prove that the number $\mathcal{I}_\varepsilon - f(b)G(b)$ lies within the interval bounded by the numbers

$$[f(a_1) - f(b)][\inf_{x \in [a, b]} G(x)] \quad \text{and} \quad [f(a_1) - f(b)][\sup_{x \in [a, b]} G(x)].$$

Deduce that the number $I - f(b)G(b)$ lies within the interval bounded by the numbers

$$[f(a) - f(b)][\inf_{x \in [a, b]} G(x)] \quad \text{and} \quad [f(a) - f(b)][\sup_{x \in [a, b]} G(x)].$$

(d) Consider the continuous function H defined on the interval $[a, b]$ by $H(x) = [f(a) - f(b)]G(x)$; prove that there exists a number c in the interval $[a, b]$ such that $H(c) = I - f(b)G(b)$. Deduce the following equality:

$$I = f(a) \int_a^c g(t) dt + f(b) \int_c^b g(t) dt.$$

(This equality is called the *second mean value theorem for integrals*.)

32. (Cauchy-Schwarz, Hölder and Minkowski inequalities)

(a) Show that if f is Riemann integrable on $[a, b]$, then so is $|f|^p$ for $p \geq 0$. In the following write

$$\|f\|_p = \left(\int_a^b |f|^p(x) dx \right)^{1/p}, \quad p > 1.$$

(b) Starting from Hölder's inequality for sums, obtain *Hölder's inequality* for integrals (*Cauchy-Schwarz inequality* for $p = 2$):

$$\left| \int_a^b (f \cdot g)(x) dx \right| \leq \|f\|_p \|g\|_q,$$

if f, g are Riemann-integrable functions on $[a, b]$, $p > 1, q > 1$, and $\frac{1}{p} + \frac{1}{q} = 1$.

Note that the equality occurs when $|f|^p / \|f\|_p^p = |g|^q / \|g\|_q^q$. Deduce that

$$\|f\|_p = \sup \left\{ \int_a^b (f \cdot g)(x) dx \mid \|g\|_q \leq 1, g \text{ is Riemann integrable} \right\}$$

(c) Deduce *Minkowski's inequality* for integrals:

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p.$$

if f, g is Riemann integrable on $[a, b]$ and $p \geq 1$.

(d) Show that the previous inequality reverses direction if $0 < p < 1$.

(e) Verify that if f is a continuous convex function on $\mathbb{R}$ and φ an arbitrary continuous function on $\mathbb{R}$, then Jensen's inequality

$$f\left(\frac{1}{c} \int_0^c \varphi(t) dt\right) \leq \frac{1}{c} \int_0^c f(\varphi(t)) dt$$

holds for $c \neq 0$.

33. Let f be a function of class C^2 on $\mathbb{R}$ and x a point in $\mathbb{R}$ such that $f''(x) \neq 0$.

(a) Show that there exists $\eta > 0$ such that for all $h \in [-\eta, \eta] \setminus \{0\}$, there exists a unique number $\theta \in (0, 1)$ with

$$f(x + h) = f(x) + hf'(x + \theta h).$$

The function defined on $[-\eta, \eta] \setminus \{0\}$ that associates h with this unique θ is denoted by θ_x .

(b) Show that

$$\lim_{h \rightarrow 0} \theta_x(h) = \frac{1}{2}.$$

34. Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function. Use the fundamental theorem of calculus and Darboux theorem to prove the intermediate value theorem.

35. Let f be a continuous function on $[-1, 1]$. Prove that $\lim_{h \rightarrow 0^+} \int_{-1}^1 \frac{h}{h^2 + x^2} f(x) dx = \pi f(0)$. (Hint: consider the integrals on a carefully-chosen small neighborhood of 0 dependent of h and on its complement separately, and use the mean value theorem.)

- 36.** Let $a, b > 0$. Calculate the interior area of the ellipse, i.e., the region enclosed by the curve

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

- 37.** We propose to calculate the area of the region delimited by the two curves:

$$f(x) = \frac{x^2}{2} \quad \text{and} \quad g(x) = \frac{1}{1+x^2}.$$

- (1) Show that the two curves intersect at the points of abscissa $x = 1$ and $x = -1$.
- (2) Show that $f(x) < g(x)$ for all $x \in (-1, 1)$.
- (3) Deduce that the bounded region is located above the graph of f and below the graph of g between the two vertical lines of equation $(x = 1)$ and $(x = -1)$.
- (4) Deduce that the desired area A is expressed:

$$A = \int_{-1}^1 g(x) dx - \int_{-1}^1 f(x) dx.$$

- (5) Conclude that $A = \frac{\pi}{2} - \frac{1}{3}$.

- 38.** Find the following limits

$$(1) \lim_{n \rightarrow \infty} \frac{\sqrt[n]{(n+1) \cdots (n+n)}}{n}$$

$$(2) \lim_{n \rightarrow \infty} \prod_{k=1}^n \left(2 + \cos \left(\frac{k\pi}{n} \right) \right)^{\pi/n}$$

$$(3) \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{n+k}$$

$$(4) \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{2}{2n+2k-1}$$

- 39.** We partition the interval $[1, 2]$ into n subintervals by the points $\{1+k/n\}_{k=0}^n$. We denote by U_n and L_n the corresponding upper and lower Darboux sums, respectively. Show that

$$U_1 = 1, \quad L_1 = 1 - \frac{1}{2}, \quad U_2 = 1 - \frac{1}{2} + \frac{1}{3}, \quad L_2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4}, \dots$$

and generally the sequence $U_1, L_1, U_2, L_2, \dots, U_n, L_n, \dots$ is identical with the sequence of the partial sums of the series

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{1}{2n-1} - \frac{1}{2n} + \cdots$$

Then show that

$$\lim_{n \rightarrow \infty} \left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{1}{2n-1} - \frac{1}{2n} \right) = \log 2.$$

40. For any pair of reals $\{a, b\}$ with $a \leq b$ assume that a map $I_{a,b} : R([a, b]) \rightarrow \mathbb{R}$ is given and satisfies the following conditions:

(1) $I_{a,b}$ is a linear functional on $R([a, b])$, i.e.

$$I_{a,b}(\lambda f + g) = \lambda I_{a,b}(f) + I_{a,b}(g), \quad \forall f, g \in R([a, b]), \lambda \in \mathbb{R};$$

(2) $I_{a,b}(1) = b - a$;

(3) $f \geq 0 \Rightarrow I_{a,b}(f) \geq 0$;

(4) $I_{a,b}(f) = I_{a,c}(f) + I_{c,b}(f)$ for any $f \in R([a, b])$ and $c \in [a, b]$.

Our aim is to show that $I_{a,b}(f) = \int_a^b f(x)dx$.

(1) Show that $f \leq g \Rightarrow I_{a,b}(f) \leq I_{a,b}(g)$ and $|I_{a,b}(f)| \leq I_{a,b}(|f|)$.

(2) Show that if $f, g \in R([a, b])$ differ only at a finite number of points $x_1, \dots, x_n$ in $[a, b]$, then $I_{a,b}(f) = I_{a,b}(g)$.

(3) Show that $I_{a,b}(f) = \int_a^b f(x)dx$ first for any step function, then for any $f \in R([a, b])$.

41. Let $f \in R([a, b])$ and define $F(x) = \int_a^x f(t)dt$ for $x \in [a, b]$.

(1) Show that F is continuous on $[a, b]$.

(2) Assume that f admits a right (resp. left) limit at $x_0 \in [a, b]$. Show that F is right (resp. left) differentiable at x_0 and its right (resp. left) derivative $F'(x_0+)$ (resp. $F'(x_0-)$) is equal to $f(x_0+)$ (resp. $f(x_0-)$).

(3) How about the converse to the previous assertion? Namely, if F is right (resp. left) differentiable at x_0 , does f admit a right (resp. left) limit at x_0 ? Justify your answer by a proof or by a counter-example.

(4) Assume now that f is continuous on $[a, b]$ and that $u : I \rightarrow [a, b]$ is a differentiable function, where I is an interval. Show that the function G defined by

$$G(x) = \int_a^{u(x)} f(t)dt$$

is differentiable on I .

42. We propose to show the **second mean value theorem**: Let $f : [a, b] \rightarrow \mathbb{R}_+$ be decreasing and $g : [a, b] \rightarrow \mathbb{R}$ be integrable. Then there exists $c \in [a, b]$ such that

$$\int_a^b f(x)g(x)dx = f(a+) \int_a^c g(x)dx,$$

where $f(a+) = \lim_{x \searrow a} f(x)$. In the following we assume that $f(a+) = f(a) > f(b)$, otherwise, there is nothing to prove. Let

$$G(x) = \int_a^x g(t)dt.$$

- (1) Let $m = \min_{x \in [a,b]} G(x)$ and $M = \max_{x \in [a,b]} G(x)$. Show that the desired statement is equivalent to

$$(*) \quad mf(a) \leq \int_a^b f(x)g(x)dx \leq Mf(a).$$

Further, replacing g by $-g$, it suffices to show the second inequality above.

- (2) Show the second inequality of $(*)$ if f is a step function.
 (3) Show that for any $\varepsilon > 0$ there exists a decreasing step function such that

$$0 \leq f \text{ and } \int_a^b (f(x) - (x))dx < \varepsilon.$$

- (4) Deduce the second inequality of $(*)$.
 (5) Show that if $f : [a, b] \rightarrow \mathbb{R}$ is monotone, then there exists $c \in [a, b]$ such that

$$\int_a^b f(x)g(x)dx = f(a+) \int_a^c g(x)dx + f(b-) \int_c^b g(x)dx.$$

- 43.** Suppose that $F(x)$ is differentiable on $[a, b]$ and $F'(x) = f(x)$. Show that $f(x)$ is integrable if and only if there exists an integrable function $g(x)$ such that for all $x \in [a, b]$ it holds that

$$F(x) = F(a) + \int_a^x g(t)dt.$$

- 44.** Let f be a decreasing nonnegative continuous function on $(0, +\infty)$. For any integer $n \geq 1$, we define

$$u_n = \sum_{k=1}^n f(k) \text{ and } I_n = \int_1^n f(x)dx.$$

- (1) Show that

$$\int_n^{n+1} f(x)dx \leq f(n) \leq \int_{n-1}^n f(x)dx, \quad n \geq 2.$$

- (2) Show that there exist $\alpha, \beta > 0$ such that $I_{n+1} - \alpha \leq u_n - \beta \leq I_n$ for all $n \geq 2$.
- (3) Deduce that $\{u_n\}_{n=1}^\infty$ and $\{I_n\}_{n=1}^\infty$ either converge or diverge at the same time.
- 45.** Let $(x_n)_{n \geq 1} \subset [a, b]$ be a sequence converging to x_0 . Let f, g be two bounded functions on $[a, b]$ that differ only at x_n for $n \geq 0$. Show that if one of them is integrable, so the other does. In this case, we have

$$\int_a^b f(x)dx = \int_a^b g(x)dx.$$

- 46.** Let $T > 0$ and f be a continuous function on $\mathbb{R}$. Show that the following assertions are equivalent:
- (1) f is T -periodic, i.e. $f(x+T) = f(x)$ for any $x \in \mathbb{R}$;
- (2) the function $F(x) = \int_x^{x+T} f(t)dt$ is constant on $\mathbb{R}$.
- 47.** Let (f_n) be a sequence in $C([a, b])$ converging uniformly to f on $[a, b]$. Let (c_n) be a sequence in $[a, b]$ converging uniformly to c . Define

$$F_n(x) = \int_{c_n}^x f_n(t)dt \quad \text{and} \quad F(x) = \int_c^x f(t)dt, \quad x \in [a, b].$$

- (1) Show that $\lim_{n \rightarrow \infty} \int_c^{c_n} (f_n(t) - f(t))dt = 0$.
- (2) Show that (F_n) uniformly converges to F on $[a, b]$.
- 48.** Let $\{u_i\}_{i=1}^N \subset [0, 1]$ be a sequence. For any $(a, b) \subset [0, 1]$, let $N(a, b)$ be the number of u_i 's in (a, b) . Show that the following statements are equivalent:
- (a) For any $(a, b) \subset [0, 1]$, it holds that $\lim_{N \rightarrow \infty} \frac{N(a, b)}{N} = b - a$.
- (b) For every continuous function $f : [0, 1] \rightarrow \mathbb{R}$, it holds that

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{k=1}^N f(u_k) = \int_0^1 f(x)dx.$$